Algorithmic Trading

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Notation List

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset relation / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function from X to Y
$\Longrightarrow,\ \Longleftrightarrow$	implies / if and only if
$:=$	defined to be equal to
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\mathbb{F}$	field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$
a, b, c, λ	scalars
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0}$	vectors and the zero vector
$\mathcal{U}, \mathcal{V}, \mathcal{W}$	vector spaces or subspaces, depending on context
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$	finite ordered list, often a basis
$\operatorname{span}(E),\ \dim(\mathcal{V})$	span of E / dimension of $\mathcal{V}$
$\mathcal{U} + \mathcal{W},\ \mathcal{U} \oplus \mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$T : \mathcal{V} \to \mathcal{W},\ \mathcal{L}(\mathcal{V}, \mathcal{W})$	linear map / space of linear maps
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$[\mathbf{v}]_\beta,\ [T]_{\gamma \leftarrow \beta}$	coordinate vector / matrix representation
$A, B, C,\ \operatorname{Mat}(m, n, \mathbb{F})$	matrices / $m \times n$ matrices over $\mathbb{F}$
$A^{-1},\ A^T,\ A^*,\ \operatorname{tr}(A), \det(A)$	standard matrix operations

Declared overrides

Symbols used in a sense other than the guide's house register, with the scope of each override.

σ^2	variance of one period's value shock, not annualised; Grossman–Miller entries
σ	absolute volatility of an arithmetic Brownian midprice, in price units; Avellaneda–Stoikov
$p^{\pm},\ P^{\pm}(\delta)$	arrival and fill probabilities, not prices; static limit-order depth
$\lambda^a,\ \lambda^b$	Poisson execution intensities, not price-impact slopes; Avellaneda–Stoikov
$\phi,\ A,\ b,\ \kappa,\ \psi;\ \nu,\ \mu$	running and terminal inventory penalties, permanent- and temporary-impact coefficients, bid–ask spread; trading rate and others' net order flow, per unit time; stochastic-control execution
$\lambda_{\text{jump}},\ \lambda_{\text{invest}}$	Poisson arrival rates of value jumps and of investors, not price-impact slopes; latency-arbitrage arms race
$p,\ \alpha$	probability of the high value, not a price; fraction of informed traders; Glosten–Milgrom half-spreads and market collapse
$A,\ C;\ Q_t,\ q_t$	adverse-selection and transaction-cost spread components (full-spread amounts), not matrices; trade sign, +1 for a buy, not a quantity; spread components, spread measures, Roll's estimator and component identification

Preface

This compendium follows the sections of the institutional quantitative finance learning guide, pinned in `guide/`: one chapter per guide section, holding typed, labelled entries that cite their sources at each claim. Notes written before these notes followed the guide's study method are compiled in the final part, Notes awaiting migration, until each is rewritten as entries in its section's chapter.

This content builds upon the foundational works of Rishi K. Narang [s7-narang], Raja Velu [s3-velu], and Marcos López de Prado [alpha-afml], among others, whose insights form the backbone of our discussion.

Chapter 1

Research engineering and validation

Pathways drawn on: `model-governance-controls` (four draft entries from the migration: operational controls for automated trading, from Harris's 2002 account of reliability, audit trails and network failures, with the U.S. market-access rule, Rule 15c3-5, checked in its current text; and the sources of strategy risk and their monitoring, from Narang's 2024 chapter on risks inherent to quant strategies; and the research and production chain of a strategy, from López de Prado's 2018 account of its stations and stages; none of the pathway's readings yet studied, and no EU or U.K. rule checked). The first entry came from a legacy execution file mapped to this pathway, the other two from Narang's legacy risk-assessment file, whose contagion block went to `funding-crowding`, and the last from the production-chain block of Narang's legacy overview file, which follows López de Prado. `backtest-inference` (four draft entries from the migration: hypothesis-first research, backtest evaluation diagnostics, parameter overfitting, and cost and short-availability assumptions, from Narang's chapter on research in the 2024 edition, with its heuristics qualified; none of the pathway's readings yet studied, so multiple testing and Sharpe-ratio uncertainty are only in its backlog). Those entries came from a legacy file mapped to the section overview, whose outcome, replaying a research decision from versioned data and control records, they do not address. `point-in-time-data` (three draft data entries from the migration: the reference data a research platform must hold, the levels of market data, and derived daily, intraday and binned statistics, from Velu, Hardy and Nehren's 2020 taxonomy of trading data, which the legacy notes followed without naming it; none of the pathway's readings yet studied). They came from a legacy ML data-taxonomy file, routed here outside its map candidates because the pathway's outcome, a panel whose every field was available at decision time, is what reference and market data serve. No older notes remain mapped to this section's pathways.

1.1 Strategy research and backtest evaluation

Method 1.1 Hypothesis-first strategy research: observation, theory, deduced consequences and attempted falsification, with Narang's four sources of ideas

Statement. Narang describes quant research as following the scientific method [s7-narang, chap. "Research", sec. 9.1, pp. 161–163]:

1. observe something in the market that might be explainable, a pattern;
2. form a theory that explains it;
3. deduce the theory's consequences, for example that a trend should give a better-than-random forecast from past returns alone;
4. test the theory by looking for evidence against those consequences (Popper's falsification), not for proof. A theory not yet disproved is accepted only for the moment. At best the researcher gains enough confidence that the tests were rigorous to "warrant risking some capital on the validity of this theory" (p. 163).

Because markets change, through regulation, investor psychology and competition for alpha, quants must keep researching rather than treat a validated theory as permanent (p. 163).

Narang's four common sources of ideas [s7-narang, sec. 9.2, pp. 163–165]:

- *observing the markets*, the source closest to the scientific method; his example is Donchian's two-week breakout rule for futures;
- *academic literature*: many quant firms spend significant time on journals, working papers and conference presentations; literature from astronomy, physics or psychology is also used;

- *migration* of researchers or portfolio managers between firms, despite noncompete and nondisclosure agreements. Investors who have seen inside one firm can also carry ideas to another;
- *lessons from discretionary traders*, such as "ride winners and cut losers" formalised as a stop-loss policy.

For data-driven (machine-learning) alphas he reports, as paraphrases of practitioners, that feature selection and engineering draw on the same sources (p. 165).

Conditions. A practitioner's account of how ideas arise and are screened, in a 2024 book, not a statistical protocol. It does not say how many hypotheses were tried, how significance is adjusted for them, or what evidence suffices to risk capital: those are the subject of this pathway's readings (see the backlog of `backtest-inference`).

Practice. Interpretation: stating the theory and its deduced consequences before testing fixes what would count as evidence against it. That is the step that separates a test from a search. Narang's out-of-sample discussion makes the same point from the other side: refitting after looking at out-of-sample data, or choosing models already known to have survived a period, "burns" that data (sec. 9.3.4, pp. 183–185; parameter overfitting).

Pitfalls. The legacy block reduced the method to three steps (observe and theorise, seek information, test and risk capital). It dropped the deduction of consequences and the aim of falsification, which Narang treats as the core (p. 162). Of discretionary traders' rules, Narang notes that systematic versions of chart patterns have mostly failed. That may be because the theory is invalid or because human practice is less rule-based than it seems, and "not all successful traders have skill" (p. 165). An idea's source is not evidence for it. Ideas that move with staff raise contractual and legal questions; Narang describes a dispute over noncompete agreements, which was not checked further.

See also. backtest evaluation diagnostics (the tests that follow); parameter overfitting (what testing guards against); cost and short-availability assumptions.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/` under the pathway key `research-engineering-validation`.

Index. scientific method; falsification; hypothesis-driven research; theory before data; idea generation; sources of alpha; academic literature; researcher migration; noncompete; discretionary traders; stop-loss; trend following; Donchian; feature engineering

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Research", pp. 161–165 and 183–185; the catalogue does not establish which edition it means). Stability: draft. Currency: practitioner account as of 2024.

Method 1.2 Backtest evaluation diagnostics: the profit curve, return, variability and lumpiness, drawdowns, predictive power, win rates, risk-adjusted ratios, fit with other strategies, and time decay

Statement. Narang lists statistics quants use to judge a model, in sample and out of sample, illustrated on his own daily S&P 500 strategy. Its signal is an adjustment to the equity risk premium, the S&P 500's earnings yield less the 10-year Treasury note's yield, computed each day: a higher stock yield is read as a signal to be long, a lower one as a signal to be short. It ran on daily closes from June 1982 to December 2000, was built in the mid-1990s and was never traded [s7-narang, chap. "Research", sec. 9.3.2, p. 168]. It does not trade every day: it gives a signal only when the model judges the opportunity attractive enough to trade at all, and its signal is zero on 65 percent of days (sec. 9.3.2.6, p. 173). Narang does not state the adjustment or the threshold for a non-zero signal. Each diagnostic, with his example's value:

- *cumulative profit graph*: shows whether, how smoothly and with what downside the strategy made money. His example is "lumpy", with years of inactivity (late 1989 to early 1995), sharp losses and steep gains, which he reads as "real problems" (sec. 9.3.2.1, p. 168);
- *average rate of return*: 746 percent cumulative, 12.1 percent a year, before costs and fees (sec. 9.3.2.2, p. 168);
- *variability of returns*: "in general, the less the variability for a given level of returns, the better" (sec. 9.3.2.3, p. 169); 21.2 percent annualised. His firm also uses *lumpiness*, the share of total return earned in periods significantly above average, as a measure of consistency, while noting that consistency is not always a primary goal;
- *worst peak-to-valley drawdowns* and recovery times: lower drawdowns are better and long recoveries are disliked; several drawdowns are measured, not only the worst. His example's worst is −39.7 percent, short before the October 1987 crash (sec. 9.3.2.4, pp. 169–170);
- *predictive power*: the R^2 of the target on the signal, 0.0085 for his example in Exhibit 9.2 (sec. 9.3.2.5, p. 171). Returns are also bucketed by quantiles of the forecast: returns rising monotonically across buckets indicate reliable predictive power, and non-monotonic buckets may mean the strategy works by accident (p. 172);

- *percentage of winning trades or periods*, a measure of consistency. His example has a zero signal 65 percent of the time, wins 19 percent and loses 16 percent, so about 54 percent of non-zero days win (sec. 9.3.2.6, pp. 172–173);

- *ratios of return to risk*, as Narang defines them (sec. 9.3.2.7, pp. 173–174): the Sharpe ratio, average periodic return above the risk-free rate divided by its variability; the information ratio, the same with "any arbitrary benchmark return—most often zero" in place of the risk-free rate (0.57 in the example); the Sterling ratio, average return over the variability of below-average returns (0.87); the Calmar ratio, "average return divided by the worst peak-to-valley drawdown" (0.31); and the Omega ratio, "the sum of all positive returns divided by the sum of all negative returns" (1.26). Taken literally, both of Narang's quotients have a negative denominator, but he reports both ratios as positive. Interpretation: the explicit conventions are

$$\text{Calmar} = \frac{\bar{R}}{|\text{MDD}|}, \qquad \Omega(0) = \frac{\sum_t \max(R_t, 0)}{\left|\sum_t \min(R_t, 0)\right|},$$

  with $\bar{R}$ the average (annualised) return, MDD the worst peak-to-valley drawdown, a negative number, and R_t the periodic returns. With his example's $\bar{R} = 12.1$ percent and MDD $= -39.7$ percent, the first gives $12.1/39.7 \approx 0.3048$, which rounds to 0.30, while the literal quotient gives -0.30. Narang reports a Calmar ratio of 0.31 (pp. 173–174); the recomputation from his printed inputs does not reproduce that second digit, and the passage does not explain the difference. For returns of $+2$, $+1$, -1.5 and -0.5 percent, the second gives $3/2 = 1.5$, while the literal sum quotient gives -1.5. The Sterling and Omega values cannot be recomputed without the strategy's returns;

- *relationship with other strategies*: a correlation with the existing book, or better, the existing strategy's results with and without the new idea (sec. 9.3.2.8, p. 174);

- *time decay*: the strategy's returns when trades start one, two, three or more days after the signal. This shows its sensitivity to timely information and, Narang says, "some information about how crowded the strategy is", since crowding would mean faster moves to a new equilibrium (sec. 9.3.2.9, pp. 174–176).

Conditions. One author's list and one illustrative strategy, not a validated scorecard. Narang says "a significant amount of discretion is used" in judging models (p. 176). None of the diagnostics is adjusted for the number of strategies or parameter sets tried, and none comes with a standard error (see the backlog).

Practice. Narang's time-decay example uses analyst downgrades of Microsoft announced before the open on 28 April 2006. The stock opened about 11.1 percent down and underperformed the S&P 500 by about 11.4 percent that day. A simulation must not assume a fill before the open. Selling at the close on 28 April, 1 May or 2 May made money over two weeks; selling from 3 to 12 May lost (pp. 174–175). Delay can also help: entering his S&P strategy one day late raised the total return from 746 to 870 percent (12.9 percent a year), which he takes as a bad sign, because the signal is early (pp. 175–176). Interpretation: most of his example figures agree within rounding: $12.1/21.2 = 0.57$, $19/35 = 0.54$, and 746 and 870 percent over about 18.6 years compound to about 12.2 and 13.0 percent a year (checked). The Calmar ratio is the exception: $12.1/39.7 \approx 0.3048$ from his printed inputs, against his reported 0.31; the difference is unexplained.

Pitfalls. Interpretation, qualifying the heuristics:

- *Smooth profits.* Narang reads lumpiness as a problem in his example and says consistency is not always the goal. A smooth curve can come from stale or managed valuations rather than skill (relative performance evaluation), and from short-volatility payoffs whose losses the sample did not contain. Narang's own convertible-arbitrage example (1990–1997, then 1998) shows this (p. 170). The legacy rule "if the profit profile is not smooth ... the model may have issues" is his reading of one example, not a test.

- *Lower volatility.* His rule holds return fixed ("for a given level of returns"); the legacy "the less variable ... the better" dropped that condition. His "67 percent of the time within one standard deviation" assumes roughly normal annual returns (about 68 percent for a normal distribution).

- *Drawdowns.* A worst drawdown from a favourable sample understates risk. Narang's convertible-arbitrage example is "sample bias", and his remedy, shuffling the historical returns to get many paths, still draws only from the sample (pp. 170–171). Shuffling single returns also destroys serial dependence, which drives drawdown depth; that point is not in Narang.

- *The R^2 of 0.05.* "A superb R^2 in our industry is 0.05 (out of sample)", with two colleagues' remarks (0.15 too good to be legitimate, "rich off a 0.02"), is a practitioner's judgement with no horizon, asset, sample or design stated (p. 171). An R^2 depends on the horizon, on whether the regression is a time series or a cross-section, and on the target's noise. Treat it as an order of magnitude for one context, not a benchmark; his own example is a daily, single-index time series. The legacy "exceedingly high" paraphrases "superb". Out-of-sample magnitudes are stubbed.

- *Bucket study.* On p. 172 the text gives the two most bearish quintiles average next-day returns of −2.35 and −0.19 percent. In Exhibit 9.3 (checked by page image) those figures label the signal buckets on the horizontal axis, and the bars, the next-day returns, run from about −0.03 to +0.13 percent. The monotonic ordering holds in the figure; the text's return figures do not match it.

- *Ratios.* Narang's information ratio with a zero benchmark is return over volatility, not active return over tracking error, and his Sterling ratio, the variability of below-average returns, is a downside-deviation measure. Other sources define both differently; they were not checked. The legacy "Calmer" is the Calmar ratio.

- *Time decay and crowding.* A fast decay shows that the strategy depends on timely information. It does not identify crowding: in Narang's Microsoft example the move is a public announcement priced at the open, and a signal that is early improves with delay. His crowding inference is an interpretation without evidence in the chapter (stub).

See also. hypothesis-first research (what is being tested); parameter overfitting (sensitivity to parameters, the next diagnostics); cost and short-availability assumptions (what the returns assume); relative performance evaluation (the same statistics applied to managers, and smoothed returns).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/` under the pathway key `research-engineering-validation`.

Index. backtest evaluation; cumulative profit curve; equity curve; average return; volatility; lumpiness; consistency; drawdown; peak-to-valley drawdown; recovery time; resampling; R-squared; predictive power; quantile study; decile buckets; monotonicity; hit rate; winning days; Sharpe ratio; information ratio; Sterling ratio; Calmar ratio; Omega ratio; strategy diversification; marginal contribution; time decay; signal delay; alpha decay; crowding

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Research", pp. 168–176, with Exhibits 9.2 and 9.3 on pp. 171–172 by page image; the example's ratios and growth rates recomputed; the signal's no-trade state (p. 173) and the Calmar and Omega sign conventions corrected 2026-10-15 in the slice-25 follow-up; the recomputed Calmar ratio, 0.3048 against the reported 0.31, distinguished 2026-10-15 in the slice-26 follow-up). Stability: draft. Currency: practitioner account as of 2024; the S&P example covers 1982–2000.

Pitfall 1.3 Parameter overfitting: an isolated optimum, a model that explains its sample too closely, and the balance against underfitting

Statement. *The error.* A parameter set is chosen because it performed best in a sample, although nearby values perform very differently. A model with many parameters or signals explains the past "so closely . . . that it might be useless for predicting the future" [s7-narang, chap. "Research", secs. 9.3.2.10–9.3.3, pp. 176–179].

How it arises. Training searches parameters and keeps the best (p. 167). In Narang's terms, overfitting is "expecting too much from the data you have", treating an unrepresentative sample as the population. An isolated peak, his point D in Exhibit 9.5, is likely a "spurious coincidence in the fitting period" (pp. 177–178). Re-using out-of-sample data, by refitting after seeing it or by building models already known to work in that period, turns it into in-sample data ("burning data", pp. 183–184).

Checks. Vary each parameter slightly. In Narang's P/E example, a cheap threshold of 12 against 11 and an expensive threshold of 50 against 49 should give similar results, and if they do not, he would distrust and use neither (sec. 9.3.2.10, p. 176). Plot outcome against parameter value (Exhibit 9.5, p. 178). Compare the out-of-sample R^2 with the in-sample R^2: a ratio of about half or better "is considered a good thing" (p. 183).

Remedies. Choose a value with a margin of safety on a broad plateau, his point B, over the plateau's highest point near a cliff (C) or a lonely peak (D) (pp. 177–179). Prefer parsimonious models, since models with many parameters or signals carry more risk of overfitting (p. 179). Separate research from strategy selection, or withhold data from researchers, to limit burning (pp. 184–185).

Conditions. In the third edition Narang presents overfitting as one side of a balance. Overfitting is a Type I error and underfitting a Type II error; as easy alpha became scarce from about the early 2010s, and as data and methods improved, narrower and more fitted models became reasonable (pp. 180–182). How much fitting is warranted depends on the competition for alpha and on the strength of the prior relative to the data; he believes the industry may overestimate the value of more fitting, especially beyond the shortest horizons (pp. 182–183). Einstein's "as simple as possible, but not simpler" is his caution against oversimplifying (p. 179).

Practice. Interpretation: smooth performance across nearby parameters is a sign of robustness to the parameter, not evidence that the strategy has an edge. A whole family of rules, each stable in its own parameters, can be the survivor of a search over many families. The adjustment for the number of trials (deflated Sharpe ratio, multiple-testing control) is this pathway's subject and is stubbed. Narang's "flat" criterion is local: the legacy "a good model has a flatter curve with no jumps" overstates it, since his preferred point B lies on a plateau in a curve that has cliffs.

Pitfalls. The legacy block equated parsimony with fewer assumptions and therefore less overfitting. Narang adds the opposite risk, underfitting, and the role of priors, which the legacy omitted. His claim that simpler theories are better because more easily tested is attributed to Popper (p. 179) and was not checked at the original.

See also. backtest evaluation diagnostics (the statistics being optimised); hypothesis-first research (theory before fitting).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/` under the pathway key `research-engineering-validation`.

Index. overfitting; underfitting; Type I error; Type II error; parameter sensitivity; parameter stability; robustness; plateau; parameter surface; parsimony; Occam's razor; in-sample; out-of-sample; out-of-sample R-squared ratio; burning data; look-ahead bias; data snooping

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Research", pp. 166–167 and 176–185, with Exhibit 9.5 on p. 178 by page image). Stability: draft. Currency: practitioner view as of 2024.

Pitfall 1.4 Backtest cost and short-availability assumptions: costs set too high or too low, and shorts the strategy could not have borrowed

Statement. *The error.* A backtest assumes trading costs, or the ability to short, that the strategy would not have had. A backtest has no record of what the strategy would have cost, because it did not trade, so costs, especially market impact, must be assumed [s7-narang, chap. "Research", sec. 9.3.5, p. 185].

How it arises, costs. Narang's example: a model right 55 percent of the time, making $0.01 a share, and wrong 45 percent, losing $0.01, earns $0.10 per 100 shares if trading is free. At an all-in cost of $0.01 a share it breaks even on its winners and loses $0.02 on its losers, −$0.90 per 100 shares (pp. 185–186; recomputed). Overestimating costs makes a quant hold positions longer than is likely optimal; underestimating them makes the portfolio turn over too fast and "bleed" costs (p. 186).

How it arises, shorts. In a U.S. market-neutral or long/short equity backtest, the best short candidates may be on hard-to-borrow lists, stocks the broker generally restricts from shorting because it cannot locate shares to borrow. The live strategy then cannot place those shorts and substitutes inferior ones, so the short book's backtested value may not exist (p. 186).

Checks and remedies. Narang advises erring towards overestimating costs if one must err, though "it is always preferable to get the cost estimation approximately right" (p. 186). He notes that making a model aware of past hard-to-borrow status is not trivial, because such historical data are hard to come by (p. 186). Interpretation: compare the backtest's cost assumptions with pre-trade cost forecasts and with realised costs measured after trading (transaction cost analysis). Test how the result changes as assumed costs vary and when hard-to-borrow names are excluded or charged their fees.

Conditions. Narang's example ignores how often the position turns over and how costs scale with size; it shows only that a thin edge per share can be smaller than the cost. His short-availability point concerns U.S. equities. He says that shorting without locating shares "would be considered a naked short sale, which is illegal in the United States" (p. 186). That is broader than the rule: Regulation SHO bars a broker-dealer from accepting a short-sale order, or selling short for its own account, without a borrow, an arrangement to borrow or reasonable grounds to believe the security can be borrowed, with exceptions such as bona-fide market making, and imposes close-out duties for fails (securities lending for short sales). Its wording was not adopted.

Practice. Interpretation: Narang discusses whether shares can be borrowed, not what borrowing costs. A short backtest also needs the lending fee or rebate, which is high for securities on special (lending fees and rebates), and allowance for recall and buy-in risk (securities lending for short sales). Neither is in the legacy block or in Narang's passage.

Pitfalls. The legacy block said only that hard-to-borrow lists "must be taken into consideration"; it did not note that historical lists are hard to obtain, so the check is often impossible without point-in-time borrow data (stub). "Overestimation . . . hold positions for longer than optimal" and "underestimation . . . high turnover and bleed" match Narang. The legacy dropped his advice on which way to err.

See also. transaction cost analysis (cost forecasts and measurement); transaction costs (components); securities lending for short sales (locate and borrow); lending fees and rebates (borrow cost); backtest evaluation diagnostics.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/` under the pathway key `research-engineering-validation`.

Index. backtest assumptions; transaction cost assumptions; cost overestimation; cost underestimation; turnover; cost bleed; market impact; short availability; hard-to-borrow list; locate; naked short sale; borrow cost; short book; market-neutral backtest

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Research", pp. 185–186; the example recomputed; the legal point compared with

the Regulation SHO text already cited in `mech:securities-lending-short-sale`, not rechecked). Stability: draft. Currency: practitioner account as of 2024; U.S. rules as cited in the linked entry.

1.2 Point-in-time reference and market data

Data 1.5 Trading reference data: universe, symbology, corporate actions, static and exchange data, condition codes, special days, futures and options reference, event calendars, related and composite instruments

Statement. Velu, Hardy and Nehren call reliable reference data "the key foundation of a robust quantitative strategy development" and list what a research platform must hold [s3-velu, chap. 1, sec. 1.4.1, pp. 26–35]:

- the *trading universe*, which changes daily with listings and delistings; knowing when a stock stopped trading avoids survivor bias (p. 26);
- *symbology mapping* (ISIN, SEDOL, RIC, Bloomberg tickers) and *ticker changes*: the mapping "needs to persist as point-in-time data and allow for historical 'as of date' usage, requiring the implementation of a bi-temporal data structure", handling changes backward and forward (pp. 26–27);
- *corporate actions*: dividends (announcement and payment dates), splits, reverse splits, rights offers, mergers, spin-offs, changes in free float or shares outstanding, suspensions; prices and volumes must be adjusted backward for splits, and suspensions leave gaps that affect backtests (pp. 27–28);
- *static data*: country, sector (GICS, ICB, TOPIX's 33 sectors), primary exchange, quotation currency and quote factor, with USD price = local price × FX × quote factor and cent-quoted currencies such as GBp (pp. 28–29);
- *exchange data*: holiday calendars, session hours with auctions, lunch breaks and daylight-saving shifts, disrupted days, tick sizes, trade and quote lots, limit-up and limit-down rules and short-sale restrictions, whose values must be kept historically "in order to run realistic historical backtests" (pp. 29–30);
- *condition codes* on trades and quotes, mapped so that, for example, off-exchange trades reported for regulatory purposes are excluded from accessible liquidity and from daily volume, high and low (p. 31);
- *special-day calendars*: half days, local holidays and exam days, month and quarter ends, index rebalance days with volume skewed to the close, and expiry days (pp. 31–32);
- *futures and options reference data*: expiry calendars, the most liquid contract, roll dates, contract size and quotation, session phases, and option chains (pp. 32–34);
- *event calendars* for macro releases and company events, *related tickers* (composite tickers, dual listings, ADRs and GDRs), *composite assets* with their constituents, cash components and divisors, and *latency tables* between data centres (pp. 34–35).

Conditions. Every table that changes over time must be stored with the dates on which each value applied and was known, so that a backtest sees the universe, identifiers, corporate actions, tick sizes and rules in force at each decision time. Velu, Hardy and Nehren's reasons for special days are practical: a single special day moves a 30- or 60-day ADV, and low-order autoregressive models underperform on and after such days, so modellers "often remove special days and model normal days first" and model special days separately (p. 32). Futures liquidity moves between contracts at the roll (about 1.5 million E-mini contracts a day in the front month against 30,000 in the next before expiry, reversing in the last days, p. 33), so rolling-window statistics must blend contracts across roll dates.

Practice. Velu, Hardy and Nehren's ABC→DEF example (p. 27) shows the three queries a symbology table must answer: a 30-day history as of a date before the change (old ticker), after it (both tickers spliced), and a forward window for a position held across it. Examples they give: European equity volume 30–40 percent lower on U.S. holidays; Indonesia's longer Friday lunch break; U.S. daylight saving starting about two weeks before Europe's (pp. 29–30). These are illustrations as of the book's writing, not current calendars.

Pitfalls. The legacy notes merged Velu, Hardy and Nehren's dividend points, attributing the price jump to the announcement: they separate the drop in price when the dividend is paid from the volatility that may accompany the announcement (p. 27); interpretation, the price drop occurs on the ex-dividend date. The legacy said merger and spin-off data are "hence used in" merger arbitrage, where the source says such situations "can also be exploited" (p. 28), and dropped the requirement to keep exchange-rule histories. The book describes what to collect; it does not test how much results change when it is ignored. That evidence (delisting returns, history rewriting, survivorship) is in the pathway's readings.

See also. market data levels; derived market statistics; financial data types; ETF trick (rolled futures series); index reconstitution impact; securities lending and short sales; CME Globex trading.

Further reading. The pathway's readings on bitemporal history (`s8-bitemporal`), CRSP files, delisting bias and vendor history rewriting are unread (backlog). The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. reference data; trading universe; survivorship bias; symbology; identifiers; ticker changes; bitemporal data; as-of date; corporate actions; split adjustment; dividends; static data; quote factor; holiday calendar; trading sessions; tick size; lot size; limit up limit down; short-sale restrictions; condition codes; special days; index rebalance; expiry; futures roll; option chain; event calendar; ADR; composite assets; latency tables

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `s3-velu`, CRC Press, © 2020 in the copy, chap. 1, pp. 26–35, printed page = PDF page −17). Stability: draft. Currency: examples as of about 2018–2020; exchange rules and calendars change and were not checked against current sources.

Data 1.6 Market data levels: Level I trades and quotes, Level II depth, Level III order messages, and the special order types that complicate them

Statement. Market data are sold "in various levels of granularity (and price!)", from exchanges as direct feeds or from vendors [s3-velu, chap. 1, sec. 1.4.2, p. 35]. *Level I* holds trades and top-of-book quotes, enough to rebuild the best bid and offer; quotes update roughly ten times as often as trades in liquid markets, and trade status and qualifiers (cancelled, late, odd lot, auction, intermarket sweep, average-price, venue) decide whether a print should update the last price and cumulative volume (pp. 35–36). *Level II* adds every lit order-book update at every level and venue; many users keep five or ten levels (p. 36). *Level III*, message data, gives each order an identifier tracked through execution, cancellation or amendment; once messages from all exchanges are "consolidated into one single dataset ordered by timestamp", a full book can be built at any moment (pp. 36–37). Their example of a U.S. Level III record has timestamp (milliseconds after midnight), ticker, order identifier, message type (add buy, add sell, partial execution, partial cancel, full execution, delete, cross, non-displayed execution), shares, price (the last four digits decimals, so "divide by 10000"), participant identifier and market centre (Table 1.10, p. 37).

Conditions. Exchange timestamps are matching-engine times; databases also store a reception time, and the latency between them must be accounted for when opportunities are short-lived (p. 36). Special order types change identifiers and priority in venue-specific ways: price-sliding orders ranked at the locking price as hidden and displayed one tick away; pegged and midpoint-peg orders; reserve orders whose displayed part is replenished with a new timestamp; discretionary orders; intermarket sweep orders (pp. 37–38). Velu, Hardy and Nehren advise weighing the value of raw data against simpler binned data, and note that most practitioners outside high-frequency trading build strategies on binned data (pp. 36 and 39).

Practice. With Level III data one can model inter-arrival times, arrival and cancellation rates by distance from the touch, queue position and imbalance, and then ask about the impact of market orders, a limit order's chance of moving up the queue, the probability of earning the spread and the short-horizon direction of prices (p. 38).

Pitfalls. The legacy notes said to divide the price field by 1000; the table says 10000 (p. 37). They dropped the consolidation condition for a full book, and the reception-time latency. The Table 1.10 layout is the book's illustration; a real feed's specification, which changes, governs.

See also. trading reference data (condition codes); order exposure decisions (hidden and reserve orders); static limit-order depth; information-driven bars.

Further reading. The pathway's TAQ measurement reading (`wide-holden-jacobsen`) is unread (backlog). The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. market data; Level I; Level II; Level III; message data; best bid and offer; depth of book; trade qualifiers; matching-engine timestamp; reception timestamp; order identifiers; pegged orders; reserve orders; intermarket sweep orders

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `s3-velu`, chap. 1, pp. 35–39; Table 1.10 checked by page image). Stability: draft. Currency: U.S. feed examples as of about 2018–2020.

Data 1.7 Derived market statistics: daily activity data, intraday microstructure statistics, pre-computed aggregates and binned data

Statement. Velu, Hardy and Nehren group the statistics most strategies derive from binned or daily data [s3-velu, chap. 1, sec. 1.4.3, pp. 39–42]. *Daily activity*: open, high, low, close and previous close (adjusted for corporate actions), the last trade before the close, volume with its lit and dark split, auction volume and price, VWAP, short interest, days-to-cover and borrow utilisation, futures open interest and basis, index-level data, options open interest, Greeks and implied volatility, and cross-asset rates, CDS spreads and the dollar index (pp. 39–40). *Intraday microstructure*: number and frequency of trades and quote updates, top-of-book size, depth, spread, trade size and ticking time, whose daily distributions can start an algorithm's day and be updated "through online Bayesian updates" (p. 41). *Pre-computed aggregates*: X-day ADV and auction volume, volatility, beta (plain or asymmetric) and correlation matrices (p. 41). *Binned data* discretise the day into bins

of a few seconds to 30 minutes with the same features; volume and spread profiles "are rarely built with a granularity finer than one minute" to avoid noise from frictions (p. 42).

Conditions. Aggregates "need to fully support the peculiarities described in the Reference Data section": special days skew distributions, and asynchronous markets distort beta and correlation (p. 41). Binned data allow discrete time-series methods and reduce dimension (390 one-minute rows per U.S. stock-day against millions of messages), but relationships stable on daily closes become noisier at higher frequency, and illiquid names leave empty bins (p. 42). Interpretation: each derived field must carry the time it became available, for example short interest only after its publication date, and pre-computed X-day aggregates must be computed through the decision date, not the end of the sample.

Practice. Velu, Hardy and Nehren's uses are judgements to test, not results: the high–low range "usually gives a better indication of market sentiment than the simple close-to-close return"; "a large short interest usually indicates a bearish view from institutional investors"; days-to-cover sizes a potential short squeeze; futures basis against one's dividend estimates may show mispricing (pp. 39–40).

Pitfalls. The legacy notes headed the daily-activity list "Binned Data", which in the source is a different passage (p. 42), and dropped the reference-data caveat on aggregates.

See also. Garman–Klass volatility (range-based estimators); spread measures; securities lending fees and rebates (utilisation, hard to borrow); activity-sampled bars; trading reference data.

Further reading. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. daily statistics; OHLC; previous close; VWAP; auction volume; short interest; days to cover; utilisation; open interest; implied volatility; intraday statistics; top-of-book size; ticking time; ADV; beta; correlation matrix; binned data; minute bars; volume profile; asynchronous trading

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `s3-velu`, chap. 1, pp. 39–42). Stability: draft. Currency: practitioner descriptions as of about 2018–2020.

1.3 Operational controls for automated trading

Method 1.8 Operational controls for automated trading: reliability and recovery, unknown order state, audit trails, and the U.S. market-access rule

Statement. Harris describes the operational demands on automated trading systems of markets, "as well as brokers and dealers", as of 2002 [`alpha-harris`, chap. "Floor Versus Automated Trading Systems", sec. 27.7.1, pp. 549–551]. The systems must validate users, accept and process orders, report trades and report order status, reliably, securely and fast:

- *reliability*: trading stops when the computer or network is down, and traders will not use unreliable systems. Builders therefore eliminate single points of failure and, since failures are inevitable, invest in recovering from interruptions: fault-tolerant hardware; redundant computers and network connections through different vendors and physical paths; backup power; a remote "hot disaster recovery site" to which activity can switch at once; significant excess capacity for surges in demand; quality controls in software development; redundant data backups; and thorough testing by independent quality-assurance groups (pp. 549–550, page image of p. 550);
- *security*: encryption and user authentication, so that hackers cannot tamper with systems and impersonators cannot trade in others' names (p. 550);
- *speed*, because traders without the latest information decide badly (pp. 550–551).

An *audit trail* records the submission and disposition of every order, with detailed information on everything that happens to it. Regulators use it to decide whether traders have violated trading rules; floor rules require time stamps when orders are received and filled, sequential recording and immediate reporting; automated systems produce complete audit trails of all activity within them (sec. 27.1.1, pp. 544–545).

When a connection fails, Harris's examples show three distinct risks (box, p. 550, page image). A trader whose accepted limit order may or may not have filled cannot safely trade elsewhere: if she does so without knowing, she risks trading twice. An arbitrageur who has bought one leg cannot sell the other, and is left unhedged. A speculator cannot exit a losing position; a stop order placed immediately after the purchase would have represented him while he could not communicate.

In the United States, Rule 15c3-5 requires a broker or dealer with market access, or that provides others with access to an exchange or alternative trading system, to establish, document and maintain risk-management controls and supervisory procedures [`cfr-240-15c3-5`, 17 CFR 240.15c3-5(b)]. These are reasonably designed, among other things:

- to prevent orders that exceed pre-set credit or capital thresholds ((c)(1)(i));

- to prevent erroneous orders, by rejecting orders that exceed price or size parameters, order by order or over a short period, "or that indicate duplicative orders" ((c)(1)(ii));
- to restrict access to trading systems and technology to pre-approved, authorised persons and accounts ((c)(2)(iii));
- to assure that surveillance personnel receive immediate post-trade execution reports ((c)(2)(iv)).

The controls must be under the firm's direct and exclusive control, with a limited contractual allocation of some regulatory controls to broker-dealer customers ((d)); they must be reviewed regularly, at least annually, with an annual certification by the chief executive officer ((e)).

Conditions. Interpretation, a synthesis of the passages above and not a procedure stated in them: after a failure the states to separate are these.

- *Unknown order state*: an order may be unacknowledged, live, partly filled, filled, cancelled or rejected. Venue messages report these transitions (acknowledgements, rejections, cancellations and fills; automated trading architecture), and the trader's view is stale until they arrive.
- *Unknown fill state and position*: what matters for risk is the quantity filled and the resulting position, including the unhedged leg of a multi-leg trade.
- *Reconciliation*: establishing state from authoritative records, such as venue or broker execution reports and post-trade reports, before acting.
- *Retry*: resubmitting before reconciliation is how a trader trades twice. For U.S. broker-dealers with market access, rejecting orders that indicate duplicative orders is a required pre-trade control, but such a filter is a backstop, not a reconciliation.
- *Cancellation*: a cancel sent while state is unknown may cross with a fill, so its outcome also needs confirmation.
- *Recovery*: redundancy, backups and disaster-recovery sites restore service; they do not by themselves establish what happened to orders in flight.

Practice. Interpretation: the legacy advice to prefer safe state reconciliation over blind resubmission follows from Harris's "trading twice" and is consistent with the duplicative-order control of Rule 15c3-5, which states the control, not the reconciliation procedure. Harris's stop-order example is a way to leave a resting instruction that protects a position while disconnected; it is not a substitute for knowing the order's state. The pathway's checkpoint maps such controls to rules and to the Knight Capital failure points; the Knight Capital order (`wide-knight-capital`) is unread.

Pitfalls. Keep three kinds of statement apart.

- *Description*: Harris's lists are descriptions of practice in 2002 markets, not requirements.
- *Law*: Rule 15c3-5 is U.S. law for broker-dealers with market access, checked in its current eCFR text. It does not mention kill switches, reconciliation or "drop copies"; the immediate post-trade execution reports it requires are the nearest provision.
- *Not checked*: the EU's algorithmic-trading rules (`wide-mifid-rts6`) could not be retrieved, because EUR-Lex answered with a bot challenge, so no EU obligation is stated here. U.K. supervisory statements, industry guidance (`w2-fia-automated-controls`), Regulation SCI and audit-trail rules such as the U.S. consolidated audit trail were not checked.

Unsupported legacy items, kept only in the artefact:

- drop copies, replayable feeds, tamper-resistant logs and validated orders among reliability measures (Harris names validating users);
- "bounded delays" for each stage, which Harris states only as a need for speed;
- the claim that reliability is "a market-quality variable, not only an IT variable" (Harris says only that traders will not use unreliable systems and that trading stops when systems are down);
- in the audit trail, allocation, linking private messages to public market state, and testing best execution, front running, allocation and unauthorised trading. Harris names front running, inappropriate order exposure, fraudulent trade assignment and prearranged trading as the floor scandals that audit trails help deter.

See also. automated trading architecture, for venue messages, recorded messages and model accuracy checks, and floor against automated trading systems, owned by the section overviews `execution-trading-costs` and `execution-market-making`; best execution.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/model-governance-controls/`. The adopting release of Rule 15c3-5 is `wide-sec-15c3-5`, not read.

Index. operational resilience; reliability; single point of failure; redundancy; disaster recovery site; excess capacity; software quality assurance; security; authentication; audit trail; unknown order state; unknown fill state; reconciliation; resubmission; duplicate orders; cancellation; recovery; legging risk; market access rule; Rule 15c3-5; pre-trade controls; post-trade execution reports

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, chap. 27, pp. 543–554, page images of pp. 550 and 552; and the eCFR text of 17 CFR 240.15c3-5 as of 13 October 2026). Stability: draft. Currency: Harris's practice as of 2002; Rule 15c3-5 current as of 13 October 2026; no EU or U.K. rule checked.

1.4 Sources of strategy risk and their monitoring

Definition 1.9 Sources of risk in a quant strategy: model risk (inapplicability, misspecification, implementation errors), regime change and exogenous shocks, kept apart from funding contagion and operational failure

Statement. Narang names four risks that are specific to quant strategies, or larger for them than for discretionary ones [s7-narang, chap. "Risks Inherent to Quant Strategies", secs. 10.1–10.4 and 10.6, pp. 191–211]. The first three are:

- *model risk*: "the risk that the strategy does not accurately describe, match, or predict the real-world phenomenon it is attempting to exploit". It can arise in any part of the strategy, from back-testing software and data-feed handlers to the alpha, risk, transaction-cost and portfolio-construction models and the execution algorithms. Small errors may build slowly and then "explode on a busy trading day" (sec. 10.1, pp. 191–192). Its three common sources:
  - *inapplicability of modelling*, in two forms: using quantitative modelling on a question it cannot answer, and misapplying an otherwise valid technique. His example of the second is VaR built on correlations and historical volatility, which assumes normal distributions and linear relationships that market data often lack (sec. 10.1.1, pp. 192–193);
  - *misspecification*: a model that badly describes the world. The common case is a model that works most of the time and fails in rare events. His example is quants who in August 2007 found they had modelled the liquidity risk of large-cap U.S. stocks from their own holdings alone, when the aggregate size of similar holdings being liquidated mattered more (sec. 10.1.2, pp. 193–195);
  - *implementation errors*: errors in programming or in the architecture of systems, "perhaps [the] most common" and usually the least painful (sec. 10.1.3, pp. 195–196). His examples are a reversed sign in execution software, Knight's August 2012 loss and AXA Rosenberg's risk-model coding error, which he reports without sources, and an unnamed firm whose execution server restarted before its alpha server and, seeing no signals, sold 80 percent of the portfolio, which then had to be bought back;
- *regime change*: models rest on relationships in historical data, and when a structural shift changes an instrument's behaviour or the relationships between instruments "dramatically and quickly", the strategy typically suffers (sec. 10.2, pp. 196–200). His examples are the Schwab and Merrill Lynch relationship from 1996 to 2007, and the value-against-growth spread in 2007–2008, whose reversals, he says, came with reversals in sectors, currencies, bonds and commodities, so that regime changes "tend to occur across many levels simultaneously" (p. 199);
- *exogenous shocks*: big moves driven by information outside the market, which models built on market data cannot explain. His examples are 11 September 2001, the start of the Iraq war in 2003, the Bear Stearns rescue in March 2008 and the SEC's 2008 change to short-sale rules for financial stocks. He calls the risk difficult to do anything about "other than with discretionary overrides" (sec. 10.3, pp. 200–202).

The fourth, contagion or common-investor risk, is in common-investor contagion.

Conditions. One practitioner's classification (third edition, 2024), whose examples run from 1996 to 2012. Narang concedes that whether a regulatory intervention or a rumour-driven collapse is endogenous or exogenous to the market is arguable (p. 201). His company figures (Knight, AXA Rosenberg) are not sourced in the chapter and were not checked.

Practice. Interpretation: ask which of five causes is at work, because each points to a different control. The categories overlap: a regime change can expose a misspecification, forced selling by others can look like a

regime change, and one loss can have several causes, so the list is a checklist of questions, not a set of exclusive labels.

- A *model error* (inapplicability or misspecification) is found by validation and by monitoring for unexpected behaviour (strategy monitoring).
- A *regime change* is a change in the world after the model was built. It is not overfitting, in which the model was fitted to an unrepresentative sample (parameter overfitting), though the two can look alike in live results.
- An *exogenous shock* is information the model never sees.
- *Funding contagion* is loss from other holders' forced selling (common-investor contagion). Narang's own value-against-growth example mixes it with regime change: he attributes the 2007 reversal partly to quant unwinds (p. 198).
- An *operational failure*, such as software, architecture or restart order, is what Narang files under implementation errors. Testing, change control and the controls in operational controls for automated trading make such failures less likely and limit their cost; they mitigate it, they do not guarantee its prevention, and better modelling does not address it.

Pitfalls. *Discretionary overrides.* The legacy "may require discretionary overrides" repeats Narang's remedy for exogenous shocks. Interpretation: two kinds of intervention need keeping apart. An *ad hoc* intervention, improvised outside the validated design, changes the strategy's behaviour in a way that was never tested, and is in effect an unvalidated model change. An *approved contingency response*, such as a pre-specified de-risking rule, a position limit or a kill switch whose triggers, authority and effects were tested and approved before use, is part of the validated design, not a departure from it. Not every override is the first kind. Narang's own account of August 2007 says that the main discretion many quants used, unwinding when models misbehave, spread the losses among managers with overlapping positions (p. 208). His story of one firm that, by delay, held its positions and recovered is a single anecdote, not evidence that holding is better (p. 208). Who may override, on what evidence, with what record and review is a governance question for this pathway's readings (stub). The legacy's "models loaded in a wrong sequence" compresses the server-restart example, in which servers, not models, came back in the wrong order.

See also. common-investor contagion (the fourth risk, owned by `funding-crowding`); strategy monitoring (how the risks are watched); operational controls for automated trading (operational failure); parameter overfitting (contrast with regime change); arbitrage risks (Harris's model risk for arbitrageurs, a special case).

Further reading. The legacy blocks it replaces, verbatim, are in `risk-assessment.tex` in the folder `notes/artefacts/` under the pathway key `model-governance-controls`.

Index. model risk; inapplicability of modelling; misspecification; implementation error; coding error; architecture error; VaR assumptions; fat tails; nonlinear dependence; regime change; structural break; relationship breakdown; exogenous shock; event risk; discretionary override; Knight Capital; AXA Rosenberg

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Risks Inherent to Quant Strategies", pp. 191–202 and 208–211; the company and market examples not checked against other sources; the treatment of overrides, operational controls and overlapping categories corrected 2026-10-15 in the slice-26 follow-up). Stability: draft. Currency: practitioner account as of 2024; examples 1996–2012.

Method 1.10 Monitoring a running quant strategy: exposure, profit and loss, execution and systems monitors

Statement. Narang describes four kinds of monitoring tool that quants use alongside their risk models [s7-narang, chap. "Risks Inherent to Quant Strategies", sec. 10.5, pp. 209–210]:

- *exposure monitors*: start from current positions and group them by whatever exposures concern the manager. Examples are asset classes; valuation, momentum or volatility; or, for equities, gross and net exposure by sector and industry, capitalisation bucket and style factor, with proprietary or vendor tools. VaR-based tools measure overall gross risk "at least according to that measure". Exposures out of line with limits or expectations can be an early warning of a model problem or difficult markets;
- *profit and loss monitors*: compare current prices with the previous close and chart intraday performance, so that unexpected gains or losses prompt a check. Others look at *how* money is made: whether the strategy is holding losers longer or selling winners sooner than usual, and its *hit rate*, the share of positions on which it makes money, against what its designer expects;
- *execution monitors*: orders being worked and recently completed, with sizes and prices; fill rates of limit orders, particularly for passive execution; and, at some managers, slippage and market impact against expectations;

- *systems performance monitors*: software and infrastructure errors, CPU performance, the speed of each automated stage and the latency of messages to and from exchanges. Narang calls this "perhaps the most important for sniffing out systems errors and some types of model risk" (p. 210).

Conditions. A description of tools, not a standard or a set of thresholds: "the art is in how to use them" (p. 209). The expected values against which deviations are judged, such as normal exposures, hit rates and slippage, come from the strategy's design and its backtest (backtest evaluation diagnostics).

Practice. Narang reports one manager who, in August 2007, noticed intraday profit and loss falling with nearly every tick, researched it and cut risk earlier than most (p. 210). Interpretation: each monitor watches a different source of risk (sources of strategy risk). Exposures and the pattern of profit and loss point to model error and regime change; systems monitors point to implementation and operational failure; execution monitors point to costs. A monitor only detects. The response, whether a limit, a kill switch or an approved override, belongs to the controls this pathway designs (operational controls for automated trading).

Pitfalls. Execution monitoring here is the manager's view across a strategy. The indicators for supervising many execution algorithms in flight are in decision support for trading algorithms and are not repeated here. The legacy block matches Narang's four monitors; it omitted the VaR measure, the "how money is made" monitors and the early-warning purpose.

See also. sources of strategy risk (what is monitored); decision support for trading algorithms (execution indicators); operational controls for automated trading (the controls that act on alerts); backtest evaluation diagnostics (expected hit rates and exposures).

Further reading. The legacy blocks it replaces, verbatim, are in `risk-assessment.tex` in the folder `notes/artefacts/` under the pathway key `model-governance-controls`.

Index. risk monitoring; exposure monitoring; gross and net exposure; factor exposure; VaR; profit and loss monitoring; intraday P&L; hit rate; winners and losers; execution monitoring; fill rate; slippage; market impact; systems monitoring; latency monitoring; early warning

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Risks Inherent to Quant Strategies", sec. 10.5, pp. 209–210). Stability: draft. Currency: practitioner account as of 2024.

1.5 The research and production chain of a strategy

Method 1.11 Strategy production lifecycle: López de Prado's research stations and the embargo, paper-trading, graduation, re-allocation and decommission stages

Statement. López de Prado argues that a firm, not a lone researcher, should produce machine-learning strategies, under what he calls the meta-strategy paradigm: "It takes almost as much effort to produce one true investment strategy as to produce a hundred", because of data curation, high-performance computing, software, feature analysis, execution simulators and backtesting [`alpha-afml`, chap. 1, sec. 1.2.2, p. 5]. He describes the stations of the production chain (sec. 1.3.1, pp. 6–8):

- *data curators* collect, clean, index, store, adjust and deliver all data, as experts in market microstructure and data protocols such as FIX;
- *feature analysts* turn raw data into informative signals and catalogue findings, but do not develop strategies;
- *strategists* turn features into strategies, each strategy being "the experiment designed to test the validity" of a theory that must identify "the economic mechanism that causes an agent to lose money to us"; "features may be discovered by a black box, but the strategy is developed in a white box";
- *backtesters* assess the strategy under several scenarios and evaluate "the probability of backtest overfitting by taking into account the number of trials"; their results go to management and are not shared;
- the *deployment team* integrates the code into production, keeps it "logically identical to the prototype" and optimises it for latency;
- *portfolio oversight*, which follows a deployed strategy through five stages (pp. 8–9):
  1. *embargo*: running on data after the backtest's end date; promotion if performance is consistent with the backtest;
  2. *paper trading* on a live feed, capturing data, calculation and execution delays, "for as long as it is needed to gather enough evidence that the strategy performs as expected";
  3. *graduation*: managing a real position, alone or in an ensemble, with risk, returns and costs attributed;

4. *re-allocation*: allocations reassessed "frequently and automatically" in a diversified portfolio, "in general" following a concave function, small at graduation, rising while the strategy performs and shrinking as performance decays;
5. *decommission*, when performance stays below expectations long enough "to conclude that the supporting theory is no longer backed by empirical evidence".

He prefers releasing new variations to run in parallel with old ones (p. 9).

Conditions. This is one author's prescription, presented as what "every successful quantitative firm I am aware of" does (p. 5), not a regulatory standard or a tested result. None of the stages carries a quantitative criterion: how much consistency promotes a strategy from embargo, how long paper trading lasts, the shape of the concave allocation function and the length of underperformance that triggers decommission are all left to the firm. His view that "the only true alpha left is microscopic" and needs "heavy ML tools", while the chances of an individual finding macroscopic alpha with simple tools such as econometrics are "quickly converging to zero" (p. 6), is an opinion without data in the passage.

Practice. Interpretation: the stages map onto this pathway's controls. Embargo and paper trading come after deployment in López de Prado's sequence (pp. 8–9), so they are validation before live-capital trading, that is before graduation, not before production; graduation, re-allocation and decommission are change and approval decisions that need an owner, evidence and a record (strategy monitoring supplies the evidence; operational controls the pre-trade and kill-switch layer). The backtester's count of trials is the subject of parameter overfitting and backtest diagnostics. An embargo period is one further sample, of whatever length the backtest left out; a promotion decided on it is another trial, and its strength depends on that length, which the source does not set.

Pitfalls. The legacy block followed López de Prado closely but dropped his decommission condition, the loss of empirical support for the theory, and his "in general" on the concave allocation. Its stations and stages are proposals for organising work, not evidence that the organisation works. The legacy alpha-models file's machine-learning item, on the cost of producing strategies and macroscopic against microscopic alpha, is the same author's sec. 1.2.2 and 1.3.1 (pp. 5–6), not Narang's; it is represented here.

See also. hypothesis-driven strategy research (theory before testing, in Narang's version); components of a quant strategy; sources of strategy risk (implementation errors in deployment).

Further reading. The legacy blocks, verbatim, are in `systematic-investing-overview.tex` and `alpha-models.tex` under the section key `portfolio-construction-risk` in `notes/artefacts/`.

Index. meta-strategy paradigm; production chain; data curators; feature analysts; strategists; backtesters; deployment team; portfolio oversight; embargo; paper trading; graduation; re-allocation; decommission; strategy lifecycle; microscopic alpha; macroscopic alpha

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-afml`, Wiley 2018, ISBN 9781119482086, chap. 1, secs. 1.2.2 and 1.3.1, pp. 5–9; printed page = PDF page −27). Stability: draft. Currency: one practitioner's prescription as of 2018.

Chapter 2

ML and alternative data

Pathways drawn on, in the guide's order: `statistical-learning-foundations` (eleven draft entries from the migration, from López de Prado's chapters on financial data structures, labelling, sample weights and fractional differentiation, a required reading of the pathway read in those chapters only: activity-sampled and information-driven bars, the ETF trick and futures roll, event-based sampling with the CUSUM filter, triple-barrier labelling, meta-labelling, label concurrency and uniqueness, the sequential bootstrap, sample weights, and fractional differencing with its fixed-width window and minimum order; the source's own errors are marked and the stationarity claims qualified; none of the pathway's readings yet studied); `financial-ml-evaluation` (one draft entry, on the time-validity of labels, sample weights and fitted transforms, drawn from the same chapters; none of the pathway's readings yet read); `reinforcement-learning-trading` (one draft entry from the migration: optimal execution with order flow as stochastic control, the benchmark against which the pathway compares learned policies, from Lehalle and Laruelle's account of Cartea and Jaimungal, with the contrast between control and reinforcement learning from the survey of Hambly, Xu and Yang, a required reading checked in its arXiv version; none of the pathway's readings yet studied); and `alternative-data-sources` (one draft entry: López de Prado's four types of financial data, with Velu, Hardy and Nehren's fundamental and analyst data and their point on data uniqueness; none of the pathway's readings yet read). The entries came from the legacy AFML data chapter, whose reference- and market-data definitions went to `point-in-time-data` in Chapter 1 and whose PCA allocation went to `portfolio-construction`, and from a legacy execution file. No legacy notes for this section remain.

2.1 Financial data structures: bars, multi-product series and event sampling

Method 2.1 Activity-sampled bars: time, tick, volume and dollar bars, what each corrects, and what the evidence on their returns shows

Statement. *Bars* are the rows of a table into which unstructured trade data are regularised; standard bar methods "transform a series of observations that arrive at irregular frequency (often referred to as 'inhomogeneous series') into a homogeneous series derived from regular sampling" [`alpha-afml`, chap. 2, sec. 2.3.1, p. 26]. Each bar records a timestamp, VWAP, open, close, high, low and volume. *Time bars* sample at fixed clock intervals; *tick bars* each time a fixed number of transactions has taken place (for example 1,000); *volume bars* each time a fixed number of the security's units has traded; *dollar bars* each time a fixed market value has traded, in the security's currency (secs. 2.3.1.1–2.3.1.4, pp. 26–28).

Conditions. López de Prado's case against time bars is that markets do not process information at a constant rate, so time bars "oversample information during low-activity periods and undersample information during high-activity periods", and that time-sampled series "often exhibit poor statistical properties, like serial correlation, heteroscedasticity, and non-normality of returns" (p. 26, citing Easley, López de Prado and O'Hara 2012). Tick bars synchronise sampling with "a proxy of information arrival" (p. 27), but auction prints and order fragmentation distort the tick count: an opening or closing auction is published as one trade "for an outsized amount" that "could be the equivalent of thousands of ticks", and one buy of 10 lots against ten resting orders of 1 lot prints as ten ticks; matching engines may also "split one fill into multiple artificial partial fills" (p. 27). Volume bars avoid the fragmentation problem. Dollar bars keep a constant traded value when prices move a lot: after a 100 percent rise, $1,000 buys half the shares (p. 28). After split adjustment, issuance and buybacks still change tick and volume counts, and dollar bars "tend to be robust" to those actions; the bar size can be set dynamically from free-float capitalisation or outstanding debt (p. 28).

Practice. The evidence cited is of different kinds and does not settle which clock is best. López de Prado quotes Mandelbrot and Taylor (1967), "Price changes over a fixed number of transactions *may* have a Gaussian

distribution", says that "multiple studies have confirmed" that sampling by trading activity gives "returns closer to IID Normal" (citing Ané and Geman 2000), and reports Clark (1973) as finding volume sampling closer to IID Gaussian than tick sampling (p. 27). Ané and Geman's abstract claims that normality "can be recovered through a stochastic time change" and that "the cumulative number of trades is a better stochastic clock than the volume for generating virtually perfect normality in returns" [ane-geman-2000, abstract], which ranks trades above volume, the reverse of the ranking reported for Clark. Neither statement is about independence: "IID" is López de Prado's gloss. His own support for dollar bars is one series: on E-mini S&P 500 futures, the exponentially weighted number of tick and volume bars per day for a fixed bar size "will vary wildly over the years", and that of dollar bars less (Figure 2.1, p. 28). His exercises ask the reader to compare bar types by weekly bar-count stability, serial correlation, variance of monthly variances and Jarque–Bera statistics (Exercise 2.1, pp. 40–41): these are the tests to run on one's own data.

Pitfalls. Better distributional properties are an empirical, market- and period-specific finding, not a property guaranteed by the sampling rule. The legacy notes said dollar bars are robust to "splits, reverse splits, issuance of new shares, buying back"; López de Prado assumes splits are already adjusted and says only that dollar bars "tend to be robust" to issuance and buybacks (p. 28). The legacy "Sampling as a function of trading activity creates returns closer to IID Normal" attached López de Prado's gloss to Ané and Geman without the qualification above. Interpretation: an activity clock moves the volatility clustering into the bar durations, so a bar series has irregular calendar timing that must be kept for any time-of-day feature, for joins with other data and for the timing of decisions.

See also. information-driven bars; event-based sampling; derived market statistics (Velu's binned data); volatility components; intraday impact profile.

Further reading. The replication evidence on transaction clocks is a backlog item of the pathway. The legacy file and its figure, a reproduction of López de Prado's Figure 2.1, are kept in `notes/artefacts/statistical-learning-foundations/`.

Index. bars; time bars; tick bars; volume bars; dollar bars; inhomogeneous series; subordinated process; transaction clock; volume clock; auction prints; order fragmentation; normality of returns

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-afml`, Wiley 2018, chap. 2, pp. 26–28 and 40–41, printed page = PDF page −27; Figure 2.1 viewed as a page image; the Ané–Geman abstract read on RePEc, the paper itself not read). Stability: draft. Currency: methods timeless; the evidence cited is from 1967–2012.

Method 2.2 Information-driven bars: the tick rule, tick, volume and dollar imbalance bars, and tick, volume and dollar runs bars, with causal thresholds

Statement. Information-driven bars "sample more frequently when new information arrives to the market", information being meant in the microstructure sense of persistent imbalances in signed volume, associated with informed traders [alpha-afml, chap. 2, sec. 2.3.2, p. 29]. For ticks (p_t, v_t) the *tick rule* sets $b_t = b_{t-1}$ if $\Delta p_t = 0$ and $b_t = \Delta p_t / |\Delta p_t|$ otherwise, so $b_t \in \{-1, 1\}$, with b_0 "set to match the terminal value b_T from the immediately preceding bar" (p. 29). Each bar closes at the first T at which a statistic θ_T of the ticks since the bar opened reaches its expectation as set at the bar's start (pp. 29–32):

Bar	θ_T	bar closes at the first T with
tick imbalance (TIB)	$\sum_{t=1}^{T} b_t$	$\lvert\theta_T\rvert \geq E_0[T]\,\lvert 2P[b_t = 1] - 1\rvert$
volume or dollar imbalance (VIB, DIB)	$\sum_{t=1}^{T} b_t v_t$	$\lvert\theta_T\rvert \geq E_0[T]\,\lvert 2v^+ - E_0[v_t]\rvert$
tick runs (TRB)	$\max\{\sum_{t\lvert b_t=1} b_t,\ -\sum_{t\lvert b_t=-1} b_t\}$	$\theta_T \geq E_0[T]\max\{P[b_t = 1], 1 - P[b_t = 1]\}$
volume or dollar runs (VRB, DRB)	$\max\{\sum_{t\lvert b_t=1} b_t v_t,\ -\sum_{t\lvert b_t=-1} b_t v_t\}$	$\theta_T \geq E_0[T]\max\{v^+,\ v^-\}$

with v_t the units (VIB, VRB) or the currency amount (DIB, DRB) traded, $v^+ = P[b_t = 1]E_0[v_t|b_t = 1]$ and $v^- = P[b_t = -1]E_0[v_t|b_t = -1]$, so that $E_0[v_t] = v^+ + v^-$ and $E_0[\theta_T] = E_0[T](v^+ - v^-) = E_0[T](2v^+ - E_0[v_t])$ for the volume imbalance (p. 30).

Notation. $E_0[T]$ is the expected number of ticks per bar, $P[b_t = 1]$ the unconditional probability that a tick is a buy. Imbalance bars offset buys against sells; runs bars count each side "without offsetting them" and allow sequence breaks, so a run is the larger one-sided total in the bar, not the longest unbroken sequence (p. 31).

Conditions. The expectations are estimated as exponentially weighted moving averages "from prior bars": of T, of b_t (for $2P[b_t = 1] - 1$), of $b_t v_t$, of the proportion of buy ticks and of buy and sell volumes (pp. 29–32). The threshold is therefore fixed when a bar opens and uses only closed bars, so bar formation is causal. López de Prado gives no EWMA span, no warm-up rule for the first bar and no b_0 for the very first tick: these are implementation choices to record. Interpretation, derived here: the factorisation $E_0[\sum_{t \leq T} x_t] = E_0[T]\,E_0[x_t]$ needs T independent of the tick signs and sizes, or a Wald-type identity for IID ticks, neither of which the text states; and for runs bars $E_0[\max\{X, Y\}] \geq \max\{E_0 X, E_0 Y\}$, so the runs "expectation" is a heuristic threshold, not the expectation of θ_T.

Practice. López de Prado's interpretation is that TIBs "are produced more frequently under the presence of informed trading" and can be understood "as buckets of trades containing equal amounts of information" (p. 30); VIBs and DIBs, like volume and dollar bars, address tick fragmentation and outliers and, because the bar size adjusts, corporate actions (p. 31); runs bars target large traders who sweep the book, use icebergs or slice orders (p. 31). No test of these claims is given. In a toy order flow computed for this entry (`afml-checks.py`), all four bar types closed more often during a one-sided burst than elsewhere, but on a perfectly balanced flow the tick-imbalance threshold $E_0[T]|2P[b_t = 1] - 1|$ is near zero, bars close after a tick or two, and the EWMA of T then shrinks further: the scheme needs a floor or cap that the text does not give.

Pitfalls. The legacy notes wrote the runs statistics with a minus sign inside a one-element maximum, which makes the tick-runs θ_T equal to T and the volume-runs θ_T the total volume; the source has two arguments, buy side against sell side (pp. 31–32). They also dropped the absolute values in the imbalance thresholds, so that "$E_0[T]\,|\,2P[b_t = 1] - 1$" read as a conditional expectation, and used v^+ without defining it. The tick rule classifies a tick by the last price change, not by the aggressor; where aggressor flags exist they are the better sign.

See also. activity-sampled bars; event-based sampling; informed trader types; order exposure decisions (icebergs and slicing).

Further reading. López de Prado's chap. 19 on microstructural features, not read here. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. information-driven bars; tick rule; tick imbalance bars; volume imbalance bars; dollar imbalance bars; tick runs bars; volume runs bars; dollar runs bars; exponentially weighted moving average; causal threshold; order-flow imbalance; informed trading

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 2, pp. 29–32; the thresholds and their failure on a balanced flow checked on a toy flow). Stability: draft. Currency: timeless.

Method 2.3 The ETF trick: a basket, spread or rolled futures position as the value of $1 invested, with holdings, costs, tradable volume and the single-futures roll

Statement. The ETF trick models "a basket of securities as if it was a single cash product", transforming a multi-product dataset "into a single dataset that resembles a total-return ETF", so that a strategy can always assume it trades "cashlike products (non-expiring cash instruments)" [alpha-afml, chap. 2, secs. 2.4–2.4.1, pp. 32–33]. It addresses spreads whose weights change (the spread can converge with no price change), spreads that go negative, constituents that do not trade at the same times, and execution costs (p. 33). With raw opens $o_{i,t}$ and closes $p_{i,t}$, the USD value of one point $\varphi_{i,t}$ (including FX), volume $v_{i,t}$ and carry, dividend or coupon $d_{i,t}$ (which can also charge margin or funding costs), allocations ω_t rebalanced or rolled on bars B, and $K_0 = 1$ (p. 34):

$$h_{i,t} = \begin{cases} \dfrac{\omega_{i,t} K_t}{o_{i,t+1}\varphi_{i,t} \sum_i |\omega_{i,t}|} & t \in B \\ h_{i,t-1} & \text{otherwise} \end{cases}, \qquad \delta_{i,t} = \begin{cases} p_{i,t} - o_{i,t} & (t-1) \in B \\ \Delta p_{i,t} & \text{otherwise,} \end{cases}$$

$$K_t = K_{t-1} + \sum_i h_{i,t-1}\varphi_{i,t}(\delta_{i,t} + d_{i,t}).$$

Every instrument must have been tradable at $t-1$ and t (p. 33). With τ_i the cost of trading \$1 of instrument i, the rebalance cost $c_t = \sum_i (|h_{i,t-1}|p_{i,t} + |h_{i,t}|o_{i,t+1})\varphi_{i,t}\tau_i$ for $t \in B$ is kept out of K_t and treated as a negative dividend; the cost of buying or selling one unit is $\tilde{c}_t = \sum_i |h_{i,t-1}|p_{i,t}\varphi_{i,t}\tau_i$; and the tradable basket volume is $v_t = \min_i v_{i,t}/|h_{i,t-1}|$ (p. 34).

Notation. $h_{i,t}$ holdings in securities or contracts; $\delta_{i,t}$ the change in market value per unit from $t-1$ to t. Dividing by $\sum_i |\omega_{i,t}|$ "de-levers" the allocation. Interpretation, derived: $\sum_i |h_{i,t}|o_{i,t+1}\varphi_{i,t} = |K_t| \sum_i |\omega_{i,t}| \operatorname{sgn}(o_{i,t+1}\varphi_{i,t}) / \sum_j |\omega_{j,t}|$, which is $|K_t|$ when every $o_{i,t+1}\varphi_{i,t} > 0$, and K_t itself only when also $K_t \geq 0$. With a negative price or conversion factor the sign of a holding no longer follows the sign of its weight and the sum is not a gross exposure.

Conditions. *Timing.* Holdings decided at t are sized at the next bar's open $o_{i,t+1}$ because at a roll "we may not know $p_{i,t}$ of the new contract" (p. 34). Interpretation: this is an accounting convention, not an order. The quantity $h_{i,t}$ depends on $o_{i,t+1}$, which is unknown when the decision is taken at t, so weights ω_t that use only information up to t are necessary but not sufficient for the series to be executable: an order placed at t cannot carry that quantity. The convention is reproduced only by an order whose quantity is set by its own fill price, for example a currency-amount (notional) order for $\omega_{i,t}K_t / \sum_i |\omega_{i,t}|$ in each leg, filled in full in a single-price opening auction at exactly $o_{i,t+1}$ and with $\varphi_{i,t}$ known (or the FX conversion fixed at that time); whether a venue accepts such orders, and fills them without rounding to lots, partial fills or a price other than the published open, must be checked for each instrument. A quantity sized at t on $p_{i,t}$ and filled at the open leaves a gross exposure different from K_t when the open gaps (checked on a two-leg toy with opens 1.5 percent above and 1.6 percent below the closes: 1.0026 against $K_t = 1$), and a trader who first observes the opening

print can no longer trade at it. The gap between $p_{i,t}$ and $o_{i,t+1}$ on the old holdings is not booked, which is the intent at a roll (two contracts) and an approximation at a same-instrument rebalance, and $\varphi_{i,t}$ is applied to a $t+1$ price. *Positivity and capital.* López de Prado says the series will be "strictly positive (at worst, infinitesimal)" because profits and losses are reinvested at rebalances (pp. 33–34). Interpretation, checked: that holds for long-only baskets with positive prices, but a short leg can lose more than K between rebalances (a short worth \$1 that rises 120 percent leaves $K = -0.2$). The recursion has no rule for exhausted capital: at $K_t = 0$ the next holdings are zero, and at $K_t < 0$ the formula for $h_{i,t}$ reverses every position (the long leg becomes short) while their gross value is $|K_t|$. A series that reaches $K_t \leq 0$ therefore stops describing the strategy; a backtest must stop it or state an explicit financing, margin or loss rule, which the method does not supply. *Costs.* c_t charges the full value of old and new holdings, so it overstates the cost when holdings change little; López de Prado notes that cost functions need not be linear and can be simulated (p. 34). v_t is undefined for a zero holding, which must be skipped.

Practice. *Single futures roll.* For one contract the ETF trick is a one-legged spread; López de Prado's more direct method forms the cumulative series of roll gaps, each the new contract's open minus the old contract's previous close, and subtracts it from the price series, backward (the end matches the raw series) or forward (the start matches) (sec. 2.4.3, pp. 36–37, Snippet 2.2). Rolled prices serve for PnL and mark-to-market; "raw prices should still be used to size positions and determine capital consumption", and rolled prices "can indeed become negative, particularly in futures contracts that sold off while in contango" (p. 37). The non-negative alternative computes the return as the rolled price change divided by the *previous raw price* and compounds $(1 + r)$ (p. 37, Snippet 2.3). Interpretation: the result stays non-negative while $1 + r \geq 0$ on every bar, which holds on non-roll bars when raw prices are positive but not if raw prices themselves turn negative. A two-leg toy (`afml-checks.py`) is a consistency check only: the change in K equals the holdings' profit and loss recomputed from the same prices, and gross exposure at the booked open equals K (here $K > 0$ and all prices positive). The timing and capital cases above are computed in `slice32-followup-checks.py`.

Pitfalls. The legacy notes wrote the open in $h_{i,t}$ as $o_{i,i+1}$, dropped the reason for using the next open, called the roll denominator the "previous roll price" rather than the previous raw price, and stated positivity without its hedge. The allocation vector itself is a separate choice; one way to set it is in PCA risk-distribution weights.

See also. PCA risk-distribution weights; CME Globex trading (contract months); futures hedges and basis (stack and roll); transaction costs; ETF creation and redemption (real ETFs, as against this virtual one).

Further reading. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`, with the toy check in `afml-checks.py`.

Index. ETF trick; multi-product series; spread; basket; \$1 investment series; holdings; de-levering; rebalance cost; bid–ask spread of a virtual ETF; tradable volume; futures roll; roll gap; back-adjusted futures; rolled price series; non-negative rolled series

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 2, pp. 32–37 and Snippets 2.2–2.3; a two-leg toy computed; timing, gross exposure and exhausted capital restated and computed 2026-10-16). Stability: draft. Currency: timeless.

Method 2.4 Event-based sampling: downsampling for reduction and the symmetric CUSUM filter, with its update, reset and threshold rules

Statement. Fitting on every bar is usually a poor idea, because some ML algorithms do not scale with sample size and because they "achieve highest accuracy when they attempt to learn from relevant examples" [`alpha-afml`, chap. 2, sec. 2.5, p. 38]. *Sampling for reduction* (downsampling) takes bars at a constant step (linspace sampling), which is simple but has an arbitrary step and depends on the seed bar, or uniformly at random; neither selects the most informative observations (sec. 2.5.1, p. 38). *Event-based sampling* keeps the bars at which some event occurs and lets the model learn whether a prediction function exists there; if not, the event is redefined (sec. 2.5.2, p. 38). The *CUSUM filter*, a quality-control method for detecting a shift in a mean, sets for observations y_t

$$S_t = \max\{0,\, S_{t-1} + y_t - E_{t-1}[y_t]\}, \quad S_0 = 0,$$

and recommends an action at the first t with $S_t \geq h$, the filter size; the symmetric filter adds $S_t^- = \min\{0, S_{t-1}^- + y_t - E_{t-1}[y_t]\}$ and uses $S_t = \max\{S_t^+, -S_t^-\}$ (sec. 2.5.2.1, pp. 38–39). For sampling, "we will sample a bar t if and only if $S_t \geq h$, at which point S_t is reset" (p. 39).

Conditions. $S_t = 0$ whenever $y_t \leq E_{t-1}[y_t] - S_{t-1}$: the zero floor skips downward deviations, so the one-sided filter detects run-ups from the last reset (p. 39). Without a reset, $S_t \geq h$ if and only if some partial sum $\sum_{i=\tau}^{t}(y_i - E_{i-1}[y_i]) \geq h$ (López de Prado prints $E_{i-1}[y_t]$); interpretation, after a reset τ must start after it. López de Prado describes the inputs as "IID observations arising from a locally stationary process" (pp. 38–39), but his Snippet 2.4 sets $E_{t-1}[y_t] = y_{t-1}$ and cumulates first differences of a level series such as a price, which treats y as a level with martingale-like increments; the operational definition is the cumulated deviation

$y_t - E_{t-1}[y_t]$ under whatever forecast is used. In the snippet the comparisons are strict ($S^- < -h$, $S^+ > h$), only the side that triggered is reset, and at most one event fires per bar, the negative side checked first (p. 39).

Practice. A filter event needs "a full run of length h", so a series hovering near a level does not fire repeatedly, unlike Bollinger bands; S_t can be built on other features, such as structural-break statistics or microstructure measures (p. 40). Checked on toy series (`afml-checks.py`): a series hovering within ± 0.05 of a level fires no event at $h = 0.5$, and a run of exactly h fires under the text's $\geq$ but not under the snippet's $>$. López de Prado's exercises use returns, or absolute returns, as y_t with $h = 0.05$ (p. 40); interpretation, with the snippet's differencing that needs a cumulative return series as input. Lam and Yam's CUSUM trading rule and Fama and Blume's filter rule are different uses: they trade on the signal, whereas here the filter only selects observations (p. 39).

Pitfalls. The legacy notes wrote the zero condition with $=$ instead of $\leq$, copied the misprint $E_{i-1}[y_t]$ in the run condition, and omitted the sampling and reset rule. The threshold h and any volatility scaling of it (López de Prado's labelling exercise sets h to the standard deviation of daily returns, p. 55) are parameters: estimating them on the full sample before a backtest uses future data; see time-validity of labels and weights.

See also. triple-barrier labelling (which the sampled events seed); activity-sampled bars; information-driven bars.

Further reading. Structural-break and entropy features (López de Prado, chaps. 17–18) are not covered. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. sampling for reduction; downsampling; linspace sampling; uniform sampling; event-based sampling; CUSUM filter; symmetric CUSUM; filter size; reset; quality control; filter rule

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 2, pp. 38–40, Snippet 2.4; toy series computed). Stability: draft. Currency: timeless.

2.2 Labels and sample weights for overlapping financial outcomes

Method 2.5 Triple-barrier labelling: the fixed-horizon method and its limits, volatility-scaled barriers, first touch, barrier configurations, and labelling side against size

Statement. *Fixed-horizon labelling* gives an observation X_i, drawn from bars $t = 1, \ldots, T$, the label $y_i = -1$, 0 or 1 as the return $r_{t_{i,0},t_{i,0}+h} = p_{t_{i,0}+h}/p_{t_{i,0}} - 1$ is below $-\tau$, within $\pm\tau$, or above τ, with $t_{i,0}$ "the index of the bar immediately after X_i takes place" and τ a constant [`alpha-afml`, chap. 3, sec. 3.2, p. 43]. López de Prado gives three reasons to avoid it: time bars have poor statistical properties; one τ is applied whatever the volatility, so a τ of 1 percent labels most observations 0 at night and few at the open; and it ignores the path, labelling as winners positions that "would have been stopped-out" (p. 44). The *triple-barrier method* labels an observation by the first of three barriers touched: an upper (profit-taking) and a lower (stop-loss) horizontal barrier, set as multiples of a volatility estimate, and a vertical barrier, an expiration limit counted from the position's start; the upper gives 1, the lower -1, and the vertical either the sign of the return or 0, López de Prado preferring the sign (sec. 3.4, p. 45). With $t_{i,1} \leq t_{i,0} + h$ the first-touch time, the label's return is $r_{t_{i,0},t_{i,1}}$, which uses the prices at $t_{i,0}$ and $t_{i,1}$ and compounds the one-bar returns $r_{t-1,t}$ with $t_{i,0} < t \leq t_{i,1}$; the barriers need not be symmetric (p. 45).

Conditions. *Volatility.* The barrier width is a target such as an exponentially weighted standard deviation of daily returns (Snippet 3.1, default span 100 days, pp. 44–45), computed from data up to the event, so it is known at the decision time; observations whose target is below a minimum return are skipped (Snippet 3.3, p. 48). *Configurations.* Writing $[pt, sl, t1]$ for which barriers are active, López de Prado calls $[1, 1, 1]$, $[0, 1, 1]$ and $[1, 1, 0]$ useful (the last "somewhat unrealistic", holding as long as it takes); $[0, 0, 1]$, $[1, 0, 1]$ and $[1, 0, 0]$ less realistic ($[0, 0, 1]$ is the fixed-horizon method, still useful on activity bars; $[1, 0, 0]$ "could mean being locked on a losing position for years"); and $[0, 1, 0]$ and $[0, 0, 0]$ illogical, the last producing no label (p. 46). *Side and size.* Without a model for the side of the bet, the profit-taking and stop-loss barriers cannot be told apart, so "either there are no horizontal barriers or ... the horizontal barriers must be symmetric" (sec. 3.5, p. 48); with a known side, the barriers may be asymmetric and the label becomes a meta-label (meta-labelling).

Practice. López de Prado's code compares bar closes with the barriers using strict inequalities and takes the earliest of the three touch times (Snippets 3.2–3.3, pp. 45–48); the label's return is measured to the close at $t_{i,1}$, so it can overshoot the barrier. His vertical barrier is set in calendar days (Snippet 3.4, p. 49). Interpretation, checked on toy paths (`afml-checks.py`): with closes alone a bar cannot touch both horizontal barriers, but with highs and lows one bar can, and the text gives no rule for which came first; an implementation must choose one (stop-loss first, finer data, or drop the label) and record it. The code also starts the path at the event's own timestamp, whereas the definition starts it at the next bar: the entry price should be the first one tradable after the features are known.

Pitfalls. The legacy notes called the vertical barrier "physical", the $[1, 0, 0]$ position "loose" and the $[1, 0, 1]$ losses "immediate" rather than intermediate, dropped the path limitation of fixed horizons, the preference for

the sign at the vertical barrier and the side-versus-size rule. A label uses the prices from $t_{i,0}$ to $t_{i,1}$: labels near the end of a training sample use prices from the test period; see time-validity of labels and weights.

See also. event-based sampling (the events that seed the barriers); meta-labelling; label concurrency and uniqueness; sample weights.

Further reading. López de Prado, chap. 10 (bet sizing from predicted probabilities), not read. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. labelling; fixed-time horizon method; triple-barrier method; profit taking; stop loss; vertical barrier; expiration limit; first touch; dynamic thresholds; daily volatility estimate; barrier configurations; side and size

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 3, pp. 43–50, Snippets 3.1–3.5; toy paths labelled; the returns a label uses restated 2026-10-16). Stability: draft. Currency: timeless.

Method 2.6 Meta-labelling: a secondary model that decides whether to act on a primary model's side and how much to bet, with its claimed benefits stated as claims

Statement. When a primary model (any model: ML, econometric, technical, fundamental or a human) sets the side of a bet, *meta-labelling* trains a secondary model only to decide its size, including zero [`alpha-afml`, chap. 3, secs. 3.6 and 3.8, pp. 50 and 53–54]. Triple-barrier events are labelled with the primary side $s_i \in \{-1, 1\}$: the meta-label is 1 if the side-adjusted return $s_i\, r_{t_{i,0},t_{i,1}}$ is positive and 0 otherwise, and the horizontal barriers may be asymmetric because profit-taking and stop-loss are now distinguishable (Snippets 3.6–3.7, pp. 50–51). The secondary model predicts the meta-label; when it predicts 1, "we can use the probability of this secondary prediction to derive the size of the bet, where the side (sign) of the position has been set by the primary model" (p. 51).

Conditions. López de Prado's recipe is to build a primary model with high recall, even at low precision, and to apply meta-labelling to its predicted positives so as to filter out false positives (sec. 3.7, pp. 52–53). He states that meta-labelling "is particularly helpful when you want to achieve higher F1-scores" and "will increase your F1-score" (pp. 52–53), that it lets ML sit on top of a white-box model, that "the effects of overfitting are limited" because ML decides only the size, that it allows separate long and short models, and that it concentrates ML on sizing (p. 53). These are claims without evidence in the chapter; his Exercise 3.5 asks the reader to compare the primary and secondary models' precision, recall and F1 (p. 55). Interpretation, derived here: filtering predicted positives can only keep or reduce recall, so F1 rises only if the precision gained outweighs the recall lost; it is not guaranteed.

Practice. Interpretation: the meta-labels must be built from the primary model's *out-of-sample* predictions (for example walk-forward, or cross-fitted with purging), because in-sample primary signals carry the primary model's overfitted hit rate, which the secondary model would learn. López de Prado's quantamental recipe says instead to "fit your model again on the same training set, but this time using the meta-labels" (sec. 3.8, p. 53), with no out-of-sample condition. For a discretionary portfolio manager he suggests features ranging from market information to stress and sleep (p. 54). Under heavy class imbalance he drops rare labels recursively (Snippet 3.8, p. 54) or uses balanced class weights (sample weights).

Pitfalls. The legacy notes' steps (train a binary primary model, choose a high-recall threshold, possibly from ROC curves, concatenate its features and predictions, and call the final prediction positive "only where both are true"), and their pipeline figure, follow a Hudson & Thames blog example, not López de Prado [`ht-meta-labeling-toy`, "Model Architecture" section]. In López de Prado the primary side is ± 1 and the secondary output sets a size; "both true" is the special case of sizing with a 0/1 threshold. The legacy's five feature groups for the secondary model, beyond the concatenation, and its three limitations of meta-labelling were found in neither source; the first ("if model has overfit the data, meta-labelling will not add much value") contradicts López de Prado's own claim, and the third (it trades recall for precision and needs many trades) is consistent with his precision–recall trade-off (p. 52). They are not adopted, and kept in the artefact.

See also. triple-barrier labelling; time-validity of labels and weights; parameter overfitting; alpha signal combination.

Further reading. Evidence on whether meta-labelling improves after-cost results is a backlog item of the pathway. The legacy file and its TikZ figure are kept in the pathway's artefact folder, `notes/artefacts/statistical-learning-foundations/`.

Index. meta-labelling; primary model; secondary model; side and size; bet sizing; precision; recall; F1 score; false positives; confusion matrix; quantamental; class imbalance

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 3, pp. 50–55; the Hudson & Thames page read for provenance only). Stability: draft. Currency: timeless; the Hudson & Thames page as retrieved 2026-10-16.

Definition 2.7 Overlapping outcomes, label concurrency and average uniqueness

Statement. A label y_i is a function of the prices over $[t_{i,0}, t_{i,1}]$; when $t_{i,1} > t_{j,0}$ for $i < j$, y_i and y_j "will both depend on a common return", and the labels are not IID whenever some consecutive outcomes overlap, $t_{i,1} > t_{i+1,0}$ [`alpha-afml`, chap. 4, sec. 4.2, p. 59]. Restricting horizons to remove overlaps ties the sampling

frequency to the label horizon, or, with path-dependent labels, to the first barrier touched, which López de Prado calls "a terrible solution" (pp. 59–60). Two labels are *concurrent* at t when both depend on the return $r_{t-1,t} = p_t/p_{t-1} - 1$ (sec. 4.3, p. 60). With returns indexed by their ending time t, a label from $t_{i,0}$ to $t_{i,1}$ depends on exactly the returns with $t_{i,0} < t \le t_{i,1}$, so

$$1_{t,i} = \begin{cases} 1 & t_{i,0} < t \le t_{i,1}, \\ 0 & \text{otherwise}, \end{cases}$$

the *concurrency* is $c_t = \sum_i 1_{t,i}$, the *uniqueness* of label i at t is $u_{t,i} = 1_{t,i}/c_t$, and its *average uniqueness* is

$$\bar{u}_i = \frac{\sum_t u_{t,i}}{\sum_t 1_{t,i}},$$

the reciprocal of the harmonic mean of c_t over the label's life, between 0 and 1 (sec. 4.4, p. 61).

Conditions. $\bar{u}_i$ "requires information that is not available until a future time", the end time $t_{i,1}$; López de Prado says this is not a problem because the $\bar{u}_i$ "are used on the training set in combination with label information, and not on the testing set", and are not used to forecast the label (p. 62). The no-leakage claim therefore holds only under that use; see time-validity of labels and weights. *Index conventions.* López de Prado writes $1_{t,i} = 1$ "if and only if $[t_{i,0}, t_{i,1}]$ overlaps with $[t-1, t]$" (p. 60); read with closed intervals that may touch at an endpoint, that would also mark the return ending at $t_{i,0}$ and the one starting at $t_{i,1}$, which his own toy matrix (p. 65) does not, so the rule above is the positive-length overlap. Any indexing that marks the same returns is equivalent: indexing returns by their starting time s, as $r_{s,s+1}$, a label marks $t_{i,0} \le s < t_{i,1}$. López de Prado's toy example (labels on $r_{0,3}$, $r_{2,4}$ and $r_{4,6}$, sec. 4.5.3, p. 65) is consistent in this sense: its printed matrix has rows for the returns ending at $t = 1, \ldots, 6$, and Snippet 4.6 codes it as spans {0:2, 2:3, 4:5}, the start-time indexing with each end shifted down by one; those coded values are not the labels' end times. His production code is a different convention: Snippets 4.1 and 4.3 mark every bar from the event's timestamp $t_{i,0}$ to $t_{i,1}$ inclusive (`count.loc[tIn:tOut]`, `indM.loc[t0:t1]`), and Snippet 4.10 sums `ret.loc[tIn:tOut]` with `ret` the log return ending at each bar (pp. 60–61, 64 and 69). Applied to event times and returns indexed by ending time, that also counts the return ending at $t_{i,0}$, from before the label starts, so a label that starts where another ends acquires a return it never uses and the two appear concurrent. Including the start timestamp is therefore not an equivalent reindexing; the inclusive code is right only when its inputs are already shifted as in Snippet 4.6.

Practice. Checked on the toy example (`afml-checks.py`, and by brute force over the returns each label compounds in `slice32-followup-checks.py`): with the rule above, concurrency $(1, 1, 2, 1, 1, 1)$ for $t = 1, \ldots, 6$ and average uniqueness $(0.833, 0.75, 1)$, López de Prado's matrix and values (pp. 65–66). The inclusive rule applied to the same events, rows $t = 0, \ldots, 6$, gives concurrency $(1, 1, 2, 2, 2, 1, 1)$ and average uniqueness $(0.75, 0.5, 0.833)$: labels 2 and 3, adjacent at $t = 4$, then share the return $r_{3,4}$, which label 3 does not use. Interpretation: the mean of $\bar{u}_i$ over the sample is the number of effectively independent outcomes per observation, the quantity behind the bootstrap and bagging choices in the sequential bootstrap.

Pitfalls. The legacy notes wrote $r_{t-1,t} = p_t/p_{t-1} = 1$ for -1, and dropped the training-set condition from "hence there is no information leakage".

See also. triple-barrier labelling; sequential bootstrap; sample weights.

Further reading. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. overlapping outcomes; non-IID labels; concurrent labels; concurrency; uniqueness; average uniqueness; label span; indicator matrix

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 4, pp. 59–62 and 64–66; the toy example recomputed; the indicator convention restated and the inclusive code convention computed 2026-10-16, pp. 60–61, 64–65 and 69). Stability: draft. Currency: timeless.

Method 2.8 Bootstrapping overlapping labels: redundancy, oversampling, and the sequential bootstrap

Statement. Drawing I times with replacement from I items misses a given item with probability $(1 - I^{-1})^I \to e^{-1}$, so about $1 - e^{-1} \approx 0.63$ of the items are drawn [`alpha-afml`, chap. 4, sec. 4.5, p. 62]. When the mean average uniqueness $I^{-1} \sum_i \bar{u}_i \ll 1$, bootstrapped in-bag observations are redundant with one another and similar to out-of-bag ones: bagged trees become "very similar copies of a single overfit decision tree" and out-of-bag accuracy "will be grossly inflated" (pp. 62–63). Dropping overlapping outcomes loses too much information; better is to sample only a fraction near the mean uniqueness (for example `max_samples` in scikit-learn's bagging classifier) (p. 63). The *sequential bootstrap* draws with changing probabilities: the first draw is uniform, $\delta_i^{(1)} = I^{-1}$; after the draws φ so far, label j's uniqueness given those draws is $u_{t,j}^{(2)} = 1_{t,j}(1 + \sum_{k \in \varphi} 1_{t,k})^{-1}$, its average

$\bar{u}_j^{(2)} = \sum_t u_{t,j}^{(2)} / \sum_t 1_{t,j}$, and the next draw has probabilities

$$\delta_j^{(2)} = \frac{\bar{u}_j^{(2)}}{\sum_{k=1}^{I} \bar{u}_k^{(2)}},$$

updated after every draw until I draws are made; repeats remain possible but less likely (sec. 4.5.1, pp. 63–64).

Conditions. López de Prado says the sequential sample "will be much closer to IID", which "can be verified by measuring an increase" in mean uniqueness (p. 64). His evidence is a Monte Carlo on random spans (10 observations, 100 bars, label spans of 1 to 4 bars from Snippet 4.7's `maxH=5`, 10^6 runs): median mean uniqueness 0.6 for the standard bootstrap against 0.7 for the sequential one, with an ANOVA test of the difference in means (Figure 4.2, pp. 66–68). That shows higher average uniqueness on those synthetic spans, a proxy for less redundancy; it is not a guarantee for every sample, nor evidence of better out-of-sample accuracy. The cost is one pass over all labels per draw.

Practice. Checked (`afml-checks.py`): on López de Prado's toy example, after label 2 is drawn the probabilities are $(5, 3, 6)/14$, as printed (p. 66), and after labels 2 and 3 they are $(5, 3, 3)/11$ (his Exercise 4.6). A 400-run Monte Carlo with his settings gives medians 0.600 and 0.700, with the sequential sample less unique than the standard one in 22 percent of runs (an independent 3,000-run check found 23 percent).

Pitfalls. López de Prado writes that with at most $K \leq I$ non-overlapping outcomes the probability of missing an item is $(1 - K^{-1})^I$, "approximated as $(1 - I^{-1})^{I\frac{K}{I}} \approx e^{-K/I}$", so that the share of unique observations drawn is $1 - e^{-K/I} \leq 1 - e^{-1}$ (p. 62). The approximation is wrong: $(1 - K^{-1})^I \approx e^{-I/K}$; for $I = 1000$, $K = 100$ the exact value is 4.3×10^{-5} and $e^{-I/K} = 4.5 \times 10^{-5}$, against $e^{-K/I} = 0.905$. Interpretation, checked by simulation: treating the K non-overlapping outcomes as the effective units, almost all of them appear in a bootstrap sample, but its I rows hold only about $K(1 - e^{-I/K})$ distinct units, 0.100 per row against 0.632 under IID draws. The conclusion that assuming IID draws oversamples survives; the formula does not. The legacy notes also wrote "overlapping" for non-overlapping, used $\bar{u}_j^{(2)}$ in its own normaliser in place of $\bar{u}_k^{(2)}$, and, like the source, omitted the superscript in $u_{t,j}^{(2)}$ inside $\bar{u}_j^{(2)}$.

See also. label concurrency and uniqueness; sample weights; parameter overfitting.

Further reading. Cross-validation under overlapping labels, with purging and embargo (López de Prado, chap. 7), is a backlog item of the pathway; Rao, Pathak and Koltchinskii (1997) is the sequential resampling scheme López de Prado distinguishes from his (p. 63), not read. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. bootstrap; bagging; out-of-bag accuracy; redundancy; oversampling; average uniqueness; sequential bootstrap; draw probabilities; max_samples; Monte Carlo

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 4, pp. 62–68; toy example, Monte Carlo and the overlap approximation recomputed). Stability: draft. Currency: timeless.

Method 2.9 Sample weights for overlapping labels: return attribution, time decay on cumulative uniqueness, and class weights

Statement. *Return attribution.* For labels that depend on the sign of the return (± 1, or $\{0, 1\}$ for meta-labels), López de Prado weights observation i by the absolute sum of the returns attributed to it,

$$\tilde{w}_i = \Big| \sum_{t=t_{i,0}+1}^{t_{i,1}} \frac{r_{t-1,t}}{c_t} \Big| \text{ with } r_{t-1,t} = \ln(p_t / p_{t-1}), \qquad w_i = \tilde{w}_i \frac{I}{\sum_{j=1}^{I} \tilde{w}_j}, \qquad \sum_i w_i = I,$$

(log returns, "so that they are additive", in Snippet 4.10, unlike the simple return used to define concurrency), because overlapping outcomes would otherwise be over-weighted and "labels associated with large absolute returns should be given more importance than labels with negligible absolute returns"; the weights sum to I because libraries assume a default weight of 1 [`alpha-afml`, chap. 4, sec. 4.6, pp. 68–69, Snippet 4.10]. *Time decay.* Weights are multiplied by $d[x] = \max\{0, a + bx\}$ on $x \in [0, \sum_i \bar{u}_i]$, cumulative uniqueness rather than calendar time, with $d[\sum_i \bar{u}_i] = 1$ for the newest observation; for $c \in [0, 1]$, $d[0] = c$, giving $b = (1 - c)/\sum_i \bar{u}_i$; for $c \in (-1, 0)$, $d[-c\sum_i \bar{u}_i] = 0$, giving $b = [(c + 1)\sum_i \bar{u}_i]^{-1}$; in both cases $a = 1 - b\sum_i \bar{u}_i$ (sec. 4.7, p. 70). *Class weights* correct for under-represented labels, "particularly critical" when the important classes are rare, such as liquidity crises; López de Prado's default is `class_weight='balanced'`, or `'balanced_subsample'` for bagging (sec. 4.8, pp. 71–72).

Conditions. Return attribution "will not work if there is a 'neutral' (return below threshold) case", for which lower returns would need higher weights; López de Prado advises dropping neutral cases (p. 69). Decay runs on cumulative uniqueness because chronological decay "would reduce weights too fast in the presence of redundant observations" (p. 70). Cases: $c = 1$ no decay; $0 < c < 1$ linear decay with every weight positive; $c = 0$ decay to zero; $c < 0$ zero weight for the oldest observations, those with $x \leq -c\sum_i \bar{u}_i$, a share $|c|$ of cumulative

uniqueness (pp. 70–71). In scikit-learn, class weights that do not sum to the number of classes act like a change in the regularisation parameter (p. 71).

Practice. Checked (`afml-checks.py`): attributed weights normalised by $I/\sum_j \tilde{w}_j$ sum to I; the decay constants meet $d[\sum \bar{u}] = 1$ and $d[0] = c$ for $c \geq 0$, and $d[-c\sum \bar{u}] = 0$ for $c < 0$, for $c = 1, 0.5, 0, -0.5$. Interpretation: López de Prado's implementation evaluates d at the cumulative sums $\bar{u}_1, \bar{u}_1 + \bar{u}_2, \ldots$ (Snippet 4.11), so the oldest observation sits at $x = \bar{u}_1 > 0$ and receives slightly more than c; and his printed sum runs from $t = t_{i,0}$ (p. 69), as does Snippet 4.10, which adds the return ending at the event's own bar; the lower limit $t_{i,0} + 1$ above sums exactly the returns the label uses, as in label concurrency. On a toy path the two give attribution sums $(0.0247, 0.0145, 0.0190)$ and $(0.0149, 0.0146, 0.0288)$ for López de Prado's three labels (`slice32-followup-checks.py`).

Pitfalls. The legacy notes dropped the factor I (their weights summed to 1 while claiming I), and their "lower returns should be assigned higher weights" reversed the rule by dropping its neutral-class context. Two slips are López de Prado's: the text says $d[1] = c$ where its own boundary condition, $a + b \cdot 0 = c$, gives $d[0] = c$; and "the oldest portion cT of the observations receive zero weight" should be a share $-c$ of cumulative uniqueness, not of T (pp. 70–71). Weights built from $t_{i,1}$, c_t and realised returns are functions of future outcomes and belong to training only; see time-validity of labels and weights.

See also. label concurrency and uniqueness; sequential bootstrap; meta-labelling.

Further reading. King and Zeng (2001) on rare events, cited by López de Prado (p. 71), not read. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. sample weights; return attribution; absolute return weighting; neutral labels; time decay; cumulative uniqueness; piecewise-linear decay; class weights; balanced class weights; rare events

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 4, pp. 68–72, Snippets 4.10–4.11; weights and decay constants recomputed; the summation's lower limit corrected 2026-10-16). Stability: draft. Currency: timeless; scikit-learn behaviour as described in 2018.

Pitfall 2.10 Time-validity of labels, sample weights and fitted transforms: what is known when, and what may be used only in training

Statement. Several of López de Prado's constructions use information from after the observation they describe. A triple-barrier label and its return use prices up to $t_{i,1}$ [`alpha-afml`, chap. 3, p. 45]; average uniqueness "requires information that is not available until a future time" and is safe only because it is "used on the training set ... and not on the testing set" (chap. 4, p. 62); attribution weights use realised returns over the label's life (p. 69). These are legitimate as training targets and weights. Interpretation, derived here: they leak when (i) a training label's span reaches into the test period, so the training set contains test-period prices; (ii) concurrency c_t is computed with labels from both sets; (iii) a quantity fitted on the full sample (a CUSUM threshold, a barrier multiple, a fractional-differencing order d, an EWMA warm-up, a meta-label primary model) is then used in a backtest of that same sample; or (iv) weights or labels are used as features.

Conditions. The decision-time inputs, by contrast, are causal as López de Prado defines them: bar thresholds use EWMAs of prior bars (chap. 2, pp. 29–32), and the barrier width uses volatility up to the event (chap. 3, pp. 44–45). His fractional-differencing order is chosen on the whole E-mini history (chap. 5, Snippet 5.4, p. 85), and his quantamental recipe refits the secondary model "on the same training set" as the primary (p. 53); both need re-estimation inside each training window when the result is to be evaluated out of sample.

Practice. Interpretation, a checklist for one train/test split: compute labels, spans, concurrency, uniqueness and weights from training data; drop (purge) training observations whose spans overlap the test period, and leave a gap (embargo) after it; fit d, thresholds and barrier multiples on training data and apply them unchanged to the test period; build meta-labels from out-of-sample primary predictions; and record each choice with its data window.

Pitfalls. The legacy notes reported "no information leakage" for uniqueness without its training-set condition, and kept the fractional-differencing order as a property of the series rather than an estimate. Purging and embargo are López de Prado's chap. 7, not read here; the description above is the entry's.

See also. label concurrency and uniqueness; sample weights; meta-labelling; fixed-width fractional differencing; strategy production lifecycle (embargo before live trading, a different use of the word); parameter overfitting.

Further reading. The pathway's readings on information timing, beginning with `alpha-jansen` and the point-in-time language-model papers, and López de Prado's chap. 7 are in the backlogs of `financial-ml-evaluation` and `statistical-learning-foundations`.

Index. look-ahead bias; leakage; label span; training set; test set; purging; embargo; out-of-sample predictions; parameter estimation window; time-valid inputs

Status. Verification: partly checked 2026-10-16 (the López de Prado passages cited, chaps. 2–5, as for the linked entries; the checklist is the entry's own and unchecked against chap. 7). Stability: draft. Currency:

timeless.

2.3 Fractionally differentiated features

Model 2.11 Fractional differencing weights: the binomial expansion of $(1-B)^d$, the recurrence, integer and fractional cases, signs and decay

Statement. With the backshift operator $B^k X_t = X_{t-k}$, a real order d gives the formal binomial series [alpha-afml, chap. 5, sec. 5.4, p. 77]

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \qquad \tilde{X}_t = \sum_{k=0}^{\infty} \omega_k X_{t-k}, \qquad \omega_k = (-1)^k \frac{\prod_{i=0}^{k-1}(d-i)}{k!},$$

so $\omega = (1, -d, d(d-1)/2!, -d(d-1)(d-2)/3!, \dots)$, generated by $\omega_0 = 1$ and $\omega_k = -\omega_{k-1}(d-k+1)/k$ (secs. 5.4.1–5.4.2, pp. 77–78). For a positive integer d the weights vanish beyond $k = d$ and "memory beyond that point is cancelled": $d = 1$ gives $(1, -1, 0, \dots)$, returns; $d = 0$ gives $(1, 0, \dots)$, the original series (p. 78).

Conditions. For $0 < d < 1$ every weight after ω_0 lies in $(-1, 0)$; for $1 < d < 2$, $\omega_1 < -1$ and $\omega_k > 0$ for $k \geq 2$ (p. 78, Figures 5.1–5.2). For $k > d$ the ratio $|\omega_k/\omega_{k-1}| = |(d-k+1)/k| < 1$; the initial weights alternate in sign while $k < d+1$, and beyond that the weights keep one sign, negative when $\text{int}[d]$ is even and positive when it is odd (sec. 5.4.3, p. 80). López de Prado concludes that the weights "converge asymptotically to zero, as an infinite product of factors within the unit circle" (p. 80). Interpretation, checked: the ratio tends to 1 (0.9986 at $k = 1{,}000$ for $d = 0.4$), so the product argument alone is not a proof; the decay is a power law, $|\omega_k| \approx k^{-(1+d)}/|\Gamma(-d)|$, which matches within 0.05 percent at $k = 10^3$ and to four digits at $k = 10^4$. The weights therefore fall slowly and the partial sums approach $\sum_k \omega_k = 0$ slowly (0.017 after 10^4 terms at $d = 0.4$), which is why truncation matters (fixed-width window).

Practice. Checked (`afml-checks.py`): the recurrence equals the closed form for $d = 0, 0.4, 1, 1.5, 2$; $d = 0$ and $d = 1$ give $(1, 0, \dots)$ and $(1, -1, 0, \dots)$; all weights after the first lie in $(-1, 0)$ at $d = 0.4$. López de Prado's motivation: integer differencing makes price series stationary "at the expense of removing all memory", while "stationarity is a necessary property for inferential purposes" and memory "is the basis for the model's predictive power" (sec. 5.2, p. 75); supervised learning needs stationary features, but "stationarity does not ensure predictive power" (p. 76). His literature review attributes the idea to Hosking (1981), whose fractionally differenced processes "exhibit long-term persistence and antipersistence, hence enhancing the forecasting power compared to the standard ARIMA approach" (sec. 5.3, p. 76), a claim reported, not checked here.

Pitfalls. The legacy notes wrote the general weight with $(d-1)$ for $(d-i)$, which at $d = 0.4$ gives $1, 0.6, 0.18, \dots$ instead of $1, -0.4, -0.12, \dots$; they dropped the variable from the limits and from "$\forall k > d$"; and they turned López de Prado's "this alternation of weight signs is necessary to make $\{\tilde{X}_t\}$ stationary" (p. 80) into a sufficient condition ("makes … stationary"). For $0 < d < 1$ the weights do not alternate after ω_1, and no sign pattern by itself makes a series stationary: that depends on the series (fixed-width window, Conditions).

See also. fixed-width fractional differencing; return measures (integer differencing of log prices).

Further reading. Hosking (1981) and the fractional-integration literature López de Prado lists (p. 76) are not read. The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. fractional differencing; fractional differentiation; backshift operator; binomial series; long memory; differencing weights; weight recurrence; integer differencing; stationarity and memory; ARFIMA

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 5, pp. 75–80; the weights, signs and decay rate computed). Stability: draft. Currency: timeless.

Method 2.12 Fractionally differentiating a finite series: expanding window and weight loss, the fixed-width window (FFD), and choosing the minimum order d with a unit-root test

Statement. A finite series $X_1, \dots, X_T$ cannot use infinitely many weights. With an *expanding window*, $\tilde{X}_T$ uses $\omega_0, \dots, \omega_{T-1}$ and earlier points fewer, so they carry different amounts of memory; López de Prado drops the early points whose relative weight loss exceeds a tolerance τ, such as 0.01 [alpha-afml, chap. 5, sec. 5.5.1, p. 80]. Even so the series drifts down, "because values … are computed on an expanding window" that keeps adding negative weights (pp. 80–82, Figure 5.3). The *fixed-width window* (FFD) drops the weights once $|\omega_k|$ falls below a threshold τ: with l^* the last lag with $|\omega_{l^*}| \geq \tau$, $\tilde{\omega}_k = \omega_k$ for $k \leq l^*$ and 0 beyond, and $\tilde{X}_t = \sum_{k=0}^{l^*} \tilde{\omega}_k X_{t-k}$, the same weights for every point, which avoids the drift (sec. 5.5.2, pp. 82–83). The *minimum order* d^* is the smallest d for which the FFD series passes a stationarity test; López de Prado's recipe is to take the cumulative sum of a feature (so that some differencing is needed), compute FFD(d) for $d \in [0, 1]$, and keep the smallest d whose ADF p-value is below 5 percent (secs. 5.6–5.7, pp. 84 and 88).

Notation. Interpretation, with one consistent index (the source's is not): the point computed from the first t observations misses the weights $\omega_t, \dots, \omega_{T-1}$, a relative weight loss $\lambda_t = \sum_{k=t}^{T-1} |\omega_k| / \sum_{k=0}^{T-1} |\omega_k|$, and is

kept if $\lambda_t \leq \tau$. FFD values exist for $t = l^* + 1, \ldots, T$.

Conditions. López de Prado's evidence: E-mini S&P 500 log prices (rolled with the ETF trick, daily) have an ADF statistic of -0.34 undifferenced and -46.9 as returns; the 95 percent critical value -2.86 is crossed near $d = 0.35$, where the FFD series still has correlation 0.995 with the original, against 0.03 for returns (pp. 84–85, Figure 5.5); across 87 liquid futures, every log-price series passes at $d < 0.6$, most at $d < 0.3$, and some need no differencing (Table 5.1, pp. 85–87). He reads this as "most studies have over-differentiated the series" (p. 85) and the FFD series as "a driftless blend of level plus noise ... no longer Gaussian ... however it is stationary" (p. 83). Interpretation: these are unit-root test outcomes on single samples, not demonstrations of stationarity. Snippet 5.4 runs the ADF test with one lag and a constant only, on FFD with $\tau = 0.01$, a short window (11 weights at $d = 0.4$); the ADF test is built against a unit root, not against fractional integration, and under fractional-integration theory a series integrated of order 1 and differenced by d is integrated of order $1 - d$, covariance-stationary only for $d > 1/2$. The truncated weights also do not sum to zero, so a small multiple of the level remains (0.264 times X_{t-l^*} at $d = 0.4$, $\tau = 0.01$; 0.036 at $\tau = 10^{-5}$, computed here).

Practice. Checked (`afml-checks.py`): at $d = 0.4$ the FFD window keeps 11, 55 and 1,458 weights for $\tau = 10^{-2}, 10^{-3}, 10^{-5}$; for an expanding window on 200 points, the weight loss of the point using t observations falls from 0.48 at $t = 1$ to 0.0135 at $t = 100$, so $\tau = 0.01$ keeps points from $t = 118$. On 2,000 points of a simulated random walk, the one-lag ADF test on FFD(d) with $\tau = 10^{-4}$ rejects a unit root at 5 percent already at $d = 0.2$ (statistic -5.3) and at $d = 0.4$ (-10.2), although the theory above says such a series is not stationary for $d < 1/2$: the outcome reflects the test, its lag choice and the truncation. Interpretation: choose d, τ and the test on training data only, and apply them unchanged out of sample; López de Prado chooses d on the full E-mini history (time-validity of labels and weights).

Pitfalls. The legacy notes copied two of the source's index slips: the weight-loss sum running to T rather than $T - 1$, with the dropped points named $\tilde{X}_1, \ldots, \tilde{X}_{l^*}$ although λ_l refers to point $T - l$; and the FFD range $t = T - l^* + 1, \ldots, T$, which should be $l^* + 1, \ldots, T$ (pp. 80 and 82). They stated "it is stationary" as a general property of FFD output, and omitted the minimum-d procedure on which the stationarity claims rest. Their two figures are reproductions of López de Prado's Figures 5.3(b) and 5.4 (pp. 81 and 83), kept in the artefact, not here.

See also. fractional differencing weights; ETF trick (the rolled series López de Prado differentiates); time-validity of labels and weights.

Further reading. Unit-root and fractional-integration testing (for example MacKinnon and Nielsen 2014, listed by López de Prado) are a backlog item. The legacy file and its two figures are kept in `notes/artefacts/statistical-learning-foundations/`.

Index. fractional differentiation; expanding window; weight loss; fixed-width window; FFD; truncation threshold; minimum differencing order; ADF test; unit root; stationarity; memory preservation; over-differencing

Status. Verification: partly checked 2026-10-16 (against the text layer and page images of a local copy of `alpha-afml`, chap. 5, pp. 80–88, Snippets 5.2–5.4 and Table 5.1; windows, weight loss and a simulated ADF illustration computed). Stability: draft. Currency: the futures evidence as of 2018.

2.4 Optimal execution as stochastic control: the benchmark for learned policies

Model 2.13 Optimal execution with order flow as stochastic control: the Cartea–Jaimungal objective, dynamics and dynamic-programming equation, as presented by Lehalle and Laruelle

Statement. A trader sells an inventory Q_0 by time T, choosing a trading rate ν_t [`alpha-laruelle`, app. A, sec. A.7.3, pp. 281–282, page images]. The state is the midprice S_t, the net order flow μ_t of all other traders, the remaining inventory Q_t and the cash account X_t, with

$$dS_t = b\,(\mu_t - \nu_t)\,dt + \sigma\,dW_t, \qquad \hat{S}_t = S_t - \Big(\frac{\psi}{2} + \kappa\nu_t\Big), \qquad dQ_t = -\nu_t\,dt, \qquad dX_t = \hat{S}_t\,\nu_t\,dt$$

(p. 282): permanent impact linear in the trading rate, and temporary impact $\kappa\nu_t$ with full immediate decay (p. 281). The reward of a given policy ν is

$$H_t(\nu) = \mathbb{E}_t\Big[X_T^\nu + Q_T^\nu\Big(S_T - \frac{\psi}{2} - A\,Q_T^\nu\Big) - \phi\int_t^T (Q_s^\nu)^2\,ds\Big].$$

This is the book's objective (p. 281) with the half-spread added to the terminal payoff, so that it agrees with the book's own terminal condition (Interpretation; the printed objective is discussed under Pitfalls). The value function is the best such reward over admissible policies, $H^* = \sup_\nu H(\nu)$, finite in the parameter regime given under Conditions. By dynamic programming it formally solves (Interpretation: the book's Proposition 6, p. 282,

with the corrections listed under Pitfalls)

$$0 = \Big(\partial_t + \tfrac{1}{2}\sigma^2\partial_{SS} + \mathcal{L}^\mu\Big)H^* - \phi q^2 + \sup_\nu\Big\{\nu\Big(S - \frac{\psi}{2} - \kappa\nu\Big)\partial_x H^* + b(\mu-\nu)\,\partial_S H^* - \nu\,\partial_q H^*\Big\},$$

with $H^*(T,x,S,\mu,q) = x + q(S-\psi/2) - Aq^2$, the terminal payoff above, where $\mathcal{L}^\mu$ is the generator of the flow. Defining $h := H^* - x - q(S-\psi/2)$, so that $h(T,\mu,q) = -Aq^2$, gives

$$\partial_t h + \mathcal{L}^\mu h + b\mu q - \phi q^2 + \sup_\nu\big\{-\kappa\nu^2 - (bq + \partial_q h)\nu\big\} = 0, \qquad \nu^* = -\frac{bq + \partial_q h}{2\kappa}$$

(p. 283, page image; the half-spread cancels in this decomposition). With $h = h_0 + qh_1 + q^2h_2$, the coefficient h_2 solves a Riccati equation that does not involve μ, h_1 carries the expected future flow, and h_0 follows by integration (pp. 283–284, page images). The authors' point is that the trader can act on a view of future flows (p. 284). Interpretation, derived and checked numerically: with $b + 2h_2 = -2\xi$, the optimal rate is

$$\nu_t^* = \frac{\xi(t)}{\kappa}\,Q_t - \frac{h_1(t,\mu_t)}{2\kappa}, \qquad h_1(t,\mu) = b\int_t^T e^{-\frac{1}{\kappa}\int_t^s \xi(u)\,du}\,\mathbb{E}_{t,\mu}[\mu_s]\,ds,$$

so the trader sells in proportion to inventory and sells more slowly when other traders are expected to keep buying.

Notation. Units, for a sale: Q in shares; ν and μ in shares per unit time, $\nu > 0$ selling and $\mu > 0$ net buying by others; S, ψ and $\hat{S}$ in currency per share; b in currency per share^2, the price change per share of net flow; σ in currency per share per square root of time; κ in currency × time per share^2; ϕ in currency per share^2 per unit time; A in currency per share^2. Then h is in currency, h_1 in currency per share, h_2 and ξ in currency per share^2, and $\gamma = \sqrt{\phi/\kappa}$ per unit time; $A - b/2$ and $\sqrt{\kappa\phi}$ are both in currency per share^2, as the formula for ζ requires. Signs: the trader's own selling lowers S permanently by b per share; each fill is $\psi/2 + \kappa\nu$ below the midprice; others' net buying raises S. $H_t(\nu)$ is the reward of one fixed policy; the legacy notes called it the value function, which is H^*.

Conditions. The trader observes S, μ, Q and X at each time. The flow μ is Markov with a known generator and does not respond to the trader. The parameters b, κ, σ, ψ, ϕ and A are constant: the book writes b_t in the price dynamics but a constant b in the equation.

Finite value. Interpretation, own derivation, checked numerically: with the cash account starting at $X_0 = 0$, so that cash is measured from the start of execution (otherwise add X_0 to the right-hand side), integrating by parts, the realised reward of any policy is

$$\begin{aligned} &X_T + Q_T\Big(S_T - \frac{\psi}{2} - AQ_T\Big) - \phi\int_0^T Q_t^2 dt \\ &\quad = Q_0S_0 - \frac{\psi Q_0}{2} - \frac{bQ_0^2}{2} + b\int_0^T Q_t\mu_t\,dt + \sigma\int_0^T Q_t\,dW_t - \kappa\int_0^T \nu_t^2 dt - \phi\int_0^T Q_t^2 dt - \Big(A - \frac{b}{2}\Big)Q_T^2, \end{aligned}$$

because the trader's own permanent impact contributes $b(Q_T^2 - Q_0^2)/2$ whatever the path. A simple sufficient regime for a finite value and for the formulas used here is $\kappa > 0$, $\phi > 0$ and $A > b/2$, with admissible policies progressively measurable with respect to the observed state and $\mathbb{E}\int_0^T \nu_t^2 dt < \infty$, and a flow with $\mathbb{E}\int_0^T \mu_t^2 dt < \infty$. Then $\mathbb{E}\int Q\,dW = 0$; $|\mathbb{E}\int Q\mu\,dt| \le \|Q\|\,\|\mu\|$ with $\|Q\| \le \sqrt{T}Q_0 + T\|\nu\|$ (norms in $L^2(\Omega\times[0,T])$), so the expected reward is bounded above by a concave quadratic in $\|\nu\|$; and $|\zeta| < 1$, so ξ is finite and positive on $[0,T]$. The regime is sufficient, not necessary, and no verification theorem is proved here: the quadratic ansatz is a formal solution. Outside it the value can be infinite. With $T = Q_0 = 1$, $\mu = \sigma = \psi = 0$, $b = 0.1$, $\kappa = A = 0.01$, $\phi = 0.001$ and a constant buying rate $\nu = -M$, the reward's M^2 coefficient is $b/2 - \kappa - A - \phi/3 \approx 0.0297 > 0$: buying raises the price at which the enlarged position is marked at T faster than the penalties grow. Simulated rewards divided by M^2 converge to this coefficient.

Constraints and completion. The unconstrained problem is a formal benchmark. Economically valid one-directional execution requires $\nu \ge 0$, and $Q_t \ge 0$. The unconstrained optimum can ask the seller to buy back, for instance when strong expected buying flow makes h_1 large, and the half-spread term then credits the purchase (Pitfalls). Clipping ν^* at zero does not solve the constrained problem: the constraint changes the value function itself, whose equation then takes the supremum over $\nu \ge 0$ and has a different h. A finite A charges the inventory left at T, which is sold at $S_T - \psi/2 - AQ_T$. That is a quadratic penalty, not mandatory completion; completion is forced only in the limit $A \to \infty$.

Penalty and variance. The running penalty $\phi\int Q^2$, which the book calls risk aversion (p. 282), is not in general the variance of implementation shortfall. For a deterministic schedule Q the random part of the reward

is $b\int Q_t\mu_t\,dt+\sigma\int Q_t\,dW_t$. Its variance is

$$b^2\,\mathrm{Var}\Big(\int Q_t\mu_t\,dt\Big)+\sigma^2\int Q_t^2\,dt+2b\sigma\,\mathrm{Cov}\Big(\int Q_t\mu_t\,dt,\int Q_t\,dW_t\Big),$$

and the covariance is nonzero when flow and price shocks are correlated. Thus $\sigma^2\int Q^2dt$ is only the contribution of the Brownian innovations. It is the whole variance when $b=0$ or the flow path is deterministic; only then does $\phi=\lambda\sigma^2$ reproduce a mean–variance penalty. Example: $Q_t=1-t$, $\mu_t=B_t$ a Brownian motion, $T=1$, $\sigma=0$ give variance $b^2\int_0^1\int_0^1(1-s)(1-t)\min(s,t)\,ds\,dt=b^2/20$ (exact; checked by quadrature and Monte Carlo). For a policy that adapts to the flow, ϕ is a penalty on holding inventory, not a variance.

Benchmark limit. With $\mu\equiv 0$ and $A\to\infty$ the optimal holding is $Q_t=Q_0\sinh(\gamma(T-t))/\sinh(\gamma T)$ with $\gamma=\sqrt{\phi/\kappa}$. This is the continuous-time form of the Almgren–Chriss schedule, with κ in the role of η and ϕ of $\lambda\sigma^2$ (Interpretation; matched numerically to four decimals).

Practice. The legacy note's parameter intuition holds under the model's assumptions, with qualifications (Interpretation; own numerical paths with $\mu\equiv 0$, $Q_0=1$, $T=1$, base case $\kappa=0.01$, $\phi=0.001$, $A=0.5$, in consistent units):

- a larger ϕ front-loads the sale: tenfold ϕ leaves 0.45 rather than 0.50 at $T/2$;
- a larger A mostly cuts what is left at T, 0.2 rather than 1.9 percent for tenfold A, and barely changes the early path;
- a larger κ slows and smooths trading: tenfold κ leaves 0.58 at $T/2$, and with finite A it also leaves 17 percent unsold at T.

With flow, the rate also depends on the expected μ. Lehalle and Laruelle suggest Almgren–Chriss when price formation is well understood at a 5–30-minute scale, and this framework when one knows a continuous-time version of price formation focused on trading flows (p. 284).

As the control-theoretic benchmark for the pathway's learned policies (legacy block *Control vs Learning*): stochastic control solves for a policy given a specified model. Simple models are tractable but may oversimplify the market; realistic ones are often intractable [wide-hambly-xu-yang, sec. 1, p. 2]. Reinforcement learning learns decisions from repeated interaction with a system, without exemplary supervision or a complete model of the environment (sec. 2, p. 3). When the transition law and rewards are unknown the problem becomes one of learning: *model-based* methods learn them explicitly, *model-free* methods implicitly (sec. 2.2, p. 7). The survey motivates RL for execution by Almgren–Chriss's reliance on its dynamics and linear impact, and by its pre-planned schedule, which ignores real-time conditions (sec. 4.2, p. 33, page image). It reports one study that combined the Almgren–Chriss solution with Q-learning, and one double-DQN study that added a final one-second step to force liquidation (p. 34); these studies were not checked. On data, most RL algorithms need more samples than the relevant historical data provide, and financial series are non-stationary, hence the call for sample-efficient methods or good market simulators (sec. 5, p. 49). Interpretation: training against a simulator is interaction, but the simulator must be validated, which is the pathway's checkpoint. Replaying historical order books does not show how the market would have responded to the agent's own orders, so own impact must still come from a model. Learning from logged historical decisions without interaction is a different setting, not covered in the passages read.

Pitfalls. Inconsistencies in the appendix (pp. 281–284, page images; corrections are Interpretation, checked algebraically and numerically):

- Proposition 6 defines the value function maximising the sale as "$H^*:=\min_\nu H(\nu)$" while its equation takes a sup; the value is $\sup_\nu H(\nu)$.
- The printed objective is $H_t(\nu)=\mathbb{E}_t[X_T^\nu+Q_T^\nu(S_T-AQ_T^\nu)-\phi\int_t^T(Q_s^\nu)^2ds]$ (p. 281). Its terminal payoff lacks the half-spread that the book's terminal condition $x+q(S-\psi/2)-Aq^2$ (p. 282) charges, and its ansatz and h-equations follow the terminal condition. This entry therefore uses the corrected payoff of the Statement. There every share pays $\psi/2$, whether sold before or at T: the total charge is the constant $\psi Q_0/2$, and the spread does not affect the control. Under the printed payoff the natural decomposition is a different one, $g:=H-x-qS$ with $g(T,\mu,q)=-Aq^2$, which gives

$$\partial_t g+\mathcal{L}^\mu g+b\mu q-\phi q^2+\sup_\nu\Big\{-\kappa\nu^2-\Big(bq+\partial_q g+\frac{\psi}{2}\Big)\nu\Big\}=0,\qquad \nu^{\mathrm{pr}}=-\frac{bq+\partial_q g+\psi/2}{2\kappa}.$$

  Here $g_2=h_2$, and $g_1+\psi/2$ solves the h_1 equation with terminal value $\psi/2$, so

$$\nu_t^{\mathrm{pr}}=\nu_t^*-\frac{\psi}{4\kappa}\,e^{-\frac{1}{\kappa}\int_t^T\xi(u)\,du}\,:$$

shares held to T escape the half-spread, so selling slows, most near T. Both policies were checked against a direct discretised optimisation with $\mu = \sigma = 0$. With $\kappa = 0.01$, $\phi = 0.001$, $A = 0.05$, $b = 0.02$, $\psi = 0.01$ and $T = Q_0 = 1$, the inventory left at T is 0.195 under the corrected payoff and 0.245 under the printed one; direct and feedback paths agree to four decimals.

- The equation prints $-\partial_q H^*$ inside the supremum. The inventory dynamics $dQ = -\nu\, dt$ give $-\nu\, \partial_q H^*$, and the book's own next step on p. 283 is consistent only with $-\nu\, \partial_q H^*$.
- Notation: p. 282 says "add the bid–ask spread ϕ" where the spread is ψ and ϕ is the running penalty. The equation writes k for κ and b_t for b, and the terminal conditions for h_0 and h_1 are printed at t rather than T.
- The half-spread enters $\hat{S}$ linearly in ν, so a negative rate (buying back) would be credited the half-spread. The spread model is economically coherent only for one-directional selling, which the formal benchmark does not impose (Conditions).
- The printed closed form $h_2 = -\xi - b/2$, $\xi(t) = \sqrt{\kappa\phi}\,(1+\zeta e^{2\gamma(T-t)})/(1-\zeta e^{2\gamma(T-t)})$, $\zeta = (A-b/2+\sqrt{\phi\kappa})/(A-b/2-\sqrt{\phi\kappa})$ gives $h_2(T) = A - b$, not $-A$. The Riccati solution has $\xi(t) = \sqrt{\kappa\phi}\,(1+\zeta e^{-2\gamma(T-t)})/(1-\zeta e^{-2\gamma(T-t)})$ with $\zeta = (A - b/2 - \sqrt{\kappa\phi})/(A - b/2 + \sqrt{\kappa\phi})$; equivalently, the printed expression with the sign of ξ reversed. Checked against numerical integration.
- The boxed h_1 "when $b = 0$" omits the factor b: with $b = 0$ the source term vanishes and $h_1 \equiv 0$, so the flow drops out. The general solution is the h_1 above.

The legacy notes attribute the model to Cartea and Jaimungal. Their papers and book (`crypto-cartea`) were not consulted for it; this entry follows Lehalle and Laruelle's account. The legacy taxonomy's "learn the policy or value function from simulated or historical interaction" conflates simulator interaction with learning from historical data. Its "hybrid execution" (control structure for constraints, machine learning for forecasts, fills and regime detection) is not in the passages read; it is kept in the artefact and stubbed in the pathway's backlog.

See also. mean–variance execution, the discrete deterministic benchmark, owned by the section overview `execution-trading-costs`; price impact (permanent, temporary and transient impact); trading envelope (where a schedule sits in a trading algorithm); implementation shortfall.

Further reading. The legacy blocks it replaces, verbatim, and the numerical checks: `notes/artefacts/reinforcement-learning-trading/`.

Index. optimal execution; stochastic control; Cartea–Jaimungal; order flow; Hamilton–Jacobi–Bellman equation; value function; policy reward; trading rate; permanent impact; temporary impact; terminal inventory penalty; running inventory penalty; Riccati equation; Feynman–Kac; Almgren–Chriss limit; reinforcement learning; model-based RL; model-free RL; market simulator

Status. Verification: partly checked 2026-10-15 (against the text layer and page images of a local copy of `alpha-laruelle`, 2018 World Scientific printing, app. A, sec. A.7.3, pp. 281–284, with the HJB reduction, the Riccati and Feynman–Kac solutions and the parameter paths checked numerically; follow-up the same day: the terminal payoff made consistent, the finite-value regime, the variance decomposition and the units, with independent numerical checks; and `wide-hambly-xu-yang` in the arXiv version 2112.04553v4 of 28 February 2023, not the journal version, pp. 2–4, 7, 32–35 and 49, page images of pp. 32–33). Stability: draft. Currency: model timeless; the survey's account of RL as of February 2023.

2.5 Types of financial data

Definition 2.14 Four types of financial data: fundamental, market, analytics and alternative data, with their timing, cost and uniqueness

Statement. López de Prado orders four "essential types of financial data" by increasing diversity [`alpha-afml`, chap. 2, sec. 2.2, Table 2.1, pp. 23–24]:

- *fundamental data*: assets, liabilities, sales, costs and earnings, macro variables; "mostly accounting data, reported quarterly" and "reported with a lapse" (p. 23);
- *market data*: prices, yields and implied volatilities, volume, dividends and coupons, open interest, quotes and cancellations, aggressor side; "all trading activity that takes place in an exchange . . . or trading venue" (p. 24);
- *analytics*: analyst recommendations, credit ratings, earnings expectations, news sentiment; "derivative data, based on an original source", processed for the user and "not readily available from an original source" (p. 25);

- *alternative data*: satellite and CCTV images, web searches, social media, metadata; "primary information, that is, information that has not made it to the other sources" (p. 25).

He takes the division of alternative data into data produced by individuals, by business processes and by sensors from Kolanovic and Krishnamachari (2017) (p. 25).

Conditions. *Timing.* Fundamental data must be used only from the moment each point was released; López de Prado's example is Bloomberg, which indexes fundamental data by the last date in the report, "which precedes the date of the release (often by 1.5 months)" (p. 23). It is also backfilled (missing values assigned values "unknown at that time") and reinstated (corrections overwriting the first release), so a dataset that keeps only final values assigns information to dates when it was not known; some vendors keep every release, for example three values for a quarterly GDP figure (p. 24). *Cost and use.* Analytics may be costly, their methodology "biased or opaque", and "you will not be the sole consumer"; alternative data raise cost and privacy concerns (p. 25). Velu, Hardy and Nehren add that the value of an alternative dataset depends on the user's resources and "maybe more importantly—on the uniqueness of the data set. As more investors get access to it, the harder it becomes to extract meaningful alpha" [`s3-velu`, chap. 1, sec. 1.4.4, pp. 43–44].

Practice. López de Prado's judgements are opinions, not findings: fundamental data being "extremely regularized and low frequency" leaves "rather unlikely that there is much value left", though it may help combined with other data (p. 24); FIX market data is "appealing" because it is "not trivial to process" and abundant, "over 10TB being generated on a daily basis" (p. 25, no source or market given); and "a dataset may be useful if it annoys your data infrastructure team", since competitors may not have tried, gave up or processed it wrongly (p. 25). Velu, Hardy and Nehren list fundamental and analyst inputs with their own hedges (pp. 42–43): key ratios (EPS, P/E, P/B) for cross-sectional comparison; analyst recommendations, individually "noisy" but in aggregate a consensus whose changes "can have a direct impact on price"; earnings estimates and surprises; holdings disclosed on Form 13F, where sudden changes "might indicate changes in sentiment by sophisticated investor"; insider purchases and sales, "thought to be an indicator of future stock price moves"; and credit rating downgrades, which "generally have a negative impact on equity prices". These are claims to test with the pathway's evidence readings, not results.

Pitfalls. The legacy notes generalised the Bloomberg indexing lag to all fundamental data, turned "over 10TB" into "∼10TB" and an appealing feature into a burden, moved the "annoys your data infrastructure team" line under cost and privacy, dropped the crowding point about analytics, and stated Velu, Hardy and Nehren's holdings, insider and rating claims without their hedges. They also merged López de Prado's four types with Velu, Hardy and Nehren's reference, market, derived and fundamental data without saying so; the two taxonomies are kept apart here. Filing lags for holdings and insider data are a `point-in-time-data` matter (backlog).

See also. trading reference data; market data levels; derived market statistics; activity-sampled bars.

Further reading. The pathway's required readings, beginning with `wide-kolanovic-big-data` and `rf-monk-prins-rook-2019`, and the evidence on how alternative-data edges decay are unread (backlog). The legacy file is kept in `notes/artefacts/statistical-learning-foundations/`.

Index. fundamental data; market data; analytics; alternative data; release lag; backfilling; reinstated values; data vintages; FIX messages; BWIC; primary information; data uniqueness; alpha decay; analyst recommendations; earnings surprise; Form 13F; insider trades; credit ratings

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-afml`, chap. 2, pp. 23–25, and `s3-velu`, CRC Press, copyright page © 2020 in the local copy (the catalogue records a 2021 copyright), ISBN 9781498737166, chap. 1, pp. 42–44, printed page = PDF page −17). Stability: draft. Currency: practitioner descriptions as of 2018–2020; the 10TB figure undated and unsourced.

Chapter 3

Rates: yield curves, term premia and derivatives

Pathways drawn on: the section overview `rates-credit-mortgages` (five draft entries from the migration: interest-rate measures, reference rates and compounding, with time-value identities derived here; forward rates and FRAs; duration and convexity; U.S. day count and quotation conventions; and OTC clearing and collateral, from Hull's 11th Global edition with the New York Fed's reference-rate methodology and the CPSS–IOSCO principles; none of the overview's readings has been studied). The entries came from legacy Hull notes mapped to the options overview and from an unsourced legacy time-value file; its return measures went to `performance-manager-evaluation`. No legacy notes for this section remain.

3.1 Interest rates, discounting and rate risk

Definition 3.1 Interest-rate measures: reference rates, compounding conventions, zero rates, bond yields and par yields, with time-value identities derived here

Statement. An interest rate states what a borrower promises to pay; the extra amount over a risk-free rate for credit risk is the *credit spread* [s7-hulloptions, chap. 4, "Interest Rates", sec. 4.1, p. 98]. Hull's types of rate (pp. 98–99):

- *Treasury rates*, at which a government borrows in its own currency; "it is usually assumed" that a developed country will not default on such obligations;
- *overnight rates*, from banks lending reserves to one another overnight; in the United States the federal funds rate, which the Federal Reserve "may intervene" to move;
- *repo rates*, secured: securities are sold and bought back at a slightly higher price, the difference being the interest; overnight and term repos.

The *n-year zero* or spot rate is earned on an investment with no intermediate payments; a bond's theoretical price discounts each cash flow at its own zero rate; its *yield* is the single rate that reprices it to its market price; its *par yield* is the coupon that prices it at par (secs. 4.5–4.6, pp. 104–106). An amount A at rate R compounded m times a year grows to $A(1+R/m)^{mn}$ in n years, and to Ae^{Rn} with continuous compounding; with $m=1$ the rate is the *equivalent annual* rate (sec. 4.4, pp. 101–103).

Notation. R_m a rate compounded m times a year, R_c the equivalent continuous rate: $R_c = m\ln(1+R_m/m)$ and $R_m = m(e^{R_c/m}-1)$ (eqs. (4.3)–(4.4), p. 104). Interpretation, derived here and not from a source read: the *effective annual rate* (EAR) is the annual-compounding equivalent, $\text{EAR} = (1+R_m/m)^m - 1 = (1+r_p)^m - 1$ for a periodic rate $r_p = R_m/m$, and $e^{R_c}-1$ in the continuous case; present value is $PV = FV_N(1+R_m/m)^{-mN}$ or $FV_N e^{-R_c N}$. For a level payment A at a periodic rate $r>0$ over N periods:

$$FV_N^{\text{ord}} = A\,\frac{(1+r)^N-1}{r}, \qquad FV_N^{\text{due}} = FV_N^{\text{ord}}(1+r), \qquad PV^{\text{ord}} = A\,\frac{1-(1+r)^{-N}}{r}, \qquad PV^{\text{perp}} = \frac{A}{r},$$

where an ordinary annuity pays at the end of each period (first payment one period from now) and an annuity due at the start (first payment now).

Conditions. The annuity factors need $r \neq 0$; as $r \to 0$ they tend to N (each payment simply summed), and the perpetuity value A/r is the limit of PV^{ord} as $N \to \infty$, which exists only for $r>0$. Checked: at $r=5$ percent and $N=10$, the ordinary and due future values of 100 a period are 1,257.79 and 1,320.68 both by formula and by summing the payments; the future-value factor tends to 10.46, 10.0045, 10.000005 for

$r = 10^{-2}, 10^{-4}, 10^{-7}$; perpetuity partial sums are 1,825.59, 1,999.88 and 2,000.00 after 50, 200 and 1,000 periods against $A/r = 2{,}000$. Monthly compounding at 6 percent gives an EAR of 6.1678 percent, continuous 6.1837 percent. Hull's examples check: 10 percent semiannual is 9.758 percent continuous (p. 104); the two-year bond on his Table 4.2 zero curve prices at 98.39, yields 6.76 percent and has a par yield of 6.87 percent (pp. 105–106).

Practice. *Reference rates.* Hull's description of the effective federal funds rate as the volume-weighted *average* of brokered transactions (p. 99) is out of date. The New York Fed changed the EFFR's data source and methodology with the rate for 1 March 2016; it is now a volume-weighted *median*, the rate at the 50th percentile of dollar volume, of overnight federal funds transactions reported on the FR 2420 [`exp-sofr-data`, EFFR and OBFR methodology]. SOFR, "a broad measure of the general cost of financing Treasury securities overnight", is likewise a volume-weighted median, of tri-party repo, GCF Repo and FICC-cleared bilateral (DVP) Treasury repo transactions, with some specials filtered out [`exp-sofr-data`, SOFR description]; Hull calls it an important volume-weighted median of overnight repo rates (p. 99). Derivatives traders do not use Treasury rates as risk-free rates, because tax and regulatory factors make them artificially low; they use rates derived from overnight rates (sec. 4.3, p. 101), and the new overnight reference rates are backward-looking (p. 101).

Pitfalls. The legacy table called EAR the "expected annual rate"; it is the effective annual rate, a conversion of compounding, not an expectation. The legacy repeated Hull's EFFR definition, superseded in 2016 as above. "Treasury rates of developed countries are regarded as risk-free" holds for default in their own currency, as Hull states it, not for price risk, and not as the rates used to discount derivatives. The legacy "future value (continuous)" used a "stated" rate: it must be the continuously compounded rate, not a rate quoted with another frequency. Compounding conventions must not be mixed within one calculation.

See also. forward rates and FRAs; duration and convexity; U.S. day count and quotation conventions; securities lending and short sales (repo and lending kept apart).

Further reading. The pathway `curve-term-structure` owns bootstrapping and curve building beyond Hull's illustration. The legacy files are kept verbatim in the section folders of `notes/artefacts/`.

Index. interest rate; credit spread; Treasury rate; overnight rate; federal funds rate; effective federal funds rate; EFFR; repo rate; SOFR; reference rate; compounding frequency; continuous compounding; effective annual rate; EAR; present value; future value; annuity; annuity due; perpetuity; zero rate; spot rate; bond price; bond yield; par yield

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 4, pp. 98–106, printed page = PDF page −1; the New York Fed's reference-rate methodology page, `exp-sofr-data`, retrieved 2026-10-15; the time-value identities derived and checked numerically, with no source for them). Stability: draft. Currency: conventions timeless; the reference-rate methodology as published on 15 October 2026.

Model 3.2 Forward rates and forward rate agreements: forward rates from zero rates, the instantaneous forward, FRA payoff, value and timing, and theories of the term structure

Statement. With continuously compounded zero rates R_1, R_2 for maturities $T_1 < T_2$, the forward rate for the period between them is [`s7-hulloptions`, chap. 4, "Interest Rates", sec. 4.8, pp. 109–110, eqs. (4.5)–(4.6)]

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} = R_2 + (R_2 - R_1)\frac{T_1}{T_2 - T_1}, \qquad R_F(T) = R + T\frac{\partial R}{\partial T} = -\frac{\partial}{\partial T}\ln P(0,T),$$

the second being the instantaneous forward rate, with $P(0,T) = e^{-RT}$ the zero-coupon bond price. A *forward rate agreement* exchanges a fixed rate R_K for a reference rate observed at a future time, both applied to a principal L for a period of length τ; the principal is not exchanged (sec. 4.9, pp. 110–112). Derived from Hull's definitions (pp. 111–112), the party receiving fixed receives $\tau(R_K - R_M)L$ at the end of the period, where R_M is the realised reference rate (negative when it pays), and the FRA is worth the *present value* of $\tau(R_K - R_F)L$, with R_F the current forward rate; the payer of fixed holds the negative (p. 112).

Notation. Rates in R_F's formula are continuously compounded; in the FRA, R_K, R_F and R_M are expressed with the compounding of the period τ (quarterly for a three-month rate, p. 111). $P(0,T)$ discount factor.

Conditions. The FRA value rests on the result that an FRA can be valued by assuming the forward rate is realised; it needs the discount rate for the payment date (Hull's Example 4.3 discounts at the 2-year risk-free rate). Interpretation: this is a single-curve illustration, in which the forward reference rate and the discount rate come from consistent curves; current practice separates projection and discounting curves and, for backward-looking compounded overnight rates such as SOFR, the floating amount is known only at the end of the period (Hull p. 101). Settlement timing changes the amount: Hull's LIBOR FRA, known at the start of the period, was paid at the start as the present value of the end-of-period amount (p. 111, n. 4). Hull presents three term-structure theories, none established as a result: *expectations* theory (forward rates equal expected future zero rates), *market segmentation* (no necessary relation between maturities) and *liquidity preference* (forward rates above expected future zero rates, consistent with upward-sloping curves being more common); Hull calls

expectations and segmentation conjectures and finds liquidity preference "most appealing", consistent with the empirical observation (sec. 4.12, pp. 117–119).

Practice. Checked: from zero rates of 3, 4, 4.6, 5 and 5.3 percent for 1–5 years, forward rates are 5, 5.8, 6.2 and 6.5 percent (Table 4.5, p. 109); the instantaneous forward of an illustrative curve $R(T) = 0.03 + 0.005T$ at $T = 2$ is 5.0 percent by finite difference of $-\ln P$ and by $R + TR'$. Hull's Example 4.3, receiving 5.8 against a forward of 5.0 percent (semiannual) on $100 million for six months, discounted at 4 percent continuous for 2 years, gives $369,247; Hull prints $369,200 (p. 112). Locking in forward rates by borrowing and lending at different maturities, and the Orange County yield-curve plays, are Hull's illustrations of the risk of betting against forwards (pp. 110–111).

Pitfalls. Hull's own example has the payment direction reversed: Party A receives the fixed 3 percent and pays LIBOR, so with LIBOR at 3.5 percent A pays, rather than receives, $125,000 (p. 111); his formula for the FRA's value (p. 112) has the correct sign; the author's errata were not checked. The legacy wrote the FRA value as $\tau(R_K - R_F)L$ without discounting; that is the payoff if the forward is realised, not its present value. The legacy liquidity-preference line said forward rates "are" greater than expected future spot rates; Hull presents this as what the theory implies. Evidence on term premia, which decides between the theories, belongs to `curve-term-structure` (backlog).

See also. interest-rate measures; duration and convexity; cost-of-carry forward price (valuing a forward by assuming the forward is realised).

Further reading. Multi-curve FRA and swap valuation under current conventions, and the term-premium evidence, are in the overview's backlog.

Index. forward rate; forward interest rate; instantaneous forward rate; zero-coupon bond price; discount factor; forward rate agreement; FRA; reference rate; settlement timing; single-curve; expectations theory; market segmentation theory; liquidity preference theory; yield curve play

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 4, pp. 101 and 109–119; Table 4.5 and Example 4.3 recomputed; the instantaneous forward checked by finite difference). Stability: draft. Currency: formulas timeless; LIBOR-based conventions historical, SOFR conventions as Hull states them in 2021.

Model 3.3 Duration and convexity: Macaulay and modified duration, convexity, the second-order price approximation, and the parallel-shift assumption

Statement. For a bond paying c_i at t_i with price $B = \sum_i c_i e^{-yt_i}$ at continuously compounded yield y, the duration is the present-value-weighted average time of payment [s7-hulloptions, chap. 4, "Interest Rates", sec. 4.10, pp. 112–115, eqs. (4.7)–(4.13)]:

$$D = \sum_i t_i \frac{c_i e^{-yt_i}}{B}, \qquad \frac{\Delta B}{B} \approx -D\,\Delta y, \qquad C = \frac{1}{B}\frac{d^2B}{dy^2} = \frac{\sum_i c_i t_i^2 e^{-yt_i}}{B}, \qquad \frac{\Delta B}{B} \approx -D\,\Delta y + \tfrac{1}{2}C(\Delta y)^2.$$

With y compounded m times a year, $\Delta B \approx -BD^*\Delta y$ with modified duration $D^* = D/(1+y/m)$ (pp. 114–115); dollar duration is BD^* (p. 115). Convexity is from sec. 4.11 (pp. 116–117).

Notation. D Macaulay duration (years); D^* modified duration; C convexity (years squared); Δy a change in the bond's own yield. For continuous compounding $D^* = D$.

Conditions. The duration relation is a first-order approximation, accurate for small changes; convexity adds the second-order term. For a portfolio, the weighted-average duration assumes all yields change by about the same amount, which for bonds of widely different maturity means a small parallel shift of the zero curve; matching durations protects against small parallel shifts only, and matching convexity as well against larger parallel shifts, but neither against non-parallel shifts (pp. 115–117).

Practice. Checked on Hull's Table 4.6 bond (3-year 10 percent coupon, yield 12 percent continuous): $B = 94.213$, $D = 2.653$ years and, computed here, $C = 7.570$. For $\Delta y = +0.001$ the exact price is 93.9634 against 93.9631 from duration, as in Hull's Example 4.4 (p. 114). For $\Delta y = +0.02$ the exact price is 89.354 against 89.214 from duration and 89.357 with convexity; for -0.02, 99.357 against 99.212 and 99.355. With the yield expressed semiannually (12.3673 percent), $D^* = 2.499$, and a 10 bp rise gives 93.978 both exactly and by modified duration (Example 4.5, p. 115).

Pitfalls. The legacy wrote the duration and convexity relations as equalities and wrote $\Delta B = \frac{dB}{dy}\Delta y$ without the approximation; Hull marks them as approximate. A change in a bond's yield is not the same as a shift in the zero curve. Interpretation, derived and checked here: under a parallel shift Δ of the zero curve the bond's yield moves by exactly Δ, for every Δ, if and only if the zero rates at all of the bond's payment dates equal its yield. A flat curve is therefore sufficient but not necessary: a zero-coupon bond has $y = R(T)$ and moves one for one on any curve, and the curve may take any shape between payment dates. For a coupon bond on a curve that is not level across its payment dates, the yield moves by only about Δ: 0.9996Δ for Hull's Table 4.2 bond and 0.986Δ for a 10-year 5 percent annual-coupon bond on a curve rising 40 bp a year from 1.4 percent. Duration against the bond's own yield, $-B^{-1}\,\partial B/\partial y$, and sensitivity to a parallel zero-curve shift,

$-B^{-1}\,\partial B/\partial\Delta$, differ by that factor: 8.13 against 8.01 years for the 10-year bond. Modified duration applies only when y uses the stated compounding; mixing a continuously compounded D with an m-compounded Δy misstates the change.

See also. interest-rate measures; forward rates and FRAs; minimum-variance hedge ratio (duration-based futures hedges, Hull chap. 6, not read).

Further reading. Key-rate and factor durations, which handle non-parallel moves, are not covered here; Hull's duration-based futures hedging (sec. 6.4) was not read. The legacy file is kept verbatim in `notes/artefacts/volatility-options/`.

Index. duration; Macaulay duration; modified duration; dollar duration; convexity; yield change; parallel shift; non-parallel shift; immunisation; bond price sensitivity

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 4, pp. 112–117; Table 4.6 and Examples 4.4–4.5 recomputed; convexity and the ±200 bp comparisons computed here; the yield response to parallel zero-curve shifts computed 2026-10-16 in `notes/artefacts/rates-credit-mortgages/yield-shift-check.py`). Stability: draft. Currency: timeless.

Convention 3.4 U.S. day count and price quotation: actual/actual, 30/360 and actual/360; Treasury bill discount quotes; Treasury bond prices in thirty-seconds; clean and dirty prices

Statement. A day count X/Y gives interest between two dates as the days between them, counted by X, over the days in the reference period, counted by Y, times the interest for the reference period [s7-hulloptions, chap. 6, "Interest Rate Futures", sec. 6.1, pp. 152–153]. In the United States, Hull lists actual/actual (in period) for Treasury bonds, 30/360 for corporate and municipal bonds and actual/360 for money-market instruments. Treasury bills are quoted as a discount rate on face value: $P = \frac{360}{n}(100 - Y)$, with P the quoted price, Y the cash price per \$100 and n the days to maturity. Treasury bonds are quoted in dollars and thirty-seconds per \$100 face, and the quote is the *clean* price; the cash or *dirty* price adds interest accrued since the last coupon (p. 154).

Conditions. These are U.S. conventions as Hull gives them; conventions differ by country and instrument (money-market instruments on actual/365 in Australia, Canada and New Zealand; euro and sterling bonds usually actual/actual, p. 153). Futures quotes have their own conventions (sec. 6.2, p. 155).

Practice. Checked with Hull's examples: \$4 earned March 1 to July 3 on an 8 percent \$100 bond is $\frac{124}{184} \times 4 = 2.6957$ on actual/actual and $\frac{122}{180} \times 4 = 2.7111$ on 30/360 (p. 153); a 90-day bill at a cash price of 99 is quoted at $\frac{360}{90}(100 - 99) = 4$ (p. 154); 120-05 is $120\frac{5}{32}$, \$120,156.25 for \$100,000 face; accrued interest of $\frac{54}{181} \times 5.50 = 1.64$ makes a clean price of 155.50 a dirty price of 157.14 (p. 154).

Pitfalls. The legacy wrote "P quoted price, Y cash price" correctly but described the discount rate only as "interest as a percentage of final face value"; Hull adds that the true return for the period is on the price paid, 2.064 percent rather than 2.022 percent in his example (p. 154). Hull notes that SOFR compounding assumes 360 days a year (p. 100); day counts for SOFR and other reference rates are set per product, not by the bond conventions above.

See also. interest-rate measures; fixed-income market structure.

Further reading. The legacy file is kept verbatim in `notes/artefacts/volatility-options/`.

Index. day count; actual/actual; 30/360; actual/360; Treasury bill; discount rate; Treasury bond; thirty-seconds; clean price; dirty price; accrued interest

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 6, pp. 152–155; examples recomputed). Stability: draft. Currency: U.S. conventions as of Hull's 2021 account; no current market convention document checked.

Mechanism 3.5 OTC derivatives clearing and collateral: central counterparties, bilateral ISDA/CSA clearing, variation and initial margin, and post-crisis reforms as Hull describes them

Statement. An OTC trade can be presented to a central counterparty (CCP) or cleared bilaterally [s7-hulloptions, chap. 1, "Introduction", sec. 1.2, p. 25]. A CCP accepting a trade between A and B becomes counterparty to both and takes on the credit risk of each; its members post initial margin and daily variation margin and contribute to a guaranty fund, and non-members clear through members (chap. 2, sec. 2.5, pp. 54–55). In the CPSS–IOSCO principles a CCP "interposes itself between counterparties", ensuring the performance of open contracts through novation or an open-offer system, and has "the potential to reduce significantly risks to participants" through multilateral netting and risk controls [exp-pfmi, para. 1.13, p. 9]. Bilaterally cleared trades are covered by a master agreement, usually ISDA's, with a credit support annex (CSA) requiring collateral, typically valued daily (p. 55).

Conditions. *Reforms, as Hull reports them* (pp. 26 and 55–56): standardised OTC derivatives between U.S. financial institutions must, "whenever possible", trade on swap execution facilities; most parts of the world require CCP clearing of most standardised derivatives between financial institutions; trades must be reported to a repository; and from 2016 regulations required initial and variation margin for bilaterally cleared transactions between financial institutions, with initial margin held by a third party. Hull notes that "financial institutions" include banks, insurers, pension funds and hedge funds, while most non-financial corporations and some FX

transactions are exempt (p. 56, n. 4). None of these rules was checked against its primary text here (backlog). *Interest.* Initial margin in cash usually earns interest; futures variation margin does not, being daily settlement, while OTC variation margin in cash does, because OTC trades are not settled daily (p. 57). Securities used as collateral are subject to haircuts (p. 57).

Practice. Interpretation: clearing and collateral mitigate counterparty risk rather than eliminate it. Margin is sized to cover most, not all, losses on a default (Hull's clearing-house example aims at about 99 percent certainty, p. 54), a guaranty fund mutualises the rest, and the CCP itself concentrates risk; Hull cites Duffie and Zhu for the dependence of the benefit on the number of CCPs and the share of trades cleared (p. 57, n. 5). Collateral also creates funding and liquidity demands: LTCM had to post collateral on both legs of its trades in 1998 and, being highly leveraged, had to close out positions (Business Snapshot 2.2, p. 56).

Pitfalls. The legacy said the clearing house "takes care of the credit risk"; Hull's phrase is that it does so "by requiring ... margin" (p. 25), which reduces the risk without removing it. The legacy reform list dropped Hull's "whenever possible" and "in most parts of the world" qualifiers. Futures-specific margin rules (maintenance margin, margin calls, interest on futures balances) are in futures margin and settlement.

See also. futures margin and settlement; derivative contracts; common-investor contagion (margin calls in deleveraging).

Further reading. The primary rules (Dodd–Frank swap clearing and execution, reporting, the uncleared-margin framework) are in the overview's backlog. The legacy file is kept verbatim in `notes/artefacts/volatility-options/`.

Index. OTC derivatives; central counterparty; CCP; novation; guaranty fund; default fund; bilateral clearing; ISDA master agreement; credit support annex; CSA; collateral; initial margin; variation margin; haircut; swap execution facility; trade repository; uncleared margin rules

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 1, pp. 25–26, and chap. 2, pp. 54–57; and `exp-pfmi`, April 2012, para. 1.13, p. 9, read in slice 27). Stability: draft. Currency: Hull's account of the reforms as of 2021; no primary rule checked.

Chapter 4

Systematic equities

Pathways drawn on: `equity-long-short-construction` (one draft entry from the migration: Narang's approaches to combining alpha signals into one forecast, linear, conditional, rotational and machine-learning, with the case for not combining them at all; none of the pathway's readings yet studied). The entry came from a legacy file on Narang's alpha models mapped to the portfolio-construction overview; signal combination is this pathway's outcome, so it was routed here.

4.1 Combining alpha signals

Method 4.1 Combining alpha signals into one forecast: linear weights, conditional and rotational models, machine learning, or separate factor portfolios

Statement. Narang treats blending as the problem of turning several alpha models' forecasts for an instrument into one composite forecast, which then goes to portfolio construction [s7-narang, chap. 3, "Alpha Models: How Quants Make Money", sec. 3.5, pp. 64–68]. He names four approaches:

- *linear models*, "by far the most common": each factor contributes additively, its weight usually set by multiple regression on past returns. In his example a yield factor forecasting +20 percent at weight 0.7 and a trend factor forecasting −10 percent at weight 0.3 give $0.7 \times 20 + 0.3 \times (-10) = +11$ percent (pp. 65–66; checked). Equal weights, equal-risk weights and discretionary weights are special cases (p. 66);
- *nonlinear models*: *conditional* models, where one factor's reading sets another's weight, for example acting on value only when momentum agrees, which in his example gives no signal, or +20 percent if the trend agreed (Exhibit 3.10, pp. 66–67, page image); and *rotation* models, whose weights are refitted on recent data and "usually" favour the factors that have done best recently, "a form of trend following in the timing of alpha factors" (p. 67);
- *machine-learning* models that choose the weights, which he calls "both more common and more successful" than machine learning used to forecast markets (pp. 67–68);
- not combining at all: building one portfolio per alpha model and combining those portfolios, as in signal-portfolio allocation (p. 64).

Signal mixing produces one forecast per instrument; portfolio construction then sizes positions from forecasts, risk and costs (p. 68).

Conditions. Narang's argument for blending is that two partly right forecasts can together be right more often (Exhibit 3.9, p. 65), and that "any reasonably intelligent combination of alphas" is likely to beat any one of them over time (p. 64). Interpretation: that holds when the forecasts each carry information and their errors are not highly correlated; combining a weak forecast with a strong one, or two near-duplicates, need not help, and weights fitted to past returns carry estimation error. His Exhibit 3.9 is a diagram, not evidence.

Practice. Interpretation: "linear" describes the form of the combination, not the factors. A linear model assumes each factor's contribution does not depend on the others (no interaction), which is what Narang means by inclusion being "independent" (p. 65); it does not need the factors to be statistically independent. Correlated factors can be combined linearly, though highly correlated ones make regression weights unstable. A conditional rule is an interaction; a rotation rule is a time-varying weight. Combining signals into one forecast before optimising, or optimising on the signals separately, is the mixing-against-integrating choice that this pathway's checkpoint asks to compare (stub).

Pitfalls. The legacy "linear models: require independence of factors" misreads Narang's additivity as statistical independence. Rotation by recent performance is a bet that factor performance trends; Narang gives

no evidence for it, and refitting weights on recent data counts as further trials (parameter overfitting). His claim that machine learning works better for weighting than for forecasting is a judgement without data in the chapter. He refers to Chapter 7 for portfolio construction techniques (p. 64), but they are in Chapter 6; Chapter 7 is on execution. The legacy block's third item, on the cost of machine-learning strategies and "microscopic alpha", comes from López de Prado rather than Narang and is in the strategy production lifecycle.

See also. alpha models; alpha strategy specification (conditioning variables); signal-portfolio allocation; rule-based sizing (equal and equal-risk weights); mean–variance optimisation.

Further reading. The pathway's readings on forecast scaling and signal combination (Grinold, Qian and others) are unread; stubs in the pathway note. The legacy file, verbatim, is in `alpha-models.tex` under the section key `portfolio-construction-risk` in `notes/artefacts/`.

Index. signal combination; alpha blending; signal mixing; composite forecast; linear model; multiple regression; equal weighting; equal risk weighting; conditional model; interaction; rotation model; factor timing; machine learning; factor portfolios; mixing versus integrating

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 3, sec. 3.5, pp. 64–68, with Exhibits 3.9 and 3.10 on pp. 65 and 67 by page image; the linear example checked). Stability: draft. Currency: practitioner account as of 2024.

Chapter 5

Option pricing, volatility models and hedging

Pathways drawn on: the section overview `volatility-options` (two draft entries from the migration, from Hull's *Options, Futures, and Other Derivatives*, the overview's first reading, in its 11th Global edition: derivative contracts with their payoffs, rights and convergence, and the cost-of-carry forward price; no option-pricing, volatility or hedging material has yet been written, and none of the overview's readings has been studied). The entries came from a legacy file of Hull notes mapped to this overview; the same notes' rates, hedging and clearing material went to the rates, commodity and execution chapters. No legacy notes for this section remain.

5.1 Forwards, futures and options, and the cost of carry

Definition 5.1 Derivative contracts: spot, forward, futures and options; payoffs, rights and obligations, exercise styles, futures specification and convergence

Statement. A derivative "involves two parties agreeing to a future transaction"; its value depends on the values of underlying variables, very often traded asset prices [s7-hulloptions, chap. 1, "Introduction", p. 23]. Hull's basic contracts:

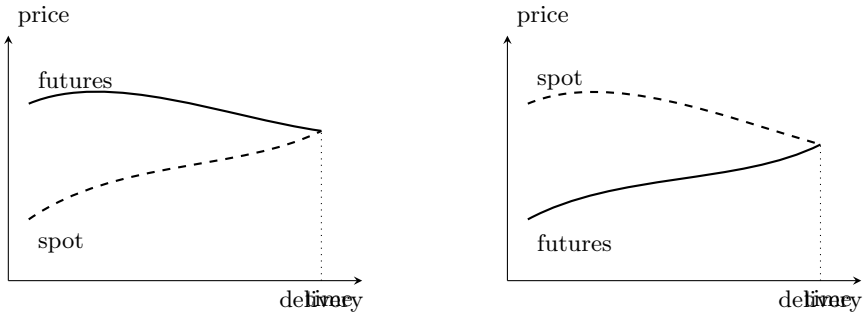
- a *spot contract* buys or sells an asset almost immediately; a *forward contract* buys or sells it at a certain future time for a certain price, the delivery price K, and trades over the counter. The long side agrees to buy and the short side to sell (sec. 1.3, p. 28). At maturity the payoff per unit is $S_T - K$ to the long and $K - S_T$ to the short, where S_T is the spot price; because it costs nothing to enter a forward, the payoff is also the total gain or loss (p. 29);
- a *futures contract* is the same kind of agreement, standardised and normally traded on an exchange, with the exchange clearing house standing between the parties (sec. 1.4, p. 30). Most positions are closed out by the opposite trade before delivery, but "it is the possibility of final delivery that ties the futures price to the spot price" (sec. 2.1, pp. 47–48);
- an *option*: a *call* "gives the holder the right to buy" and a *put* "the right to sell" the underlying asset by a certain date for the strike price. *American* options can be exercised at any time up to expiration, *European* options only on the expiration date. The holder has a right, not an obligation, and pays for it; forward and futures holders are obliged to buy or sell and pay nothing to enter apart from margin (sec. 1.5, p. 31).

The exchange specifies a futures contract's asset (with acceptable grades), contract size, delivery place and period, last trading day and price quotation; when alternatives are allowed, the short side usually chooses and files a notice of intention to deliver (sec. 2.2, pp. 48–50). For most contracts the exchange also sets daily price limits, after which trading normally stops for the day, though the exchange can change them; it also sets position limits for speculators (p. 50).

Notation. S_T spot price at maturity; K delivery (forward) or strike (option) price; for a European call the payoff is $\max(S_T - K, 0)$ and for a put $\max(K - S_T, 0)$ (interpretation of Hull's definitions; the writer's payoffs are the negatives). *Profit* is payoff less the premium paid, with the premium's time value ignored, as Hull does in his footnote (p. 32, n. 5).

Conditions. *Convergence.* As the delivery period approaches, the futures price converges to the spot price of the deliverable asset (sec. 2.3, pp. 50–51; Figure 5.1). If futures traded above spot during delivery, shorting futures, buying the asset and delivering would profit; if below, buyers who want the asset would go

long and take delivery. Interpretation: the argument needs the asset delivered to be the asset priced as spot, at the delivery location and grade, and delivery to be possible; cash-settled contracts converge through their final settlement rule instead, and when the short side chooses grade, place or timing, the futures price tracks the alternative it is expected to choose; Hull treats the timing choice in sec. 5.13 (p. 144). *Forward and futures* differ in standardisation, a single date against a range, settlement at the end against daily settlement, and credit risk (Table 2.3, p. 65); a forward gain is realised at maturity, a futures gain day by day (pp. 64–65).



(a) futures above spot (b) futures below spot

Figure 5.1: Convergence of the futures price to the spot price of the deliverable asset as delivery approaches; schematic after Hull, Figure 2.1, p. 51.

Practice. Checked with Hull's examples: a long six-month forward on £1 million at 1.2230 is worth +\$77,000 if spot ends at 1.3000 and −\$23,000 at 1.2000 (p. 29). One Apple December call contract on 100 shares, strike \$340, bought for \$2,030, pays $100 \times (400 - 340) = \$6{,}000$ if exercised at \$400, a profit of \$3,970; one September put written at \$12.70 (\$1,270 received), strike \$290, exercised at \$250, costs \$4,000, a loss of \$2,730 (p. 32). Daily settlement and margin are in futures margin and settlement; pricing in the cost of carry.

Pitfalls. The legacy definitions reversed the option rights: they gave a call the right to sell and a put the right to buy. Hull's is the opposite (p. 31). The legacy "derivatives market is much larger than the stock market in terms of underlying assets" is Hull's statement (p. 23), but the size is a notional principal, not a value: Hull gives \$558.5 trillion of OTC notional against a gross market value of about \$11.6 trillion in December 2019 (pp. 27–28). Daily price limits, delivery terms and grades are contract-specific exchange rules, not universal features; Hull's examples are CME Group and ICE contracts as of 2021, and Hull notes that whether limits are on balance good is "controversial" (p. 50).

See also. cost-of-carry forward price; futures margin and settlement; synthetic short through options (put–call combinations as forwards); OTC clearing and collateral.

Further reading. The legacy files are kept verbatim in the section's folder of `notes/artefacts/`, with the original figure.

Index. derivative; spot contract; forward contract; futures contract; option; call option; put option; American option; European option; payoff; profit; premium; long position; short position; delivery price; strike price; contract specification; notice of intention to deliver; daily price limit; position limit; convergence; delivery; forward versus futures; notional principal

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, Pearson 2022, ISBN 9781292410654, chap. 1, pp. 23–33, and chap. 2, pp. 47–51 and 64–65, printed page = PDF page −1; Hull's examples recomputed; the convergence figure redrawn). Stability: draft. Currency: definitions timeless; contract examples and market sizes as of 2019–2021.

Result 5.2 Cost-of-carry forward price: no-arbitrage forward prices with income, yield and storage, the value of an existing forward, and the limits for consumption assets

Statement. For an *investment asset*, one held solely for investment by at least some traders, with continuously compounded risk-free rate r to maturity T [`s7-hulloptions`, chap. 5, "Determination of Forward and Futures Prices", secs. 5.1–5.6, pp. 124–132]:

$$F_0 = S_0 e^{rT} \text{ (no income)}, \qquad F_0 = (S_0 - I)e^{rT} \text{ (income with present value } I\text{)}, \qquad F_0 = S_0 e^{(r-q)T} \text{ (yield } q\text{)}.$$

If F_0 is above the bound, buying the asset with borrowed money and selling forward locks in a profit; if below, shorting the asset (or, if shorting is impossible, selling it from holdings) and buying forward does (pp. 127–131). A long forward with delivery price K agreed earlier is worth $f = (F_0 - K)e^{-rT}$, for investment and consumption assets alike (sec. 5.7, p. 133). Special cases: a stock index with dividend yield q (index arbitrage, sec. 5.9, pp. 135–137); a currency, with the foreign risk-free rate r_f as the yield, $F_0 = S_0 e^{(r-r_f)T}$, interest rate parity (sec. 5.10, pp. 137–140); a commodity that is an investment asset, with storage cost net of income U in

present value, $F_0 = (S_0+U)e^{rT}$, or u as a proportion of price, $F_0 = S_0e^{(r+u)T}$ (sec. 5.11, pp. 141–142). The cost of carry c, storage plus financing less income, gives $F_0 = S_0e^{cT}$ for an investment asset (sec. 5.12, pp. 143–144).

Notation. S_0 spot price; F_0 forward (or futures) price; r the T-year zero rate, continuously compounded; q average yield over T; U, u storage cost net of income; y convenience yield; c cost of carry (r for a non-dividend stock, $r-q$ for an index, $r-r_f$ for a currency, $r-q+u$ for a commodity with income and storage, p. 144). Here S_0 and F_0 for a currency are dollars per unit of foreign currency (p. 138).

Conditions. The derivation assumes that *some* participants, such as large dealers, trade without transaction costs, pay the same tax rate on all net trading profits, borrow and lend at the same risk-free rate, and take arbitrage opportunities as they arise (sec. 5.3, p. 126). For a *consumption asset*, held to be used rather than as an investment, the second arbitrage fails: owners will not sell the commodity to buy futures, so only $F_0 \le (S_0+U)e^{rT}$ or $F_0 \le S_0e^{(r+u)T}$ holds. The convenience yield y is defined to close the gap, $F_0e^{yT} = (S_0+U)e^{rT}$, so that $F_0 = S_0e^{(c-y)T}$; it is zero for investment assets (pp. 142–144). Forward and futures prices are equal in theory when the short rate is constant or a known function of time; with random rates, futures exceed forwards when the asset is positively correlated with rates and fall below them when negatively correlated, differences Hull finds small for contracts of a few months (sec. 5.8, p. 135). Index arbitrage failed on 19–20 October 1987, when order delays and program-trading restrictions left December S&P 500 futures up to 18 percent below the index (Business Snapshot 5.4, p. 137).

Practice. Checked with Hull's examples: $40e^{0.05\times0.25} = 40.50$ (p. 128); $(900-40e^{-0.03/3})e^{0.04\times0.75} = 886.60$ (p. 131); $(50-2.162)e^{0.08\times10/12} = 51.14$ (p. 132); with 2 percent paid once in six months, $q = 2\ln 1.02 = 3.96$ percent and $F_0 = 25e^{(0.10-0.0396)\times0.5} = 25.77$ (p. 132); $f = (26.28-24)e^{-0.05} = 2.17$ (p. 133); $0.7500e^{(0.01-0.03)\times2} = 0.7206$ USD per AUD (p. 139); $(450+2e^{-0.07})e^{0.07} = 484.63$ (p. 141). When short sales are impossible the arbitrage against a forward that is too low still works through holders who sell the asset and buy forward (pp. 129 and 131).

Pitfalls. The legacy block stated both commodity arbitrages as if each always worked; for consumption commodities only the upper bound holds, and the legacy convenience-yield sentence ("reflects market expectation on future availability") is Hull's (p. 143) but applies to consumption assets only. The legacy "when a forward contract is first entered, the value is close to zero" is Hull's opening sentence (p. 133); Hull adds that K is set equal to the forward price, so $f = 0$ at inception, and the legacy dropped that. Its currency formula defined S_0 as "price of local currency in foreign currency"; in Hull both S_0 and F_0 are the domestic price of one unit of foreign currency, with r domestic and r_f foreign, and many market quotes run the other way (p. 138). Borrowing constraints, borrow fees and recall, which make shorting costly, are in lending fees and rebates.

See also. derivative contracts; arbitrage (carrying costs and bounds in Harris's terms); minimum-variance hedge ratio; futures hedges and basis.

Further reading. The theory of storage and convenience yields belongs to the commodity pathways; a backlog item in `commodity-hedging` names it. The legacy file is kept verbatim in the section's folder of `notes/artefacts/`.

Index. forward price; futures price; cost of carry; investment asset; consumption asset; known income; known yield; value of a forward contract; index arbitrage; interest rate parity; storage cost; convenience yield; short selling; forward and futures prices; delivery options

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 5, pp. 124–144; seven examples recomputed). Stability: draft. Currency: theory timeless; examples and market episodes as given.

Chapter 6

Portfolio construction and risk

Pathways drawn on, in the guide's order: the section overview `portfolio-construction-risk` (three draft entries from the migration, from Narang's chapters introducing quant trading and on alpha models: the components of a quant strategy, with a redrawn schematic, the taxonomy of alpha models, and the choices that specify an alpha strategy; none of the overview's readings yet studied); `portfolio-construction` (six draft entries from the migration, from Narang's chapters on risk models, portfolio construction and transaction cost models, with one of the pathway's readings, Boyd et al. on Markowitz at seventy, read in part for method mechanics: rule-based position sizing, mean–variance optimisation with its limits and penalties, the allocation of signal portfolios, the factor risk model, and transaction cost functions as inputs, with their units; and, from López de Prado's chapter on financial data structures, weights that allocate risk across principal components; none of the pathway's readings yet studied); and `performance-manager-evaluation` (two draft entries from the migration: relative performance evaluation, market-adjusted and risk-adjusted returns, market timing and return smoothing, from Harris; and time- and money-weighted, annualised, real and leveraged return measures, from the GIPS 2020 definitions with the formulas derived here; of the pathway's readings only the GIPS standards were read, in part). The entries came from Narang's systematic-investing notes, whose other parts went to the systematic-equities, research-engineering, execution and microstructure chapters, from a legacy execution file whose performance-evaluation blocks belong here, and from a legacy time-value file. No legacy notes for this section remain.

6.1 The structure of a quant strategy and its alpha models

Definition 6.1 Components of a quant trading strategy: alpha, risk and transaction cost models feeding portfolio construction, which interacts with execution, built on data and research

Statement. Narang's schematic of a live, production trading strategy has three modules, "an alpha model, a risk model, and a transaction cost model", which "feed into a portfolio construction model, which in turn interacts with the execution model" [s7-narang, chap. 2, "An Introduction to Quantitative Trading", sec. 2.2, Exhibit 2.1, p. 17, page image]:

- the *alpha model* predicts the future of the instruments considered, to generate returns;
- the *risk model* helps limit exposure to factors "unlikely to generate returns but could drive losses", for example a trend follower's limit on directional exposure to commodities (pp. 17–18);
- the *transaction cost model* estimates the cost of the trades needed to move from the current to the desired portfolio (p. 18);
- the *portfolio construction model* balances profit, risk and cost to choose the best portfolio to hold; the difference between the target and the current portfolio is the list of trades (Exhibit 2.2, p. 18);
- the *execution model* carries out those trades, using inputs such as their urgency and the dynamics of liquidity, "in an efficient and low-cost manner" (p. 18).

The schematic covers only the production system; data, without which black boxes are "inert and useless", and research complete it (Exhibit 2.3, p. 19). Figure 6.1 redraws Exhibit 2.1.

Conditions. Narang calls a trader a quant when both what to own and how much are decided systematically as standard practice, allowing emergency overrides (sec. 2.1, p. 16). The structure "is by no means universal": many strategies run without a transaction cost, portfolio construction or execution model, risk constraints can be built into the alpha model, and some links are recursive, such as executions feeding the cost model (p. 19).

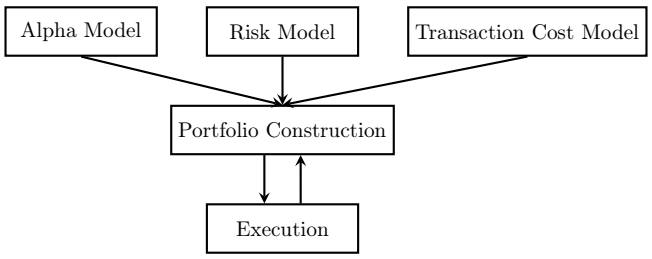


Figure 6.1: The modules of a live quant trading strategy, after Narang's Exhibit 2.1 (p. 17); data and research, his Exhibit 2.3, are not drawn.

Practice. Each module has its entries: forecasts in alpha models and their specification, combined in signal combination; risk in the factor risk model; costs in cost functions; the decision in rule-based sizing and mean–variance optimisation; execution in order exposure decisions; and the research and production chain around them in the strategy production lifecycle. Checked: Narang's Exhibit 2.2 trades follow from target minus current weights ($-25-(-30)=+5$, $25-20=+5$, $25-40=-15$, $-25-(-10)=-15$ percent).

Pitfalls. The legacy diagram drew only a downward arrow from portfolio construction to execution; Narang's exhibit and text show the two interacting, and the redrawn figure adds the return arrow. The legacy description of the risk model as "directional exposure limit on an asset class" generalises his single example. As in the factor risk model, the risk model informs the decision; the portfolio construction model selects and sizes.

See also. sources of strategy risk (model risk can arise in every module); strategy monitoring.

Further reading. The legacy overview, its chapter wrapper and its original diagram are kept verbatim in the section's folder of `notes/artefacts/`, as `systematic-investing-overview.tex`.

Index. quant trading system; black box; alpha model; risk model; transaction cost model; portfolio construction model; execution model; target portfolio; trade list; data; research; production system

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 2, secs. 2.1–2.2, pp. 16–19, with Exhibit 2.1 on p. 17 by page image; Exhibit 2.2 arithmetic checked). Stability: draft. Currency: practitioner framework as of 2024.

Definition 6.2 Alpha models: theory-driven and data-driven, and Narang's eight classes of theory-driven alpha by input

Statement. Narang defines alpha, "not conventional" by his own account, as "skill in timing the selection and/or sizing of portfolio holdings"; an alpha model is the software that does this timing systematically [s7-narang, chap. 3, "Alpha Models: How Quants Make Money", pp. 23–24]. Alpha models are *theory-driven*, starting from an economic explanation that is then tested, or *data-driven*, looking for patterns without a prior theory (sec. 3.1, pp. 25–28). Theory-driven alphas fall into eight classes, sorted by input (Exhibit 3.2, p. 29):

- *price data*: *trend* or momentum, explained by slow consensus-building or by "greater fools" (pp. 29–30); *mean reversion*, explained by temporary liquidity imbalances, such as index additions, and overshooting, with statistical arbitrage on similar stocks as its best-known form (pp. 33–35); and *technical sentiment*, from option volumes and skews, trading volume and the order book (pp. 35–37);
- *fundamental data*: *value/yield*, ratios with price in the denominator such as earnings yield, read as being paid to hold riskier instruments (pp. 38–41); *growth/sentiment*, including analysts' revisions (pp. 42–43); *supply/demand* (pp. 43–44); *quality*, from leverage, revenue diversity, management, fraud risk and quality sentiment (pp. 44–46); and *tactical/events*, one-off events such as elections or the pandemic (pp. 46–47).

Data-driven strategies are "far less widely practiced", more common at high frequency where data are abundant, and in his view mostly unsuccessful as forecasters of markets (sec. 3.3, pp. 47–52).

Conditions. The classes are Narang's framework for organising strategies, not a result: he offers explanations for trend and reversion and cites little evidence: two papers for value/yield (Fama and French, n. 5; Ball, n. 6; pp. 37–40) and his reading of research on volume as a contrarian signal (p. 37), and otherwise examples and anecdotes. Trend and reversion coexist "largely" because they operate at different horizons, longer trends with shorter oscillations (Exhibit 3.5, pp. 35–36). He stresses that every class has "relatively long periods of poor returns" because the behaviours are "unstable and episodic" (p. 32), and that some strategies once called alpha, such as longer-term trend, are now treated by many as risk premia or "smart beta" (p. 24).

Practice. Interpretation: whether a class is used as an alpha to forecast, a premium to harvest or a risk to limit depends on the strategy, as in the factor risk model. His relative-value example combines value and financing: "when done on a relative basis … this strategy is also known as a carry trade", earning the yield spread (p. 40); with the euro at 4.25 percent and the dollar at 2 percent, the carry is 2.25 percent (checked).

Pitfalls. The legacy block listed five classes (trend, mean reversion, value/yield, growth, quality); the third edition has eight, adding technical sentiment, supply/demand and tactical/events. The legacy sentence that a moving-average crossover "has less than one point of return for every point of downside risk" attaches to one rule what Narang says of "the typical successful trend follower" (p. 32), with no data. The legacy "buying undervalued security and selling overvalued security is a carry trade" drops the context of his sentence, in which "this strategy" is the yield strategy, so that the long earns the higher yield; interpretation, not checked against the factor readings: value (cheapness against fundamentals) and carry (the return if prices do not change) are distinct signals in later work, though they can coincide. His performance figures for Seykota, Mint and Renaissance (pp. 31–32) are anecdotes with press or web sources. "In general, strategies that use data-mining techniques to forecast markets do not work" (p. 49) is his judgement. Evidence for each class belongs to its specialist pathway (stubs in the overview's note).

See also. alpha strategy specification; signal combination; hypothesis-driven strategy research (theory-driven research as a method); parameter overfitting.

Further reading. The legacy file, verbatim, is in `alpha-models.tex` in the section's folder of `notes/artefacts/`.

Index. alpha; alpha model; theory-driven; data-driven; data mining; trend following; momentum; mean reversion; statistical arbitrage; technical sentiment; value; yield; earnings yield; carry; quant long/short; growth; estimate revisions; supply and demand; quality; tactical events; smart beta; risk premia

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 3, secs. 3.1–3.3, pp. 23–52, with Exhibit 3.2 on p. 29 by page image; the carry example checked). Stability: draft. Currency: practitioner taxonomy as of 2024; anecdotes from 1970 to 2023.

Method 6.3 Specifying an alpha strategy: forecast target, time horizon, bet structure, investment universe, model specification, conditioning variables and run frequency

Statement. Narang lists the choices that turn an alpha idea into a strategy [s7-narang, chap. 3, sec. 3.4, pp. 52–63]:

1. *forecast target*: direction, magnitude and/or duration of a move, possibly with a confidence. Signal strength is the expected return and/or the confidence, and "in general, though certainly not always" a stronger signal means a bigger bet (pp. 52–53);
2. *time horizon*, from microseconds to a year or more, with his classes high-frequency, short-term (a day to two weeks), medium-term and long-term (pp. 53–55);
3. *bet structure*: an instrument by itself (intrinsic) or relative to others, in pairs or larger clusters, grouped statistically or heuristically (pp. 55–57);
4. *investment universe*: geography, asset class, instrument class and exclusions, preferring liquidity, plentiful clean data and predictable behaviour (pp. 57–59);
5. *model specification*: the precise definition and parameters, and how often the model is refitted (pp. 59–60);
6. *conditioning variables*: modifying conditioners such as stops and profit targets, and secondary conditioners that require agreement between signals (pp. 60–61);
7. *run frequency*: from monthly to continuous, trading noise and cost against large, market-moving rebalances and missed prices (pp. 61–62).

With two forecast types, four horizons, two bet structures and four asset classes, one idea gives $2 \times 4 \times 2 \times 4 = 64$ variants (pp. 62–63; checked).

Conditions. Narang qualifies several items himself. Very large signals are rare, so the forecast–outcome relation may be less reliable there (p. 53). Differences between strategies are larger at short horizons, because small changes are amplified over many trades (pp. 54–55). Pairs are rarely precisely comparable, statistical groupings "can be fooled by the data" and heuristic ones are imprecise and rigid; grouping techniques suffer when regimes change (pp. 56–57). Refitting raises the risk of overfitting (p. 60). Most directional trend-following strategies make money on a minority of their trades, "often less than 40 percent" (pp. 60–61).

Practice. Interpretation: forecast strength becomes position size only through calibration and the portfolio decision. Sizing by strength presumes that strength is calibrated to realised returns (rank-based or equal sizing need not); size also depends on each instrument's risk, its correlation with other positions, costs and constraints, so "bigger signal, bigger bet" holds for a calibrated forecast with other inputs equal (rule-based sizing, where Narang gives the case against unequal weights; mean–variance optimisation; cost functions). Specification choices made after looking at results are trials to be counted (parameter overfitting).

Pitfalls. "Relative alpha strategies tend to exhibit smoother returns during normal times than intrinsic alpha strategies, but … unique problems related to incorrect groupings during stressful periods" (p. 57) is a

conditional generalisation with no data in the chapter; the legacy kept it. Narang's statement that machine-learning techniques are "designed to provide" a "scientifically valid way of testing ... without overfitting" (p. 60) is not adopted. Interpretation: searching many specifications is a source of overfitting, which no technique removes by its name (backtest diagnostics). His biotech example is that "some quants exclude" such stocks (p. 58), which the legacy turned into a rule. The legacy omitted the conditioning variables. "Relative value", he notes, is industry jargon for any relative bet structure, mostly unrelated to value (p. 57).

See also. alpha models; signal combination; hypothesis-driven strategy research; factor risk model (relative bets as risk control).

Further reading. The legacy file, verbatim, is in `alpha-models.tex` in the section's folder of `notes/artefacts/`.

Index. forecast target; signal strength; confidence; time horizon; high-frequency; short-term; medium-term; long-term; bet structure; intrinsic; relative; pairs; clusters; grouping; relative value; investment universe; liquidity; model specification; parameters; refitting; conditioning variables; stop-loss; profit target; time stop; run frequency

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 3, sec. 3.4, pp. 52–63, with Exhibits 3.7 and 3.8 read from the text layer; the permutation count checked). Stability: draft. Currency: practitioner account as of 2024.

6.2 Portfolio construction models: rules and optimisers

Method 6.4 Rule-based position sizing: equal position, equal risk, alpha-driven and decision-tree weighting, with Narang's case against unequal weights

Statement. Narang divides quantitative portfolio construction models into two families: *rule-based* models, heuristics "derived from human experience, such as by trial and error", which can be "exceedingly simple or rather complex"; and *optimisers*, which search for the best value of an objective function [`s7-narang`, chap. 6, "Portfolio Construction Models", pp. 99–100] (the second family is in mean–variance optimisation). He names four rule-based types (sec. 6.1, p. 100):

- *equal position weighting*: every position the alpha model deems worth holding gets the same size; signal strength is ignored "except insofar as the signal is strong enough to be worthy of a position at all", and the instruments are implicitly assumed homogeneous in risk (sec. 6.1.1, pp. 100–101);
- *equal risk weighting*: sizes "inversely to their volatilities (or whatever other measure of risk, such as drawdown, is preferred)", so that each position is equalised "by the amount of risk that the allocation contributes to the portfolio" (sec. 6.1.2, p. 102);
- *alpha-driven weighting*: size grows with the forecast, under limits from the risk model on single positions and groups, for example 3 percent a position and 20 percent a sector (sec. 6.1.3, pp. 103–104);
- *decision-tree weighting*: "a defined set of rules in a particular order to determine position sizing", named on p. 100 only; the third edition has no section on it.

Conditions. *Why some quants weight equally.* Narang gives two problems with *unequal* weighting, which the equal-weighters avoid (pp. 101–102). First, it assumes enough statistical power to forecast not only direction but also the relative size or probability of moves. Second, it leads to "a few large bets on the 'best' forecasts and many smaller bets on the less dramatic forecasts", and so to excess idiosyncratic risk: the strongest momentum or reversal signals often come where prices have already moved most, sometimes on real information (his "adverse selection bias"), and large forecasts can come from bad data, such as a price quoted in pence instead of pounds. Winsorising forecasts can partly address this; equal weights need no such adjustment. *Equal risk weighting* relies on a backward-looking risk measure: his example is bank stocks, stable for a decade and then more volatile than many technology stocks in 2008 (p. 103). *Alpha-driven weighting* relies on the alpha model being right about size as well as direction; trend followers using it "can suffer sharp drawdowns relatively frequently", holding the largest position as the trend reverses (p. 104). Costs can enter any rule, for example by setting a forecast to zero when it is below its expected transaction cost (sec. 6.1.4, p. 104; cost functions). Liquidity can cap any weight (p. 102).

Practice. Interpretation, checked numerically: his Exhibit 6.1 (p. 102, page image) gives volatilities of 2.5 and 2.0 percent and "volatility-adjusted" weights of 39 and 61 percent. Those are inverse-*variance* weights, $(1/2.5^2)/(1/2.5^2 + 1/2.0^2) = 39.0\%$; inverse-volatility weights would be 44.4 and 55.6 percent, which give each stock the same $w_i\sigma_i$ (1.11) while the table's weights do not (0.98 and 1.22). Neither scheme in general equalises the contribution to *portfolio* risk, $w_i(\Sigma w)_i / w^\top \Sigma w$, defined when the portfolio variance $w^\top \Sigma w$ is positive, and dependent on correlations. For two assets inverse-volatility weights do equalise it at any correlation, but not for more. With three assets of 20 percent volatility, where the first two correlate at 0.8

and the third is uncorrelated, equal (inverse-volatility) weights give risk contributions of 39, 39 and 22 per-cent. Equal contributions need weights of about 29.93, 29.93 and 40.15 percent, $a = 1/(2+\sqrt{1.8})$ for the first two and $a\sqrt{1.8}$ for the third (checked). Narang's "contributes to the portfolio" therefore describes equal-risk-contribution weighting, which uses the full covariance matrix, more than the inverse-volatility rule he gives (stub `meth:equal-risk-contribution`).

Pitfalls. The legacy block put Narang's second argument *against* unequal weighting, "few large bets on 'best' forecast, many smaller bets", under equal weighting, where it reads as a property of the equal-weight scheme. It describes what equal weighting avoids. The legacy decision-tree details ("constraints may include percentage limits", "model size grows dramatically") are not in the third edition's chapter; they may come from an earlier edition (not checked). Narang's choice between rules is his observation: rule-based users are mostly futures traders with "intrinsic" forecasts, and optimiser users are mostly equity market-neutral traders with relative forecasts (sec. 6.4, p. 123). He gives no data for this.

See also. mean–variance optimisation (the other family); signal-portfolio allocation (rule-based signal portfolios mixed by an optimiser); cost functions (costs as inputs); common-investor contagion (risk targeting under rising volatility).

Further reading. Equal risk contribution and volatility targeting belong to the risk-parity pathway, whose readings are unread; a stub here names it. The legacy file it replaces is kept verbatim in the section's folder of `notes/artefacts/`, as
`portfolio-construction-models.tex`.

Index. portfolio construction model; rule-based weighting; equal weighting; equal position weighting; equal risk weighting; inverse volatility; inverse variance; risk contribution; equal risk contribution; alpha-driven weighting; position limits; sector limits; decision tree; winsorisation; adverse selection; bad data; intrinsic and relative alpha

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 6, pp. 99–104 and 123, with Exhibit 6.1 on p. 102 by page image; the weights and risk contributions computed; the ERC weights' rounding corrected in the slice-28 follow-up). Stability: draft. Currency: practitioner account as of 2024; the risk-contribution arithmetic is timeless.

Model 6.5 Mean–variance portfolio optimisation: objectives, inputs, constraints and solvers, and what the legacy claimed about them

Statement. Markowitz's problem, as written by Boyd, Johansson, Kahn, Schiele and Schmelzer, chooses the weights $w \in \mathbb{R}^n$ given expected returns μ and a covariance matrix Σ in one of two forms [wide-boyd-markowitz-seventy, author version of 5 Jan. 2024, sec. 1.1, eqs. (1)–(2), pp. 4–5]:

$$\max_w \ \mu^\top w \ \text{ s.t. } \ w^\top \Sigma w \le \sigma_{\text{tar}}^2, \ \mathbf{1}^\top w = 1; \qquad \max_w \ \mu^\top w - \gamma\, w^\top \Sigma w \ \text{ s.t. } \ \mathbf{1}^\top w = 1.$$

Both trace the same trade-off curve of Pareto-optimal portfolios as σ_{tar} or $\gamma > 0$ varies, and both have closed forms, though the first "can be infeasible when σ_{tar} is too small". The closed form of the second is $w^\star = \frac{1}{2\gamma}\Sigma^{-1}(\mu + \nu^\star \mathbf{1})$ with $\nu^\star = (2\gamma - \mathbf{1}^\top\Sigma^{-1}\mu)/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$, the optimal dual variable of the budget constraint; its Σ^{-1} hints at sensitivity to the inputs. Adding $w \ge 0$ (long only) makes the second a quadratic programme (pp. 4–5). Interpretation, checked numerically: the closed form assumes Σ positive definite, so that Σ^{-1} exists and the objective is strictly concave with one maximiser, and $\gamma > 0$. The term $\nu^\star\mathbf{1}$ shifts every expected return by the same amount, chosen so that $\mathbf{1}^\top w^\star = 1$; with a three-asset example, $\gamma = 1$ and $\gamma = 5$, the weights sum to one and the first-order condition $\mu - 2\gamma\Sigma w^\star + \nu^\star\mathbf{1} = 0$ holds to rounding. The first form is feasible only when σ_{tar}^2 is at least the minimum attainable variance under the budget constraint, $1/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$. A singular Σ, for example two perfectly correlated assets, has no inverse and leaves directions of zero variance along which the objective may be unbounded or the maximiser not unique. It needs separate treatment, such as a better-conditioned estimate or explicit constraints, chosen and justified for the problem; substituting a pseudoinverse into the formula is not a justified default (Interpretation). Maximising the Sharpe ratio is a third objective, also solvable by convex optimisation "at least when the constraints are simple" (sec. 3.2, p. 20). Narang describes the same model in words: inputs of expected returns, volatilities and a correlation (covariance) matrix; output the efficient frontier; a common objective of "the highest possible return for a unit of risk"; and in practice also the portfolio's size in currency, a desired risk level "usually measured in terms of volatility or expected drawdown", and other constraints such as a hard-to-borrow list, which "are not required by the optimizer" [s7-narang, chap. 6, sec. 6.2, pp. 104–106].

Notation. μ, Σ, w, γ (risk aversion), σ_{tar} (target volatility) as in Boyd et al.; the Sharpe ratio $\mu^\top w/\sqrt{w^\top\Sigma w}$ (excess returns, no benchmark) is unchanged when w is scaled by any positive number.

Conditions. *Which objective.* Mean–variance maximises return for a variance target, or return minus a variance penalty; maximising the Sharpe ratio selects one direction on the frontier (with cash and leverage, the tangency portfolio, Narang's Exhibit 6.7, p. 122, page image). These are different problems with different solutions. Expected drawdown is not part of the mean–variance model: Narang lists it among practical inputs

that set a risk level (p. 105) and names a return-to-drawdown objective as a different choice (p. 124). The legacy "desired risk level (volatility or expected drawdown)" and "maximising return of portfolio relative to volatility" put both inside one model. *Unconstrained optimisation.* Narang calls it "a quirk of unconstrained optimizers" that they "often" put everything in the instrument with the highest risk-adjusted return (sec. 6.2.2.1, p. 114); the legacy made this "might". It is not a property of mean–variance: the unconstrained solution is proportional to Σ^{-1} terms and generally holds every asset, and his own unconstrained two-stock example has its maximum Sharpe ratio at about 85 percent CVX and 75 percent NFLX (p. 106; Exhibit 6.2, p. 107, page image). Interpretation: a single-asset answer arises when only expected return is maximised under a budget with no shorting, a bounded problem whose optimum is the asset with the highest expected return. Ignoring risk without such bounds does not give one asset: with shorting allowed and expected returns not all equal, the budget-constrained return $\mu^\top w$ is unbounded, since any direction d with $\mathbf{1}^\top d = 0$ and $\mu^\top d > 0$ can be scaled without limit (checked with an example). Box limits on each weight make it bounded again, with the solution at the limits. Because the Sharpe ratio is scale-free, the point he marks is one of a ray of equally good portfolios within his ±100 percent search box (checked: $\Sigma^{-1}\mu$ and three times it have the same ratio).

Practice. *Inputs.* Expected returns come from the alpha model; a sign-only forecast can be used by setting all positive and all negative forecasts equal (p. 108). For volatility Narang calls GARCH, with Bollerslev's 1986 paper, "the most widely used such technique (particularly in equity trading)", and adds that a simpler EWMA "may perform just about as well" (pp. 108–110); he cites no evidence for the prevalence. Boyd et al. list EWMA, dynamic conditional correlation and iterated EWMA as covariance estimators that work well in practice (sec. 3.4, p. 22). For correlations Narang shows instability: S&P 500 against Nikkei 225 rolling one-year correlations from +0.01 to +0.66 (pp. 110–111, Exhibit 6.4 by page image), and XOM–CVX steadier than XOM–NFLX (Exhibit 6.5, p. 112), and says grouping similar instruments, "in the definition of *relative* in a relative alpha strategy and/or in the specification of the risk model", improves stability (p. 111). *Constraints* limit positions and groups, and can end up driving the portfolio: with 100 names and a 1.5 percent cap the best position is only 1.5 times the average (sec. 6.2.2.2, p. 114). Risk penalties can rise faster than linearly, and costs enter as per-stock or generic impact functions (pp. 114–115); Boyd et al. use a spread term plus a $|z|^{3/2}$ impact term (sec. 2.4, eq. (10), p. 17), the form in cost functions. Industry-neutral constraints can turn a positive forecast into a short, and the substitution effect can prefer a cheaper correlated stock (sec. 6.2.3, pp. 120–121). The output is a target portfolio; the trades are its difference from the current one (sec. 6.3, pp. 122–123). *Solvers.* Convex formulations, including nondifferentiable terms, are solved reliably and fast; integer shares, cardinality limits and minimum position sizes make the problem nonconvex and much slower (Boyd et al., secs. 1.4 and 4.3, pp. 9–10 and 27–28). Narang's "10,000 assets … in about one second" (p. 120) matches Boyd et al.'s Table 2 only with a factor model: 0.65–0.89 seconds for 10,000 assets and 50–100 factors with one commercial solver (sec. 5.6, p. 40). *Size limits* (slice 29). Narang's risk-model chapter classifies limits by how size is limited, how risk is measured and what is limited [s7-narang, chap. 4, "Risk Models", sec. 4.1, pp. 73–76]. A hard constraint draws a line, such as no position above 3 percent; because "why is a 3.01 percent position so much worse?", some quants use a penalty that lets a position pass the level only for a much larger expected return, increasingly hard the further past it (pp. 73–74). Boyd et al. describe the same device as softening: the hard constraint $f \le f_{\max}$ is removed and a penalty $\gamma(f - f_{\max})_+$ is subtracted from the objective, with γ a priority parameter; softening preserves convexity, and limits that are strict or mandated, such as long-only or no borrowing, stay hard (sec. 4.2, pp. 25–27). A penalty therefore replaces a limit; it does not permit breaching a separate hard cap on the same quantity. Limit levels come from theory, set and adjusted by testing (his 5 percent limit revised to 3 percent), or from data (p. 74); tuning them on a backtest invites overfitting. Limits can apply to positions, groups such as sectors, factor exposures (factor risk model) and leverage (p. 75).

Pitfalls. *Estimation error.* Boyd et al. agree that the basic method is sensitive to its inputs, and answer with regularisation and robust optimisation: long-only or leverage limits, Black–Litterman's shrinkage of the mean towards market-implied returns, and covariance shrinkage (sec. 1.3, pp. 8–9; sec. 3.4, p. 22). *Black–Litterman.* Narang credits it with a default portfolio and a blend of "an investor's expectations with a degree of confidence about those expectations with the historical precedent evident in the data" (sec. 6.2.2.3, p. 117). In Boyd et al.'s account it acts on the expected returns, towards the market-implied return, not on correlations. The legacy "adjusts historically observed correlation levels by utilising investor's forecast" is in neither source and is not adopted. Black and Litterman's paper is unread (stub); Narang's note 5 dates it 1982, his text and the catalogue 1992. *Resampled efficiency.* Narang presents Michaud's method as reordering the observed returns by Monte Carlo "to reduce the estimation error inherent in the inputs" (sec. 6.2.2.5, pp. 118–119). His own remark that average return and volatility "remain the same across all these alternate histories" concerns one series, the S&P 500: reordering a single series leaves its mean and variance unchanged. That does not show that every optimiser input is unchanged. Interpretation, checked with a small example: applying one common permutation to the rows of a multivariate return sample also leaves its covariances unchanged, but permuting each asset's returns independently changes them; for $x = (1, 2, 3, 4)$ and $y = (1, 2, 3, 4)$, reversing y alone keeps both means (2.5) and variances (1.25) but turns their covariance from +1.25 into −1.25. What Narang describes, then, does not settle what Michaud's procedure changes. His example concerns the robustness of a

trading strategy, not optimiser inputs. What Michaud's procedure does and whether it improves out-of-sample results were not checked (stub). *Machine learning.* The legacy "data-mining approaches: supervised learning or genetic algorithms, as MVO involves searching many possible portfolios" is not in the third edition. There Narang says "we do not see many strictly 'machine learning' models for portfolio construction" and that the state of the art remains Markowitz-type optimisation (sec. 6.2.2, pp. 113–114). Interpretation: three tasks are being run together. *Forecasting* estimates inputs such as expected returns or covariances, and supervised learning is commonly used for it. *Learning allocation weights* maps data directly to portfolio weights by training a model on a portfolio objective; that is possible and has its own literature, which was not read here. *Optimisation* chooses weights for given inputs and constraints; for a convex problem, solvers reach the optimum, so the size of the search space is no reason to use a generic heuristic such as a genetic algorithm, though such heuristics have a place in nonconvex problems. The legacy item's reason, that mean–variance optimisation "involves searching many possible portfolios", supports none of these, and the item has no support in the source read. Not adopted. *Narang's details not adopted.* He describes convex functions as "smooth and continuous" with "a single highest or lowest point" (p. 120); convex functions can have kinks, such as $|x|$ or the costs above, and flat sets of minima. He calls GARCH a stochastic-volatility model and says a negative autoregression "implies ... trending behavior" (pp. 108–110); neither is adopted. His 1,998,000 distinct correlations for 2,000 instruments (p. 113) should be $2000 \times 1999/2 = 1{,}999{,}000$ (checked). His grouping evidence is two pairs over one sample, not a test of a method; factor and shrinkage estimators are stubbed. *Volatility and covariance (slice 29).* Narang names two "generally accepted" measures of risk: longitudinal volatility, and dispersion, the cross-sectional standard deviation of returns or the correlation among instruments, a higher dispersion making the market "less risky" (sec. 4.1.2, pp. 74–75). Interpretation: these are not alternatives. A covariance risk model contains both volatilities, on its diagonal, and correlations, off it, and a portfolio's variance $w^\top \Sigma w$ needs both; cross-sectional dispersion in one period is a different statistic from covariance over time, and a high reading can reflect large idiosyncratic moves rather than safety. His inference is not adopted. *VaR-based leverage.* He describes VaR models as setting leverage inversely to measured risk, criticises the practice, and says "most VaR models calculate what a daily single standard deviation move in portfolio returns will be" (pp. 75–76). That last description is not adopted: VaR is usually defined as a loss quantile at a stated confidence level; its estimation belongs to `risk-tail-systemic`, whose readings are unread (stub there). The deleveraging mechanism is in common-investor contagion.

See also. rule-based position sizing; signal-portfolio allocation; cost functions; mean–variance execution (the same trade-off over an execution schedule); backtest evaluation diagnostics (Sharpe and Calmar ratios as Narang defines them).

Further reading. Markowitz's paper, Black and Litterman, Michaud, shrinkage and factor-model readings of the pathway: not read (stubs). The legacy file it replaces is kept verbatim in the section's folder of `notes/artefacts/`, as
`portfolio-construction-models.tex`.

Index. mean–variance optimisation; Markowitz; modern portfolio theory; efficient frontier; risk aversion; target volatility; Sharpe ratio maximisation; tangency portfolio; unconstrained optimisation; constrained optimisation; position limits; industry neutrality; substitution effect; expected returns; sign forecasts; GARCH; EWMA; correlation instability; estimation error; regularisation; robust optimisation; Black–Litterman; resampled efficiency; convex optimisation; mixed-integer; solve time; target portfolio

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 6, pp. 99–125, with Exhibits 6.2, 6.4–6.7 by page image; and the authors' version of 5 January 2024 of `wide-boyd-markowitz-seventy`, secs. 1.1–1.4, 2.4, 3.2, 3.4, 4.3 and 5.6, pp. 3–10, 17, 19–22, 27–28 and 39–40, not the journal pagination; Exhibit 6.1 weights, pair count and Sharpe scale invariance computed; the closed form's conditions, the unbounded case, the permutation example and the three tasks behind the machine-learning item corrected 2026-10-15 in the slice-28 follow-up; in slice 29, chap. 4, secs. 4.1–4.1.3, pp. 73–76, and Boyd et al. sec. 4.2, pp. 25–27, for size limits, penalties and risk measurement). Stability: draft. Currency: formulation timeless; solver timings as of 2024.

Method 6.6 Signal-portfolio allocation: weighting simulated factor portfolios rather than positions, as Narang describes Grinold and Kahn's approach

Statement. Narang describes an optimiser "directly aimed at building a portfolio of signals, whereas most optimizers try to size positions", attributing it to Grinold and Kahn's *Active Portfolio Management* [`s7-narang`, chap. 6, sec. 6.2.2.4, pp. 117–118]:

1. build one *factor portfolio* per type of alpha forecast (momentum, value, growth), usually rule-based, "very often equal-weighted or equal risk-weighted";
2. simulate each historically to obtain a return series;
3. treat those series as the instruments of an optimiser, with the risk model, transaction cost model, portfolio size and risk target as inputs;

4. set each position as the sum, over factors, of its weight in the factor portfolio times that factor's signal times the factor portfolio's weight.

His example: 100 equally weighted stocks, factor weights 60 and 40 percent, signals $+1$ and -1 for one company, give $1\% \times (+1) \times 60\% + 1\% \times (-1) \times 40\% = +0.2\%$ (p. 118, page image; checked).

Conditions. The number of factor portfolios is "typically much more manageable, usually not more than about 20, corresponding to the number of individual factors in the alpha model" (p. 117). That number is Narang's practitioner statement, with no source; it depends on how many signals a model has. The factor-portfolio returns are simulated, so the optimiser's inputs inherit the backtests' assumptions and their overfitting risk (parameter overfitting).

Practice. Interpretation: the method trades a large position-level estimation problem for a small one over a handful of return series, "an easier hurdle to clear in terms of the amount of data needed" (p. 117). The position weights that result are linear in the factor weights, so position limits and costs at stock level are not enforced unless they are added back.

Pitfalls. Whether this is Grinold and Kahn's own procedure, and in what form, was not checked: their book (`alpha-grinold`, a reading of the section overview) is unread. The legacy "what is optimised is then a handful of factor portfolios" and "not more than 20" repeat Narang's account as fact.

See also. mean–variance optimisation; rule-based position sizing (how each factor portfolio is weighted); backtest evaluation diagnostics.

Further reading. Grinold and Kahn, *Active Portfolio Management* (`alpha-grinold`), not read. The legacy file it replaces is kept verbatim in the section's folder of `notes/artefacts/`, as `portfolio-construction-models.tex`.

Index. factor portfolio; signal portfolio; Grinold and Kahn; alpha factor weighting; strategy allocation; simulated factor returns; position aggregation

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 6, pp. 117–118, with p. 118 by page image; the example computed; Grinold and Kahn not read). Stability: draft. Currency: practitioner account as of 2024.

Model 6.7 PCA risk-distribution weights: allocating a target variance across the principal components of a covariance matrix

Statement. Given an $N \times N$ covariance matrix V of an invariant (returns, yield changes, volatility changes) and a target distribution of risk R across V's principal components, López de Prado computes the allocation ω as follows [`alpha-afml`, chap. 2, sec. 2.4.2, p. 35]. Decompose $VW = W\Lambda$ with eigenvalues in descending order. For any ω, with $\beta = W'\omega$ its projection on the orthogonal basis,

$$\sigma^2 = \omega' V \omega = \beta' \Lambda \beta = \sum_{n=1}^{N} \beta_n^2 \Lambda_{n,n}, \qquad R_n = \frac{[W'\omega]_n^2 \Lambda_{n,n}}{\sigma^2}, \qquad \sum_n R_n = 1.$$

To deliver a chosen R at risk σ, set $\beta_n = \sigma \sqrt{R_n / \Lambda_{n,n}}$ and $\omega = W\beta$; rescaling ω rescales σ and keeps R (p. 35). With no R given, Snippet 2.1 puts all risk on the smallest-eigenvalue component, so ω is the last eigenvector scaled to σ (p. 36).

Notation. ω is in the units of the invariant (for example weights on returns); σ is the target standard deviation (`riskTarget`); $R_n \geq 0$.

Conditions. López de Prado frames V as the covariance of an IID multivariate Gaussian process with mean μ (p. 35); interpretation, the mean plays no part, and the construction needs only V. It needs $\Lambda_{n,n} > 0$ for every component given risk: a zero eigenvalue cannot carry risk, and a near-zero one, where estimation error is largest, produces very large weights, which the snippet does not guard against. The decomposition does not fix signs: each β_n with $R_n > 0$ can be $\pm\sigma\sqrt{R_n/\Lambda_{n,n}}$, so with distinct eigenvalues 2^k allocations share one risk distribution, k being the number of components given risk (two for the snippet's default); the snippet's positive roots, and the arbitrary signs of the computed eigenvectors, are conventions. With repeated eigenvalues the eigenvectors, and so the allocations, are not unique at all. The weights are those of an estimated V; with an estimated covariance, the lowest-variance component is the one most likely to be noise.

Practice. Checked on a random positive-definite 4×4 matrix (`afml-checks.py`, in the artefact folder of `statistical-learning-foundations`): the construction delivers the target $R = (0.1, 0.2, 0.3, 0.4)$ and σ exactly, and flipping the sign of one β_n with $R_n > 0$ leaves R unchanged. López de Prado's Figure 2.2 contrasts an inverse-variance portfolio, whose risk is spread over nearly all components, with the PCA portfolio, whose risk is all in the lowest-variance component (pp. 35–36). In his chapter the weights feed the ETF trick for a spread or hedge.

Pitfalls. The legacy notes wrote $\sigma^2 = (\Lambda^{1/2}\beta)'(\Lambda^{1/2}\beta)'$, with a stray transpose, and $R_n = [W'n]_n^2 \Lambda_{n,n}\sigma^{-2}$ for $[W'\omega]_n^2$. Allocating all risk to the smallest eigenvalue is a hedging choice, not a return forecast; it says nothing about expected return.

See also. ETF trick; factor risk model; mean–variance portfolio optimisation.

Further reading. López de Prado cites López de Prado and Leinweber (2012) and Bailey and López de Prado (2012) for other hedging weights (p. 35), not read here. The legacy block is kept in `notes/artefacts/statistical-learning-foundations/financial-data-structures.tex`.

Index. principal components; PCA weights; spectral decomposition; eigenvalues; risk distribution; risk contribution; risk budgeting; hedging weights; smallest-eigenvalue portfolio

Status. Verification: partly checked 2026-10-16 (against the text layer and a page image of a local copy of `alpha-afml`, chap. 2, pp. 35–36, Snippet 2.1; the construction checked numerically). Stability: draft. Currency: timeless.

6.3 Risk models: factor exposures and their control

Model 6.8 Factor risk model: factor returns, exposures, residuals and the factor covariance, with economic and statistical factors and their use in limiting unintended exposures

Statement. In Narang's framing a *risk* is an exposure that accompanies the pursuit of alpha but is not intentionally forecast: "risk exposures are those that are not intentionally sought out by the nature of whatever forecast the quant is making in the alpha model". A risk model measures such exposures, highlights them and provides input to the portfolio construction model, which decides whether to accept, limit or eliminate each [s7-narang, chap. 4, "Risk Models", pp. 71–73]. Risk *management*, in his words, "is about the intentional selection and sizing of exposures to improve the quality and consistency of returns" (p. 71). The factor model that underlies most risk models, as Boyd, Johansson, Kahn, Schiele and Schmelzer write it, decomposes the n asset returns into k factor returns and idiosyncratic returns [wide-boyd-markowitz-seventy, author version of 5 Jan. 2024, sec. 3.3, eq. (14), pp. 20–21]:

$$r = Ff + \epsilon, \qquad \Sigma = F\Sigma_f F^\top + D, \qquad \sigma^2 = (F^\top w)^\top \Sigma_f (F^\top w) + w^\top D w,$$

with F the $n \times k$ loading (exposure) matrix, known when the portfolio is built, f the factor returns with covariance Σ_f, and ϵ the idiosyncratic returns with diagonal covariance D (positive entries), uncorrelated with f. The means of ϵ are the assets' alphas.

Notation. F, f, Σ_f, ϵ, D, w as in Boyd et al.; $F^\top w$ is the portfolio's vector of factor exposures; σ^2 the portfolio variance. Usually $k \ll n$, so Σ is low rank plus diagonal.

Conditions. *Assumptions.* The decomposition is exact only under its assumptions: factor and idiosyncratic returns uncorrelated, and idiosyncratic returns uncorrelated across assets (D diagonal). Interpretation, checked numerically: under them, an equally weighted portfolio of n assets with unit loading on one factor of 15 percent volatility and residuals of 30 percent has volatility 33.5, 17.8, 15.3 and 15.0 percent for $n = 1$, 10, 100 and 1,000, so the residual part diversifies away and the factor part does not. This is what Narang means by systematic risks that "cannot be diversified away" (p. 77). A diagonal D, with bounded residual variances and equal weights, is sufficient for this, not necessary. What matters is how much residual covariance remains when it is summed over pairs (Interpretation, checked numerically). With a constant residual correlation of 0.2 between every pair and residual variance 0.09, equal-weighted residual variance is $0.09\,(0.2 + 0.8/n) = 0.018 + 0.072/n$, which stays above 0.018 however large n is: that common correlation behaves like an omitted factor. With the same correlation of 0.2 only within disjoint pairs of assets, it is $0.108/n$, which still vanishes as n grows. In these examples correlation confined to small groups still diversifies, while a correlation shared by every pair leaves a floor. *Two kinds of factor.* Theory-driven or economic factors are named in advance on an economic argument: for equities the market, sectors and market capitalisation; for bonds, interest rates (pp. 77–78). Boyd et al. give market, industry and style portfolios such as Fama–French for equities and yield-curve, rate and spread factors for bonds, which are "interpretable". Empirical or statistical factors come from historical returns by methods such as principal component analysis; apart from the first, "which typically is close to the market return, these factors are less interpretable" (Boyd et al., p. 20). Narang says the first principal component of Treasury yields usually corresponds to the level of rates, and in equities the market comes first and then sectors (p. 78). Interpretation: a principal component is defined only up to sign and can change order and composition from one sample to the next, so its economic label is an after-the-fact reading, not a property of the method. Narang notes that statistical models can be "fooled" into factors that do not persist or into spurious exposures (p. 78).

Practice. *Uses* (pp. 72 and 82–83): measuring which systematic exposures a portfolio takes and how large they are; explaining profits and losses; and, inside an alpha model, forecasting the residuals of returns against the risk model's factors. *Limiting exposures.* Narang's general rule is "to eliminate any unintentional exposures, since there should be no expectation of being compensated sufficiently for accepting them" (p. 76); his examples are a value strategy's incidental sector and small-cap bets (pp. 72 and 77) and a view that CVX would beat XOM, expressed as a long-only CVX trade that also bets on the market, which buying CVX and shorting an

equivalent amount of XOM "substantially eliminated" (p. 76). Interpretation: equal dollar amounts hedge the market exactly only if the two stocks have equal market betas; otherwise the pair keeps a net market exposure of the beta difference times the dollar size, so the hedge is approximate, and exact neutrality needs beta-weighted sizes under a stated risk model. Exposure limits, such as a ±5 percent net market exposure, apply to risks the alpha model does not forecast (p. 75). He prefers, "to the extent possible", to build exposure decisions into the alpha model's bet structure, for example forecasting stocks relative to their sectors, rather than relying on one risk model in an optimiser (pp. 76–77 and 80–81). Boyd et al. express neutrality to factor i as zero covariance of the portfolio with that factor, $(\Sigma_f F^\top w)_i = 0$, a linear constraint (sec. 3.3, eq. (15), p. 21). Interpretation, checked numerically: with correlated factors this differs from zero exposure, $(F^\top w)_i = 0$. For three assets with loadings $(1, 0.5)$, $(1, -0.5)$, $(0.8, 1)$, factor variances 0.04 and 0.01 and factor covariance 0.012, the weights $(0.5, -0.5, 0)$ have exposures $(0, 0.5)$ but a covariance of 0.006 with the first factor. *Choosing a model.* Most quants use theory-driven models for their interpretability; empirical ones promise adaptiveness but lag a regime change and trade adaptiveness against estimation error. With two years of daily data each new day changes two of about 520 observations, while shorter windows lose statistical power. Intraday data is one partial remedy. The two kinds can be combined, and commercial models (BARRA, Northfield, Qontigo/Axioma, Wolfe) are mostly theory-driven and for equities (pp. 78–80). Boyd et al. add that a factor model can estimate a large covariance better than direct estimation and cuts the solve cost from $O(n^3)$ to $O(nk^2)$ (p. 21).

Pitfalls. *Risk model against decision.* The legacy opening, "risk model concerns the intentional selection and sizing of exposures", moves to the model what Narang says of risk *management*; in his black-box schematic (Exhibit 4.1, p. 83, page image) the risk model is an input to the portfolio construction model, which selects and sizes. *Factor, exposure, signal or premium?* The legacy listed market, size, value and momentum as "the most common factors" of a risk model, with size and value described as risks. That list is not in Narang's risk chapter, which names market, sector and market-capitalisation risk; value is his example of an alpha (pp. 71–72), and whether such a factor is a risk to limit, a signal to forecast or a premium expected to be paid depends on the strategy's forecasts, not on the factor. The same factor can play several roles. *Unintended is not the same as uncompensated.* Narang's rule is that "there should be no expectation of being compensated sufficiently" for unintentional exposures (p. 76): not that they earn nothing, but that any expected reward is not enough for the risk. Whether a given exposure is compensated, and by how much, is an empirical question about risk premia, not settled by its being unintended (Interpretation; not checked against the factor-premia readings). *Empirical, not adaptive by default.* Updating an empirical model with new data is not evidence that its forecasts improve; Narang himself describes the lag and the loss of power (p. 79). The legacy "empirical risk models … used by statistical arbitrage traders" repeats his "most commonly found among" (p. 78) as a definition.

See also. mean–variance optimisation (where exposure limits, penalties and the covariance enter); rule-based position sizing; sources of strategy risk (regime change and model risk); common-investor contagion (VaR-based leverage under rising volatility).

Further reading. Factor-model estimation and covariance shrinkage remain in the pathway's backlog. The legacy file it replaces is kept verbatim in the section's folder of `notes/artefacts/`, as `risk-models.tex`.

Index. risk model; factor model; factor exposure; factor loading; factor returns; idiosyncratic return; residual return; specific risk; factor covariance; low rank plus diagonal; systematic risk; diversification; theory-driven risk model; empirical risk model; statistical risk model; principal component analysis; factor neutrality; zero beta; market neutrality; sector neutrality; unintended exposure; risk premium; commercial risk models; regime change

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 4, "Risk Models", pp. 71–84, with Exhibit 4.1 on p. 83 by page image; and the authors' version of 5 January 2024 of `wide-boyd-markowitz-seventy`, sec. 3.3, pp. 20–21; the diversification, correlated-residual and neutrality examples computed; the residual-diversification condition, the CVX–XOM hedge and Narang's compensation wording corrected 2026-10-15 in the slice-29 follow-up). Stability: draft. Currency: formulation timeless; Narang's practitioner account as of 2024.

6.4 Transaction cost models as inputs to portfolio construction

Model 6.9 Transaction cost functions for portfolio construction: flat, linear, piecewise-linear and quadratic forms, with units and size conventions

Statement. Narang's transaction cost model quantifies "the cost of making a trade of a given size" so that the portfolio construction model can weigh it against the alpha and risk models when choosing the portfolio to hold [s7-narang, chap. 5, "Transaction Cost Models", p. 85]. It is "not designed to minimize the cost of trading, only to inform the portfolio construction engine"; minimising cost is the execution algorithm's job (p. 86), and cost is reduced in two phases, first by the portfolio construction model using the cost estimate and then by execution (sec. 5.3, p. 96). In his framing there are "only two reasons to make a trade", a better reward from the alpha model or a lower risk from the risk model, and a trade's benefit must "clear the hurdle of the cost of

transacting" (p. 85). He names four basic forms, each describing the total cost of a trade against its size, with commissions and fees as the baseline (sec. 5.2, pp. 91–96, Exhibits 5.1–5.4 by page image):

- *flat*: the same cost whatever the size; reasonable when the size traded is nearly always about the same and liquidity is fairly constant, "rarely correct" otherwise (pp. 92–93);
- *linear*: cost rising with a constant slope in size, overestimating at small sizes and underestimating at large ones in his exhibit (pp. 93–94);
- *piecewise-linear*: linear in ranges, with a steeper line beyond each breakpoint; "rather popular among quants as a happy medium between simplicity and accuracy" (p. 94);
- *quadratic*: cost rising ever faster with size; in his words the most "accurate" of the four in the exhibits and the most computationally intensive (p. 95).

Notation. Interpretation, for a single equity traded in shares: let q be the signed trade size in shares ($q > 0$ a buy, $q < 0$ a sale), $V > 0$ a volume scale in shares such as average daily volume, $P > 0$ the price in currency per share, and $C(q)$ the *total* expected cost of the trade in currency. For $q \neq 0$ the *per-share* cost is $c(q) = C(q)/|q|$ in currency per share, and $c(q)/P$ is the same cost as a fraction of price (times 10^4 for basis points); $c(0)$ is not defined. With coefficients $a \geq 0$ in currency, $b, b_0, h_i \geq 0$ in currency per share, $0 < \kappa_1 < \cdots < \kappa_m$ in shares, $k \geq 0$ in currency per share squared and $\eta \geq 0$ dimensionless, the forms are

$$C_{\text{flat}}(q) = \begin{cases} a, & q \neq 0, \\ 0, & q = 0, \end{cases} \qquad C_{\text{lin}}(q) = b|q|, \qquad C_{\text{quad}}(q) = b|q| + kq^2,$$

$$C_{\text{pwl}}(q) = b_0|q| + \sum_{i=1}^{m} h_i\big(|q| - \kappa_i\big)_+, \qquad C_{\text{pow}}(q) = \eta P|q|\Big(\frac{|q|}{V}\Big)^{1/2},$$

with $(x)_+ = \max(x, 0)$. Every form has $C(0) = 0$ and $C \geq 0$, and depends on q only through $|q|$. All but the flat form are continuous; the piecewise-linear form has slope $b_0 + \sum_{i:\kappa_i<|q|} h_i$, rising at each breakpoint κ_i, and equals $\max_j(s_j|q| - d_j)$ with $s_j = b_0 + \sum_{i\leq j} h_i$ and $d_j = \sum_{i\leq j} h_i\kappa_i$ ($s_0 = b_0$, $d_0 = 0$). A maximum of lines with increasing slopes alone does not give $C(0) = 0$: its intercepts must take this form. Checked with exact fractions for $b_0 = 0.01$, $h = (0.02, 0.05)$ and $\kappa = (10{,}000, 50{,}000)$: $C = 0$ at $q = 0$; 50, 100, 700, 1,300 and 2,100 at $|q| = 5{,}000$, 10,000, 30,000, 50,000 and 60,000, the same for buys and sales; continuous at both breakpoints; equal to the maximum of lines with slopes 0.01, 0.03, 0.08 and intercepts 0, 200, 2,700. The forms assume a cost symmetric between buys and sales, never negative (no rebates or favourable price drift), expected rather than realised, for one instrument and one period, independent of the trade's duration and of other trades. Contracts or currency notional need their own size and price units, such as a contract multiplier; the per-share conversions above do not carry over unchanged. A total cost linear in size is a constant per-share cost; a quadratic total cost is a per-share cost linear in size, $c(q) = b + k|q|$ for $q \neq 0$. A per-share commission of, say, \$0.001 (Narang's example, p. 87) is therefore a linear, not a flat, cost; only a charge per ticket is flat. A per-share cost proportional to $(|q|/V)^{1/2}$, as in the square-root law, is a total cost proportional to $|q|^{3/2}$. Numerical illustration (own, checked): with C_{pow}, $\eta = 0.5 \times 2\% = 0.01$, $P = \$50$ and $V = 10^6$ shares, 10,000 shares bought or sold cost 10 bp per share, \$500 in total; 20,000 shares cost 14.1 bp, \$1,414; doubling the size multiplies the per-share cost by $2^{1/2}$ and the total by $2^{3/2} \approx 2.83$. Using the peak-impact law as a cost scale is a simplification: the evidence entry gives an execution shortfall per share of about two-thirds of peak impact.

Conditions. The ranking in the legacy block is not an established result. Narang's exhibits are schematic, with unlabelled axes and no data; the "true" cost curve in each is drawn convex and starting at zero (Exhibits 5.1–5.4, pp. 93–95, page images), although his text puts fixed charges under it as a floor (p. 91). Against an assumed convex curve, the closeness of each form follows from the shape drawn, not from estimation. His own summary is conditional: simpler models suit traders whose sizes are roughly constant or small, more complex ones those who may trade significant or variable quantities quickly, and "any of the four models described here can be valid in the right set of circumstances" (sec. 5.3, p. 96). His statement that a quadratic shape "is generally accepted by the quant community" (p. 92) is unsourced, and his note 2 qualifies it: "there is good evidence that, at least for U.S. equities, cost per share scales with an exponent of 1.5, or in other words, as the square root of trade size" (n. 2, p. 97, page image). That sentence conflates two conventions: a per-share cost scaling as the square root of size is a total cost with exponent 1.5 (Notation). Both a quadratic and a 3/2-power total cost are convex, so "the cost gets higher ever more quickly" as size grows (p. 92) does not distinguish them; which exponent fits is an empirical question owned by `price-impact-execution` (square-root impact, direct estimation, whose fitted temporary-impact exponent of 0.6 per share implies, at a fixed duration, a total exponent of about 1.6).

Practice. Narang describes models built separately for each instrument, "highly empirical", evolving with the quant's own recorded executions and refreshed to keep up with market conditions; a less liquid small-cap

stock costs more to trade than a mega-cap (pp. 91–92 and 96). His Exhibit 5.4 contrasts the expected cost, specified before trading, with what is "observed empirically after trading"; the differences come from changes in liquidity, volatility and the mix of traders (p. 95). Interpretation: the realised cost observed after trading is measured against a benchmark price, and what is observed is a cost, not the impact component; how ex-ante cost forecasts are tested against realised costs is in transaction cost analysis. Interpretation, not from the sources read: inside an optimiser the cost enters the objective jointly with the alpha forecast and the risk penalty, so a trade is not tested against a separate hurdle one by one; and trades can also be forced by flows, margin or constraints, which Narang's two reasons leave out. The cost of underestimating or overestimating costs (too much turnover, or positions held too long, p. 86) is the subject of backtest cost and short assumptions.

Pitfalls. *No universal ranking of accuracy or computation.* The legacy "better estimate", "better accuracy than flat or linear models" and "most computationally intensive, but also most accurate" repeat Narang's comparisons with his drawn curve as if they held generally; they are not adopted. Which form is most accurate depends on the instrument, the size range traded and the true shape, which Narang's own note 2 says may not be quadratic. Computational cost depends on how the form enters the optimiser, not on how many terms it has. Interpretation, not checked against the pathway's readings: a convex piecewise-linear cost can be written with linear constraints, a quadratic cost fits a quadratic programme, and a 3/2-power cost is convex but not quadratic; a flat cost per trade, the "simplest" form, is discontinuous at zero and needs integer variables to switch trades on and off. Narang's own wording is hedged ("this perhaps more accurate one is more computationally intensive", n. 2, p. 97). Since slice 28 one reading supports part of this: Boyd et al. use a spread term plus a $|z|^{3/2}$ impact term inside a convex problem [`wide-boyd-markowitz-seventy`, author version of 5 Jan. 2024, sec. 2.4, eq. (10), p. 17], and find integer, cardinality and minimum-size constraints nonconvex and much slower to solve (sec. 4.3, pp. 27–28); see mean–variance optimisation. *Units.* A cost function stated without its convention (total or per unit; shares, currency or fraction of volume) cannot be compared with an impact law; say which. *Benefit and cost.* The trade-off is between expected benefit and expected cost over the same horizon; comparing a forecast return with a cost in different units or horizons misstates the hurdle.

See also. transaction costs (the components, including Narang's commissions and fees, slippage and impact); square-root impact and direct estimation of impact (the empirical shape); mean–variance execution (linear temporary impact, a quadratic total cost for a given schedule); transaction cost analysis (forecasts against realised costs); backtest cost and short assumptions.

Further reading. Boyd et al.'s Markowitz paper, a reading of the pathway, was read in part in slice 28 (secs. 2.4 and 4.3 cited above); the multi-period paper and the optional branch on no-trade regions are unread and named in the pathway note's backlog. The legacy file it replaces, verbatim, is `transaction-cost-models.tex` in the folder `notes/artefacts/` under the section key `portfolio-construction-risk`.

Index. transaction cost model; t-cost model; cost function; flat cost; fixed cost; linear cost; piecewise-linear cost; quadratic cost; power-law cost; 3/2 power; total cost; per-unit cost; cost per share; trade size; portfolio construction; trading hurdle; cost-aware rebalancing

Status. Verification: partly checked 2026-10-15; units, parameter conditions and the piecewise-linear form corrected 2026-10-15 in the slice-27 follow-up; Boyd et al. secs. 2.4 and 4.3 cited for the computational remarks in slice 28 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. 5, "Transaction Cost Models", pp. 85–97, with Exhibits 5.1–5.4 on pp. 93–95 and note 2 on p. 97 by page image; the numerical illustration computed; of the readings of `portfolio-construction`, only parts of `wide-boyd-markowitz-seventy` read, in slice 28). Stability: draft. Currency: practitioner account as of 2024; the forms are timeless.

6.5 Performance evaluation

Method 6.10 Relative performance evaluation: total returns, market-adjusted and risk-adjusted returns, market timing, and smoothed returns

Statement. Portfolio performance depends on management and on every factor that moves the prices of its holdings, most of which nobody can foresee, so good results may reflect skill or luck [`alpha-harris`, chap. "Performance Evaluation and Prediction", sec. 22.1, pp. 443–444]. Relative evaluations are therefore more valuable than absolute ones: an absolute evaluation computes the total return; a relative one compares it with a benchmark chosen to show how the portfolio would have done without active management (sec. 22.2, p. 445). Harris's measures (secs. 22.2.1–22.2.2.3, pp. 445–450):

- *total return* over an interval, including income and paying out commissions, fees and margin flows; with capital added or withdrawn, either the *internal rate of return*, approximately a time- and value-weighted average of the returns between flows, or the *holding-period return* of one unit invested at the start with distributions reinvested, which differ when flows are not pro rata;
- *market-adjusted return*, the portfolio return minus a market index return;

- *risk-adjusted excess return* or realised alpha, the portfolio return minus the portfolio's beta times the market return, with the beta measured concurrently because it changes when the manager trades;
- *market-timing return*, beta times the market return minus the market return; so that

$$r^p = \underbrace{(r^p - \beta r^m)}_{\text{excess return}} + \underbrace{(\beta r^m - r^m)}_{\text{market timing}} + \underbrace{r^m}_{\text{market}}$$

(p. 450). Harris notes that the decomposition generalises to other risk factors, such as size and growth.

Conditions. Returns of illiquid assets depend on estimated values; managers who adjust them slowly, especially downwards, report smooth returns with positive serial correlation; the low volatility inflates t-ratios, and forced revaluation later produces large losses, an artificial peso problem [alpha-harris, same chap., box "Smooth Sailing, Rapids Ahead", p. 448, and sec. 22.5.2.1, pp. 469–470]. Holding-period returns compound: +100 percent then −50 percent is zero, not +50 percent (box, p. 447).

Practice. Interpretation: transaction cost analysis applies the same idea over an order's horizon, subtracting beta times the index move from the arrival cost (transaction cost analysis); in both, the residual is not proof of skill.

Pitfalls. Interpretation: Harris's realised alpha $r^p - \beta r^m$ omits the risk-free rate; the legacy regression form $r^p_t - r^f_t = \alpha + \beta(r^m_t - r^f_t) + u_t$ (Jensen's alpha) differs from it by $(1-\beta)r^f$, for example −0.9 percentage points at $\beta = 1.3$ and $r^f = 3$ percent (checked); the regression also estimates a constant beta, whereas Harris uses concurrent betas. The legacy decomposition $r^p = r^m + (r^p - \beta r^m) + (\beta - 1)r^m$ is Harris's identity. The legacy list called the IRR "money-weighted, useful when capital flows are controlled by the investor" and added "trade-weighted" fill-quality measures; Harris calls the IRR approximately time- and value-weighted and does not discuss who controls flows, and fill quality belongs to transaction cost analysis. Return smoothing is in Harris a problem of fraudulent or careless valuation; the formal treatment the pathway asks for (`wide-getmansky-lo-makarov`) is unread.

See also. transaction cost analysis (the execution analogue, owned by `execution-trading-costs`).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. performance evaluation; absolute performance; relative performance; total return; internal rate of return; IRR; holding-period return; market-adjusted return; risk-adjusted return; realised alpha; Jensen's alpha; beta; market timing; performance attribution; return smoothing; stale valuations; serial correlation; peso problem

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 443–450 and 469–470; the decomposition and the Jensen comparison checked numerically). Stability: draft. Currency: timeless (methods); presentation standards named by Harris (AIMR-PPS) as of 2002.

Definition 6.11 Return measures: time-weighted and money-weighted returns, cumulative, geometric-mean and annualised returns, log returns, gross and net, real and leveraged returns

Statement. GIPS 2020 defines a *time-weighted return* (TWR) as "a method of calculating period-by-period returns that reflects the change in value and negates the effects of external cash flows", and a *money-weighted return* (MWR) as "the return for a period that reflects the change in value and the timing and size of external cash flows" [wide-gips-2020, Glossary, pp. 83 and 78]. An *external cash flow* is capital entering or leaving a portfolio; dividend and interest income are not external cash flows (p. 75). For portfolios in composites, firms must value at least monthly and on the date of all large cash flows, calculate sub-period returns at large cash flows if daily returns are not calculated, with exceptions for private-market investment portfolios, and "geometrically link periodic and sub-period returns" (provisions 2.A.23–2.A.24, p. 11). When calculating money-weighted returns, firms must "calculate annualized since-inception money-weighted returns" using daily external cash flows (2.A.29, p. 12), but "returns for periods of less than one year must not be annualized" (2.A.12, p. 9); accordingly a composite money-weighted report presents the annualised since-inception return through the most recent annual period end, and, for a track record shorter than a full year, the *non-annualised* since-inception return through the initial annual period end (5.A.1.a–b, p. 33). A *gross-of-fees* return is "reduced by any transaction costs", and a *net-of-fees* return is "the gross-of-fees return reduced by investment management fees" (Glossary, pp. 76 and 78).

Notation. Interpretation, derived here: with sub-period returns $r_1, \ldots, r_n$, each measured between external cash flows, the cumulative TWR is $R = \prod_{i=1}^{n}(1 + r_i) - 1$; the geometric mean per sub-period, for equal sub-periods, is $(1 + R)^{1/n} - 1$; and a return R over d days annualises mathematically to $(1 + R)^{365/d} - 1$. That formula is arithmetic; whether an annualised figure may be *presented* is a separate rule, and under GIPS it may not be for periods shorter than a year (2.A.12). The MWR is the internal rate of return x solving $\sum_t CF_t(1 + x)^{-t} = 0$, with contributions negative and withdrawals and the final value positive, from the investor's side. A continuously compounded (log) return is $\ln(1 + R)$. A real return is $(1 + r_{\text{nom}})/(1 + \pi) - 1$ for

inflation π. With own capital V_0 and borrowing V_B at rate r_B invested at return r for one period, the return on own capital is $r_L = [r(V_0 + V_B) - r_B V_B]/V_0 = r + (V_B/V_0)(r - r_B)$.

Conditions. The TWR removes the effect of the investor's timing of contributions and withdrawals, which suits judging a manager who does not control them; the MWR includes it, and GIPS allows firms to present it only when they control the external cash flows and the composite or pooled fund is closed-end, fixed-life, fixed-commitment or significantly illiquid (1.A.35, pp. 5–6). The IRR need not exist or be unique: the polynomial in $(1 + x)^{-1}$ can have several positive roots when cash flows change sign more than once, or none. Checked: flows of -1, $+2.5$, -1.54 have IRRs of 10 and 40 percent; flows of -1, $+1$, -1 have none, since their net present value is at most -0.75. Annualising a short-period return assumes it would recur; GIPS forbids annualising returns for periods of less than one year (2.A.12, a calculation-methodology provision that also governs what is presented), and gives no reason; interpretation, that assumption is the obvious one. For periods of a year or more, annualising is a change of units. The leveraged formula assumes one period, borrowing at a fixed rate r_B with no margin calls, fees or forced sales, and r and r_B measured over the same period.

Practice. Checked: \$100 grows 10 percent to \$110, \$100 is added, and the \$210 falls 10 percent to \$189. The TWR is $1.10 \times 0.90 - 1 = -1.0$ percent *cumulative over both periods*, unaffected by the addition, a geometric mean of -0.50 percent per sub-period. The MWR solves $-100 - 100(1 + x)^{-1} + 189(1 + x)^{-2} = 0$, giving $x = -3.71$ percent *per sub-period*, or -7.29 percent compounded over both. On matching horizons the MWR is lower, -7.29 against -1.0 percent over the two periods and -3.71 against -0.50 percent per sub-period, because more money was invested in the losing period. Sub-period returns of $+10$, -10 and $+5$ percent compound to 3.95 percent, a geometric mean of 1.30 percent per period. Four percent over 200 days annualises mathematically to 7.42 percent, a figure GIPS would not allow a firm to present (2.A.12). With 8 percent nominal and 3 percent inflation the real return is 4.85 percent, not 5. With $V_0 = 100$, $V_B = 50$, $r = 10$ and $r_B = 4$ percent, $r_L = 13$ percent. Harris's IRR and holding-period return, and the effect of non-pro-rata flows, are in relative performance evaluation.

Pitfalls. The legacy time-weighted formula, $\sqrt[1/n]{\prod(1 + HPR_i)} - 1$, is malformed: the $(1/n)$-th root would raise the product to the power n. The geometric mean is $(\prod(1 + HPR_i))^{1/n} - 1$, and the TWR for the whole period is the product minus one, without the root; the sub-periods must be split at external cash flows. The legacy real return, $(1 + r_{\text{Real}}) = (1 + r_{\text{Nom}})(1 + r_{\text{Risk}})/(1 + \pi)$, has an unexplained "risk" factor that does not belong to an inflation adjustment and is not adopted. The legacy treated pre-tax and after-tax returns only as labels; after-tax returns depend on jurisdiction, investor and realisation timing and were not checked. The source of the legacy's return table is not recorded; its formulas were checked here, not against a source.

See also. relative performance evaluation; interest-rate measures (compounding and annualising rates).

Further reading. The pathway's checkpoint asks for a GIPS-style composite presentation; composite construction and disclosures go beyond the definitions read here. The legacy file is kept verbatim in `notes/artefacts/rates-credit-mortgages/`.

Index. time-weighted return; TWR; money-weighted return; MWR; internal rate of return; IRR; multiple IRRs; external cash flow; geometric linking; cumulative return; geometric mean return; annualised return; log return; continuously compounded return; gross-of-fees return; net-of-fees return; real return; inflation; leveraged return; GIPS

Status. Verification: partly checked 2026-10-15 (against the text layer of the GIPS standards for firms, 2020 edition, CFA Institute, official PDF downloaded 2026-10-15, provisions 1.A.35 and 2.A.23–2.A.30 and the Glossary, printed page = PDF page -18; the formulas derived and checked numerically; provisions 2.A.12 and 5.A.1 read and the cash-flow example recomputed on matching horizons 2026-10-16). Stability: draft. Currency: GIPS 2020, effective 1 January 2020.

Chapter 7

Funding, collateral and liquidity

Pathways drawn on: `funding-crowding` (one draft entry from the migration: common-investor contagion, crowding and forced exits, with Narang's 2024 account of August 2007 kept as a practitioner's explanation; none of the pathway's readings yet studied, so its evidence is only in its backlog); `money-markets-repo-lending` (draft entries from the migration: street name and beneficial ownership, securities lending for short sales, and lending fees and collateral rebates, from Harris, the Duffie–Gârleanu–Pedersen working paper, the ICMA repo FAQ, the ISLA best-practice handbook and Regulation SHO as published in October 2026; none of the pathway's readings yet studied); and `synthetic-equity-cfd-trs` (one draft entry from the migration: the synthetic short through options, from Harris and own derivation; none of the pathway's readings yet studied). The entries came from a legacy execution file on securities lending and synthetic shorts, whose lending mechanics belong to the money-markets pathway rather than to its map candidate. The contagion entry came from the legacy risk-assessment file of Narang's systematic-investing notes, mapped to `model-governance-controls`; its forced-liquidation subject belongs here.

7.1 Crowding and common-investor contagion

Definition 7.1 Common-investor (contagion) risk: crowded positions and forced exits through what else their holders own, with Narang's account of August 2007

Statement. Contagion, or common-investor, risk is the risk a strategy bears "not because of the strategy itself but because other investors hold the same strategies" [s7-narang, chap. "Risks Inherent to Quant Strategies", sec. 10.4, p. 202]. Narang splits it in two:

- *crowding*: how many others hold the same positions;
- the *ATM effect*: what else those holders own that could force them out. When highly levered investors take losses on an illiquid holding and face margin calls, they sell what they can, the liquid strategy, to raise cash. Losses in one strategy thus cause the liquidation of an unrelated one (p. 202).

His examples are August 1998, in which LTCM's losses on Russian debt forced it to sell positions such as Royal Dutch against Shell, whose spread widened from 8–10 percent to over 20 percent (pp. 202–203), and August 2007 (sec. 10.4.1, pp. 203–209).

Conditions. *Narang's account of August 2007, a practitioner's explanation.* He names four drivers (pp. 203–205):

1. the size and popularity of quant long/short equity strategies from 2004 to 2007. He estimates "likely" hundreds of billions of capital and about $1 trillion of gross positions, mostly in large-cap U.S. stocks; despite diverse models, the overlap was enough to crowd individual trades;
2. weak returns before August, with value lagging growth from about the end of May 2007, and multi-strategy and proprietary desks that cut whatever had done poorly;
3. the practice, "in my view, a critically important one", of banks' proprietary desks and multi-strategy funds cross-collateralising strategies explicitly or implicitly. When credit strategies lost heavily in July 2007, the liquid quant books were sold to raise cash;
4. risk targeting: leverage set inversely to measured risk, usually VaR. Falling volatility from 2003 to 2006 called for one and a half to two times more leverage, so a four-times gross strategy became six or eight times. Higher returns were also sought by levering up. When volatility and correlations jumped together in late July 2007, VaR rose and many players had to cut leverage at once.

In his sequence, multi-strategy funds and desks began deleveraging the most liquid book, U.S. quant long/short. Statistical arbitrageurs first supplied liquidity, then took losses and joined the selling. By 8–9 August the losses had spread across strategies and to Japan, while U.S. indices were roughly flat for the first ten days of August (pp. 205–209). He calls the losses "solely the result of market impact" (p. 208), reports losses of −5 to −45 percent at funds (p. 208), and concludes that "for the first time, crowding became a risk of quant trading strategies" (p. 209). Phrases such as "it is likely", "it appears" and "in my view" mark these as his inferences: the chapter gives no data on positions, leverage or flows.

Practice. Interpretation: the entry separates two channels. In *common positions*, crowding, the same trades are held by many, so any one holder's exit moves the others' prices. In *common funding*, the ATM effect, an unrelated loss forces a holder to sell. Cross-collateralisation is what joins the second channel to a liquid strategy, and volatility targeting can make many holders sell at the same moment. The same mechanics in arbitrage, where staying power fails under leverage, are in arbitrage risks. Leverage and margin terms are in futures margin and settlement and lending fees and rebates.

Pitfalls. The account is one practitioner's reconstruction. It is not causal evidence that the four drivers produced the losses, nor a measure of their weights; the pathway's evidence readings (stubs) are unread. Narang's Pulte Homes illustration (Exhibit 10.6, p. 206, checked by page image) does not support all its words. By the table, the stock fell 22.0 and then 12.5 percent (31.75 percent together) and then rose 15.6 percent, which recovers about a third of the lost price, not "about half". Over his 44-day and four-day periods that is roughly four to five times the earlier pace, not "about 20 times". He also calls the episode "an interesting illustration of the quadratic nature of market impact" (p. 206). Interpretation: the table cannot test that. Aggregate returns and average daily volume over a crisis period do not identify a metaorder impact exponent: they give no metaorder sizes or execution schedules, the volume mixes many traders' buying and selling, and the price that would have prevailed without the liquidation is not observed. The illustration is therefore evidence neither for a quadratic nor against the square-root law for metaorders (square-root impact), which is estimated from identified metaorders. His "as much as 15 percent in market impact" on 9 August is likewise unsourced in the chapter. A separate question stays open in `backtest-inference`'s stub `ev:signal-decay-crowding`: whether faster decay of a delayed signal identifies crowding. Common positions and forced liquidation, the subject here, do not answer it. The legacy "quant liquidation criss" is the 2007 crisis; its four causes match Narang's list.

See also. arbitrage risks (staying power under leverage); sources of strategy risk (the other three quant-specific risks); square-root impact (impact of large liquidations); futures margin and settlement (margin calls).

Further reading. The legacy block it replaces, verbatim, is in `risk-assessment.tex` in the folder `notes/artefacts/` under the pathway key `model-governance-controls`.

Index. contagion risk; common-investor risk; crowding; crowded trade; ATM effect; forced liquidation; deleveraging; fire sale; cross-collateralisation; margin call; risk targeting; volatility targeting; VaR-based leverage; quant liquidation crisis; quant meltdown; August 2007; LTCM; August 1998; Royal Dutch Shell

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-narang`, third edition, Wiley 2024, chap. "Risks Inherent to Quant Strategies", sec. 10.4, pp. 202–209, with Exhibit 10.6 on p. 206 by page image and its arithmetic recomputed; no evidence source read; the reading of the Pulte illustration against metaorder impact corrected 2026-10-15 in the slice-26 follow-up). Stability: draft. Currency: practitioner account as of 2024 of events in 1998 and 2007.

7.2 Securities lending and short selling

Definition 7.2 Street name and beneficial ownership: holder of record, beneficial owner, custody, permission to lend, and who votes lent shares

Statement. Clients hold securities in *street name* when their broker or a depository holds them on their behalf. The issuer then registers the broker or depository as the *holder of record*, and the client becomes a *beneficial owner* holding a claim on the broker rather than legal title [`alpha-harris`, chap. "Brokers", secs. 7.2.2.2 and 7.5.6.1, pp. 148 and 169–170]. Four things are kept apart:

- *registration*: whose name is on the issuer's register (Harris notes that most securities held by the Depository Trust Company are registered in the name of its nominee, Cede & Co., p. 169);
- *beneficial ownership*: the client's economic claim, recorded in the broker's books;
- *custody*: the broker's or depository's safekeeping, with duties to collect and allocate dividends and interest and to process corporate actions (p. 148);
- *permission to lend*: in Harris's U.S. account, a margin agreement lets the broker hypothecate the client's securities as loan collateral and lend them to short sellers; clients generally do not know whether their securities were lent and generally receive no lending fee, which is why some large institutions hold securities in their own name (pp. 169–170).

Traders hold in street name to avoid losing certificates, to use securities as margin collateral and to have them available for settlement (p. 148).

Conditions. Voting. Only holders of record have the legal right to vote; a beneficial owner can only instruct its broker. A securities loan transfers legal title, and with it the voting rights attached to equities [al-icma, Repo FAQ 13]. When a lent share is sold short, its buyer may register it and acquire the vote, so the beneficial owners in a broker's books can together believe they own more shares than the broker holds of record. Harris's example: the broker then votes its record shares in proportion to its clients' instructions, and since many clients do not instruct, it usually has enough shares unless it has lent too many [alpha-harris, sidebar "Can Investors Vote More Than 100 Percent of the Shares Outstanding?", p. 169]. Several parties may hold economic claims; only record shares carry legal votes. Because a loan transfers votes, lenders keep an express right of recall so that they can recover securities to vote or respond to corporate actions [al-icma, Repo FAQ 13]; the borrower pays the lender amounts equal to dividends ("cash-in-lieu") [al-duffie, sec. 2, p. 8].

Practice. Record dates decide who may vote. Hu and Black call borrowing shares shortly before a meeting's record date and returning them soon after "record date capture": the borrower holds votes without economic ownership and the lender economic ownership without votes [cfd-hu-black-2007, working-paper version, sec. 2.2.3, p. 5]. They report that in the United States Federal Reserve Board Regulation T bars lending shares for voting purposes "in many circumstances"; that statement was not checked against the regulation. Decoupling through equity swaps and options, and beneficial-ownership disclosure, are a separate matter (Pitfalls).

Pitfalls. The legacy caveat said stock lending "can separate economic exposure from voting rights" and that brokers "need entitlement records because more investors may believe they own voting shares than there are record shares available". Both match Harris's sidebar, which describes how brokers vote, not a legal requirement; proxy-processing rules and current broker practice were not checked. Do not infer duplicate legal votes from several economic claims. Empty voting and hidden ownership through equity swaps, and the 2023 U.S. beneficial-ownership reporting rules, belong to `synthetic-equity-cfd-trs` and are stubbed there. Harris describes U.S. practice as of 2002.

See also. securities lending for short sales; lending fees and rebates; broker conflicts.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/synthetic-equity-cfd-trs/`.

Index. street name; holder of record; shareholder of record; beneficial owner; Cede & Co.; custody; hypothecation; margin agreement; permission to lend; lent shares; voting rights; record date; record date capture; proxy voting by brokers; over-voting; cash in lieu of dividends; manufactured dividend; recall for voting

Status. Verification: partly checked 2026-10-15 (against the text layer of `alpha-harris`, 2002 Oxford printing, pp. 148 and 169–170; `al-icma`, ICMA ERCC Repo FAQ 13 as retrieved 15 October 2026; `al-duffie`, working-paper version of 24 September 2001, sec. 2, p. 8; `cfd-hu-black-2007`, ECGI working-paper version, pp. 1 and 5). Stability: draft. Currency: U.S. practice as described in 2001–2007; ICMA as of October 2026.

Mechanism 7.3 Securities lending for short sales: locate, borrow, delivery, recall, replacement borrow and buy-in, with broker and institutional layers kept apart

Statement. A short seller sells securities it does not own and must deliver them at settlement; the standard route is to borrow them for delivery through a stock loan. In Duffie, Gârleanu and Pedersen's description of U.S. equity lending in 2001 [al-duffie, sec. 2, pp. 5–8]:

1. *locate*: the would-be short seller asks its broker for a "locate"; the broker looks in its own inventory, in customer accounts that permit lending, then to custodian banks or other lenders, such as insurers, index funds and pension funds. Hard-to-locate stocks may get partial fills or delays (pp. 5–6);

2. *borrow*: given a locate, the loan can be done the same day; through a broker, the broker borrows from the outside lender and lends to the outside borrower (p. 7). A broker may pay a lender to hold located shares until the short sale ("pay-for-hold", p. 6);

3. *delivery*: the borrowed shares are delivered to the buyer at settlement;

4. *recall*: loans are normally open, renewed daily; the lender may recall, and the borrower must return the shares, typically by borrowing them from another lender (*replacement borrow*) or by its broker recalling them from another borrower (pp. 7–8);

5. *buy-in*: if no lendable shares can be found the borrower must buy the stock; if the borrower fails to deliver within standard settlement time, the lender may buy it with the cash collateral, the borrower remaining liable for extra costs. A buy-in interrupts the short sale (p. 8).

A securities loan transfers legal title in exchange for collateral, and lenders keep an express right of recall [al-icma, Repo FAQ 13].

Conditions. *U.S. rules (Regulation SHO, 17 CFR part 242, as published on eCFR for 13 October 2026).* A broker-dealer may not accept a short-sale order in an equity security, or short for its own account, unless it has borrowed the security or arranged in good faith to borrow it, or has reasonable grounds to believe it can be borrowed for delivery when due, and has documented compliance, with stated exceptions including bona-fide market making [cfr-242-203, § 242.203(b)]. A clearing-agency participant with a fail to deliver on a short sale must close it out by borrowing or buying securities of like kind and quantity by the beginning of regular trading hours on the settlement day after settlement date, with longer periods for some cases [cfr-242-204, § 242.204(a)]. A locate is therefore a belief or arrangement before the sale, not a completed borrow; delivery and close-out are later obligations. Harris's 2002 "affirmative determination" (p. 155) predates Regulation SHO. Other jurisdictions' rules were not checked.

Market practice, not law. ISLA's best-practice handbook, which yields to the Global Master Securities Lending Agreement and to local codes, recommends that a recall notice give the ISIN, quantity and settlement date, reach the borrower at the latest one hour before the relevant market close minus two business days or, if greater, the standard settlement cycle; that the borrower acknowledge; and that, if the return fails to settle or the borrower does not respond, the lender calculate and claim the costs under GMSLA 9.3, after which the loan is closed and collateral returned [al-isla, "Loan Recall to buy-in/close out" and "Legal", pp. 2–3 and 24–25, page images; sections last updated 28 January 2022].

Practice. Keep apart, Interpretation using the sources above:

- *broker–customer arrangements*: the customer's margin agreement, its short-sale margin and what the broker pays it, and the broker's permission to lend customer securities;
- *the institutional loan*: lender (or agent lender) and borrower under a master agreement, with collateral, fee or rebate, marking to market, recall and claims;
- the *availability* of a security to borrow (general collateral or special, lending fees and rebates) from *operational* risks: a locate that cannot be turned into a borrow, a delivery failure, a recall, a failed return, a buy-in.

Factors said to make shares hard to locate include small capitalisation, low float, exclusion from indices, illiquidity, concentrated ownership and events such as IPOs, mergers, spinoffs and acquisitions; the authors had no data on locate delays (p. 6).

Pitfalls. The legacy mechanism's "brokers source inventory from street-name holdings, stock-loan desks, brokers, or institutions, earning lending fees or rebate spreads" matches Harris (pp. 155 and 158) and Duffie, Gârleanu and Pedersen. The legacy taxonomy listed recall risk and settlement risk beside general collateral and specials; they are operational risks, not availability classes, and are separated here. Its "failed locate … can break the short-sale trade" runs two steps together: under Regulation SHO the locate precedes the order, and it is a delivery failure that triggers close-out. DGP's description of retail short-sale margin and of a price test in 2001 is historical; current margin rules and short-sale price tests were not checked.

See also. lending fees and rebates; street name and beneficial ownership; synthetic short through options; arbitrage risks.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/synthetic-equity-cfd-trs/`.

Index. securities lending; short sale; locate; borrow; pre-borrow; pay-for-hold; delivery; fail to deliver; close-out; Regulation SHO; Rule 203; Rule 204; recall; replacement borrow; buy-in; short squeeze; open loan; agent lender; GMSLA; ISLA best practice; hard to locate

Status. Verification: partly checked 2026-10-15 (`al-duffie`, working-paper version of 24 September 2001, sec. 2, pp. 5–8, text layer; `al-icma` as retrieved 15 October 2026; `al-isla`, 29-page compilation dated 31 March 2025, pp. 2–3 and 24–25 by OCR of page images; 17 CFR 242.203(b) and 242.204(a) on eCFR, version of 13 October 2026; `alpha-harris`, pp. 155 and 158). Stability: draft. Currency: Regulation SHO as of October 2026; ISLA practice as last updated 2022; market description as of 2001–2002.

Definition 7.4 Lending fees and collateral rebates: cash and non-cash collateral, general collateral and specials, reinvestment returns, broker spreads and customer margin

Statement. A securities borrower pays the lender a fee for the use of the security. Against non-cash collateral (other securities) the fee is paid outright. Against cash collateral, the lender reinvests the cash and *rebates* an agreed part of the reinvestment return to the borrower, usually deducting the fee, so that the fee is implicit in the rebate rate [al-icma, Repo FAQ 13]. In Duffie, Gârleanu and Pedersen's U.S. description, cash collateral is normally 102 percent of the market value of the shares for domestic securities (105 percent for international); an overnight rebate rate r pays $r/360$ times the cash collateral per day; and the shortfall of the rebate below a market rate such as the federal funds rate is the lender's benefit [al-duffie, sec. 2, p. 7]. ISLA describes the rebate as a negotiated spread in basis points over or under a benchmark rate [al-isla, "Cash Rebate", pp. 7–8].

With r_m a market overnight rate and r_b the rebate rate on cash collateral, the implicit annualised lending fee is approximately $r_m - r_b$ (Interpretation); the definitions above do not fix the benchmark.

Conditions. *Availability classes.* Easy-to-borrow securities carry small fees; in Harris's 2002 U.S. account the fee is "only about 10 basis points per year", and hard-to-borrow securities are said to be *on special*, with fees that depend on scarcity [`alpha-harris`, sec. 7.3.1.4, pp. 156–157]. Fees are highest for closely held securities that interest short sellers (sec. 7.3.1.7, p. 158). Duffie, Gârleanu and Pedersen model lending fees as the outcome of search for lenders and bargaining: if lendable securities are hard to find, the price is initially elevated and expected to decline, and may exceed even the most optimistic buyer's valuation because owners expect lending fees (abstract; a model result). *General collateral* is used here for the easy-to-borrow class; the term's meaning in repo, collateral not specific to an issue, is related but not identical (Interpretation).

Practice. Who receives what, in the sources' U.S. descriptions:

- the *lender* earns the fee, or the reinvestment return minus the rebate, and bears reinvestment risk on cash collateral (Interpretation);
- the *broker* intermediating the loan normally offers its outside borrower a lower rebate than it receives (DGP, p. 7), a broker spread;
- *retail short sellers* typically receive no interest on their short proceeds, and retail customers whose stocks are lent typically receive no fee (DGP, p. 7; Harris, pp. 157 and 170); large clients demand a rebate (Harris, p. 157);
- *customer margin* is a separate obligation of the customer to its broker, not collateral in the institutional loan; DGP give the Regulation T requirement of 2001 as an additional 50 percent of market value, which was not checked as current (p. 7, footnote 9).

Pitfalls. The legacy definition said "the short seller posts cash collateral" and "the lender may rebate part of the collateral interest". In the sources, cash collateral passes from the borrower (often the broker) to the lender; the retail short seller's obligation is margin to its broker. Its "the harder the borrow, the lower the rebate and the higher the effective shorting cost" matches the sources for cash-collateral loans; rebates can in principle be negative when the fee exceeds the reinvestment rate, as the legacy taxonomy said, but no source read gives a negative-rebate example. Harris's rebate benchmarks (federal funds or LIBOR) are of 2002 and are not current.

See also. securities lending for short sales; synthetic short through options; futures margin.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/synthetic-equity-cfd-trs/`.

Index. lending fee; borrow fee; rebate rate; short interest rebate; cash collateral; non-cash collateral; collateral reinvestment; easy to borrow; hard to borrow; on special; specials; general collateral; broker spread on rebates; retail short proceeds; customer margin; Regulation T; search and bargaining in securities lending

Status. Verification: partly checked 2026-10-15 (`al-icma` as retrieved 15 October 2026; `al-duffie`, working-paper version of 24 September 2001, abstract and sec. 2, p. 7; `al-isla`, pp. 7–8 by OCR, section last updated 15 January 2021; `alpha-harris`, pp. 155–158 and 170). Stability: draft. Currency: market practice as described in 2001–2002 and by ICMA and ISLA in 2021–2026.

7.3 Synthetic short positions

Result 7.5 Synthetic short through options: long put plus short call is a short forward, not short stock, and does not remove the cost of scarce borrow

Statement. Take a European put and a European call on the same underlying, with the same strike K, maturity T and contract multiplier, both settled the same way. Long the put and short the call pays at T

$$\max(K - S_T, 0) - \max(S_T - K, 0) = K - S_T$$

for every terminal price S_T: a short forward position at delivery price K. With no arbitrage, a constant continuously compounded rate r and a continuous dividend yield q, put–call parity gives the initial value

$$P - C = Ke^{-rT} - S_0 e^{-qT} = e^{-rT}(K - F), \qquad F = S_0 e^{(r-q)T},$$

so the package costs $e^{-rT}(K - F)$ upfront, positive or negative, and is a zero-cost short forward only when $K = F$.

Adding a *short* zero-coupon bond that repays K at T (borrowing Ke^{-rT} now) turns the package's payoff into $K - S_T - K = -S_T$. For a stock paying no dividends ($q = 0$) that is the terminal value of one share sold short, and the initial values agree: $P - C - Ke^{-rT} = -S_0$. So long put, short call and short bond are equivalent

to short stock; equivalently, long put and short call are equivalent to short stock plus a long bond paying K, a short sale whose proceeds are invested. With dividends the package less the bond is a terminal claim $-S_T$, worth $-S_0 e^{-qT}$ now, while a share sold short is worth $-S_0$ and owes the dividends paid before T; the two differ by those dividend liabilities.

Harris's sidebar says that a long put, a short call with the same strike and maturity, "plus a long position in a bond with the same maturity", is economically equivalent to a short position in the underlying [alpha-harris, chap. "Brokers", sidebar "Synthetic Short Positions", p. 157, page image]. As worded, with a long bond paying K the payoff is $2K - S_T$; the bond belongs on the other side, as above. The discrepancy is recorded here and Harris's wording is quoted as printed.

Notation. S_0, S_T: underlying price now and at maturity; P, C: European put and call premia per unit of underlying (multiply by the contract multiplier for one contract); F: forward price; r: financing rate; q: dividend yield. With discrete dividends of present value D, replace $S_0 e^{-qT}$ by $S_0 - D$.

Conditions. The identity needs matching strike, maturity, underlying and multiplier, the same settlement (both cash or both physical), European exercise, and frictionless borrowing and lending at r. Parity example (own computation, Black–Scholes prices with $S_0 = 100$, $K = 95$, $r = 5\%$, $q = 2\%$, $\sigma = 25\%$, $T = 1$): $C = 13.6847$, $P = 6.0317$, $P - C = -7.6531 = 95e^{-0.05} - 100e^{-0.02}$; $F = 103.0455$ and $e^{-rT}(K - F) = -7.6531$. The script `synthetic_short_checks.py` beside the artefact reprices both options by numerical integration over the risk-neutral density, and checks the payoffs leg by leg at $S_T = 60, 95, 100, 140$: $K - S_T$ for the package, $-S_T$ with a short bond, $2K - S_T$ with a long bond (Interpretation).

Practice. Against a short sale, Interpretation using the numbers above:

- *package against short sale, no dividends*: take $q = 0$, European options priced at frictionless parity, and borrowing or lending at r. Financing the premium $P - C = Ke^{-rT} - S_0$ at r, the package ends with $K - S_T - (P - C)e^{rT} = S_0 e^{rT} - S_T$. Sell one share short, hold the whole proceeds S_0 as cash collateral credited at a constant rebate r_b (the lending fee implicit in r_b, no other fee, no extra margin, no recall), and buy back at T: $S_0 e^{r_b T} - S_T$. The two agree only when $r_b = r$, and the package is ahead by $S_0(e^{rT} - e^{r_b T})$ otherwise. With $S_0 = 100$, $K = 95$, $r = 5\%$, $\sigma = 25\%$, $T = 1$: $C = 15.0471$, $P = 5.4138$, $P - C = -9.6332$; at $S_T = 100$ the package ends with 5.1271 and the short sale with 5.1271, 3.0455 or -0.9950 for $r_b = 5\%$, 3% or -1%, gaps of 0, 2.0817 and 6.1221 at every S_T (checked with day-by-day accounts in the script). The synthetic therefore replicates a short sale that earns the full market rebate, which is Harris's point: when options dealers receive the rebate and competition makes them price it in, the synthetic short "in effect" obtains short interest rebate (p. 157);

- *borrow cost*: a market maker who sells the put or buys the call is left long the stock's delta and hedges by selling the stock, short if it holds none (Interpretation), so a scarce borrow can be priced into the options and parity with a frictionless r need not hold. That is an expectation, not a finding of the sources read; the package does not guarantee access to scarce borrow or remove its cost;

- *upfront and dividends*: the premium $P - C$ is paid or received at the start; dividends enter through F, whereas a short seller pays them to the lender as they fall due;

- *costs*: option spreads (option spread measurement) and the absence of listed options on many securities limit use; Harris notes that transaction costs "can significantly reduce the economic benefits" and that the strategy works only where options trade (p. 157).

Pitfalls. Listed U.S. equity options are typically American and physically settled; that was not checked here against exchange or OCC documents. With American options the payoff identity holds only to maturity: the short call can be assigned early, notably before an ex-dividend date, which delivers a real short stock position that must then be borrowed or covered; the long put may be worth exercising early. Parity for American options is an inequality, not an equality. The legacy principle said the package "replicate[s] a short forward" and "can replace scarce borrow, subject to funding, dividends, exercise, and option-liquidity costs"; the first is exact for European options, and the second is qualified above. Hull's treatment of put–call parity, a pathway reading, was not available and was not read; the derivation is standard and was checked numerically. Hu and Black note that a put and call with the same exercise price and expiration are economically equivalent to selling shares [cfd-hu-black-2007, working-paper version, p. 5, footnote 6].

See also. securities lending for short sales; lending fees and rebates; option market-maker spreads; arbitrage; arbitrage risks.

Further reading. The legacy block it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/synthetic-equity-cfd-trs/`.

Index. synthetic short; put–call parity; long put short call; short forward; forward price; conversion and reversal; dividend yield; short interest rebate in options; American early exercise; assignment risk; ex-dividend early exercise; synthetic short with futures

Status. Verification: partly checked 2026-10-15 (the text layer of `alpha-harris`, 2002 Oxford printing, p. 157; `cfd-hu-black-2007`, ECGI working-paper version, p. 5, footnote 6; the page image of p. 157 rechecked in a follow-up the same day; the identities derived and checked numerically by the script beside the artefact; `s7-hulloptions` not read). Stability: draft. Currency: timeless identity; market remarks as of 2002.

Chapter 8

Market microstructure and market making

The first two sections, from slices 33 and 34, cover orders and their instructions, matching and trade pricing, the limit order book, limit orders as options, queues and imbalance measures (for the overview `execution-market-making`, `market-making-lob` and `lob-deep-learning`), and order flow in time, from stylised facts and realised variance to deseasonalisation, market clocks, point processes and Hawkes processes (for `market-simulation` and `market-making-lob`); they draw on Harris, Hasbrouck, Lehalle and Laruelle, Bouchaud et al., and two catalogue papers read in preprint, Cont, Kukanov and Stoikov and Bacry, Mastromatteo and Muzy, the latter a required reading of `market-making-lob`; none of these pathways' readings yet studied. Pathways drawn on: `market-making-lob` (draft entries from the migration: supplemental inventory, information and quoting models, Glosten–Milgrom, a practitioner account of dealer quoting, and textbook accounts of inventory-based quoting; of the pathway's required readings only Avellaneda–Stoikov and Glosten–Milgrom, through its 1983 working paper, are drawn on); and the section overview `execution-market-making` (draft entries from the migration, from Harris, Hasbrouck and Lehalle and Laruelle; none of its readings yet studied); and `hft-market-design` (draft entries from the migration: high-frequency trading as a set of strategies sharing speed tools, from the SEC's 2010 concept release and Lehalle and Laruelle; the latency-arbitrage arms race, from Budish, Cramton and Shim, a required reading checked in its published version; and a stylised model in which a narrower spread need not lower other traders' costs, from Lehalle and Laruelle; none of the pathway's readings yet studied). The section beginning with fundamental value covers informed trading, informative prices, informed-trader types, the winner's curse, market efficiency, inside information and its deterrence, and the price formation process, from Harris and Lehalle and Laruelle; the section beginning with arbitrage covers arbitrageurs, order anticipators and bluffers, from Harris. The section beginning with spread measures measures and decomposes spreads, from Harris and Hasbrouck (the latter a required reading of this section's overview route, not yet studied); the section beginning with fundamental and transitory volatility covers volatility, returns by time scale, the relation of volatility to spreads, and bubbles and crashes, from Harris, Hasbrouck and Lehalle and Laruelle. A later section compares floor and automated trading systems, from Harris (2002). The last section, for `fx-dealer-market-making`, covers spot FX market structure, internalisation and last look, from the BIS Markets Committee's 2020 report, the FX Global Code, a provider disclosure and three of the pathway's readings; none of its readings yet studied. The section after it, for `options-execution`, covers option market-maker spreads, the measurement of option execution costs and retail option order routing, from six of the pathway's readings, one read in full and five through their abstracts; none yet studied. Notes for this section that predate the study method are in Chapter 13, in the part Notes awaiting migration.

8.1 Orders, limit order books and queues

Mechanism 8.1 Order lifecycle and messages: order states, insert, modify, cancel and execute, private and public channels, and the instructions that change priority

Statement. Lehalle and Laruelle's account starts from a book that is "most often empty" at the start of the day; orders are stored while they do not match a resting order, and two orders match when their prices are compatible, each match producing one public trade message and two private messages, one to the owner of each order [`alpha-laruelle`, chap. 3, sec. 3.3.1, p. 217]. Members usually reach the matching engine through two channels: a private one, on which they send their orders and receive the messages that concern them, and a public one, on which everyone connected sees the transactions and the state of the book; the private channel is generally the faster (p. 218). At any time a trader can insert, modify or cancel an order and attach tags to it,

such as a time in force (immediate or cancel, good till day, good till time) or a displayed quantity, which creates an iceberg order (p. 218). An order then leaves the book by execution against an incoming order that crosses it, by cancellation, or by expiry; an iceberg's hidden part usually has lower time priority than displayed orders at the same price submitted later, and an immediate-or-cancel order that matches nothing leaves no visible trace [catalogue-0528, chap. 3, sec. 3.2.2, p. 55]. The states an order passes through are those a broker or venue reports for it. Harris calls an order *pending* after submission and before the broker accepts it (for example while an account or a borrow is confirmed) and *working* once accepted and until filled or cancelled [alpha-harris, chap. 4, sec. 4.2, p. 71]. In the FIX 4.4 message standard, OrdStatus (tag 39) "identifies current status of order" with the values new, partially filled, filled, done for day, cancelled, pending cancel, stopped, rejected, suspended, pending new, calculated, expired, accepted for bidding and pending replace; there is no value for *replaced*. ExecType (tag 150) "describes the specific ExecutionRpt ... while OrdStatus (39) will always identify the current order status": a confirmed replacement is reported with ExecType 5 (Replace) and an OrdStatus such as new or partially filled, and a fill with ExecType F (Trade) [fix44-orchestra, OrdStatus and ExecType code sets, FIX.4.4_EP311].

Notation. *Insert* adds displayed or hidden quantity at a price; *modify* changes price, quantity or instructions; *cancel* removes resting quantity; *execute* is a match against an incoming marketable order. A replacement (cancel-replace) is reported by its execution type, not as an order state. *Working* (resting) and *part-filled* orders have quantity still governed by their instructions; filled, cancelled, expired and rejected orders are terminal.

Conditions. *Sessions.* Whether a book starts empty depends on the venue and on the orders' durations. A day order is valid for the trading day on which it is submitted and expires at the close if unfilled, and most brokers assume a day order when no instruction is given; a good-till-cancel order stays valid until cancelled (some brokers cancel such orders after a month or two), and good-until orders until a stated date [alpha-harris, chap. 4, sec. 4.9, pp. 82–84, Table 4-3]. FIX 4.4's TimeInForce treats an absent value as "Day (or session)" and also defines good till cancel, good till date, at the opening, immediate or cancel, fill or kill, good till crossing and at the close [fix44-orchestra, TimeInForce code set]. Orders carried across sessions therefore rest in the book when it reopens, which a replay must load; the current rules of a named venue (for example how CME Globex carries good-till-cancel and good-till-date orders) are a backlog item, not stated here. Priority after a modification is a venue rule, not a general one. On CME Globex under FIFO matching an order loses its priority and is requeued if its quantity is increased, its price changed or its account changed [ex-cme-matching, page "Supported Matching Algorithms", updated 11 February 2025]; a reduction in quantity is not on that list. Under pro-rata matching, which Bouchaud et al. describe as common in short-rate futures, time does not break ties at a price, and each resting order receives a share of an incoming order in proportion to its size [catalogue-0528, chap. 3, sec. 3.2.1, p. 54]. A rule that applies to one venue, order type or matching algorithm must be stated with that venue and its rule date. Lehalle and Laruelle note in passing that "inserting a new order will have a better rank than a modified order" on reaching a matching engine [alpha-laruelle, chap. 1, sec. 1.2.1, p. 70], without naming a venue.

Practice. Limit and stop instructions let a trader who is absent take part: many traders use limit orders to tell their brokers the conditions under which they would trade, and brokers then watch the market on their behalf [alpha-harris, chap. 4, sec. 4.4.5, pp. 77–78]. A stop order becomes an ordinary order once triggered and "remains valid even if price crosses back over the stop price", so it can fill well beyond its stop price in a fast market (Harris's cotton example: a stop at 70 cents sold at 66.90; sec. 4.5, p. 78). For a replay, an execution record therefore needs the acknowledgement, every state change and the channel timestamps of each order, not only its fills; the public feed alone cannot show which resting order was the replayer's own.

Pitfalls. Slice 33 listed "replaced" among FIX 4.4's order states, from a vendor dictionary that itself marked the value as replaced in FIX 4.3; it is an execution type, not a state, and was removed in the slice-33 follow-up, as was an unqualified "an order book is empty at the start of the day". The legacy notes said a modification "loses priority when price worsens or size increases". On CME's FIFO algorithm any price change loses priority, an improvement included; the legacy rule matches no venue rule read here. Their five "order states" (pending, accepted, working, part-filled, terminal) are a simplification of the reported states above, and they named minimum-quantity and cancel-on-disconnect instructions, which were not checked against any venue's rules and are left as a backlog item.

See also. limit order book; order instructions; matching and trade pricing; CME Globex trading conventions (matching algorithms); order exposure decisions (hidden and iceberg orders); operational controls (unknown order state); market data levels (Level III order messages).

Further reading. The pathway's required readings on replaying order books (Huang, Lehalle and Rosenbaum; Noble, Rosenbaum and Souilmi) are not read here. The legacy file is kept verbatim in `notes/artefacts/execution-market-making/`.

Index. order lifecycle; order state; OrdStatus; pending new; partially filled; insert; modify; cancel; execute; private channel; public feed; time in force; immediate or cancel; iceberg order; loss of priority; requeue; stop order; absent traders

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`,

2018 World Scientific printing, pp. 217–218; catalogue-0528, Cambridge 2018, pp. 54–55; alpha-harris, 2002 Oxford printing, pp. 77–78; the CME page as cited in slice 19, not reread; corrected 2026-10-16 against the FIX Trading Community's FIX 4.4 Orchestra repository as rendered by Orchimate (OrdStatus, ExecType and TimeInForce code sets) and alpha-harris pp. 71 and 82–84). Stability: draft. Currency: FIX 4.4 (2003) code sets; Harris's broker practice as of 2002; the CME rule as of 2025; other venues not checked.

Definition 8.2 Order instructions: what an order specifies, the standard order types, validity, quantity, display and handling instructions, and how they interact

Statement. Orders are trade instructions: they always specify the instrument, quantity and side, and may add conditions, most often limits on price, and also how long the order is valid, when it can be executed, whether partial fills are acceptable, where to present it, how to search for the other side and even with whom to trade [alpha-harris, chap. 4, sec. 4.1, pp. 68–69]. Traders who cannot watch the market must write contingencies into their orders, and lose when old orders trade before they can be cancelled (p. 69). Harris's standard instructions are:

- *market*: trade at the best price currently available; small market orders pay the spread, may receive price improvement from a dealer, large ones have market impact, and all bear execution-price uncertainty from quote changes between submission and execution (sec. 4.3, pp. 71–73);
- *limit*, including the *marketable* limit order: trade only at the limit price or better (sec. 4.4, pp. 73–78; limit order book);
- *stop* and *stop limit*: a stop price activates the order when the market reaches or passes it, after which a stop order is an ordinary market order and a stop limit order an ordinary limit order; "a stop instruction provides for the activation of an order", whereas a limit price sets the terms of trade (sec. 4.5.1, p. 79); stop orders demand liquidity when it is least available and add momentum (sec. 4.5.2, p. 79);
- *market-if-touched*: a market order activated when price falls to the touch price for a buy (rises, for a sell), contrarian, and very rare (sec. 4.6, p. 80, Table 4-2);
- *tick-sensitive*: a buy downtick order can fill only on a downtick or zero downtick, a sell uptick order only on an uptick or zero uptick; they cannot have market impact, act as limit orders with a limit that follows the last different price, and are most attractive when the tick is large (sec. 4.7, p. 81);
- *market-not-held*: the broker may use discretion over timing and price and is not accountable for failing to trade, used mostly for large orders (sec. 4.8, p. 82);
- *validity*: day (the most common, and the default at most brokers), good-till-cancel, good-until a date (good this week, good this month), immediate-or-cancel, good-after, market-on-open and market-on-close (sec. 4.9, pp. 82–84, Table 4-3);
- *quantity*: all-or-none, which must fill at once, and minimum-or-none (minimum acceptable quantity), each of whose fills must exceed a stated size (sec. 4.10, pp. 83–84);
- *other*: spread orders, with a limit on the price difference stated as a premium to the buy or sell side; display instructions that cap the displayed size (undisclosed, hidden, reserve or iceberg orders); substitution orders that let the broker choose among securities; and special settlement instructions (sec. 4.11, pp. 84–86).

Notation. FIX 4.4 carries these in separate fields: OrdType (40) for market, limit, stop, stop limit, market-if-touched, pegged and others; TimeInForce (59) for day, good till cancel, at the opening, immediate or cancel, fill or kill, good till crossing, good till date and at the close; and ExecInst (18) for handling instructions such as not held, all or none, do not increase or reduce, and the pegs (market, primary, mid-price, last and others) [fix44-orchestra, OrdType, TimeInForce and ExecInst code sets, FIX.4.4_EP311]. A label such as "hidden pegged IOC" is shorthand for values in several fields.

Conditions. *Interactions.* An order type and its instructions combine, and the combinations are not all equivalent across sources. Harris lists immediate-or-cancel, good-on-sight and fill-or-kill together as orders valid only on submission, "in some markets" called by the other names, and in his Euronext example a fill-or-kill limit order for 60,000 shares, sent against a 10,000-share display, fills in full from the hidden size behind it, while he says that any unfilled remainder of that order would have been cancelled, which is immediate-or-cancel behaviour in FIX's terms (pp. 83–85); Bouchaud et al. and FIX distinguish them: an immediate-or-cancel order executes what it can and cancels the rest, while fill-or-kill (all-or-nothing) executes in full or not at all [catalogue-0528, chap. 3, sec. 3.2.2, p. 55]. A marketable limit order without such an instruction leaves its unfilled remainder resting at its limit, and "traders who do not want their unfilled orders to stand in the book must attach a fill-or-kill or an immediate-or-cancel instruction" [alpha-harris, chap. 6, sec. 6.4, p. 126]. Orders with size restrictions usually rank behind unrestricted orders at the same price (p. 117). A stop limit order activated by

a gap through both prices does not trade, and a stop market order that does trade may fill far beyond its stop (pp. 78–80; order lifecycle). Market-on-open and market-on-close orders participate in the opening or closing auction where there is one (p. 83; matching and trade pricing).

Practice. The order type is the trader's choice of how to trade off price, certainty and exposure; which orders a venue accepts, and how each behaves at that venue, are its rules, to be read from its rulebook with a date (a backlog item for named venues). Interpretation: in a replay, each order's type, time in force and quantity and display instructions are needed to know whether its remainder rests, where it ranks and whether it could have traded.

Pitfalls. The legacy taxonomy treated "IOC/FOK" as one category with two meanings; Harris uses the names interchangeably and FIX does not, so the source of a definition must be named. Its "pegged" orders are not in Harris's 2002 list; they are FIX 4.4 ExecInst values, and pegging references are venue rules. Harris's special-settlement example assumes U.S. regular-way settlement of T+3 (p. 85), as of 2002; it is not current. The legacy rule on tick-sensitive orders is a description of an order instruction, not a market rule; short-sale tick rules are a different matter, not covered here.

See also. order lifecycle; limit order book; limit orders as options; matching and trade pricing; order exposure decisions (display and hidden orders); CME Globex trading conventions.

Further reading. The legacy file is kept verbatim in `order-types-instructions.tex` in the folder `notes/artefacts/execution-market-making/`.

Index. order; order instruction; market order; limit order; marketable limit order; stop order; stop limit order; market-if-touched; tick-sensitive order; downtick; zero downtick; market-not-held; not held; time in force; day order; good-till-cancel; good-till-date; immediate or cancel; fill or kill; market-on-open; market-on-close; all-or-none; minimum acceptable quantity; display instruction; reserve order; spread order; substitution order; special settlement; pegged order; OrdType; TimeInForce; ExecInst

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, chap. 4, pp. 68–86, and chap. 6, pp. 117 and 126, and `catalogue-0528`, Cambridge 2018, p. 55; and the FIX 4.4 code sets as rendered by Orchimate, retrieved 2026-10-16). Stability: draft. Currency: Harris's instructions and settlement as of 2002; FIX 4.4 (2003); no current venue's order types checked.

Definition 8.3 Limit order book: best quotes, limit-price placement, visibility, the three event types and how prices change, and quote-driven requests for quotes

Statement. *Order type and aggressiveness are different things.* Harris's market order is "an instruction to trade at the best price currently available" [alpha-harris, chap. 4, sec. 4.3, p. 71]; a limit order trades only at its limit price or better, and a *marketable* limit order is one that can execute on arrival because its price reaches the opposite best quote, "like market orders, except that they limit the price concessions" (sec. 4.4.1, p. 73). Bouchaud et al. use a convention for event categories instead: "orders that match upon arrival are called market orders" and those that do not are limit orders, so in their counts a marketable limit order is a market order [catalogue-0528, chap. 3, sec. 3.1.3, p. 45]. A crossing limit order can do both: the part that matches executes at once, and "any portion of x that does not match instead becomes active at the price p_x" (sec. 3.1.2, p. 45), unless an instruction such as immediate or cancel removes the remainder (p. 55). In this entry "market order" follows Bouchaud et al.'s convention only where stated. A *limit order book* is the set of all limit orders for an asset on one platform at a given time, its best buy price is the bid $b(t)$, its best sell price the ask $a(t) > b(t)$, with mid-price $m(t) = \frac{1}{2}[a(t) + b(t)]$ and spread $s(t) = a(t) - b(t)$, and the book is a set of queues, one per price level [catalogue-0528, chap. 3, secs. 3.1.3–3.1.4, pp. 45–47]. In that convention, limit orders supply liquidity and market orders demand it: patient traders meet impatient ones (p. 43). Harris classifies a limit order by its price relative to the market [alpha-harris, chap. 4, sec. 4.4.1, Table 4-1, pp. 73–74]: *marketable* (a buy at or above the best offer), *making a new market* (between the best bid and offer), *at the market* (at the best bid or offer), and *behind* or, further away, *far from* the market; aggressiveness falls in that order. Open-book markets display their books to all traders, closed-book markets do not, and some let traders hide their orders [alpha-harris, chap. 5, p. 109].

Conditions. Book dynamics are driven by three kinds of event: insertions add liquidity, while cancellations and transactions (by market or marketable limit orders) remove it, and trades can occur only at the best limits [alpha-laruelle, chap. 3, sec. 3.3.1, p. 219]. The bid rises when a buy limit order arrives inside the spread and falls when sells exhaust the volume at the bid; a sell market order larger than the bid queue walks down the book and trades at several prices [catalogue-0528, chap. 3, sec. 3.1.8, p. 51; Table 3.1, p. 53]. Request-for-quote systems differ: a trader asks several market makers, each free to answer, and modern systems aggregate the answers into a book-like view; Lehalle and Laruelle note fixed-income markets' move towards all-to-all RFQs that look like central limit order books [alpha-laruelle, chap. 3, p. 216]. Interpretation: an aggregated RFQ screen shows conditional, request-specific prices, not firm resting orders, so its depth is not comparable with a book's.

Practice. Bouchaud et al. measure prices relative to the same-side best quote, $d = b(t) - p_x$ for a buy and $p_x - a(t)$ for a sell, to aggregate order flow across traders (sec. 3.1.6, p. 50). The spread and mid-price are

measured as in spread measures, which this entry does not repeat.

Pitfalls. Slice 33 first wrote that an order matching on arrival "is" a market order; that is Bouchaud et al.'s event convention, not an order type, and was corrected in its follow-up. The legacy taxonomy's "far from the market: low immediacy cost" reverses Harris's point: such orders give a better price if they fill but rarely fill. Its "top-of-book" and "depth book" visibility levels are data products, defined in market data levels. A displayed book is not all liquidity: in Bouchaud et al.'s 2015 NASDAQ sample, 3 to 5 percent of market orders matched a hidden order inside the spread (Table 4.1, p. 60).

See also. order lifecycle; limit orders as trading options; queue-reactive intensities; spread measures; tick size; floor and automated trading systems; fixed-income market structure (RFQ protocols).

Further reading. The legacy file is kept verbatim in `notes/artefacts/execution-market-making/`.

Index. limit order book; market order; limit order; bid; ask; mid-price; marketable limit order; at the market; behind the market; making a new market; open book; closed book; insertion; cancellation; transaction; walking the book; quote-relative price; request for quote; RFQ; central limit order book

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `catalogue-0528`, Cambridge 2018, pp. 43–53 and 60; `alpha-harris`, 2002 Oxford printing, pp. 73–74 and 109; `alpha-laruelle`, 2018 World Scientific printing, pp. 216–219; order type against aggressiveness restated 2026-10-16 against Harris pp. 71 and 73 and Bouchaud et al. pp. 45 and 55). Stability: draft. Currency: definitions timeless; the hidden-order shares are NASDAQ 2015.

Mechanism 8.4 Matching and trade pricing: order precedence, the matching procedure, pro-rata allocation and its remainders, uniform, discriminatory and derivative pricing, and call-auction uncrossing

Statement. Rule-based order-matching systems arrange trades from submitted orders in the same sequence, in calls and in continuous trading: they rank orders by their *order precedence rules*, match the highest-ranked buy with the highest-ranked sell, trade if the buyer will pay at least what the seller asks, carry the remainder of the larger order to the next match, and price each trade by their *trade pricing rules*; in a call the system makes one attempt after the call, in continuous trading one whenever an order arrives [`alpha-harris`, chap. 6, secs. 6.2–6.2.2, pp. 116–119]. *Precedence* is hierarchical: price priority first everywhere, market orders ranking highest; then at least one secondary rule: time precedence (floor time, giving only the first order at a price precedence, or strict time), display precedence (displayed before undisclosed orders at a price, the parts of a partly displayed order treated separately), size precedence (large first in some markets, small first in others) or pro-rata allocation among orders at parity; orders with size restrictions usually rank lower (sec. 6.2.1, p. 117). Some equity exchanges also apply *public order precedence*, forbidding members to trade ahead of a public order at the same price (sec. 6.1.1.3, p. 115). *Pricing*: single-price auctions use the *uniform* rule, one clearing price for all trades; continuous two-sided auctions use the *discriminatory* rule, each trade at the standing order's limit price; crossing networks use the *derivative* rule, a price taken from another market (sec. 6.2.3, p. 120; secs. 6.3–6.5, pp. 120–136).

Notation. For a candidate price p, supply $S(p) = \sum_i q_i^s \mathbf{1}\{s_i \le p\}$, demand $D(p) = \sum_j q_j^b \mathbf{1}\{b_j \ge p\}$ (market orders counted at every price), matched volume $\min\{S(p), D(p)\}$ and excess demand $D(p) - S(p)$; the single-price auction clears where the volume is greatest (Harris, sec. 6.3.2, pp. 121–122). Bouchaud et al. write the auction price as $\arg\max_p Q(p)$ [`catalogue-0528`, chap. 3, sec. 3.2.3, eq. (3.8), p. 56]. The *surplus* at an auction price is the quantity left unmatched there [`alpha-laruelle`, chap. 2, sec. 2.1.1, p. 122].

Conditions. *Call auctions in European markets.* Lehalle and Laruelle describe four successive rules (pp. 121–122): maximum executable volume; then minimum surplus; then market pressure (all surplus on the buy side: the highest candidate price; all on the sell side: the lowest); then the price nearest a reference price, generally the last traded price; if orders do not cross, no price is formed. Some venues end the call at a random time after its theoretical end; their Table 2.2 gives maxima of 0 seconds for the LSE and Euronext Paris, 30 for Deutsche Börse, 60 for Borsa Italiana and 300 for U.S. markets, with variation by day and stock (pp. 122–123); those are as of their study, not current rules. *Surplus and price priority.* The total trader surplus of a trade is the buyer's valuation minus the seller's whatever the price; the price only divides it. If traders bid and offer their valuations, matching by price priority puts every buyer who values the asset above the clearing price with every seller who values it below, so no other arrangement of the same orders yields more total surplus (sec. 6.3.3, pp. 122–125). *Incidence.* For a given set of orders, large impatient traders prefer discriminatory pricing, which fills the first parts of their orders at better prices, and standing limit-order traders prefer uniform pricing; but traders bid more aggressively under uniform pricing, so the comparison overstates the effect, and continuous markets cannot enforce uniform pricing because large traders split their orders (sec. 6.4.2, pp. 129–130). Continuous trading lets traders trade when they want to, a benefit the surplus does not measure (p. 131).

Practice. Checked (`slice34-checks.py`) on Harris's nine-order example (Table 6-1): demand and supply give matched volumes 1, 7, 6 and 4 at 19.8, 20.0, 20.1 and 20.2, so the call clears 7 at 20.0 by the first rule alone; total surplus is 1.6 in the call against 1.0 for the same orders traded continuously (11 units), Harris's Tables 6-7 and 6-9; Bif's market buy averages 20.15 under discriminatory pricing (p. 128). *Pro-rata allocation* fills orders

at parity in proportion to size: 18 against resting 10 and 20 gives 6 and 12 (p. 117), and POSIT's 3,000 to sell against 5,000 and 10,000 to buy gives 1,000 and 2,000 (p. 134). When the proportions are not whole lots a remainder must be allocated by a further rule: 10 against three resting orders of 3 gives $\frac{10}{3}$ each, floors of 3 and one lot left over, and 7 against 1, 1 and 100 floors the small orders to nothing; which rule applies (time, size, rounding, minimum allocation) is venue-specific and is a backlog item, together with CME's own allocation steps (CME Globex trading conventions). *Matching curves.* In Lehalle and Laruelle's study of auction feeds for FTSE 100, FTSE MIB, SBF 120, DAX and OMXS 30 stocks from 1 January to 30 March 2011, the indicative volume relative to the final volume rises late and convexly in opening calls and early and concavely in closing calls; in the German closing auction, whose book is not shown, the indicative volume exceeds the final one in about a quarter of cases, which they read as orders placed to test the book and then cancelled (pp. 123–127). Indicative prices and volumes are therefore signals, not commitments; CME publishes an indicative opening price in its pre-open (CME Globex).

Pitfalls. The legacy notes' frequent batch auctions ("within-batch time priority is muted, reducing pure latency value") come from Budish, Cramton and Shim's proposal, whose batch-auction sections were not read for latency arbitrage; the claim is left as a backlog item of `hft-market-design`. Their surplus proposition, "trade the largest Q with $b_Q \geq s_Q$", is Harris's result for truthful limit prices (an exchange argument), not a property of any auction whatever traders submit. Public order precedence is described for U.S. equity exchanges as of 2002, when Harris notes that decimalisation had greatly weakened it (p. 115); it is not stated as a current rule. "Discriminatory pricing favours aggressive size" holds only for a given set of orders.

See also. limit order book; order instructions; order lifecycle (priority after modification); CME Globex trading conventions (matching algorithms; pre-open); dark pools and crossing networks (derivative pricing); latency arbitrage; passive order split (queue position).

Further reading. The legacy file is kept verbatim in `matching-algorithms.tex` in the folder `notes/artefacts/execution-market-making/`.

Index. order matching; rule-based order-matching system; order precedence; price priority; time precedence; floor time precedence; display precedence; size precedence; pro-rata allocation; parity; allocation remainder; public order precedence; trade pricing rule; uniform pricing; discriminatory pricing; derivative pricing; single-price auction; call auction; continuous auction; supply and demand schedules; clearing price; auction surplus; maximum executable volume; market pressure; reference price; random end; matching curve; indicative price; trader surplus

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, chap. 6, pp. 115–136; `alpha-laruelle`, 2018 World Scientific printing, chap. 2, pp. 121–127; and `catalogue-0528`, Cambridge 2018, p. 56; Harris's examples and the allocation remainders recomputed exactly). Stability: draft. Currency: mechanisms timeless; venue settings (random ends, public order precedence) as of 2002–2012.

Result 8.5 Standing limit orders are trading options: what sets their value, the two risks of using them, and why fills are adversely selected

Statement. Standing limit orders are options to trade: a sell limit order is a call that lets others buy when they want to, a buy limit order a put that lets them sell, and the strike is the limit price [`alpha-harris`, chap. 4, sec. 4.4.2, p. 75]. They are options but not option contracts: their writers grant them free to the market and can cancel them, and anyone can exercise one with a market or marketable limit order (p. 75). Their value to others rises with how close the limit price is to the market, how long the order will stand, and volatility; to limit that value, traders place orders further from the market when it is volatile, so spreads are wide in volatile markets (pp. 75–76). Bouchaud et al. make the same point, adding that limit orders "typically earn profits when prices mean-revert but suffer losses when prices trend" [`catalogue-0528`, chap. 3, sec. 3.1.3, p. 46]. The legacy formalisation, derived here: for an order of size q at limit L, a contra trader with value V_t exercises only when that is worth while, so the exercise value is $q(V_t - L)^+$ for a sell order and $q(L - V_t)^+$ for a buy order, while the order is live.

Conditions. The option view needs a contra trader whose value differs from the limit price and who can act before the writer cancels; the payoff is the exerciser's surplus, not the writer's loss on a market that later moves. Two risks follow [`alpha-harris`, chap. 4, sec. 4.4.4, p. 77]: *execution uncertainty*, when prices move away and the order fails to trade (Harris's example: a buyer who chased the price paid 9 cents more than a market order would have, p. 77), and *ex post regret*, when prices move through the limit and the order fills just before the market moves against the new position. Adverse selection, trading with better-informed traders, is "the most important cause of ex post regret". Placing orders further away lowers regret and the fill probability; placing them closer does the reverse.

Practice. Interpretation, from the passages above: fills are not a random sample of the times an order was available, because informed or faster traders exercise it when it has become cheap, so an evaluation of a passive strategy must measure the price move after its own fills (as in the realised spread of spread measures), not only its fill rate. In a continuous order-driven market the equilibrium spread makes the marginal trader

indifferent between limit and market orders; that result, with its conditions, is spread determinants, which this entry does not restate. *Order choice.* Market orders trade at once and pay the spread; limit orders get good prices if they execute and risk failing to trade, after which a trader may pay more than a market order would have cost; "if the spread is wide, market orders will be expensive ... if the spread is narrow, market orders may be more attractive" [alpha-harris, chap. 14, sec. 14.4, p. 304]. Information changes the comparison: Harris's Table 14-1 shows uninformed traders losing to informed ones whichever order they use, a limit order filling when informed traders want the other side and failing when they want the same side (p. 304). Interpretation, derived from the legacy proposition: relative to crossing now, posting gains the spread S less the expected adverse move A if it fills (probability p) and costs the expected chase M if it does not, so posting is better when $p(S-A) > (1-p)M$; with $p = 0.6$, $S = 1$, $M = 0.8$, an adverse move of 0.3 favours posting (+0.10) and one of 0.7 crossing (−0.14) (`slice34-checks.py`). The terms are conditional expectations at the decision time, and A rises with exactly the information that makes a fill likely, so a rule based on spread width alone ignores the trader's own information about whether the quote is stale. Aggressive orders buy certainty and reveal urgency; passive and hidden orders reveal less and bear queue, non-execution and adverse-selection risk (order exposure decisions).

Pitfalls. The legacy proposition gave the payoff "to the market" without saying it is exercised at the other trader's discretion and that the writer may cancel; its "timing option" mechanism added cancellation speed and monitoring speed, which match Harris's argument about volatility and time to cancel (spread determinants) but were not stated as such in the pages read here.

See also. limit order book; order instructions; order exposure decisions; trading aggressiveness; spread determinants; static limit-order depth; latency arbitrage; Glosten–Milgrom.

Further reading. The legacy blocks are kept verbatim in `notes/artefacts/execution-market-making/` (limit-order-book file) and `notes/artefacts/market-making-lob/` (queue file).

Index. limit order as option; free trading option; call option; put option; option value of a limit order; execution uncertainty; ex post regret; adverse selection; chasing the price; equilibrium spread; timing option

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 75–77, and `catalogue-0528`, Cambridge 2018, p. 46; the payoff expression derived here; order choice added in slice 34 from Harris p. 304, with the posting threshold derived and computed). Stability: draft. Currency: timeless.

Definition 8.6 Queue imbalance at the best quotes: two normalisations, its predictive power for the next price move, and its limits

Statement. With V_b and V_a the volumes at the best bid and ask, Bouchaud et al. define the *queue imbalance*

$$I = \frac{V_b}{V_b + V_a} \in (0,1), \qquad \text{equivalently} \qquad QI = \frac{V_b - V_a}{V_b + V_a} = 2I - 1 \in (-1,1),$$

invariant to rescaling both volumes; I near 1 (a much larger bid queue) suggests net buying pressure and an upward move [catalogue-0528, chap. 7, sec. 7.2, eq. (7.1), p. 119]. For a large-tick stock whose spread is one tick, the price changes only when one best queue empties, so the next move is a race between the two queues (sec. 7.1, p. 118). Estimating from random times the share of races that end with an up move, they find "a strong, monotonic correlation between the queue imbalance and the direction of the next price move" (p. 120). Lehalle and Laruelle state the same fact in queue-reactive terms: "when the first queue is smaller than its opposite queue, it will most probably be consumed" [alpha-laruelle, chap. 3, sec. 3.3.2, pp. 223–224].

Conditions. *Sample.* Bouchaud et al.'s figure uses two large-tick NASDAQ stocks, INTC and CSCO, in 2015 (chap. 4, p. 58; Table 4.1, p. 60); Lehalle and Laruelle's statistics are for a large-tick stock with an average spread of about 1.3 ticks (p. 227). *Horizon.* The prediction is for the next mid-price move, and Lehalle and Laruelle add that "the amplitude of this future price move is far lower than a tick" in most cases, so that no rewarding arbitrage can be built on it alone (p. 224). *Large and small ticks.* The race argument needs a spread pinned at one tick; for small-tick stocks the spread can be filled inside and the argument does not apply as stated.

Practice. Checked (`order-flow-checks.py`): $QI = 2I - 1$ to machine precision, and both are unchanged when both volumes are scaled. Interpretation: as a model feature it must be bound to a stated forecast horizon and validated on later days and other stocks, since hidden orders, replenishment after depletion and the next queue's size all enter after the next move.

Pitfalls. The legacy notes defined queue imbalance twice, once as $(B_t - A_t)/(B_t + A_t)$ and once in words, with a second claim ("a large opposite queue can make the next price move less likely to reverse immediately") found in neither source; it is not adopted. Queue imbalance is a state of displayed volume, distinct from the event-based order-flow imbalance and from signed trade volume.

See also. queue-reactive intensities; order-flow imbalance; tick size (large and small ticks).

Further reading. The pathway's readings on imbalance-based placement (Lehalle and Mounjid) and on why imbalance predicts prices (Pulido, Rosenbaum and Sfendourakis, an optional branch) are not read here;

Bouchaud et al.'s models of the race (secs. 7.3–7.5) are not summarised. The legacy blocks are kept verbatim in `notes/artefacts/market-making-lob/` and `notes/artefacts/lob-deep-learning/`.

Index. queue imbalance; volume imbalance; best-queue imbalance; race to the bottom; next price move; large-tick stock; worst-kept secret of high-frequency trading

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `catalogue-0528`, Cambridge 2018, pp. 58, 60 and 118–120, and `alpha-laruelle`, 2018 World Scientific printing, pp. 223–224 and 227; the identity computed). Stability: draft. Currency: evidence from NASDAQ 2015 and an unnamed European large-tick stock.

Model 8.7 Queue-reactive order-book intensities: the Poisson baseline and why it fails, intensities conditioned on queue sizes, the arrival–departure ratio, and the reset that limits mean reversion

Statement. Conditioning a variable means specifying the contexts in which it is "less random"; for an order book, the probability of an insertion at the first limit when it is large, rather than of any insertion anywhere [`alpha-laruelle`, chap. 3, sec. 3.3.2, pp. 219–220]. The baseline models three intensities, of insertions λ^+, cancellations λ^c and transactions λ^x, as homogeneous Poisson processes independent of the book; each is estimated by starting at many times t, recording the waiting time $\tau - t$ to the next event of that type and averaging, which estimates $1/\lambda$ (p. 221). Simulated with those constant intensities, the book's queue sizes are "very far away from the reality" (Figure 3.11, p. 222). Huang, Lehalle and Rosenbaum's *Model I* keeps one intensity per event type and per size of the queue concerned, measured in average event sizes, by filing each waiting time under the queue size at the start time; Models II and III also condition on the sizes of the neighbouring queues, for example $\lambda^c(4, 2, 1)$ for a first bid of 4 with a second bid of 2 and a best ask of 1 (pp. 222–223).

Notation. The *arrival–departure* (A/R) ratio of a queue is its insertion intensity over the sum of its cancellation and transaction intensities; Lehalle and Laruelle read a ratio above one as a queue that grows on average and below one as one that depletes (p. 224). AES: average event size, the average size of an insertion, cancellation or trade.

Conditions. In their figures the first queue's ratio is below one at every size and the second's above one, and the book is stable only because, when the first queue empties, the second is promoted and starts to deplete (pp. 224–225; the book is ergodic, n. 13). Two regularities explain why conditioning helps: a first queue smaller than the opposite one is usually consumed, and a large first queue brings more cancellations to the second (pp. 223–224). The statistics are for a large-tick stock with a spread of about 1.3 ticks; with small ticks a queue carries less meaning (p. 227). Interpretation, checked on a birth–death toy (`order-flow-checks.py`): with intensities that do not depend on size, a ratio of 1.2 gives an upward drift of $+0.09$ per event and 0.8 a drift of -0.11; with size-dependent intensities the ratio has to be read at each size.

Practice. Liquidity dynamics fitted this way give simulated prices whose volatility is "far too low": a newly promoted first queue is large, which pulls the price back, so mean reversion is too strong (pp. 225–226). Lehalle and Laruelle suggest Hawkes processes to add memory, or a reset: with some probability θ, when a first queue depletes, the book is redrawn from the database as if participants accepted the new price (p. 226). The *starting time* of the waiting-time estimator matters: from every second, from just after any event, or just after a trade gives different estimates (p. 221, n. 12). Checked (`order-flow-checks.py`): for a clustered (Hawkes) sample the mean wait from uniform start times is 0.594 against a mean inter-event time of 0.405. Interpretation, derived: for a stationary renewal process with finite second moment the mean wait from an arbitrary time is $E[X^2]/(2E[X])$, which equals $E[X]$ exactly when the coefficient of variation is 1; that does not single out exponential gaps (gaps of 0.5 with probability 0.8 and 3 with probability 0.2 have $E[X] = 1$, $E[X^2] = 2$, and a simulated mean wait of 1.0006). The Hawkes sample's gaps have a coefficient of variation of 1.37 and $E[X^2]/(2E[X]) = 0.579$, close to its measured wait.

Pitfalls. The queue-reactive model is a conditional Markov model with exponential waiting times; the guide's catalogue notes that it misses clustered latency races and a trader's longer-lived impact, so a simulator built on it must reproduce both book statistics and policy costs before its fills are trusted. The legacy "latency race boundary" (latency matters most when a trader can change queue position, less with wide ticks, deep queues or orders not competing for priority) is interpretation; Lehalle and Laruelle say a small tick shifts attention "from queuing issues to limit price issues" (chap. 1, p. 89), and Bouchaud et al. that activity at the best quotes is faster for large-tick stocks, reflecting the importance of queue position (chap. 4, p. 59).

See also. queue imbalance; Hawkes processes for order flow; point processes and intensities; static limit-order depth; passive order split (queue position across venues).

Further reading. Huang, Lehalle and Rosenbaum (2015) and Cont, Stoikov and Talreja (2010), required readings of `market-making-lob`, are not read here: the account is Lehalle and Laruelle's. The legacy blocks and the order-flow checks are in `notes/artefacts/market-making-lob/` and `notes/artefacts/market-simulation/`.

Index. queue-reactive model; Model I; conditioning; intensity; insertion; cancellation; transaction; homo-

geneous Poisson; waiting time; average event size; arrival–departure ratio; queue promotion; ergodic order book; mean reversion; book reset; latency and queue position

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 3, pp. 219–227, and chap. 1, p. 89; `catalogue-0528`, p. 59; the drift and waiting-time illustrations computed). Stability: draft. Currency: the model as described in 2018.

Definition 8.8 Order-flow imbalance: Cont, Kukanov and Stoikov's event-based measure at the best quotes, against signed trade volume, and the family of imbalance features

Statement. For consecutive best-quote states $(P_n^B, q_n^B, P_n^A, q_n^A)$, Cont, Kukanov and Stoikov define the contribution of the nth event to supply and demand at the best quotes,

$$e_n = \mathbf{1}\{P_n^B \geq P_{n-1}^B\} q_n^B - \mathbf{1}\{P_n^B \leq P_{n-1}^B\} q_{n-1}^B - \mathbf{1}\{P_n^A \leq P_{n-1}^A\} q_n^A + \mathbf{1}\{P_n^A \geq P_{n-1}^A\} q_{n-1}^A,$$

so that at an unchanged bid e_n is the change in bid size, a higher bid contributes the new bid size and a lower bid minus the old one, and the ask side enters with opposite signs; the *order-flow imbalance* over $[t_{k-1}, t_k]$ is $\mathrm{OFI}_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n$, which "encompasses trades, limit orders and cancelations" [`catalogue-0503`, arXiv v3, sec. 2.1, p. 4]. The *trade imbalance* is $\mathrm{TI}_k = \sum(b_n - s_n)$, buyer- minus seller-initiated volume (sec. 4.1, p. 16).

Conditions. *Evidence.* On one month (April 2010) of consolidated NYSE TAQ quotes and trades for 50 S&P 500 stocks drawn at random, with 10-second intervals and regressions per half-hour, $\Delta P_k = \beta\, \mathrm{OFI}_k + \epsilon_k$ is linear with an average R^2 of 65 percent, against 32 percent for TI_k; with both, the trade imbalance is significant in only 31 percent of subsamples (pp. 8 and 16), and the slope is inversely proportional to depth (abstract, p. 1). Timestamps were rounded to the second and quotes aggregated into an NBBO; Level 1 data only (p. 8). *Names.* Bouchaud et al. call signed market-order volume $\sum \varepsilon_n v_n$ the "order-flow imbalance" [`catalogue-0528`, chap. 11, sec. 11.4, p. 222]; that is Cont et al.'s trade imbalance. A paper's "OFI" must be read from its formula.

Practice. Checked on a toy sequence (`order-flow-checks.py`): a market sell of 200 that is replenished, a cancellation of 300 at the ask and a market buy of 100 that empties the ask give OFI $= +400$ and TI $= -100$; the two can differ in sign. The legacy list of imbalance features (trade imbalance, best-quote imbalance as in queue imbalance, multi-level depth imbalance, event imbalance by queue, and pressure aggregated across venues) is a menu of constructions, not results; multi-level and cross-asset OFI are the subject of Cont, Cucuringu and Zhang (a required reading of `lob-deep-learning`, not read here). Normalising by volume, average daily volume or depth before comparing stocks is interpretation; Cont et al.'s depth result suggests depth.

Pitfalls. The legacy notes defined "order flow imbalance" in words as net buyer- minus seller-initiated activity, then gave signed trade volume as its formula: that is trade imbalance, not the event-based OFI that the pathway's readings use. Their caveat that imbalance "can predict the next quote move but fail over longer horizons" was not found in these sources: Cont et al. report contemporaneous fits, not forecasts, and on a subsample of stocks the results held for intervals from 10 quote updates to 10 minutes, the fit improving with the interval (p. 8). Any OFI feature used to forecast must be computed only from events before the forecast time.

See also. trade-sign classification; queue imbalance; information-driven bars (imbalance bars, a different use); Kyle (impact linear in net order flow); order-flow impact regression.

Further reading. Kolm, Turiel and Westray, and Cont, Cucuringu and Zhang, required readings of `lob-deep-learning`, are not read here. The legacy file is kept verbatim in `notes/artefacts/lob-deep-learning/`.

Index. order-flow imbalance; OFI; event-based imbalance; trade imbalance; signed volume; best bid and offer; Level 1 data; TAQ; price impact of order book events; depth; imbalance features; multi-level OFI

Status. Verification: partly checked 2026-10-16 (against the text layer of `catalogue-0503` in its arXiv version 3 of 13 April 2011, pp. 1, 4, 8 and 16, downloaded 2026-10-16; the journal version was not checked; and `catalogue-0528`, Cambridge 2018, p. 222; the toy sequence computed). Stability: draft. Currency: evidence from April 2010 U.S. data.

8.2 Order flow in time: stylised facts, clocks and point processes

Evidence 8.9 Stylised facts of high-frequency prices and order flow: heavy tails, volatility clustering, long memory of order signs, activity clustering and intraday patterns, with their samples

Statement. Bouchaud et al. report, each with its own sample:

- *heavy tails*: unconditional return distributions decay as a power law; between about a minute and a few hours a Student-t with tail parameter μ near 3 fits "reasonably well", and their figure of normalised daily

returns of U.S. stocks, CDS spreads and implied volatilities for 2011–2013 shows tails with exponent about 3.5 [catalogue-0528, chap. 2, sec. 2.2.1, pp. 28–30];

- *volatility clustering*: returns are nearly uncorrelated but far from IID; writing $r_t = \sigma_t \xi_t$, log-volatility has a variogram growing like $\ln(1+\tau)$ up to a cut-off of years, so bursts occur on all time scales (sec. 2.2.2, pp. 30–31);
- *long memory of order signs*: the autocorrelation of market-order signs in event time decays roughly as $\ell^{-\gamma}$ with $\gamma < 1$, close to $\ell^{-0.5}$ for four NASDAQ stocks in 2015; with $C(\ell) \approx 0.5\,\ell^{-1/2}$, $C(10{,}000) \approx 0.005$ (chap. 10, sec. 10.1, pp. 187–189);
- *activity clustering*: event arrivals cluster beyond what intraday patterns explain; counts in windows have variance well above their mean (chap. 9, secs. 9.1.2 and 9.3.3, pp. 165 and 174–175);
- *intraday patterns*: market-order and limit-order volume follow a J-shaped profile over the day, the busiest periods about four times the quietest, for PCLN, TSLA, CSCO and INTC on NASDAQ in 2015 (chap. 4, sec. 4.2.1, p. 61); volume at the best quotes behaves differently for large- and small-tick stocks (same page).

Conditions. Long memory of signs and clustering of activity are different facts: Poisson arrivals can carry persistent signs, and uncorrelated signs can arrive in clusters (chap. 10, pp. 187–188). Each fact is reported for a sample and a method; "universality" of the tail exponent is Bouchaud et al.'s reading of many studies, whose estimates vary (p. 29). Hasbrouck notes the same intraday seasonality, volume and volatility elevated at the start and end of sessions [s6-hasbrouck, note 1 to chap. 4, p. 180].

Practice. For a simulator, these are the targets the pathway's outcome names: a reactive simulation is checked against them, not tuned until it matches one of them by construction. They describe data; none is by itself a trading signal.

Pitfalls. The legacy taxonomy listed "long memory" of "signs, durations and volatility" together; the passages read support long memory of order signs and of volatility and clustering of activity, each with a different measure. Durations' own persistence was not checked here.

See also. realised variance and the signature plot; intraday deseasonalisation; Hawkes processes for order flow; propagator model (sign persistence and impact); activity-sampled bars (heavy tails under different clocks); intraday and overnight returns.

Further reading. Vyetrenko et al.'s realism metrics and Noble, Rosenbaum and Souilmi's reality-gap tests, readings of `market-simulation`, are not read here. The legacy file is kept verbatim in `notes/artefacts/market-simulation/`.

Index. stylised facts; heavy tails; fat tails; tail exponent; Student-t; volatility clustering; long memory; order-sign autocorrelation; activity clustering; clustering ratio; intraday seasonality; J-shaped profile; U-shaped profile

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `catalogue-0528`, Cambridge 2018, pp. 28–31, 58–61, 165, 174–175 and 187–189, and `s6-hasbrouck`, 2007 Oxford printing, p. 180). Stability: draft. Currency: samples of 2011–2015.

Method 8.10 Realised variance and covariance at high frequency: the signature plot, microstructure noise and the Epps effect

Statement. For a log price $X = \log S$ observed over $[0,T]$ with step Δ and returns $r_\Delta(j) = X_{j\Delta} - X_{(j-1)\Delta}$, Lehalle and Laruelle state that "in the Itô semi-martingale framework" the realised variance $\hat{V}_R(\Delta) = \sum_{j=1}^{T/\Delta} r_\Delta(j)^2$ converges in probability to the integrated variance $\int_0^T \sigma_s^2\,ds$ as $\Delta \to 0$, and the realised covariance $\hat{C}_R(\Delta) = \sum_j r^1_\Delta(j) r^2_\Delta(j)$ to $\int_0^T \rho_s \sigma^1_s \sigma^2_s\,ds$ [alpha-laruelle, app. A, secs. A.13.1–A.13.2, pp. 316–318]. Interpretation, standard and checked below: those limits hold for continuous Itô processes; for a semimartingale with jumps $\hat{V}_R$ converges to the quadratic variation $\int_0^T \sigma_s^2\,ds + \sum_{s\le T}(\Delta X_s)^2$ and $\hat{C}_R$ to $\int_0^T \rho_s\sigma^1_s\sigma^2_s\,ds + \sum_{s\le T}\Delta X^1_s \Delta X^2_s$, the last term from co-jumps. If observed prices are $X = M + \varepsilon$ with M a semimartingale and ε noise, $\hat{V}_R$ adds the noise terms $\sum \eta_\Delta(j)^2$ and a cross term, and on Rio Tinto data (1 January to 19 June 2012) the estimator diverges as $\Delta \to 0$ (pp. 316–318). The covariance estimator needs synchronous data, which quotes and trades are not; the *Epps effect* is Epps's (1979) finding that "correlations among price changes … are found to decrease with the length of the interval for which the price changes are measured", shown for Total and France Telecom on Euronext Paris, 23–27 July 2012 (pp. 318–319, Figure A.26).

Notation. Bouchaud et al.'s *volatility signature plot* is $\sigma^2(\tau) = V(\tau)/\tau$ against the lag τ, with $V(\tau)$ the price variogram: flat for uncorrelated returns, rising with trends and falling with mean reversion or with mean-reverting high-frequency noise [catalogue-0528, chap. 2, secs. 2.1.2–2.1.3, pp. 25–27]. Lehalle and Laruelle plot $\hat{V}_R(\Delta)$ against Δ; the two plots carry the same information in different units.

Conditions. *Jumps.* Checked (`order-flow-checks.py`): a unit-jump Poisson path with five jumps on $[0,1]$ has zero diffusion variance, yet its realised variance is 9 on a 10-step grid (two jumps in one interval) and 5, the jump count, from 10^4 steps on; three common jumps of $+1$ and -0.5 give a realised covariance of -1.5. *Noise.* The finite-grid expectation below is exact for a driftless Brownian M with constant volatility and observation noise i.i.d., centred and independent of M; with drift it adds the squared drift increments, with autocorrelated noise (as in Bouchaud et al.'s mean-reverting noise) the noise term is no longer $2n\sigma_\eta^2$, and for a general semimartingale only the limits above are claimed. Explanations offered for the Epps effect are lead–lag between related assets, asynchronous transactions, and tick and other microstructure effects (Lehalle and Laruelle, p. 318). Checked (`order-flow-checks.py`): in the simulated model, $X = M + \varepsilon$ with M a driftless Brownian motion of volatility σ and ε i.i.d. centred noise of standard deviation σ_η independent of M, $E[\hat{V}_R] = \sigma^2 T + 2n\sigma_\eta^2$ exactly for $n = T/\Delta$ returns (each noise difference has variance $2\sigma_\eta^2$ and the cross terms have mean zero), and simulated values track it from 13 to 23,400 returns a day; two Brownian motions with correlation 0.6, observed at independent random times and sampled at the previous tick, show a mean correlation over 40 paths of 0.12 at a step of 5 time units against 0.63 at 600, from asynchrony alone.

Practice. Bouchaud et al.'s E-mini S&P 500 signature plot is almost flat from one second to about 20 trading days, decreasing by about 20 percent, which indicates weak mean reversion (sec. 2.1.4, pp. 27–28): for that contract, noise inflates short-horizon volatility only modestly. Whether the plot rises at fine steps depends on the price used (trades bounce between bid and ask; mid-quotes much less) and on the instrument. Lehalle and Laruelle list range-based, multi-scale and bid–ask-model estimators as alternatives, and Barndorff-Nielsen–Shephard, Hayashi–Yoshida and Zhang for covariance (pp. 317–318), none read here.

Pitfalls. The legacy model said realised variance "can diverge as sampling becomes too fine" and the Epps definition credited tick effects; both hold under stated noise and asynchrony, not for every series. Its remedy for covariance, "refresh-time or previous-tick sampling", is not in the passages read: previous-tick sampling itself produces the Epps effect in the check above, and refresh-time sampling is left to the backlog.

See also. fundamental and transitory volatility; Roll's estimator (bounce as negative autocovariance); Garman–Klass volatility; stylised facts.

Further reading. The legacy blocks are kept verbatim in `notes/artefacts/market-simulation/`, with the simulations in `order-flow-checks.py`.

Index. realised variance; integrated variance; realised covariance; signature plot; volatility signature plot; variogram; microstructure noise; bid–ask bounce; Epps effect; asynchronous trading; previous-tick sampling

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, pp. 316–319, and `catalogue-0528`, Cambridge 2018, pp. 25–28; noise and asynchrony simulated; the limits restricted to continuous processes, the jump and co-jump limits and the noise-model assumptions added 2026-10-16, with a jump counterexample computed). Stability: draft. Currency: estimators timeless; illustrations from 2012 data.

Method 8.11 Intraday deseasonalisation: additive and multiplicative decompositions, centred moving averages, seasonal coefficients and a factorised volume model

Statement. A series is decomposed into a trend C_t, a seasonal component S_t and an irregular part ε_t, additively $X_t = C_t + S_t + \varepsilon_t$ or multiplicatively $X_t = C_t S_t + \varepsilon_t$ [`alpha-laruelle`, app. A, sec. A.11.1, p. 299]. A centred moving average of odd length $2k+1$ is $MA_t = (X_{t-k} + \cdots + X_{t+k})/(2k+1)$; of even length $2k$ it weights the end points by $\frac{1}{2}$, $MA_t = (\frac{1}{2}X_{t-k} + X_{t-k+1} + \cdots + X_{t+k-1} + \frac{1}{2}X_{t+k})/(2k)$, and it cannot be computed for the first and last k observations (pp. 299–300). For period p and $t = (i-1)p + j$, with MA of length p estimating the trend: additively, $\hat{S}_j = \frac{1}{n}\sum_i (X_{ij} - MA_{ij})$, centred as $S_j = \hat{S}_j - \frac{1}{p}\sum_k \hat{S}_k$; multiplicatively, $\hat{S}_j = \frac{1}{n}\sum_i X_{ij}/MA_{ij}$, normalised as $S_j = \hat{S}_j / (\frac{1}{p}\sum_k \hat{S}_k)$, so that the coefficients sum to p; means may be replaced by medians (pp. 300–302).

Conditions. The moving average's length must equal the period of the seasonal variation or a multiple of it, so that it carries no seasonality (p. 300). The multiplicative estimate assumes ε_{ij} small relative to the trend (p. 301). Interpretation, checked (`order-flow-checks.py`): a centred average uses k later observations, so a shock at t changes the estimates for the preceding k times; seasonal coefficients estimated this way on a sample cannot be used at a decision time inside that sample without look-ahead.

Practice. Lehalle and Laruelle's volume model for slot n, day d, month m and year y is $V(y,m,d,n,\omega) = (f(n) + g(d,n))F(m)G(y)v(n,\omega)$, with f the mean slot profile, g special-day effects such as expiries or macroeconomic news, F and G monthly and yearly effects and v random; with $E[H] = 1$ and $\sum_m F(m) = 1$, $E[V_M(y,m,\cdot)] = G(y)F(m)$ and $E[V_Y(y,\cdot)] = G(y)$ (sec. A.11.2, pp. 302–303). Their Total SA example (2 January 2007 to 1 June 2012) shows a bump at noon on Fridays from derivative expiries (pp. 303–304). Bouchaud et al.'s alternative for point processes is a *business time* $\hat{t}(t) = \int^t \bar{\phi}(u)\,du$ built from the average activity profile, which flattens it [`catalogue-0528`, chap. 9, sec. 9.3.1, p. 173]; see market clocks. Checked on a synthetic series with 13 slots and 60 days: multiplicative coefficients recovered to within 0.005.

Pitfalls. The legacy notes wrote the even-length average as A_t and asked for the window to "match" the period; Lehalle and Laruelle require it to equal the period or a multiple. The practitioner note "remove

predictable clock-time effects before comparing venues, estimating impact, or training intraday features" is sound as a warning that a noon lull, a close auction, an expiry day or a news window can look like a signal, but the removal must use profiles estimated from earlier data (interpretation).

See also. stylised facts (intraday patterns); market clocks; time-validity of labels and fitted transforms; execution benchmarks (volume curves).

Further reading. The legacy blocks are kept verbatim in `notes/artefacts/market-simulation/`.

Index. seasonality; deseasonalisation; additive model; multiplicative model; trend; centred moving average; seasonal coefficient; intraday volume profile; volume model; special days; expiry effect; business time

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, pp. 298–304, and `catalogue-0528`, Cambridge 2018, p. 173; a synthetic recovery and the look-ahead computed). Stability: draft. Currency: timeless; the illustration from 2007–2012 data.

Definition 8.12 Market clocks and durations: clock, event, trade, quote, volume and business time, and when each is the right index

Statement. For event times $0 < \tau_1 < \tau_2 < \cdots$ the *duration* is $\Delta\tau_i = \tau_i - \tau_{i-1}$ [alpha-laruelle, app. A, sec. A.12.1, Definition 5, p. 306]. An analysis can be indexed by an event counter (trades, quote revisions, or both) or by wall-clock time at fixed intervals; Hasbrouck's rule is that for a single precisely time-stamped security an event index is "probably the first choice", because the process is more likely to be covariance stationary in event time, while comparisons across securities may favour wall-clock time, volatility per minute being more comparable than volatility per trade [s6-hasbrouck, sec. 9.7.3, p. 90]. *Business time* rescales clock time by the average activity profile $\bar{\phi}$, $\hat{t}(t) = \int^t \bar{\phi}(u)\,du$, so that intraday activity is flat [catalogue-0528, chap. 9, sec. 9.3.1, p. 173]. Volume and dollar clocks, one step per fixed quantity or value traded, are the sampling rules of activity-sampled bars.

Conditions. With a wall-clock index, several events in one interval must be aggregated or transformed (Hasbrouck, p. 90). Propagator models are almost always written in trade time and Hawkes processes in physical time; hybrids exist [alpha-laruelle, app. A, sec. A.12.3, p. 311]. For a homogeneous Poisson process the durations are i.i.d. exponential with mean $1/\lambda$ (p. 306), so high intensity means short expected durations; for clustered processes the expected duration from a given time depends on the recent history.

Practice. Lehalle and Laruelle note that replacing physical time by event time "would change a lot the meaning of the obtained models" (chap. 3, p. 227). Interpretation: latency, risk limits, news and benchmark windows are defined in clock time, and an event-time or business-time model must be mapped back before it is used for them; a transformation that flattens activity also hides the periods of intense activity a risk manager cares about, so diagnostics are kept in both clocks.

Pitfalls. The legacy definition called the gap between any events a "trade duration"; Hasbrouck uses duration analysis for other waiting times too, such as a limit order's time to execution, whose observations are censored when the order is cancelled first [s6-hasbrouck, sec. 15.2.3, pp. 163–164]. Its "quote time" and "business time" entries are covered above; "business time" there meant activity-adjusted time "after removing intraday seasonality", which matches Bouchaud et al.'s construction only when $\bar{\phi}$ is estimated from data before the period it rescales.

See also. point processes and intensities; activity-sampled bars; intraday deseasonalisation; propagator model (trade time).

Further reading. Engle and Russell's autoregressive conditional duration model is named by Bacry et al. (`wide-bacry-hawkes`, p. 1) but not read. The legacy file is kept verbatim in `notes/artefacts/market-simulation/`.

Index. duration; inter-arrival time; event time; trade time; quote time; wall-clock time; calendar time; business time; volume time; covariance stationarity; time deformation

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, pp. 227, 306 and 311, `s6-hasbrouck`, 2007 Oxford printing, pp. 90 and 163–164, and `catalogue-0528`, Cambridge 2018, p. 173). Stability: draft. Currency: timeless.

Definition 8.13 Point processes for order-book events: counting process, conditional intensity, the homogeneous Poisson baseline and the time-change theorem

Statement. A simple point process is an increasing sequence of event times $(\tau_i)_{i\geq 1}$, for example trades or limit-order arrivals; its *counting process* is $N_t = \sum_{i\geq 1} \mathbf{1}\{\tau_i \leq t\}$ [alpha-laruelle, app. A, sec. A.12.1, Definitions 3–4, pp. 305–306]. For N adapted to a filtration $\mathcal{F}_t$, the left-continuous *intensity* is $\lambda(t \mid \mathcal{F}_t) = \lim_{h\downarrow 0} h^{-1} E[N_{t+h} - N_t \mid \mathcal{F}_t]$, or equivalently $\lim_{h\downarrow 0} h^{-1} P(N_{t+h} - N_t > 0 \mid \mathcal{F}_t)$; it depends on the filtration, taken to be the process's own history (Definition 6, p. 306). A homogeneous Poisson process has $P(N_{t+h} - N_t = 1 \mid \mathcal{F}_t) = \lambda h + o(h)$, an intensity that ignores the history, and i.i.d. exponential durations (p. 306). With the integrated intensity $\Lambda(\tau_{i-1}, \tau_i) = \int_{\tau_{i-1}}^{\tau_i} \lambda(s)\,ds$, the *time-change theorem* states that $\tilde{N}_\tau = N_{t_\tau}$, with $\int_0^{t_\tau} \lambda(s)\,ds = \tau$, is a unit-rate Poisson process (Theorem 1, pp. 307–308).

Notation. Lehalle and Laruelle model order-book events with three intensities, λ^+ for limit-order insertions, λ^c for cancellations and λ^x for transactions [alpha-laruelle, chap. 3, p. 221]. Bouchaud et al. write $\phi(t)$ for the intensity and define the clustering ratio $r(\tau) = V[N(t+\tau) - N(t)]/E[N(t+\tau) - N(t)]$, equal to 1 for a Poisson process, above 1 for clustering and below 1 for inhibition [catalogue-0528, chap. 9, secs. 9.1–9.1.2, pp. 164–165].

Conditions. The intensity is conditioned on information strictly before t: interpretation, the intensity must be predictable (left-continuous), which is what makes it usable for prediction and simulation. Lehalle and Laruelle print the theorem's condition as $\int_0^\infty \lambda(s)\,ds = 1$ (p. 308, checked by page image); the integrated intensity must instead diverge, $\int_0^\infty \lambda(s)\,ds = \infty$, for the rescaled times to cover the half-line (interpretation; the standard statement is in the point-process texts they cite, not read here). Applied as a goodness-of-fit test, a correctly specified model's rescaled durations are i.i.d. unit exponential.

Practice. Checked (`order-flow-checks.py`): durations of a simulated exponential-kernel Hawkes process are far from exponential (Kolmogorov–Smirnov $p < 10^{-260}$), while the same durations rescaled by the true compensator have mean 1.005 and are not rejected as unit exponential ($p = 0.35$). The Poisson baseline is diagnostic: as queue-reactive intensities shows, context-free intensities do not reproduce queue sizes.

Pitfalls. The legacy definitions wrote $\mathcal{F}_t$ as "the information available before t" without the left-continuity that makes it so, and stated the time change without its conditions. The legacy statement that a Poisson baseline "cannot reproduce realistic queue-size distributions because event rates are strongly state-dependent" is Lehalle and Laruelle's finding for one large-tick stock (Figure 3.11), not a general theorem.

See also. Hawkes processes for order flow; queue-reactive intensities; market clocks and durations.

Further reading. Daley and Vere-Jones, cited by Lehalle and Laruelle for point processes, is not read. The legacy file is kept verbatim in `notes/artefacts/market-making-lob/`.

Index. point process; counting process; conditional intensity; stochastic intensity; filtration; homogeneous Poisson process; exponential durations; integrated intensity; compensator; time-change theorem; time rescaling; clustering ratio; insertion intensity; cancellation intensity; transaction intensity

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 221 and 305–308, page images of pp. 306 and 308; and `catalogue-0528`, Cambridge 2018, pp. 164–165; the time change computed). Stability: draft. Currency: timeless.

Model 8.14 Hawkes processes for order flow: the linear self-exciting intensity, its conditions, branching ratio and stationarity, multivariate and marked forms, inhibition through a non-linear link, and what calibrations show

Statement. A linear Hawkes process has intensity

$$\lambda(t) = \lambda_0(t) + \int_{-\infty}^{t} \nu(t-s)\,dN_s = \lambda_0(t) + \sum_{\tau_i < t} \nu(t-\tau_i),$$

with a deterministic baseline $\lambda_0 \geq 0$ and a kernel $\nu \geq 0$ that "expresses the positive influence of the past events" [alpha-laruelle, app. A, sec. A.12.2, Definition 7, p. 308]. Bacry, Mastromatteo and Muzy's multivariate definition, $\lambda_t^i = \mu^i + \sum_j \int \varphi_{ij}(t-t')\,dN_{t'}^j$, requires each kernel component to be non-negative, causal ($\varphi_{ij}(t) = 0$ for $t < 0$) and integrable, and the process has asymptotically stationary increments if the matrix of kernel norms $\|\Phi\| = \{\int |\varphi_{ij}|\}$ has spectral radius below 1 [wide-bacry-hawkes, arXiv v2, sec. 2.1, Definition 1 and Proposition 1, pp. 3–4]. In one dimension, with norm $g = \int_0^\infty \nu(u)\,du < 1$, the mean intensity is $\bar{\lambda} = \lambda_0/(1-g) \geq \lambda_0$; g is the *branching ratio*, the expected number of events directly triggered by one event, and a family of events has mean size $1/(1-g)$ [catalogue-0528, chap. 9, secs. 9.2.2 and 9.2.6, pp. 167–168 and 172]. For $\nu(u) = \alpha e^{-\beta u}\mathbf{1}\{u \geq 0\}$ and constant λ_0, $g = \alpha/\beta$ and $E[\lambda(t)] = \lambda_0/(1-\alpha/\beta)$ (Lehalle and Laruelle, pp. 309 and 311).

Notation. Parameterisations differ: Lehalle and Laruelle's $\alpha e^{-\beta t}$ has norm α/β; Bacry et al.'s $\alpha\beta e^{-\beta t}$ (p. 5) and Bouchaud et al.'s $g\omega e^{-\omega t}$ (eq. (9.21), p. 170) have norms α and g. A fitted "α" must be read with its kernel.

Conditions. *Excitation only.* A non-negative kernel is necessary and sufficient for the intensity to stay non-negative for every realisation, and always gives self-excitation and clustering [catalogue-0528, chap. 9, sec. 9.5, p. 181]. *Inhibition* needs a non-linear form, $\lambda(t) = F\big(\lambda_0 + \int \nu(t-u)\,dN_u\big)$ with $F \geq 0$, such as the positive part or $\ln(1+e^x)$; if F's slope never exceeds 1 the same stability condition holds (Brémaud and Massoulié, 1996, as reported, p. 181), but such models lose most analytical results, while simulation and calibration remain possible (Bacry et al., sec. 2.2.4, p. 8). Checked (`order-flow-checks.py`): with $\lambda_0 = 0.5$ and a kernel $-0.8e^{-2u}$, the linear intensity just after an event is -0.3, not an intensity; with the positive-part link the simulated rate is 0.37 and the clustering ratio 0.62, below 1 as for repelling events. *Stationarity.* With $g \geq 1$ the process explodes or, at $g = 1$, is critical; the stationary formulas then fail (Bouchaud et al., p. 168). *Multivariate and marked flow.* Market orders can trigger market orders, limit orders and cancellations, and a K-type process has $\bar{\phi} = (I-g)^{-1}\phi_0$ with $g_{ij} = \int \Phi_{ij}$ (sec. 9.5, pp. 182–183); Lehalle and Laruelle sketch

classes of agents (banks, brokers, high-frequency traders) with buy and sell intensities excited across classes under $\|\varphi\|_1 < 1$ [alpha-laruelle, app. A, sec. A.12.3, pp. 315–316].

Practice. *Calibrations.* Lehalle and Laruelle fit a one-dimensional exponential Hawkes process to trades in France Telecom on the Chi-Delta dark pool on 23 July 2012, with $\lambda_0 = 140$, $\alpha = 250$, $\beta = 300$, a branching ratio of 0.83 (Figure A.24, p. 310). Bouchaud et al. report maximum-likelihood fits across many contracts with g close to 1 and power-law kernels, the short time scale falling from seconds in 1998 to milliseconds in 2012, and TSLA mid-price changes whose effective feedback approaches 1 as the window grows (pp. 174–175 and 178). Bacry et al. report, from Bacry and Muzy's model of EuroStoxx and Euro-Bund futures over 800 trading days in 2009–2012, slightly negative self-kernels in the feedback of price moves on market-order flow: an up move raises sell intensity and lowers buy intensity, an inhibitory effect [wide-bacry-hawkes, arXiv v2, sec. 5.1, p. 27]. Checked by simulation: an exponential-kernel process with branching ratio 0.6 has mean rate 2.47 against the theoretical 2.50 and a clustering ratio of 6.4 in windows of 50 against a long-window limit of 6.25.

Pitfalls. The legacy taxonomy said a marked Hawkes model "lets one class of event excite or inhibit another"; in the linear model, kernels are non-negative and only excitation is possible. Calibration "does not prove the existence of a genuine causal self-excitation mechanism": two-point correlations do not imply causality, and near-critical fits may mean either fragile markets or a framework too restrictive (Bouchaud et al., pp. 169, n. 4, and 184). Linear Hawkes processes do not reproduce strong volatility clustering and fat tails without extensions such as price feedback (p. 185). Time-stamp errors distort maximum-likelihood fits of power-law kernels (p. 178).

See also. point processes and intensities; queue-reactive intensities (Hawkes memory as a remedy); stylised facts (activity clustering); propagator model (event-time alternative).

Further reading. `wide-bacry-hawkes` read in its arXiv version 2 (17 May 2015), secs. 1–2.2.4 and the passages of secs. 5.1 and 6.1 cited; the journal version was not checked. Hawkes (1971), Brémaud and Massoulié (1996, 2001) and Bacry and Muzy are not read. The legacy file is kept verbatim in `notes/artefacts/market-making-lob/`.

Index. Hawkes process; self-exciting process; mutually exciting process; kernel; exponential kernel; power-law kernel; branching ratio; kernel norm; stationarity; spectral radius; criticality; endogeneity; multivariate Hawkes; marked point process; inhibition; non-linear Hawkes; clustering ratio; maximum likelihood

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, pp. 304–311 and 315–316, page image of p. 308; `catalogue-0528`, Cambridge 2018, pp. 163–185; and `wide-bacry-hawkes`, arXiv v2, pp. 1–8 and 27–30, downloaded 2026-10-16; the excitation, inhibition and stationarity cases simulated). Stability: draft. Currency: calibrations from 1998–2015 data.

8.3 Inventory risk and the price of immediacy

Model 8.15 Grossman–Miller market making: the price of immediacy

Statement. Competitive, risk-averse market makers supply immediacy to a liquidity trader who arrives before the trader with the offsetting need, and the price concession they charge compensates exactly for carrying the inventory until that trader arrives [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 21]. In the simplified three-date version, n identical market makers, liquidity trader LT1 holding i units and liquidity trader LT2 holding $-i$ units (arriving at date 2) trade an asset worth $S_3 = \mu + \epsilon_2 + \epsilon_3$ at date 3, and the date-1 equilibrium is [crypto-cartea, same chap., pp. 23–24]

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1}, \qquad q_1^{MM} = q_1^{LT1} = \frac{i}{n+1},$$

so LT1 sells $\frac{n}{n+1}\, i$ at once and each market maker holds $\frac{i}{n+1}$ until date 2, when LT2 arrives, the price is efficient, $S_2 = \mathbb{E}[S_3 \mid \epsilon_2] = \mu + \epsilon_2$, and every agent exits with zero inventory [crypto-cartea, same chap., pp. 22–24]. The liquidity discount is $|S_1 - \mu| = \gamma\sigma^2 \, |i| \, /(n+1)$; when LT1 buys ($i < 0$) the market makers earn the same amount as a premium [crypto-cartea, same chap., p. 24]. Figure 8.1 shows the holdings and trades.

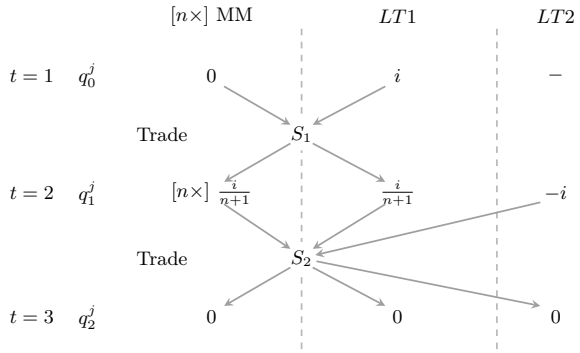


Figure 8.1: Holdings q_t^j and trading in the Grossman–Miller model, after [crypto-cartea, Figure 2.1, p. 23].

Notation. Local override for this entry, the fee result and the liquidity measures: σ^2 is the variance of one period's value shock, in squared price units and not annualised; γ is the coefficient of absolute risk aversion; q_t^j is agent j's holding on leaving date t, and S_t the date-t trade price.

Conditions. Dates $t \in \{1, 2, 3\}$; shocks ϵ_2, ϵ_3 independent $N(0, \sigma^2)$, each announced between dates $t-1$ and t; every agent has CARA utility $U(X) = -\exp(-\gamma X)$; market makers start with no inventory; the arrival of LT2 at date 2 is certain; no trading or holding costs [crypto-cartea, same chap., p. 21]. Interpretation: the concession charges for risk, not for information; no agent is informed, which separates this model from Kyle's. Idea of the proof: solve backwards; with CARA utility and normal shocks each agent's demand is $(\mathbb{E}[\text{next price}] - \text{price})/(\gamma\sigma^2)$, and market clearing at each date gives the prices [crypto-cartea, same chap., pp. 22–23]; the full working is in the derivations artefact listed under *Further reading*.

Practice. The discount rises with the size of the liquidity need $|i|$, risk aversion γ and volatility σ^2, and falls with competition n; as n grows the premium goes to zero, S_1 tends to μ, and LT1's immediate trade tends to i [crypto-cartea, same chap., p. 24]. Competition is itself set by participation costs: in Grossman and Miller's earlier entry stage, the number of market makers falls monotonically as the participation cost c rises, which raises the discount [crypto-cartea, same chap., p. 25]. Per-share fees and two empirical liquidity measures follow from the same equilibrium: see trading fees and price impact and return reversal. Interpretation: the quoted inside spread implied by the model, a bid at $\mu - p$ and an ask at $\mu + p$ with $p = \gamma\sigma^2 |i| / (n+1)$, is $2p$.

Pitfalls. The model solves a single Walrasian clearing rather than a trading mechanism [crypto-cartea, same chap., pp. 26–27], and the offsetting trader arrives for sure; the arrival time and its uncertainty, which drive inventory risk in continuous-time quoting models, are absent. Interpretation: reading every concession as risk compensation overstates inventory costs where order flow is informed.

See also. Kyle single-auction model (contrast: risk-neutral market makers, price impact from information); Avellaneda–Stoikov quoting (generalisation: inventory risk in continuous time with random order arrival); static limit-order depth (the same immediacy supplied through posted orders).

Further reading. [grossman-miller-1988], the original model, including participation costs (not yet read for this entry); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Grossman–Miller; immediacy; liquidity premium; liquidity discount; inventory risk; risk transfer; dealer competition; CARA–normal equilibrium

Status. Verification: partly checked 2026-10-14 (every formula and the comparative statics against `crypto-cartea`, pp. 21–27, read from page images of a scanned copy; the original paper not consulted). Stability: draft. Currency: timeless (theory).

Result 8.16 Grossman–Miller with per-share trading fees

Statement. If every trader pays a fee η per share bought or sold, the date-2 price is unchanged, $S_2 = \mu + \epsilon_2$, but for a seller ($i > 0$) the date-1 price falls to [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 25–26]

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1} - 2\Big(\frac{n}{n+1}\Big)\eta, \qquad q_1^{LT1,*} = \frac{i}{n+1} + 2\Big(\frac{n}{n+1}\Big)\frac{\eta}{\gamma\sigma^2}.$$

The initiating liquidity trader pays his own fee and, through the lower price, the fraction $n/(n+1)$ of the market makers' round-trip fees 2η, and keeps more inventory after date 1 than without fees [crypto-cartea, same chap., p. 26].

Notation. As in the Grossman–Miller model.

Conditions. The model of the Grossman–Miller entry, with fees proportional to shares rather than to dollar volume, positions remaining after date 2 liquidated at date 3, and desired trades large enough that every participant prefers trading to not trading [crypto-cartea, same chap., p. 25]. Idea of the proof: fees cancel at

date 2 because remaining positions are sold either then or at date 3; at date 1 they reduce the market makers' demand by the round-trip fee, $q_1^{MM} = (\mu - S_1 - 2\eta)/(\gamma\sigma^2)$, and clearing $i = nq_1^{MM} + q_1^{LT1}$ gives the price [crypto-cartea, same chap., pp. 25–26].

Practice. Liquidity traders whose desired trade is small relative to the fee do not trade at all [crypto-cartea, same chap., p. 25]. Participation costs and fees act differently: participation costs work through the number of market makers, while fees work mainly through the discount and the immediacy obtained, since each market maker pays 2η per trade but recovers $2n\eta/(n+1)$ through the discount [crypto-cartea, same chap., p. 26].

Pitfalls. Real exchange fees are often proportional to dollar volume and differ by side and by maker or taker [crypto-cartea, same chap., p. 25]; the closed form assumes the same per-share fee for everyone.

See also. Grossman–Miller model (the base case, $\eta = 0$).

Index. trading fees; exchange fees; fee incidence; participation cost; no-trade region; liquidity discount

Status. Verification: partly checked 2026-10-14 (the formulas and fee incidence against `crypto-cartea`, pp. 25–26, from page images of a scanned copy in which several signs are faint; the printed results agree with the clearing condition on the same page). Stability: draft. Currency: timeless (theory).

Result 8.17 Grossman–Miller liquidity measures: linear price impact and negative autocovariance

Statement. Read as trading in a limit-order book, where LT1's market orders walk the book posted by the market makers, the Grossman–Miller date-1 price is linear in the quantity LT1 trades [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 26–27]:

$$S_1 = \mu + \lambda\, q^{LT1}, \qquad \lambda = -\frac{1}{n}\,\gamma\sigma^2, \qquad q^{LT1} = i\,\frac{n}{n+1}.$$

With public news ϵ_1 before LT1 trades and efficient price $\mu_t = \mathbb{E}[S_3 \mid \epsilon_1, \dots, \epsilon_t]$, the price changes $\Delta_1 = S_1 - S_0$ and $\Delta_2 = S_2 - S_1$ satisfy [crypto-cartea, same chap., pp. 27–28]

$$\operatorname{Cov}[\Delta_1, \Delta_2] = -\lambda^2 \operatorname{Var}[q^{LT1}] < 0.$$

Notation. As in the Grossman–Miller model; here q^{LT1} is the quantity LT1 trades at date 1, not a holding, and λ is the slope of price on that quantity.

Conditions. The model of the Grossman–Miller entry, with shocks $\epsilon_1, \epsilon_2, \epsilon_3$ i.i.d. normal with variance σ^2, and liquidity demand independent of news [crypto-cartea, same chap., pp. 27–28]. Idea of the proof: $S_1 = \mu_1 + \lambda q^{LT1}$ and $S_2 = \mu_2$, so the order-pressure term enters Δ_1 and reverses in Δ_2 [crypto-cartea, same chap., p. 28].

Practice. A more liquid market has a smaller $|\lambda|$, from more competition, less risk aversion or lower volatility [crypto-cartea, same chap., p. 27]. As λ goes to zero, so does the autocovariance, and the price process converges to the efficient-price martingale [crypto-cartea, same chap., p. 28]. The two measures coincide here only because the model is essentially static; in dynamic models they capture different things, permanent effects of order flow on the efficient price as against the temporary effect in each trader's execution price [crypto-cartea, same chap., p. 28].

Pitfalls. Interpretation: an empirical regression of price changes on signed order flow, or a negative return autocovariance, mixes this inventory effect with information and bid–ask bounce; this model alone cannot attribute it. Compare the information slope of Kyle's model.

See also. Grossman–Miller model (source of the equilibrium); Kyle single-auction model (contrast: a price-impact slope from information).

Index. price impact; Kyle's lambda (contrast); liquidity measure; autocovariance of price changes; return reversal; temporary impact; transitory price pressure

Status. Verification: partly checked 2026-10-14 (both formulas and the comparative statics against `crypto-cartea`, pp. 26–28, from page images of a scanned copy). Stability: draft. Currency: timeless (theory).

8.4 Quoting with limit orders

Model 8.18 Static limit-order depth with exponential fill probability

Statement. A small, risk-neutral market maker with costless inventory and infinite patience chooses depths $\delta^{\pm}$ from the midprice S_t for a sell order at $S_t + \delta^+$ and a buy order at $S_t - \delta^-$; a buy (sell) market order arrives with probability p^+ (p^-) and reaches depth δ with probability $e^{-\kappa^{\pm}\delta}$, and expected profit per trade is maximised by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 29, eq. (2.3)]

$$\max_{\delta^+,\delta^-} \left\{ p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^- \right\}, \qquad \delta^{\pm,*} = \frac{1}{\kappa^{\pm}}.$$

The optimal depth equals the mean depth of the book on that side [crypto-cartea, same chap., p. 30]. This is a static simplification set up in place of the dynamic Ho–Stoll dealer model, which the authors give as the usual first reference [crypto-cartea, same chap., p. 28]. Figure 8.2 shows the depths and fill probabilities.

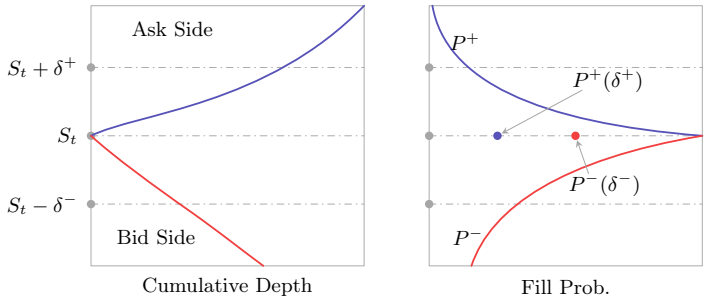


Figure 8.2: Limit-order depths and probability of execution, after [crypto-cartea, Figure 2.2, p. 30].

Notation. Local override for this entry: $p^{\pm}$ are probabilities of market-order arrival, not prices, and $P^{\pm}(\delta)$ is the probability that an arriving market order reaches depth δ, not a bond or put price.

Conditions. Everyone agrees on the value S_t (no information asymmetry); other market makers' orders form a fixed book that does not react; the market maker has no inventory or trading costs and liquidates at the midprice [crypto-cartea, same chap., pp. 28–29]. Idea of the proof: the problem separates by side, and the first-order condition $p^{\pm}e^{-\kappa^{\pm}\delta^{\pm}}(1-\kappa^{\pm}\delta^{\pm}) = 0$ gives the depth; the arrival probabilities scale profit but not the optimal depth (Interpretation, own derivation).

Practice. The model isolates the trade-off between margin per trade and probability of execution [crypto-cartea, same chap., p. 30]; quoting deeper earns more per fill and fills less often.

Pitfalls. The authors list what makes it unrealistic: the arrival and fill probabilities are special, constant and exogenous; the market maker's decision does not affect other traders; and the objective is static [crypto-cartea, same chap., p. 30]. Interpretation: there is no inventory penalty and no adverse selection, so the depth does not respond to position or to informed flow; Avellaneda–Stoikov adds inventory.

See also. Avellaneda–Stoikov quoting (generalisation: dynamic, with inventory and the same exponential intensity); Grossman–Miller model (the immediacy this quoting supplies).

Further reading. [ho-stoll-1981], the dynamic dealer model this simplifies (not yet read for this entry; see the pathway backlog); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. limit-order placement; quote depth; fill probability; probability of execution; exponential order-book depth; static market making; Ho–Stoll (static simplification)

Status. Verification: partly checked 2026-10-14 (the problem, the solution and the stated limitations against `crypto-cartea`, pp. 28–30, from page images of a scanned copy; the Ho–Stoll paper not consulted). Stability: draft. Currency: timeless (theory).

Model 8.19 Avellaneda–Stoikov inventory-aware quotes

Statement. A dealer with CARA utility, inventory q and horizon T quotes around a reservation (indifference) price rather than the midprice s [s6-stoikov, sec. 2.2, eq. (8), p. 219]:

$$r(s,q,t) = s - q\gamma\sigma^2(T-t).$$

With symmetric exponential execution intensities $\lambda^a(\delta) = \lambda^b(\delta) = Ae^{-k\delta}$, an expansion of the value function in q with a first-order approximation of the order-arrival terms gives the same reservation price and a bid–ask spread, independent of inventory, of [s6-stoikov, sec. 3.2, eqs. (29)–(30), p. 221]

$$\delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right).$$

Notation. Local override for this entry: the midprice follows $dS_u = \sigma\, dW_u$, so σ is an absolute volatility in price units per square root of the model's time unit, not the annualised volatility of returns; s is the current midprice; $\delta^a = p^a - s$ and $\delta^b = s - p^b$ are the quote distances; λ^a, λ^b are Poisson execution intensities, not price-impact slopes [s6-stoikov, secs. 2.1 and 2.4, pp. 218–219].

Conditions. Arithmetic Brownian midprice with no drift and a zero interest rate; exponential utility of terminal wealth plus inventory marked at the midprice; one-share limit orders, updated continuously at no cost, filled at Poisson rates that decrease with distance from the midprice [s6-stoikov, secs. 2.1–2.4, pp. 218–219]. The exponential intensity follows from a constant market-order frequency, a power-law order-size distribution

and a logarithmic temporary impact, with $A = \Lambda/\alpha$ and $k = \alpha K$; a power-law impact gives a power-law intensity instead [s6-stoikov, sec. 2.5, eq. (12), p. 220]. Exact quotes solve the HJB equation and two implicit equations for the distances [s6-stoikov, sec. 3.1, eqs. (15)–(19), pp. 220–221]; the closed form above is an approximation [s6-stoikov, sec. 3.2, p. 221]. A long position ($q > 0$) puts the reservation price below the midprice, so both quotes shift down to sell [s6-stoikov, sec. 2.2, p. 219].

Practice. In 1000 simulations with $s = 100$, $T = 1$, $\sigma = 2$, $\gamma = 0.1$, $k = 1.5$, $A = 140$ and $dt = 0.005$, the inventory strategy earned a mean profit of 65.0 with standard deviation 6.6, against 68.4 and 12.7 for a strategy quoting the same average spread of 1.49 symmetrically around the midprice; final inventory had standard deviation 2.9 against 8.4 [s6-stoikov, sec. 3.3, Table 1, p. 222]. The symmetric strategy earns slightly more because it trades more, but its P&L is much more dispersed [s6-stoikov, sec. 3.3, p. 222].

Pitfalls. Interpretation: the closed form is a control template, not a production model; real books add queue position, tick size, discrete inventory limits, latency, rebates, hedging instruments and fill probabilities that change with regime. The approximate spread does not depend on inventory only because the intensities are exponential [s6-stoikov, sec. 3.2, p. 221]. Interpretation: the formula has no inventory limit; a hard limit needs the bounded-inventory solution read in this pathway (`wide-gueant-lehalle-tapia`, not yet read).

See also. Static limit-order depth (special case: no inventory risk, static); Grossman–Miller model (the same inventory-risk premium in a three-date equilibrium).

Further reading. Own working from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Avellaneda–Stoikov; reservation price; indifference price; inventory skew; optimal bid–ask spread; market making with limit orders; exponential fill intensity; HJB market making

Status. Verification: partly checked 2026-10-14 (the reservation price, intensity, spread and simulation results against the author-hosted journal PDF of `s6-stoikov`, pp. 217–222; the HJB equation and expansion steps read but not re-derived). Stability: draft. Currency: as published in 2008; review annually with the catalogue record.

8.5 Informed trading and informative prices

Definition 8.20 Fundamental value, market value and noise: what informative prices are informative about

Statement. The *market value* of an instrument is the price at which traders can buy or sell it. Its *fundamental value* (intrinsic value) is the expected present value of all present and future benefits and costs of holding it: the value everyone would agree on if all knew everything known about the instrument, analysed it properly and perceived its benefits and costs alike [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.1, pp. 222–223]. The difference between the two is *noise*; prices are informative when they are near fundamental values and completely informative when they equal them [alpha-harris, same sec., p. 223]. Writing V_t for fundamental value and P_t for the price, noise is $\epsilon_t = P_t - V_t$. Fundamental values use only information available now, so they are the best estimates of perfect-foresight values, not those values; and their changes are unpredictable, since a predictable change would mean they did not yet reflect the information behind the prediction [alpha-harris, same sec., p. 223]. Relative to a price, information is of three kinds:

- *public*: available to any trader, such as public news and past prices and volumes [alpha-harris, same chap., pp. 223 and 240];
- *private*: held only by some traders, whether learnt first or produced by their own research [alpha-harris, same chap., pp. 223 and 228];
- *stale*: already in the price, meaning it cannot be used to forecast future price changes; traders who act on it are *pseudo-informed*, and in fact uninformed [alpha-harris, same chap., sec. 10.5.2.1, p. 229].

Notation. V_t fundamental value, P_t market price, $\epsilon_t = P_t - V_t$ noise; the sign is a convention, since Harris defines noise only as the difference.

Conditions. Fundamental values are not observed, so whether a price is informative or noisy cannot easily be determined [alpha-harris, same chap., p. 223]. Prices become informative in two ways: when values are common knowledge no one trades at other prices; otherwise informed traders' trading moves prices towards their estimates, which is the more important route because values are rarely common knowledge [alpha-harris, same chap., sec. 10.6.1, pp. 236–237]. Information gets into the price when all traders understand its significance or when informed traders push the price towards their estimates [alpha-harris, same chap., p. 229].

Practice. Whether information is stale can rarely be checked directly. Harris's guide: information that is unique and that others could not have anticipated is probably not stale; information acted on slowly, widely known, cheap to obtain, or reasonably anticipated probably is. His example: Occidental Petroleum rose 10 percent the day after its long-ill chairman died and fell back the next day, because his death was expected and

its date was not material [alpha-harris, same chap., p. 229 and box, p. 230]. Fischer Black would call stock prices informative if they lay between half and twice their fundamental values [alpha-harris, same chap., box, p. 223].

Pitfalls. Two conditioning sets are in use. Harris's fundamental value conditions on everything known to anyone, private information included. Hasbrouck calls the expectation of the payoff conditional on *all public information* the fundamental value or efficient price [s6-hasbrouck, sec. 3.2, p. 26]: the m_t of the random-walk decomposition. Harris's noise is therefore not Hasbrouck's pricing error $s_t = p_t - m_t$. A price equal to the public-information expectation has a zero pricing error but nonzero noise whenever private information has not yet reached it (Interpretation). The legacy notes defined fundamental value as "implied by all payoff-relevant information", which reads as the perfect-foresight value that Harris distinguishes from it. They also described private information as "costly or restricted"; private information can be produced lawfully from public data by research [alpha-harris, chap. "Insider Trading", p. 585].

See also. fundamental and transitory volatility (the statistical version, conditioned on public information); informed trading and aggregation (how prices become informative); market efficiency; inside information; outside spread (value traders' estimates in use).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. fundamental value; intrinsic value; market value; noise; informative prices; perfect-foresight value; random walk; public information; private information; stale information; pseudo-informed traders; efficient price

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 222–230, 236–237, 240 and 585, and `s6-hasbrouck`, 2007 Oxford printing, p. 26). Stability: draft. Currency: timeless (concepts).

Result 8.21 Informed trading makes prices informative: the price as an aggregate of value estimates

Statement. Informed traders buy when the price is below their estimate of fundamental value and sell when it is above. Their buying pushes prices up and their selling pushes them down, so their trading moves prices towards their estimates [alpha-harris, chap. "Informed Traders and Market Efficiency", secs. 10.2–10.3, pp. 223–224]. Harris's algebraic illustration [alpha-harris, same chap., box, p. 225]: N traders hold unbiased forecasts $f_i = V + e_i$ of the value V, each wants the position $D_i = a(f_i - P)$, and the security is in zero net supply. Market clearing, $\sum_i a(f_i - P) = 0$, gives

$$P = \frac{1}{N}\sum_{i=1}^{N} f_i = V + e_M, \qquad e_M = \frac{1}{N}\sum_{i=1}^{N} e_i.$$

With independent errors, $e_M \to 0$ as N grows, by the law of large numbers: prices are most informative when many informed traders collect information independently. Interpretation: if the errors are also independent of V with common variance σ^2, then $\mathbb{E}[P \mid V] = V$ and $\mathrm{Var}(P - V) = \sigma^2/N$.

Conditions. When forecasts differ in accuracy, the best combination is a weighted average that gives more weight to the more accurate. Well-informed traders tend to be wealthier and take larger positions, so prices approximate such an optimally weighted average: the market is a statistical calculator [alpha-harris, same chap., p. 224 and box, p. 226]. Interpretation: with trader-specific slopes a_i the same clearing condition gives $P = \sum_i a_i f_i / \sum_i a_i$, still unbiased when each f_i is; with independent errors of variances σ_i^2, $\mathrm{Var}(P - V) = \sum_i a_i^2 \sigma_i^2 / (\sum_i a_i)^2$, minimised by $a_i \propto 1/\sigma_i^2$ at $\left(\sum_i \sigma_i^{-2}\right)^{-1}$, below every σ_i^2. Many mistaken traders, or one large one, make prices less informative, until their losses drive them out [alpha-harris, same chap., p. 224]. Informed traders do not trade to make prices informative: price impact is their cost, and they want prices to adjust only after they have built their positions [alpha-harris, same chap., p. 224]. They profit only from uninformed traders; competition among them pushes prices closer to values and lowers their profits until the least successful just cover their costs [alpha-harris, same chap., secs. 10.6.3 and 10.6.5, pp. 237–239].

Practice. Trade aggressively when the information will soon be common knowledge or many others will act on it, since the first to trade profit most and a flood of orders makes liquidity suppliers wary; trade slowly (stealth trading) when the advantage will last, so that others cannot infer that the trader is well informed [alpha-harris, same chap., sec. 10.4, pp. 225–226]. The legacy notes added a third speed, *inactive*, for information already in the price; that is stale information, on which no trading pays [alpha-harris, same chap., p. 229], rather than a trading speed. The most profitable traders have estimates that are both precise and *orthogonal*, uncorrelated with other traders'; they face little competition for liquidity when they trade. Precision and orthogonality trade off: a precise but crowded estimate and an imprecise but orthogonal one can be equally profitable [alpha-harris, same chap., sec. 10.6.2.1, p. 237].

Pitfalls. Aggregation removes independent errors, not shared ones. Interpretation: if the errors have common variance σ^2 and pairwise correlation ρ (which requires $\rho \geq -1/(N-1)$), then $\mathrm{Var}(e_M) = \frac{1}{N^2}[N\sigma^2 + N(N-1)\rho\sigma^2] = \sigma^2[\rho + (1-\rho)/N] \to \rho\sigma^2$ for $\rho \geq 0$, so a shared data source, model or stale signal leaves an error

that no number of traders removes; Harris notes that traders who use the same information and methods have highly correlated estimates [alpha-harris, same chap., p. 237]. Harris also says the market's forecast error is smaller than the average individual error whenever the individual errors are not identical, and that the price estimates value better than any individual trader can [alpha-harris, same chap., box, p. 225, page image]. Interpretation, by error measure: the signed errors average exactly, $e_M = \frac{1}{N}\sum_i e_i$, so "smaller" needs a loss. For one realisation, squared error gives $e_M^2 \le \frac{1}{N}\sum_i e_i^2$, since the gap is $\frac{1}{N}\sum_i (e_i - e_M)^2$, with equality only when all e_i are equal, as Harris's condition says. Absolute error gives $|e_M| \le \frac{1}{N}\sum_i |e_i|$ by the triangle inequality, with equality whenever the errors share a sign, so $e = (1, 2)$ is not identical yet gives 1.5 on both sides. In expectation, with independent, zero-mean, nondegenerate errors, both inequalities are strict, since errors of opposite sign then occur with positive probability. "Better than any individual" holds in expected squared error only when accuracies are equal: with variances 0.01 and 1 the equal-weighted average has variance 0.2525, worse than the better trader's, while inverse-variance weights give 0.0099. The legacy proposition assumed the price is the average forecast; in the box this follows from linear demands with a common slope and zero net supply, an assumption Harris says simplifies the arithmetic without changing the qualitative results. The legacy mechanism had informed traders buy when $P_t < V_t$: they act on estimates, and lose when their estimates are poor [alpha-harris, same chap., p. 223].

See also. fundamental value; market efficiency (why prices cannot be fully informative); Kyle single-auction model and Glosten–Milgrom model (equilibrium models of informed trading); price formation process; informed-trader types; order anticipation (front runners accelerate informed traders' impact).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. informed traders; informative prices; information aggregation; forecast averaging; law of large numbers; correlated errors; orthogonal estimates; stealth trading; aggressive trading; statistical calculator; price impact as a cost; inverse-variance weighting

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 222–226, 229 and 236–239; the box on p. 225 also against its page image, 2026-10-15, slice 6). Stability: draft. Currency: timeless (theory).

Definition 8.22 Informed-trader types: value traders, news traders, information-oriented technical traders and arbitrageurs

Statement. Harris classifies informed traders by what they estimate [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.5, pp. 226–235, Table 10-3, p. 235]:

- *value traders* estimate total fundamental value from all available information; slow, because careful review guards against biased estimates;
- *news traders* estimate changes in value from new information, assuming the price already reflects everything else; fast on public news, slower on information they produce by their own research;
- *information-oriented technical traders* identify price patterns left by informed traders' systematic mistakes, such as under- or overreaction; fast;
- *arbitrageurs* estimate relative values of instruments that share valuation factors; fast.

Conditions. Value and news traders profit in different circumstances. Price departs from value either because value changes and price lags, where news traders, first to learn the news, profit; or because uninformed traders push price away from value, where value traders profit [alpha-harris, chap. "Value Traders", sec. 16.1, pp. 338–339]. Their mistakes put them on opposite sides. News traders who misjudge the size of a change leave underreaction for value traders to trade on, or overreaction for them to trade against (Table 10-1, p. 230); pseudo-informed traders acting on stale news lose to value traders (box, p. 229). Value traders who mistake a price move driven by news for uninformed pressure lose to news traders [alpha-harris, chap. "Informed Traders and Market Efficiency", p. 239].

Practice. Organisation follows speed: large value traders have pyramid-shaped organisations with layers of review, news traders flat organisations that decide quickly [alpha-harris, same chap., pp. 227–229]. Technical traders who trade on the predictable impact of *uninformed* traders are not informed traders: they act as dealers if they supply liquidity, or as order anticipators if they front-run [alpha-harris, same chap., p. 231]. Information-oriented technical strategies decay as informed traders learn from their mistakes [alpha-harris, same chap., p. 231].

Pitfalls. The legacy taxonomy described technical traders as exploiting "predictable errors by other traders". In Harris that holds for errors by informed traders; errors by uninformed traders belong to sentiment-oriented technical traders, who are order anticipators (order anticipation). The legacy principle *value traders versus news traders* added dealer inventory pressure as a separate source of value-trader profit. In Harris it is a channel of uninformed demand (see outside spread).

See also. informed trading and aggregation; outside spread (value traders as liquidity suppliers); arbitrage; order anticipation; fundamental value (stale information).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. informed traders; value traders; news traders; information-oriented technical traders; arbitrageurs; pseudo-informed traders; underreaction; overreaction; trading speed

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 226–235 and 239, Tables 10-1 and 10-3 included, and pp. 338–339). Stability: draft. Currency: timeless (concepts).

Definition 8.23 Winner's curse: what winning a common-value trade reveals, and how bids adjust

Statement. Buyers who compete for an item of common but unknown value suffer the *winner's curse* when they pay more than it is worth. The highest bidder tends to be the one who most overestimated the value, so bids near one's own estimate lose on average when they win. Buyers avoid it by lowering their bids to reflect what winning would tell them about their estimate; sellers whose offers are taken symmetrically raise their offers [alpha-harris, chap. "Value Traders", sec. 16.2.1.2, pp. 341–342]. Interpretation: let $\widehat{V}$ be the buyer's estimate, c an all-in cost, and $\omega(b) = \mathbb{E}[\widehat{V} - V \mid \widehat{V},\ \text{win at bid } b]$ the expected overestimate given one's own estimate and winning at bid b. The expected profit of winning is $\widehat{V} - \omega(b) - b - c$, so the highest bid that breaks even solves $b = \widehat{V} - \omega(b) - c$.

Conditions. The required adjustment is larger [alpha-harris, same sec., p. 342]:

- the more bidders there are, because outbidding many implies a larger overestimate;
- the harder the value is to estimate, because extreme estimates are then further from the truth.

Against market power: with only one or very few competitors a bidder may underbid in the hope of winning cheaply, so bids are more aggressive against several competitors than against one or two; this matters only with very few [alpha-harris, same sec., p. 342]. In a two-sided single-price auction the curse matters little, because traders do not trade at their own bids and offers and the clearing price is near the median estimate. In continuous markets they do, and aggregation is sequential, so traders who misestimate lose when they push prices [alpha-harris, same sec., pp. 342–343].

Practice. Oil companies overpaid for drilling leases until they allowed for the curse; their geologists were right on average over all tracts but optimistic about the tracts they won [alpha-harris, same chap., box, p. 342]. Value traders face the curse whenever they trade on a large apparent mispricing and widen their outside spreads in response; Harris attributes part of late-1990s Internet-stock volatility to prices bouncing within wide outside spreads [alpha-harris, same chap., p. 343 and box]. Do not compete against foolish bidders: one cannot profit bidding against strategies that ensure losses [alpha-harris, same chap., p. 342].

Pitfalls. The legacy proposition wrote the zero-profit bid as $b^\star = \widehat{V} - \omega_N - c$ with ω_N conditioned only on winning among N buyers. Conditioning must also be on one's own estimate, and the event of winning depends on the bid itself, and in equilibrium on rivals' strategies. The expression is therefore a fixed-point condition, not a closed form. It also gives the highest acceptable bid, not the profit-maximising one, which shades further when competitors are few (Interpretation). The legacy severity list added that common data reduce dispersion while independent noisy estimates increase selection risk; that is not in the passages checked (Interpretation: shared information makes estimates move together, which reduces how much winning reveals).

See also. outside spread (the curse widens value traders' spreads); informed trading and aggregation (correlated and independent estimates); Glosten–Milgrom model (adverse selection, the analogous inference a dealer makes from the side of a trade).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. winner's curse; common value; bid shading; adverse selection; number of bidders; valuation uncertainty; market power; single-price auction; oil lease auctions

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 341–343). Stability: draft. Currency: timeless (concept).

Definition 8.24 Market efficiency: weak, semistrong and strong forms, and the definition with information costs

Statement. In an *efficient market*, prices reflect all information that traders can acquire and profitably trade upon. The definition builds in the costs of acquiring information and of acting on it, and the impact of informed trading on prices [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.6, p. 240]. Harris sets it beside the three traditional forms [alpha-harris, same sec., p. 240]:

- *weak form*: prices reflect all information in past prices, so no one can predict price changes from past prices alone;

- *semistrong form*: prices reflect all public information, which includes all public news and past prices and volumes in all securities and contracts;
- *strong form*: prices reflect all public and private information as soon as it is known; since informed traders could never profit, only markets in instruments whose values are common knowledge are strong-form efficient.

Conditions. Prices never fully reflect all the information traders could collect: information that is too expensive or too inconsequential is not acquired [alpha-harris, same sec., p. 240]. Harris's market paradox: if prices were fully informative, informed trading would not pay, so no one would make them informative. Prices and values change constantly, so gaps open and informed trading closes them: prices are informative but not always [alpha-harris, same chap., sec. 10.6.4, pp. 238–239]. Lehalle and Laruelle make the same point with Grossman and Stiglitz (1980), not consulted: market dynamics must contain enough inefficiency to reward those who improve the information in prices [alpha-laruelle, Introduction, p. 4]. How informative prices are depends on the costs of information and of trading, and on there being uninformed traders to lose to informed ones [alpha-harris, same chap., secs. 10.6.3 and 10.7, pp. 238 and 243–244].

Practice. Informative prices are not free. What uninformed traders lose to informed traders pays for much of the information in prices, so policies that frustrate informed traders can make prices less informative and markets more liquid; restrictions on insider trading are an example [alpha-harris, same chap., sec. 10.6.7, p. 241]. Harris concludes that these restrictions narrow spreads and make prices less informative in the short run [alpha-harris, chap. "Insider Trading", sec. 29.4, pp. 597–598]. The trade-off is not universal: publishing material fundamental information raises both liquidity and informativeness [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.8, pp. 241–242]. Trading halts before material releases protect uninformed traders, especially those who offer liquidity, at the cost of a few minutes' delay in price formation, a trade Harris judges worth making [alpha-harris, same chap., box, p. 241]. Most published studies find markets weak-form efficient, and semistrong-form efficient for public information that is easy to obtain and interpret [alpha-harris, same chap., p. 240].

Pitfalls. Comparing market structures needs the change in benefits, not their totals, and the benefits of informative prices and of liquidity are both hard to measure [alpha-harris, same chap., sec. 10.6.7, p. 241]. The legacy notes put volumes in the weak-form information set; Harris's weak form uses past prices, and his semistrong set adds volumes. They also called the cost-sensitive definition *economic* efficiency, a name Harris does not use. Their trade-off said informed trading transfers wealth from "slower" liquidity traders and that efficiency-raising policies reduce "displayed" liquidity. Harris speaks of uninformed traders and liquidity suppliers, and of liquidity in general. A second legacy statement of the paradox said markets persist between its limits because information is costly, temporary and unevenly distributed. Harris's resolution rests on prices and values changing constantly, with information costs limiting what prices reflect (pp. 238–240); "unevenly distributed" is implicit in his informed and uninformed traders rather than stated as part of the resolution.

See also. fundamental value; informed trading and aggregation; inside information (the main example of the trade-off); spread determinants (asymmetric information widens spreads); order anticipation (front running of informed traders lowers informativeness in the long run).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Grossman and Stiglitz (1980), cited by Lehalle and Laruelle, not consulted.

Index. market efficiency; efficient markets; weak-form efficiency; semistrong-form efficiency; strong-form efficiency; information costs; market paradox; Grossman–Stiglitz paradox; liquidity and informative prices; public information; trading halts

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 237–244 and 597–598, and `alpha-laruelle`, 2018 World Scientific printing, p. 4). Stability: draft. Currency: timeless (concepts); the empirical summary as of 2002.

Definition 8.25 Inside information: material nonpublic information and who may not trade on it (United States, as summarised in 2002)

Statement. For economic analysis, Harris defines *inside information* as material information about the value of a security that is not available to public traders, and *material* information as information that would cause prices to change if it were widely known; *insider trading* is trading on it [alpha-harris, chap. "Insider Trading", sec. 29.1, p. 584]. Inside information keeps its status however many people pass it on, and loses it only when it is publicly available, through a broadly distributed press release or a public filing [alpha-harris, same sec., p. 585]. The legal definitions come from statutes, regulations and case law, are complex, and call for an attorney [alpha-harris, same sec., p. 584]. The SEC's summary, as quoted by Harris from its web page of January 2002, covers trading in breach of a fiduciary duty or other relationship of trust and confidence while in possession of material nonpublic information, and tipping, trading by those tipped, and misappropriation [alpha-harris, same sec., box, p. 585]. Those who may not trade on such information include:

- *corporate insiders*: senior managers, directors, large shareholders, key employees, and anyone involved in corporate activities that will materially affect the firm's value [alpha-harris, sec. 29.1.1, p. 587];
- employees of law, banking, brokerage and printing firms given the information to serve the corporation, and government employees who learn it at work [alpha-harris, sec. 29.1, box, p. 585];
- *tippees*: friends, family and business associates of insiders, and the friends of their friends [alpha-harris, same sec., pp. 584–585];
- *misappropriators*, who take confidential information from an employer or other source. Chiarella's conviction was reversed in 1980 because he owed no fiduciary duty to the firms whose stock he traded; Harris says that under the later law he would be convicted for misappropriation [alpha-harris, same sec., box, p. 586]. Winans was convicted under the misappropriation doctrine [alpha-harris, same sec., box, p. 587].

Analysts whose private information comes only from analysing public information are not trading on inside information [alpha-harris, same sec., p. 585]. Harris's box "Inside information, skilled speculation, or good advice?" shows why the line is hard to draw from the trade alone: the same well-timed purchase before a takeover may rest on a tip, on independent analysis of public information, or on analysis that the trader then sold to the target's management; and a trader who had a tip will claim the second [alpha-harris, sec. 29.2, box, p. 590].

Conditions. Jurisdiction and date: United States law as Harris summarises it in 2002. Most countries prohibit insider trading, but only a few, mainly the United States, Canada and Great Britain, regularly and seriously enforce their laws [alpha-harris, chap. "Insider Trading", p. 584]. Regulation FD (2000) requires an issuer that discloses material nonpublic information to anyone to disclose it publicly [alpha-harris, same chap., box, p. 586].

Practice. Corporations restrict their employees' trading: many require corporate approval before a trade, may refuse it before a release, or allow trading only in a window after quarterly earnings [alpha-harris, same chap., sec. 29.1.1, p. 588]. Exchange surveillance looks for large price changes on large volume before significant news [alpha-harris, same chap., sec. 29.2, p. 588]. Insider trading is hard to prove: a sophisticated speculator can claim independent analysis of public data, and trading on information that is not revealed in a discrete event is essentially undetectable from prices and volumes [alpha-harris, same chap., pp. 589–590]. Like all informed trading, insider trading hurts liquidity suppliers and raises spreads for uninformed traders [alpha-harris, same chap., sec. 29.3.1.2, pp. 591–592].

Pitfalls. Harris's categories are legal categories summarised for economists, not a legal test; check the law in force. The reporting deadlines, prearranged-plan rules and penalties he gives [alpha-harris, same chap., pp. 587–591] are as of 2002, are not repeated here, and were not checked against current law. The legacy notes used "temporary insider" and a "relationship, benefit or confidence" test for tipping. Neither term appears in the passages checked. They also named auditors, consultants and exchange staff, whom the SEC summary does not list. A legacy principle, *signal tradability requires permission*, said that data provenance, confidentiality duties, firm restricted lists, compliance approvals and execution capacity decide whether a predictive signal may be used. The passages checked support confidentiality duties (misappropriation), employer approval of employees' own trades (p. 588), and the dependence of informed trading profits on liquidity [alpha-harris, chap. "Informed Traders and Market Efficiency", box, p. 236]. Firm restricted lists, compliance approval of signals and the provenance of purchased data are not in them. A legacy caveat, *skilled speculation versus inside information*, named alternative data "with clean rights" and channel checks as legitimate sources; the passages checked do not discuss either, and their legal status was not checked.

See also. fundamental value (private and public information); market efficiency (insider-trading restrictions in the information–liquidity trade-off); spread determinants (enforced insider-trading rules as a secondary determinant); insider-trading profitability, detection and deterrence.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inside information; insider trading; material nonpublic information; MNPI; materiality; corporate insiders; tippees; misappropriation; Chiarella; Regulation FD; trading windows; preclearance; market surveillance; signal permission

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 236 and 584–592, the box on p. 590 added in slice 6; the SEC web page Harris quotes was not consulted). Stability: draft. Currency: dated, United States law as summarised in 2002; not checked against current law or other jurisdictions.

Result 8.26 Insider-trading profitability, detection and deterrence (United States, as described in 2002)

Statement. Harris's account of when trading on inside information pays and when it is caught [alpha-harris, chap. "Insider Trading", secs. 29.2–29.3, pp. 588–595]:

- *competition among insiders*: one insider with information that will not soon be released can trade slowly and extract the most; many insiders who learn it at once must race, which quickly reveals the information and lowers their total profit [alpha-harris, sec. 29.3.1.2, p. 592];
- *discreteness of release*: trading on information revealed in a discrete event can be traced back from it, whereas trading on information that firms do not reveal for months or years is essentially undetectable from prices and volumes [alpha-harris, sec. 29.2, p. 590]; restrictions are therefore effective mainly against information that will soon be public [alpha-harris, sec. 29.3.2.1, p. 595];
- *deterrence*: a rational criminal trades when the expected benefit exceeds the expected cost, the probability of being caught times the penalty, so a low probability of detection requires a large penalty [alpha-harris, sec. 29.2, p. 590].

Interpretation: with expected gross gain G, execution and leakage costs K, a risk charge R, detection probability q and penalty F, the trade is deterred when $G \leq K + R + qF$. If the penalty were only disgorgement of the gain, $F = G$, the condition becomes $(1 - q)G \leq K + R$, which a small q rarely satisfies; hence fines on top of disgorgement.

Conditions. United States as Harris describes it in 2002. Penalties then included disgorgement, fines and, rarely, prison; they were heavier for traders closer to the source. He gives maximum criminal penalties of ten years and a million dollars, a maximum civil penalty of three times the profit gained or loss avoided, and an SEC bounty of up to 10 percent of civil penalties collected [alpha-harris, sec. 29.2, pp. 590–591 and box]. These figures are dated and were not checked against current law. R is a cost only for a risk-averse trader; a risk-neutral trader would not subtract it (Interpretation).

Practice. Exchange surveillance identifies most cases. It looks for large price changes on large volume and asks the listed firm whether news explains them; trading before releases of significant unexpected news often indicates insiders or confederates [alpha-harris, sec. 29.2, p. 588]. Investigators then match a list of those who knew with a list of those who profited, in stocks, options or bonds, through links such as family, neighbours or classmates [alpha-harris, same sec., pp. 588–589 and box]. In the Colt Industries case, deep out-of-the-money options bought by friends and relatives of a law-firm employee were traced through the buyers' home addresses [alpha-harris, same sec., box, p. 589].

Pitfalls. These footprints are leads, not proof. Relationships between the two lists do not prove insider trading, and a sophisticated speculator will claim independent analysis of public data [alpha-harris, same sec., pp. 589–590]. The legacy footprint list also named widened spreads, quote fading and dark-pool activity before news. These are not among the signals in the passages checked, although wider spreads in the presence of insiders follow from adverse selection [alpha-harris, sec. 29.3.1.2, p. 591]. The legacy profitability list named market liquidity as a factor; Harris makes that point for informed trading generally (chap. "Informed Traders and Market Efficiency", box, p. 236). Its "traceability" factor is a paraphrase of Harris's investigative method, not his term.

See also. inside information (definition and who is covered); informed trading and aggregation (aggressive against stealth trading); market efficiency (restrictions in the information–liquidity trade-off); bluffing and manipulation (the same proof problem).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. insider trading; deterrence; detection probability; disgorgement; penalties; market surveillance; competition among insiders; discrete information events; abnormal volume; options activity before news

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 236 and 588–595). Stability: draft. Currency: dated: United States law and enforcement practice as described in 2002; not checked against current law.

Definition 8.27 Price formation process: immediate impact and its decay

Statement. The *price formation process* (PFP) is how trades and information become prices. Lehalle and Laruelle report a consensus that it consists of the impact of each trade combined with the variation of the asset's "fair price", after Bouchaud et al. (2002), not consulted [alpha-laruelle, Introduction, p. 27]. They split it into two phases [alpha-laruelle, same, Figure 8, pp. 27–29]:

1. *immediate impact*: trades move the price through the balance of buyers and sellers, and of liquidity-providing and liquidity-consuming orders, together with moves of the spread;
2. *decay*, or the *resilience* of the book: liquidity returns, or does not, according to the distance between the best quotes and investors' views of the fair price.

Pressure inconsistent with participants' consensus generates only temporary impact; pressure consistent with it is not pushed back, and the impact is permanent [alpha-laruelle, same, pp. 3 and 28]. The PFP is mainly driven by information [alpha-laruelle, same, p. 7].

Conditions. The more investors with views of the fair price take part, the faster impact decays. Market makers with inventory constraints contribute to the decay without any view of the fair price, but only if their inventory exceeds the turnover of a price jump [alpha-laruelle, Introduction, p. 28]. Post-trade transparency speeds the decay by warning traders that the last price is out of line with their views [alpha-laruelle, same, p. 28]. A design tension runs through it: the more information traders have, the better the PFP should be, but each trader fears information leakage, front-running and excessive impact [alpha-laruelle, same, pp. 19–20].

Practice. Market makers buy from early sellers and sell to later buyers, so that most temporary oscillations shrink to a spread related to the asset's risk [alpha-laruelle, Introduction, pp. 3–4]. Volatility is the random, unexpected component of the PFP, and the more efficient the microstructure, the lower intraday volatility [alpha-laruelle, chap. 2, sec. 2.5.1, pp. 171–172]. In a fragmented market, a consolidated tape organises how information is shared [alpha-laruelle, Introduction, p. 7]; the legacy fragmentation and outage blocks are assigned to `market-structure-execution-quality`. The authors see high-frequency traders' growing role as a possible threat to the decay, since they need not hold a view of the fair price and have small inventory limits [alpha-laruelle, same, p. 28]; that question belongs to `hft-market-design`.

Pitfalls. The impact of a single large order depends on what others are doing. If many others buy at the same time, the impact of a large buy is permanent; if most sell, it can be almost invisible. Impact is measured only by averaging over many market configurations [alpha-laruelle, Introduction, p. 3]. "Clean" and "noisy" PFP, the legacy terms, are not defined by the authors, although they speak of a "less noisy PFP" [alpha-laruelle, chap. 2, sec. 2.5.1, p. 177]. Interpretation: the noise is the transitory component of price changes.

See also. informed trading and aggregation (the information side); component identification (permanent and transitory price effects of trades); Kyle single-auction model (permanent impact from information); Grossman–Miller model (temporary concession for immediacy); returns by time scale; price impact (reaction and prediction impact, and the two readings of impact).

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Bouchaud et al. (2002), cited for the decomposition, not consulted.

Index. price formation process; PFP; fair price; immediate price impact; impact decay; resilience; temporary impact; permanent impact; post-trade transparency; information leakage; consolidated tape

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, Introduction, pp. 3–7, 19–20 and 27–29, and chap. 2, pp. 171–172 and 177; Figure 8 not inspected). Stability: draft. Currency: timeless (concept); the regulatory context is European, 2007–2018.

8.6 Arbitrageurs, order anticipators and bluffers

Definition 8.28 Arbitrage: hedge portfolio, basis, fair value, arbitrage spread and bounds; pure and speculative arbitrage

Statement. Arbitrageurs trade instruments with correlated prices. An *arbitrage opportunity* arises when those prices diverge from their normal relation: the arbitrageur buys the relatively cheap instruments, sells the relatively expensive ones, and profits if the prices converge [alpha-harris, chap. "Arbitrageurs", sec. 17.1, p. 348]. The positions form a *hedge portfolio*, whose legs are scaled by hedge ratios against a reference instrument, the *arbitrage numerator*, chosen to minimise the portfolio's risk. Its *carrying costs* include financing, dividends paid on short positions, storage and spoilage, offset by dividends, lending fees and interest received [alpha-harris, same sec., pp. 348–349]. With B_t the *basis*, the difference in prices between the instruments in the hedge portfolio, and $B_t^\star$ its *fair value*, the basis if all were correctly priced relative to each other, which depends on carrying costs and must be estimated, the *arbitrage spread* is

$$A_t = B_t - B_t^\star$$

[alpha-harris, same sec., p. 349]. Arbitrageurs trade only when the basis lies outside the *arbitrage bounds*, the values on either side of fair value at which they are just willing to trade [alpha-harris, same sec., p. 349]. Interpretation: if one unit of basis converges fully to fair value, the trade earns $|A_t|$ gross; with all-in round-trip cost K_t and a charge R_t for residual basis risk it is attractive only when $|A_t| > K_t + R_t$, so the bounds are $B_t^\star \pm (K_t + R_t)$.

Conditions. Harris distinguishes two types by the behaviour of the hedge portfolio's value [alpha-harris, same chap., sec. 17.3, pp. 350–363]:

- *pure arbitrage*: the value is strictly mean reverting, because some mechanism forces convergence. The

mechanisms are *shipping* (the same instrument in two places, bounded by shipping and financing costs), *delivery* (contract and cash prices converging at maturity, the basis set by carrying costs before it) and *conversion* (the same risk in different forms, bounded by the costs of converting and carrying). Cyclic arbitrages such as currency cross-rates are a kind of conversion arbitrage.

- *speculative arbitrage*, also called risk arbitrage: the value is nonstationary, because instrument-specific factors make it wander, and arbitrageurs bet that short-run mean reversion dominates. Kinds: spreads (maturity, calendar, yield-curve and credit spreads), pairs trading, statistical arbitrage with factor models, and merger arbitrage, the usual sense of "risk arbitrage".

Arbitrage is understood two ways [alpha-harris, same chap., secs. 17.6 and 17.9, pp. 375–378]. In the *price characterisation*, arbitrageurs are informed traders in relative values who enforce the law of one price [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.5.4, pp. 232–235]. In the *quantity characterisation*, they discipline slow prices, ship common-factor risk between markets and repackage it, manufacturing derivative positions; the arbitrage spread is the payment for these services.

Practice. Arbitrageurs are dealers or value traders in the hedge portfolio: fast dealer-type arbitrageurs trade short-term reversals, and value-type arbitrageurs take large positions when sure of fair value [alpha-harris, chap. "Arbitrageurs", sec. 17.2, pp. 349–350]. Dealers span time, matching buyers and sellers who arrive at different times in one market; arbitrageurs span place, matching those who arrive together in different markets [alpha-harris, same chap., sec. 17.7, p. 376]. Free entry narrows arbitrage spreads until they just cover the costs of the most efficient arbitrageurs. These are often traders with a natural interest in one leg, such as grain operators near a delivery point [alpha-harris, same chap., sec. 17.8, pp. 377–378].

Pitfalls. Full convergence is the assumption behind the $|A_t|$ in the condition above; with partial or slow convergence the gross gain is the expected narrowing net of extra carry (Interpretation; see arbitrage risks). The legacy notes listed four parallel types: pure, speculative, risk and statistical arbitrage. In Harris risk arbitrage is another name for speculative arbitrage, and statistical arbitrage one of its kinds. They also made cross-rate arbitrage a fourth pure mechanism beside shipping, delivery and conversion; Harris treats it as a cyclic conversion arbitrage. Even pure arbitrage can lose if convergence is slow or financing fails [alpha-harris, same chap., secs. 17.3.1 and 17.9, pp. 350 and 378].

See also. arbitrage risks; informed-trader types; outside spread (value traders' no-trade band, the analogue of arbitrage bounds); dealer inventory risk (dealers span time). The pathways `statistical-arbitrage`, `commodity-spatial-arbitrage` and `fx-cip-dollar-funding` develop particular arbitrages.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. arbitrage; arbitrageurs; hedge portfolio; arbitrage numerator; hedge ratio; basis; fair value of the basis; arbitrage spread; arbitrage bounds; carrying costs; pure arbitrage; speculative arbitrage; risk arbitrage; merger arbitrage; shipping arbitrage; delivery arbitrage; conversion arbitrage; cyclic arbitrage; pairs trading; statistical arbitrage; law of one price

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 232–235 and 347–379). Stability: draft. Currency: timeless (concepts); examples to 2000.

Result 8.29 Arbitrage risks: implementation, basis, model and carrying-cost risk, and staying power

Statement. Harris names four arbitrage risks [alpha-harris, chap. "Arbitrageurs", sec. 17.4, pp. 363–373]:

- *implementation risk*: transaction costs larger than expected, from adverse execution prices with market orders or failed executions with limit orders. It is worst when one leg has filled and another has not, when markets move fast, and when the arbitrage spread is small relative to short-term volatility;

- *basis risk*: the basis moves against the position, because specific factors diverge or common factors are mispriced. Total basis risk scales with the size of the position, and adverse moves force liquidation, at the worst time in a pure arbitrage, when the trader cannot finance the losses;

- *model risk*: misunderstanding the value relation, so that the trader takes the wrong side, carries unrecognised risks or uses wrong hedge ratios. The commonest error is treating a move of the basis towards a changed fair value as an opportunity;

- *carrying-cost risk*: higher financing, borrowing, storage or dividend costs than expected; unexpected price rises whose losses on short legs must be funded at once; slow convergence; and forced buy-ins of borrowed securities, which tend to come when the basis has moved against the arbitrageur.

Risky arbitrage needs compensation, so arbitrage bounds are wide when arbitrage is risky [alpha-harris, same sec., p. 364].

Conditions. Arbitrage opportunities have two causes [alpha-harris, same chap., sec. 17.5, pp. 373–375]. In the *slow price adjustment* scenario a common factor changes and some prices lag. Arbitrage then moves prices to their new values, and the arbitrageur should trade quickly, demanding liquidity. In the *uninformed liquidity demand* scenario values are unchanged and uninformed traders push prices apart in different markets. The arbitrageur then supplies liquidity, competing with value traders and dealers, and need not trade aggressively. Opportunities do not arise when an instrument-specific value changes: then the fair value of the basis has moved, and a pairs trade against it loses [alpha-harris, same chap., pp. 358–360 and 368].

Practice. Never leverage to the limit of capital: those who do have no staying power when the basis moves against them [alpha-harris, same chap., secs. 17.4.2 and 17.4.4.2, pp. 368 and 372]. LTCM, highly leveraged in mean-reverting positions, failed in 1998 although many positions later paid [alpha-harris, same chap., box, p. 369]. In illiquid markets, trade the illiquid leg first, or interleave small orders in both legs, so as never to be far from hedged [alpha-harris, same chap., pp. 366–367]. To judge whether a move in the spread is an opportunity, ask which leg moved and why. In Harris's GE against S&P 500 futures example, a widening caused by the index falling while GE lags is more likely an opportunity than one caused by GE rising, which may be firm-specific news [alpha-harris, same chap., box and Table 17-2, pp. 361–362].

Pitfalls. The legacy risk list also named *staying power* and *specific-factor* risk. In Harris staying power is the consequence of basis and carrying-cost risk under leverage, and specific factors are a source of basis risk and model error, not separate categories. The legacy principle *common versus specific factor test* said residualisation decides whether a spread is an opportunity or new information. The passages support the test; "residualisation" is the legacy term for removing common-factor exposure and is not Harris's.

See also. arbitrage; order anticipation (short squeezes hit arbitrageurs); bubbles and crashes (limits to arbitrage). The pathway `funding-crowding` develops forced liquidation and crowding.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. arbitrage risk; implementation risk; legging risk; basis risk; model risk; analytic risk; carrying-cost risk; staying power; leverage; forced buy-in; slow convergence; slow price adjustment; uninformed liquidity demand; common and specific factors; LTCM

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 356–375). Stability: draft. Currency: timeless (concepts); examples to 2000.

Definition 8.30 Order anticipation: front running, quote matching, sentiment-oriented technical trading, squeezes and stop gunning

Statement. *Order anticipators* profit from information about other traders' orders, not about fundamental values, by trading before them [alpha-harris, chap. "Order Anticipators", pp. 245 and 256–257]:

- *front runners* trade on orders others have decided to arrange. Ahead of aggressive traders they profit from the orders' price impact. Ahead of passive traders they are *quote matchers*, or penny jumpers, who step in front of standing limit orders and keep the limit order as an option to exit: long stock plus the implicit put of a large buy order is a call [alpha-harris, sec. 11.1, pp. 245–250];
- *sentiment-oriented technical traders* predict trades uninformed traders have not yet decided to make, from cash-flow timing, hedging needs, asset exchanges, gamblers' interests and speculators' mistakes [alpha-harris, sec. 11.2, pp. 251–254];
- *squeezers* corner one side of a market so that those who must close positions have to trade with them [alpha-harris, sec. 11.3, pp. 254–256].

Gunning the market, pushing prices to trigger clustered stop orders and trading with them, is a manipulation of stop orders that Harris treats alongside these strategies [alpha-harris, sec. 11.4, p. 256].

Conditions. Effects [alpha-harris, same chap., secs. 11.1.3–11.1.4, 11.2.1 and 11.3, pp. 250–256]. Front running accelerates the impact of the traders it front-runs. It makes prices more informative in the short run when it front-runs informed traders, but in the long run it lowers their profits and investment in information, and so informativeness. It makes prices less informative when it front-runs uninformed traders. It generally makes markets less liquid: front runners' profits are other traders' costs, they offer liquidity back at worse prices, and they drive away the orders they feed on. Front runners benefit large traders only in narrow conditions: better traders who do not compete with each other and do not squeeze. Sentiment-oriented technical trading makes prices less informative, especially in bubbles, and loses when value traders push prices back, so it pays best in hard-to-value instruments. Squeezes make prices less informative and greatly raise others' costs.

Practice. Time precedence with a meaningful minimum price increment makes quote matchers pay to step ahead, to the traders on the other side rather than the passive traders they front-run; a small tick weakens this protection. Passive traders defend themselves by hiding or splitting orders or switching to market orders, which raises their costs and reduces displayed size [alpha-harris, same chap., pp. 250 and 257]. Traders protect themselves from front runners by guarding order information, from sentiment traders by trading quickly,

and from squeezers by keeping several ways to close [alpha-harris, same chap., p. 257]. NYSE and AMEX specialists, as Harris describes them in 2002, had privileged access to the order flow and used it in the same ways [alpha-harris, chap. "Specialists", sec. 24.5, pp. 501–508]:

- speculating on short-term forecasts from the order flow;
- quote matching against a heavy side of their book;
- *cream skimming*, stepping in front of the book to fill only the more profitable, usually small retail, marketable orders;
- anticipating stop orders in their books;
- *stopping stock*, guaranteeing an execution price and then deciding when to fill, a look-back timing option that harms the limit-order traders who would have filled the order.

Public order precedence, the public liquidity preservation principle and audit trails constrain these privileges [alpha-harris, same chap., sec. 24.2.2, pp. 498–499].

Pitfalls. Not all front running is illegal: it is illegal when the information is obtained improperly, through a confidential brokerage relationship or eavesdropping, but an observant floor trader may legally infer a broker's order [alpha-harris, chap. "Order Anticipators", sec. 11.1.1, pp. 246–247]. Squeezers and gunners defend themselves as speculators, which makes the laws against them hard to enforce [alpha-harris, same chap., pp. 256–257]. The legacy taxonomy listed quote matching and stop gunning as categories parallel to front running. In Harris quote matching is front running of passive traders, and gunning a manipulation of stop orders. The specialist strategies describe a floor-based market structure of 2002 and are dated.

See also. bluffing and manipulation; informed-trader types (information-oriented against sentiment-oriented technical trading); arbitrage risks (short squeezes); bubbles and crashes (order anticipators accelerate the rise); spread determinants (tick size); HFT strategies (order anticipation as one of the SEC's 2010 strategy types, and pinging as a tool). The pathway `futures-delivery-manipulation` develops squeezes at delivery.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. order anticipators; front running; quote matching; penny jumping; sentiment-oriented technical traders; squeezes; corners; short squeeze; gunning the market; stop orders; time precedence; tick size; specialists; cream skimming; stopping stock; look-back option; parasitic traders

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 245–258 and 498–508). Stability: draft. Currency: concepts timeless; the specialist privileges and legal remarks as of 2002.

Definition 8.31 Bluffing and market manipulation: rumourmongers, price manipulators and the asymmetric-impact condition

Statement. *Bluffers* try to fool other traders into trading unwisely and then profit from them. Harris names two techniques [alpha-harris, chap. "Bluffers and Market Manipulation", p. 259 and Table 12-3, p. 274]:

- *rumourmongers* spread false information, or true information presented so that it will be misread;
- *price manipulators* arrange trades at prices, volumes and times meant to change others' opinions of value. The trades may be genuine arm's-length trades or *wash trades* with confederates that create artificial activity.

Market manipulation occurs when bluffers or their victims change prices from what they would otherwise be. Bluffing is profitable when the price impact of the bluffer's purchases differs from that of its sales: if buys move prices more, buy first and sell later, and conversely [alpha-harris, same chap., sec. 12.3, pp. 270–273]. In Harris's example, liquidity suppliers raise prices 10 cents per 100 contracts for 1,000-contract orders but only 5 cents for 500-contract orders. Buying 4,000 contracts in 1,000s and selling them in 500s then averages 12.50 bought against 12.875 sold, a profit of $1.5 million on a 1,000-unit contract, and leaves the price at 12 rather than 10 [alpha-harris, same sec., p. 270, page image]. With the same impact per unit for all trades, the same round trip loses $3 million (p. 271; both figures rechecked, Interpretation).

Conditions. Bluffers imitate informed speculators: both promote their positions and trade to show confidence. The difference is that speculators trade on well-founded opinions about value, whereas bluffers manufacture the signals; bluffs make prices less informative and are more likely followed by reversals [alpha-harris, same chap., sec. 12.2.1, pp. 265–267]. Value traders foil bluffs by trading against prices pushed from value, unless bluffers have more capital and force them out. Bluffs therefore succeed most in securities value traders do not follow or cannot trade: illiquid, hard to value or hard to borrow. Long-side bluffs are probably commoner than short-side ones [alpha-harris, same chap., sec. 12.2.2, pp. 268–269].

Practice. Liquidity suppliers who set prices from net order flow must make buys and sells of every size, speed and timing move prices equally per unit traded; otherwise bluffers can exploit them [alpha-harris, same chap., sec. 12.3, pp. 271–273]. Momentum traders and liquidity suppliers, who react to the trades they see, are the most vulnerable [alpha-harris, same chap., sec. 12.4, p. 274]. Transaction costs limit bluffing, which trades intensively [alpha-harris, same chap., p. 273]. A legacy principle, *impact symmetry disciplines bluffing*, restated this rule. It added that impact should depend on "net pressure and risk", of which Harris names only net quantity, and that a zero-net sequence then cannot create a higher average sale price than purchase price. That conclusion holds in Harris's example, where each trade's impact is immediate and permanent. In impact models with decay and changing liquidity, equal treatment of buys and sells does not exclude manipulation (see absence of price manipulation).

Pitfalls. A price reversal does not prove manipulation, because good speculators also misjudge values. Conviction usually requires evidence that the bluffer spread information known to be false or arranged wash trades, or testimony on intent; careful bluffers therefore pass as speculators [alpha-harris, same chap., sec. 12.2.3, pp. 269–270]. Harris's legality column (Table 12-3): false rumours and wash trades are illegal in most jurisdictions and easy to prosecute if caught; misleadingly framed true information and genuine trades meant to mislead are of questionable legality and hard to prove. The legacy taxonomy made wash traders a third type beside rumourmongers and price manipulators; in Harris wash trading is a technique of price manipulators.

See also. order anticipation (gunning stop orders); insider-trading detection (the same proof problem); informed-trader types; Kyle single-auction model (Interpretation: impact linear and symmetric in signed flow, as in Harris's no-profit case); absence of price manipulation (manipulation through one's own impact in a model, not through others' inferences). The pathways `crypto-exchange-market-quality` and `prediction-integrity` develop wash trading and manipulation in those markets.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. bluffers; market manipulation; rumourmongers; price manipulators; painting the tape; wash trades; momentum traders; asymmetric price impact; value traders call a bluff; manipulation enforcement

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 259–275, and the page image of p. 270; pp. 271–273 rechecked in slice 7). Stability: draft. Currency: timeless (concepts); legal remarks as of 2002.

8.7 Information and price impact

Model 8.32 Kyle single-auction model: informed trading and linear price impact

Statement. One risk-neutral insider who knows the asset value $v \sim N(\mu, \sigma_v^2)$, price-insensitive noise traders with net order $u \sim N(0, \sigma_u^2)$ independent of v, and competitive risk-neutral market makers who see only the net order flow $y = x + u$ and set $S(y) = \mathbb{E}[v \mid y]$ [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 31–32] have a linear equilibrium [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 93, eqs. (4.3)–(4.5)]:

$$x(v) = \beta(v - \mu), \qquad S(y) = \mu + \lambda y, \qquad \beta = \frac{\sigma_u}{\sigma_v}, \qquad \lambda = \frac{\sigma_v}{2\sigma_u}.$$

The insider's ex ante expected profit is $\frac{1}{2}\sigma_v\sigma_u$ [catalogue-0496, same chap., p. 97, eq. (4.17)], and the posterior variance of v after the auction is exactly half the prior, whatever the order flow: price reveals half of the insider's information [catalogue-0496, same chap., p. 98, eq. (4.20)].

Notation. Source notation: `catalogue-0496` writes p_0 for μ, Σ_0 for σ_v^2, σ_μ for σ_u and P for the price; `crypto-cartea` writes σ for σ_v. Here λ is a price-impact slope, in price per share.

Conditions. A single auction; the insider does not observe u before trading; normality of v and u; market makers earn zero expected profit given the order flow, so price is semi-strong efficient [catalogue-0496, same chap., pp. 91–92]; [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 32]. There is no other linear equilibrium, but nonlinear equilibria are not excluded [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 97]. Idea of the proof: given a linear price rule the insider's expected profit $x(v - \mu) - \lambda x^2$ is quadratic, so $\beta = 1/(2\lambda)$; given a linear strategy, v and y are jointly normal and the projection $\mathbb{E}[v \mid y]$ is linear in y; the two conditions fix β and λ [catalogue-0496, same chap., pp. 93–96].

Practice. Doubling the noise-trader standard deviation doubles the insider's order and expected profit and leaves the ex ante price unchanged [catalogue-0496, same chap., p. 97]. Market makers are risk-neutral, so any premium compensates their informational disadvantage and is borne by the price-insensitive liquidity traders [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 31]. Interpretation (own derivation): the insider trades only part of the mispricing, because a larger order reveals more through y; λ rises with σ_v and falls with σ_u.

Pitfalls. A linear price rule implies a negative price for a large enough sell imbalance, so it is at best an approximation of actual price behaviour [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 96]. The joint-normality assumption drives the predictions on trade size and volume [catalogue-0496, same chap., p. 98]. The textbook account in `crypto-cartea` prints the equilibrium price as $\mu + 2(x+u)\sigma_u/\sigma$, so $\lambda = 2\sigma_u/\sigma$ [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 33]; this is the reciprocal of the value in `catalogue-0496`, eq. (4.5), and of the value implied by its own preceding step $\beta = (2\lambda)^{-1}$. The authors' errata correct the price to $\mu + (x+u)\sigma/(2\sigma_u)$, so $\lambda = \sigma/(2\sigma_u)$, in agreement with `catalogue-0496` [crypto-cartea-errata, entries for p. 33]. Do not copy λ from the printed page.

See also. Grossman–Miller model (contrast: a premium for risk, not for information); Grossman–Miller liquidity measures (contrast: an inventory price-impact slope).

Further reading. [s6-kyle], the original model and its dynamic version (not yet read for this entry); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Kyle model; Kyle's lambda; price impact; informed trading; insider; noise traders; adverse selection; semi-strong efficiency; information revelation; toxic order flow

Status. Verification: partly checked 2026-10-14 (the equilibrium, profit and posterior variance against `catalogue-0496`, pp. 91–98, from page images of a scanned 1998 printing; the setup against `crypto-cartea`, pp. 31–33, whose printed λ conflicts as noted under *Pitfalls*; Kyle's paper not consulted; the p. 33 misprint confirmed against the authors' errata page 2026-10-15). Stability: draft. Currency: timeless (theory).

8.8 Adverse selection and the bid–ask spread

Model 8.33 Glosten–Milgrom sequential-trade model: spreads from adverse selection

Statement. In a pure dealership market, investors arrive one at a time, randomly and anonymously, see the specialist's bid b_t and ask a_t, and buy one unit at the ask, sell one unit at the bid, or leave; some arrivals are informed about the asset's future value V, the rest trade for liquidity reasons [glosten-milgrom-1983, sec. II, pp. 3–4]. An arrival with information $\mathcal{F}_t$ has reservation price $Z_t = \rho_t \mathbb{E}[V \mid \mathcal{F}_t]$ and buys if $Z_t > a_t$, sells if $Z_t < b_t$; competitive, risk-neutral specialists with information $\mathcal{S}_t$ earn zero expected profit on each purchase and each sale, so the quotes solve [glosten-milgrom-1983, p. 6]

$$a_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t > a_t], \qquad b_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t < b_t].$$

Then $a_t \geq \mathbb{E}_t[V] \geq b_t$, with strict inequalities whenever informed buying above and informed selling below $\mathbb{E}_t[V]$ have positive probability: adverse selection alone produces a spread, with no inventory cost, risk aversion or monopoly power [glosten-milgrom-1983, Prop. 1, pp. 7–9]. Each transaction price is the specialist's updated expectation of V, so transaction prices form a martingale with respect to the specialist's and the public information [glosten-milgrom-1983, Prop. 2, p. 10].

Notation. $V > 0$ is the asset's value, realised at a future date, with finite variance; ρ_t is the arriving investor's liquidity (time-preference) parameter, with $\rho = 1$ for the specialist; $\mathcal{F}_t$ and $\mathcal{S}_t$ are the information of the arrival and of the specialist; $\mathbb{E}_t$ is expectation given the information both share [glosten-milgrom-1983, pp. 4–7]. The paper writes A_t, B_t for a_t, b_t, F_t, S_t for $\mathcal{F}_t, \mathcal{S}_t$.

Conditions. Every participant is risk-neutral with utility $\rho x V + c$ over shares x and consumption c; ρ is independent across traders and carries no information about V [glosten-milgrom-1983, pp. 4–5]. Uninformed participation is exogenous; trades are single units; short positions in cash or stock cost nothing; zero profit on each side is motivated by Bertrand competition between equally informed specialists, and existence of the quotes is not shown in general [glosten-milgrom-1983, pp. 4–6]. If the specialist sees only public information, the ask is the smallest a with $a \geq \mathbb{E}[V \mid \mathcal{H}_t, Z_t > a]$ and the bid the largest b with $b \leq \mathbb{E}[V \mid \mathcal{H}_t, Z_t < b]$ [glosten-milgrom-1983, p. 7]. Idea of the proof of Proposition 1: conditional on a buy, the ask equals the expectation of V given the buy; since buying selects arrivals whose own expectation exceeds a threshold and ρ is uninformative, that conditional expectation is at least $\mathbb{E}_t[V]$ [glosten-milgrom-1983, p. 8].

Practice. The spread is larger, for given information, when insiders' information is finer, when the ratio of informed to uninformed arrivals rises, or when the dispersion of the uninformed liquidity parameter around its median falls [glosten-milgrom-1983, Prop. 5, pp. 15–16]. If buy and sell probabilities stay above $\theta_a, \theta_b > 0$, prices converge and the root-mean-square spread over the first k trades is at most $\sqrt{\mathrm{Var}(V)/(\theta_a \theta_b k)}$ [glosten-milgrom-1983, Prop. 3, p. 11]; with Poisson arrivals the average spread is then of the order of one over the square root of volume, because trades reveal information [glosten-milgrom-1983, pp. 18–19]. The specialist and the arrivals come to agree on the expected value of V [glosten-milgrom-1983, Prop. 4, p. 13]. The market maker need not know which traders are informed: since well-informed traders rarely reveal themselves, dealers treat every order as possibly informed and revise prices with the side of each trade [alpha-harris, chap. "Dealers", pp. 288–289]. The two-point version with closed-form half-spreads is a separate result.

Pitfalls. If the liquidity motive is weak relative to the insiders' advantage, the zero-profit quotes may

preclude trade, and the market shuts down until enough public information neutralises the insiders' advantage [glosten-milgrom-1983, p. 9]; if the bounded-probability condition fails, the spread need not shrink: the market may close, or every trade may occur at the bid [glosten-milgrom-1983, p. 12]. Proposition 5 compares spreads for given information; it does not say that stocks with more insiders have wider spreads on average, since more insiders also reveal information faster [glosten-milgrom-1983, p. 17]. The authors list unit lots, the absence of risk aversion and inventory costs, and a fixed underlying value as limitations [glosten-milgrom-1983, sec. IV, pp. 24–25]. The locators are to the 1983 discussion paper, not the 1985 journal article that the catalogue names.

See also. Glosten–Milgrom half-spreads (special case: two values, closed form); market collapse (special case: price-sensitive liquidity traders); adverse-selection spread component (special case: a textbook example); Kyle single-auction model (contrast: batch order flow and a strategic insider); Grossman–Miller model (contrast: a spread for inventory risk, without information).

Further reading. [s6-glosten], the published version (not consulted for this entry); the legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Glosten–Milgrom; sequential trade model; adverse selection; bid–ask spread; informed trading; competitive specialist; zero-profit quotes; martingale prices; semi-strong efficiency; market breakdown; toxic order flow

Status. Verification: partly checked 2026-10-15 (every statement against the scanned 1983 discussion paper, pp. 1–25, read from page images with OCR for running text; the 1985 journal version not consulted; the Harris passage from the text layer of a local copy). Stability: draft. Currency: timeless (theory).

Result 8.34 Glosten–Milgrom half-spreads with two values

Statement. Let the asset's value be V_H with probability p and $V_L < V_H$ otherwise, let a fraction α of traders be informed, and let competitive risk-neutral market makers quote $a = \mu + \Delta_a$ and $b = \mu - \Delta_b$ around $\mu = \mathbb{E}[V \mid \mathcal{F}]$. Zero expected profit on each side gives the half-spreads [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35]

$$\Delta_a = \frac{\alpha p}{\alpha p + (1-\alpha)/2}\,(V_H - \mu) = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{1-p}}.$$

With trading over time, $p_t = \mathbb{P}(V = V_H \mid \mathcal{F}_t)$ and $\mu_t = \mathbb{E}[V \mid \mathcal{F}_t]$ are updated by Bayes' rule from the order flow, the same formulas hold with p_t, μ_t, and each execution price equals the expected value given the history and the side of the trade, so the price at execution times is a martingale [crypto-cartea, same chap., p. 36].

Notation. Local override for this entry and market collapse: p is the probability of the high value, not a price; α is the fraction of informed traders; Δ_a, Δ_b are the ask and bid half-spreads; $\mathcal{F}$ is public information, including the order flow. Source notation: `crypto-cartea` writes v for V.

Conditions. A simplified, essentially static version of the model: one-unit orders; informed traders know V and buy one unit if $a < V_H$ and $V = V_H$, sell one unit if $b > V_L$ and $V = V_L$; liquidity traders are price-insensitive and buy or sell with probability $\frac{1}{2}$ each, whatever the value; market makers are risk-neutral, carry inventory at no cost and compete to zero expected profit, using only public information [crypto-cartea, same chap., pp. 34–35]. Idea of the proof: a buy comes from a liquidity trader with weight $(1-\alpha)/2$, who pays Δ_a, and from an informed trader with weight αp, who costs $V_H - \mu - \Delta_a$; setting the weighted profit to zero gives Δ_a, and the bid is symmetric [crypto-cartea, same chap., p. 35].

Practice. Reading asymmetric information as toxicity, α is its prevalence, and p with $V_H - \mu$, or $1-p$ with $\mu - V_L$, its magnitude on the buy or sell side; market makers widen each half-spread as the prevalence and magnitude of toxicity on that side rise [crypto-cartea, same chap., pp. 35–36]. Interpretation (own derivation): as $\alpha \to 0$ both half-spreads vanish, and as $\alpha \to 1$ the quotes reach V_H and V_L, where only informed traders would trade.

Pitfalls. This is a textbook simplification: the paper allows a general value distribution and price-sensitive uninformed traders [glosten-milgrom-1983, sec. II, pp. 3–6]. With price-insensitive liquidity traders a monopolist would quote $a = V_H$, $b = V_L$; the formulas rely on competition between market makers [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35].

See also. Glosten–Milgrom model (generalisation); market collapse (extension: price-sensitive liquidity traders); adverse-selection spread component (contrast: a parametrisation by the dealer's pricing error).

Index. Glosten–Milgrom half-spread; ask half-spread; bid half-spread; toxicity; prevalence of informed trading; Bayesian quote updating; zero-profit market making

Status. Verification: partly checked 2026-10-15 (the setup, both formulas, the dynamic version and the interpretation against `crypto-cartea`, pp. 34–36, read from page images of a scanned copy; the zero-profit algebra re-derived). Stability: draft. Currency: timeless (theory).

Result 8.35 Glosten–Milgrom with price-sensitive liquidity traders: market collapse

Statement. If liquidity trader i trades only when the half-spread is at most his urgency c_i, with c_i distributed with cumulative distribution function F across liquidity traders, the zero-profit half-spreads are

defined implicitly by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37]; [crypto-cartea-errata, entry for p. 37]

$$\Delta_a = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_a))/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_b))/2}{1-p}}.$$

Widening a half-spread drives out liquidity traders; if urgency is small, the only solutions may be the extreme ones, $\Delta_a = V_H - \mu$ and $\Delta_b = \mu - V_L$: the market collapses, only informed traders, who are then indifferent, might trade, and any trade reveals the value [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37].

Notation. As in the two-value half-spreads; c_i is the cash-equivalent gain to liquidity trader i from completing the desired trade.

Conditions. The two-value model of the half-spreads entry, with only the fraction $1 - F(\Delta)$ of would-be liquidity buyers or sellers trading at half-spread Δ [crypto-cartea, same chap., p. 37]. "Small" urgency means that no liquidity trader's urgency exceeds the expected value of the asset: $F(\mu - \epsilon) = 1$ for some $\epsilon > 0$ [crypto-cartea, same chap., p. 37, n. 5].

Practice. The paper gives the same mechanism: when the proportion of insiders is high or the liquidity motive of the uninformed weak, the zero-profit quotes can preclude trade, a lemons problem as in Akerlof, and the market reopens when public information neutralises the insiders' advantage [glosten-milgrom-1983, pp. 9 and 17].

Pitfalls. The book prints $F(\Delta_a)$ in the bid formula; the authors' errata correct it to $F(\Delta_b)$, as given above [crypto-cartea-errata, entry for p. 37]. Do not copy the bid formula from the printed page.

See also. Glosten–Milgrom half-spreads (the price-insensitive case, $F \equiv 0$); Glosten–Milgrom model (generalisation, with market shutdown).

Index. market collapse; market breakdown; market shutdown; lemons problem; price-sensitive liquidity traders; urgency; liquidity dry-up; adverse selection

Status. Verification: partly checked 2026-10-15 (the formulas, the collapse condition and its footnote against `crypto-cartea`, p. 37, from page images of a scanned copy; the correction against the authors' errata page, read from the equation source of the page; the shutdown passages against the 1983 discussion paper). Stability: draft. Currency: timeless (theory).

Result 8.36 Spread components: the adverse-selection component, its two views, and the total spread

Statement. The bid–ask spread is split, for analysis, into a *transaction-cost* (transitory) component C, which pays the dealer's normal costs of doing business, and an *adverse-selection* (permanent) component A, which recovers from uninformed traders what the dealer loses to informed ones [alpha-harris, chap. "Bid/Ask Spreads", pp. 298–299]. A dealer first estimates the value V_0 from all current information, then the values V^B and V^S she would estimate if the next trader were a buyer or a seller; $A = V^B - V^S$, and she quotes [alpha-harris, same chap., Figure 14-1, p. 301]

$$a = V^B + \tfrac{1}{2}C, \qquad b = V^S - \tfrac{1}{2}C, \qquad a - b = A + C.$$

If the dealer learns nothing more from the next trade than its side, the conditional estimate becomes her new value: after a buy both quotes rise by $A/2$ [alpha-harris, same chap., Figure 14-2, pp. 301–302]. In a simple example based on the Glosten–Milgrom model, a dealer believes that the security is worth V unconditionally, that the next trader is equally likely to buy or sell and is informed with probability α, and that it is worth $V + e$ if an informed trader buys and $V - e$ if an informed trader sells [alpha-harris, chap. "Bid/Ask Spreads", p. 320]. Information view: conditional on a buy the dealer's value estimate is $(1 - \alpha)V + \alpha(V + e) = V + \alpha e$, conditional on a sell $V - \alpha e$, so the adverse-selection component is $2\alpha e$ [alpha-harris, same chap., p. 320]. Accounting view: expected profit on the next trade is $\frac{1}{2}(a - b) - \alpha e$, so breaking even requires $a - b = 2\alpha e$ [alpha-harris, same chap., pp. 320–321]. Both views give the same component [alpha-harris, same chap., p. 300]; in the example $A = 2\alpha e$ and $C = 0$.

Notation. A and C are amounts of the full spread, in price units; a, b are the ask and bid. Source notation: Harris names the components in words and writes V_0^B, V_0^S for V^B, V^S (Figure 14-1); in his appendix he writes P for α, E for e (the dealer's pricing error when the trader is informed), and A, B for the ask and bid a, b, so his A there is not the component.

Conditions. Normal costs of doing business are ignored; the example assumes that dealers know the value when informed traders trade, which the result does not need [alpha-harris, same chap., pp. 300 and 320]. The two views agree when the loss from trading with an informed trader equals the pricing error, as when the dealer cannot restore inventory before prices adjust; if the dealer can rebalance but does not recognise the informed trade, informed traders keep trading with him until the price adjusts, and his cumulative loss is still the original pricing error [alpha-harris, same chap., p. 302]. Low barriers to entry and competition from

public limit orders, which undercut wide dealer quotes, limit monopoly spreads [alpha-harris, same chap., p. 298]; in a competitive dealer market free entry and exit drive dealers to normal profits, so the total spread is this component plus a transaction-cost component that covers operating costs, financing, any monopoly profit and any premium for inventory risk [alpha-harris, same chap., pp. 298–299]. V^B and V^S lie equally far from V_0, at $V_0 \pm A/2$, when informed buyers and sellers are equally likely and their expected pricing errors equal [alpha-harris, same chap., p. 301]. A three-way split (order processing, inventory, adverse selection) separates the inventory-risk premium out of C; Hasbrouck likewise notes that a spread impounds noninformational costs and inventory effects as well as asymmetric information [s6-hasbrouck, sec. 5.5, p. 53]. In the structural sequential-trade version with a cost c per trade, $a = \mathbb{E}[V \mid \text{buy}] + c$ and $b = \mathbb{E}[V \mid \text{sell}] - c$; the bid and ask sequences are martingales but trade prices are not [s6-hasbrouck, sec. 5.4.2, p. 49]. There c is a per-side amount, so Harris's C corresponds to $2c$ (Interpretation).

Practice. The component rises with the probability that the next trader is informed, with the materiality of the information and with trade size, because dealers believe large traders are often well informed [alpha-harris, chap. "Dealers", pp. 289–291 and 295]. Price changes from the adverse-selection component are permanent, while the transaction-cost component causes bid–ask bounce [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300]. Volatile instruments have wider spreads, both through inventory risk for risk-averse dealers and as a proxy for asymmetric information [alpha-harris, same chap., pp. 312–313]; the other determinants are in spread determinants. Uninformed traders lose to informed ones whichever order they use: their limit orders fill quickly when informed traders want the other side and fail to fill when informed traders compete on the same side, and their market orders pay spreads widened by adverse selection, so that the component is in effect a fee dealers charge market-order traders for bearing adverse-selection risk [alpha-harris, same chap., sec. 14.3, pp. 303–304, Table 14-1].

Pitfalls. The legacy notes said that A is twice the dealer's mark-to-market loss against a correctly signed informed trade. In the example that holds only if $\alpha = 1$: against an informed buyer the dealer's loss relative to V is the whole pricing error e, while $A/2 = \alpha e$ is the expected revision given a buy, that is, the expected loss per buy, informed and uninformed buyers together, of a dealer whose ask is at V (Interpretation: own derivation from the example, pp. 300–302 and 320). The two views equate a probability-weighted loss with a conditional revision, not A with the loss on one informed trade. This is a textbook example, not the Glosten–Milgrom paper, whose quotes are expectations conditional on a trade at the quote (the model); the legacy note that attributed $a = V + \alpha e$, $b = V - \alpha e$ to Glosten and Milgrom followed this example. The accounting view fixes only the width $a - b$; the level $V \pm \alpha e$ comes from the information view (Interpretation). Harris prints the dealer's profit against an informed buyer as $A - (V - E)$; his stated result requires $A - (V + E)$ [alpha-harris, chap. "Bid/Ask Spreads", p. 321].

See also. Glosten–Milgrom model (generalisation); Glosten–Milgrom half-spreads (contrast: a parametrisation by the value distribution); dealer quote adjustment (practice: how dealers set and revise quotes); spread measures (effective and realised spreads, which measure the components after the fact); component identification (estimating A and C from prices and trade signs).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. adverse selection spread component; permanent spread component; transaction cost spread component; transitory spread component; order-processing cost; total spread; dealer quote decomposition; Glosten–Milgrom theorem (Harris); information view; accounting view; spread decomposition; bid–ask bounce; uninformed traders lose

Status. Verification: partly checked 2026-10-15 (the example, both derivations, the conditions and the spread determinants against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 289–304, 312–313 and 320–321; p. 321 and Figure 14-1, p. 301, also from page images; the structural version against the text layer of a local copy of `s6-hasbrouck`, 2007 Oxford printing, pp. 49 and 53). Stability: draft. Currency: timeless (theory).

8.9 Dealer inventory and quote adjustment

Definition 8.37 Dealer inventory imbalance and the two inventory risks

Statement. A dealer's *target inventory* is the position it wants to hold, and its *inventory imbalance* is actual inventory minus target [alpha-harris, chap. "Dealers", p. 283]. Dealers face two inventory risks, distinguished by whether future price changes are correlated with the imbalance [alpha-harris, same chap., p. 285]:

- *diversifiable inventory risk*: price changes independent of the imbalance; gains and losses are equally likely, they do not cause losses in the long run, and dealing in many instruments reduces their effect on total profit [alpha-harris, same chap., pp. 285–286];

- *adverse-selection risk*: price changes inversely correlated with the imbalance, because informed traders take the side their information favours, make order flow one-sided and push the dealer's inventory away from target before prices move [alpha-harris, same chap., p. 286].

Conditions. If long and short positions are equally costly, a dealer that does not also speculate, hedge or invest has a target of zero; where shorting costs more, targets are positive; speculating dealers set targets by their views [alpha-harris, same chap., pp. 283–284]. Large imbalances are costly to finance and expose the dealer to losses if prices move, and an imbalance beyond the dealer's capital forces liquidation [alpha-harris, same chap., p. 284]; a dealer can also hedge an imbalance with a correlated instrument [alpha-harris, same chap., p. 285, Table 13-2]. Interpretation: the costs of carrying inventory thus group into price risk, the financing of long and the borrowing of short positions, the use of capital, and the basis risk and costs of hedges; recall risk and risk-limit use, which the legacy notes also listed, are not in the passages checked.

Practice. Dealers avoid both risks by keeping inventory under control, so most attend to *market values*, the prices that produce two-sided order flow, more than to fundamental values, which they discover as a by-product [alpha-harris, same chap., pp. 286–287]. Adverse selection is still the most important determinant of dealer profitability [alpha-harris, same chap., p. 286]. Inventory risk does not need informed traders: diversifiable inventory risk is present when no one is better informed, risk-averse dealers require a premium for it, mainly in highly volatile instruments, and that premium is part of the transaction-cost spread component [alpha-harris, chap. "Bid/Ask Spreads", pp. 299 and 313]; see spread components.

Pitfalls. "Diversifiable" here means unrelated to the dealer's imbalance; within one instrument it is still a risk that a risk-averse dealer is paid to bear, as in Grossman–Miller (Interpretation). A falling inventory after buys does not by itself say which risk is present; separating them needs the subsequent price path (Interpretation).

See also. dealer quote adjustment (how dealers control the imbalance); Grossman–Miller model (a model of the first risk); Glosten–Milgrom model (a model of the second); inventory-based quotes (models of quotes as functions of the imbalance).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory imbalance; target inventory; inventory risk; diversifiable inventory risk; adverse selection risk; inventory spread component; inventory costs; financing cost; market value; fundamental value; dealer risk management

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 283–287, 299 and 313). Stability: draft. Currency: timeless (concepts).

Method 8.38 Dealer quote adjustment for inventory and suspected informed trading

Statement. Dealers control inventory and adverse selection with four levers: the level of both quotes, their skew, the width of the spread and the sizes shown.

1. *Inventory.* To reduce inventory, lower bid and ask, decrease the bid size and raise the ask size; to increase it, do the opposite; an impatient dealer can take another trader's offer or hit another trader's bid, or hedge with a correlated instrument [alpha-harris, chap. "Dealers", pp. 284–285, Table 13-2].

2. *After a suspected informed trade.* After selling to a suspected well-informed buyer, raise the ask and lower its size to discourage further informed buying, raise the bid and its size to rebuild inventory, or buy at another trader's ask or buy a correlated instrument; after buying from a suspected informed seller, do the reverse [alpha-harris, same chap., pp. 287–288, Table 13-3].

3. *Before the next trade.* Set the ask at the value expected if the next trader buys and the bid at the value expected if the next trader sells, adjusting more when informed trading is more likely and the information more material [alpha-harris, same chap., pp. 288–290]; see the adverse-selection component.

4. *Size.* Avoid displaying large size, so that traders must ask for it and reveal themselves [alpha-harris, same chap., p. 291].

Conditions. Dealers rarely know who is informed: informed traders hide behind brokers and claim to be uninformed, and rules of thumb (large, active, impatient traders are often informed; small retail traders rarely) are useful but unreliable, so every order is treated as possibly informed [alpha-harris, same chap., pp. 288–289]. Spreads widen with asymmetric information, volatility and weak utilitarian trading interest [alpha-harris, chap. "Bid/Ask Spreads", pp. 312–313] (spread determinants).

Practice. Searching for prices that bring two-sided order flow is price discovery: it finds market values [alpha-harris, chap. "Dealers", p. 284]. Demanding liquidity to rebalance gives a negative realised spread [alpha-harris, same chap., p. 284]. After a buy, raising quotes is right if the buyer was informed and a mistake if not, when value traders sell to the dealer and realised spreads fall; doing nothing is the reverse; buying at another trader's ask may limit losses against an informed buyer but is a big mistake against an uninformed one

[alpha-harris, same chap., pp. 291–292, Table 13-4]. Poorly informed dealers more often mistake uninformed for informed clients, so they turn inventory quickly and earn smaller realised spreads; well-informed dealers can hold larger positions longer and earn larger realised spreads [alpha-harris, same chap., pp. 292–294]. Value traders trade when prices diverge from fundamental value, often after such mistakes or after risk-averse dealers obtain large concessions for taking big positions, and in doing so restore dealers' inventories [alpha-harris, same chap., pp. 291 and 294]. Some dealers avoid informed traders by trading only with retail or known clients, refusing anonymous size, or slicing large orders over time, which lets them offer more liquidity to the flow they accept [alpha-harris, same chap., pp. 290–291].

Pitfalls. Informed traders can bluff, sending small orders on one side to move a dealer before trading on the other; against traders they cannot identify, dealers should adjust prices at the same rate per quantity whatever the side, speed, size or sequence of orders [alpha-harris, same chap., pp. 292 and 294]. Interpretation: refusing flow protects the dealer but reduces the immediacy available to others; one-sided flow turns dealing into warehousing risk. These are heuristics without an objective function; Avellaneda–Stoikov derives the inventory skew from one.

See also. dealer inventory risk (prerequisite); adverse-selection spread component (the spread set before the next trade); Avellaneda–Stoikov quoting (a model of the inventory skew); Glosten–Milgrom model (a model of the revision after each trade); inventory-based quotes (textbook models of quotes as functions of inventory); spread determinants (why spreads differ across instruments).

Further reading. The legacy text it replaces, tables included, is kept verbatim in `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory control; quote skew; quote shading; quote size; dealer tactics; price discovery; market value; realised spread; informed-trader identification; refusing flow; value traders; bluffing; toxic flow

Status. Verification: partly checked 2026-10-15 (every step and table summary against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 283–294 and 312–313). Stability: draft. Currency: timeless (practice description from 2002; venue examples dated).

Model 8.39 Inventory-based dealer quotes: reservation prices linear in inventory (Stoll; Ho–Stoll as reported by O'Hara)

Statement. In Hasbrouck's version of Stoll's one-period model, a dealer with CARA utility $U(W) = -e^{-\gamma W}$ holds n shares of a security with payoff $X \sim N(\mu_X, \sigma_X^2)$, borrowing and lending at a zero rate, and posts a bid B for one share that leaves him indifferent to its execution; equating certainty equivalents gives [s6-hasbrouck, sec. 11.2, eq. (11.2), p. 110]

$$B = \mu_X - (2n+1)\,\frac{\gamma\sigma_X^2}{2}.$$

If the dealer could first trade at a notional price P, his subjective valuation, his optimal holding would be $n^* = (\mu_X - P)/(\gamma\sigma_X^2)$, and at that optimum $B = P - \gamma\sigma_X^2/2$ [s6-hasbrouck, same sec., pp. 110–111]. The same indifference for a one-share sale gives $A = \mu_X - (2n-1)\gamma\sigma_X^2/2$ (Interpretation: own derivation; the source gives only the bid), so the quotes are

$$B = r - \tfrac{1}{2}\gamma\sigma_X^2, \qquad A = r + \tfrac{1}{2}\gamma\sigma_X^2, \qquad r = \mu_X - \gamma\sigma_X^2\, n = P - \gamma\sigma_X^2\,(n - n^*)\,:$$

the midpoint of the quotes falls linearly as inventory rises above target, and the spread $\gamma\sigma_X^2$ does not depend on inventory.

Notation. n is the dealer's inventory and n^* his target; r is the reservation (indifference) midpoint; A, B are the dealer's ask and bid. The source writes α for the risk aversion γ, and μ_x, σ_x^2 in eq. (11.2). The legacy notes wrote $r_t = m_t - \kappa(q_t - q^\star)$ with spread s_t; here $\kappa = s = \gamma\sigma_X^2$, $q^\star = n^*$ and $m = P$.

Conditions. One period, a normally distributed payoff, exponential utility, one-share trades, and no information asymmetry: the quotes compensate only for being pulled off the optimal portfolio [s6-hasbrouck, same sec., p. 110]. Idea of the proof: with CARA utility and normal wealth the certainty equivalent is $\mu_W - \gamma\sigma_W^2/2$; set it equal with and without the trade, and solve for the price [s6-hasbrouck, same sec., p. 110]. With two correlated securities the bid for the first is $B_1 = \mu_1 - \frac{1}{2}\gamma\sigma_1\big[(1+2n_1)\sigma_1 + 2\rho n_2\sigma_2\big]$, lower when the securities are positively correlated, but at the optimum it is again $P_1 - \gamma\sigma_1^2/2$ and the correlation drops out, a consequence of CARA utility [s6-hasbrouck, same sec., eq. (11.3), p. 111].

Practice. O'Hara's account of the multiperiod model of Ho and Stoll (1981): a monopolistic, risk-averse dealer facing Poisson buy and sell arrivals whose rates are linear in his ask and bid adjustments around a fixed true price p, quoting $p + a$ and $p - b$ to maximise expected utility of terminal wealth over a finite horizon; the problem has no known closed form, and Ho and Stoll approximate it with a first-order expansion and quadratic utility [catalogue-0496, sec. 2.3, pp. 29–35]. As O'Hara reports them, three properties follow: the spread is the risk-neutral monopoly spread, set by the slopes of the arrival functions, plus a risk adjustment that rises with relative risk aversion, transaction size, the stock's variance and the time remaining; transaction uncertainty by

itself does not affect the spread; and the spread does not depend on inventory, the dealer instead shifting both quotes (with true price 50, quotes 48 and 52 move to 47 and 51 after a purchase at the bid) by an amount set by risk aversion, the stock's risk and his wealth [catalogue-0496, same sec., pp. 35–36]. Garman's fixed-quote dealer is ruined with probability one, within days for realistic parameters, which is why quotes must depend on inventory; in Amihud and Mendelson's risk-neutral model with inventory bounds both quotes fall as inventory rises, but the spread *widens* with distance from the preferred position and the quotes are not symmetric about the true value [s6-hasbrouck, secs. 11.1.1–11.1.2, pp. 107–109].

Pitfalls. Neither original paper was consulted: the model is Hasbrouck's version of Stoll (1978), and the Ho–Stoll properties are O'Hara's report, not checked against the paper, `ho-stoll-1981`, which the route's `catalogue-0527` does not identify by title. A spread independent of inventory is a feature of these models, not a general result (Amihud and Mendelson above). O'Hara notes that a fixed true price is fundamental to the Ho–Stoll analysis, and that its finite horizon, with liquidation certain at the end, implies spreads narrowing predictably through the day, which seems unrealistic [catalogue-0496, sec. 2.3, p. 36]. Empirically, quotes rising after a customer purchase fit inventory control and adverse selection alike; NYSE specialist quote changes are explained mostly by trades, with inventories adding little, and dealers manage positions more through non-public quotes, interdealer trading and discretion over participation than through posted quotes [s6-hasbrouck, secs. 11.4–11.5, pp. 115–117]; see component identification.

See also. Avellaneda–Stoikov quoting (generalisation: continuous time, with the spread also independent of inventory in its approximation); Grossman–Miller model (contrast: risk compensation set by market clearing, not by a quoting dealer); dealer inventory risk (prerequisite); dealer quote adjustment (practice).

Further reading. [ho-stoll-1981], the multiperiod model (not consulted: no accessible copy); the legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory-based quoting; inventory skew; reservation price; Stoll dealer model; Ho–Stoll; Amihud–Mendelson; Garman; dealer risk aversion; CARA dealer; inventory control; preferred inventory

Status. Verification: partly checked 2026-10-15 (the Stoll version, eqs. (11.2)–(11.3), and the inventory evidence against the text layer of a local copy of `s6-hasbrouck`, 2007 Oxford printing, pp. 106–117; the Ho–Stoll account against page images of `catalogue-0496`, 1998 Wiley printing, pp. 29–36; the ask and the reservation price derived here; neither Stoll (1978) nor Ho and Stoll (1981) consulted). Stability: draft. Currency: timeless (theory); the evidence is from 1990s NYSE data.

8.10 Measuring and decomposing the spread

Definition 8.40 Spread measures: quoted, inside, relative, effective and realised spreads

Statement. With best bid b_t, best ask a_t and midpoint $m_t = \frac{1}{2}(a_t + b_t)$:

- the *quoted spread* is $a_t - b_t$; the *inside spread* is the highest bid's distance from the lowest ask across all dealers, and is no wider than the narrowest dealer's spread [alpha-harris, chap. "Dealers", p. 280];
- the *relative spread* is the spread as a fraction of price, the only basis on which spreads of different instruments compare [alpha-harris, chap. "Bid/Ask Spreads", p. 312];
- for a trade at p_t with sign $Q_t = +1$ if buyer-initiated and -1 if seller-initiated, the *effective spread* is twice the signed difference from the midpoint at the time of the trade, and the *realised spread* twice the signed difference from the midpoint a set interval Δ later, commonly 5, 10, 15 or 60 minutes [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 424–425]:

$$ES_t = 2Q_t(p_t - m_t), \qquad RS_t = 2Q_t(p_t - m_{t+\Delta}), \qquad ES_t = RS_t + 2Q_t(m_{t+\Delta} - m_t).$$

The last term, the signed midpoint revision after the trade, is sometimes used as an estimate of the trade's price impact [s6-hasbrouck, sec. 14.2.2, eq. (14.2), p. 148].

Notation. Q_t is the trade sign. Harris's effective and realised spreads are doubled, full-spread amounts; Hasbrouck's *effective cost* $p_t - m_t$ and *realised cost* $p_t - m_{t+5}$ are per-side amounts written for a buy order, with the midpoint five minutes after the order's receipt, as in the SEC's Rule 605 [s6-hasbrouck, same sec., p. 148]. Harris calls the undoubled signed difference the *liquidity premium* [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 425]. The legacy notes wrote ϵ_t for Q_t, s_t for the quoted spread and PI_t for the impact term, and measured the relative spread against m_t.

Conditions. The effective spread equals the quoted spread when every purchase is at the ask and every sale at the bid; it is smaller with price improvement and larger when large orders fill outside the quotes. Realised and effective spreads are equal if the midpoint does not move over the interval; usually it moves with the trade, so realised spreads are smaller [alpha-harris, same chap., p. 425]. When the analyst did not take part in the trades, signs are inferred, usually by the Lee–Ready rule, which overstates costs for traders who do not only take liquidity and cannot see orders filled in several trades [alpha-harris, same chap., p. 423]. Delayed trade

reports bias effective costs down, and hidden orders and market-maker improvement distort the inferred sign [s6-hasbrouck, sec. 14.3.3, pp. 150–151].

Practice. Quoted minus effective spread measures the price improvement dealers give; effective minus realised spread measures dealers' losses to well-informed traders [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 425–426]. The realised cost is the revenue of a dealer who sold to the customer at p_t and covered at the later midpoint [s6-hasbrouck, sec. 14.2.2, p. 148]. The effective spread is the least noisy benchmark estimator, because its benchmark is contemporaneous; the realised spread grows noisier the longer the interval [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 427]. The quoted spread measures the cost of immediacy for small orders only; depth and resiliency are other aspects of liquidity [alpha-harris, chap. "Bid/Ask Spreads", p. 312].

Pitfalls. Harris also uses *realised spread* for the difference between the prices at which a dealer actually sells and buys on a round trip, which can be negative when quotes move against the inventory [alpha-harris, chap. "Dealers", p. 280]; that is a dealer's outcome, not the midpoint-based estimator above. Mixing doubled and per-side conventions changes figures by a factor of two. Interpretation: effective minus realised spread is the permanent component only if the interval is long enough for prices to adjust and short enough to exclude unrelated news, and quote moves for inventory that have not yet reversed fall into it too; see component identification.

See also. spread components (what the measures estimate); Roll's estimator (alternative: a spread from trade prices alone); outside spread (contrast: an undisplayed willingness to trade).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. quoted spread; inside spread; relative spread; half-spread; effective spread; realised spread; realized spread; effective cost; realized cost; liquidity premium; price impact; price improvement; Rule 605; Lee–Ready; trade signing; transaction cost measurement

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 280–281, 312 and 423–427, and `s6-hasbrouck`, 2007 Oxford printing, pp. 148 and 150–151). Stability: draft. Currency: as of the 2002 and 2007 printings for the reporting rules (Rule 605 named by Hasbrouck); definitions timeless.

Result 8.41 Roll's serial-covariance spread estimator

Statement. Let trade prices be $P_t = V_t + \frac{1}{2}SQ_t$, where the fundamental value V_t is a random walk with independent innovations ε_t of mean zero and variance σ^2, S is the spread, and the trade sign $Q_t = \pm 1$ is a buy with probability one-half, independently of earlier signs and of ε_t and ε_{t+1}. Then [alpha-harris, chap. "Volatility", box, p. 415]

$$\operatorname{Var}(\Delta P_t) = \sigma^2 + \tfrac{1}{2}S^2, \qquad \operatorname{Cov}(\Delta P_t, \Delta P_{t-1}) = -\tfrac{1}{4}S^2, \qquad S = 2\sqrt{-\operatorname{Cov}(\Delta P_t, \Delta P_{t-1})},$$

and all autocovariances of order two or more are zero [s6-hasbrouck, sec. 3.4, p. 30]. Roll's estimator replaces the covariance by its sample value [alpha-harris, chap. "Volatility", p. 415].

Notation. S is the full spread, in price units; changes are between successive trades. Hasbrouck writes $p_t = m_t + cq_t$ with half-spread $c = S/2$, efficient-price innovation u_t of variance σ_u^2, and γ_0, γ_1 for the variance and first autocovariance of price changes, so $\gamma_0 = 2c^2 + \sigma_u^2$, $\gamma_1 = -c^2$, $c = \sqrt{-\gamma_1}$ and $\sigma_u^2 = \gamma_0 + 2\gamma_1$ [s6-hasbrouck, sec. 3.4, eqs. (3.3)–(3.5), pp. 29–30]. Harris calls the estimate an effective spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434].

Conditions. The spread is constant and noninformational: competitive dealers quote $m_t \pm c$ to cover a cost c per trade [s6-hasbrouck, sec. 3.4, p. 29]. Idea of the proof: $\Delta P_t = \varepsilon_t + \frac{1}{2}S(Q_t - Q_{t-1})$; every cross-product has zero expectation except $\mathbb{E}Q_{t-1}^2 = 1$, which gives $-\frac{1}{4}S^2$ for the first autocovariance [alpha-harris, chap. "Volatility", p. 415]. The variance splits into a fundamental part σ^2 and a transitory part from bid–ask bounce [alpha-harris, same chap., pp. 415–416].

Practice. The estimator is used where quotes are unavailable [s6-hasbrouck, sec. 3.4, p. 30]; Harris advises using it only then [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434]. For one NYSE stock (PCO, October 2003), $\hat{\gamma}_1 = -0.0000294$ gives $c = \$0.017$ and a spread of \$0.034, against a time-weighted average quoted spread of \$0.032 [s6-hasbrouck, sec. 3.4, p. 30].

Pitfalls. In that sample the assumptions fail: the spread ranged from \$0.01 to \$0.49, successive trade signs had correlation about 0.34, and midpoint changes were positively correlated with the latest trade sign [s6-hasbrouck, sec. 3.4, p. 30]. The biases go both ways. Positively autocorrelated signs, $\operatorname{Corr}(q_t, q_{t-1}) = \rho > 0$, give $\gamma_1 = -c^2(1 - 2\rho)$ and bias $\hat{c}$ down; signs positively correlated with the efficient-price innovation, as with informed trading, give $\gamma_1 = -c(c + \rho\sigma_u)$ and bias $\hat{c}$ up [s6-hasbrouck, sec. 4.4, Exercises 4.2–4.3, pp. 38–39]. A positive sample autocovariance is not uncommon, and estimation error alone can produce one when the model is correct; the square root is then undefined [s6-hasbrouck, sec. 4.5, p. 39]. Harris says violations typically make it underestimate the spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434],

and that reversals lasting beyond one trade make Roll's variance components understate transitory volatility [alpha-harris, chap. "Volatility", p. 416]. The legacy notes listed informed trading among causes of a less negative covariance; under Hasbrouck's Exercise 4.3 it makes the covariance more negative. Price discreteness and hidden liquidity, also listed there, are not discussed in the passages checked.

See also. component identification (generalisation: the generalized Roll model, with an information component); fundamental and transitory volatility (generalisation: random-walk decompositions with a general pricing error); spread measures (when quotes are available); Grossman–Miller liquidity measures (contrast: negative autocovariance from inventory risk).

Further reading. The legacy blocks it replaces, with the full derivation, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Roll's 1984 paper was not consulted.

Index. Roll model; Roll estimator; serial covariance spread estimator; bid–ask bounce; negative autocovariance; transitory volatility; implicit spread; effective spread estimate

Status. Verification: partly checked 2026-10-15 (the model and derivation against the page image of `alpha-harris`, p. 415, and its text layer, pp. 416 and 434; the parametrisation, example, biases and estimation against the text layer of `s6-hasbrouck`, pp. 29–30 and 38–39; the original paper not consulted). Stability: draft. Currency: timeless (method); the example is from 2003 data.

Result 8.42 Separating transitory and permanent spread components: bounce, revisions and identification

Statement. The transaction-cost component makes prices bounce between bid and ask, so its price changes reverse; the adverse-selection component moves the dealers' value estimates, so its price changes are permanent and unpredictable; both depend on whether a buyer or a seller initiated the trade, and the usual decomposition explains current and future price changes with the signs of trades [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300 and 302–303]. In Hasbrouck's generalized Roll model the efficient price moves with the trade and with public news, and trades occur at a cost c either side of it [s6-hasbrouck, sec. 8.2, eq. (8.1), pp. 67–68]:

$$m_t = m_{t-1} + \lambda q_t + u_t, \qquad p_t = m_t + c q_t, \qquad \text{ask} - \text{bid} = 2(c+\lambda),$$

$$\gamma_0 = c^2 + (c+\lambda)^2 + \sigma_u^2, \qquad \gamma_1 = -c(c+\lambda),$$

with all higher autocovariances zero [s6-hasbrouck, sec. 8.3, eq. (8.3), p. 68]. Two moments cannot identify the three parameters λ, c, σ_u^2: trade prices alone do not separate the components. Only the variance of efficient-price changes, $\lambda^2 + \sigma_u^2 = \gamma_0 + 2\gamma_1$, is identified without restrictions [s6-hasbrouck, same sec., pp. 68–69].

Notation. $q_t = \pm 1$ is the trade sign; λ is the adverse-selection (information) cost and c the noninformational cost, both per side; u_t is nontrade public information, uncorrelated with q_t; γ_0, γ_1 are the variance and first autocovariance of Δp_t. In Harris's terms the adverse-selection component is 2λ and the transaction-cost component $2c$ (Interpretation).

Conditions. Timing: after the trade at $t-1$ the efficient price is m_{t-1}; public news u_t arrives; dealers set bid and ask symmetrically about $m_{t-1} + u_t$; the trade q_t arrives and the efficient price becomes m_t [s6-hasbrouck, sec. 8.2, p. 68]. Setting $\lambda = 0$ gives Roll's model; setting $\sigma_u^2 = 0$ or $c = 0$ identifies the rest, but such restrictions are economically strong [s6-hasbrouck, sec. 8.3, p. 69].

Practice. With trade signs, analyses indicate that in most markets the adverse-selection component is larger than the transaction-cost component [alpha-harris, chap. "Bid/Ask Spreads", p. 303]. Inventory control predicts the same short-run quote response as adverse selection: after a customer buys, the dealer's quotes rise either way. The two separate in the long run, since information effects are permanent while inventory effects vanish once offsetting trades restore the dealer's position; for NYSE specialists, most quote dynamics are attributable to trades, with inventories adding little explanatory power [s6-hasbrouck, sec. 11.4, p. 115]. Identification therefore needs more than prints: trade signs and subsequent quote revisions and, to separate inventory, dealer positions.

Pitfalls. An estimated permanent component includes any inventory-driven quote change that has not reversed within the horizon studied, and the transitory component includes any mean reversion from uninformed order imbalances (Interpretation). Where the dealer is counterparty to every one-unit trade, $q_t = -\Delta I_t$ for inventory I_t [s6-hasbrouck, sec. 11.3.2, p. 112], and a regression on current and lagged trades and inventories is then collinear; the NYSE data avoid this because the specialist does not take part in every trade [s6-hasbrouck, sec. 11.4, p. 115].

See also. Roll's estimator (special case: $\lambda = 0$); spread components (the theory being estimated); spread measures (a model-free split by effective and realised spreads); inventory-based quotes (the inventory alternative); propagator model (impact that decays over many trades rather than one); order-flow impact regression (the Glosten–Harris extension with trade size).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread decomposition; transitory component; permanent component; bid–ask bounce; generalized Roll model; identification; trade signs; inventory versus information; price revision; adverse selection estimation

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, pp. 299–303, and `s6-hasbrouck`, pp. 67–69, 112 and 115). Stability: draft. Currency: timeless (method); the NYSE evidence is from 1990s data.

Result 8.43 Determinants of bid–ask spreads: primary factors, order-driven equilibrium and observable proxies

Statement. Three primary factors set spreads [alpha-harris, chap. "Bid/Ask Spreads", sec. 14.6.1, pp. 311–313]:

- *asymmetric information* widens them, most when informed traders know material information;
- *volatility* widens them: it raises the value of the timing option that standing limit orders and dealer quotes give away when they cannot be adjusted at once, makes diversifiable inventory risk costly to risk-averse dealers, and proxies for asymmetric information;
- *utilitarian trading interest* narrows them: active markets spread dealers' fixed costs over more trades, make inventory easier to lay off, draw public limit orders, and dilute informed flow.

In a continuous order-driven market the equilibrium spread makes the marginal trader indifferent between limit and market orders; it would be zero if traders were identical and equally informed, could cancel instantly and free, and cared neither for time nor risk [alpha-harris, same chap., secs. 14.4.1–14.4.2, pp. 304–310]. Relaxing those assumptions gives Harris's Table 14-2 [alpha-harris, same chap., p. 311]: information asymmetry (increases spreads; great importance), time to cancel limit orders (increases; high), volatility (increases; high), limit-order management costs (increases; moderate), the value of traders' time (increases; moderate), the commission difference between limit and market orders (one for one), and risk aversion (increases; low).

Conditions. With costly limit-order management, the equilibrium spread equals the expected cost of managing limit orders plus twice the value of the timing option [alpha-harris, same chap., p. 307]; with only a commission difference, it equals that difference [alpha-harris, same chap., p. 306]. If all traders were equally uninformed, uncertainty would narrow spreads, because value estimates are less volatile than values; uncertainty widens spreads through asymmetry [alpha-harris, same chap., p. 310]. The primary factors interact: asymmetry widens spreads, which deters uninformed traders and lowers volume, which widens them further [alpha-harris, same chap., p. 312]. Public limit-order traders without dealers' costs can undercut dealers, who survive by exploiting their view of the order flow [alpha-harris, same chap., sec. 14.5, pp. 310–311].

Practice. Observable *secondary* determinants stand in for the primary ones [alpha-harris, same chap., sec. 14.6.2, pp. 314–317]. For asymmetric information: disclosure rules, market-condition reports, analyst and press coverage, economy-wide commodity contracts, diversified portfolios and diversified firms, established rather than emerging industries, firm age, enforced insider-trading rules, and expected material news such as earnings. For volatility: financial and operating leverage and uncertain growth for stocks; perishability, storage costs, low inventories and weather for commodities; political, inflation and interest-rate uncertainty for currencies; inflation, rates and credit quality for bonds. For utilitarian interest: trading activity (volume, number of trades), firm size, debt issue size and on-the-run status, and contracts that replicate risks many traders want to hedge. When asymmetric information is severe or utilitarian interest small, spreads can be so wide that the market fails; small firms then finance through banks and venture capitalists, whose due diligence, restrictions and supervision rights need a private relationship [alpha-harris, same chap., sec. 14.6.3, p. 317]. Empirically, daily spreads rise with daily volatility, the spread rewarding market makers for the risk they bear, and across small-tick instruments spreads follow $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with N_i the daily number of trades [alpha-laruelle, chap. 2, secs. 2.2 and 2.3.1, pp. 144–145 and 150–151]; see spread–volatility regression. When the spread is stuck at one tick, liquidity providers respond by changing size at the best quotes rather than the spread [alpha-laruelle, same chap., sec. 2.2, p. 148]. In crashes liquidity suppliers withdraw: in 1987 buyers waited until they believed all sellers were satisfied, and some Nasdaq dealers took their phones off the hook [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", pp. 561 and 563].

For large-tick instruments, Lehalle and Laruelle add a bounce term inside the same scaling by trading activity, $\psi_i = a + (b_1\sigma_i + b_2\,\eta \cdot \text{tick})/\sqrt{N_i}$, as in the spread–volatility regression (eq. (2.1), page image of p. 145). Here $\eta = N_c/(2N_a)$ is half the ratio of price continuations to alternations in traded prices: the more bid–ask bounce, the more alternations and the smaller η, which is one half when moves are random [alpha-laruelle, chap. 2, sec. 2.2, eq. (2.1), pp. 144–145]. When the spread is stuck at one tick, liquidity providers who cannot step inside it increase the size at the best bid and offer, so a larger book plays the role of a narrower spread (p. 148). Interpretation: the tick is the constraint; bounce and book size are observed outcomes; matching rules (time priority or pro-rata allocation) govern who is filled within the larger queue (CME Globex trading conventions). The legacy principle *Large Tick Futures* (slice 19) stated that futures "often" have economically large ticks, with

bounce, reversals and larger displayed quantities "relative to comparable equities". The mechanism matches these passages, which are not specific to futures; the claim about futures ticks and the comparison with equities were not checked.

Pitfalls. Secondary factors are correlated with more than one primary factor and with each other, so an observed effect can have the opposite sign to the prediction (emerging industries attract volume), and tests need controls [alpha-harris, same chap., p. 314]. Volatility has a second, opposite effect through gambling and hedging interest, which raise activity [alpha-harris, same chap., pp. 316–317]. Trading activity is the best observable predictor of spreads, but Harris keeps it secondary to preserve the distinction between theoretical and observable determinants [alpha-harris, same chap., p. 312]. These are qualitative predictions; the importance ratings in Table 14-2 are Harris's judgement, not estimates. The legacy notes' "availability of close risk substitutes" is not Harris's wording: his factor is risk replication for hedgers. A legacy principle, *volatility is a liquidity price*, listed smaller displayed size and faster cancellations among liquidity providers' responses to volatility. The passages checked support wider spreads and, in crashes, withdrawal. Lehalle and Laruelle report book size moving inversely to the spread over the day [alpha-laruelle, chap. 2, sec. 2.2, p. 148]; that is co-movement, not a measured response to volatility.

See also. spread components (the components these factors move); market collapse (a model of market failure); outside spread (the spread of value traders); dealer quote adjustment; spread–volatility regression (an empirical test of the volatility determinant); fundamental and transitory volatility.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread determinants; asymmetric information; volatility; utilitarian traders; trading activity; equilibrium spread; order-driven market; timing option; limit-order management cost; commission difference; secondary determinants; spread proxies; market failure; liquidity and capital structure; volatility as a liquidity price; liquidity withdrawal; tick size and depth

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 303–318, Table 14-2 included, and pp. 561 and 563; and of `alpha-laruelle`, 2018 World Scientific printing, pp. 144–151, with eq. (2.1) and p. 148 added in slice 19, eq. (2.1) corrected from the page image of p. 145 in its follow-up). Stability: draft. Currency: timeless (theory); institutional examples from 2002; the European evidence from 2008–2016.

Definition 8.44 Outside spread: the prices at which value traders will trade

Statement. A value trader's *outside spread* is the interval between the prices at which he is willing to buy and to sell; value traders rarely quote these prices, so as not to reveal their value estimates or give free trading options [alpha-harris, chap. "Value Traders", sec. 16.2, p. 340]. Writing $\widehat{V}$ for his value estimate, he buys only below b^o and sells only above a^o, and $s^o = a^o - b^o$. Interpretation: if every cost and risk of a one-unit trade could be summarised in an all-in half-cost $c > 0$, the expected surplus would be $\widehat{V} - P - c$ from buying and $P - \widehat{V} - c$ from selling. He would trade only outside the no-trade band $[\widehat{V} - c, \widehat{V} + c]$, so $b^o = \widehat{V} - c$ and $a^o = \widehat{V} + c$. The spread widens with [alpha-harris, same chap., pp. 340–344]:

- adverse selection from news traders who know of value changes he has not yet learnt;
- the winner's curse: the risk that his estimate is wrong, larger when values are hard to estimate and when many others estimate them, with losses from the market impact of trading on a wrong estimate;
- the costs of research, which must be recovered also on the instruments analysed that offer no trade.

Conditions. Value trading is profitable only when price differs from fundamental value, which happens in two ways. New information changes the value: news traders, by definition the first to learn it, profit and push the price to the new value. Or uninformed traders push the price away from value: value traders profit and push it back. Value traders therefore profit only by trading against uninformed traders' demands for liquidity [alpha-harris, same chap., sec. 16.1, pp. 338–339]. They do so directly, through limit orders or blocks, or indirectly through dealers whose inventories the uninformed demand has unbalanced [alpha-harris, same chap., secs. 16.1.1–16.1.2, pp. 339–340]. Outside spreads are much wider than dealer spreads: value traders trade when order flow is one-sided and cannot learn their mistakes from it, take larger positions held until prices revert, so bear more inventory risk and financing cost, trade less volume over which to spread fixed costs, and earn at best half the outside spread when price returns to value, while a dealer often earns the whole spread on a round trip [alpha-harris, same chap., sec. 16.3, pp. 344–345].

Practice. Value traders buy when dealers lower their quotes to shed inventory taken from uninformed sellers, and so supply depth to the market and resiliency against uninformed price moves; dealers take larger positions where value traders follow the market closely [alpha-harris, same chap., pp. 340, 345–346]. When their value estimates differ enough, value traders' outside spreads overlap and they trade with each other [alpha-harris, same chap., p. 343]. Not buying does not oblige a trader to sell: uncertain estimates mean he should speculate only when he believes an instrument is very mispriced, and the larger the apparent mispricing,

the more likely it is a mistake [alpha-harris, same chap., box, p. 343]. In continuous markets the winner's curse bites because traders trade at the prices they quote, and the market aggregates estimates sequentially; in a two-sided single-price auction it matters little [alpha-harris, same chap., pp. 342–343].

Pitfalls. The outside spread is not observed in quotes or trades (Interpretation: it is inferred from where value traders act). The legacy notes listed leverage and borrowing among its financing costs; Harris names the cost of financing larger positions, without that detail [alpha-harris, same chap., p. 344]. A single half-cost c understates the problem: adverse selection and the winner's curse are not per-unit costs but depend on what a trade at a given price reveals and on how many others are estimating the value, which is why Harris treats them as risks that widen the spread (Interpretation). The legacy notes gave value traders three sources of profit: information changes, uninformed liquidity demand and dealer inventory pressure. In Harris, information changes are where news traders profit; value traders gain there only by correcting news traders' under- or overreaction [alpha-harris, chap. "Informed Traders and Market Efficiency", Table 10-1, p. 230, and p. 239]. Dealer inventory pressure is the indirect channel of uninformed demand [alpha-harris, chap. "Value Traders", pp. 339–340].

See also. spread measures (contrast: displayed and transaction spreads); spread determinants; dealer quote adjustment (value traders restore dealers' inventories); fundamental value; informed trading and aggregation; winner's curse; informed-trader types (value against news traders).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. outside spread; value traders; winner's curse; research costs; resiliency; depth; liquidity supplier of last resort; news traders; no-trade band; value-trading profits; uninformed liquidity demanders

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 338–346, and pp. 230 and 239 for the profit sources). Stability: draft. Currency: timeless (concepts).

8.11 Volatility, liquidity and crashes

Definition 8.45 Fundamental and transitory volatility: the random-walk and transitory components of prices

Statement. *Volatility* is the tendency of prices to change unexpectedly. *Fundamental volatility* comes from unanticipated changes in instrument values; *transitory volatility* comes from trading by impatient uninformed traders, which pushes prices away from values until they revert. Total volatility is the sum of the two [alpha-harris, chap. "Volatility", pp. 410, 413–414]. In statistical form, the trade price is a random-walk efficient price plus a stationary pricing error [s6-hasbrouck, sec. 8.6, p. 72]:

$$p_t = m_t + s_t, \qquad m_t = m_{t-1} + w_t, \qquad \Delta p_t = w_t + \Delta s_t,$$

where s_t has mean zero, is covariance-stationary, and may be serially correlated and correlated with w_t. Fundamental volatility is the variance σ_w^2 of w_t. If the price changes have the moving-average representation $\Delta p_t = \theta(L)\varepsilon_t$, then $\sigma_w^2 = \theta(1)^2\sigma_\varepsilon^2$, the same for every structural model consistent with the data, while the variance of the pricing error has only a lower bound [s6-hasbrouck, same sec., eqs. (8.10)–(8.13), pp. 73–74]. Over long horizons $\sigma_w^2 \approx \mathrm{Var}(p_t - p_{t-k})/k$, because microstructure effects matter less as the interval grows [s6-hasbrouck, sec. 8.3, p. 69].

Notation. p_t is the trade price, m_t the efficient (fundamental) price, $s_t = p_t - m_t$ the pricing error, w_t the efficient-price innovation; $\theta(1)$ is the sum of the moving-average coefficients. Harris writes V_t for value and ε_t for its change. The legacy notes wrote $P_t = M_t + u_t$; Hasbrouck's u_t is public news in the generalized Roll model, so this entry uses s_t.

Conditions. Fundamental volatility comes only from *unexpected* changes in the factors that set values. Predictable price changes that pay holders for carrying costs and risk, such as a zero-coupon bond's creep towards par, are not volatility; an unexpected change in rates is [alpha-harris, chap. "Volatility", pp. 411–412]. When news is common knowledge prices can change without trading; when few know it, prices move on high volume as the informed trade; and the price revisions that make up the adverse-selection spread component contribute to fundamental volatility [alpha-harris, same chap., p. 411]. Transitory volatility includes bid–ask bounce, produced by the transaction-cost (transitory) spread component, and the price effects of large orders and cumulative uninformed imbalances that reverse when value traders and arbitrageurs act; it includes both the price change and its reversal. Value traders, arbitrageurs and dealers do not cause it but resolve it [alpha-harris, same chap., pp. 413–414]. Roll's model is the special case $s_t = \frac{1}{2}SQ_t$ with trade signs independent of w_t, which gives the two-term variance $\sigma^2 + \frac{1}{2}S^2$ (see Roll's estimator).

Practice. Fundamental factors differ by instrument: supply and demand, interest rates and storage costs for commodities; rates and credit quality for bonds; management, resources and technologies, product and input

markets and rates for stocks; inflation, macroeconomic policy and trade and capital flows for currencies. Volatility is higher for commodities that are costly to store or perishable, especially with long production pipelines, variable demand or inelastic demand; for firms whose value depends on unresolved uncertainties, such as research outcomes or the growth behind high price–earnings ratios; for instruments exposed to political risk; and for the equity of highly leveraged firms [alpha-harris, same chap., pp. 411–413]. Because transitory price changes revert, negative serial correlation of price changes is a strong indicator of transitory volatility: over successive trades for bid–ask bounce, and over minutes to months for uninformed order imbalances [alpha-harris, same chap., p. 414]. A variance ratio $V_{M,N} = [\mathrm{Var}(p_t - p_{t-M})/M]/[\mathrm{Var}(p_t - p_{t-N})/N]$ equals one for a random walk; when microstructure effects dominate short horizons, $V_{M,N} > 1$ for $M < N$ [s6-hasbrouck, sec. 8.7, eq. (8.14), pp. 74–75]. Transitory volatility moves with uninformed traders' transaction costs and is small in liquid markets [alpha-harris, chap. "Volatility", p. 414]. Regulators can have no lasting effect on fundamental volatility but can change transitory volatility, in either direction [alpha-harris, same chap., p. 410]. Circuit breakers slow fundamental price changes; their net effect on transitory volatility is indeterminate, lowering it when they restrain uninformed traders, order anticipators and bluffers and raising it when they restrain informed traders, dealers and arbitrageurs, and empirical studies have been inconclusive. Margins and transaction taxes cannot discriminate between these trader types [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", secs. 28.3.2–28.3.4, pp. 575–577].

Pitfalls. Negative serial correlation is not proof of transitory volatility: successive spot prices of a perishable commodity are prices of different items, for different delivery dates, so their negative correlation can be fundamental [alpha-harris, chap. "Volatility", box, p. 413]. Its absence is not proof of a random walk either: lagged price adjustment makes price changes positively autocorrelated [s6-hasbrouck, sec. 8.6, Exercise 8.2, p. 74], and negative autocorrelation at some lags can offset positive autocorrelation at others to give a variance ratio near one [s6-hasbrouck, sec. 8.7, p. 75]. Roll's variance components understate transitory volatility when reversals take longer than one trade [alpha-harris, chap. "Volatility", p. 416]. The split of $\mathrm{Var}(\Delta p_t) = \sigma_w^2 + \mathrm{Var}(\Delta s_t) + 2\,\mathrm{Cov}(w_t, \Delta s_t)$ beyond σ_w^2 is not identified; assigning the covariance term to transitory volatility, as the legacy notes did, is a convention, and Harris's "sum of the two" holds when the components are uncorrelated (Interpretation). The legacy notes also listed *episodic* and *continuing* volatility as further sources. Harris uses episodic volatility for episodes of large price changes in short intervals [alpha-harris, same chap., p. 410], a description rather than a third source; continuing volatility does not appear in the passages checked.

See also. Roll's estimator (special case and estimator); component identification (the same split for spreads: the generalized Roll model); Garman–Klass volatility (measuring total volatility from ranges); bubbles and crashes (episodes mixing both components); spread determinants (volatility as a determinant); fundamental value (Harris's value, conditioned on all information, not only public); returns by time scale.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. volatility; fundamental volatility; transitory volatility; episodic volatility; total volatility; random-walk decomposition; efficient price; pricing error; permanent and transitory components; variance ratio; negative serial correlation; bid–ask bounce; storage costs; perishable commodities; political risk; leverage; circuit breakers; regulating volatility

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 410–417 and 575–577, and `s6-hasbrouck`, 2007 Oxford printing, pp. 69 and 72–75). Stability: draft. Currency: timeless (concepts and method).

Method 8.46 Garman–Klass volatility from open, high, low and close prices

Statement. Lehalle and Laruelle estimate intraday volatility over a fixed interval from its open, high, low and close prices p^O, p^H, p^L, p^C and its average price $\bar{p}$ [alpha-laruelle, chap. 2, sec. 2.5.1, p. 172]:

$$\mathrm{Vol}_{GK} = \frac{\sqrt{\frac{1}{2}\left(p^H - p^L\right)^2 - (2\log 2 - 1)\left(p^C - p^O\right)^2}}{\bar{p}}.$$

1. Choose the interval. Lehalle and Laruelle use 5 minutes to show intraday seasonality and 15 minutes to compare markets [alpha-laruelle, same chap., secs. 2.2 and 2.5.1, pp. 146 and 172].

2. Record the open, high, low and close of each interval and apply the formula.

3. Average across intervals or days; the comparison of FTSE 100 and IBEX 35 components used a 20-day moving average of daily averages [alpha-laruelle, same chap., sec. 2.5.1, p. 172].

Notation. The superscripts O, H, L, C avoid a clash with the transaction-cost component C. The legacy notes used log ratios, $\sigma_{GK} = \sqrt{\frac{1}{2}(\log H/L)^2 - (2\log 2 - 1)(\log C/O)^2}$; since $\log(p^H/p^L) \approx (p^H - p^L)/\bar{p}$ for small ranges, the two forms agree to first order (Interpretation).

Conditions. Volatility is not observed: each estimate presumes a model. Under an arithmetic Brownian price the natural estimator sums squared trade-to-trade changes, but on high-frequency data that estimator is polluted by bid–ask bounce, and with jumps they must be removed first. Garman–Klass is a measure not polluted too much by the bounce. Implied volatility measures something else, the uncertainty priced into options [alpha-laruelle, same chap., sec. 2.2, pp. 146–147].

Practice. Lehalle and Laruelle use it for the intraday seasonality of volatility, with volume, spread and book size, on Royal Dutch Shell, 2011–2016 (Figure 2.12, pp. 146–148), and to compare the intraday volatility of FTSE 100 and IBEX 35 components after MiFID, finding no significant effect of the UK's fragmentation on intraday volatility [alpha-laruelle, same chap., sec. 2.5.1, pp. 172–174].

Pitfalls. The coefficient $2\log 2 - 1$ and the estimator's properties come from Garman and Klass (1980), which was not consulted; Lehalle and Laruelle give the formula without derivation or assumptions. The estimate is per interval and relative to price; scaling it to a daily or annual figure needs an assumption about how variance adds across intervals (Interpretation). When the high and low are trade prices they include part of the spread, so on short intervals the range still carries some bounce (Interpretation).

See also. fundamental and transitory volatility (what the total contains); spread–volatility regression (a use of interval volatilities); Roll's estimator (the bounce that trade-based estimators pick up); returns by time scale (open-to-close and overnight returns).

Further reading. The original estimator [garman-klass-1980], not consulted. The legacy block, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Garman–Klass; range-based volatility; open high low close; OHLC; intraday volatility; realised volatility; bid–ask bounce; volatility estimation; intraday seasonality

Status. Verification: partly checked 2026-10-15 (the formula against the page image of `alpha-laruelle`, 2018 World Scientific printing, p. 172; its use against the text layer, pp. 146–148 and 172–174; the original paper not consulted). Stability: draft. Currency: timeless (method); the applications use 2007–2016 European data.

Definition 8.47 Intraday, overnight and close-to-close returns: volatility and correlation by time scale

Statement. For day t with opening price p_t^O and closing price p_t^C, Lehalle and Laruelle define the *intraday* and *overnight* returns [alpha-laruelle, chap. 2, sec. 2.5.1, eqs. (2.2)–(2.3), p. 175]:

$$r_t^{\text{intra}} = \log p_t^C - \log p_t^O, \qquad r_t^{\text{on}} = \log p_t^O - \log p_{t-1}^C,$$

and compute the correlation between two stocks separately on each series, over a six-month sliding window, averaging the pairwise correlations across an index's components [alpha-laruelle, same sec., pp. 174–175]. Interpretation: the close-to-close (*extraday*) return is exactly their sum, $r_t^{\text{cc}} = r_t^{\text{on}} + r_t^{\text{intra}}$. Volatility likewise has an intraday and an extraday scale: investors with extraday constraints bear extraday volatility, and intraday volatility only through its effect on that scale [alpha-laruelle, same chap., sec. 2.5, p. 169].

Notation. Superscripts O and C for open and close, as in Garman–Klass, avoid a clash with the transaction-cost component C.

Conditions. The two returns come from different market mechanisms: overnight the market is organised as an auction fixing, during the day as a continuous double auction. The authors hold that the opening fixing is less contaminated by microstructure noise than the closing fixing, and so better reflects investors' consensus on the price [alpha-laruelle, same chap., sec. 2.5.1, p. 177]. Interpretation: $\operatorname{Var}(r^{\text{cc}}) = \operatorname{Var}(r^{\text{on}}) + \operatorname{Var}(r^{\text{intra}}) + 2\operatorname{Cov}(r^{\text{on}}, r^{\text{intra}})$, so extraday risk mixes the two scales and their covariance.

Practice. On the components of the CLES 60, DAX, CAC 40 and IBEX 35, 1991–2012, correlation rises mainly on the overnight side, and overnight correlation explains well the spikes in extraday correlation, to which it is closer; intraday correlation acts as a baseline that overnight correlation keeps bouncing above [alpha-laruelle, same sec., Figures 2.27–2.28, pp. 174–177]. Intraday correlation spiked in summer 2010 and in August 2011, and for three of the indices was in October 2010 at its highest level since 1991 [alpha-laruelle, same sec., pp. 174 and 176]. The authors read intraday correlation as an intermediate marker in a contagion process by which microscopic causes have macroscopic effects [alpha-laruelle, same sec., p. 174].

Pitfalls. The authors attribute part of the March–October 2010 rise in extraday correlation to rising intraday correlation, the consequence of a less noisy PFP, and offer growing high-frequency activity as one possible explanation [alpha-laruelle, same sec., p. 177]: a conjecture, not a test. The legacy notes said daily correlation spikes come from different mechanisms at different horizons; the authors find the spikes mainly overnight. The legacy notes tied overnight correlation to auction and news revaluation and overnight volatility to information arrival. The auction is in the source; news arriving overnight is not stated in the passages checked (Interpretation). A legacy principle said intraday frictions matter for daily risk when they change the closing or auction price or the path used by benchmark investors. The source asks whether changes in intraday price formation show at the extraday scale [alpha-laruelle, same chap., sec. 2.5, p. 169], and compares the noise in the two fixings (p. 177); benchmark paths are not discussed there. Interpretation: the returns depend on

which prices are used, auction prints or last trades, and the open and close of a fragmented market may differ by venue.

See also. Garman–Klass volatility (volatility within intraday intervals; the FTSE 100 and IBEX 35 comparison of the same section); fundamental and transitory volatility (variance across horizons); price formation process. The legacy Flash Crash and systemic-risk blocks of the same section of the source are assigned to `hft-market-design` and `market-structure-execution-quality`.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. intraday return; overnight return; close-to-close return; extraday volatility; intraday volatility; overnight correlation; intraday correlation; correlation spikes; opening auction; closing auction; contagion; microstructure and macro prices

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 169 and 174–177, figure captions included; no page images). Stability: draft. Currency: as of the 1991–2012 data; definitions timeless.

Method 8.48 Spread–volatility regression in levels and logs

Statement. To test how the spread moves with volatility, Lehalle and Laruelle proceed in five steps [alpha-laruelle, app. A, sec. A.10, pp. 292–298]:

1. Compute the spread ψ_i and the volatility σ_i over the same intervals; their example uses 15-minute intervals of Rio Tinto from January 2012.
2. Fit $\psi_i = a_0 + a_1\sigma_i + \epsilon_i$ by ordinary least squares: $\hat{a}_1 = \sum_i(\sigma_i - \bar{\sigma})(\psi_i - \bar{\psi}) / \sum_i(\sigma_i - \bar{\sigma})^2$, $\hat{a}_0 = \bar{\psi} - \hat{a}_1\bar{\sigma}$.
3. Inspect the distributions: both variables looked log-normal.
4. Fit $\log\psi_i = a_0 + a_1\log\sigma_i + \epsilon_i$, which in levels is the multiplicative model $\psi = A_0\sigma^{a_1}e^{\epsilon}$, $A_0 = e^{a_0}$.
5. If $\epsilon \sim N(0, \sigma_\epsilon^2)$, then $\mathbb{E}[\psi] = A_0\sigma^{a_1}e^{\sigma_\epsilon^2/2}$, so predict levels with $\hat{\psi} = \hat{A}_0\sigma^{\hat{a}_1}e^{\hat{\sigma}_\epsilon^2/2}$, which they call the unbiased predictor.

A positive $\hat{a}_1$ is consistent with the spread rewarding market makers for the risk they take: the higher the volatility, the more they ask [alpha-laruelle, chap. 2, sec. 2.3.1, p. 150].

Notation. ψ is the spread, σ the volatility over the interval, a_0, a_1 the intercept and slope, σ_ϵ^2 the residual variance. A_0 is the level intercept, not the adverse-selection component A.

Conditions. The least-squares assumptions are those of the simple linear model: the regressor measured without error; errors with mean zero, constant variance and no correlation with each other or with the regressor; normal errors for the t tests [alpha-laruelle, app. A, sec. A.10.1, pp. 293–295]. Across small-tick instruments, Lehalle and Laruelle report the relation $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with N_i the average number of trades per day and a, b shared within an asset class; for large-tick instruments, $\psi_i = a + (b_1\sigma_i + b_2\,\eta\,\text{tick})/\sqrt{N_i}$, where $\eta = N_c/(2N_a)$ is half the ratio of price continuations to alternations, smaller the more the price bounces [alpha-laruelle, chap. 2, sec. 2.2, pp. 144–145].

Practice. For Rio Tinto, the level fit gave $\hat{a}_0 = 2.4894$, $\hat{a}_1 = 0.0485$, $R^2 = 0.2404$; the log fit $\hat{a}_0 = 0.5609$, $\hat{a}_1 = 0.2252$, $R^2 = 0.1553$; the back-transformed predictor, with $\hat{\sigma}_\epsilon = 0.2509$, explains 0.1097 of the variance of the spread in levels [alpha-laruelle, app. A, Figures A.14, A.17 and A.18, pp. 296–298]. Daily spreads rise with daily volatility on BNP Paribas and BP, 2008–2012; points off the line in August 2011 coincide with the short-selling ban, when market makers could not meet their commitments in financial stocks [alpha-laruelle, chap. 2, sec. 2.3.1, Figure 2.13, pp. 150–151].

Pitfalls. The fit is descriptive: in the example volatility explains a quarter or less of the spread's variance. Unless the comparison controls for them, other variables drive both series. Tick size: when the spread is stuck at one tick, liquidity providers add size at the best quotes instead, so in large-tick assets volatility may show in depth rather than in the spread [alpha-laruelle, chap. 2, sec. 2.2, p. 148]. Time of day: spread, volatility, volume and book size share an intraday seasonality, high after the open, so pooling intervals across the day mixes the seasonal pattern with the relation [alpha-laruelle, same sec., pp. 145–148]. Trading activity: the cross-sectional relation scales volatility by $1/\sqrt{N_i}$ [alpha-laruelle, same sec., pp. 144–145]. Harris's general warning applies: observable determinants are correlated with each other and with more than one primary factor, so tests need controls [alpha-harris, chap. "Bid/Ask Spreads", p. 314]. The legacy notes also listed venue and depth as controls; the passages checked do not discuss them as regression controls. Causation runs both ways, since a wider spread adds bounce to trade-based volatility (Interpretation). The predictor is unbiased only with known a_0, a_1 and σ_ϵ^2; with estimates plugged in it is approximately unbiased (Interpretation).

See also. spread determinants (the theory being tested: volatility as a primary determinant); Garman–Klass volatility (an interval volatility); spread measures (which spread to regress).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread and volatility; spread–volatility regression; log-log regression; log-normal; retransformation bias; unbiased predictor; ordinary least squares; tick size; number of trades; bid–ask bounce ratio; intraday seasonality; market depth

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 144–151 and 292–298, figure captions included; `alpha-harris`, p. 314). Stability: draft. Currency: as of the 2008–2016 data used; method timeless.

Mechanism 8.49 Bubbles and crashes: how mispricing builds and unwinds

Statement. A *bubble* is a rise of prices substantially above fundamental values, quickly or over a long period; a *crash* is a very fast fall, often but not always after a bubble. Since fundamental values are not common knowledge, bubbles and crashes can occur at all [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", sec. 28.1, p. 556]. The typical sequence [alpha-harris, same chap., sec. 28.1.1, pp. 556–557]:

1. Buyers grow overly optimistic about new technologies or markets, often without recognising what prices already reflect, and push prices up.

2. Momentum traders and order anticipators buy and accelerate the rise; early buyers grow more confident and early sellers buy back to stop their losses.

3. Value traders and arbitrageurs cannot sell enough (no position to sell, too little capital to carry a short, short selling hard) or will not (they expect uninformed buying to continue, or doubt their own value estimates).

4. Sellers start trading aggressively: long-term holders, early buyers taking gains, contrarians, value traders, arbitrageurs. Momentum buyers lose interest and optimists lose confidence.

5. Late buyers sell to stop losses, margin calls and stop orders force more selling, and order anticipators sell ahead of them: prices fall quickly. Panic can take prices below value, and a bounce follows.

The price accelerator (Harris, after a manuscript by Treynor): rising prices move wealth from pessimists with short positions to optimists with long ones; the richer optimists may buy more, and the losing pessimists may buy to cover margin calls, so prices rise further; the sellers are mild optimists and mild pessimists [alpha-harris, same chap., box, p. 556].

Conditions. Most bubbles and crashes involve both fundamental and transitory volatility. A bubble usually starts on good news, a fundamental change; pseudo-informed traders then overreact, the transitory part. It usually bursts on bad news, but the volume and price response is out of proportion to the news because of the overpricing that preceded it; the reversion towards value counts as transitory volatility, since it occurs only because uninformed traders inflated prices, and panic below value adds more [alpha-harris, same chap., sec. 28.1.2, pp. 557–558].

Practice. Harris's 1987 account shows the mechanisms together: portfolio insurance traded like stop orders, and uncertainty about how much of it was waiting kept buyers out; NYSE order printers could not keep up, so traders did not know whether their orders had filled; index futures led the cash market down and traded at a discount of more than 10 percent once index arbitrageurs stopped; and buy-side traders could not reach Nasdaq dealers [alpha-harris, same chap., sec. 28.2.2, pp. 559–563]. Crashes rarely have a single cause: traders become confused about risk, attend to the wrong issues and, when events move fast, panic [alpha-harris, same chap., sec. 28.2.8, p. 571].

Pitfalls. A large fall without news of commensurate importance does not show that the market failed. Most crashes correct earlier pricing errors; if they were failures, prices would rebound fully once traders saw them below value, but after most crashes prices rebound little [alpha-harris, same chap., box, p. 559]. Attribution therefore looks at the reversal after the event, at what preceded it, and at overloaded systems, forced selling and broken links between markets, as in 1987 (Interpretation, from the cases on pp. 557–563). Bubbles are hard to identify in real time, and analysts generally understand the causes of extreme volatility better after the fact [alpha-harris, same chap., p. 555]. The legacy list named policy among the stories that start a bubble; Harris's general account names technology and new markets, and his Japanese example turns on loose monetary policy [alpha-harris, same chap., sec. 28.2.7, pp. 570–571].

See also. fundamental and transitory volatility (the two components in each phase); outside spread (why value traders wait); spread determinants. The legacy flash-crash, liquidity-spiral and shock-propagation blocks are assigned to `hft-market-design` and remain in Chapter 13.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Treynor's manuscript "The Canonical Bubble" was not consulted.

Index. bubble; crash; market break; meltdown; price accelerator; momentum traders; order anticipators; limits to arbitrage; margin calls; stop-loss orders; dead cat bounce; portfolio insurance; 1987 crash; crash attribution; market failure

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 555–563 and 570–571). Stability: draft. Currency: timeless (mechanism); the case material is historical, to 2002.

8.12 High-frequency trading: strategies, speed races and the costs of other traders

Definition 8.50 High-frequency trading: proprietary strategies, the speed tools they share, and why the label names no single mechanism

Statement. In the SEC's 2010 concept release the term "high frequency trading" is "relatively new and is not yet clearly defined". It typically refers to professional traders acting in a proprietary capacity whose strategies generate many trades a day, organised as proprietary trading firms, proprietary desks of broker-dealers or hedge funds [sec-concept-release-2010, sec. IV.B, p. 45]. Five characteristics are often attributed to them: (1) high-speed programs for generating, routing and executing orders; (2) co-location and individual data feeds to cut latency; (3) very short holding periods; (4) many orders cancelled shortly after submission; (5) ending the day close to flat (p. 45; restated in [alpha-laruelle, sec. 2.4.3, p. 163]). Rather than define HFT, the release treats *strategies* and *tools* separately. One firm may run many strategies at once, some generating many trades and some not, and "conceivably" some benefit market quality and long-term investors while others harm them [sec-concept-release-2010, sec. IV.B, p. 46]. To present issues for comment, "without limiting the broad request", it describes four broad types of strategy often associated with proprietary firms [sec-concept-release-2010, sec. IV.B.1, pp. 48–57]:

- *passive market making*: mainly non-marketable resting bids and offers, profiting from the spread and from liquidity rebates, and taking liquidity only to liquidate a position quickly. Layering many orders can give cancellation rates of 90 percent or more, with orders often lasting a second or less (pp. 48–49);
- *arbitrage*: capturing price discrepancies between related products or markets, such as an ETF against its basket, mostly by taking liquidity, with hedged positions that may last several days. The release asks how far such strategies trade price differences between trading centres and depend on latency between direct and consolidated data feeds (pp. 51–52);
- *structural*: exploiting structural vulnerabilities in the market or in other participants. Examples are using the fastest data to trade with participants still offering executions at stale prices, and setting a new NBBO with a small order and then triggering guaranteed-match trades on the other side (pp. 52–53);
- *directional*: a significant, unhedged position on an expected intraday move. Some, such as trading a temporary departure from fundamental value, "often may contribute" to price discovery. The release singles out two: *order anticipation*, detecting large buyers or sellers and trading ahead of them, and *momentum ignition*, orders and trades, perhaps with false rumours, meant to spark a rapid price move or trigger stop orders, followed by unwinding the position (pp. 53–57).

Many of these strategies are not new; what is new is the technology for identifying and executing them (p. 48). The tools discussed separately are co-location and individual trading-centre data feeds. Speed matters "both in the absolute sense of achieving very small latencies and in the relative sense of being faster than competitors" (sec. IV.B.2, pp. 57–58).

Conditions. Interpretation: speed and automation are means shared by these strategies, not a strategy, and the economic mechanism differs between them. Passive market making is paid the spread for supplying immediacy and bears inventory and adverse-selection risk (Avellaneda–Stoikov, dealer quote adjustment). Arbitrage enforces relations between prices (arbitrage). Trading against stale quotes earns rents from reacting first to public information (latency-arbitrage arms race). Order anticipation profits from information about others' orders rather than about values (order anticipation). Momentum ignition is a form of bluffing (bluffing and manipulation). Effects on liquidity, price discovery and other traders' costs must therefore be judged strategy by strategy. The release's four types organise a request for comment on U.S. equities in January 2010; they are not a universal classification, and they overlap: trading against stale prices is filed under "structural" and cross-venue price differences under "arbitrage".

Practice. Lehalle and Laruelle describe high-frequency market makers in European equities around 2009–2012 as arbitraging between the venues that quote the same stock, which keeps those prices consistent. Maker–taker fee schedules on MTFs reward this, and trading very small orders very often lowers their inventory risk [alpha-laruelle, secs. 2.4 and 2.4.3, pp. 158 and 163]. Venues court them with low latency and co-location, "especially since the lower the latency, the smaller the inventory risk"; the authors read one venue's low "flow reactivity" as its high-frequency market makers offering less liquidity when activity rises (p. 158). Probing orders are a tool, not a strategy. The release defines a "pinging" order as an immediate-or-cancel order used to find

undisplayed liquidity, and distinguishes its use in a normal search for liquidity from its use to detect and trade in front of large interest [sec-concept-release-2010, p. 47, n. 69]. An order anticipator that detects a large buyer first buys, and is largely indifferent to whether the large trader is buying or selling; a long-term investor seeking liquidity wants the opposite side, and both parties benefit from that search [sec-concept-release-2010, p. 55]. Liquidity seeking by routers is in smart order routing. Narang (slice 30) groups high-frequency strategies as contractual and non-contractual market making, arbitrage and *fast alpha*. Fast alpha applies the price-based alphas of slower strategies, momentum, mean reversion and technical sentiment, to prices, volumes and order-book information that change during the day; its statistical relationships carry risk because "there is no structural reason" they must persist, and intraday trend following is less common than reversion because impact and slippage work against it [s7-narang, chap. 15, "High-Frequency Trading", sec. 15.4, pp. 291–293]. Execution algorithms also adjust their aggressiveness to the order book's imbalance, a "micro price" (chap. 7, pp. 133–134), and one that leaves a detectable footprint lets other participants react "to increase the market impact and slippage" it incurs (p. 130).

Pitfalls. The legacy taxonomy listed five strategies: passive market making, statistical arbitrage, latency arbitrage, order anticipation and liquidity detection. Compared with the release:

- "statistical arbitrage" on short-horizon relative value or lead–lag signals straddles the arbitrage and directional types;
- "latency arbitrage" corresponds to the structural example of trading against stale prices, and across venues to the release's arbitrage question;
- "liquidity detection" is a tool that order anticipators and ordinary liquidity seekers both use, not a strategy;
- momentum ignition, which the release names, was missing.

The legacy closing sentence, that one firm may run several strategies whose risk and social value differ, matches p. 46, where the difference in social value is a possibility to assess, not a finding. A strategy label does not establish illegality. The release treats order anticipation that involves no breach of duty or misappropriation as a question of policy; front running in breach of a duty, false rumours and manipulative conduct were already prohibited (pp. 53–55 and 57). These legal remarks are as of January 2010 and are not statements of current law. The release's estimate that HFT was 50 percent or more of U.S. equity volume came from press reports (p. 45, n. 67). Lehalle and Laruelle's "more than 80% of the deals in the US, 40% in Europe, and 35% in Japan" is reported as hearsay ("They are said to be", p. 163) without source or date. Neither is a measured share. From the legacy execution-model file (slice 30): its "microstructure alphas" on bets of seconds or less correspond to Narang's fast alpha; its claim that machine learning can discern other players' execution patterns, more easily the worse their execution models, and that such patterns are "somewhat stable", is not in the third edition and has no support in the passages read. Its "HFT shark strategy" (sending very small orders to detect a large iceberged order, then trading in front of it and exiting "with a quick and relatively riskless profit") is also not in the third edition; it describes pinging used for order anticipation, as the release does (p. 47, n. 69; p. 55). Three qualifications. (i) "Front-run" is used loosely: Narang defines front-running as a fiduciary using knowledge of a customer's order before acting on it, and argues that most high-frequency traders, who see orders only once they reach the market, do not front-run in that sense, except contractual market makers, who see customer flow and for whom it is "theoretically possible" (p. 312) [s7-narang, Appendix, sec. A.2.1, pp. 312–314]; FINRA's rule bars members from trading with "material, non-public market information concerning an imminent block transaction" [finra-rule-5270, para. (a)], and inferring a hidden order from fills is not obviously that information on the rule's wording (Interpretation; the term is not defined there, FINRA guidance was not checked, and the rule's supplementary material notes that other rules, such as Rules 2010 and 5320, may apply to front-running other orders). Whether pinging for order anticipation is lawful elsewhere was not checked. (ii) The profit is not riskless: the anticipator bears the risk that the inferred order is not there, is smaller or ends, and that the price does not move (Interpretation; order anticipation). (iii) The block is a historical description, kept with its original wording in the artefact, not a supported mechanism.

See also. latency-arbitrage arms race; spread against investor cost; order anticipation and bluffing and manipulation (owned by `execution-market-making`); arbitrage; smart order routing (owned by `market-structure-execution-quality`).

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Menkveld's survey (`wide-menkveld-hft-review`), the pathway's first reading, was not available and is unread.

Index. high-frequency trading; HFT; fast alpha; microstructure alpha; shark strategy; front-running; proprietary trading firms; passive market making; liquidity rebates; layering; cancellation rates; arbitrage; ETF arbitrage; structural strategies; stale quotes; directional strategies; order anticipation; momentum ignition; pinging; immediate-or-cancel orders; liquidity detection; co-location; individual data feeds; SEC concept release 2010

Status. Verification: partly checked 2026-10-15 (against the SEC's concept release, Release 34-61358 dated 14 January 2010, official PDF, sec. IV.B, pp. 45–58, text layer; and the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 158 and 163; in slice 30, `s7-narang`, third edition, Wiley 2024, chap. 7, pp. 129–134, chap. 15, pp. 291–293, and Appendix, pp. 311–314, text layers, and FINRA Rule 5270 as displayed on 15 October 2026). Stability: draft. Currency: U.S. equities as of January 2010; European venues as of the 2018 printing; Narang's account as of 2024; FINRA Rule 5270 as displayed on 15 October 2026, the only current rule checked; no market share checked and the historical figures dated as given.

Result 8.51 Latency arbitrage in a continuous limit order book: stale quotes are sniped, the speed race dissipates the rents, and investors pay through the spread (Budish, Cramton and Shim)

Statement. Budish, Cramton and Shim's model [`wide-budish-cramton-shim`, secs. VI.A–VI.B, pp. 1576–1583]. A security x trades on a continuous limit order book. Its value is perfectly correlated with a public signal y, which jumps as a compound Poisson process with rate λ_{jump} and absolute jump size J, and x can always be liquidated at that value. Investors arrive at rate λ_{invest}, each needing to buy or sell one unit, and take liquidity. Risk-neutral trading firms make and take liquidity, and messages that arrive together are processed serially in random order. In equilibrium exactly one unit is quoted on each side, a bid at $y - s/2$ and an ask at $y + s/2$; the two quotes may belong to one trading firm or to two different firms (Proposition 1(i), p. 1583). The authors describe play with one firm in the liquidity-provider role for simplicity, noting economically equivalent equilibria in which the role rotates or is random (pp. 1579–1580). When y jumps by more than $s/2$, the firm whose quote is stale sends a cancel while the other $N-1$ firms send orders to trade against ("snipe") it, so the stale quote is sniped with probability $(N-1)/N$ (Proposition 1(iii), p. 1583). The spread equates investors' payments with the rents from sniping:

$$\lambda_{\text{invest}} \frac{s}{2} = \lambda_{\text{jump}} \Pr\Big(J > \frac{s}{2}\Big) \mathbb{E}\Big[J - \frac{s}{2} \;\Big|\; J > \frac{s}{2}\Big]$$

(eq. (3), p. 1582), which does not depend on N. The spread is positive with symmetric information and no inventory costs. Sniping costs grow linearly with the quantity quoted, while the benefit of quoting more grows less than linearly, so the spread widens unit by unit and the book is thin (Proposition 2, pp. 1583–1585). This assumes investors trade their whole quantity at once, which the authors note is not optimal for them. When firms can rent speed at c_{speed} per unit time, the zero-profit conditions for the liquidity provider and the snipers (eqs. (5)–(6), p. 1589) give

$$\lambda_{\text{invest}} \frac{s^*}{2} = N^* c_{\text{speed}}$$

(eq. (7), p. 1590): the total prize, investors' cost through the spread and the total spent on speed are equal, so the rents are dissipated by the race (Proposition 3, pp. 1590–1591; page images of pp. 1589–1590). The race is a prisoner's dilemma: welfare would be higher by $N^* c_{\text{speed}}$ if the firms could commit not to invest (Proposition 4, p. 1591). The prize rises with the frequency of jumps and with a mean-preserving spread of their distribution, and does not depend on the cost of speed, the speed of the fastest firms or the speed difference (Proposition 5, p. 1592).

Notation. $\lambda_{\text{jump}}, \lambda_{\text{invest}}$ are Poisson arrival rates, not impact slopes; s is the full bid–ask spread in price units; N is the number of fast trading firms; c_{speed} is a rental cost per unit time. Starred values are equilibrium values with endogenous entry.

Conditions. The assumptions are deliberately strong, a "best case" for the continuous book: a perfectly correlated public signal, costless liquidation, no asymmetric information or inventory costs, unit investor demand, prices on a continuum, and equally fast fast firms (pp. 1576–1579). With a discrete tick the liquidity-provider role is strictly preferred and firms race for the top of the book instead (n. 19, p. 1580). The authors do not claim that markets are less liquid than before electronic trading and HFT, which they say the evidence contradicts. Their claim is that markets are less liquid than under a design without sniping, the frequent batch auctions they propose (n. 5, pp. 1554–1555; abstract, p. 1548). Evidence (sec. I, pp. 1552–1553, summarising sec. V, which was not read): ES futures and SPY, 2005–2011, exchange direct-feed data. In 2011 the median return correlation was 0.1016 at 10 ms and 0.0080 at 1 ms. The median duration of ES–SPY arbitrage opportunities fell from 97 ms in 2005 to 7 ms in 2011, while the median profit, about 0.08 index points per unit traded, was roughly constant, and their frequency moved with volatility. The ES–SPY prize averaged $75 million a year. The authors call extrapolation to other instruments "common sense" and decline to give a total.

Practice. With speed costs that differ across firms, the marginal fast firm earns zero profit and infra-marginal firms earn strictly positive profits; total sniping rents then exceed total spending on speed (n. 27, p. 1590). Lehalle and Laruelle describe a related competitive margin. High-frequency market makers extended coverage to less liquid stocks once their hedging was built and their technology, a fixed cost, was paid; growing competition between them "forced them to diversify their sources of revenue". Below a liquidity threshold, hedging and short-selling costs keep them out [`alpha-laruelle`, sec. 2.4.2, pp. 161–162]. This is their interpretation of coverage statistics for European indexes, 2009–2012, not a test.

Pitfalls. The legacy principle said that speed advantages are valuable only while competitors cannot replicate them, and that once speed diffuses, durable profits must come from inventory control, better models, lower costs or privileged order flow. In this model and its ES–SPY evidence, competition does not shrink the arbitrage prize; it raises the speed needed to win it, and under free entry the rents go into speed spending, which investors bear through the spread. What erodes is a firm's *relative* lead: at the margin, profit net of speed costs is zero (Interpretation of eqs. (5)–(7), checked algebraically and numerically). Lower costs of speed do sustain profits (n. 27). Inventory control, better models and privileged order flow are not in the passages read and remain unsupported legacy assertions, kept in the artefact. The empirical constancy is for one pair, 2005–2011. The pathway's evidence on races at a UK venue (`exp-aquilina-budish-oneill-races`) and on HFT profits and competition (`wide-baron-hft-competition`) is unread.

See also. HFT strategies; spread against investor cost; Glosten–Milgrom (adverse selection from private information, against sniping on public information); arbitrage; smart order routing (latency and mirrored liquidity).

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. The frequent-batch-auction analysis is in secs. VII–VIII of the paper, not read for this entry.

Index. latency arbitrage; stale-quote sniping; continuous limit order book; serial processing; arms race; speed technology; co-location; prisoner's dilemma; rent dissipation; free entry; ES–SPY arbitrage; correlation breakdown; frequent batch auctions; Budish–Cramton–Shim

Status. Verification: partly checked 2026-10-15 (against the authors' open-access published version, *Quarterly Journal of Economics* 130(4), 2015, hosted by the University of Maryland, text layer of pp. 1547–1556 and 1576–1592, page images of pp. 1589–1590; quote ownership in Proposition 1(i), p. 1583, rechecked in slice 16; eqs. (3) and (5)–(7) rederived and checked numerically; and the text layer of `alpha-laruelle`, 2018 printing, pp. 161–162). Stability: draft. Currency: model timeless; evidence ES and SPY, 2005–2011.

Result 8.52 A narrower spread need not mean lower costs: Lehalle and Laruelle's stylised model of a fast trader quoting inside the book

Statement. Lehalle and Laruelle's stylised book [`alpha-laruelle`, sec. 2.4.3, pp. 164–165, Figure 2.21, page images]. Two traders, A and B, rest at the best bid and ask, a spread ψ apart, and take turns becoming impatient. Each pays ψ when crossing and nothing when the other crosses, so each pays $\psi/2$ per trade on average. A high-frequency trader then quotes just inside B, so that A, crossing, pays $\psi/2+\epsilon$; it at once steps in front of A, and B pays $\psi/2+\epsilon$ too. The effective spread falls to a little over $\psi/2$, but A and B now pay $\psi/2+\epsilon$ on every trade, because each must cross every time, while the fast trader pays nothing. In the authors' words, "the effective spread is reduced on an overall basis, but ... costs for other participants can remain unchanged or slightly increased" (p. 164). In the calibrated version, ψ_B is the spread before HFT and ψ_{HFT} the tighter one an HFT quotes. An investor hitting an HFT pays $\psi_B/2+\epsilon$ with $\epsilon=\psi_{\mathrm{HFT}}/2$, and if HFTs are in a fraction q of trades the spread becomes $\psi_N=q\psi_{\mathrm{HFT}}+(1-q)\psi_B$ (p. 165). An investor posting one order meets an HFT with probability r and then pays $(\psi_{\mathrm{HFT}}+\psi_B)/2$. Otherwise, with equal probability, the investor crosses and pays ψ_B or stays passive and pays 0, so

$$\mathbb{E}[C]=\frac{\psi_B+r\,\psi_{\mathrm{HFT}}}{2},\qquad \mathrm{Var}(C)=r\Big(\frac{\psi_{\mathrm{HFT}}+\psi_B}{2}\Big)^2+(1-r)\frac{\psi_B^2}{2}-\mathbb{E}[C]^2$$

(Figure 2.22, p. 166, page image). The average cost of N independent trades has the same mean and variance $\mathrm{Var}(C)/N$ (p. 168). Meeting HFTs more often raises the expected cost and lowers its dispersion. The dispersion falls a lot for an investor with few trades, such as a retail investor, and little for one with many, such as a pension fund (pp. 167–169, Figures 2.23–2.24).

Notation. ψ is a spread in basis points of price; C is one order's cost. Interpretation: in the figure each trader's cost is measured from its own resting quote, so a passive fill costs 0 and a crossing the full spread. C is therefore a cost against waiting at one's own quote, not the half-spread against the midpoint of spread measures.

Conditions. Calibration: $\psi_B=5.41$ bp (April to November 2007) and $\psi_N=4.73$ bp (April to November 2011), for CAC 40 stocks on the primary market, Euronext Paris, only, with $q=40$ percent assumed (pp. 165–166, page image of p. 166). Interpretation, recomputed with exact arithmetic: these imply $\psi_{\mathrm{HFT}}=(\psi_N-(1-q)\psi_B)/q=3.71$ bp. $\mathbb{E}[C]$ then runs from 2.705 bp at $r=0$ to 4.56 bp at $r=1$, which matches the ends of Figure 2.24, and is 3.26 bp at $r=30$ percent and 3.63 bp at $r=50$ percent. The model attributes the whole fall in the primary-market spread between 2007 and 2011 to HFT and holds ψ_B fixed, as if the earlier spread would otherwise have persisted. It fixes q rather than measuring it, ignores trading moved to other venues, and assumes that an investor who meets an HFT must cross while others cross half the time. The two spreads are observations; their attribution to HFT and the cost model are the authors' stylised interpretation, not an estimate of HFT's effect.

Practice. The legacy caveat's advice is consistent with this: judge fast liquidity provision by what other

traders pay (effective and realised spreads, spread measures), by depth, adverse selection and behaviour under stress, not by quoted spreads or message counts. The criteria are the legacy note's (Interpretation). The SEC's 2010 release asked these as open questions. Are orders lasting a second or less "phantom liquidity that disappears when most needed", or do many firms quoting together give a stable market with many size updates but few changes in the NBBO price? Do proprietary firms, which have replaced specialists and market makers with affirmative and negative obligations, supply liquidity in times of stress [`sec-concept-release-2010`, sec. IV.B.1, pp. 48–51]? In Budish, Cramton and Shim's model the cost of being sniped is priced into the spread and grows with the depth shown (latency-arbitrage arms race). Lehalle and Laruelle add that HFTs' preference for small trades makes other traders split orders and bear more reporting and technology costs, "fixed losses" unevenly distributed (p. 165); this is an argument, not a measurement.

Pitfalls. The text says that raising the probability of meeting an HFT from 30 to 50 percent "increases the expected cost by approximately 0.5bp" (p. 168). The book's own formula and calibration give $0.2 \times 3.71/2 = 0.37$ bp, and Figure 2.24's markers sit at about 3.27 and 3.64 bp, which agrees with 0.37 (page image; Interpretation). Its "by 20%" means 20 percentage points. The legacy caveat's "high message rates can improve displayed prices" and "can also increase cancellation risk and stale-book uncertainty" are, in the passages read, a stylised mechanism and open questions, not measured effects. The pathway's empirical readings on liquidity and price discovery (`wide-hendershott-jones-menkveld`, `wide-brogaard-hft-discovery`) are unread. The model's conclusion concerns expected cost per trade under its assumptions; it is not a finding that HFT raised investors' costs.

See also. HFT strategies; latency-arbitrage arms race; spread measures and spread determinants (owned by `execution-market-making`); smart order routing (displayed liquidity is only probable); order anticipation (quote matching steps ahead of passive traders).

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. effective spread; execution cost; HFT and spreads; quote improvement; stepping ahead; cost uncertainty; retail and institutional costs; phantom liquidity; fleeting quotes; cancellation rates; affirmative obligations; market quality; CAC 40 spreads 2007–2011

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, sec. 2.4.3, pp. 163–169, page images of pp. 164–166 and 168, formulas and calibration recomputed; and the SEC concept release, pp. 48–51). Stability: draft. Currency: model timeless; calibration Euronext Paris, 2007 and 2011; release as of January 2010.

8.13 Floor and automated trading systems

Definition 8.53 Floor against automated trading systems: two technologies for the same order-driven auction, compared on fairness, access, capacity, speed, information and cost (as of 2002)

Statement. Floor-based oral auctions and automated rule-based auctions are both order-driven markets that match buy and sell orders by very similar rules. They differ mainly in technology: in oral auctions traders exchange information in person, in automated markets computers arrange the trades [`alpha-harris`, chap. "Floor Versus Automated Trading Systems", p. 544]. This entry groups Harris's comparison under six headings; his own sections are fairness, the convenience of distributed access, system capacity, negotiation speed, exchange of information on trading floors, market data reporting and cost (secs. 27.1–27.7, pp. 544–552), and his Table 27-1 lists eleven issues (p. 552, page image):

- *fairness*: automated systems apply their rules exactly and produce complete audit trails; oral auctions depend on traders' skill and honesty and need surveillance. Floor traders see and react first, and in pits even size and voice matter, while automated markets favour traders with computer-generated orders (pp. 544–546);
- *convenience*: distributed access from traders' desks (pp. 546–547);
- *capacity*: electronic matching scales; oral auctions become disorderly "fast markets" when too many traders take part (p. 547);
- *speed*: automated systems complete and report trades faster, but oral negotiation of size is faster than splitting firm orders, which undisclosed orders and negotiation messaging partly address (pp. 547–548);
- *information exchange*: floors win when traders need to know who the counterparty is and whether more size is coming, to avoid informed or price-discriminating traders, and when brokers know of latent interest (pp. 548–549). Automated systems report market data faster and more accurately (p. 549);

- *cost*: both have high initial creation costs; screen-based systems have low operating costs, their main operating costs being telecommunications and data backup. Floor systems are labour-intensive, needing floor brokers, reporters, officials and runners, and well-trained floor brokers are expensive (sec. 27.7, pp. 549–552).

Conditions. Automated trading also needs heavy investment in reliability, security and speed (pp. 549–551; operational controls for automated trading). Automated markets exploit economies of scale, which drives exchange mergers. Because of the order-flow externality, the survival of large floor markets does not prove that floor technology is better (p. 553). Harris expects automated systems to serve active markets and small traders, and floors large institutions that need to exchange information (pp. 552–553).

Practice. Harris's examples of competition, such as the Paris Bourse against London and the DTB against LIFFE, did not decide the question, because other events, a repealed French stamp tax and German banks repatriating their market, also moved trading (p. 543).

Pitfalls. Both systems have high initial creation costs in absolute terms (p. 549), and Table 27-1 rates the automated system's software costs as relatively lower than the cost of building a floor and training its staff (p. 552); the two statements are compatible. The legacy mechanism's "high fixed costs for software, data networks, security, and recovery, but low marginal labour cost per order" agrees with the text. Its claim that floors remain valuable "when negotiation, discretion, and soft information dominate raw matching speed" paraphrases sec. 27.5. The market facts (the NYSE and most U.S. options exchanges still floor-based, the firms and systems named) are as of 2002 and are not current.

See also. order exposure decisions (negotiating size and undisclosed orders); block trades; best execution.

Further reading. The legacy block it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/model-governance-controls/`.

Index. floor trading; oral auction; automated trading system; electronic trading; fairness; fair access; audit trail; distributed access; scalability; fast market; negotiation of size; undisclosed orders; information exchange; market data reporting; trading system costs; economies of scale; order-flow externality

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, chap. 27, pp. 543–554, page image of p. 552 with Table 27-1). Stability: draft. Currency: historical, as of 2002.

8.14 Spot FX dealing: market structure, internalisation and last look

Definition 8.54 Spot FX market structure: venues and access channels, trading protocols, firm and non-firm liquidity, and principal and franchise liquidity

Statement. The FX market is fragmented and has no consolidated tape of prices or volumes [exp-fx-algos, sec. 2.2, p. 22]. Cartea, Jaimungal and Walton add that there is no central exchange framework. The most common G10 currencies may trade on more than 20 electronic crossing networks (ECNs), and most major banks also offer trading through their own platforms, applications or APIs [dm-cartea-jaimungal-walton-2019, arXiv version, sec. 1, p. 2]. Four things should be kept apart:

- *Venues and access channels.* The primary electronic venues (EBS Market, Refinitiv Spot Matching, and CME for some pairs) supply the reference prices most algorithms use. Their market share had dwindled as alternative ECNs and liquidity pools emerged; clients reach liquidity by voice or chat, APIs, single-dealer and multi-dealer platforms [exp-fx-algos, pp. 12, 22–23].

- *Trading protocols.* In a "risk transfer" the client requests a price, by request for quote (RFQ) or request for stream (RFS), and trades its full ticket at the price received, paying the spread for immediacy; RFQ systems differ in whether the side is revealed, who receives the request and whether quotes are executable or indicative (pp. 5 and 43). Orders may instead be worked by execution algorithms across venues (sec. 2).

- *Firm and non-firm liquidity.* Anonymous pools based on a central limit order book typically match "firm liquidity", without the option to reject. Disclosed pools allow liquidity to be partitioned by counterparty and typically include "non-firm liquidity", with a short window in which the provider may reject ("last look"); some ECNs offer both (Box D, p. 23).

- *Principal and franchise liquidity.* In principal pools the provider takes the position on its market-making desk. In franchise pools it uses one client's flow to offset another's, *internalising* rather than hedging externally (Box D, p. 23).

Internalisation is a dealer's risk-management choice, not a venue. A dealer may *internalise* a client trade by warehousing the risk until offsetting flow arrives, or *externalise* it by hedging in the open market [cfd-butz-oomen-2019, accepted version, abstract and sec. 1, p. 1]; see FX internalisation horizon.

Conditions. The BIS figures are a central-bank study group's survey and interviews as of 2020, not measurements of the whole market. The largest dealers report internalisation ratios "as high as 90%", but the term has several definitions, from matching only between users of one platform to including flow routed to a dealer's own desk or to mid-matching venues, so reported ratios are not comparable (Box D, p. 23). Visible depth can mislead: trading interest posted on several venues at once can create a "liquidity mirage", which the report expects last look to exacerbate (p. 27).

Practice. Interpretation: routing in FX must weigh firmness and internalisation as well as price; the BIS describes routers that consider quoted size, queue length, fill ratios and fill probability, and that assume lower price impact for internalised orders, an assumption checked through transaction cost analysis (p. 24; compare smart order routing, from European equities). Transaction cost analysis in FX compares traded prices with benchmarks at order arrival, during and after execution, and measures slippage, market impact and the ratio of rejected trades, without a consolidated tape to supply the benchmark (Box B, p. 11).

Pitfalls. The legacy principle listed prime-brokered access and bilateral relationships among the places liquidity sits, and its taxonomy listed "fixing windows" as a venue type. Prime brokerage is not in the passages read. Fixing windows are a matter of benchmark timing, not a venue; they belong to the pathway `fx-funding-settlement` and are left to it. The legacy taxonomy's RFQ "for size or less liquid pairs" is a plausible use, but the source describes RFQ as a protocol for full-size risk transfer without restricting it to those cases. Venue names and market shares are as of 2020.

See also. FX internalisation horizon; last look; smart order routing and best execution (equity sources, not FX evidence); dealer inventory risk.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/fx-dealer-market-making/`. King, Osler and Rime (`wide-king-osler-rime`), the pathway's survey of FX market structure, was not read.

Index. FX market structure; spot FX; fragmentation; consolidated tape; primary venues; EBS; ECN; single-dealer platform; multi-dealer platform; request for quote; request for stream; risk transfer; firm liquidity; non-firm liquidity; disclosed pools; anonymous pools; principal liquidity; franchise liquidity; internalisation; externalisation; liquidity mirage

Status. Verification: partly checked 2026-10-15 (against `exp-fx-algos`, BIS Markets Committee report of October 2020, pp. 5, 11–13, 21–24, 27–28 and 43, text layer; `dm-cartea-jaimungal-walton-2019` in arXiv 1806.04460v1, p. 2; and `cfd-butz-oomen-2019` in its accepted version, pp. 1–2). Stability: draft. Currency: market structure as described in 2018–2020.

Model 8.55 FX internalisation horizon: how long a dealer takes to offset client risk internally, and what position skewing costs (Butz and Oomen)

Statement. Butz and Oomen model internalisation as a queue: each client trade either increases or decreases the dealer's risk position, and the *internalisation horizon* is the time a risk-increasing trade waits until risk-decreasing trades offset it. To control the build-up of risk the dealer *skews* its prices to attract risk-reducing flow. By Little's law the average horizon is proportional to the ratio of the dealer's risk limit to the rate of client flow [`cfd-butz-oomen-2019`, accepted version, sec. 1, p. 2]. Precisely, with φ_n the stationary distribution of the dealer's position n (in units of trade size) and $\lambda_n^{\pm}$ the arrival rates of buys and sells at position n, the expected queue length and the arrival rate of risk-increasing trades are

$$L = \sum_n \varphi_n |n|, \qquad \lambda = \sum_{n \geq 0} \varphi_n \lambda_n^+ + \sum_{n \leq 0} \varphi_n \lambda_n^-$$

(eq. (8), p. 7, page image). The average horizon of a *risk-increasing* trade is L/λ. Risk-reducing trades are internalised at once, with zero wait, so the average over *all* trades is $W = \frac{1}{2} L/\lambda$ (eq. (9), p. 7). In the skew specifications, λ_0 is the baseline intensity of buys and of sells, not λ. Under a binary skew $\lambda_n^+ + \lambda_n^- = 2\lambda_0$, with a risk-reducing tilt α beyond the threshold R (Example 1). Under an exponential skew $\lambda_n^+ = \lambda_{-n}^- = \lambda_0 e^{-n/(2R^2)}$, so that the position is normal with standard deviation R (Example 2, eqs. (5)–(7), pp. 5–6, page images). Proposition 1 then gives the all-trade average

$$W = \frac{R}{\lambda_0} \frac{c_R}{4} \text{ (binary)}, \qquad W \approx \frac{R}{\lambda_0} \frac{c_R}{\sqrt{2\pi}} \text{ (exponential)}, \qquad c_R \to 1 \text{ as } R \text{ grows},$$

with c_R given in the proposition (eqs. (10)–(11), p. 8, page image). The average is the same whatever rule the dealer uses internally to match offsetting trades, provided its position starts and ends at zero (p. 8).

Notation. Positions are counted in unit trade sizes; in the calibration a unit is \$1 million. R is the risk limit (or skew scale) in those units. λ_0 is the baseline intensity of buys, and of sells, in units per minute, and λ the resulting arrival rate of risk-increasing trades. L/λ is the conditional horizon of a risk-increasing trade and $W = L/(2\lambda)$ the all-trade average, both in minutes.

Conditions. The calibration is top-down, from the BIS 2016 triennial survey rather than dealer data.

Spot turnover of $1,652 billion a day is reduced to customer flow, a representative tier-1 dealer is given a 6.6 percent share, and its risk limit is set so that its position stays within $25 million 95 percent of the time, $R = \$25\text{mn}/1.96 = \$12.75\text{mn}$. For EUR/USD, $\lambda_0 = \frac{1}{2} \times \$111\text{mn/min} \times 6.6\% = \$3.66\text{mn/min}$ of each type, giving the all-trade average $W \approx 12.75/3.66 \times 1/\sqrt{2\pi} \approx 1.39$ minutes, which the authors report as the average internalisation horizon (pp. 8 and 11, page image of p. 11). Interpretation, recomputed: a risk-increasing trade waits about twice as long on average, $L/\lambda \approx 2.78$ minutes. Summing the exact stationary distribution of Example 2 with these values gives $L/\lambda = 2.774$ and $W = 1.387$ minutes, so the approximation with $c_R \approx 1$ is close. Horizons are single-digit minutes for liquid G10 currencies and tens of minutes for most Asian and emerging-market currencies and some less liquid G10 ones (p. 11, Figure 6), and vary predictably over the day.

Practice. Smaller dealers, or those with less risk appetite, can shorten horizons by skewing more aggressively. That has a direct cost, less spread captured on risk-decreasing trades, and an indirect one, "information leakage or signalling risk associated with skewing". Internalisation therefore runs on a continuum from passive to aggressive (p. 2). The authors' message is that a client's transaction costs and liquidity access depend on its own trading and on the dealer's risk management; clients should tell externalisers from internalisers, and passive from aggressive internalisers (abstract). The BIS report gives the user's side: internalisation can reduce information leakage and market impact, though dealers' skewing may reveal information and negate some of that benefit [exp-fx-algos, p. 27].

Pitfalls. The authors worked in Deutsche Bank's electronic FX spot division, as the paper discloses in a footnote on its abstract page. The horizons are model outputs from survey aggregates, not observed internalisation times; the accepted manuscript was read, not the published version. Generic dealer-inventory theory (dealer inventory risk, dealer quote adjustment, from Harris's equity and floor examples) motivates skewing but is not FX evidence.

See also. spot FX market structure; inventory-based quotes; dealer quote adjustment.

Further reading. The legacy item it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/fx-dealer-market-making/`.

Index. internalisation; externalisation; internalisation horizon; Little's law; queuing theory; position skewing; risk limit; client flow; passive internaliser; aggressive internaliser; tier-1 dealer; BIS triennial survey

Status. Verification: partly checked 2026-10-15 (against `cfd-butz-oomen-2019` in the accepted version on LSE Research Online, abstract and pp. 1–2, 8 and 11, page images of pp. 8 and 11, calibration recomputed; pp. 5–7 with page images, the conditional and all-trade horizons separated in a correction the same day; and `exp-fx-algos`, p. 27). Stability: draft. Currency: model timeless; calibration from the 2016 BIS survey.

Mechanism 8.56 Last look in spot FX: the provider's final accept or reject window, its validity and price checks, and how it differs from firm liquidity

Statement. The FX Global Code defines *last look* as a practice in electronic trading "whereby a Market Participant receiving a trade request has a final opportunity to accept or reject the request against its quoted price". If used, it should be a risk control that checks *validity*, the operational details and sufficient credit, and/or *price*, whether the requested price remains consistent with the current price. The provider has sole discretion, leaving the client with market risk if the request is rejected [exp-fx-code, Principle 17, pp. 22–23, page images]. Under the Code providers should disclose:

- whether and how price changes in either direction affect the decision;
- the expected or typical period of time for making the decision, the duration of the last-look window;
- the purpose of last look.

They should not use it to gather information with no intention of accepting, and should not trade on the information in the request during the window, whether by pricing on electronic platforms or by hedging. The guidance does not apply to an arrangement that has *all* of three features (p. 23, page image):

1. an explicit understanding that the provider will fill the client's request without taking on market risk in connection with it, by first entering into offsetting transactions in the market;
2. the volume traded in the last-look window is passed on to the client in its entirety;
3. this understanding is appropriately documented and disclosed to the client.

Such "cover and deal" arrangements are the Code's only exception to the rule against trading in the window.

Conditions. The Code is voluntary: it "does not impose legal or regulatory obligations on Market Participants nor does it substitute for regulation" (p. 1); the edition read is the update of December 2024. Contractual practice is in each provider's disclosures. In one example, completed on 18 September 2025, Deutsche Bank reports for its wholesale FX business:

- for spot FX, a price check of up to 10 milliseconds, which may be shorter where latency and location matter less, and no additional hold time beyond the time the price and validity checks take; the validity check's own duration is not bounded in the sheet and depends on networks, systems, client connection, liquidity, time of day and product, and the window may differ for products other than spot;
- a validity check covering counterparty-exposure limits, credit checks and currency-pair permissioning;
- symmetric and asymmetric price checks, with the asymmetric check as its default, since it causes fewer rejections;
- that technical problems or disrupted markets may lengthen the window, and that acceptance rates, price-check durations and pricing may differ between customers;
- no trading during the window except under agreed cover-and-deal arrangements.

[db-fxgc-lp-cover-2025, sec. C.4]. These are one provider's self-reported terms, not market facts.

Practice. Keep apart, Interpretation using the sources above:

- a *firm* quote, matched without rejection, as in anonymous order books, from a *conditional* (non-firm) quote subject to last look [exp-fx-algos, Box D, p. 23];
- a *price-check rejection* from a *validity* rejection for credit, exposure or permissioning, and both from a *technical failure* that lengthens or breaks the process;
- the *decision time*, the duration of the last-look window, which the Code asks to be disclosed. Within it, the time spent on the price and validity checks differs from any *additional hold* beyond them, which this provider says it does not apply;
- the client's measures: fill and reject rates, and markouts, prices after the trade compared with the trade price. FX transaction cost analysis includes the ratio of rejected trades and benchmarks after execution [exp-fx-algos, Box B, p. 11].

In theory, with risk-neutral providers who recover losses to latency arbitrageurs from other traders, last look lowers the quoted spread on a single venue. With two venues an equilibrium may keep both or only one, and a last-look venue may even quote wider spreads than a firm one [dm-cartea-jaimungal-walton-2019, arXiv version, abstract]. This is a model prediction, not an empirical finding.

Pitfalls. The legacy definition said last look "reduces stale-quote risk for the provider" and creates "execution uncertainty and potential information-leakage concerns for the client". The first is the model's premise; the second matches the Code's concern with information use and the client's market risk on rejection. The legacy caveat's list (credit, prime-broker limits, rejection rates, hold times, skewed streams, information leakage), and its claim that these "can dominate fill quality", is not established by the passages read. Credit appears as part of the validity check; prime-broker limits were not checked; no source read measures how much each item contributes, so "can dominate" is dropped. The latency-arbitrage evidence of Budish, Cramton and Shim concerns ES and SPY, not FX (latency-arbitrage arms race). Oomen's papers on last-look design and on aggregators (`dm-oomen-last-look-2017`, `dm-oomen-aggregator-2017`) were not available and are unread.

See also. spot FX market structure; latency-arbitrage arms race; best execution.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/fx-dealer-market-making/`.

Index. last look; FX Global Code; Principle 17; validity check; price check; symmetric last look; asymmetric last look; hold time; reject rate; fill rate; markout; cover and deal; firm liquidity; non-firm liquidity; information leakage; latency arbitrage

Status. Verification: partly checked 2026-10-15 (against `exp-fx-code`, FX Global Code updated December 2024, pp. 1 and 22–23, page images of pp. 22–23; the Deutsche Bank cover sheet completed 18 September 2025, sec. C.4; `exp-fx-algos`, pp. 11, 23 and 27; and `dm-cartea-jaimungal-walton-2019`, arXiv v1, abstract and p. 2). Stability: draft. Currency: the Code as of December 2024; provider practice as of September 2025.

8.15 Options dealing: market-maker spreads, cost measurement and retail routing

Result 8.57 Option market-maker spreads: discrete hedging, volatility uncertainty, inventory and informed trading, priced across linked contracts

Statement. An option's value depends on the underlying price, volatility, rates, dividends, strike and maturity, so the quotes on one underlying form a linked set of contracts, and a market maker who sells one option holds risks that it can hedge in the underlying or in other options. Four findings, each from a different sample or model, separate the risks that this linkage leaves the market maker:

- *Hedging and volatility risk.* For a sample of Chicago Board Options Exchange options, the risks of being unable to rebalance an option hedge continuously and of uncertainty about the underlying's return volatility "each account for a statistically and economically significant proportion" of quoted bid–ask spreads, risks the authors call unique to options [ex-jameson-wilhelm-1992, abstract].

- *Inventory risk.* Decomposing the price impact of option trades into inventory-risk and asymmetric-information components, both are large and the inventory component is larger; order imbalances attributable to inventory risk have "five times larger impact on option prices than previously thought", and past imbalances predict option returns better than other common predictors [wide-muravyev-order-flow, abstract].

- *Informed trading and the hedging instrument.* In Huh, Lin and Mello's model, market makers who hedge against adverse selection widen spreads in both the option and the stock, more in the option, and more when hedging with the underlying than with other options; hedging also changes informed traders' strategies. Their empirical tests are reported as consistent with the model's key implication [ex-huh-lin-mello-2015, abstract].

- *Who supplies quotes.* In the U.S. listed-options market that Bryzgalova, Pavlova and Sikorskaya study, orders are executed on exchanges, and retail orders are typically filled by a wholesaler's affiliated market maker [ex-bryzgalova-pavlova-sikorskaya-2023, published version, pp. 2 and 11]; the routing is in retail option order routing.

Conditions. The first three are abstract-level checks of the published papers; their samples, periods, contracts and estimates were not read. Jameson and Wilhelm's evidence is for CBOE equity options in a sample period not read (the paper was published in 1992); Huh, Lin and Mello's result is a model prediction whose empirical support is stated only as consistency with one implication. None of them says how large each risk is in today's markets.

Practice. Interpretation, keeping the risks apart:

- *hedging cost*: the cost and residual risk of trading the hedge, discretely, in the underlying or other options;

- *volatility risk*: exposure to changes in volatility, which a delta hedge does not remove;

- *inventory risk*: the position a market maker accumulates from net order flow, priced through quote adjustment as in equity dealer models (dealer inventory risk, dealer quote adjustment);

- *information risk*: trading against better-informed traders, in the option or through the hedge (adverse-selection spread component).

A linked set of prices, an implied volatility surface, does not mean liquidity is uniform across it: liquidity is measured contract by contract (option spread measurement), and the liquidity of an option differs from that of the instrument used to hedge it.

Pitfalls. The legacy principle's "liquidity therefore lives on a surface, not in one instrument" is kept as interpretation only: the papers read link spreads to hedging, not liquidity to a surface. The legacy caveat's claim that fragmentation is "constrained by clearing, exercise, assignment, margin, and position-risk management" was not checked: no clearing-house or exchange rule was read, and Battalio, Hatch and Jennings's study of multiply listed equity options contrasts June 2000, when quotes were often ignored, with January 2002, when the market more closely resembled a national market; it does not address clearing [ex-battalio-hatch-jennings-2004, abstract]. Its "economics are dominated by cross-contract risk" is not measured by any source read. Of the taxonomy, "hedging cost transfers into option spreads" is supported in direction by the hedging findings above; "high uncertainty widens quotes" is supported cross-sectionally by Jameson and Wilhelm's volatility-uncertainty result, and "reduces size" is not checked. Generic dealer models from equity markets motivate but do not establish option claims, nor do the futures EFRP (exchange of options for options), portfolio trading or ETF entries.

See also. spread determinants; inventory-based quotes; option spread measurement; retail option order routing.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/options-execution/`.

Index. option market making; options bid–ask spread; discrete hedge rebalancing; volatility uncertainty; inventory risk; option order imbalance; asymmetric information; informed trading in options; hedging with the underlying; hedging with options; implied volatility surface; multiply listed options

Status. Verification: partly checked 2026-10-15 (for `ex-jameson-wilhelm-1992`, the abstract only; for `wide-muravyev-order-flow`, the abstract only; for `ex-huh-lin-mello-2015`, the abstract only; for `ex-battalio-hatch-jennings-2004`, the abstract only; these four from the publishers' records as served by RePEc, Crossref and OpenAlex, with the full texts unread; and for `ex-bryzgalova-pavlova-sikorskaya-2023`,

the published open-access version, pp. 2 and 11). Stability: draft. Currency: findings as of their samples, from the CBOE before 1992 to U.S. options in 2019–2021.

Method 8.58 Measuring option execution costs: quoted against effective spreads, premium units, execution timing, and segmentation by moneyness, expiry and trade type

Statement. Option spreads are usually measured relative to the option's premium. In Bryzgalova, Pavlova and Sikorskaya's U.S. data, the *quoted spread* is the best ask minus the best bid across all exchanges relative to the midpoint at the time of the trade, and the *effective spread* twice the absolute percentage deviation of the trade price from that midpoint, both frequency-weighted [ex-bryzgalova-pavlova-sikorskaya-2023, published version, Table I, p. 15, page image]. These are the equity measures of spread measures applied per contract. Two corrections apply in options:

- *Benchmark and timing.* Option price changes are predictable at high frequency, and many traders time executions, buying when the option's fair value is close to the ask and selling when it is close to the bid. Effective spreads of such traders are less than 40 percent of conventional measures, the overall average effective spread is one-quarter smaller than conventional estimates, and price-impact measures are also affected [wide-muravyev-pearson, abstract]. Interpretation: the midpoint then overstates the cost of a timed trade.

- *Units.* A spread in percent of premium is not a spread in volatility units: a cheap out-of-the-money option can have a large percentage spread and a small spread in implied volatility (Interpretation).

Conditions. The segmentation figures are from Table I of Bryzgalova, Pavlova and Sikorskaya, all U.S. listed-option trades reported to OPRA from 4 November 2019 to 30 June 2021, split into SLIM trades, those executed in single-leg non-ISO price-improvement auctions and used as the authors' retail proxy, and non-SLIM trades, the whole remaining pool, institutional and retail, single- and multi-leg (pp. 7, 11–12 and 15–16, page images):

- *moneyness*: at-the-money trades are about 70 percent of trades in both groups, with quoted spreads of 8.7 percent (SLIM) and 8.4 percent (non-SLIM), far below the 28.7 and 21.3 percent of slightly out-of-the-money trades (-1 to -0.1); they are not the lowest, since slightly in-the-money trades (0.1 to 1) have 8.6 and 5.9 percent. Table I writes moneyness for calls as (MidpointPrice − Strike)/Strike, with the opposite sign for puts. Interpretation: since it is compared with the strike, the midpoint here is the underlying's, not the option-premium midpoint used for the spreads; the caption does not say so explicitly;

- *expiry*: options with less than a week to expiration carry about half of SLIM volume and 41.5 percent of non-SLIM volume; their quoted spreads are 12.6 and 13.3 percent and their effective spreads 6.6 and 10.1 percent. Spreads are not monotone in maturity: 3–12 month options have SLIM quoted spreads of 18.5 percent but non-SLIM ones of 10.2 percent;

- *trade size*: SLIM trades below $250 have quoted spreads of 23.5 percent; trades above $20,000 have effective spreads above quoted spreads, consistent with price impact (p. 17);

- *totals*: quoted and effective spreads average 13.7 and 6.7 percent for SLIM and 11.6 and 8.6 percent for non-SLIM trades.

Practice. Report spreads per contract with the market, sample period, contract set and units, and separate:

- quoted from effective spreads, and both from timing-adjusted measures against a fair-value benchmark;

- outright from multi-leg trades: Bryzgalova, Pavlova and Sikorskaya restrict their measure to single-leg auctions and note that multi-leg auctions are a noisier retail measure (p. 12), so their results do not isolate the execution costs of multi-leg strategies: the non-SLIM pool mixes multi-leg trades with the rest, and no figure is reported for multi-leg executions as such;

- the option's liquidity from the hedge's: a narrow option spread can still carry a costly hedge, and the reverse.

Pitfalls. The legacy taxonomy's "at-the-money contracts are usually deepest" is supported only as trade share in one U.S. sample: at-the-money trades are about 70 percent of trades, and their percentage quoted spreads are far below those of slightly out-of-the-money trades but not the lowest of the moneyness bins. Depth, size at the quote, was not measured in the passages read. "Near expiries trade differently" is consistent with the expiry split but is not a finding. "Spreads and multi-leg trades may be more relevant than outrights" is not checked: no source read measures complex-order books or exchange complex-order rules.

See also. spread measures; option market-maker spreads; retail option order routing; best execution.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/options-execution/`.

Index. option quoted spread; option effective spread; relative spread in premium; spread in volatility units; execution timing; fair-value benchmark; moneyness; at-the-money; out-of-the-money; weekly options; time to expiry; single-leg; multi-leg; complex orders; option price impact

Status. Verification: partly checked 2026-10-15 (`ex-bryzgalova-pavlova-sikorskaya-2023`, published open-access version, pp. 7, 11–12 and 14–17 with page images of Table I, rechecked in a follow-up the same day; abstract of `wide-muravyev-pearson` from the publisher's record via RePEc and OpenAlex, full text unread). Stability: draft. Currency: U.S. options, November 2019 to June 2021; Muravyev and Pearson's sample not read.

Mechanism 8.59 Retail option order routing: payment for order flow, wholesalers and price-improvement auctions (U.S., 2019–2021)

Statement. In U.S. equities wholesalers internalise retail orders off exchange. In the authors' description of U.S. options, whose data are exchange-listed option trades reported to OPRA, "all options orders in the United States must be executed on exchanges", so internalisation was thought specific to equities (p. 11; "all options in the United States trade on exchanges", p. 2). Bryzgalova, Pavlova and Sikorskaya argue that much retail option flow is also effectively internalised [ex-bryzgalova-pavlova-sikorskaya-2023, published version, p. 2]. A retail broker routes the order to a wholesaler in exchange for payment for order flow (PFOF) and a promise of price improvement on the NBBO. The wholesaler's affiliated market maker brings the order, paired with its own contra side, to an exchange's price-improvement auction, where other participants ("responders") may offer a better price within a time window. Fees are stacked against responders: on most exchanges the affiliated market maker pays $0.05 a contract, while another market maker pays $0.50 to break up a paired order; in that responder-execution case the wholesaler receives a net rebate of $0.30 a contract for bringing the order (p. 11, footnote 15).

Conditions. Sample and measure: OPRA trades from 4 November 2019 to 30 June 2021, with the retail proxy built from the single-leg non-ISO price-improvement auction flag that OPRA introduced on that date (pp. 11–12). Some large SLIM trades are likely institutional; the authors check robustness on trades below $20,000 (p. 17). Retail trading reached over 60 percent of total option volume, and nearly 90 percent of PFOF came from three wholesalers (abstract), named as Citadel, Susquehanna and Wolverine, by the second quarter of 2021 (pp. 2 and 11). The fees are as the authors report them for their sample; no exchange fee schedule was read. The exchange-only statement concerns listed options. Over-the-counter (conventional) options, customised bilateral contracts, also exist, including ones with terms identical or substantially similar to listed options; FINRA's background notes that such trades are not generally centrally cleared nor reported to FINRA, the SEC, an exchange or the Consolidated Audit Trail [finra-rn-22-14, Background and Discussion]. That notice was a request for comment on a proposed regulatory-only reporting requirement, not an effective rule, and its later status was not checked.

Practice. What retail flow pays, in the authors' measures: SLIM trades had an average quoted spread of 13.7 percent and effective spread of 6.7 percent of the premium (Table I), so they received price improvement but paid spreads far above those in equities (p. 14). Aggregate trading costs, measured from trade price to midquote for all SLIM trades, were $6.4 billion over the sample, against about $900 million of commissions; at a 10-day holding horizon the aggregate retail portfolio lost $2.1 billion (pp. 4–5).

Pitfalls. The legacy mechanism said buy-side traders "typically consume liquidity" because coherent quoting needs specialised models and hedging infrastructure. The routing above shows retail market orders filled by wholesalers' market makers, and Muravyev and Pearson find that many traders time their liquidity-taking [wide-muravyev-pearson, abstract]; neither establishes who supplies or takes liquidity across the options market, or why, so the general claim and its explanation are not adopted. Payment for order flow and broker conflicts in equities are in broker conflicts; they do not establish option facts.

See also. best execution; broker conflicts; option spread measurement; option market-maker spreads.

Further reading. The legacy block it draws on, verbatim, is in `derivations.tex` in the folder `notes/artefacts/options-execution/`.

Index. payment for order flow; PFOF; wholesaler; retail option trading; price improvement auction; price improvement mechanism; SLIM; paired order; responder fee; affiliated market maker; internalisation in options; OPRA; retail trading costs

Status. Verification: partly checked 2026-10-15 (`ex-bryzgalova-pavlova-sikorskaya-2023`, published open-access version, abstract and pp. 2, 4–5, 11–12, 14–17, with page images of Table I, rechecked in a follow-up the same day; abstract of `wide-muravyev-pearson`; `finra-rn-22-14`, background only). Stability: draft. Currency: U.S. options, November 2019 to June 2021; exchange fees as the authors report them, not checked against any current schedule.

Chapter 9

Execution, trading costs and market structure

Pathways drawn on: the section overview `execution-trading-costs` (draft entries from the migration: execution benchmarks, implementation shortfall and the choice of an algorithm's benchmark, from Harris, Kissell and Lehalle and Laruelle, with Perold's paper not yet read; the architecture of automated trading tools, trading algorithms and their envelopes, tools for deciding and monitoring, decisions on order exposure, portfolio trading and the feedback from hedging rules, from Lehalle and Laruelle, Kissell and Harris; transaction costs, transaction cost analysis and the aggressiveness trade-off, from Harris, Kissell and Lehalle and Laruelle, with Narang's commissions and fees, slippage and impact mapped onto them; Narang's execution-algorithm choices, portfolio bids and trading infrastructure merged into the order-exposure, portfolio-trading and architecture entries; mean–variance optimal execution, from Lehalle and Laruelle and the Almgren–Chriss preprint; and block trades, from Harris; none of its readings yet studied); `market-structure-execution-quality` (draft entries from the migration: best execution and broker conflicts, from Harris and Kissell, as of 2002 and 2014; smart order routing, fee-adjusted venue choice and passive order splits, from Lehalle and Laruelle; none of the pathway's readings yet studied); and `price-impact-execution` (draft entries from the migration: what price impact is, the propagator model, the absence of price manipulation, the square-root law, the intraday and post-execution impact profiles, direct estimation of impact, the Glosten–Harris regression and index-reconstitution pressure, from Webster's handbook, Bouchaud, Bonart, Donier and Gould, Lehalle and Laruelle, Harris and Hasbrouck, and from four required readings checked in preprint versions, Almgren and Chriss, Bouchaud, Gefen, Potters and Wyart, Bacry, Iuga, Lasnier and Lehalle, and Almgren, Thum, Hauptmann and Li; none of the pathway's readings yet studied). Also drawn on is `etf-index-closing-execution` (one draft entry from the migration: ETF creation and redemption, from Lettau and Madhavan, in their working-paper version, Pan and Zeng and Harris; none of the pathway's readings yet studied). So is `futures-execution` (draft entries from the migration: CME Globex trading conventions, futures block trades and EFRPs, futures margin and settlement, from CME's rulebook, advisory notices and client documentation as retrieved on 15 October 2026 and Harris, and index-futures price discovery, from Hasbrouck's working paper; none of the pathway's readings yet studied). So is `otc-fixed-income-trading` (draft entries from the migration: fixed-income market structure, U.S. and EU bond transparency rules, and the evidence on bond transparency and trading costs, from the Bessembinder–Spatt–Venkataraman survey, three readings' abstracts, FINRA's rulebook and ESMA documents as retrieved on 15 October 2026; none of the pathway's readings yet studied). Slice 34 added a section on dark trading and fragmentation for `market-structure-execution-quality`: dark pools and crossing networks, venue market share, fragmentation and concentration metrics, a model of smart order routers and market shares, and European evidence after MiFID, from Lehalle and Laruelle and Harris, and extended the smart-order-routing, fee, order-exposure and block-trade entries. Slice 33 added two short sections: trade-sign classification, for `price-impact-execution`, from Harris and Hasbrouck; and tick size, its effects on market quality and Harris's discretised spread, for `market-structure-execution-quality`, from Lehalle and Laruelle, Bouchaud et al. and Harris (none of the pathway's readings yet studied). All notes for this section that predated the study method have been migrated (the last in slice 24); impact material written for the previous section is in Chapter 13, in the part Notes awaiting migration.

9.1 Execution benchmarks and implementation shortfall

Definition 9.1 Execution benchmark: a reference price for judging an execution, or the trading target of an algorithm

Statement. The word has two uses.

- A *benchmark price* P_B against which realised execution is measured. With average execution price $\bar{P}$ and Side $= +1$ for a buy and -1 for a sell, the cost in basis points is Side $(\bar{P} - P_B)/P_B \times 10^4$, positive when the execution was worse than the benchmark; the profit-and-loss form multiplies it by -1 [s7-kissellscience, chap. 3, eqs. (3.11)–(3.14), pp. 105–106]. Harris states it in currency: trade size times the signed difference, trade price minus benchmark for a purchase and benchmark minus trade price for a sale, $C = dq(\bar{P} - P_B)$ with $d = \pm 1$; across all buyers and sellers in a trade the estimated costs sum to zero, so liquidity suppliers tend to show negative costs [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.2, pp. 422–423]. Common benchmark prices are the quotation midpoint when the manager decided to trade (Perold's implementation shortfall, below), the price when the order arrived in the market, the volume-weighted average price (VWAP) of the day or of the order's own trading interval, a participation-weighted price, and the open, the close, the next day's open or close, or the average of the day's open, high, low and close [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 424–426]; [s7-kissellscience, chap. 3, pp. 106–108].

- The *target* of an algorithm class. A VWAP algorithm trades in proportion to the intraday volume curve; a TWAP algorithm trades equal amounts in equal time intervals, about 1/390 of a full-day US order per minute, often skipping the opening and closing auctions; a percentage-of-volume (POV) algorithm trades a fixed fraction of market volume until the order is done; arrival-price and implementation-shortfall algorithms balance expected cost against risk at a chosen risk aversion, the latter adding real-time adaptation, and their meaning differs across brokers [s7-kissellscience, chap. 1, pp. 16–18]. A liquidity seeker has no reference price: it is relative-price oriented, captures liquidity across pools and minimises information leakage [alpha-laruelle, sec. 3.4.1, Table 3.2, p. 230].

Notation. What each benchmark price fixes, from the passages checked (the order's own fills are s_j shares at p_j; the unfilled remainder is treated in implementation shortfall):

- *Decision price*: the quotation midpoint when the manager decided to trade [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 426]; in practice most managers use the previous day's close [alpha-harris, same chap., box "The Plexus Iceberg of Transaction Costs", p. 437]. Known before the order is worked.

- *Arrival price* P_0: the price when the order is entered in the market, the bid–ask midpoint in market hours, the last trade or official close otherwise [s7-kissellscience, chap. 3, p. 102]. Known when trading starts.

- *VWAP*: total traded value over total volume, each trade weighted by its size [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 425]. Full-day VWAP has no official value; vendors choose the eligible trades and may drop delayed or negotiated ones. Interval VWAP covers the order's own trading horizon, computed from tick data; VWAP to the end of the day judges orders finished early [s7-kissellscience, chap. 3, pp. 107–108]. Known only afterwards, and it includes the order's own trades.

- *Participation-weighted price* at rate ρ: the VWAP of the first (shares traded)$/\rho$ shares of market volume from the arrival time [s7-kissellscience, chap. 3, eqs. (3.15)–(3.16), p. 108]. Known only afterwards.

- *Open, close, later closes*: single daily prices; next-day and later prices are used to separate temporary from permanent impact [s7-kissellscience, chap. 3, p. 106].

Kissell's benchmark formulas use the average execution price without fees, so they are gross of explicit costs; his implementation shortfall adds fees [s7-kissellscience, chap. 3, eqs. (3.4)–(3.16), pp. 98–108]. Signs are not standard across the industry: some report a cost, positive when worse, others a gain, positive when better [s7-kissellscience, chap. 3, pp. 97 and 108].

Conditions. Harris compares the benchmark prices on four properties [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.4, pp. 427–432, Table 21-1, p. 433]:

- *noise*, which grows with the time between trade and benchmark: the contemporaneous midpoint is least noisy, open and close most;

- *split orders*: a benchmark moved by the order's own impact understates its cost. A trader who is the only buyer of the day matches VWAP whatever the impact (pp. 428–429);

- *momentum and contrarian trading*, which bias open and, less, VWAP benchmarks; *informed trading*, which makes the close and, less, VWAP understate costs;

- *gaming*: a broker with discretion over timing can game any benchmark that depends on when it trades, for example by spreading fills through the day to match VWAP or trading at the close. A benchmark fixed before the broker receives the order cannot be gamed, so implementation shortfall cannot (pp. 430–431).

Practice. When data cost is no object, implementation shortfall is the best estimator; retail traders use the effective spread, and sponsors who receive only daily average prices use VWAP [alpha-harris, same chap., pp. 426 and 431–432]. VWAP results are not comparable across stocks of different volatility, and converge to zero as the order becomes a large share of volume [s7-kissellscience, chap. 3, p. 107]. Single-trade estimates are noisy; averaging reduces noise but not bias or gaming, and comparisons across brokers are meaningful only for similar trading problems [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 432].

Pitfalls. The legacy TCA notes (slice 13) restated this entry's benchmark list, the VWAP formula, Harris's estimator properties and gaming; they are merged here. Two of their glosses go beyond the sources: that the post-trade midpoint "measures adverse selection" (in Harris the effective minus the realised spread measures dealers' losses to informed traders, spread measures) and that brokers game by "selecting only favourable venues" (Harris's gaming is by timing discretion and by mimicking the benchmark, pp. 430–431, 440). VWAP is an average market price, which includes the order's own trades, not a no-impact counterfactual. Interpretation: a benchmark price is a reference, not an executable price. A midpoint benchmark charges an aggressive order at least the half-spread, as the effective spread does. Decision and arrival prices are known when the order is planned; VWAP, participation-weighted and closing prices are known only after it, so an algorithm can target them only through forecasts, such as a forecast volume curve. Performance against the benchmark an algorithm targets is not its cost: a VWAP-tracking execution can match VWAP and still have paid spread and impact against arrival. Kissell calls TWAP "a constant participation rate" while describing equal shares per minute, which is a constant rate in time, not in volume [s7-kissellscience, chap. 1, p. 17]. His participation-weighted formula (3.16) has no side factor and is correct as written only for buys. Lehalle and Laruelle's "benchmarks" in Table 3.1 are algorithm classes; PoV and liquidity seekers are not benchmark prices (see benchmark choice). No passage checked defines a TWAP benchmark price.

See also. implementation shortfall (the decision-price benchmark with its unfilled remainder); choosing an algorithm's benchmark; spread measures (contemporaneous and post-trade midpoint benchmarks); intraday impact profile (impact measured from the arrival price); mean–variance execution (the schedule behind arrival-price algorithms).

Further reading. The legacy table it replaces in part, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. execution benchmark; benchmark price; decision price; arrival price; VWAP; interval VWAP; full-day VWAP; VWAP to end of day; TWAP; participation-weighted price; PWP; percentage of volume; POV; PoV; opening price; closing price; transaction cost analysis; TCA; split orders; gaming; benchmark noise; liquidity seeker; signed benchmark cost; signed difference; zero-sum transaction costs

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 422–437 and 440, Table 21-1 on p. 433 by page image, pp. 422–423 and 440 added in slice 13; `s7-kissellscience`, Academic Press printing with copyright 2014, ISBN 978-0-12-401689-7, chap. 1, pp. 16–18, and chap. 3, pp. 97–108, pp. 100, 103 and 105 by page image; and `alpha-laruelle`, 2018 World Scientific printing, p. 230). Stability: draft. Currency: definitions timeless; vendor and consultant practice as of 2002 and 2014.

Definition 9.2 Implementation shortfall: paper portfolio minus real portfolio, with execution, delay and opportunity components

Statement. Implementation shortfall is the paper return of a portfolio assumed to trade the whole order at the manager's decision price, minus the actual return, using actual fills and fees [s7-kissellscience, chap. 3, eqs. (3.1)–(3.4), pp. 97–99]. For an intended order of S shares, fills of s_j shares at p_j, decision price P_d, price P_n at the end of the measurement period and fees,

$$\mathrm{IS} = \underbrace{\textstyle\sum_j s_j p_j - \sum_j s_j P_d}_{\text{execution}} + \underbrace{\big(S - \textstyle\sum_j s_j\big)(P_n - P_d)}_{\text{opportunity}} + \text{fees},$$

which Kissell calls Perold's formulation [s7-kissellscience, same chap., pp. 100–101, eq. (3.6), p. 104]. With the arrival price P_0, Wagner's expanded form splits it into delay, trading and opportunity parts:

$$\mathrm{IS} = S(P_0 - P_d) + \textstyle\sum_j s_j\,(\bar{P} - P_0) + \big(S - \sum_j s_j\big)(P_n - P_0) + \text{fees}, \qquad \bar{P} = \textstyle\sum_j s_j p_j \big/ \sum_j s_j$$

[s7-kissellscience, same chap., pp. 101–104, eq. (3.7)]. The trading-related part is the arrival cost [s7-kissellscience, eq. (3.9), p. 105]. Harris states the same estimator with the decision-time midpoint: the filled size times the signed gap between average fill and benchmark, plus the unfilled size times the gap between a later price and the benchmark [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 426].

Notation. Quantities are signed, $S, s_j > 0$ for buys and < 0 for sells [s7-kissellscience, chap. 3, eq. (3.9), p. 105]; IS is in currency and positive when the real portfolio did worse. The legacy TCA notes wrote it with unsigned quantities and a side $d = \pm 1$: for parent size Q, filled q at average $\bar{p}$, decision price p_0 and later price p_T, $\mathrm{IS} = dq(\bar{p} - p_0) + d(Q - q)(p_T - p_0) +$ explicit costs, which is Perold's form with $S = dQ$ and

$\sum_j s_j = dq$; on the example below it gives 3,960 for the buy and for the sell (checked). Fees are explicit costs: commissions, taxes, clearing and settlement charges, ticket charges and rebates (p. 98). The spread and impact are inside the fill prices.

Conditions. When the order fills completely, IS reduces to $S(\bar{P} - P_d)$ + fees and does not depend on P_n (pp. 99–100). An unfilled or cancelled remainder is charged at P_n, so its cost depends on when it is measured: at the end of trading [s7-kissellscience, chap. 3, p. 102], or a day, a week or a month later; the Plexus Group uses 30 trading days after the decision [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 436–437]. The remainder counts only for an order that could realistically have been filled: Harris's 4-million-share sell order, 50 days' volume, would show a missed-trade cost of $714,600 that the trader never had the opportunity to avoid (p. 436).

Practice. The split shows who is responsible: delay belongs to the manager or desk, trading-related cost to the broker or algorithm, and the arrival cost is the main measure of broker and algorithm performance [s7-kissellscience, chap. 3, pp. 101–104]. When the decision price is unknown, only the market-activity part from P_0 is measured (p. 104, eq. (3.8)). Plexus's components, manager timing, trader timing, market impact, commission and missed trades, sum to the total; Plexus estimated its clients' costs at 12 basis points of commission, 20 of impact, 53 of timing and 16 of missed trades [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", box, p. 437]. Interpretation, checked with exact arithmetic: a buy of 10,000 shares decided at a midpoint of 50.00, arriving at 50.10, filling 3,000 at 50.20 and 3,000 at 50.30, cancelling the rest, with $P_n = 50.60$ and fees of 60, has paper gain 6,000 and real gain 2,040, so IS = 3,960: Perold $1{,}500 + 2{,}400 + 60$; Wagner delay 1,000, trading 900, opportunity 2,000, fees 60. The mirror-image sell gives the same figures with signed quantities, and Kissell's worked examples ($2,600, and $3,080 split as 1,250, 1,000, 750 and 80) reconcile.

Pitfalls. Interpretation: Kissell's intermediate actual return for a partial fill adds the fees instead of subtracting them (p. 100); his final formula is right. His basis-point arrival cost (3.10) divides by the signed $(\sum s_j)P_0$, which reverses the sign for sells; (3.11), with the side factor, does not (in the sell above, -30.06 against $+30.06$ basis points). Count each cost once: with a midpoint decision or arrival price, the half-spread is already inside the execution component, so adding an effective-spread term double-counts it, and fees enter only as the explicit term. Splitting the trading-related part into spread and impact, as the overview's checkpoint asks, needs a further benchmark; transaction costs gives an accounting split into half-spread paid and midpoint drift, which is not a causal measure of impact. The Almgren–Chriss shortfall $XS_0 - \sum n_k \tilde{S}_k$ is the trading-related part for a complete sale measured from the pre-trade price, with half-spread and fees inside the temporary impact; its mean and variance are model forecasts, not a realised shortfall, and it has no opportunity term because the model forces completion. Perold's paper itself (`wide-perold-shortfall`) was not available: the attribution of the formula to Perold, and of the expanded form to Wagner (1975, 1991), is Kissell's; Harris credits Perold with popularising the decision-midpoint estimator and Treynor, writing seven years earlier, with the paper-portfolio method (p. 424).

See also. execution benchmarks (the other benchmark prices); mean–variance execution (expected shortfall and its variance); spread measures; price impact.

Further reading. `wide-perold-shortfall`, the overview's first reading, not yet read. The derivations artefact: `notes/artefacts/market-making-lob/derivations.tex`.

Index. implementation shortfall; IS; paper portfolio; paper return; decision price; arrival price; arrival cost; execution cost; delay cost; trading-related cost; opportunity cost; missed trade opportunity cost; expanded implementation shortfall; Wagner; Perold; Treynor; Plexus; fees; commissions; partial fill; unfilled order; cancellation; missed quantity

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 424, 426 and 434–437, and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 3, pp. 97–105, pp. 100, 103 and 105 by page image; formulas and examples checked with exact arithmetic; `wide-perold-shortfall` not consulted). Stability: draft. Currency: timeless (accounting); the Plexus figures as of 2002.

Method 9.3 Choosing an algorithm's benchmark from order size, market context and the risk to hedge (Lehalle and Laruelle)

Statement. Lehalle and Laruelle tie each benchmark of a trading algorithm to the investor needs in a region: in the US, where high-frequency traders are everywhere, stealth; in Asia, comparing trading efficiency across local markets, given their differences; in Europe, both avoiding detection by high-frequency traders and learning how efficiently a set of order books is reached, to adjust routing quickly. "On paper" European investors should therefore target implementation shortfall, Asian ones VWAP or PoV, US ones liquidity seekers. Size and market conditions adjust these "drastic rules": liquidity seekers suit orders without minimum participation constraints, IS small orders, VWAP larger and PoV very large ones [alpha-laruelle, sec. 3.4.1, pp. 227–228]. Their Table 3.1 summarises the remarks (p. 229):

Benchmark	Region	Order size	Market context	Risk hedged (do not miss)
PoV (target)	Asia	more than 10% of ADV	possible negative news	the rapid spread of unexpected news, especially if the investor has the information
VWAP/TWAP (target)	Asia, Europe	5–15% of ADV	unusual volume negligible	the slow spread of information
Implementation shortfall (measure)	Europe, US	0–6% of ADV	possible price opportunities	an unexpected price move
Liquidity seeker	US	any	price oscillates around fair value	a liquidity burst or relative price move

ADV is average daily consolidated volume. Sources: rows from [alpha-laruelle, sec. 3.4.1, Table 3.1, p. 229], checked by page image; links to this compendium's definitions. The table is a summary of the entries, and the risk column is condensed from the source's wording.

Conditions. Lehalle and Laruelle's guidance is practitioner judgement: the passage gives no data for it, and the table "should not be read as 'all conditions in the columns have to be fulfilled"', only as possible reasons for a benchmark (p. 228). The size bands overlap as printed (0–6%, 5–15%, above 10%).

Practice. The hedged-risk column matters because an algorithm performs a risk–reward arbitrage: once its risk is under control it is free to adjust the prices and quantities of its orders, so optimal execution combines risk control, by following an "optimal" trading rate, with capturing price and liquidity opportunities. The choice also depends on the stock's liquidity and on the investment style behind the trade: hedging trades are more exposed to execution-specific impact, long-term position building to "systemic" impact that algorithms cannot do much about [alpha-laruelle, sec. 3.4.1, p. 228]. Their Table 3.2 adds the type of stock and of trade suited to each benchmark, and the main features of its algorithm (p. 230):

Benchmark	Stock	Trade	Main features of the algorithm
PoV	medium to large market depth	long-duration position	follows current market flow; very reactive, can be very aggressive; more price-opportunity driven if the range between maximum and minimum percent is large
VWAP/TWAP	any market depth	hedging order; long-duration position; unwinding tracking error (delta hedging of a fast-changing inventory)	follows the "usual" market flow; good if the market moves with unexpected volume in the order's direction; can be passive
Implementation shortfall	medium liquidity depth	alpha extraction; hedge of a non-linear position (typically gamma hedging); inventory-driven trade	finishes very fast if the price is good and liquidity sufficient; "cuts losses" if the price moves too far away
Liquidity seeker	poor, fragmented market depth	alpha extraction; opportunistic position building; order already split or scheduled	relative-price oriented (between pools or securities); captures liquidity everywhere; stealth, using fragmentation to limit leakage

Sources: rows from [alpha-laruelle, sec. 3.4.1, Table 3.2, p. 230], checked by page image. How each algorithm pursues its benchmark, through a trading envelope and tactics, is in trading envelope; what each benchmark measures is in execution benchmarks.

Pitfalls. The legacy notes said VWAP and PoV help Asia compare "fragmented market conventions"; the source speaks of differences between local markets, not fragmentation. They gave Europe only routing discipline, omitting avoidance of detection by high-frequency traders. Their risk column dropped "especially if I have the information" for PoV. The legacy version of Table 3.2 (slice 12) called VWAP/TWAP a "volume-curve schedule", which fits VWAP but not TWAP, and dropped PoV's "can be very aggressive", the VWAP/TWAP condition on unexpected volume in the order's direction, and the liquidity seeker's relative-price orientation. Interpretation: the legacy caveat that a benchmark "justifies the comparison, not the whole execution policy" is consistent with the risk–reward reading on p. 228 but is not in the source. The algorithm a trade is given does not settle how its cost is measured; an order worked by a VWAP algorithm can still be judged against arrival price, and matching VWAP does not show low cost (execution benchmarks). The regional pattern is dated by the 2018 printing.

See also. execution benchmarks; implementation shortfall; mean–variance execution (the risk-control layer and its trading curve); block trades. Fragmentation, dark trading and venue rules are the subject of the specialist pathway `market-structure-execution-quality`, which has no entries yet.

Further reading. The legacy blocks and tables, verbatim: `notes/artefacts/market-making-lob/`

derivations.tex.

Index. benchmark choice; algorithm selection; trading algorithm benchmark; PoV; percentage of volume; VWAP; TWAP; implementation shortfall; liquidity seeker; regional benchmark preference; ADV; hedged risk; stealth trading; Lehalle–Laruelle Table 3.1; Lehalle–Laruelle Table 3.2; algorithm features

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of alpha-laruelle, 2018 World Scientific printing, sec. 3.4.1, pp. 227–230, Table 3.1 on p. 229 and Table 3.2 on p. 230 by page image; Table 3.2 added in slice 12). Stability: draft. Currency: practitioner guidance as of the 2018 printing.

9.2 Execution scheduling and block orders

Model 9.4 Mean–variance optimal execution: the Almgren–Chriss trade-off between impact and price risk

Statement. An order is split over time to balance two forces: market impact pushes the schedule to trade slowly, price risk pushes it to trade fast [alpha-laruelle, app. A, sec. A.7, p. 274]. In Almgren and Chriss's discrete model a holding X is liquidated over N intervals of length $\tau = T/N$; x_k is the holding at $t_k = k\tau$, with $x_0 = X$ and $x_N = 0$, and $n_k = x_{k-1} - x_k$ is sold in the kth interval [alpha-almgren, author preprint of Dec. 2000, sec. 1.1, p. 7]. The price follows $S_k = S_{k-1} + \sigma\tau^{1/2}\xi_k - \tau g(n_k/\tau)$, with permanent impact g; the kth sale executes at $\tilde{S}_k = S_{k-1} - h(n_k/\tau)$, with temporary impact h that does not enter later prices (same preprint, secs. 1.2–1.3, eqs. (1)–(2), pp. 8–9). The shortfall $XS_0 - \sum_k n_k\tilde{S}_k$ has mean $E(x)$ and variance $V(x) = \sigma^2 \sum_k \tau x_k^2$ (eqs. (4)–(5), p. 10), and an efficient schedule minimises

$$E(x) + \lambda V(x), \qquad \lambda \text{ the risk aversion}$$

(sec. 2.1, eq. (15), p. 13). With linear impact, $g(v) = \gamma v$ and $h = \epsilon\,\mathrm{sgn}(n_k) + (\eta/\tau)n_k$, the unique minimiser for $\lambda \geq 0$ is

$$x_j = \frac{\sinh\big(\kappa(T - t_j)\big)}{\sinh(\kappa T)}\, X, \qquad \frac{2}{\tau^2}\big(\cosh(\kappa\tau) - 1\big) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \tfrac{1}{2}\gamma\tau,$$

and $\theta = 1/\kappa$ is the trade's half-life, independent of T [alpha-almgren, same preprint, secs. 2.2–2.3, eqs. (16)–(17), pp. 13–15, page images of pp. 11 and 14].

Notation. Lehalle and Laruelle's discrete version, used by the legacy notes, is for a buy order of $v^\star$ over N slices: v_n shares in slice n, market volume V_n, volatility σ_n in currency per slice, and execution price $\tilde{S}_n(v_n) = S_n + \kappa\sigma_n(v_n/V_n)^\gamma$, with κ and γ stock-dependent constants [alpha-laruelle, app. A, sec. A.7.1, eqs. (A.16)–(A.19), pp. 275–276]. Here κ and γ are impact parameters, not Almgren and Chriss's decay rate κ and permanent-impact slope γ; λ is risk aversion in both.

Conditions. The price is a martingale over the horizon, with no view on its trend [alpha-laruelle, app. A, sec. A.7.2, p. 277]; volumes and volatilities per slice are taken as known in advance (p. 276); the schedule is static, which Almgren and Chriss show is optimal under their price dynamics (preprint, n. 10, p. 13); the trades do not change sign (p. 13). In Lehalle and Laruelle's version:

- *Temporary impact only.* The concession $\kappa\sigma_n(v_n/V_n)^\gamma$ is in currency per share and applies only to slice n's fills: an instantaneous execution cost, not a transient displacement of later prices. The slice costs v_n times it; the expected impact cost of the order is $\kappa\sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma$, in currency, not per unit time (eq. (A.20), p. 276). There is no permanent-impact term and no spread or fee term, unlike Almgren and Chriss's g and ϵ.
- *Expectation only* ($\lambda = 0$). The schedule $v_n^{\exp} = v^\star\,(V_n/\sigma_n^{1/\gamma})/\sum_\ell V_\ell/\sigma_\ell^{1/\gamma}$ (Proposition 2, eq. (A.22), p. 277), uniform if V_n and σ_n are constant (Proposition 3).
- *Variance only.* Trade everything in the first slice (Proposition 4, eq. (A.25), p. 278); in Almgren and Chriss this gives zero variance "because of the way that we have discretized time" (preprint, eqs. (12)–(13), p. 12).
- *Mean–variance*, $C = \mathbb{E}(W) + \lambda\,\mathrm{Var}(W)$ (eq. (A.26), p. 279). For $\gamma = 1$ the first-order conditions give a recurrence in the remaining quantity, solved by shooting on the first slice (eq. (A.28), pp. 279–280); larger λ trades faster, smaller λ more linearly (Figure A.11, pp. 280–281).

Practice. A trading algorithm has two layers: a risk-control layer, at a time scale around 10 minutes, that bounds the trading rate from the schedule; and an opportunistic layer, such as smart order routers and liquidity seekers, that chooses the participation rate within those bounds and may react to alpha signals [alpha-laruelle, app. A, sec. A.7, p. 274]. The model's parameters are estimated daily or weekly (p. 274). Interpretation, checked numerically for this slice: the expectation-only schedule equalises marginal impact cost across slices; for $\gamma = 1$ with constant σ_n and V_n, the mean–variance schedule in the legacy notation equals Almgren and Chriss's sinh form with $\cosh k = 1 + \lambda\sigma V/(2\kappa)$ and Lehalle and Laruelle's recurrence; Lehalle and Laruelle's model is Almgren

and Chriss's linear case without the schedule-independent terms (the permanent $\frac{1}{2}\gamma X^2$ and the fixed cost ϵX of a one-sided order) and without the $\tilde{\eta}$ correction, with $\kappa\sigma_n/V_n$ in place of η/τ and σ_n^2 in place of $\sigma^2\tau$.

Pitfalls. Indexing. Lehalle and Laruelle's price step (A.16) prices slice n after shock n, and their variance (A.24) uses the quantity remaining from slice n on, yet they state the terminal conditions as $x_0 = v^\star$, $x_N = 0$; they also label Proposition 5 as "variance minimization" although it is the mean–variance result, and drop κ from the intermediate Lagrange condition, which only rescales the multiplier. The legacy notes use the Almgren–Chriss timing instead: slice n at S_n, then $S_{n+1} = S_n + \sigma_n\xi_{n+1}$, so the remaining quantity x_n after slice n bears the next shock and the risk term $\lambda\sum_n x_n^2\sigma_n^2$ is consistent (Interpretation). Comparative statics. The legacy objective said higher risk aversion front-loads the schedule "while higher impact sensitivity back-loads it". Interpretation: the minimiser depends on λ/κ only, since $C - v^\star S_0 = \kappa\big[\sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma + (\lambda/\kappa)\sum_n x_n^2\sigma_n^2\big]$; larger κ moves the schedule towards the expectation-only one, not beyond it; in the cases checked the remaining quantity never exceeded the expectation-only path, which is not a proof. Shape. The legacy schedule taxonomy called the mean–variance path convex; that holds for constant parameters (the sinh form; Lehalle and Laruelle, p. 279), but with time-varying V_n and σ_n the path follows the volume profile and need not be convex (all 200 random cases checked were not). The parameters κ, γ of a scheduling model are inputs; the empirical exponents of Almgren et al. (0.6, against size relative to volume during execution, with a 30-minute post-trade horizon) are estimates of a related but not identical quantity. The legacy taxonomy's "dark pools and limits" in the tactical overlay are not in the passages checked.

See also. price impact (temporary against transient impact; the Almgren–Chriss impact functions); direct estimation of impact; intraday impact profile (participation and duration); absence of price manipulation (conditions on impact models used in schedules); block trades. The pathway `price-impact-execution` develops impact models and optimal schedules beyond this entry.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. The journal version of Almgren and Chriss (2001) was not compared with the preprint.

Index. optimal execution; Almgren–Chriss; mean–variance; implementation shortfall; risk aversion; execution schedule; trade half-life; expectation-minimising schedule; VWAP-like schedule; temporary impact cost; participation; trading algorithm layers

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, app. A, sec. A.7, pp. 273–281; and `alpha-almgren` in the authors' preprint of December 2000, pp. 7–15, page images of pp. 9, 11 and 14; formulas and comparative statics checked numerically). Stability: draft. Currency: timeless (model).

Definition 9.5 Block trade: an order too large to fill by normal procedures, and the four problems of arranging it

Statement. A block trade results from an order too large to fill easily through normal trading procedures, typically more than a day's normal volume; most block traders think of a block as more than a quarter of a day's average volume in an actively traded stock, and in thinly traded instruments a few thousand shares can be one [`alpha-harris`, chap. "Block Traders", sec. 15.1, p. 322]. Block initiators face four problems [`alpha-harris`, same chap., sec. 15.2, pp. 323–327]:

- *latent demand*: willing counterparties have not issued orders, because orders are costly to manage or because they have not yet thought about trading;
- *order exposure*: shopping the block lets front runners, same-side traders and opposite-side traders act on it, accelerating its price impact (Table 15-1, p. 325);
- *price discrimination*: liquidity suppliers fear that more size will follow, so they want the true size before offering liquidity;
- *asymmetric information*: suppliers suspect large, especially anonymous, traders of being informed, and demand concessions as in the adverse-selection spread.

Conditions. Exchanges classify blocks by fixed sizes for statistics, the NYSE at 10,000 shares or more regardless of activity or price; Harris suggests classifying by a fraction of average daily volume instead [`alpha-harris`, same chap., sec. 15.1 and box, p. 323]. Interpretation: the legacy suggestion to measure a block by typical depth or participation capacity follows the same logic but is not in the passage.

Practice. Solving the problems requires credible information about the initiator: that it wants to trade, exposure only to trustworthy traders, its full size, and evidence that it is uninformed. Since exchanges carry only price and size, most blocks are arranged by telephone upstairs through block brokers and dealers with reputations to protect [`alpha-harris`, same chap., secs. 15.2.2–15.2.5, pp. 324–327]. Crossing networks, hidden orders and order splitting are the alternatives when orders cannot be exposed (p. 325). *Sunshine trading* (slice 34) is the opposite: announcing identity, intentions, full size and reasons for trading. It works when the sunshine trader is well known and known to be uninformed and honest, so in large markets at best for the

largest traders, and fails when others suspect information or hidden size ("wolves in sheep's clothing"), because such motives cannot be verified on an exchange floor or screen [alpha-harris, same chap., sec. 15.3, pp. 327–328]; by revealing their trades, sunshine traders also give free trading options to order anticipators (p. 328). Block dealers and brokers audit their clients' motives: uninformed traders have utilitarian reasons unrelated to fundamental values, and block traders are most confident when the client has an easily verified reason to trade that instrument at that time; discretion over which instrument or when suggests strategic choice (Table 15-2: a firm liquidating a diversified portfolio to fund a project, probably uninformed; a legatee buying a single stock, probably informed) [alpha-harris, same chap., sec. 15.4.4, pp. 330–331]. Full and timely trade reporting favours block brokers over block dealers: a dealer who must report quickly can be front-run while liquidating the position, and reports help others infer whether the initiator was informed, which raises the importance of the motive audit; the weakest reporting is in dealer markets (sec. 15.4.3.1, p. 330). U.S. equity trades were reported within 90 seconds, and the London Stock Exchange had moved from seven days for the largest blocks to 90 minutes (box, p. 331, as of 2002). Harris also describes how blocks met the exchange book in U.S. listed equities in 2002 [alpha-harris, same chap., sec. 15.6.1, pp. 333–334]:

- brokers and dealers who were members of an exchange had to print their block trades in exchange-listed securities at an exchange if the trade took place while the market was open, presenting the matched orders so that the exchange's precedence rules protected all traders;
- a *size precedence* rule let matched crosses of more than 25,000 shares outsize the book at the trade price, jumping orders with time precedence, while orders standing at better prices were filled at the block price, since price priority was enforced universally;
- Harris calls this a compromise between block traders, who want their crosses unbroken, and standing limit-order traders, who want a return on the liquidity they offer;
- block traders often printed crosses at regional exchanges, whose books were thinner.

Pitfalls. The legacy summary, "disclose enough to find size, but not enough to be exploited", is a fair reading of the tension between the latent-demand and order-exposure problems (Interpretation), but Harris's own resolution adds disclosure of identity and true size to trusted counterparties. The legacy price-discrimination item ("suppliers fear selling too cheaply or buying too richly") paraphrases the fear that more size will follow. The legacy motive-audit process (slice 13) lists funding, diversification, hedging, benchmark change and speculation, and "comparing the initiator's past behaviour"; Harris's examples are funding, diversification, covering a short, hedging and currency needs, and "benchmark change" and past behaviour are not in the passage. The legacy reporting caveat says that delayed reporting gives informed block traders room to exploit stale prices; Harris's argument is that timely reporting hurts facilitating dealers, and elsewhere that delayed reporting helps traders building or unwinding positions at the expense of other traders (chap. "Buy-Side Traders", p. 389). Some motive audits may be unreliable where prices are too volatile for block traders to build reputations for them (p. 331). The legacy market rule *Equity Block Prints and Size Priority* (slice 18) merged three things that should be kept apart: where a trade is *reported* or printed; its *execution priority* against orders in the book (size against time precedence); and *trade-through protection* of better-priced quotes on other venues, which in the United States is now Rule 611 (best execution). Its "trade-reporting facility" and its size-priority exceptions are stated as current practice; only Harris's 2002 exchange rule was checked, and current venue rules are a backlog item of `market-structure-execution-quality`. Its design trade-off (protect the book too much and blocks leave; protect blocks too much and displayed liquidity loses value) restates Harris's compromise.

See also. order anticipation (front running exposed orders); order exposure decisions; dark pools and crossing networks; spread components (the adverse-selection concession); mean–variance execution (scheduling an order through the open market instead).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. block trade; block order; upstairs market; latent demand; order exposure; shopping the block; price discrimination; asymmetric information; block brokers; block dealers; block positioners; trading motive audit; utilitarian traders; trade reporting; delayed reporting

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 322–327, and pp. 328–331 and 389 added in slice 13, and pp. 333–334 added in slice 18; sunshine trading, pp. 327–328, added in slice 34). Stability: draft. Currency: timeless (concepts); the NYSE threshold, venues and reporting deadlines as of 2002.

9.3 Execution algorithms, order exposure and portfolio trading

Definition 9.6 Automated trading architecture: market data with real-time analytics, two-way venue connections, historical data with recorded messages, and models with an accuracy check

Statement. Lehalle and Laruelle sketch the main components of a broker's automated trading service (Figure 3.1, page image): tick-by-tick and context databases, a calendar of expiries, expected news and half-days, modelling and model selection, a trading framework with a strategic layer, local trading tactics and listeners at each venue, synchronisation, a trading station, a trading database and TCA [alpha-laruelle, sec. 3.1, pp. 193–194]. The inputs of such tools are four (sec. 3.1.1, pp. 194–197):

- *market data*: real-time feeds, consolidated across venues but quick to disaggregate, because it matters which venue currently offers a price. They hold trades, each with a price, a quantity and a timestamp that is "not always precise enough", and order books, or the sequence of their modifications, from which the book must be rebuilt on the fly. They are enriched with analytics: the spread, the side that initiated a trade, intraday volatility, an efficient spread and estimates of hidden quantities. Since raw data and analytics are needed together, "the sooner they are computed, the better", ideally on hardware as soon as raw data arrive (pp. 194–195);
- *connections to venues*, which are not natively synchronised with market data, usually being faster, and are two-way. One direction sends new orders, cancellations and modifications; the other returns acknowledgements, rejections of ill-formed messages, cancellations and partial fills. Incoming messages also yield analytics: dark pools publish only trades, so a dark buy order filled in full at a price shows resting sell quantity at that price or better (pp. 195–196);
- *historical data*: tick-by-tick or order-by-order market data; the analytics already computed in real time; recordings of the messages the firm's algorithms sent and received, to study past performance; and context such as news, corporate events, circuit breakers and venue outages (p. 196);
- *models*, each with three components. A formula takes parameters and variables: in the impact formula $\kappa\sigma(v/V)^\gamma$ the parameters κ, γ may be estimated once a day and stored, while the variables σ, V and v are, in the book's words, "known in real time". Interpretation: V, the market volume during the trading, is available only as it is observed; volume still to come requires a forecast (see the decision-time item in trading envelope). An estimation procedure computes the parameters from historical data. An accuracy computation qualifies the model and its parameters on the immediate past or in real time, so that a model can be deactivated when it is unlikely to be adequate in the current context (pp. 196–197).

Conditions. The inputs feed the components of a trading algorithm: execution parameters, a risk-control layer that sets the trading envelope, and tactics such as smart order routers (trading envelope). For latency, some autonomous tactics sit as close as possible to the venues they address, and the risk-control layer at equal distance from all its robots (sec. 3.1.2, p. 198). This describes a broker's execution service for fragmented European equities as of the 2018 printing; it is a design, not a standard or a legal requirement.

Practice. Interpretation: the recorded messages and context are what later make it possible to replay an execution and explain its performance (decision support, transaction cost analysis). The accuracy computation is a model-monitoring control: governance of models and of the systems that run them is in operational controls for automated trading.

Narang's version of the same infrastructure (slice 30): connectivity, a message protocol, and hardware and software, each built or bought [s7-narang, chap. 7, "Execution", sec. 7.2, pp. 138–140]. Most traders use brokers as trading agents, "due to regulatory and other constraints", and reach venues by direct market access through the broker's servers, which he says adds 10 to 30 milliseconds against a fraction of a millisecond from colocated servers (p. 139). The FIX protocol, begun in 1992 between Fidelity and Salomon Brothers, is the standard for order and execution messages; he calls the protocol "free and open source" and distinguishes it from the FIX engines that implement it, which differ in quality and which latency-sensitive traders often build themselves (pp. 139–140). Buying execution algorithms can cost "five or more times as much per share" as running one's own (pp. 140–141).

Pitfalls. From the legacy execution-model file: its "the software that implements the FIX protocol is free and open source" reverses Narang, for whom the protocol is free and open and the engines vary; whether a given engine is open source, and the licence of the FIX specification itself, were not checked (`s6-fix` unread). "Sponsored access" is not in Narang's chapter; for U.S. broker-dealers market access is governed by Rule 15c3-5 (operational controls). Narang's latency figures are undated estimates. The legacy principle's "the hard part is not one model; it is keeping the decision chain coherent under fragmented venues and live latency" is a synthesis, not a statement in the source. Several legacy items are not in the passages read: holidays, auctions, index events and expected halt days in the calendar (the figure shows expiries, expected news and half-days); book imbalance and queue sizes among the fast analytics; and public book updates as part of the venue connection, where the source keeps the private order messages apart from the market-data feed and notes that the two are not synchronised. Venue names, data and latencies are as of the 2018 printing.

See also. trading envelope; decision support; smart order routing (dark probes, latency); operational controls for automated trading (owned by `model-governance-controls`).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/model-governance-controls/`.

Index. trading architecture; trading tools; market data; consolidated feed; real-time analytics; venue connection; acknowledgement; rejection; partial fill; dark-pool probe; historical data; recorded messages; context database; calendar; model parameters; model variables; estimation; model accuracy; model deactivation

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, secs. 3.1–3.1.2, pp. 193–198, page image of p. 194 with Figure 3.1; in slice 30, `s7-narang`, third edition, Wiley 2024, chap. 7, pp. 138–141). Stability: draft. Currency: Lehalle and Laruelle's practitioner description as of the 2018 printing; Narang's account as of 2024, with his latency figures undated.

Definition 9.7 Trading algorithm: execution parameters, a risk-control layer that sets a trading envelope, and tactics inside it

Statement. Lehalle and Laruelle describe a trading algorithm, instantiated from a class such as VWAP, TWAP, implementation shortfall, target close or liquidity seeking, as three components [alpha-laruelle, sec. 3.1.2, pp. 197–198, p. 198 by page image]:

- *execution parameters*: side and quantity, the benchmark (VWAP, TWAP, percent of volume, entry price, close price), start and end times, bounds on the trading rate (maximum and minimum percent), aggressiveness, access to dark pools;
- a *risk-control layer* that computes the *trading curve*, a minimum and maximum trading rate, from models of market impact, market risk and alpha and from "expected market contexts", balancing impact against market risk and its adverse selection;
- *trading robots* or liquidity-seeking tactics, such as a smart order router, that navigate inside the envelope set by the risk-control layer using the immediate state of liquidity; the risk-control layer assigns them short-term goals.

Because the analytics behind the curve are noisy, the curve is thick: a *trading envelope*, inside which no trajectory is a priori more efficient than another for risk control. An algorithm trading faster than its curve stops posting; one trading too slowly trades aggressively, paying the spread [alpha-laruelle, sec. 3.4.1, pp. 231–232, page image of p. 231]. Envelopes are shown for a VWAP and an implementation-shortfall algorithm at several risk levels (Figures 3.13–3.14, pp. 232–233).

Notation. Interpretation: the envelope bounds the cumulative executed quantity, or equivalently its rate, as a function of time, as plotted in Figure 3.2 (p. 199) and Figure 3.13 (p. 232). It applies to the parent order; child-order size, price and venue are the tactics' decisions.

Conditions. How each algorithm class pursues its benchmark, from the passages checked:

- *Benchmark against policy.* VWAP and TWAP name both a price statistic and a policy (execution benchmarks). Kissell's VWAP policy trades in proportion to the intraday volume curve and his TWAP policy evenly in time; both are static schedules fixed for the day; his POV policy trades a set fraction of market volume, so it reacts to realised volume but does not guarantee completion, and some brokers add a minimum rate to finish by the end time [s7-kissellscience, chap. 1, pp. 16–17]. Arrival-price and implementation-shortfall algorithms solve a cost–risk optimisation, the latter adding real-time adaptation to volumes and prices; their meaning differs across brokers, as does the scale of the risk-aversion parameter [s7-kissellscience, same chap., pp. 17–19].
- *Information at decision time.* Model parameters, such as κ and γ in the impact term $\kappa\sigma(v/V)^{\gamma}$, are estimated about once a day from history; the variables σ, V and v are fed in real time [alpha-laruelle, sec. 3.1.1, p. 197]. Interpretation: V, the market volume during the trading, is known only as it occurs, so a schedule fixed at the start uses a forecast of it, and a policy that adapts uses only the volume observed so far.
- *Participation.* Interpretation: neither source checked says whether the market volume a participation rate ρ applies to includes the order's own fills. If it does, the order trades $\rho/(1-\rho)$ of other traders' volume (25 percent at $\rho = 20$ percent); if not, ρ of it, which is $\rho/(1+\rho)$ of the total (16.7 percent). Kissell's participation-weighted price, the volume-weighted price of the first (shares traded)$/\rho$ shares, counts the order's own fills in the window (p. 108). The measurement window, the denominator and auction volume (often skipped by TWAP, p. 17) need stating.

Practice. Benchmarks can be combined by constraints: minimum or maximum participation rates make an algorithm behave like PoV when volume is very high or low; a limit price and a "would level" make it behave like implementation shortfall; a smart order router makes it resemble a liquidity seeker. Such hard combinations are bang-bang controls, applied fully once a threshold is crossed; they can be gamed if the threshold is detected,

or adversely selected, finishing too early because of the constraints, if the conditions are not chosen accurately. Building the logic on envelopes avoids this. Customising then has two steps: inject the desired risk profile and any volume or price constraints into the envelope, and plug in tactics suited to the investment style under defined market conditions, such as "go into dark pools if volatility increases" [alpha-laruelle, sec. 3.4.1, pp. 231–233, page images]. The equations for optimal envelopes are cited there to Bouchard et al. (2011), Lehalle (2012) and Guéant et al. (2012), not consulted.

Pitfalls. Lehalle and Laruelle refer to Appendix A.6 for the trading curve (p. 197), but A.6 is the Harris tick-size model; the scheduling appendix is A.7 (Interpretation, from the table of contents and p. 266). The caption of Figure 3.14 reads "for a VWAP (left) or a (right) trading algorithm", with the second name missing; the text calls the pair a VWAP and an IS (p. 232). The legacy notes said a too-fast algorithm "pauses or posts passively"; the source says it stops posting. Their PoV "capped by urgency", VWAP "would-level" constraints and IS with "SOR and dark-pool probes" rearrange the source's examples, in which the would level turns an algorithm into IS and the SOR into a liquidity seeker. The "minimum-impact curve" among the legacy benchmark paths is not in the passages checked. The legacy "dark probes" are supported: orders sent to dark pools come back as probes of their hidden books, and a firm records the messages its algorithms send and receive, with market data and context, to study their performance [alpha-laruelle, sec. 3.1.1, p. 196]; the legacy decision loop's full list of observed events (quotes, trades, fills, rejects, cancellations, venue responses) is a paraphrase, not a list in the source. Interpretation: the envelope is a risk-control construct; following it says nothing about the cost against a benchmark measured afterwards.

See also. mean–variance execution (a model for the trading curve); choosing an algorithm's benchmark; execution benchmarks; pre-trade and monitoring tools; price impact (the models the risk-control layer uses, developed by `price-impact-execution`).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. trading algorithm; algorithm layers; execution parameters; risk-control layer; trading curve; trading envelope; tactics; trading robots; smart order router; SOR; liquidity seeking; benchmark customisation; minimum participation; maximum participation; limit price; would level; bang-bang control; static schedule; adaptive algorithm; participation rate; POV; PoV; VWAP algorithm; TWAP algorithm; risk aversion parameter

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, secs. 3.1.1–3.1.2, pp. 196–199, and sec. 3.4.1, pp. 231–233, page images of pp. 198, 231 and 233; and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 1, pp. 16–19, and chap. 3, p. 108). Stability: draft. Currency: practitioner descriptions as of 2014 and 2018; broker definitions vary.

Method 9.8 Decision support for trading algorithms: pre-trade analytics, monitoring indicators and what-if scenarios

Statement. Before launch, pre-trade analytics use historical data, models and parameters to estimate and compare expected cost components (bid–ask spread, usual traded volume, usual quantity on the book), expected trading profiles and expected context (news, volatility levels), broken down by sector, currency and country; on portfolios they check consistency and speed some lines up and slow others down from the expected balance of market risk and impact [alpha-laruelle, sec. 3.1.3, pp. 198–199]. During trading, monitoring indicators let a trader, who may watch several hundred algorithms, find where poor performance concentrates; performance indicators estimate the result if trading stopped now or continued under unchanged conditions, from a decomposition of the average price into immediate price, market moves and impact (pp. 200–201, page image of p. 201). What-if scenarios give real-time estimates of changed execution conditions, such as finishing now by taking the visible liquidity or converting an implementation-shortfall algorithm to percentage of volume (p. 201). Pre-trade, monitoring and post-trade tools should be compatible: for an indicator such as the bid–ask spread, pre-trade gives its expected average, monitoring its current value, updated expectation and likelihood, and post-trade its value during trading and its influence on performance, with breakdowns by sector, currency, country and side (p. 202). Monitoring indicators are built for human decisions, such as changing execution parameters on the fly, and can also be used by models to identify causes of poor performance (p. 200). Because users must share the indicators' values with the algorithms, the algorithms themselves publish them as *indicator publishers*, publishing views on their own state in real time to users and companion algorithms (p. 202, page image).

Conditions. Some what-if outcomes are simple formulas, others are estimates (p. 201). Pre-trade figures are forecasts made before the order. The average-price decomposition "follows" the scheduling appendix's (A.26) (p. 201), but (A.26) is an ex-ante objective, $\mathbb{E}(W)+\lambda\,\mathrm{Var}(W) = v^\star S_0 + \kappa\sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma + \lambda\sum_n x_n^2\sigma_n^2$, labelled immediate buy, market impact and market risk [alpha-laruelle, app. A, sec. A.7.2, p. 279, page image]: its third term is a variance penalty, not a price move. Interpretation: the realised counterpart, from the book's change of variable (p. 278), is the identity $W = v^\star S_0 + \sum_n x_n\sigma_n\xi_n + \kappa\sum_n \sigma_n v_n^{\gamma+1}/V_n^\gamma$, dividing by $v^\star$ for the average price; it holds exactly inside the model (checked by simulation), with x_n the quantity remaining from

slice n on, in the book's timing. On real executions the impact term is a model value and the market-move term what is left, so the split is model-dependent. The refined version, adding planned price, unexpected market moves and trading efficiency, is given in words only (p. 201), so its additivity cannot be checked.

Practice. Interpretation: pre-trade estimates belong with the expected cost and risk of a schedule (mean–variance execution); realised performance belongs with implementation shortfall. The average-price decomposition is the post-trade side of transaction cost analysis; its accounting counterpart without a model is the split in transaction costs.

Pitfalls. The legacy pre-trade taxonomy added "portfolio coupling" (common end times, exposure constraints, synchronisation); the source mentions synchronisation for portfolios with common constraints, such as the same end time or constant exposure to a factor, after the pre-trade list (p. 200), not in it. The legacy what-if note listed "slow down, switch benchmark, seek dark liquidity, or cap participation"; the source's examples are finishing now and switching from IS to PoV. The legacy attribution taxonomy (slice 13) defined the immediate price as "the reference available when the order decision is made" and market impact as "price movement caused by the order itself"; in the source the immediate price is S_0, the price of buying everything at once without impact, and the impact term is the model's value, not an observed causal effect. Its pre-, live- and post-trade principle follows p. 202. The legacy process *Indicator Publication* (slice 17) follows pp. 200–202, but its conclusion that publishing makes algorithms "stateful agents ... rather than black boxes" is its own: the book calls trading algorithms "stateless agents" with introspection (p. 202, checked on the page image) while having them publish views on their own state. The two wordings are left as found; the point the source makes is introspection and shared indicators.

See also. trading envelope; execution benchmarks; implementation shortfall.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. pre-trade analytics; expected cost; trading profile; monitoring indicators; performance indicators; what-if scenarios; average price decomposition; planned price; trading efficiency; portfolio synchronisation; post-trade analysis; immediate price; unexpected market moves; performance attribution

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, sec. 3.1.3, pp. 198–202, page images of pp. 198 and 201, and app. A, sec. A.7.2, pp. 278–279, page image of p. 279, added in slice 13; the realised identity checked by simulation; pp. 200 and 202, page image of p. 202, for indicator publication, slice 17). Stability: draft. Currency: practitioner description as of the 2018 printing.

Definition 9.9 Order exposure decisions: how a large trader shows, splits and routes an order, and the reactive–proactive continuum

Statement. Order submission strategy is the most important determinant of execution quality that traders control: market against limit orders, where to place limits, and when and how to resubmit. Large traders must also decide whether to look for the other side or wait for it, whether to show their full interest or hide it, whether to split the order over time, whether to use one broker or several to hide their total interest, and whether to trade in one market or many; display decisions are the most important a large buy-side trader makes [`alpha-harris`, chap. "Buy-Side Traders", p. 380]. *Reactive* traders trade in response to opportunities others present; *proactive* traders articulate their interests. Proactive traders expose orders to attract reactive ones, and two reactive traders never trade with each other (same chap., sec. 18.3, pp. 383–384).

Conditions. *Benefits and costs of exposure (slice 34).* Display makes buyers and sellers easy to find and attracts reactive traders with latent interest; its costs come from how others respond [`alpha-harris`, chap. "Buy-Side Traders", secs. 18.2–18.3, pp. 382–383]. *Parasitic traders*, order anticipators, use the exposed information to profit at the exposed trader's expense; *defensive liquidity suppliers* use it to step out of the way of a large trader who would otherwise price-discriminate against them, so that the large trader pays more for liquidity (p. 383). Exposure can reveal three things: why the trader trades, the price impact of future trades, and valuable trading options (sec. 18.4, p. 384). Small traders rarely need to care, unless many small orders on one side add up to a large one (p. 383). Cost is the main reason to be reactive: proactive traders issue instructions before knowing whether an opportunity will arise, which is wasteful when opportunities are rare, but reactive traders risk that proactive ones act first (p. 383). The reactive–proactive continuum is the willingness to bear search costs; the aggressive–passive continuum is the eagerness to trade. They are similar, not the same: aggressive traders tend to be proactive, and traders who can search cheaply tend to be proactive (p. 384).

Practice. Evasive strategies: brokers to preserve anonymity, several brokers so that none knows the full interest, anonymous electronic systems, careful choice of whom to show first (the art of block trading), showing only part of the order, undisplayed orders, order indications that carry no commitment, and waiting for another trader's firm order and taking it [`alpha-harris`, same chap., sec. 18.5.1, pp. 386–387]. Deceptive and offensive strategies are costly and may be illegal in some jurisdictions (pp. 387–389). *Narang's execution algorithms (slice 30).* Narang frames an algorithm's decisions as how aggressive to be, which order type to use, what size

to send and where to send it, with the goals of completeness and cheapness [s7-narang, chap. 7, "Execution", sec. 7.1, pp. 129–130]. Market orders and marketable limit orders (a buy at the best offer, a sell at the best bid) are aggressive; a buy at or below the best bid, or a sell at or above the best offer, is passive. A passive order controls its price but may not fill, or fill only in part, and suffers adverse selection when it does fill (pp. 130–133). Liquidity rebates sweeten a passive order only if it executes, and some venues invert the fees (pp. 132–133). He finds aggressiveness "generally" higher for momentum strategies and lower for mean reversion, and higher for stronger, more confident signals; a limit order inside the spread, possible only when the spread exceeds the minimum tick, is a middle ground (p. 133). Algorithms may also price off a "micro price" that reflects the order book's imbalance (pp. 133–134). A hidden order "provides no information to the market", in Narang's words, meaning that its size and price are not displayed before it trades; it loses priority to displayed orders at the same price on most exchanges, and not all exchanges allow them; iceberging posts most of a large order as hidden pieces (pp. 131 and 134–135). Harris describes display instructions (undisclosed, hidden, reserve or iceberg orders): the broker fills the displayed part first and then shows more, up to the maximum display; at Euronext in 2002 traders could discover the full size of an undisclosed order only by submitting large marketable fill-or-kill orders [alpha-harris, chap. 4, sec. 4.11.2, pp. 84–85]. Under display precedence, displayed orders rank ahead of undisclosed ones at the same price, the displayed and undisclosed parts of one order being treated separately (chap. 6, p. 117); Bouchaud et al. say an iceberg's hidden part "usually has lower time priority" [catalogue-0528, chap. 3, sec. 3.2.2, p. 55]. Whether a refreshed slice keeps or loses time priority is a venue rule not checked here. Interpretation: hiding removes the pre-trade display, not all leakage. Executions are still printed and can be read, and the SEC's 2010 release describes immediate-or-cancel "pinging" orders used to find undisplayed liquidity, including to detect and trade in front of large interest [sec-concept-release-2010, p. 47, n. 69]; that is the order-anticipation risk in order anticipation. A hidden order lowers, rather than removes, what others can infer. Splitting a large order into pieces over time trades impact against the risk of an adverse move, with piece size set from the cost model (pp. 136–137). An algorithm's detectable "footprint" lets others predict it and raise its costs (pp. 129–130). Routing is in smart order routing.

Pitfalls. The legacy principle equated proactive execution with crossing the spread; Harris separates search (reactive–proactive) from eagerness (aggressive–passive), and a proactive trader may post a limit order. Its claim that reactive execution "usually lowers leakage" fits Harris's evasive strategy of not exposing until trading (p. 387), but he states it as a strategy, not a regularity. The legacy lever list's "discretion range", "protection price", "pegging" and "minimum fill" are not in the passages checked. Systems and venues named (Autex, POSIT, Island and others) are as of 2002. From the legacy execution-model file (slice 30): aggressiveness is side-dependent, so the legacy "limit order at current best order is fairly aggressive, while limit order below current bid is passive" holds for a buy at the offer and a buy below the bid, with the mirror image for a sale. Fee and rebate levels, hidden-order priority and the availability of order types are venue rules; Narang notes that order types are created and retired often (pp. 134–135); none was checked at a current venue. The legacy item on cancelling and replacing orders, "placing a larger number of orders with no intention of execution" and, for a trader who wants to buy, entering small sell orders away from the market and cancelling them, "improving market perception", is not in the third edition's chapter, which treats cancellation as routine queue management and defends high cancellation rates (chap. 14, sec. 14.1.3, p. 270; Appendix, sec. A.2.2, pp. 314–315). What the legacy describes is placing orders without intending them to execute, in order to move others' view of the market. For trading on or subject to the rules of a registered entity under the Commodity Exchange Act (in practice futures and swaps venues; this gloss is Interpretation), U.S. law makes unlawful any practice "of the character of, or ... commonly known to the trade as, 'spoofing' (bidding or offering with the intent to cancel the bid or offer before execution)" [usc-7-6c, paragraph (a)(5)(C), 2024 edition]. Its status for securities and in other jurisdictions was not checked and is left open (a stub in `market-structure-execution-quality`); the passage is kept as a description of conduct, not a technique.

See also. block trades (order exposure as one of the four problems; sunshine trading); dark pools and crossing networks; order instructions (display and quantity instructions); order anticipation (the parasitic traders exposure attracts); trading envelope (tactics inside an algorithm). Fragmentation and dark venues are developed by `market-structure-execution-quality`.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. order submission strategy; order exposure; display decision; hidden orders; undisplayed orders; order indications; order splitting; multiple brokers; reactive trader; proactive trader; aggressive; passive; search cost; evasive strategies; deceptive strategies; offensive strategies; marketable limit order; adverse selection; liquidity rebate; micro price; iceberg order; footprint; order cancellation; spoofing

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, chap. 18, pp. 380–389; in slice 30, `s7-narang`, third edition, Wiley 2024, chap. 7, pp. 127–137, chap. 14, p. 270, and Appendix, pp. 311–315, text layers, and 7 U.S.C. §6c(a)(5), 2024 edition). Stability: draft. Currency: concepts timeless; Harris's systems and legal remarks as of 2002; Narang's venue details as of 2024, with no current venue rule checked; the U.S. Code passage as of its 2024 edition. Slice 34 (2026-10-16)

added the benefits and costs of exposure (Harris pp. 382–384), display instructions and precedence (pp. 84–85 and 117) and Bouchaud et al. p. 55.

Definition 9.10 Portfolio trading: program trades, the firm bid/offer market for packages, basket algorithms and index-fund trading

Statement. A *program trade* is the simultaneous submission of many orders; for statistics the NYSE and SEC counted trades of 15 or more coordinated transactions worth $1 million or more, about 27 percent of NYSE volume; index arbitrageurs did about 9 percent of the reported program-trade volume, about 2.4 percent of total NYSE volume [alpha-harris, chap. "Index and Portfolio Markets", p. 489]. In the *firm bid/offer market*, also called the package, basket or portfolio trading market, package dealers quote firm prices for a whole portfolio from a characterisation of it, usually relative to its end-of-day value (for example the close minus 15 cents a share), and the list is revealed only after the close [alpha-harris, same chap., sec. 23.4.1, p. 490]. A *basket algorithm* manages cost against total basket risk, using correlations, rather than each stock's own risk; constraints include cash balancing, self-financing and minimum and maximum participation [s7-kissellscience, chap. 1, pp. 18–19].

Conditions. The characterisation states whether the portfolio is an index portfolio and, if not, summary statistics: quantities, firm sizes, betas, price levels, average trading volumes, volatilities, primary listings and index membership (p. 490). Dealers price portfolios better than single stocks because informed traders are less likely to trade portfolios; clients pay them to assume the trading problem, and dealers may place parts with other clients (pp. 490–491). Index products (index futures, index options and shares in index funds) let traders trade index risk more cheaply than the components, for three reasons Harris gives: index dealers rarely face well-informed traders, because few have valuable insight into the whole market; index markets are very active, so counterparties are easy to find and dealers turn over inventory quickly; and a single transaction is arranged, cleared and settled instead of many [alpha-harris, same chap., p. 484 and sec. 23.4, p. 489]. Index prices generally lead the cash index, because index traders price only index risk (p. 489).

Practice. Index funds rebalance a value-weighted portfolio only when the index list changes, and a price-weighted one also at splits [alpha-harris, same chap., sec. 23.2, p. 486]. They can improve returns by supplying rather than demanding liquidity, substituting non-index stocks for expensive components, rebalancing only with flows and dividends, and holding a subset of the index; each tactic raises tracking error, so many funds use them sparingly (p. 487). A basket algorithm may slow a stock despite good liquidity, or speed one despite poor liquidity, if that reduces residual basket risk (Kissell, pp. 18–19).

Narang describes the same service as a *portfolio bid*: the trader describes a "blind" portfolio by its characteristics, such as the valuation ratios of longs and shorts, sector breakdown and market capitalisations; brokers quote a fee in basis points of the gross market value traded and guarantee the price, and the broker then bears the risk of trading out of the portfolio at future prices [s7-narang, chap. 7, "Execution", p. 128]. "Human" execution of quant portfolios "often looks like" a portfolio bid (p. 128).

Pitfalls. The legacy taxonomy treated basket trades and package trades as different; Harris uses package, basket and portfolio trading as names for one market. It explained tight index-basket quotes by diversified residual risk; Harris's reason is less adverse selection (Interpretation: diversification may also help, but is not in the passage). Its characterisation inputs added sector, currency, country, factor loadings and borrow constraints, and its index-rebalance controls named futures and ETFs as substitutes and timing around the effective window; Harris names non-index stocks, and the timing around reconstitution is in index reconstitution. The legacy principle *Index Markets Trade Common Risk* (slice 18) gave four reasons for lower costs: index flow is less security-specific, more active, easier to net and easier to hedge with futures, ETFs, options or baskets. The first two match Harris's adverse-selection and activity reasons; "easier to net" and "easier to hedge" are not among his reasons, and his third, a single transaction to arrange and settle, is missing from the legacy list. "Portfolio transition" is named by Kissell ("transition management", p. 16) but not defined in the passages checked. The NYSE figures are as of 2002.

See also. block trades (single-stock size); index reconstitution (price pressure from forced index trading); ETF creation and redemption (owned by `etf-index-closing-execution`); trading envelope; pre-trade analytics (portfolio consistency checks).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. program trade; program trading; package trading; basket trading; portfolio trading; firm bid/offer market; package dealer; portfolio characterisation; basket algorithm; portfolio algorithm; index fund; tracking error; index rebalancing; transition management

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, chap. 23, pp. 486–491, with pp. 484 and 489 added in slice 18, and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 1, pp. 16–19; in slice 30, `s7-narang`, third edition, Wiley 2024, chap. 7, p. 128). Stability: draft. Currency: market figures as of 2002.

Mechanism 9.11 Positive-feedback hedging: portfolio insurance sells into falls and buys into

rises, and its trades reinforce the move

Statement. Portfolio insurance is a dynamic trading strategy that replicates a portfolio plus a put option; the formula for how much to trade comes from the Black–Scholes model and depends on the portfolio's volatility, the protection period and the guaranteed minimum value. Insurers buy when prices rise and sell when they fall, the same effect as stop orders, which many used. It works when the money covered is small, but when a large fraction of the market is insured, sellers arrive together and prices can fall substantially; order anticipators sell at the first sign of a fall and buyers withdraw, so expected insurance selling can drive prices down even before it occurs [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", sec. 28.2.2, pp. 560–561]. Interpretation (the legacy proposition): if a rule holds $q(S)$ shares with $q'(S) > 0$, a fall $dS < 0$ gives $dq = q'(S)\,dS < 0$, a sale; if each trade moves the price by $\lambda\,dq$ with $\lambda > 0$, the induced move has the sign of the original one.

Conditions. Interpretation: if the induced move persists until the rule trades again and triggers the same response, the moves form a geometric series with ratio $\lambda q'(S)$, which is dimensionless (λ in currency per share, q' in shares per unit of currency); the total is $dS/(1-\lambda q'(S))$ for $\lambda q' < 1$, for example $1.67\,dS$ at $\lambda q' = 0.4$, checked by iteration. Purely temporary impact that reverts before the next rebalance gives no such amplification. Harris notes that exchange-listed puts would have made the insured amount visible through open interest and reduced the uncertainty that kept buyers away (p. 561).

Practice. Harris's rough sizing for 1987: about $100 billion insured, a typical delta of about 0.4, so a 1 percent fall would require selling about 0.4 percent, roughly 10 million shares at $40, against average daily volume of 160 million shares (box, p. 561). Insurers who raised the protected value as markets rose had to sell more on the way down (box, p. 560).

Pitfalls. Interpretation: the size of a rebalancing trade depends on how much the hedge ratio changes, the option's gamma, not on its delta. In a Black–Scholes illustration with zero rates, 20 percent volatility and one year to expiry, a 1 percent fall requires selling about 1.0, 1.6 and 1.8 percent of the insured value for protection at 80, 90 and 100 percent of the current value. Harris's box multiplies the fall by the delta, so its 10 million shares is an order-of-magnitude guess on a different basis. The legacy proposition shows only the sign of the effect, not its size or stability.

See also. bubbles and crashes (the 1987 crash, owned by `execution-market-making`); price impact; portfolio trading.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. portfolio insurance; dynamic hedging; synthetic put; positive feedback trading; stop orders; hedging feedback; 1987 crash; delta; gamma; insurance overhang

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, chap. 28, pp. 560–561, with boxes; the feedback series and the Black–Scholes illustration are own calculations). Stability: draft. Currency: mechanism timeless; the sizing is historical, for 1987.

9.4 Transaction costs and transaction cost analysis

Definition 9.12 Transaction costs: explicit, implicit and missed-trade costs, and an accounting split of shortfall

Statement. Transaction costs include all costs associated with trading [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.1, pp. 421–422]:

- *explicit costs*, which a cost accountant would identify: commissions, exchange fees, taxes, and the resources devoted to trading, such as running a buy-side desk;
- *implicit costs*, which arise because traders affect prices: the bid–ask spread paid by buying at the ask and selling at the bid, and the price impact of large orders;
- *missed trade opportunity costs*, when orders fail to fill, or fill late. Harris's example: a cotton buyer who bid 64.95 cents when the price was 65 and saw it rise to 68 lost 2.75 cents a pound against an aggressive fill at 65.25, $137,500 on 100 contracts (checked).

Explicit costs are easy to measure. Implicit and missed-trade costs need a benchmark: what prices would have been without the trade, or what the unfilled order would have cost to complete (p. 422). Kissell lists ten components, commissions, fees, taxes, rebates, spreads, delay cost, price appreciation, market impact, timing risk and opportunity cost, and classifies them as investment-related, trading-related and opportunity costs, fixed or variable, visible or hidden [s7-kissellscience, chap. 3, pp. 89–95, Tables 3.1–3.2]. Narang names three major components for a quant's cost model [s7-narang, chap. 5, "Transaction Cost Models", sec. 5.1, pp. 86–91]: *commissions and fees*, paid to brokerages, exchanges and regulators, small per trade for quants who use only a bank's infrastructure, with brokers' clearing and settlement fees usually inside the commission (pp. 86–87); *slippage*, "the change in the price between the time a trader (or quant system) decides to transact

and the time when the order is actually at the exchange for execution" (p. 87); and *market impact*, how much an order moves the market by its demand for liquidity, "normally defined as the difference between the price at the time a market order enters the exchange and the price at which the trade is actually executed" (pp. 88–89). His terms map onto the ones above as set out under Pitfalls.

Notation. Interpretation: with the signed quantities of implementation shortfall (S intended, s_j filled at p_j, decision price P_d, arrival midpoint P_0, end price P_n) and m_j the quotation midpoint just before fill j, shortfall splits additively as

$$\text{IS} = \underbrace{S(P_0 - P_d)}_{\text{delay}} + \underbrace{\textstyle\sum_j s_j(p_j - m_j)}_{\text{half-spread paid}} + \underbrace{\textstyle\sum_j s_j(m_j - P_0)}_{\text{midpoint drift}} + \underbrace{\big(S - \textstyle\sum_j s_j\big)(P_n - P_0)}_{\text{opportunity}} + \text{fees}.$$

The second term is the filled size times Harris's liquidity premium, half the effective spread at each fill (spread measures); the middle two terms are Wagner's trading-related part. Checked with exact arithmetic: a buy of 10,000 shares (decision 50.00, arrival 50.10, fills of 3,000 at 50.20 and 50.30 against pre-fill midpoints 50.15 and 50.25, rest cancelled, $P_n = 50.60$, fees 60) gives $1{,}000 + 300 + 600 + 2{,}000 + 60 = 3{,}960$, the paper-minus-real total; the mirror-image sell gives the same.

Conditions. The split is an accounting identity, not a causal attribution. The midpoint drift mixes the order's own impact with market movement and other traders' impact. Kissell defines market impact as the difference between the price path with the order and without it, which cannot be observed together, so impact cannot be measured exactly [s7-kissellscience, chap. 3, pp. 91–92]; Harris likewise needs a no-trade benchmark (p. 422). The half-spread term measures each fill against its own contemporaneous midpoint, so it misses the cumulative impact of earlier fills, which falls into the drift (Harris's split-order example, pp. 428–429). Kissell's timing risk is the uncertainty of cost, not a realised component (p. 92), and his price appreciation is the price trend that would have occurred anyway (p. 91).

Practice. Interpretation: an index or market adjustment (transaction cost analysis) removes the part of the drift predicted by the market's move; what remains is still a residual, which includes impact, stock-specific news and model error, and is impact only under the model used to remove the rest. The overview's checkpoint asks for a split into spread, impact, timing and fees: on the sources checked, spread, delay (timing before arrival), opportunity and fees are measurable as above, but "impact" is available only as a model estimate or a market-adjusted residual, not as an observed component.

Pitfalls. Count each cost once: the spread is inside fill prices, so adding a quoted or effective spread to an arrival-price shortfall counts it twice; fees enter once. The legacy taxonomy listed "adverse selection" and "delay cost" among implicit costs "relative to a benchmark"; in Harris adverse selection is a cause of the spread, and in Kissell delay is an investment-related cost before the order reaches the market. "Borrow charges" among explicit costs is not in the passages checked. Kissell describes the spread as the round-trip cost for small orders, not blocks (p. 90).

Narang's terms (slice 27). Interpretation of the mapping, with his claims kept as his:

- *Slippage* runs from the decision to the order's arrival at the exchange, so it is the delay term $S(P_0 - P_d)$ above, signed by side: his sale decided at \$100.00 that reaches the market at \$99.90 costs \$0.10 a share, and a move the other way is a gain (pp. 87–88). The word is not used uniformly (interpretation); name the benchmark and the interval.

- *Market impact* in his definition runs from arrival to fill. It is therefore the trading-related part of shortfall (half-spread paid plus midpoint drift), not impact in the causal sense of price impact. His own example shows the difference: a sale decided at \$100.00 reaches the market at \$100.05 and fills at \$100.20 in a rising market, giving slippage of −\$0.05 and "impact" of −\$0.15 (checked). That is a negative arrival-to-fill cost against a benchmark; it does not identify what the order itself did to the price, and he concedes that impact may be impossible to separate from slippage or from the stock's own move (pp. 89–90).

- *Known after execution?* Narang writes that one does not know how much a trade impacts the market "until the trade has already been completed" (p. 89). After completion the fills and the realised cost against a benchmark are known; the impact is not, because the price that would have prevailed without the order is never observed (Kissell, above; price impact). It is estimated by a model or across many orders.

- *Conditional claims.* That trend-following strategies suffer most from slippage and mean-reverting ones least, sometimes gaining (p. 88), is a statement about the expected sign of the price move over the delay relative to the trade's side, and holds only to the extent that prices continue in the direction a strategy trades or revert against it over that interval; he gives no data. Lower latency reduces slippage by shortening the delay (p. 88), and higher volatility raises it (pp. 88 and 91); interpretation: with no expected drift over the delay, a longer delay or a higher volatility widens the spread of outcomes rather than raising their

mean, and raises the expected cost only together with an adverse drift, such as an accurate short-term forecast that the market starts to price, which is his own condition (p. 88). His GOOG figure, an average daily range 16 times the average day-to-day move, is undated and unsourced, and was not used.

- *Fees.* The legacy pairing of brokerages with access, exchanges with security and regulators with infrastructure reads his list "respectively"; the text does not pair them, and on p. 87 credits exchanges with access to liquidity and with guaranteeing that both counterparties perform. Interpretation: in current market structure that guarantee is given by a central counterparty, which "interposes itself between counterparties" and ensures the performance of open contracts [exp-pfmi, para. 1.13, p. 9], whether or not the exchange operates it. His fixed charges as a floor "below which the expense of trading cannot go" (p. 91) ignore the liquidity rebates he describes on p. 90 (about 0.2 cents a share to a provider against 0.3 cents charged to a taker, the venue keeping about 0.1).

- *Clearing and settlement.* Narang's clearing, "regulatory reporting and monitoring, tax handling, and handling failure" before settlement, and his settlement, "the delivery of securities in exchange for payment in full" (pp. 86–87), are his descriptions of what brokers charge for, not standard definitions. The CPSS–IOSCO principles describe securities settlement systems that transfer securities "either free of payment or against payment", with delivery versus payment when "delivery of the security occurs if and only if payment occurs", and that may add clearing functions "such as the confirmation of trade and settlement instructions" [exp-pfmi, para. 1.12, p. 8], and central counterparties that net trades multilaterally and take margin [exp-pfmi, para. 1.13, p. 9]. The CPMI glossary's definitions of clearing and settlement were not retrieved; no definition of clearing is adopted here (stub).

See also. implementation shortfall; execution benchmarks; spread measures, an entry of the microstructure overview; price impact, an entry of the price-impact pathway; transaction cost analysis.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`; Narang's cost-components block (slice 27): `transaction-cost-models.tex` in the folder `notes/artefacts/` under the section key `portfolio-construction-risk`.

Index. transaction costs; explicit costs; implicit costs; missed trade opportunity cost; commissions; fees; taxes; rebates; spread; delay cost; price appreciation; market impact; timing risk; opportunity cost; investment-related cost; trading-related cost; midpoint drift; liquidity premium; cost decomposition; slippage; Narang's cost components; clearing fees; settlement fees; delivery versus payment; central counterparty; liquidity rebates; latency and slippage

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 420–423 and 428–429, and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 3, pp. 87–96; the split and examples checked with exact arithmetic; in slice 27, `s7-narang`, third edition, Wiley 2024, chap. 5, pp. 85–91, and `exp-pfmi`, April 2012, paras. 1.8–1.13, pp. 7–9, both text layers, Narang's examples recomputed; the reading of his sell example corrected 2026-10-15 in the slice-27 follow-up). Stability: draft. Currency: timeless (accounting); Narang's account as of 2024.

Method 9.13 Transaction cost analysis: measuring costs after the trade, predicting them before, and comparing brokers and algorithms

Statement. Traders measure transaction costs to evaluate how well they and their brokers implemented their strategies, whether they traded too aggressively or not enough, and which brokers should get future orders; they predict costs to judge whether strategies are worth implementing. Exchanges, brokers (notably to show best execution when paid for order flow), sponsors (ERISA commission duties) and regulators measure them too [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 420–421]. Lehalle and Laruelle separate post-trade analysis, usually of one day, which explains what the market context did, from TCA over months of trades, which uses averaging to cancel most market effects and find what part of performance comes from the algorithm's design; its outcome is to match an algorithm or provider to a market context or trading style, so that flow is routed accordingly, using breakdowns by volatility, jumps, liquidity and news, and comparing only stocks of similar liquidity (bid–ask spread, tick size in basis points, quantity on the book, turnover, trade size, free float, trading rate) [alpha-laruelle, sec. 3.1.3, pp. 202–203].

Notation. Kissell's measures [s7-kissellscience, chap. 3, pp. 110–113]: index cost = (index VWAP − index arrival)/index arrival $\times 10^4$ over the trading horizon (eq. (3.21)); index-adjusted cost = arrival cost $-\hat{\beta}\times$ index cost (3.22); and Z = (pre-trade estimate − arrival cost)/pre-trade timing risk (3.23), positive when the trade beat its estimate. A cost *measure* is ex post and single-valued; a cost *forecast* is ex ante, a mean with a standard deviation, the timing risk (pp. 95–96).

Conditions. Prediction: explicit information (contracted fees and displayed liquidity, an upper bound on impact for orders smaller than displayed size) and implicit information from past costs, modelled by regression on order variables (size, price placement), contemporaneous market variables (spread, displayed size, recent volume, recent returns, money flow) and general ones (average volume, volatility, market capitalisation); past costs

predict future ones only if liquidity conditions are unchanged [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.7, pp. 438–439]. Comparison: estimates from single trades are noisy, averaging removes noise but not bias or gaming, and comparing brokers is meaningful only for similar trading problems (p. 432). Kissell's Z should be near mean zero and variance one if pre-trade estimates are accurate; the test should be run by order size and strategy, on shares actually traded rather than the full order, on enough orders, and with the investor's own pre-trade model rather than the broker's (pp. 112–113).

Practice. Kissell separates cost from performance: comparing the execution price with the day's VWAP is a performance proxy, with the close a tracking-error proxy, while comparison with the open or the arrival price is a cost (p. 96). Harris advises that strategists and traders share why they trade, and that incentives reward joint performance rather than blame (pp. 439–440).

Pitfalls. Interpretation: the population, aggregation weights and benchmark must be stated: averaging by order and by value answer different questions, and both depend on which orders reached each broker. Lehalle and Laruelle's averaging cancels market effects only if they are unrelated to which algorithm was chosen; if traders send urgent or difficult orders to one provider, its average cost is worse whatever its quality, so a routing table built on unadjusted averages confounds selection with skill, and outperformance of a benchmark does not show causal superiority. The legacy failure modes (wrong exposure, wrong clock, own-impact contamination, gaming, selection) are consistent with Harris's biases (sec. 21.4) and Kissell's conditions but are a synthesis, not a list in either. Kissell's index cost is "not adjusted for the side of the trade" (p. 112); with a side-signed arrival cost this is wrong for sells: a sale at 50 basis points below arrival while the index fell 50 basis points and $\hat{\beta} = 1$ shows 100 basis points of adjusted cost instead of 0, so the index term needs the side factor (Interpretation). His approximation "Index VWAP $= \frac{1}{2}R_m$" is an approximation to the index cost, not the index VWAP. The legacy prediction variables added venue variables (hidden fill rates, fee tiers, queue loss, routing latency) and limit price and urgency, which are not in the passages checked.

See also. transaction costs; execution benchmarks; pre-trade, monitoring and post-trade tools; aggressiveness trade-off; best execution; the market adjustment parallels relative performance evaluation of managers.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. transaction cost analysis; TCA; post-trade analysis; cost measurement; cost prediction; pre-trade cost model; index-adjusted cost; market-adjusted cost; z-score; timing risk; broker evaluation; algorithm selection; routing table; selection bias; liquidity breakdowns; cost versus performance

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 420–421, 432 and 438–440; `alpha-laruelle`, 2018 World Scientific printing, pp. 202–203; and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 3, pp. 95–96 and 110–113, p. 112 by page image). Stability: draft. Currency: practitioner methods as of 2002, 2014 and 2018.

Method 9.14 Trading aggressiveness: balancing the transaction cost of trading harder against the missed-trade cost of trading softer

Statement. If the additional transaction costs of trading more aggressively are less than the missed trade opportunity costs it would save, traders earn more by trading more aggressively; if the transaction costs saved by trading less aggressively exceed the additional opportunity costs, they earn more by trading less aggressively. Most traders do not estimate these marginal costs but assume they equal average costs, trading harder or softer as their per-unit opportunity costs exceed or fall short of their per-unit transaction costs; Harris advises estimating marginal transaction costs from the costs of the last trades that fill orders [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", sec. 21.6, pp. 437–438].

Conditions. Interpretation: the objective is expected total cost, transaction cost plus missed-trade cost, of an order whose size is given; the choice variable is aggressiveness (limit placement, urgency, participation); the rule is the first-order condition of that minimisation and assumes both costs change smoothly with aggressiveness and that the missed-trade cost is measured realistically. Harris states it as advice, with no model or proof. Opportunity costs count only for orders that could have been filled, and depend on the horizon at which they are measured (pp. 435–436).

Practice. Well-informed traders, especially on short-term information, should accept substantial transaction costs rather than fail to trade, and strategists should give traders price limits that reflect the value of their information (p. 439). Interpretation: a mean–variance schedule (Almgren–Chriss) is one formal version of the same trade-off, with price risk in place of missed-trade cost.

Pitfalls. The legacy note ended "average cost is the wrong control variable for the last slice"; Harris's point is that marginal, not average, costs should drive the decision, and that the last fills estimate them. Interpretation: comparing average costs gives the right direction only if marginal and average costs move together.

See also. transaction costs; implementation shortfall (the opportunity term); mean–variance execution; order exposure decisions.

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. trading aggressiveness; marginal transaction cost; marginal opportunity cost; missed trade opportunity cost; average cost; order submission; urgency

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 435–439). Stability: draft. Currency: timeless (decision rule).

9.5 Best execution and broker conflicts

Definition 9.15 Best execution: three standards, the dimensions of execution quality, and why retail clients see commissions but not prices

Statement. A broker taking an order assumes an agency responsibility to obtain best execution, which is not well defined. To unsophisticated clients it means the best possible price for a market order and the fastest execution for a limit order, an absolute standard, which is what the term generally means in U.S. equity markets; to more sophisticated traders, the execution they pay for; to the most sophisticated, the execution they can expect given what they pay and how well they can audit it [alpha-harris, chap. "Brokers", sec. 7.4.2, pp. 160–161]. Price is one dimension of execution quality and speed another; market-order traders care most about speed, limit-order traders about price, and best execution for standing limit orders depends on whether and when they execute [alpha-harris, chap. "Internalization, Preferencing, and Crossing", sec. 25.1, p. 515]. Kissell defines best execution as the process of choosing the strategy most likely to achieve the fund's investment objective, judged on the information available when trading began, not by closeness to a benchmark price [s7-kissellscience, chap. 3, p. 88].

Conditions. In U.S. equities as Harris describes them in 2002, dealers claim best execution when they fill marketable orders at the NBBO, a standard that applies only to orders smaller than the displayed size; some markets also expect average price improvement for small market orders (p. 515). Brokers that internalise or accept payment for order flow have a conflict of interest and must regularly and rigorously examine execution quality; SEC Rules 11Ac1-5 and 11Ac1-6 required market centres to publish monthly execution-quality statistics and brokers quarterly routing reports (pp. 517–518, boxes). Retail traders can audit commissions and response times easily but not price quality, which needs trade and quote data and norms for comparison; so brokers compete on commissions, and in competitive markets the total of spread and commission for small market orders does not depend on how best execution is defined (pp. 517–518). Lehalle and Laruelle contrast two regimes as of 2018: in the U.S. the trade-through rule, they write, makes venues reroute orders marketable at a better price elsewhere, which needs a consolidated pre-trade view; in Europe the executing broker chooses how to consolidate access to liquidity and must disclose and justify a best-execution policy, raising the cost of fragmentation for intermediaries [alpha-laruelle, sec. 3.4.2, p. 234].

Practice. Retail execution quality is measured by price against the NBBO, the midpoint or the effective spread (spread measures), speed, whether and when limit orders execute, and commissions; audit firms compare each night's trades with quotes and report exceptions (Harris, pp. 515–519). Interpretation: these measures depend on order type and size and on the quotes at the time; comparisons between venues need like orders.

Pitfalls. Three obligations should not be merged: *quote protection*, *best execution* and *execution-quality reporting*. Quote protection in the U.S. is the order protection rule, 17 CFR 242.611, checked on 15 October 2026 against the eCFR point-in-time text for 13 October 2026 (adopted 70 FR 37620, 29 June 2005; last amended 89 FR 26616, 15 April 2024, whose effective and compliance dates were not checked): it requires a *trading center* to maintain written policies reasonably designed to prevent trade-throughs of protected quotations in NMS stocks, and to surveil them, with exceptions including system failure, non-regular-way trades, single-priced opening, reopening and closing transactions, crossed protected quotations, intermarket sweep orders (executed, or routed simultaneously against the full displayed size of protected quotations), orders whose price is not based on the quote, venues whose best quote within the previous second was equal to or worse than the trade-through price, and stopped customer orders (paragraphs (a)–(b)(9)). It is a duty of the venue, not the broker's best-execution duty, and it does not oblige a venue to reroute: an intermarket sweep order lets the router take the protected quotations itself. Lehalle and Laruelle's claims that U.S. clearing and settlement fees must be equal across venues, and their description of the European regime, were not checked against current law; the European statements here are as of their 2018 printing. The legacy principle (slice 14) that "U.S. routing leans on protected consolidated quotes" while "European brokers choose the consolidation layer" follows their p. 234. The legacy principle called best execution "the best auditable outcome given the client's objective, limit, urgency, available venues and information risk"; that merges Harris's third standard (auditability) with Kissell's process definition, which do not coincide. Its retail "adverse selection: post-fill price movement" measure is a realised-spread idea (spread measures), not in the best-execution passages checked. Rules 11Ac1-5 and 11Ac1-6 are now Rules 605 and 606; the expanded Rule 605 reports required from August 2026, and the 2025 withdrawal of the SEC's best-execution proposal, are the subject of this pathway's readings `ex-sec-rule605-2024` and `ex-sec-withdrawal-2025`, not read: the regulatory statements here are as of 2002.

See also. broker conflicts; transaction cost analysis; spread measures.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. best execution; execution quality; NBBO; price improvement; speed of execution; limit order execution; retail execution quality; Rule 11Ac1-5; Rule 11Ac1-6; Rule 605; Rule 606; payment for order flow; internalisation; auditability; transaction audit; order protection rule; Rule 611; trade-through; protected quotation; intermarket sweep order; best-execution policy

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 160–163 and 514–519, `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 3, p. 88, and `alpha-laruelle`, 2018 World Scientific printing, p. 234; 17 CFR 242.611 checked on 15 October 2026 against the eCFR point-in-time text for 13 October 2026, slice 14). Stability: draft. Currency: Rule 611 as verified on 15 October 2026; other U.S. rules and practice as of 2002, European regime as of 2018; superseded in part, see Pitfalls.

Definition 9.16 Broker conflicts: the principal–agent problem in order handling, from payment for order flow to front running

Statement. Brokers are agents whose interests can differ from their clients'; clients manage the problem with rewards and penalties, which require measuring performance, and measuring execution quality is hard [`alpha-harris`, chap. "Brokers", secs. 7.4–7.4.1, pp. 159–160]. Harris's conflicts (same chap.):

- *dual trading*: a broker-dealer that internalises a client order wants the opposite price to the client, and may trade ahead of clients on the same side (sec. 7.4.3, p. 161);
- *order preferencing*: routing to a preferred dealer, an own dealer subsidiary, or in exchange for reciprocal flow or payment for order flow, about 1 cent a share in U.S. stocks in 2002, independently of current quotes (sec. 7.4.4, pp. 161–162);
- *soft dollars and directed brokerage*: commissions that buy research or other services, and sponsors directing orders to particular brokers or recapturing commissions (pp. 153–155);
- *dishonest practices*: front running, inappropriate order exposure, fraudulent trade assignment of the best fills to favoured clients, prearranged trading and kickbacks, unauthorised trading and churning (sec. 7.5, pp. 163–167).

Kissell adds venue rebates: when brokers keep maker–taker rebates or pay the fees, routing may serve the broker's net fees rather than the client [`s7-kissellscience`, chap. 3, p. 90]. Interpretation (the legacy proposition, slice 14): if routing to v costs the client C_v in expectation and earns the broker h_v, the client's choice is $\arg\min_v C_v$ and a broker maximising its own net would choose $\arg\min_v (C_v - h_v)$; they differ exactly when some a, b have $C_a < C_b$ but $C_a - h_a > C_b - h_b$. This states when a conflict can arise; it is not evidence that brokers route that way, since reputation, regulation and competition for flow also enter the broker's objective.

Conditions. Payments for order flow resemble kickbacks but generally are not, since competing dealers pass the value of order flow to brokers and brokers to clients through lower commissions (Harris, pp. 166 and 516–517). Dishonesty is likeliest where it is hard to detect; audit trails, supervision and reputation deter it (pp. 163–165).

Practice. Interpretation: transaction cost analysis and audit trails turn suspected conflicts into testable comparisons, such as fill quality by venue against the NBBO or price improvement by routing destination, but a comparison shows a conflict's effect only when orders are comparable.

Pitfalls. The legacy caveat on order preferencing (slice 14) restates Harris's preferencing for service, economics or reciprocal flow; its conclusion, that review must compare all-in fills rather than routing intent, is Interpretation consistent with his best-execution standards. The legacy taxonomy grouped "allocation" separately and listed "commission recapture" as an activity incentive; Harris treats recapture as a sponsor's tool to reduce the cost of soft-dollar commissions, not a broker's incentive. Figures and rules are as of 2002.

See also. best execution; order anticipation (front running); order exposure decisions.

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. principal–agent problem; broker conflicts; dual trading; internalisation; order preferencing; payment for order flow; reciprocal order flow; soft dollars; directed brokerage; commission recapture; front running; inappropriate order exposure; fraudulent trade assignment; prearranged trading; kickbacks; churning; maker–taker rebates; audit trail; routing conflict

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 153–155, 159–167 and 516–517, and `s7-kissellscience`, Academic Press printing with copyright 2014, chap. 3, p. 90). Stability: draft. Currency: practices and figures as of 2002 and 2014.

9.6 Smart order routing across fragmented venues

Definition 9.17 Smart order routing: splitting a marketable order across venues, and why displayed liquidity is only probable

Statement. When several venues trade the same instrument, marketability is venue-specific: with 50 shares offered at 10.00 on one venue and 150 at 11.00 on another, a buy limit at 10.50 is marketable on the first only [alpha-laruelle, sec. 3.4.2, pp. 233–234]. A smart order router (SOR) decides where the child orders of a trading algorithm go; in Lehalle and Laruelle's architecture it is one of the tactics working inside the envelope set by the risk-control layer (trading envelope). Their example (Table 3.3, p. 235, page image) is a market buy of 500 shares against three European books, routed by three brokers (pp. 235–236):

- B1 has Euronext and Chi-X and splits for the best average price: 101.15;
- B2 has all three venues but sends the whole order to the single venue with the best price: the text says 102.50 on Chi-X;
- B3 has all three and splits across them: 100.75, with a free choice of venue, "surely" by fees, for the last 50 shares at 102.00.

With latency or duplicated ("mirrored") liquidity the ranking changes. If a high-frequency trader co-located at Euronext and Chi-X mirrors 150 shares on Turquoise, which the brokers reach 2 ms later than Euronext, it cancels them before B3's order arrives; B3 learns of the 150 unfilled shares about 6 ms later and, with prices rising, ends up worse than B1. What matters is not the visible liquidity but the probability of obtaining it at a price when the order arrives, given latency differences, mirrored orders and hidden liquidity (pp. 238–239).

Temporal, then spatial (slice 34). Large orders are first split through time, balancing market impact against market risk, and in a fragmented market each child order is then split across venues [alpha-laruelle, chap. 1, sec. 1.2.1, pp. 62–64]. Given a child order's limit price and quantity, a split is optimal if it obtains a price at least as good as the limit and finishes as soon as possible, the timing having been decided above it; an SOR is an "agnostic device" that takes a limit or market order and splits it with knowledge of the available liquidity, and under MiFID it must be able to prove that each decision followed the agreed execution policy, which needs records of policies, market snapshots, decisions and executions (pp. 64–66). Its efficiency is judged by results, not latency: for a given order, at a price at least as good as the main market alone, the more executed on other venues the better (pp. 70–71). Aggregating books lowers the cost of a marketable order, which would otherwise walk one venue's book (Figure 1.16, sell order of 660 shares on Euronext and Chi-X, p. 68).

Notation. Interpretation, for the routing decision at the time a child order is released: the inputs are the visible books, the venues the broker's policy allows, the latencies to each, fees, and estimates of hidden liquidity and fill rates; the decision is a quantity per venue (and, for passive orders, a price); the constraints are the child order's size, side and limit and the parent's envelope. The SOR does not decide how much the parent trades when; that is the schedule (mean–variance execution).

Conditions. The example assumes no latency issue for the base case (p. 235) and full access to each venue's displayed depth. Liquidity seeking for a marketable order must allow for hidden liquidity and for the chance that a fast trader cancels when it sees the order reach another venue first; limit orders raise further issues (p. 239). A liquidity seeker removes liquidity using estimates of hidden liquidity (including dark pools), mirrored orders and latency, and posts limit orders to finish fast; it is an "enhanced limit order" lasting from about 10 minutes in liquid stocks to a few hours in illiquid ones, while adverse selection from finishing fast is left to a scheduling layer at 10 minutes to a few hours; the hard part is estimating fill rates and real liquidity, which cluster (pp. 242–243).

Practice. *Duplicate liquidity and transparency.* High-frequency market makers quote on several venues and cancel the rest when one fills, so aggregated pre-trade depth overstates what is executable; with pegged and iceberg orders too, "pre-trade transparency is not very useful for guessing where liquidity really is", and post-trade reports say more, which made small dark pools with post-trade transparency more attractive (chap. 1, pp. 69 and 71). An order's path has latency at each step (consolidating feeds, deciding, reaching venues, matching, returning reports): about 4 ms between Paris and London over a 2012 network backbone (p. 70). *Average trade size* fell after 2007 while the number of trades rose; because a trade's size is the smaller of the two sides, falling average trade size mostly reflects smaller resting orders spread over more venues, not smaller investor orders (Figure 1.15, FTSE 100, pp. 66–67). *Consolidation.* Harris's three mechanisms join a fragmented market: traders in each segment react to information from others, liquidity demanders and suppliers route to the segments where they do best, and arbitrageurs move liquidity between segments, so that "if an order cannot move to its best market, arbitrageurs will move the best market to the order"; all three need cheap public information about every segment [alpha-harris, chap. 26, sec. 26.4, pp. 533–534]. Interpretation, derived: if an executable ask a_1 on one venue and an executable bid b_2 on another satisfy $b_2 - a_1 > k$, the all-in cost of buying on one and selling on the other, buying and selling earns $b_2 - a_1 - k$ per unit until prices move or capacity

binds; the bound needs both quotes to be executable for the size and the costs to include fees, clearing and inventory. Lehalle and Laruelle allocate across venues from "fill rate curves", the expected fill intensity against distance from the best opposite price, which differ by venue (Figure 3.17, pp. 243–244), citing Laruelle et al. (2011, 2013), not consulted. Passive splits are in passive order split; fee effects in fee-adjusted venue choice.

Pitfalls. Interpretation, recomputed from Table 3.3 by sweeping the cheapest levels first: B1 101.15 and B3 100.75 are right; the best single venue for all 500 shares is Chi-X at 102.05 (Euronext 102.10, Turquoise 102.20), so B2's "102.50" looks like transposed digits. The bracketed quantities for B1, $(25 + 50 + 100 + 100)$ and $(0 + 25 + 100 + 200)$, add to 600 and list depth up to 102.00, not what was bought (200 of the 300 shares at 102.00). The legacy sweep taxonomy's "local best price: cheapest visible venue first" is not B2's rule, which sends the order to one venue. Its routing-logic list (queue position, reject history, gateway and matching-engine delays, information cost of probing) and its four-step routing loop that updates venue scores from fills are a reasonable synthesis but not lists in the source; only fees, latency, hidden and mirrored liquidity and fill-rate estimates are. The slice-34 legacy blocks' venue-selection list (price and depth, execution probability, fees, access, information) and their routing objective, "expected cost = spread/impact + fees − rebates + latency risk + information cost", are likewise the notes' own synthesis: the terms are not in common units, and the cost of a missed or late fill is missing. Displayed depth cannot be assumed to stay available while a sequence of child orders is routed. Venue names and the HFT example describe European markets around 2018.

See also. trading envelope (routers as tactics; owned by `execution-trading-costs`); order exposure decisions (whether to show and where); best execution (routing duties and price protection); passive order split; fee-adjusted venue choice; SOR and market-share dynamics; venue market share; European fragmentation evidence; dark pools; HFT strategies (probing for liquidity against order anticipation) and latency-arbitrage arms race (owned by `hft-market-design`).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. smart order routing; SOR; marketable price; marketable order; fragmentation; venue choice; child order; sweep; latency; mirrored liquidity; duplicated quotes; hidden liquidity; fill rate curves; liquidity seeker; liquidity seeking; co-location

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, sec. 3.4.2, pp. 233–244, page images of pp. 235 and 236; the example recomputed); extended 2026-10-16 in slice 34 against chap. 1, sec. 1.2.1, pp. 62–71, and `alpha-harris`, 2002 Oxford printing, chap. 26, pp. 533–534, with the cross-venue bound derived. Stability: draft. Currency: European venues and latencies as of 2012–2018; Harris as of 2002.

Result 9.18 Fee-adjusted venue choice: a worse price with lower fees wins only when the price step is smaller than the fee difference

Statement. For a buy of Q shares, taking liquidity on venue V at price P with proportional fee f, against venue V' at $P(1+\delta)$ with fee $f-\phi$, V' is cheaper when

$$P(1+\delta)(1+f-\phi)Q < P(1+f)Q \iff \delta < \left(1 - \frac{\phi}{1+f}\right)^{-1} - 1 \simeq \phi,$$

so a worse price is worth taking only if the tick, in relative terms, is about as small as the fee difference: at a price of 100 euros and a fee difference of 0.1 basis point, the tick would have to be below 0.1 cent [`alpha-laruelle`, sec. 3.4.2, pp. 236–237, page image of p. 236]. Figure 3.16 plots the critical tick against the fee for $\rho = \phi/f$ (p. 237). Liquidity providers, who pay a fee or receive a rebate when their limit orders fill, would post first on the venue cheapest for them, and market orders would go first to the venue cheapest for takers; if these differ, both cannot persist in equilibrium (pp. 237–238).

Notation. f and ϕ are fees as fractions of traded value, paid by the taker; a rebate is a negative fee. δ is the relative price difference, at least one tick.

Conditions. Interpretation: the threshold is exactly $\delta < \phi/(1+f-\phi)$ for a buy, comparing acquisition cost. For a sale, comparing proceeds, V' at $P(1-\delta)$ with fee $f-\phi$ is better when $P(1-\delta)(1-f+\phi) > P(1-f)$, that is $\delta < \phi/(1-f+\phi)$, also about ϕ. Illustrative values $P = 100$, $f = 3$ bp, $\phi = 1$ bp: the buy threshold is 0.99980 bp and the sell threshold 1.00020 bp; at $\delta = 0.99$ bp V' wins on both sides, at 1.01 bp V wins (checked with exact arithmetic). With fees per share, c on V and c' on V', the condition is $P\delta < c - c'$ on both sides: a one-cent tick beats a fee saving of 0.2 cents. The comparison assumes both quotes are executable for the whole quantity at the time of routing, both orders take liquidity, and the trader actually pays the venue fees; the spread is inside the two prices and is not added again.

Practice. *Maker–taker and fee complexity (slice 34).* MTFs' maker–taker schedules charge liquidity takers and rebate makers, which helps high-frequency market makers be profitable [`alpha-laruelle`, chap. 1, p. 69; chap. 2, sec. 2.4.3, p. 163]. Fee schedules can defeat a router's arithmetic: clearing and settlement fees computed on netted transactions make an order's cost depend on the member's net inventory, and conditional pricing ties fees to monthly turnover or to volume on another venue, so "the marginal gain due to one order is far from

obvious" (chap. 1, pp. 68–69). Rebates attract passive flow; the economically best venue is the one with the best expected all-in fill, not the best displayed price (interpretation). Interpretation: whether a broker passes venue fees and rebates to the client decides whose cost this is; when it does not, the comparison drives the broker's net, not the client's (broker conflicts). A favourable fee-adjusted quote says nothing about the chance of filling there, or about later price moves.

Pitfalls. The legacy caveat stated the inequality correctly but called it "price priority"; it is a cost comparison between two takers' prices, not a priority rule. Current fee levels and caps are not checked here; access-fee caps belong to this pathway's reading `ex-sec-tick-access-2024`, not read.

See also. smart order routing; transaction costs (fees as explicit costs); broker conflicts.

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. fee-adjusted price; trading fees; maker–taker; taker fee; rebate; tick size; critical tick; venue choice; fee arbitrage; access fees

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, sec. 3.4.2, pp. 236–238, page image of p. 236; buy and sell thresholds derived and checked). Stability: draft. Currency: timeless (arithmetic); no current fee schedule checked; slice 34 added Lehalle and Laruelle's chap. 1, pp. 68–69, and chap. 2, p. 163, on 2026-10-16.

Method 9.19 Passive order split: ranking venues by queue, distance and trading rate, and the minimum-resting-time split

Statement. To post a buy limit of D shares below the best ask across venues, Lehalle and Laruelle compare three criteria on a stylised book (Table 3.4, p. 240, page image): rank in the queue (shares ahead at the limit price), distance of the venue's best ask from the limit, which shows where sellers are aggressive, and the recent selling rate at that price. With Q_n shares ahead and selling rate r_n on venue n, an order of size D takes on average

$$\tau_n(D) = \frac{D + Q_n}{r_n}$$

to fill (eq. (3.3)). Splitting D so that every part finishes at the same time $\tau^\star$ gives $f_n D = \tau^\star r_n - Q_n$ and, summing, $\tau^\star = (D + \sum_n Q_n)/\sum_n r_n$ [`alpha-laruelle`, sec. 3.4.2, pp. 239–241, n. 16, page images of pp. 240–241]. In their example ($D = 500$; $Q = 450, 100, 200$; $r = 10, 5, 5$ shares a second on Euronext, Chi-X and Turquoise), $\tau(500)$ is 95, 120 and 140 s, $\tau^\star = 62.5$ s, and the split is 35, 42.5 and 22.5 percent, 175, 212 and 113 shares (Table 3.5, p. 241). The criteria rank the venues differently (Table 3.6, p. 242):

Criterion	1st venue	2nd venue	3rd venue
Rank in the queue	Chi-X	Turquoise	Euronext
Distance to the best	Euronext	Turquoise	Chi-X
Trading rate	Euronext	Chi-X or Turquoise (tied)	
Minimum resting time (split)	Chi-X (212)	Euronext (175)	Turquoise (113)

Sources: rows from [`alpha-laruelle`, sec. 3.4.2, Table 3.6, p. 242], with the split sizes from Table 3.5; checked by recomputation.

Conditions. Interpretation: the formula treats the queue ahead and the selling rate as fixed and known, ignores cancellations ahead of the order and later arrivals, and fills each part at the average rate, so τ is a deterministic approximation; the source points to stochastic methods for accurate splits (Laruelle et al. 2011, not consulted). Equal finishing times minimise the time to complete the whole order, the latest part, rather than an average.

Practice. Interpretation: a venue with $f_n < 0$ cannot finish by $\tau^\star$ even with nothing posted; drop it and recompute $\tau^\star$ over the rest. Setting f_n to zero without recomputing leaves fractions that no longer sum to one. Lehalle and Laruelle's liquidity seeker combines this posting with optimal removal of liquidity and keeps the parent inside a scheduling layer (p. 242; smart order routing).

Pitfalls. The book prints eq. (3.4) as $\tau^\star = (D - \sum_n Q_n)/\sum_n r_n$ (p. 241, page image), but its own footnote gives $D = \tau^\star \sum_n r_n - \sum_n Q_n$ and Table 3.5's 62.5 s needs $D + \sum_n Q_n$; with the minus sign the example gives -12.5 s. The legacy definition has the plus sign. The text calls the Euronext rate "buy orders hit the 95.00 price level"; for a resting buy order the relevant flow is sell orders, as for the other venues. The legacy table generalised Table 3.6 without venue names ("highest adjusted rate", "residual venue"), which does not match the source's rows, and the legacy taxonomy's "resting-time target" is the split, not a separate criterion. The split sizes are rounded (212.5 and 112.5).

See also. smart order routing; order exposure decisions; queue priority and limit-order fill risk in the microstructure overview's entries.

Further reading. The legacy blocks and table, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. passive order split; limit order placement; queue position; rank in queue; distance to best; trading rate; minimum resting time; fill time; venue ranking; Lehalle–Laruelle Table 3.6

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, sec. 3.4.2, pp. 239–242, page images of pp. 240 and 241; the example and the corrected formula recomputed with exact arithmetic). Stability: draft. Currency: timeless (method); the example is stylised.

9.7 Dark trading and fragmentation

Definition 9.20 Dark pools and crossing networks: derivative pricing, the kinds of dark venue, integrated and midpoint books, why traders use them, what the evidence says, and their risks

Statement. A *dark pool* is a venue that does not publicly display its available liquidity; its main interest is to protect institutions executing large orders from opportunistic short-horizon traders [alpha-laruelle, chap. 1, sec. 1.4, p. 102]. Dark trading existed before as internal crossing, upstairs trading and OTC trading (p. 103). Lehalle and Laruelle list, for European markets after MiFID (pp. 103–104): *crossing networks*, which cross buy and sell orders at a price, typically the mid-price, derived from the primary market; *internalisation pools*, in which a broker crosses its own flow; *electronic market makers*, which trade against the operator's flow and accept or reject incoming orders at once; *exchange-based pools*, hidden liquidity on lit exchanges executed only against sufficiently large contra orders; *dark aggregators*, which split an order across pools and adjust as it fills; and *size-discovery* (large-block) pools created in response to MiFID II's caps. Crossing networks are the only order-driven markets that are not auctions: all trades take place at prices determined elsewhere, under a *derivative pricing* rule, so they discover only whether traders will trade at those prices, not the prices themselves [alpha-harris, chap. 6, sec. 6.5, p. 132].

Notation. Crossing rules: Harris's 2002 U.S. networks were calls; Instinet's after-hours cross and the NYSE's After-hours Session I crossed at the 4 p.m. close, and POSIT crossed eight times a day at the primary midpoint at a random time within seven minutes after each call, allocating excess supply or demand pro rata; some networks operate continuously (pp. 132–134). The main European MTFs' dark books, under MiFID's reference-price waiver, crossed only at the midpoint of a reliable reference market's best bid and offer, with a two-book design: an *integrated book* for displayed orders and hidden orders large in scale (LIS), and a *dark midpoint book* for smaller hidden orders, which match only at the primary midpoint; they accepted mid-pegged dark orders with limit price, minimum acceptable quantity and immediate-or-cancel options (Lehalle and Laruelle, p. 104).

Conditions. *Derivative prices* must be neither stale nor manipulated [alpha-harris, chap. 6, sec. 6.5.2, pp. 134–136]. Informed traders trade eagerly at a stale price and refuse otherwise, an adverse-selection problem: the after-hours crosses received more buy orders when prices had risen after the close, and did not cross stocks with significant after-hours news (p. 135). Anyone who will trade at a price fixed elsewhere later may try to move it: Harris's example saves $15,000 on 500,000 shares by depressing the reference price 3 cents with a 1,000-share sale (p. 135). *Price ownership.* Primary markets argue that crossing networks skim the easiest orders and use their prices without paying for them; crossing customers reply that they should not pay for price discovery in which they do not take part, and that their own orders in the primary market produce those prices (sec. 6.5.1, p. 134). *Price discovery.* Lehalle and Laruelle state that dark pools cannot create prices and influence price formation indirectly by attracting activity; the literature they cite disagrees, some empirical studies finding reduced price discovery and others, including Zhu's model, improved informativeness of exchanges (pp. 105–107); Zhu (`wide-zhu-dark-pools`) and Buti et al. are not read here.

Practice. *Why use them.* Secrecy, to reduce information leakage and impact, and price improvement, trading at the midpoint without paying the whole spread (Lehalle and Laruelle, p. 108). *Evidence, as reported.* From Buti, Rindi and Werner's U.S. study, as Lehalle and Laruelle summarise it: dark activity is concentrated in liquid stocks; for a given stock it is higher on days with high volume, higher depth and lower intraday volatility, and lower on days with large order imbalances relative to volume; it is associated with better spreads, depth and volatility and with short-term overreaction (p. 107). Lehalle and Laruelle's own one-minute data for CAC 40 and FTSE 100 stocks on Chi-X, BATS and Turquoise dark books in the first three weeks of March 2010 show dark *turnover* concentrated in the most volatile deciles (nearly a quarter of it in the top tenth), but dark *market share* not rising with volatility, and rising strongly with the spread (about three times as high in the widest-spread decile as in the narrowest), most of all for less liquid SBF 120 and FTSE 250 stocks (pp. 108–114). *Fill rates.* Harris reports that U.S. crossing networks filled less than 10 percent of the order volume submitted, and that unfilled orders were usually sent elsewhere (p. 134); a dark fill rate measured only from executions therefore misses the orders that found nothing. *Bilateral and multilateral.* On a multilateral platform anyone may supply or take liquidity; bilateral trading keeps an asymmetry that protects liquidity providers from toxic flow and is easier to sustain in illiquid instruments (Lehalle and Laruelle, p. 114).

Pitfalls. *Gaming and toxicity.* Some high-frequency traders detect large dark orders by sending small or-

ders ("fishing") or orders on the opposite side, then move the primary price; protections include large minimum quantities, restricted access, tracking participants who behave like gamers and statistical detection of consistently profitable participants; indications of interest can leak information too (Lehalle and Laruelle, p. 115). *Internal crossing.* Brokers who cross internally may expose client orders only to their own clients, which makes the search for counterparties harder, and may favour some clients over others; brokers reply that they work harder for two commissions, and exposure to the market protects clients (Harris, chap. 25, sec. 25.4, pp. 520–521; broker conflicts). The legacy finding that dark activity is concentrated in "volatile periods" holds for dark *turnover* in Lehalle and Laruelle's deciles but not for dark *market share*, and conflicts with the lower-volatility days in Buti et al.'s results as reported; it is kept only in that qualified form. Venue lists, market shares (about 5 percent of non-OTC European trades and 10–12 percent of U.S. share volume, p. 105), the LIS thresholds and the MiFID II caps are dated and not current; the 2024 EU review is a backlog item of this pathway.

See also. matching and trade pricing (derivative pricing and pro-rata allocation); order exposure decisions (hidden orders and probing); block trades; smart order routing; best execution; broker conflicts.

Further reading. Zhu (2010), Buti, Rindi and Werner and the Rosenblatt dark-pool reports, cited by Lehalle and Laruelle, are not read. The legacy file is kept verbatim in `hidden-liquidity-dark-trading.tex` in the folder `notes/artefacts/market-structure-execution-quality/`.

Index. dark pool; crossing network; derivative pricing rule; reference price; midpoint cross; POSIT; internalisation pool; electronic market maker; exchange-based pool; dark aggregator; size-discovery pool; large in scale; reference-price waiver; integrated book; midpoint book; stale price; price manipulation; price ownership; dark market share; fishing; gaming; toxic flow; bilateral trading; multilateral trading; internal crossing

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, sec. 1.4, pp. 102–115, and `alpha-harris`, 2002 Oxford printing, chap. 6, pp. 132–136, and chap. 25, pp. 520–521). Stability: draft. Currency: U.S. crossing networks as of 2002; European venues and data as of 2010–2018; no current rule stated.

Method 9.21 Venue market share: value and trade-count shares, auctions and the ex-fixing share, aggregation across stocks, estimation error, and why shares depend on the time of day

Statement. For stock k traded on N venues between t_1 and t_2, with trade ℓ at price P_ℓ^k, size V_ℓ^k, time τ_ℓ^k and venue δ_ℓ^k, Lehalle and Laruelle define the share of venue n in traded value and in trades [`alpha-laruelle`, chap. 1, sec. 1.1.1, eqs. (1.1)–(1.2), pp. 34–35, page image of p. 35]:

$$\mathfrak{M}^k(n) = \frac{\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \mathbf{1}_n(\delta_\ell^k)}{\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k}, \qquad \mathfrak{M}_T^k(n) = \frac{\sum_{1 \le \ell \le T} \mathbf{1}_n(\delta_\ell^k)}{T},$$

the value share being close to the share in shares while the price varies little. Across stocks the share can be weighted by turnover, by index weight (free float or capitalisation, not turnover) or equally, and the answers differ (pp. 35–37). The right metric depends on the question (p. 37): the all-included share for a venue's position and revenue; for the probability that a *trade* occurs on venue n, the trade-count share, with auctions and continuous trading measured separately; for the probability that a unit of *currency* trades there, the value share, again separately for auctions. Treating market share as a proportion, its estimator for asset k and venue i is the turnover on i over total turnover, $\hat{p}_k^i$, and an index-level share is $\hat{p}^i = \sum_k w_k \hat{p}_k^i$ (app. A, secs. A.8.1–A.8.2, pp. 286–287).

Notation. $m(n)$ is venue n's share of auction volume and p the auctions' share of all volume. Interpretation, derived: venue n's continuous-session share is then

$$\mathfrak{M}(n; \text{ex-fixing}) = \frac{\mathfrak{M}(n) - m(n)\, p}{1 - p},$$

since its continuous turnover is its total less its auction turnover; Lehalle and Laruelle print $(1-m(n)p)\mathfrak{M}(n)/(1-p)$ (p. 35, page image), which agrees only when $m(n) = 0$. Checked (`slice34-checks.py`): a primary market with 60 percent of all trading and all the auctions, with auctions 20 percent of volume, has 50 percent of continuous trading; the printed formula gives 60.

Conditions. *Estimation error.* Lehalle and Laruelle treat trades as independent Bernoulli trials, so $\operatorname{Var} \hat{p} = p(1-p)/n$ with a normal confidence interval $\hat{p} \pm z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, adequate when $np > 5$ and $n(1-p) > 5$ (sec. A.8, pp. 284–286), and assuming independence across stocks they derive $\operatorname{Var}(\hat{p}_k^i - \hat{p}^i) = (1-2w_k)p_k^i(1-p_k^i)/n_k^i + p^i(1-p^i)/n^i$, so that a stock with index weight below one half differs from its index more variably than the index estimate alone (sec. A.8.3, pp. 287–288, page image of p. 288). Interpretation, checked by simulation: a value share is not a count of identical trials, and a venue choice repeated over the child orders of one parent breaks independence; with venue choices repeated in runs of 20 trades the standard deviation of a 2,000-trade share is 0.047 against the binomial 0.010. The comparison of a stock with an index that contains it is mechanically correlated; a leave-one-out index is the legacy notes' remedy (interpretation). *Time of day.* Shares differ between the first and last hours (p. 35) and with auctions, which in European markets

were mostly a primary-market monopoly; in August 2012 auctions carried 11 to 26 percent of primary-market volume across eleven European indices, and closing prices fed fund NAVs and derivatives' exchange delivery settlement prices (chap. 2, sec. 2.1, Table 2.1, pp. 117–119).

Practice. *Volume curves* follow reference times: a linked market's opening (U.S. opening for European stocks; Brazil and South Africa), scheduled macroeconomic releases (the U.S. employment report at 14:30 CET), derivative expiries (the Eurostoxx 50 delivery settlement price, the mean index level from 11:50 to 12:00 CET on the third Friday) and reference-price rules (India's last-half-hour VWAP; Hong Kong's median of five snapshots in the last minute, with 17 percent of volume in the last 15 minutes); the patterns move with daylight saving time and market holidays (sec. 2.1.2, pp. 129–139; Total SA, 1 January 2006 to July 2011). Rules that define a benchmark create volume around it. An average daily profile mixes such days and should not be used unfiltered for a VWAP schedule (p. 139); see intraday deseasonalisation.

Pitfalls. The legacy market-share formula matched eq. (1.1); its "trade-count share replaces traded value by $\mathbf{1}_n$" omitted that the denominator becomes the number of trades. Its claim that fixing auctions make a primary venue "look more competitive" is right only for the all-included share. Market share measured from trades is a rough prior for routing, not a measure of available liquidity (p. 38; smart order routing).

See also. fragmentation and concentration metrics; SOR and market-share dynamics; European fragmentation evidence; intraday deseasonalisation (volume models); execution benchmarks.

Further reading. Agresti and Caffo (2000) on small-sample intervals, cited by Lehalle and Laruelle, is not read. The legacy file is kept verbatim in `cross-venue-routing-fragmentation.tex` in the folder `notes/artefacts/market-structure-execution-quality/`.

Index. market share; venue share; value share; trade-count share; ex-fixing share; fixing auctions; turnover weighting; index weighting; proportion estimator; confidence interval; binomial approximation; volume curve; reference price; exchange delivery settlement price; witching day; linked market opening; macroeconomic releases

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, pp. 33–39, chap. 2, pp. 117–139, and app. A, sec. A.8, pp. 284–288, page images of pp. 35 and 288; the ex-fixing formula derived and the estimator's dependence computed). Stability: draft. Currency: methods timeless; data and auction shares as of 2006–2013.

Method 9.22 Fragmentation and concentration metrics: the Fragmentation Efficiency Index, coverage, the Gini coefficient and two-sample Kolmogorov–Smirnov comparisons

Statement. Fragmentation exists at several levels: market operators, trading venues, order books within a venue, orders split across books, and even connection protocols; market share is the most used metric but each level needs its own [`alpha-laruelle`, chap. 1, sec. 1.1.1, pp. 33–34]. For shares $q_1, \dots, q_N$ summing to one, the *Fragmentation Efficiency Index* is normalised entropy,

$$\mathfrak{F}(q_1, \dots, q_N) = \frac{-\sum_n q_n \log q_n}{\log N} \in [0, 1],$$

0 when all liquidity is in one pool and 1 when it is spread evenly (app. A, sec. A.1, Definition 1, pp. 247–249). *Coverage* is the percentage of an index's stocks needed to make 80 percent of its turnover on a venue, 80 percent for a perfectly balanced index; the *Gini coefficient* is $G = 1 - 2B$ with B the area under the Lorenz curve, equivalently half the relative mean difference, 0 for equality and near 1 for concentration (chap. 2, sec. 2.4.1, p. 159; app. A, sec. A.9.1, pp. 288–289). The two-sample Kolmogorov–Smirnov statistic is $D_{n,m} = \sup_x |F_n^X(x) - F_m^Y(x)|$, and equality of the two distributions is rejected at level α if $\sqrt{nm/(n+m)}\, D_{n,m} > k_{1-\alpha}$, the Kolmogorov quantile (sec. A.9.2, pp. 289–291, page image of p. 291).

Conditions. The FEI depends on the venues counted: an empty new venue lowers it from $\mathfrak{F}(t)$ to $\log N / \log(N+1) \cdot \mathfrak{F}(t)$ (about 80 percent for $N = 3$, 90 for $N = 6$), and Lehalle and Laruelle suggest counting a venue only above 1 percent share (p. 250). The same formula measures different things depending on the shares fed to it (trades, value, displayed depth, sessions, universes), and none measures access, connectivity or protocol costs or hidden liquidity (interpretation, consistent with p. 34). The KS test assumes independent observations from continuous distributions; venue shares across the stocks of one index are neither identically distributed nor independent of the index.

Practice. Checked (`slice34-checks.py`): shares $(0.6, 0.25, 0.15)$ give $\mathfrak{F} = 0.854$, and 0.676 with an empty fourth venue, the predicted ratio; the Lorenz and mean-difference Gini agree (0.565 on a lognormal sample of 101 stocks); Lehalle and Laruelle's Table A.3 values follow from $G = 1 - 2B$ (for example $B = 0.18$, $G = 0.64$). Their application compares SBF 120 market-share distributions on Euronext Paris and Chi-X for 2008–2012 (Tables A.3–A.4, pp. 291–292): the KS statistics are 0.664, 0.235, 0.121 and 0.083, with p-values 8×10^{-23}, 0.003, 0.35 and 0.78. For 2010 and 2012 the test fails to reject equality at 5 percent; that does not show the distributions are the same. Checked by simulation: samples of 30 from $N(0,1)$ and $N(0.3, 1.2^2)$ are not rejected in 89 percent of trials. For coverage, they report 45 percent for the FTSE 100 on the LSE, 52 percent for the AEX, 55 for the CAC 40 and 56 for the DAX, January to June 2012 (pp. 160–161).

Pitfalls. Lehalle and Laruelle write that for 2010 and 2012 "H0 is accepted, so the market share for the SBF 120 on Euronext Paris and Chi-X have the same distribution" (p. 292); a non-significant test is a failure to reject, not acceptance, and with an index's worth of stocks its power against moderate differences can be low. They also report Gini values of 0.96 to 0.99 for primary-market turnover across the FTSE 100, AEX, CAC 40 and DAX (p. 160) alongside coverage near 50 percent; with the Lorenz definition of their appendix, coverage of 45 percent is hard to reconcile with a Gini of 0.99, and the chapter's construction (sorted in decreasing order, "two times the area between the repartition curve and the bisector", p. 160) may differ; the values are reported, not relied on. The legacy notes' Gini for "participant activity" and "time segments" extends the metric beyond the passages, which apply it to stocks and, in the appendix, to types of actors in general (p. 288).

See also. venue market share; European fragmentation evidence (coverage history); SOR and market-share dynamics.

Further reading. Shannon (1948), Gini (1912) and Kolmogorov (1933), cited by Lehalle and Laruelle, are not read. The legacy file is kept verbatim in `cross-venue-routing-fragmentation.tex` in the folder `notes/artefacts/market-structure-execution-quality/`.

Index. fragmentation; Fragmentation Efficiency Index; FEI; entropy; venue count; coverage liquidity metric; Gini coefficient; Lorenz curve; relative mean difference; liquidity concentration; Kolmogorov–Smirnov test; two-sample test; failure to reject; test power

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, pp. 33–34, chap. 2, pp. 158–161, and app. A, secs. A.1 and A.9, pp. 247–250 and 288–292, page image of p. 291; FEI, Gini and KS power computed). Stability: draft. Currency: metrics timeless; data 2008–2012.

Model 9.23 Smart order routers and market-share dynamics: Lehalle and Laruelle's toy model of liquidity capture, learning and share feedback, and why variable liquidity loses flow

Statement. An SOR splits an order of size $V(\omega)$ across K venues by a key $r = (r_1, \dots, r_K)$, $\sum_k r_k = 1$, to maximise on average the liquidity it captures,

$$C(r, \omega) = \sum_{k=1}^{K} \min\{r_k V(\omega), D_k(\omega)\},$$

with $D_k(\omega)$ the liquidity venue k provides when asked [`alpha-laruelle`, app. A, sec. A.3.1, eq. (A.5), p. 255, page image]. It learns r by trial and error, reducing the share of a venue that gives less liquidity than average and raising that of one that gives more (eq. (A.7), p. 256). If a fraction a of participants route with SORs, venue k's share evolves as $p_k(t+1) = (1-a)p_k(t) + a\, r_k(t)$ (eq. (A.8), p. 257). Coupled, with independent lognormal liquidities whose mean shares are p_k and whose standard deviation is proportional to their mean, the model takes a 95/3/2 split, with SORs starting from equal keys, to about 50/25/25 (Figure A.3, pp. 257–258). Shares end near their start if liquidity is steady and the routers' initial keys are close to the start, and nearer equal splits when liquidity is noisy and keys start near uniform (sec. A.3.5, pp. 259–260).

Notation. $\delta_{r_k V < D_k} = 1$ when venue k had more liquidity than was sent to it. a is the share of SOR users and γ the learning rate.

Conditions. The book prints three slips, all in the page images: (A.5) says "Minimize" the expected capture, which the text maximises; (A.8) is printed $p_k(1+1)$ for $p_k(t+1)$; and (A.7) is printed with $\frac{1}{K}\sum_\ell \delta_\ell - \delta_k$, which would lower the key of a venue that had spare liquidity, against the text's description, so the sign is reversed (interpretation). The optimality condition (A.6) equates the venues' average indicators $E[\delta_{r_k V < D_k}]$; derived here, the first-order condition of the maximisation weights each indicator by the order size, $E[V\mathbf{1}\{r_k V < D_k\}]$ equal across venues, which reduces to (A.6) only when V is not random. Checked (`slice34-checks.py`): at a grid optimum the size-weighted terms are nearly equal $(0.341, 0.334, 0.327)$ and the unweighted ones less so $(0.429, 0.443, 0.448)$, and a simulation with the corrected sign takes shares from $(0.95, 0.03, 0.02)$ to $(0.52, 0.27, 0.21)$. The model is a toy: liquidity is exogenous apart from its mean, there are no prices, fees or latency, and convergence needs the liquidity process to be i.i.d. or ergodic (p. 256).

Practice. Lehalle and Laruelle's narrative version (chap. 1, sec. 1.2.2, pp. 71–73): routers adjust splits from the historical performance of their executions; "the fraction of an order to send to a venue with a large variance will thus be minimized by a rational SOR", and venues under-weighted long enough lose share; with shares of 76, 17 and 7 percent in their simulation, the largest market's share fell from 87 to 56 percent with medium variance and to 40 with high variance. They add that "all studies have shown that variance is proportional with market share", so primary markets have more variance and investors send Chi-X more than its share would imply (p. 72); that statement is reported without a source and is not checked. The legacy taxonomy of drivers (initial share, liquidity level and variance, SOR adoption a, reassessment speed γ) restates the model's parameters.

Pitfalls. The legacy block, filed as an empirical finding, said that volatile fills at an alternative venue lead routers to cut its allocation and push flow back to the primary market. That direction follows from the model for a noisier alternative venue; but it is a simulation and model result, not an empirical finding, and Lehalle

and Laruelle's own (unsourced) claim is that the primary is usually the noisier venue, so in their account the same mechanism erodes the primary's share.

See also. smart order routing; venue market share; European fragmentation evidence.

Further reading. Foucault and Menkveld (2008) and Laruelle et al. (2011), cited by Lehalle and Laruelle, are not read. The legacy blocks are kept verbatim in `cross-venue-routing-fragmentation.tex` in the folder `notes/artefacts/market-structure-execution-quality/`.

Index. smart order router; SOR; liquidity capture; routing key; learning by trading; stochastic approximation; market-share dynamics; SOR adoption; liquidity variance; reinforcement; fragmentation dynamics

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, sec. 1.2.2, pp. 71–73, and app. A, sec. A.3, pp. 253–260, page images of pp. 255 and 257; the first-order condition derived and the dynamics simulated). Stability: draft. Currency: a stylised model of European fragmentation after 2007.

Evidence 9.24 European equity fragmentation after MiFID: what moved market share, which stocks fragmented, the Spanish comparison, outages and shared reference prices

Statement. Lehalle and Laruelle report, from European equity data around 2008–2013 [alpha-laruelle, chaps. 1–2]:

- *liquidity and fragmentation*: "the more liquid, the more fragmented" (chap. 1, p. 37); MTFs' shares grew faster in liquid stocks, and coverage rose for all venues and indices from 2009 to 2012 but stayed lower for less liquid stocks, which they attribute to costlier hedging and short selling for high-frequency market makers, who extend to less liquid stocks once their hedges and technology are in place, down to a liquidity threshold (chap. 2, sec. 2.4.2, pp. 161–162);
- *spreads and share*: a venue's market share moves with its spread relative to the primary market's, because routers must send flow to the best price; Chi-X's spreads became tighter than the primary's and its share rose; Turquoise's share fell sharply in 2009 when the agreements under which eight market makers had provided constant liquidity ended (sec. 2.3.2, pp. 153–155);
- *resilience*: venues where trading continues as volatility rises take more of the price formation; Chi-X approached the primary market's pattern between 2008 and 2012 (sec. 2.3.3, pp. 156–158);
- *time of day*: on CAC 40 stocks (January–February 2009 and 2012) Euronext's intraday share is highest at the start and end of the day, when its book is the only one filled after the opening auction and when routers are switched on and off (sec. 2.1.3, pp. 139–140);
- *outages*: on 13 October 2010 NYSE Euronext's cash markets halted from 16:42 CET, resuming at 17:20 without market data; MTFs' midpoint books stopped and orders pegged to the primary were cancelled or paused, most large caps kept trading on MTFs near Euronext's last price, but some stocks deviated by about 2 percent, "as if two price formation processes took place in parallel" (sec. 2.5.2, pp. 183–185);
- *Spain*: MTFs had almost no share of IBEX 35 trading from November 2007 to January 2010 (Chi-X first exceeded 0.5 percent in August 2009), because only Iberclear, owned by the primary market BME, could clear and settle until the Titulo V reform of June 2011 (sec. 2.5.1, p. 170).

Conditions. The Spanish comparison is a historical contrast, not a control: comparing 15-minute Garman–Klass volatility of FTSE 100 and IBEX 35 stocks from June 2007, Lehalle and Laruelle find no significant effect of UK fragmentation on intraday volatility, and the comparison is confounded from 2010 by the sovereign-debt crisis, to which Spain was more exposed (pp. 171–174). The spread–share and resilience findings are co-movements in their figures, with Schneider Electric's five-minute data for January to July 2012 as one example (p. 156), not estimated causal effects. High-frequency traders' shares of trades ("more than 80% ... in the US, 40% in Europe") are reported without a source (p. 163).

Practice. *Shared references.* Lehalle and Laruelle argue that European outages showed how hard fragmented trading is without information shared among agents, and that, without a consolidated tape or trade-through rule, MTFs priced off the primary's quotes, so the primary appeared to lead (introduction, p. 5; pp. 184–185). Harris's account of consolidation is the general mechanism: in fragmented markets traders adjust to information from other segments, route to the segments with the best prices, and arbitrageurs move liquidity between segments, which consolidates the market provided information about all segments is public at low cost [alpha-harris, chap. 26, sec. 26.4, pp. 533–534].

Pitfalls. The legacy case study called Spain a "fragmentation control case"; Lehalle and Laruelle use it as a comparison and themselves find no clear effect. Its claim that primary-market dominance persisted "when alternative venues could not clear and settle efficiently" is the source's explanation for Spain only. The legacy subsidised-share caveat generalised from Turquoise's 2009 episode. Venue names, shares and the regulatory framework are as of 2008–2013.

See also. venue market share; fragmentation metrics; SOR and market-share dynamics; fee-adjusted venue choice; smart order routing; Garman–Klass volatility; HFT strategies.

Further reading. The pathway's readings on fragmentation (O'Hara and Ye) and on current EU and U.S. rules are not read. The legacy file is kept verbatim in `cross-venue-routing-fragmentation.tex` in the folder `notes/artefacts/market-structure-execution-quality/`.

Index. market fragmentation; MiFID; MTF; Chi-X; Turquoise; BATS; coverage; high-frequency market makers; hedging cost; spread and market share; resilience to volatility; intraday market share; outage; Euronext outage 2010; Spain; IBEX 35; Iberclear; clearing and settlement barriers; consolidated tape; market consolidation; order-flow externality

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, introduction p. 5, chap. 1, p. 37, chap. 2, pp. 139–140, 153–163 and 170–185, and `alpha-harris`, 2002 Oxford printing, chap. 26, pp. 526–534). Stability: draft. Currency: European markets 2007–2013; no current market structure stated.

9.8 Tick size

Definition 9.25 Tick size: the price grid, tick regimes and relative tick, when the tick binds, how it prices queue jumping, and price clustering

Statement. The tick size defines the price grid on which a limit order must lie: every valid price is a multiple of it, so it is also the smallest difference between two limit prices [alpha-laruelle, chap. 1, sec. 1.3.1, p. 74]. It may apply to every stock whatever its price and liquidity (as in the United States for stocks priced above \$1) or depend on price and liquidity bands, a *tick-size regime* (p. 74). The *relative tick* $\vartheta/m(t)$, tick over mid-price, compares instruments: Priceline at about \$1,000 in 2014 had a relative tick near 10^{-5} with the same one-cent tick as a \$1 stock [catalogue-0528, chap. 3, sec. 3.1.5, pp. 48–49]. A *large-tick* instrument has a spread close to one tick most of the time, a *small-tick* one a spread of many ticks (chap. 4, p. 58).

Conditions. Being the smallest price increment, the tick is a floor on the spread, so a smaller tick lowers spreads only where that floor binds [alpha-laruelle, chap. 1, sec. 1.3.2, p. 77]. Two simple measures of how much it binds (p. 77): the *binding probability*, the share of trades, value traded or quote duration with a one-tick spread, and the *spread leeway*, the spread in ticks minus one. For S&P 500 stocks in July 2012 most stocks below \$20 had a one-tick spread most of the time, and stocks above \$150 showed no constraint (Figures 1.19–1.20, pp. 78–79).

Practice. The tick sets the price of priority: where it is very small, a trader can step ahead of resting orders by one tick for little cost, which leaves little volume at any level and makes the best quotes change very often [catalogue-0528, chap. 3, sec. 3.1.5, p. 49]; Lehalle and Laruelle add that queue jumping with a small tick "makes time priority nearly meaningless" and quotes more fleeting, while with a large tick and a one-tick spread an impatient trader must cross and a liquidity provider must manage its place in the queue [alpha-laruelle, chap. 1, pp. 88–89]. Harris reports that prices cluster on round numbers (whole numbers more than halves, halves more than odd quarters; in decimal markets multiples of 1.00, 0.50, 0.25, 0.20, 0.10 and 0.05), most when the minimum increment is a small fraction of price, the market volatile and the instrument thinly traded, and that traders place orders just above or below round numbers to step ahead of others [alpha-harris, chap. 5, box "Price Clustering", p. 91].

Pitfalls. The legacy notes said clustering reflects "communication, search, and queueing convenience"; Harris states when clustering is strongest, not why, and those reasons are not adopted. Their price- and liquidity-banded regimes match Lehalle and Laruelle's price ranges and quotation groups, which "theoretically" group stocks by liquidity (p. 74). Their claim that once the tick stops binding "further reductions mostly change queue behaviour" goes beyond the passages: Lehalle and Laruelle report spreads that did not fall, and sometimes rose, for illiquid high-priced stocks after decimalisation (tick size and market quality).

See also. Harris's discretised spread; tick size and market quality; spread determinants (large-tick bounce and depth); queue imbalance (large-tick races); order lifecycle (priority rules).

Further reading. The pathway's readings on the U.S. tick-size pilot and the 2024 tick and access-fee rule (Barardehi et al.; the SEC release) are not read here, so no current U.S. tick schedule is stated. The legacy file is kept verbatim in `notes/artefacts/execution-market-making/`.

Index. tick size; minimum price variation; price grid; tick-size regime; relative tick size; large-tick; small-tick; binding probability; spread leeway; queue jumping; time priority; price clustering; round numbers

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `alpha-laruelle`, 2018 World Scientific printing, pp. 74–79 and 88–89; `catalogue-0528`, Cambridge 2018, pp. 48–49 and 58; `alpha-harris`, 2002 Oxford printing, p. 91). Stability: draft. Currency: rules and figures as of 2002–2015; U.S. and EU tick regimes have changed since and are not checked here.

Evidence 9.26 Tick size and market quality: quoted depth, order and trade sizes, order-to-trade ratios, cumulative depth and the evidence from tick reductions

Statement. On FTSE 100 stocks in April 2012, grouped by deciles of the daily spread in ticks, Lehalle and Laruelle find far less volume at the best limits when the tick is small relative to the spread, a nearly threefold fall in the size of orders at the first limit from the first to the last decile, smaller trades, and a 50 percent rise in the number of book updates per trade [alpha-laruelle, chap. 1, sec. 1.3.2, Figures 1.21–1.24, pp. 80–84]. They read the last as a fall in the value of a quote, smaller and shorter-lived (p. 84). From the decimalisation studies they cite: spreads fell for active low-priced stocks whose spread had been one tick, with "at best no effect" and some increases for illiquid high-priced stocks, and a tick below which further reductions widen spreads (Ahn et al. found the direction of Harris's prediction right but its magnitude overestimated) (pp. 79–80).

Conditions. Spreads alone do not measure quality: a smaller quoted spread may be available for less volume (p. 81). *Cumulative depth* rebuilt from order-level data measures the cost of a marketable order of any size and includes hidden liquidity, but conditions on how liquidity takers changed their order sizes; Bacidore et al. found NYSE marketable orders cheaper at every size after decimalisation and depth thinner beyond 15 cents from the best quotes, and Goldstein and Kavajecz, with costs in basis points, found the public book less deep and large trades costlier (pp. 85–87). In a fragmented market a single book is not the market, and mirrored orders make aggregation across venues unreliable (pp. 75–76, 85). These are cross-sectional associations (FTSE 100) and before–after comparisons around single events, reported by Lehalle and Laruelle; the original studies were not read.

Practice. The legacy proxies for tick-size effects, binding probability, spread leeway, depth near the mid-price and message intensity (order-to-trade ratio, cancellations and modifications), are the measures Lehalle and Laruelle use (pp. 77 and 84); see tick size. More messages per trade need not mean more liquidity.

Pitfalls. The legacy caveat that "a consolidated tape of trades cannot fully replace quote data if only the best limit is visible" was not found in these passages and is not adopted; Lehalle and Laruelle's point is the falling value of each quote. The legacy "cumulative depth" mechanism omitted the conditioning on takers' behaviour and the fragmentation problem.

See also. tick size; Harris's discretised spread; spread measures; best execution; smart order routing (fragmented books).

Further reading. Bacidore et al. (2003), Goldstein and Kavajecz (2000), Jones and Lipson (2001), Ahn et al. (1996) and Aitken and Comerton-Forde (2005), cited by Lehalle and Laruelle, are not read. The legacy file is kept verbatim in `notes/artefacts/execution-market-making/`.

Index. tick size reduction; decimalisation; quoted depth; order size; trade size; order-to-trade ratio; message traffic; value of a quote; cumulative depth; effective cost by size; hidden liquidity; order exposure; queue jumping

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, pp. 75–90). Stability: draft. Currency: FTSE 100 data of April 2012; decimalisation studies of 1992–2005.

Model 9.27 Harris's discretised spread: a latent continuous spread rounded to the tick, its fit by maximum likelihood, and the counterfactual spread under a smaller tick

Statement. Introduced by Harris (1994) to forecast the effect of decimalisation, as presented by Lehalle and Laruelle [alpha-laruelle, app. A, sec. A.6, pp. 266–268]: for stock j, trade t on day d, the observed spread in ticks is $\Psi_{j,t} = (\text{Ask}_{j,t} - \text{Bid}_{j,t})/\text{tick}_{j,d}$, using only days with a single tick, and $\Psi_{j,t} = \Phi(\tilde{\Psi}_{j,t})$ for a latent continuous spread $\tilde{\Psi}_{j,t}$, where Φ gives the nearest integer for inputs above 1.5 and 1 otherwise. The latent spread has density $\tilde{f}_{s,\theta}$ with mean

$$E[\tilde{\Psi}_{j,t} \mid T_j, \sigma_{j,d}] = g(T_j, \sigma_{j,d}) \, \frac{P_{j,d}}{\text{tick}_{j,d}}, \qquad g = \alpha \, T_j^{\beta} \sigma_{j,d}^{\gamma},$$

with T_j mean turnover, $\sigma_{j,d}$ volatility and $P_{j,d}$ the mean traded price; g is the latent relative spread and P/tick converts it to ticks (pp. 268–269). The discrete probabilities are $f_{s,\theta}(1) = \int_0^{1.5} \tilde{f}_{s,\theta}$ and $f_{s,\theta}(i) = \int_{i-0.5}^{i+0.5} \tilde{f}_{s,\theta}$ for $i \geq 2$, and $(\alpha, \beta, \gamma, \theta)$ maximise $\sum_{j,d,k} N(j,d,k) \ln f(k)$ over the counts $N(j,d,k)$ of k-tick spreads (pp. 269–270).

Notation. Counterfactual: for a new tick of old tick/δ, the latent spread in *new* ticks is $\delta\tilde{\Psi}$, rounded by the same rule, so $f_{s,\theta,\delta}(1) = \int_0^{1.5/\delta} \tilde{f}_{s,\theta}$ and $f_{s,\theta,\delta}(k) = \int_{(k-0.5)/\delta}^{(k+0.5)/\delta} \tilde{f}_{s,\theta}$ for $k \geq 2$, and the expected spread in *old* ticks is

$$E(\Psi_{j,d} \mid T_j, \sigma_{j,d}, \delta) = \sum_k \frac{k}{\delta} \, f_{\hat{\alpha} T_j^{\hat{\beta}} \sigma_{j,d}^{\hat{\gamma}} P_{j,d}/\text{tick}_{j,d}, \hat{\theta}, \delta}(k)$$

(pp. 272–273, checked by page image): k counts new ticks and k/δ converts them to old ones. An observed spread of k old ticks is $k\delta$ new ticks.

Conditions. The latent distribution and its dependence on turnover and volatility are held fixed when the tick changes; that is the model's counterfactual assumption, which ignores changes in order exposure, depth and participants (tick size and market quality). Lehalle and Laruelle fit a gamma latent distribution to the 60 most

liquid FTSE stocks in April 2012, with standardised $\beta' = -0.46$ and $\gamma' = 2.62$ (more liquidity, lower spread; more volatility, higher), and a pseudo-R^2 of 0.8 between observed and expected relative spreads (pp. 270–271).

Practice. The expected decrease is largest for stocks whose spread is near one tick (Figure A.9, p. 272). Checked (`order-flow-checks.py`), with a gamma latent spread of shape 4 and $\delta = 5$: a latent mean of 0.6 old ticks gives an expected observed spread of 1.010 old ticks now and 0.601 after the change (−41 percent); a mean of 1.2 gives 1.302 and 1.200 (−8 percent); means of 3 and 8 change by less than 0.01 old ticks.

Pitfalls. The legacy counterfactual said "an observed k-tick spread becomes roughly k/δ old-tick units before re-rounding". That confuses the units: an observed k-tick spread is $k\delta$ new ticks, and k/δ old ticks is the value of k *new* ticks; nor does the model rescale observed spreads, it re-rounds the latent spread (for a latent mean of 8 old ticks the expected change is nil, not $8/5 - 8$). Lehalle and Laruelle's caption to Figure A.10 gives the change "in old ticks unit" while the text calls it a relative decrease (p. 273); the figure's axis is a difference in old ticks. Their Figure A.8 fit is in-sample.

See also. tick size; tick size and market quality; spread determinants; spread–volatility regression.

Further reading. Harris (1994) and the uncertainty-zone model of Dayri and Rosenbaum, cited by Lehalle and Laruelle (p. 74), are not read. The legacy blocks are kept verbatim in `notes/artefacts/execution-market-making/`; the counterfactual is computed in `notes/artefacts/market-simulation/order-flow-checks.py`.

Index. Harris model; discretised spread; latent spread; uncensored spread; spread in ticks; tick constraint; maximum likelihood; gamma distribution; turnover; tick reduction; counterfactual spread; old ticks; new ticks

Status. Verification: partly checked 2026-10-16 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, chap. 1, pp. 74–77, and app. A, sec. A.6, pp. 266–273, page images of pp. 272–273; the counterfactual computed). Stability: draft. Currency: model timeless; the fit is FTSE data of April 2012.

9.9 Signing trades

Method 9.28 Trade-sign classification: the quote rule, the tick rule and Lee–Ready, block prices, signed volume, and what an inferred sign does and does not say

Statement. Trade data rarely say which side initiated a trade, so the sign $q_t = +1$ (buyer-initiated) or -1 is inferred. If all executions occur at the posted quotes, $q_t = +1$ at the ask and -1 at the bid; more generally $q_t = \mathrm{Sign}(p_t - m_t)$ with m_t the prevailing midpoint (the *quote rule*), which leaves midpoint trades unclassified; for those, the trade takes the sign of the most recent price change (the *tick rule*) [s6-hasbrouck, chap. 9, sec. 9.7.1, p. 89]. Harris describes the combination as the Lee and Ready algorithm: above the midpoint a buyer was aggressive, below it a seller; at the midpoint, an uptick or zero uptick means a buyer, otherwise a seller [alpha-harris, chap. 21, p. 423]. For block trades, a print closer to the bid than to the ask was probably seller-initiated, otherwise buyer-initiated, presuming the initiator paid a concession [alpha-harris, chap. 15, sec. 15.5, p. 332]. Signed volume $q_t V_t$, or a concave transform such as $q_t\sqrt{V_t}$, enters impact models; Hasbrouck advises against convex transforms such as $q_t V_t^2$, which amplify extreme volumes (sec. 9.7.2, p. 89).

Conditions. Signing needs the quotes and trades in the right order: in limit-order markets the interleaving is usually accurate, but with hybrid markets, different dissemination paths, coarse time stamps or manual steps it is "likely to be suspect" (Hasbrouck, p. 89). The rule at the posted quotes works only where the book can be rebuilt and executions are against visible orders, not hidden or reserve ones (p. 89). Lee and Ready (1991) assess the quote and tick tests; Hasbrouck cites them (note 1 to chap. 9, p. 180), and the paper itself was not read here.

Practice. Checked (`order-flow-checks.py`, ten assertions on prices held as exact decimals): midpoint trades after an uptick, or after a zero tick whose last non-zero change was up, are buys, and after a downtick or a zero downtick sells, for whole-tick midpoints (10.01 in 10.00–10.02) and half-tick ones (10.005 in 10.00–10.01); a midpoint trade with no earlier price change is left unclassified; trades just above or below a half-tick midpoint (10.006, 10.004) follow the quote rule whatever the tick; a trade at the bid is a sell, and at the ask a buy, whatever the previous tick. An implementation must compare prices exactly: in binary floating point $(10.00+10.01)/2$ is $10.004999\ldots$, below 10.005, so the half-tick midpoint trade after a downtick is wrongly called a buy, as the slice-33 version of the check did. Sign persistence in event time is a stylised fact (stylised facts); Hasbrouck notes that signed size often adds little explanatory power beyond the sign itself (sec. 9.7.2, p. 90). Summed over an interval, signed volume is the trade imbalance of order-flow imbalance, not the event-based OFI.

Pitfalls. The sign identifies which side demanded immediacy at that moment, not who was better informed or more eager: a buyer who pegs the bid can accumulate aggressively while the algorithm calls every trade passive (Harris's sidebar, p. 423). Measured as transaction costs, the algorithm overstates the costs of traders who do not only take liquidity, and cannot see that several trades filled one order, so it understates that order's impact cost (p. 423). Harris reports that "analysts have determined that sellers initiate approximately 80 percent of

all large block trades in the U.S. stock markets", and reads the asymmetry as consistent with three of his four block-trading problems: latent demand (buyers can buy only from holders when short selling is hard), price discrimination (sellers can credibly reveal their full size) and asymmetric information (large buyers are presumed informed) (pp. 332–333); he gives no sample or period, and the primary evidence is a backlog item. The legacy taxonomy listed the tick rule without its use at the midpoint, and its "false persistence" from misclassification and order splitting was not found in these passages.

See also. spread measures (effective spreads need signs); information-driven bars (the tick rule in bar construction); order-flow impact regression; propagator model; block trade.

Further reading. Lee and Ready (1991) and the studies behind Harris's 80 percent are not read. The legacy file is kept verbatim in `notes/artefacts/price-impact-execution/`.

Index. trade sign; trade classification; buyer-initiated; seller-initiated; quote rule; quote test; tick rule; tick test; zero uptick; Lee–Ready algorithm; midpoint trades; block-price rule; signed volume; trade imbalance; aggressive and passive traders; block initiator asymmetry

Status. Verification: partly checked 2026-10-16 (against the text layers of local copies of `s6-hasbrouck`, 2007 Oxford printing, pp. 89–90 and 180, and `alpha-harris`, 2002 Oxford printing, pp. 332–333 and 423; the rule computed on toy trades, recomputed 2026-10-16 with exact decimal prices after a floating-point error in the first check). Stability: draft. Currency: methods timeless; the block statistic as reported in 2002.

9.10 Price impact

Definition 9.29 Price impact: unperturbed, observed and trade prices; reaction, prediction, transient and permanent impact

Statement. *Price impact* is the effect of trading on prices: trades cause price moves that would not otherwise have happened [wide-webster-price-impact, chap. 1, sec. 1.1.2, p. 5]. Webster separates three prices for a trader whose trades Q are part of all trades L [wide-webster-price-impact, chap. 2, Definition 2.2.6, pp. 53–54]:

- the *unperturbed price* S_t, the price without the trades L and their impact;
- the *perturbed* or observed price $P_t = S_t + I_t$, where I_t is the impact of L;
- the *trade price* $\widetilde{P}_{t-} = S_{t-} + I_{t-} + s_{t-}$, where s is the instantaneous transaction (spread) cost of the trader's own fill, and the subscript $t-$ marks that a fill at t executes against the state just before t.

Bouchaud, Bonart, Donier and Gould separate three impacts of a market order executed at t, at horizon ℓ, with m the mid-price and $\mathcal{F}_t$ everything that may have prompted the order [catalogue-0528, chap. 11, sec. 11.2, eqs. (11.1)–(11.4), pp. 210–211]:

$$I^{\text{react}} = \mathbb{E}[m_{t+\ell} \mid \text{exec}_t, \mathcal{F}_t] - \mathbb{E}[m_{t+\ell} \mid \text{no exec}_t, \mathcal{F}_t], \qquad I^{\text{obs}} = \mathbb{E}[m_{t+\ell} \mid \text{exec}_t] - m_t,$$

$$I^{\text{pred}} = \mathbb{E}[m_{t+\ell} \mid \text{no exec}_t, \mathcal{F}_t] - m_t, \qquad I^{\text{obs}} = I^{\text{react}} + I^{\text{pred}}.$$

Over time, impact has a *transient* part that decays and a *permanent* part. The permanent part combines reaction and prediction: it is higher for traders with predictive power than for noise trades [catalogue-0528, chap. 12, sec. 12.4, p. 240].

Notation. I^{react}, I^{obs}, I^{pred} are the reaction, observed and prediction impacts. The source writes $I^{\text{react}\cdot}_{t+\ell}(\text{exec}_t \mid \mathcal{F}_t)$ and so on; the time and conditioning arguments are dropped here. The identity $I^{\text{obs}} = I^{\text{react}} + I^{\text{pred}}$ needs every expectation conditioned on $\mathcal{F}_t$; as printed, (11.2) conditions on execution alone, and the authors state that the conditioning on $\mathcal{F}_t$ is always implicit [catalogue-0528, same sec., p. 210] (Interpretation: without it the two sides differ by $\mathbb{E}[m_{t+\ell} \mid \text{exec}_t] - \mathbb{E}[m_{t+\ell} \mid \text{exec}_t, \mathcal{F}_t]$). Webster's P is a price, not the P of other entries.

Conditions. Two readings of impact frame the literature [catalogue-0528, chap. 11, sec. 11.1, pp. 208–210]. In the efficient-market reading, traders forecast prices and trade accordingly, so trades and later price changes correlate even if trades move nothing; in Hasbrouck's words, quoted there, orders forecast prices. In the order-flow reading, trading itself moves prices, and impact exists even in a model with no information at all, such as the Santa Fe model. Reality probably lies between. If the mid-price were a martingale, observed and reaction impact would coincide; they do not, because the information behind the order raises the expected price even without the trade (p. 211). Studies with random or cash-driven trades find that, over short horizons, mechanical impact is about the same for random and allegedly informed trades; prediction shows only at longer horizons (p. 212). After a metaorder, the impact path splits into a transient part that vanishes and a permanent part $I_\infty(Q)$, which collects both the prediction behind the order and any permanent reaction to all trades [catalogue-0528, chap. 12, sec. 12.2, eq. (12.5), p. 233]. Bacry, Iuga, Lasnier and Lehalle describe the

same divide for permanent impact: a mechanical view, common among econophysicists, in which buying and selling pressure moves prices, and an informational view, favoured by economists after Kyle, in which trades reveal information. They hold the two as a dual picture rather than choosing one [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 7.1, pp. 19–20]. Within the mechanical view they report two positions, a "fair pricing" view in which permanent impact is about two-thirds of the temporary impact, and Bouchaud's view that there is no permanent impact, only the long memory of the order flow; both are cited there and were not consulted.

Practice. Reaction impact can be estimated by trading at random, by randomising execution of signals, or by conditioning on the rest of the market's order-sign imbalance; prediction impact is very hard to estimate [catalogue-0528, chap. 11, sec. 11.2, pp. 211–212]. The counterfactual cannot be observed: a trader never sees both outcomes of one trade, which Webster calls the causality challenge [wide-webster-price-impact, chap. 1, sec. 1.1.3, pp. 6–7]. Webster traces impact modelling through four communities: economists' stylised models (Kyle), mathematical finance's control problems (Almgren–Chriss, Obizhaeva–Wang, Gatheral), econophysics' empirical response functions and propagators, and books from quantitative trading desks [wide-webster-price-impact, chap. 1, sec. 1.1.5, pp. 9–15]. The model families in use differ in what they describe:

- *Almgren–Chriss*: the price follows an arithmetic random walk shifted by a permanent impact $g(v)$ of the trading rate v; each sale executes at the previous price less a temporary impact $h(v)$ that does not enter the next market price [alpha-almgren, author preprint of Dec. 2000, secs. 1.2–1.3, eqs. (1)–(2), pp. 8–9, page image]. In the linear case $g(v) = \gamma v$ and $h(n_k/\tau) = \epsilon\,\mathrm{sgn}(n_k) + (\eta/\tau)n_k$, with γ in (\$/share)/share, ϵ in \$/share (half the spread plus fees) and η in (\$/share)/(share/time) [alpha-almgren, author preprint of Dec. 2000, sec. 1.5, eqs. (6)–(7), pp. 10–11, page image]. The schedule minimises expected cost plus λ times its variance [alpha-almgren, author preprint of Dec. 2000, sec. 2, pp. 12–13].
- *Linear impact*: cost or impact proportional to the trading rate (Almgren–Chriss) or to signed size (Kyle).
- *Propagator*: each signed trade moves the price by a decaying kernel (propagator model).
- *Square-root*: metaorder impact concave in size relative to volume (square-root impact law).

Impact during execution is measured from the arrival price, signed by the order's side: in bid–ask spreads by Lehalle and Laruelle [alpha-laruelle, chap. 3, sec. 3.2.1, p. 204], relative to volatility by Almgren et al. (direct estimation); its path is in intraday impact profile.

Pitfalls. "Temporary impact" has three meanings. In Almgren and Chriss it is $h(v)$, which lowers the execution price of the trader's own sale in that period but does not enter the next market price [alpha-almgren, author preprint of Dec. 2000, sec. 1.3, eq. (2), p. 9, page image]; their prose describes a short-lived imbalance after which liquidity returns each period (p. 8). Webster accordingly reads it as the instantaneous transaction cost [wide-webster-price-impact, chap. 2, n. 3, p. 53]. Bacry et al. use "temporary market impact" for the impact at the end of a metaorder's execution, distinct from the transient curve during execution and the decay after it [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 2.2, p. 3]. Neither is a transient displacement that decays over many later trades. The legacy definition ("the price concession needed to obtain immediacy … expected to decay as liquidity replenishes") mixes the two; say which is meant (Interpretation). Calling a long-run move permanent impact needs care. Long-horizon impact is hard to measure because price variance grows with the horizon. Correlated metaorders, the same decision continued over days, can look like permanent impact. Separating mechanical from informational impact is difficult [catalogue-0528, chap. 12, sec. 12.4, p. 240]. The legacy caveats listed market and sector moves to remove. Bacry et al. remove moves of each stock's reference index (see post-execution impact profile); sector or factor moves are not in the passages checked, and Bouchaud et al. suggest conditioning on the rest of the market's order-sign imbalance (p. 212). The legacy principle *remove the information effect before calling the residual permanent impact* presumes the effect can be measured; Bacry et al., without the traders' signals, remove the market component and read it as aggregated information (p. 20). The legacy taxonomy of three lenses (theoretical, reduced-form, practitioner) is a simplification of Webster's four communities. Orders of the same size and participation can have different impact because of their timing: following a trend, reverting to a mean, or following an index rebalancing [alpha-laruelle, chap. 3, sec. 3.2, p. 204]. Separating execution mechanics from the signal behind the trade is the split between reaction and prediction impact above.

See also. propagator model; absence of price manipulation; price formation process (temporary and permanent impact by consensus, Lehalle and Laruelle); spread component identification (transitory and permanent components of the spread); Kyle single-auction model (permanent impact from information); Grossman–Miller model (a temporary concession for immediacy).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. price impact; market impact; Almgren–Chriss; permanent and temporary impact functions; mechanical and informational views of permanent impact; unperturbed price; perturbed price; trade price; instantaneous transaction cost; reaction impact; observed impact; prediction impact; transient impact; permanent impact; temporary impact; causality challenge; metaorders; price discovery and price formation

Status. Verification: partly checked 2026-10-15, against the text layers of local copies of two books (Webster's handbook, `wide-webster-price-impact`, CRC 2023 first edition, pp. 3–15 and 53–54, and Bouchaud et al., `catalogue-0528`, Cambridge 2018 first printing, ISBN 9781107156050, pp. 208–212, 233 and 240; the page image of p. 211 for eqs. (11.2)–(11.4)); and, in slice 8, the originals `alpha-almgren` (author preprint of December 2000, pp. 8–13, page images of pp. 9 and 11; journal pagination not checked) and `wide-bacry-life-cycle` (arXiv preprint of 9 December 2014, pp. 3 and 19–22); in slice 9, `alpha-laruelle`, p. 204. Stability: draft. Currency: timeless (concepts); the empirical statements as of 2018.

Model 9.30 Propagator model: transient impact of past trades

Statement. In market-order time, with $\varepsilon_n = \pm 1$ the sign of the nth market order and ξ_n price changes not induced by trades, the mid-price is [catalogue-0528, chap. 13, sec. 13.2, eq. (13.7), p. 252, page image]

$$m_t = m_{t_0} + \sum_{t_0 \le n < t} G(t-n)\,\varepsilon_n + \sum_{t_0 \le n < t} \xi_n .$$

The *propagator* $G(\ell)$ is the impact at lag ℓ of one trade, expected to decrease with ℓ; $G_\infty = \lim_{\ell\to\infty} G(\ell)$ is the permanent reaction impact. Returns then split into immediate impact, the decay of past impacts and noise, $r_t = G_1\varepsilon_t + \sum_{n<t} K(t-n)\varepsilon_n + \xi_t$ with $K(\ell) = G(\ell+1) - G(\ell)$ (same sec., eq. (13.8)). Equation (13.7) defines the family of linear transient impact models (TIM).

Notation. m_t mid-price just before event t; $r_t = m_{t+1} - m_t$; $G_1 = G(1)$; $C(\ell) = \mathbb{E}[\varepsilon_t\varepsilon_{t+\ell}]$ the sign autocorrelation; $R(\ell) = \mathbb{E}[\varepsilon_t(m_{t+\ell} - m_t)]$ the response function (observed impact).

Conditions. Why impact must be transient: if each trade had constant permanent impact G_1, then $\mathbb{E}[r_t r_{t+\ell}] = G_1^2 C(\ell)$, and the long memory of order signs would make returns strongly autocorrelated. That contradicts the nearly flat signature plots of real markets, so much of each trade's impact must relax [catalogue-0528, chap. 13, secs. 13.1–13.2, pp. 249–251]. The sign memory comes mainly from order splitting, investors working large orders in pieces because available liquidity is small; on the time scales of the power-law decay of $C(\ell)$, herding is not a relevant factor [catalogue-0528, chap. 10, secs. 10.4–10.5, p. 198]. For a power-law propagator $G(\ell) \approx \Gamma_\infty \ell^{-\beta}$ and signs with $C(\ell) \approx c_\infty \ell^{-\gamma}$, $0 < \gamma < 1$, prices are diffusive at long lags only at the fine balance $\beta = \beta_c = (1-\gamma)/2$; permanent impact ($\beta = 0$) gives trending prices, faster decay ($\beta > \beta_c$) sub-diffusive ones [catalogue-0528, chap. 13, secs. 13.2.2–13.2.3, eq. (13.17), pp. 256–257]. The simple model assumes impact independent of trade volume, time and market conditions, which the authors call gross oversimplifications, relaxed in their chap. 14 [catalogue-0528, same chap., p. 252].

Practice. The observed response and the bare propagator differ when signs are correlated: $R(\ell) = G(\ell) + \sum_{0<n<\ell} G(\ell-n)C(n) + \sum_{n>0}[G(\ell+n) - G(n)]C(n)$, which reduces to $G(\ell)$ only for uncorrelated signs. Estimating G from R and C through this relation is sensitive to finite-size effects [catalogue-0528, same chap., sec. 13.2.1, eq. (13.10), p. 253]. Webster calls the Obizhaeva–Wang model a simple form of the propagator model [wide-webster-price-impact, chap. 1, p. 10]; in its generalized form impact follows $dI_t = -\beta_t I_t\,dt + \lambda_t\,dQ_t$ [wide-webster-price-impact, chap. 2, Definition 2.8.1, p. 97]. Interpretation: that is an exponentially decaying kernel in calendar time, $G \propto e^{-\beta\ell}$ for constant β. Kernel shapes in use: *non-parametric*, from the lagged sign–return correlation $S(\ell) = R(\ell+1) - R(\ell) = \sum_{n\ge 0} C(|n-\ell|)K(n)$ inverted as $K = \mathbf{C}^{-1}S$ up to a maximum lag [catalogue-0528, chap. 13, eqs. (13.13)–(13.15), p. 254]; *power law*, which the estimates show at long lags, $G(\ell) \approx \Gamma_\infty \ell^{-\beta}$ with $\beta < 1$, decaying to zero but with divergent sum ("long-range resilience", same p. 254); and *exponential*, as in the Obizhaeva–Wang model above. The original paper by Bouchaud, Gefen, Potters and Wyart writes the price as a sum over past trades with a "bare" propagator G_0 weighting each sign by the logarithm of the trade volume, $p_n = \sum_{n'<n} G_0(n-n')\,\varepsilon_{n'} \ln V_{n'} + \sum_{n'<n} \eta_{n'}$, with η independent of the signs and uncorrelated in time [wide-bouchaud-propagator, arXiv cond-mat/0307332v2, sec. 3.1, eq. (12), p. 14, page image]. It argues that with power-law sign correlations a constant G_0 would make the response grow like $\ell^{1-\gamma}$, so G_0 must decay [wide-bouchaud-propagator, same, sec. 3.1, p. 16]; with a power-law ansatz the Hurst exponent satisfies $2H = 2 - 2\beta - \gamma$ [wide-bouchaud-propagator, same, sec. 3.2, eq. (22), pp. 16–17].

Pitfalls. The legacy block described the price as the sum of signed impulses "with different decay profiles". In the simple model every trade shares one propagator G; trade-dependent profiles need the generalised models of chap. 14, which were not read. Reading the measured response $R(\ell)$ as the impact of a single trade ignores the sign autocorrelation. Further departures in the legacy notes, now corrected: (a) they defined m_k as the mid-price *after* trade k while summing over $k' < k$, which is the price just *before* trade k in the sources; (b) they added a single noise η_k rather than the cumulative $\sum\eta$ of the sources, which would make the price stationary apart from trading instead of diffusive (Interpretation); (c) they dropped the 2004 paper's $\ln V$ volume weight, which the book's simple model also omits; (d) they said the response is estimated with C and the kernel increments, whereas R and C are measured and the kernel is estimated from them; (e) they listed agent-specific kernels,

which are not in the passages checked (the book's chap. 14 generalises by event type, per its contents, and was not read). Interpretation: Hasbrouck's generalized Roll model, in spread component identification, is the constant-propagator case applied to the efficient price with independent signs: the trade moves it permanently by λ.

See also. price impact; absence of price manipulation (restrictions on G and on the Obizhaeva–Wang parameters); returns by time scale; fundamental and transitory volatility (the variance ratio and transient components).

Further reading. The original paper [wide-bouchaud-propagator], read in its arXiv version for slice 8. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. propagator; bare impact function; transient impact model; TIM; kernel estimation; power-law propagator; fine balance; long-range resilience; order splitting; impact decay; response function; order-sign autocorrelation; long memory of order flow; signature plot; market-order time; Obizhaeva–Wang model; resilience

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `catalogue-0528`, Cambridge 2018 first printing, pp. 249–253, with the page image of p. 252; `wide-webster-price-impact`, CRC 2023, pp. 5, 10 and 97; and in slice 8 Bouchaud et al.'s book, pp. 198 and 254–257, and the original paper, `wide-bouchaud-propagator`, in its arXiv version cond-mat/0307332v2, pp. 14–17, page image of p. 14, not its journal pagination). Stability: draft. Currency: timeless (model); the empirical motivation as of 2018.

Result 9.31 Absence of price manipulation in impact models

Statement. In an impact model, *price manipulation* is a round trip, a strategy that starts and ends with no position, whose expected cost is negative. Webster quotes this definition from Gatheral (2010), which was not consulted [wide-webster-price-impact, chap. 1, n. 3, p. 7]. A usable model must exclude it. Bouchaud, Bonart, Donier and Gould write the expected cost of a closed trajectory with trading rate j and price x^* as $\mathcal{C} = \int_0^T j(u)\, x^*(u)\, du$ with $\int_0^T j(u)\, du = 0$. They show that in their latent-liquidity model it is a non-negative quadratic form, so $\mathcal{C} \geq 0$ for every schedule, generalising a result for the linear propagator model [catalogue-0528, chap. 19, sec. 19.5, pp. 361–362, page image]. In Webster's generalized Obizhaeva–Wang model, buying Δ in one jump at t and selling it at $t' > t$, without a spread, earns

$$(S_{t'} - S_t)\Delta - \tfrac{1}{2}\lambda_t(2\beta_t + \gamma'_t)(t'-t)\Delta^2 + o(t'-t),$$

so if $2\beta + \gamma' < 0$, large, quick round trips earn arbitrary expected profits from the trader's own impact. The no-manipulation condition is $2\beta + \gamma' > 0$, after Fruth, Schöneborn and Urusov (2013), not consulted [wide-webster-price-impact, chap. 2, sec. 2.7.1, Example 2.7.1, p. 94, page image].

Notation. S unperturbed price; Δ shares; impact I with $dI = -\beta I\, dt + \lambda\, dQ$, decay β per unit time, push $\lambda = e^{\gamma}$ in price per share, γ' its rate of change. As Webster's profit expression implies, the trader's buy executes at $S_t + \frac{1}{2}\lambda_t\Delta$ on average, the sell at $S_{t'} + e^{-\int_t^{t'} \beta_s\, ds}\lambda_t\Delta - \frac{1}{2}\lambda_{t'}\Delta$.

Conditions. Webster's example assumes no spread, no prior impact, jump trades and exact unwinding; the expected profit from S is zero only if S is a martingale. The condition $2\beta + \gamma' > 0$ rules out only the paradoxes that yield arbitrary profits. It does not rule out transaction-triggered manipulation, such as selling first to depress the price for later buying [wide-webster-price-impact, same sec., pp. 94–95]. In a square-root model it bounds forecast volume growth, $v'_t/v_t < 4\beta_t$: trading activity cannot double in less than a quarter of the impact half-life, so rising volume curves can create manipulation opportunities while falling ones cannot [wide-webster-price-impact, same chap., sec. 2.7.3, Example 2.7.2, p. 96].

Practice. Using an impact model that admits profitable closed trajectories in execution algorithms leads to instabilities or counter-intuitive trades [catalogue-0528, chap. 19, p. 362], and Webster asks that a model rule out manipulation paradoxes [wide-webster-price-impact, chap. 1, sec. 1.1.4, p. 8]. Interpretation, the accounting behind any check: a trader who buys and sells Q shares at average execution prices $\bar{P}_B$ and $\bar{P}_S$, impact included, and pays explicit costs F in total, ends with $Q(\bar{P}_S - \bar{P}_B) - F$ and no position, hence no mark-to-market term. With a per-share cost c on each side, $F = 2Qc$; with c quoted per round-trip share, $F = Qc$. Spread counted in $\bar{P}$ must not be counted again in c.

Pitfalls. A realised profitable round trip is not price manipulation: the profit may come from $S_{t'} - S_t$, from information, or from luck. Manipulation concerns expected profit under a model, with no information (Interpretation). Impact that treats buys and sells alike per unit is not by itself a no-manipulation guarantee. Webster's example is linear and symmetric trade by trade, yet admits manipulation when liquidity changes faster than impact decays (Interpretation). The legacy principle "impact depends on net pressure and risk, so a zero-net sequence cannot mechanically create a higher average sale price" holds in Harris's instantaneous, permanent-impact example but is not a general result. The legacy proposition stated as a test only that identity, with c undefined per side or per round trip. Strategic bluffing, which fools other traders through their inferences, is a different mechanism from manipulation through one's own impact in a model, where nobody is fooled (bluffing and manipulation).

See also. propagator model; price impact; bluffing and manipulation (Harris's asymmetric-impact bluff).

Further reading. Gatheral, "No-Dynamic-Arbitrage and Market Impact" [catalogue-0501], a required reading of `price-impact-execution`, not consulted. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. price manipulation; no-dynamic-arbitrage; round trip; closed trajectory; Obizhaeva–Wang model; transaction-triggered manipulation; volume curves; impact decay; liquidity; execution cost

Status. Verification: partly checked 2026-10-15 (against `wide-webster-price-impact`, CRC 2023, pp. 7–8 and 93–97, with the page image of p. 94; and `catalogue-0528`, Cambridge 2018 first printing, pp. 361–362, with its page image; the example rederived independently). Stability: draft. Currency: timeless (theory).

9.11 Metaorder impact

Evidence 9.32 Square-root impact law: metaorder impact concave in size relative to volume

Statement. The peak impact of a metaorder of volume Q executed over time T is well described by [catalogue-0528, chap. 12, sec. 12.3.1, eq. (12.6), p. 234, page image]

$$\mathcal{I}^{\text{peak}}(Q,T) \cong Y\sigma_T \left(\frac{Q}{V_T}\right)^{\delta}, \qquad Q \ll V_T,$$

with Y of order one ($Y \cong 0.5$ for US stocks), δ between 0.4 and 0.7, σ_T the volatility and V_T the volume traded over the execution horizon. Both sides are in percent of price. In the Capital Fund Management sample of about 500,000 futures metaorders, June 2007 to December 2010, $\delta \simeq 0.5$ for small-tick and $\simeq 0.6$ for large-tick contracts, for Q/V_T from about 10^{-5} to a few percent [catalogue-0528, same sec., Figure 12.2, p. 235]. Bacry, Iuga, Lasnier and Lehalle confirm a square-root temporary impact in daily participation on about 400,000 metaorders traded on European markets by one broker in 2010 [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, abstract, p. 1].

Conditions. The law is reported across equities, futures, FX, options and Bitcoin, before and after 2005, for small- and large-tick instruments and for different strategies and execution styles [catalogue-0528, same sec., p. 234]. It concerns *peak* impact at the end of execution, not permanent impact; the price reverts after execution (Figure 12.1, p. 234). Impact is not additive: the second half of a metaorder moves the price less than the first, and Q enters relative to the volume traded during execution, not to market capitalisation [catalogue-0528, same chap., sec. 12.3.2, pp. 235–236]. Linear models such as Kyle's predict the opposite [catalogue-0528, same chap., pp. 233 and 241]. Almgren et al., on US orders of 2001–2003, reject an exponent of one-half for temporary impact against size relative to volume during execution at the 95 percent level, estimating 0.600 ± 0.038 (direct estimation); Bacry et al. find that adding a duration factor $1/T^{\gamma}$ with $\gamma \simeq 0.25$ reinforces the square-root shape in daily participation [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, abstract, p. 1]. The exponents differ with the regressor (daily participation, trading rate, or size relative to volume during execution) and with the fitting method.

Practice. If the impact of the first fraction ϕ of a metaorder follows the same law, the execution shortfall per unit volume is two-thirds of the peak impact under square-root impact, against one-half under linear impact [catalogue-0528, same chap., sec. 12.5, eqs. (12.10)–(12.11), p. 241]. Bacry et al. debias post-execution prices with a square-root model of daily participation and report that powers from 0.4 to 0.7 give similar results [wide-bacry-life-cycle, arXiv preprint, sec. 7.2, n. 3, p. 21].

Pitfalls. The legacy taxonomy summarised the law as "concave in order size relative to market volume", omitting the volatility scale σ_T, the small-participation condition $Q \ll V_T$ and the exponent range; "square-root" names exponents from 0.4 to 0.7. The fitted exponent depends on the estimation method: Bacry et al. find smaller exponents with robust direct regressions than with log-log regressions (sec. 8, p. 22). The primary study of the futures sample, Tóth et al. (2011), is cited through Bouchaud et al. and was not consulted (`wide-toth-square-root`).

See also. price impact; propagator model; post-execution impact profile; Kyle single-auction model (linear impact, the contrast).

Further reading. Tóth et al., the futures study [wide-toth-square-root], a required reading of `price-impact-execution`, not consulted. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. square-root law; square-root impact; metaorder; peak impact; participation rate; concave impact; impact exponent; execution shortfall

Status. Verification: partly checked 2026-10-15 (against `catalogue-0528`, Cambridge 2018 first printing, pp. 233–236 and 241, page image of p. 234; and `wide-bacry-life-cycle` in its arXiv version of 9 December 2014, pp. 1, 21–22, not its journal pagination). Stability: draft. Currency: as of the samples, 2007–2010.

Method 9.33 Post-execution impact profile: debiasing later metaorders and removing index moves

Statement. To estimate how much of a metaorder's impact remains days after execution, Bacry, Iuga, Lasnier and Lehalle proceed as follows [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 7.2, pp. 20–22]:

1. For each metaorder executed on day D, take daily closes over the 41 days $d = D - 20, \ldots, D + 20$.
2. Debias for later flow: from the price moves after day D, subtract the temporary impact of the metaorders executed on the following days, using a square-root model of their aggregated daily participation fitted to the data, following Waelbroeck and Gomes (2013), cited there.
3. Split each daily log return into a systematic part and a residual,

$$\log P_d - \log P_{d-1} = \beta(\omega)\,(\log \mathcal{I}_d - \log \mathcal{I}_{d-1}) + \Delta W_d,$$

   with P_d the stock's close, $\mathcal{I}_d$ its reference index and $\beta(\omega)$ the stock's beta over $D-20$ to $D+20$ (eq. (21), p. 21, page image).
4. Cumulate from the close of $D - 1$: the idiosyncratic profile $W_d - W_{D-1}$ and the systematic profile $\beta(\log \mathcal{I}_d - \log \mathcal{I}_{D-1})$. Their sum, the post-execution profile, is an estimate of permanent impact (p. 22).

Notation. $\mathcal{I}_d$ is an index level, not impact; P_d a close price, not Webster's perturbed price. Impact itself is defined through $\epsilon\,\Delta P_t = \eta_t + \epsilon\,\Delta W_t$ for a metaorder of sign ϵ, with ΔW the move that would have occurred without it [wide-bacry-life-cycle, arXiv preprint, sec. 2.2, eq. (1), p. 3]; estimation assumes $\mathbb{E}[\epsilon\,\Delta W] = 0$, not $\mathbb{E}[\Delta W] = 0$ (p. 4).

Conditions. Lehalle and Laruelle restate the procedure, citing the 2015 published paper: each stock's reference index is its home index (CAC 40, FTSE 100, DAX), and β is estimated by least squares over $D-20$ to $D+20$ [alpha-laruelle, chap. 3, sec. 3.2.2, eq. (3.2), pp. 210 and 213–215]. Interpretation: this matches the preprint's window, so the in-sample estimation of β remains an open methodological question rather than a preprint artefact; a restatement by one of the authors is not independent confirmation. The broker's data do not say why each metaorder was placed. The authors therefore assume the clients' aggregate portfolio is close to the market portfolio, with beta one, and treat the systematic component as the information the issuers used in aggregate (p. 20). The participation of metaorders is autocorrelated beyond 20 days, at about 0.3 to 0.4 [wide-bacry-life-cycle, same sec., Figure 10, p. 21], so later executions in the same direction would otherwise be attributed to the first.

Practice. Without debiasing, prices keep trending in the direction of the execution-day jump for 20 days, with no reversal. After debiasing, the idiosyncratic profile returns to its initial level within 20 days, and the impact remaining after 20 days is entirely systematic; the authors conclude that no permanent effect is due to the metaorder itself (sec. 7.3, p. 22). They report this as compatible with two studies cited there and not consulted. Waelbroeck and Gomes (2013) found permanent impact for informed trades but none for cash-driven ones over 60 days. Brokmann et al. (2014) found that, once the issuer's alpha was removed, impact vanished after about a dozen days (p. 20).

Pitfalls. The legacy process began by classifying trades as information- or liquidity-driven. Bacry et al. could not, lacking the triggers; that split is Waelbroeck and Gomes's design, reported through Bacry et al. The legacy notes also removed market and sector moves with a factor model; the method checked removes only the reference index. Interpretation:

- The beta is estimated over a window that includes the execution and the days after it, so it is in-sample and can absorb part of the event.
- Removing the systematic part removes any impact the aggregate flow had on the index itself, which the authors read as information.
- The debiasing inherits the error of the square-root model for later flow.

The legacy model's intuition, "do not charge the execution for a market move", is one reading of the decomposition; the authors' own reading is that the systematic part carries the information behind the trades.

See also. price impact (permanent against finite-horizon impact; correlated metaorders); square-root impact law (the debiasing model); propagator model (long memory of order flow).

Further reading. Waelbroeck and Gomes (2013) and Brokmann et al. (2014), cited by Bacry et al., not consulted. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. permanent impact; post-execution profile; impact decay; debiasing; autocorrelated metaorder flow; CAPM-like decomposition; beta; idiosyncratic component; systematic component; informed and cash trades

Status. Verification: partly checked 2026-10-15 (against `wide-bacry-life-cycle` in its arXiv version of 9 December 2014, pp. 3–4 and 19–22, page image of p. 21; the 2015 journal version and its pagination not checked). Stability: draft. Currency: as of the 2010 European sample.

Evidence 9.34 Intraday metaorder impact: its path during and after execution, and its dependence on participation and duration

Statement. Measured as the price move from the arrival price, signed by the order's side and expressed in bid–ask spreads, the average impact of Crédit Agricole Cheuvreux executions on CAC 40 stocks in 2010 rises concavely over the execution: a slice executed early moves the price more than one executed late. After the execution it reverts, to a level below the one reached at the end [alpha-laruelle, chap. 3, sec. 3.2.1, Figure 3.3, pp. 204–206]. Lehalle and Laruelle summarise how impact depends on the participation rate and the duration by

$$\text{Market impact} \propto \text{Duration}^{\alpha} \times \text{Participation}^{\gamma},$$

fitted to averages over buckets of duration (in 5-minute intervals) and participation rate. The fitted surface is concave in each variable separately, with $\gamma \approx 0.5$ [alpha-laruelle, same sec., eq. (3.1), Figure 3.6, pp. 207–209, page image of p. 208]. Impact rises with both participation and duration, even separately: executions of 25 to 50 minutes lie above those shorter than 25 minutes at every participation rate (Figure 3.5, pp. 206–207).

Conditions. The fit is descriptive and the authors do not report α or uncertainty for either exponent in the passages checked. "Participation" there is the period liquidity ratio, the order's share of market volume during its execution. Bacry, Iuga, Lasnier and Lehalle measure the same dependence on another broker sample (European metaorders, 2010) by direct regression of the signed price move at the end of execution [wide-bacry-life-cycle, arXiv preprint of 9 Dec. 2014, sec. 3.2, Table 2, p. 8, page image]:

- on daily participation R with duration, the exponents are 0.54–0.59 for R and -0.23 to -0.35 for duration;
- on the trading rate r with duration, they are 0.37–0.56 for r and $+0.15$ to $+0.24$ for duration.

The sign of the duration effect thus depends on what is held fixed. At fixed daily participation a longer execution trades more slowly and costs less; at a fixed participation rate a longer execution trades more and costs more. Interpretation: with impact $\propto R^a T^b$ and $R = rT$, impact $\propto r^a T^{a+b}$, and $a + b \approx 0.2$–0.36 from the first rows, consistent with the second and with Lehalle and Laruelle's $\alpha > 0$.

Practice. Lehalle and Laruelle list stock characteristics (liquidity, volatility, spread) and order characteristics (aggressiveness, quantity, expected duration) as drivers, and timing as a further one. The same quantity has different impact depending on whether the order follows a trend, reverts to a mean, or follows an index rebalancing, and on whether high-frequency traders can detect it [alpha-laruelle, chap. 3, sec. 3.2, pp. 203–204 and 206]. After one day they take the temporary impact to have vanished, so later moves count as permanent [alpha-laruelle, same sec., p. 209].

Pitfalls. The legacy driver list added tick size, typical depth, order side, auctions and venue access (lit and dark liquidity, queue priority, routing), which are not in the passages checked. On the spread the sources disagree. Bacry et al. find a spread term with exponent 0.28 in a log-log fit (Table 2, (R.3)), while Almgren et al. find no consistent dependence of permanent impact on the spread (direct estimation). The legacy block described the decayed part as "temporary impact and liquidity replenishment". In the terms of price impact it is the decay of the transient displacement; Bacry et al. reserve "temporary impact" for the value at the end of execution. The legacy model called the exponents "calibrated from executions" and the participation exponent concave "in many datasets"; only one fitted value, $\gamma \approx 0.5$, is reported.

See also. square-root impact law; direct estimation of impact (a different design); post-execution impact profile (the daily horizon); price impact.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Moro et al. (2009), cited by Lehalle and Laruelle for the concave path, not consulted.

Index. impact path; intraday impact; arrival price; impact in spreads; participation rate; period liquidity ratio; duration; concave impact; impact decay; impact drivers; timing

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 203–209, page image of p. 208; and `wide-bacry-life-cycle` in its arXiv version of 9 December 2014, pp. 5–8, page image of p. 8). Stability: draft. Currency: as of the 2010 samples.

Method 9.35 Direct estimation of permanent and temporary impact from institutional orders (Almgren, Thum, Hauptmann and Li)

Statement. From completed orders, Almgren, Thum, Hauptmann and Li estimate two impact functions [wide-almgren-direct-impact, authors' preprint of 10 May 2005, secs. 2.2–4.3, pp. 8–21]:

1. For each order measure three prices (p. 9): the pre-trade midpoint S_0, from the last quote before the first transaction; the realised average price $\bar{S}$; and the post-trade midpoint S_{post}, from the first quote at least half an hour after the last execution, carried to the next morning if that falls after the close.

2. Define the permanent impact $I = (S_{\text{post}} - S_0)/S_0$ and the realised impact $J = (\bar{S} - S_0)/S_0$, both signed so that a cost has the sign of the order (p. 10).

3. Measure time in volume time, the fraction of an average day's volume executed so far (p. 10). The order of X shares is executed at a constant rate over T, with V the 10-day average daily volume, σ an intraday estimate of daily volatility, and Θ the shares outstanding (pp. 11–13 and 19).

4. Fit, with heteroskedastic weights from the model's error terms (eqs. (7)–(8), p. 19, page image)

$$\frac{I}{\sigma} = \gamma\, T\, \mathrm{sgn}(X) \left|\frac{X}{VT}\right|^{\alpha} \left(\frac{\Theta}{V}\right)^{\delta} + \text{noise}, \qquad \frac{1}{\sigma}\left(J - \frac{I}{2}\right) = \eta\, \mathrm{sgn}(X) \left|\frac{X}{VT}\right|^{\beta} + \text{noise}.$$

The term $I/2$ removes the effect of the order's own permanent impact on its later fills, so $J - I/2$ is the temporary impact (eq. (2), p. 15).

Notation. Here I and J are the paper's impact variables, not the impact process I_t of other entries; γ, η, α, β, δ are its coefficients and exponents, not the Almgren–Chriss γ and η of price impact.

Conditions. The sample is Citigroup US equity orders from December 2001 to June 2003: 682,562 orders before filtering, 29,509 after (pp. 5–8). The filters keep S&P 500 stocks with daily volatility below 12.5 percent. They exclude market-on-open, market-on-close and VWAP orders and orders with executions after 4:10 pm, and keep orders of at least two transactions, 1,000 shares and 0.25 percent of average daily volume. The median order lasts about half an hour; most are a few percent of daily volume or less (pp. 6–8). The model assumes a constant trading rate in volume time and no cross-impact. It makes permanent impact independent of the schedule, so stealth trading is excluded by construction (p. 13).

Practice. The estimates are $\alpha = 0.891 \pm 0.10$, $\delta = 0.267 \pm 0.22$, $\beta = 0.600 \pm 0.038$, one standard deviation (p. 20). The authors fix $\alpha = 1$, which they cannot reject, $\delta = 1/4$ and $\beta = 3/5$, rejecting the square root at the 95 percent level. They then find $\gamma = 0.314 \pm 0.041$ and $\eta = 0.142 \pm 0.0062$ [wide-almgren-direct-impact, same preprint, secs. 4.2–4.3, pp. 20–21, page image of p. 21]. The fitted model is $I = \gamma\sigma(X/V)(\Theta/V)^{1/4}$ and $J = I/2 + \mathrm{sgn}(X)\,\eta\sigma|X/VT|^{3/5}$ (p. 21). In their example of a purchase of 10 percent of daily volume in IBM ($\sigma = 1.57\%$, $\Theta/V = 263$), permanent impact is 20 bp, and realised cost is 32, 25 and 18 bp for durations of 0.1, 0.2 and 0.5 days (Table 3, p. 22). Interpretation: recomputing from the fitted coefficients gives 19.9 bp and 32.2, 24.6 and 18.4 bp, matching the table to its rounding; DRI likewise. Temporary impact needs no stock-specific factor beyond volume and volatility, and no consistent dependence on the bid–ask spread was found (p. 19).

Pitfalls. The "permanent" impact I is the move to half an hour after the last fill, chosen because the regressed values stop changing with the lag at about that point (p. 9). It is a finite-horizon measurement that can include later flow and information; compare the 20-day profile of the post-execution method. R^2 is typically below one percent, because volatility dominates the price moves of individual orders (p. 21). Filling only partly, and the exclusion of block-desk orders, could bias the sample; the authors report nearly identical coefficients with and without partly filled orders (p. 8). The fit reflects US large caps over 2001–2003; the authors expect coefficients, exponents and forms to be re-estimated (pp. 21–22). They prefer linear permanent impact because, citing Huberman and Stanzl (2004), not consulted, it is the only form free of arbitrage, while nonlinear temporary impact introduces none (p. 16); see absence of price manipulation for the conditions in other models.

See also. price impact (the Almgren–Chriss framework behind the model); square-root impact law (an exponent of 0.6 here); intraday impact profile; post-execution impact profile.

Further reading. The paper's published version, *Risk* 18(7), 2005, not compared with the preprint. The legacy blocks it informs: `notes/artefacts/market-making-lob/derivations.tex`.

Index. direct impact estimation; permanent impact; realised impact; temporary impact; volume time; participation; inverse turnover; Citigroup sample; 3/5 power law; heteroskedastic regression

Status. Verification: partly checked 2026-10-15 (against the authors' preprint of 10 May 2005 of `wide-almgren-direct-impact`, 26 pages, from a Wayback snapshot of 8 August 2017 of the author's site, pp. 1–22, page images of pp. 19 and 21; the example recomputed). Stability: draft. Currency: as of the 2001–2003 sample.

Method 9.36 Order-flow impact regression: the Glosten–Harris decomposition into permanent and transitory components

Statement. Harris presents the Glosten–Harris model as an order-flow transaction-cost estimator [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", box, p. 435, page image]. The trade price is fundamental value plus a transitory component, $P_t = V_t + \theta_t$ with $\theta_t = aq_t + bq_tS_t$. Value changes by a permanent effect proportional to signed size plus other information, $\Delta V_t = \lambda q_t S_t + \varepsilon_t$. Hence

$$\Delta P_t = \lambda\, q_t S_t + a\, \Delta q_t + b\, \Delta(q_t S_t) + \varepsilon_t,$$

estimated by regressing the price change on signed size, the change in sign and the change in signed size. Hasbrouck states the model with a size-independent permanent term as well: $m_t = m_{t-1} + u_t + q_t(\lambda_0 + \lambda_1 V_t)$, $p_t = m_t + q_t(c_0 + c_1 V_t)$ [s6-hasbrouck, sec. 9.8.1, eq. (9.26), p. 91, page image].

Notation. $q_t = \pm 1$ the trade sign (house convention); S_t (Harris's Size) or V_t (Hasbrouck) the unsigned size. Interpretation: a is half the fixed transitory spread per side and λ the permanent effect per share, as in the spread-component entries.

Conditions. Interpretation: differencing Hasbrouck's form gives $\Delta p_t = \lambda_0 q_t + \lambda_1 q_t V_t + c_0 \Delta q_t + c_1 \Delta(q_t V_t) + u_t$. Harris's regression, and the legacy one, are the special case $\lambda_0 = 0$. That follows Harris's statement that information theory suggests a permanent effect proportional to size (p. 435), but it drops a term the general model allows. Glosten and Harris's own estimation treated prices as rounded to the tick and trade signs as latent, using a nonlinear filter, and assumed size and sign exogenous with normal public news [s6-hasbrouck, sec. 9.8.1, pp. 91–92]. The original paper was not consulted.

Practice. Estimated size effects are generally concave, flattening at large sizes, and signed size often adds little beyond the sign. One explanation is time-varying liquidity: traders trade large when impact is low. The index can be trades (event time) or fixed intervals (clock time): event time favours stationarity for one security, clock time comparability across securities. Trade prices suit execution-cost questions; quote midpoints suit long-run impact and are less noisy [s6-hasbrouck, secs. 9.7.2–9.7.4, pp. 89–91]. Conditioning on an index return, or better on recent index-futures returns, matters when attributing price changes to a stock's own trades [s6-hasbrouck, sec. 9.6, p. 87].

Pitfalls. Harris writes that "the statistical properties of standard regression methods ensure that the regression error term is uncorrelated with the independent regressor variables" (p. 435). Interpretation: least squares makes the fitted residuals orthogonal to the regressors by construction, but the coefficients estimate the permanent and transitory effects only if the true ε_t is uncorrelated with them. That is an assumption, Glosten and Harris's exogeneity, not a consequence of the method. Signed order flow is chosen by traders who may trade on information or on the same news. Coefficients then mix reaction with prediction, as in price impact. The legacy caveat listed controls for market moves, time of day, volatility, spread and order splitting. Of these, the passages checked support an index return in the conditioning set (Hasbrouck, p. 87), volume time (Almgren et al.) and volatility and spread terms (Bacry et al.). Controls reduce omitted-variable bias but do not by themselves make the coefficients causal (Interpretation); random trading is the design that does (price impact, Practice).

See also. spread component identification (the generalized Roll model, the size-free special case); spread components; price impact; propagator model (impact that decays over many trades rather than one).

Further reading. Glosten and Harris (1988), not consulted. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Glosten–Harris model; order-flow regression; signed order flow; permanent and transitory components; trade size; event time; clock time; trade prices and midpoints; endogeneity; exogeneity

Status. Verification: partly checked 2026-10-15 (against `alpha-harris`, 2002 Oxford printing, p. 435 by page image; and `s6-hasbrouck`, 2007 Oxford printing, pp. 87–92, page image of p. 91). Stability: draft. Currency: timeless (method).

Evidence 9.37 Index reconstitution: price pressure from forced index-fund trading and its partial reversal

Statement. Index funds that replicate the Russell indexes rebalance when the indexes are reconstituted each June, buying additions and selling deletions near the same date. In the six years 1996–2001, Russell 3000 additions outperformed deletions by an average of 15 percent in June and underperformed them by 5 percent in July. Harris reports this from Madhavan (2002), not consulted [alpha-harris, chap. "Index and Portfolio Markets", box, p. 487, page image].

Conditions. Russell reconstitutes at the close of the last trading day in June on capitalisations as of the end of May. The impacts are large because the stocks affected are small [alpha-harris, same box]. Harris notes that changes to Standard & Poor's indexes have similar effects (same box).

Practice. The July reversal shows that part of the June move is transitory. Harris offers three explanations for the remainder: investors may value index membership, perhaps for liquidity; a momentum anomaly in small stocks; or reversal in later months [alpha-harris, same box]. The event is predictable demand that is uninformed about the firm, which makes it a natural setting for separating price pressure from information (Interpretation).

Pitfalls. The legacy finding attributed the residual to "changed ownership, attention, or information". Those are not Harris's explanations, and the box gives no test between his. "Largest when forced demand is large relative to stock liquidity" paraphrases Harris's remark that the stocks are small; the box gives no measure of demand relative to liquidity. The figures are averages over 1996–2001 and may not hold under later practice.

See also. price impact (transient and permanent components); post-execution impact profile. The pathway `etf-index-closing-execution` develops index events and the closing auction.

Further reading. Madhavan, "The Russell Reconstitution Effect" (2002), cited by Harris, not consulted. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. index reconstitution; Russell reconstitution; index additions and deletions; price pressure; reversal; index funds; forced trading

Status. Verification: partly checked 2026-10-15 (against `alpha-harris`, 2002 Oxford printing, p. 487 by page image). Stability: draft. Currency: as of 1996–2001.

9.12 Exchange-traded funds and index products

Mechanism 9.38 ETF creation and redemption: authorised participants, in-kind and cash baskets, and arbitrage that tends to keep price near value without guaranteeing it

Statement. An open-end mutual fund transacts with its investors at the end of the day at net asset value (NAV), and buys or sells the underlying assets itself when flows do not net. An exchange-traded fund (ETF) also holds a portfolio but, in the authors' description, "does not interact with capital markets directly" (p. 4): secondary trading and in-kind creations and redemptions leave the fund's dealings in the underlying to others, while cash creations and redemptions can require the fund to trade (Practice). Its sponsor contracts with *authorised participants* (APs), typically large financial institutions or specialised market makers, who can create or redeem the fund's shares in large blocks, *creation units*, in exchange for a basket of securities and/or cash [ex-lettau-madhavan-2018, NBER working paper version, pp. 4–5]. Two markets are therefore distinct:

- *creation and redemption* between APs and the fund. The fund publishes its holdings and the basket it will accept for the next business day at the end of each trading day. Transactions are either *in kind*, the AP delivering or receiving a basket identical or very similar to the holdings, or *in cash*; in the authors' description cash creation is "not typical" (p. 5);
- *secondary trading* of ETF shares on exchanges, through brokers, among investors, APs and other liquidity providers. It does not lead to transactions in the underlying securities; across U.S. ETFs secondary volume was about four times creation and redemption activity in 2014, by Investment Company Institute statistics as the authors report them (pp. 5–6).

Creation or redemption happens at the end of the day, but APs typically lock in any profit intraday. If the ETF trades at a premium to an AP's estimate of value, which need not be the NAV, the AP may deliver the creation basket for new ETF shares, which it may sell or keep. The mechanism "works through arbitrage to help keep the price ... close to the intrinsic value" of the holdings (p. 6). Harris's 2002 description is the same in kind: ETFs are trusts whose units trade like stocks, and large traders create units by depositing the index portfolio and redeem them for a pro rata share of it [alpha-harris, chap. "Index and Portfolio Markets", sec. 23.5, p. 492].

Conditions. The link between price and value is an arbitrage incentive, not a guarantee.

- *Stale reference values.* Deviations from the published NAV need not be arbitrage opportunities. The NAV and the intraday indicative value, disseminated about every 15 seconds, may rest on stale prices, as for funds holding Japanese stocks during U.S. hours or bonds that have not traded [ex-lettau-madhavan-2018, p. 6].
- *No obligation to arbitrage.* APs "are not contractually bound by legal obligations to perform ETF arbitrage" [wide-pan-zeng-etf, ESRB version, p. 1]. Lettau and Madhavan argue that one AP's withdrawal is likely to be covered by others in ETFs with many APs; in an Investment Company Institute study of 931 U.S. ETFs that they cite, the largest averaged 38 APs. Smaller niche ETFs with few APs are the exception they allow for. If all APs withdrew, the fund would trade like a closed-end fund, with possibly wider premiums or discounts (p. 18).
- *Liquidity mismatch in corporate bond ETFs.* Pan and Zeng study the case where a liquid ETF holds illiquid assets, for U.S. corporate bond ETFs only. APs are also bond dealers. Their data are AP lists from two issuers covering on average 83 percent of passive corporate bond ETF assets, 2004–2015, with dealer-identified TRACE bond trades [wide-pan-zeng-etf, abstract and pp. 1–3].
  - *Model predictions* (pp. 2–3). With small bond inventory imbalances, an arbitrage effect: APs correct mispricing, less so as volatility, bond illiquidity and trading costs rise. With large imbalances, an inventory-management effect can dominate, so that creations and redemptions follow the APs' inventories and may run against the mispricing.
  - *Empirical findings* (p. 4). A 1 percent increase in the ETF premium raises AP arbitrage by 50 basis points; a one-standard-deviation rise in market volatility, holding the premium fixed, lowers it by 10 percent; and the fall in sensitivity is larger when the ETF trades at a discount. Under extreme inventory imbalances, creations and redemptions become less sensitive to the mispricing and more sensitive to inventory. Overall, arbitrage still goes in the direction of the opportunity, but a

difference-in-differences comparison shows APs doing less (more) net redemption after large positive (negative) inventory shocks despite a discount (premium). The authors "do not find evidence of the complete 'failure' of ETF arbitrage".

Inventory pressures therefore weaken corrective arbitrage in their sample; that is not evidence that arbitrage fails outright, nor that it never runs the wrong way. The authors contrast equity ETFs, where they see no such mismatch (p. 2); their evidence does not extend to them.

Practice. In an open-end fund the fund itself trades the underlying for net flows, and the costs fall on the investors who stay. In an ETF, sellers trade with buyers at a market price, and "net selling does not require the ETF manager to interact with the capital markets": the costs are "externalised" [ex-lettau-madhavan-2018, p. 7]. Interpretation of the mechanism: secondary trading and in-kind creations and redemptions do not themselves require the fund to buy or sell the underlying, so its remaining holders bear none of those costs. Whoever trades the basket, such as an AP, bears them and passes them into the prices it quotes. Cash creations and redemptions, by contrast, can require the fund to trade the underlying and incur costs; the passages read do not say who bears them.

The authors' illustration: a mutual fund of $100 million, with spreads of 0.20 percent (0.10 percent one way), $5 million of subscriptions and $20 million of redemptions, sells $15 million at the bid and ends at $84,985,000 (recomputed: $15,000 of costs); the ETF's investors who do not sell "will hold a fund whose assets are valued at $85 million" (p. 7). Interpretation, reconciling the abbreviated ETF case: pure secondary trading leaves the fund's total assets unchanged at $100 million, with the net $15 million of shares sold now held by whoever bought them, such as an AP. Total assets of $85 million follow if that net $15 million is redeemed in kind, so that the fund delivers securities rather than selling them. The holders other than the redeeming AP, the original investors who did not sell ($80 million) and the new buyers ($5 million), then hold $85 million without bearing the $15,000.

In-kind transfers can also reduce capital-gains distributions (p. 8; Harris, p. 492, for the same tax point). Among index products Harris also lists open-end index funds, bought and redeemed at closing NAV and not tradable within the day; index futures, cheap to hold on margin but rolled at a cost; and cash-settled index options, used by speculators, hedgers and gamblers (sec. 23.5, pp. 491–492, Table 23-1 as of April 2001).

Pitfalls. "Primary market" is used two ways. Harris's Table 23-1 gives the primary market for fund shares as deposit to and redemption from the fund (p. 491), while Lettau and Madhavan use it for the market in the underlying securities in which APs acquire baskets (p. 18). This entry says "creation and redemption" for the first and "secondary trading" for exchange trading in ETF shares.

The legacy mechanism said the link "weakens when creation costs, borrow, cash substitutions, taxes, or stressed constituent liquidity are material". Stale valuation, illiquid holdings (for corporate bonds) and APs' discretion are in the passages read. Borrow costs, cash substitution as a weakening factor and taxes as a limit to arbitrage are not.

The legacy taxonomy's items were checked against Harris:

- index futures as "leveraged, cash-efficient exposure with basis and roll risk": Harris names margins and roll costs, not basis;
- index options as "convex exposure to index level and volatility": a gloss (Interpretation);
- its closing claim that product choice is "mostly" about speed, funding, tracking error, tax and operations: a synthesis.

On versions and dates: Lettau and Madhavan were read in their NBER working paper of January 2018, not the published *Journal of Economic Perspectives* version, which was behind a bot check. Madhavan was then head of ETF and index research at BlackRock, which the paper discloses (p. 1). The ETF facts are as of 2014–2018 for the United States. Current SEC rules on ETF baskets (Rule 6c-11) were not checked.

See also. portfolio trading (index products and program trades; owned by `execution-trading-costs`); index reconstitution (owned by `price-impact-execution`); arbitrage risks; HFT strategies (ETF-basket arbitrage as an example).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/etf-index-closing-execution/`.

Index. exchange-traded fund; ETF; authorised participant; creation unit; creation basket; in-kind creation; cash creation; redemption; secondary market; net asset value; intraday indicative value; ETF premium and discount; ETF arbitrage; liquidity mismatch; corporate bond ETF; transaction costs externalised; open-end index fund; index futures; index options

Status. Verification: partly checked 2026-10-15 (against `ex-lettau-madhavan-2018` in NBER Working Paper 24250, January 2018, pp. 1 and 4–8, 18, text layer; `wide-pan-zeng-etf` in ESRB Working Paper 59, December 2017, abstract and pp. 1–4, text layer, p. 4 added in a follow-up the same day; and the text layer

of alpha-harris, 2002 Oxford printing, sec. 23.5, pp. 491–492). Stability: draft. Currency: ETF practice as described for the United States, 2014–2018; Harris's products as of 2001–2002; no current rule checked.

9.13 Futures: venue conventions, off-exchange trades, margin and price discovery

Convention 9.39 CME Globex trading conventions: a centralised market with permitted exceptions, matching algorithms, the pre-open and opening match, and the E-mini S&P 500 daily settlement window

Statement. *Centralisation.* Harris reported in 2002 that most futures markets require all trades in their contracts to be arranged within their trading systems, floor or screen. The rules prevent brokers from directing orders to confederates at prices worse than the exchange's, and futures markets had historically had no block-trading procedures or cumbersome ones [alpha-harris, chap. "Block Traders", sec. 15.6.2, p. 334]. CME's current rulebook keeps the principle and names the exceptions: block trades, and exchanges for related positions, which are "permitted to be executed outside of the Exchange's centralized market in accordance with the requirements of this rule" [cme-rulebook-ch5, Rules 526 and 538]; see futures block trades and EFRPs.

Matching. On CME Globex a matching algorithm allocates quantity when an aggressor order meets one or more resting orders. The supported algorithms, identified by tag 1142, are Allocation (A), FIFO (F), FIFO with lead market maker (T), FIFO with top order and lead market maker (S), Pro-Rata (C), Configurable (K), Threshold Pro-Rata (O), Threshold Pro-Rata with lead market maker (Q) and an Institutional Prioritization algorithm (V). Under FIFO, price and time are the only criteria; an order loses its priority and is requeued if its quantity is increased, its price changed or its account changed. As one named example, SOFR futures (but not SOFR packs and bundles, which match FIFO) use a pro-rata allocation. A "top" order that betters the market has priority, then remaining quantity is allocated pro rata to resting orders at the price; orders entered in the pre-open or at the indicative opening price are matched by price and time [ex-cme-matching, page "Supported Matching Algorithms", updated 11 February 2025].

Opening. In the Pre-Open and Reserve states orders can be entered or modified, but no matching occurs, so the book can lock or cross. The trading engine computes and publishes an indicative opening price and quantity, "a probable price and quantity at which the market will open or re-open", for outright futures, futures spreads, options spreads and options series [cme-iop-wiki, updated 11 February 2025].

Daily settlement, one contract, as of 2020. In the procedure filed for trade dates from 26 October 2020, the E-mini S&P 500 (ES) and the Micro E-mini settle to the daily settlement price of the full-sized S&P 500 futures (SP), rounded to the nearest tradable tick. For SP's lead month, Tier 1 is the volume-weighted average price of trades in the E-mini contract on Globex from 14:59:30 to 15:00:00 Central Time, rounded to the nearest 0.10 index point. The amendment struck the floor trades in the full-sized contract, the multiplier of five and the word "combined". If no lead-month trades occur in the window, Tier 2 is the midpoint of the bid and ask over it; if no two-sided market is available, Tier 3 derives the price from the cash index by a carry calculation. The second month is derived from the lead month through the calendar spread. Tier 1 settles it to the VWAP of lead–second spread trades in the window "using the same methodology"; Tier 2, if no spread trades occur, applies the last spread trade price to the lead-month settlement to derive the second-month settlement, using the nearer of the spread's bid or ask if the last trade lies outside them; Tier 3, with no spread market information, uses the carry calculation. Interpretation: in Tier 1, too, the spread price is applied to the lead-month settlement to give the second-month outright, as Tier 2 states explicitly (Exhibit B, PDF p. 6). Staff may determine an alternative price if the calculations cannot be made or activity is anomalous [cme-es-settlement-2020, Exhibit B, blackline, page images].

Conditions. The matching and opening pages are living documentation, checked on 15 October 2026; the settlement procedure is stated as filed in 2020, because the current procedure page could not be retrieved. Which algorithm applies to a given contract is set per product and was checked here only for SOFR futures. Other exchanges, and other CME contracts' settlement procedures, were not checked.

Practice. Interpretation: the pathway's checkpoint needs a contract's algorithm before estimating passive fill probability. Under FIFO a resting order's fill depends on its queue position; under pro-rata allocation it depends on its size relative to the quantity at the price, which rewards larger posted size (passive order split for the FIFO queue model). When proportional shares are not whole contracts, a remainder must be allocated by a further rule; CME's own allocation steps for each algorithm were not read (slice 34; matching and trade pricing works examples of remainders).

Pitfalls. The legacy caveat *No Call Auction Norm* said many futures markets do not rely on equity-style opening and closing call auctions. For CME Globex this is wrong at the open: the pre-open with an indicative opening price, followed by the opening match, is a call-auction-like uncross. For the E-mini S&P 500 close it is right only in the narrower sense that, under the 2020 procedure, the daily settlement is a 30-second VWAP of continuous Globex trading, with midpoint and carry fallbacks, not a closing auction. Its claim that participants

instead "agree on liquid trading hours" is not in the sources checked. The legacy principle *Centralised Futures Trading* gave margining, clearing, delivery rules and fungibility as the reasons fragmentation is limited. The sources checked give exchange rules requiring on-exchange trading, with defined exceptions; the economic reasons were not checked. The legacy notes' "highly electronic" describes CME Globex today; Harris's 2002 account still includes floor trading.

See also. futures block trades and EFRPs; futures margin and settlement; spread determinants (large ticks; owned by `execution-market-making`); matching and trade pricing (precedence, pro-rata and auction rules in general).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/futures-execution/`.

Index. CME Globex; matching algorithm; FIFO; pro-rata allocation; top order; lead market maker; tag 1142; SOFR futures; pre-open; indicative opening price; opening match; call auction; daily settlement price; settlement VWAP; E-mini S&P 500; centralised market

Status. Verification: partly checked 2026-10-15 (against the CME Client Systems Wiki pages "Supported Matching Algorithms" and "Indicative Opening Price and Quantity", both updated 11 February 2025; CME Rulebook chap. 5, Rules 526 and 538; the CFTC filing of 22 September 2020, Exhibit B (blackline, page images), for the E-mini S&P 500 settlement; and the text of `alpha-harris`, 2002 Oxford printing, p. 334). Stability: draft. Currency: venue documentation as retrieved on 15 October 2026; settlement as of trade date 26 October 2020; Harris as of 2002.

Convention 9.40 Futures block trades and exchanges for related positions at CME Group: eligibility, the related position, reporting, and the prohibition of transitory EFRPs

Statement. *Block trades* (CME Rule 526, current rulebook [`cme-rulebook-ch5`, Rule 526]).

- The exchange designates the products in which blocks are permitted and their minimum quantities, and orders may not be aggregated to reach them except by eligible advisers (526.A).
- Each party must be an Eligible Contract Participant under the Commodity Exchange Act; registered or exempt commodity trading advisers and investment advisers with more than $25 million under management, and similar foreign persons, may act for their customers (526.B, I, J).
- A customer must have specified that its order be executed as a block (526.C).
- The price must be "fair and reasonable" given the block's size, other trades in the contract, related markets and the circumstances (526.D).
- Blocks do not trigger conditional orders such as stops or otherwise affect orders in the regular market (526.E).
- The seller or broker reports the block within the period and in the manner the exchange specifies, and the exchange publishes blocks separately from regular-market trades (526.F); blocks are reported to the clearing house, and records are kept (526.G–H).

Exchanges for related positions (EFRPs; CME Rule 538 [`cme-rulebook-ch5`, Rule 538]).

- An EFRP is a privately negotiated off-exchange execution of a futures or options contract together with the simultaneous, opposite execution of an equivalent quantity of the related cash product, by-product, related product or OTC derivative. The types are the exchange of futures for physical (EFP), for risk (EFR, against an OTC swap or other derivative) and of option for option (EOO).
- Each party is buyer on one side and seller on the other, and each side's two legs must be for accounts with the same beneficial ownership (538.A). A third party may facilitate the related position as principal for a customer but must show that it was passed through to the customer, except for immediately offsetting FX EFPs under 538.K: when CTAs, account controllers or others act for a customer in those, the initiating and offsetting cash legs need not be passed through, though the customer bears the profit or loss on the offsetting leg if the futures leg fails to clear (538.A). Opposing accounts must be independently controlled (538.B).
- The related position must be the underlying cash commodity, a by-product, a related product or an OTC derivative with a reasonable degree of price correlation, never a futures or option on futures. It requires a bona fide transfer of ownership or a binding contract (538.C).
- Quantities must be approximately equivalent, with hedge ratios allowed (538.E). Prices must be on the contract's increments and commercially reasonable (538.F).

- Each EFRP is submitted to the clearing house within the specified time, with the time its terms were agreed, and is accepted only once it has cleared and the first settlement variation and performance bond are confirmed (538.I). Records are kept (538.H).

Transitory EFRPs. Rule 538.C forbids an EFRP whose execution is contingent on another EFRP or related-position transaction between the parties, where the transactions offset the related positions without market risk that is material in context. CME Group's advisory notice in force from 7 August 2026 defines these as *transitory EFRPs*. It answers "No" to whether they are permitted in any CME Group product: the time between the transactions is a factor, but the test is whether they "have integrity as independent transactions exposed to market risk" [cme-ra2607-5, pp. 4 and 10, Q14]. Rule 538.K, as amended from 7 August 2026, expressly permits EFRPs in foreign currency futures whose parties immediately offset the related position, but only as EFPs whose related position is a cash transaction or EFRs whose related position is a non-deliverable forward (NDF), in foreign currencies only and never as EOOs; the cash-side confirmations must identify the counterparty, or for a CTA's client its carrying clearing member and account [cme-rulebook-ch5, Rule 538.K, page image]. The exchanges' filing describes these as physical FX cash and FX NDF related positions; the amendment changed only 538.K, leaving 538.A–J unchanged [cme-ra2607-5, pp. 1–3, Exhibit A blackline]. They are not treated as transitory, because the offsetting cash or NDF transaction is not contingent on the EFRP (Q28).

Conditions. Product thresholds and reporting deadlines are set in separate exchange notices, not checked here. The rules are those of CME Group's U.S. exchanges (CME, CBOT, NYMEX and COMEX); other venues were not checked. History, kept apart from current rules: Harris described futures block rules in 2002 as limited to eligible sophisticated traders at fair and reasonable prices. He described EFPs as originally developed for agricultural hedgers, and as used at CME in currency futures immediately after the close, where exchanges called them "transitory EFPs" because the floor traders bought or sold the cash currency at the same time [alpha-harris, chap. "Block Traders", sec. 15.6.2, p. 334]. CME Group's 2019 notice records that transitory EFRPs had been permitted in CME foreign currency, NYMEX energy and COMEX and NYMEX metals products until 4 August 2014 and have been prohibited since [cme-ra1911-5, p. 1].

Practice. Interpretation: an equity block cross interacts with the exchange book under precedence rules (block trades, Harris's 2002 equity rules). A futures block is negotiated away from the book, does not trigger the book's conditional orders and is published separately. The two regimes should not be mixed.

Pitfalls. The legacy taxonomy listed a "transitory EFP: short-lived EFP used when the book is closed or thin". The term is verified, but only as history: it matches Harris's 2002 account, and at CME Group such EFRPs have been prohibited since August 2014. The immediately offsetting FX EFRP of Rule 538.K is a different, permitted structure. Its EFP definition ("exchange cash or physical exposure for offsetting futures") matches Rule 538 in substance; its closing sentence ("move large risk off-book, subject to eligibility and reporting rules") fits Rules 526 and 538.

See also. block trades (owned by `execution-trading-costs`); CME Globex trading conventions; futures margin and settlement.

Further reading. The legacy block it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/futures-execution/`.

Index. futures block trade; Rule 526; eligible contract participant; block minimum threshold; exchange for related position; EFRP; exchange for physical; EFP; exchange for risk; EFR; exchange of option for option; EOO; related position; transitory EFRP; immediately offsetting FX EFP; Rule 538

Status. Verification: partly checked 2026-10-15 (against CME Rulebook chap. 5, Rules 526 and 538, as retrieved on 15 October 2026; CME Group advisory RA2607-5, effective 7 August 2026, pp. 1–4 and 10–17, with the Exhibit A blackline and the rulebook's pp. 27–29 as page images; RA1911-5 of 2019 for the history; and `alpha-harris`, 2002 Oxford printing, p. 334). Stability: draft. Currency: CME Group rules as of 15 October 2026; Harris as of 2002.

Mechanism 9.41 Futures margin and settlement at CME Clearing: performance bond, settlement variation each cycle, and the default waterfall

Statement. At CME Clearing, settlement for outstanding exposure is made with the clearing house in settlement cycles held "at least once each Business Day", at times the clearing house sets and possibly different by contract. Each clearing member, for its own and its customers' positions, pays or collects *settlement variation*: the difference between this cycle's settlement price and the previous cycle's, or the trade price for a new position. Payments are final once the clearing house's bank account is debited or credited [cme-rulebook-ch8, Rule 814]. Separately, clearing members post *performance bond*, initial and additional, as clearing-house staff set it, gross for each omnibus customer account and in approved collateral. The clearing house retains it "in whole or in part, as Clearing House staff may deem necessary", until the trades it covers have been offset, cash settled, delivered or otherwise closed out; if a clearing member defaults on performance bond, the collateral is taken over and sold, customer collateral being applied to customer requirements and house collateral to house requirements [cme-rulebook-ch8, Rule 820].

On a clearing member's default the Base waterfall runs as follows [cme-rulebook-ch8, Rules 802.A–802.B]:

1. the defaulting member's own resources, with default management run separately for each account class (futures, foreign futures, cleared swaps). A loss in a customer class is met first from that class's customer collateral (performance bond, excess bond and settlement gains). In a defaulted customer futures class this includes the collateral of customers who did not cause the default (802.A.2, 802.G.1). The member's own assets (its Base Guaranty Fund requirement, performance bond, CME shares or cash in lieu and other assets) are applied across account classes with priority to customer classes (802.A.5.i, iii). Collateral left after liquidating the proprietary account may be applied to customer-class losses, pro rata among defaulted customer classes (802.G.6). The restriction runs one way: segregated customer collateral may not be used for the member's proprietary obligations or for another customer account class (802.A.5.iii, 802.G);

2. CME's corporate contribution of $100 million (802.B.1.i);

3. the Base Guaranty Fund, that is, the prefunded contributions of the non-defaulting members, applied in the rule's order of priority: the subordinated amounts of members subordinated after a default-management auction, then the remaining contributions of the other members pro rata, then the seniorized amounts of winning bidders (802.B.1.ii–iii; the fund consists of members' contributions, 802.A.4, so it is counted once here);

4. assessments on the non-defaulting members, capped at 275 percent of each member's Base Guaranty Fund requirement for losses from a single defaulter and 550 percent for all defaults in a cooling-off period, divided pro rata (802.B.1.iv).

Conditions. Interpretation, keeping apart the items the legacy notes merged:

- *contract standardisation*, set in each contract's chapter (not read here);
- *initial margin*, the performance bond that covers moves between settlements and the time to close out;
- *variation settlement*, the realised gain or loss each cycle;
- *intraday calls*, which Rule 814 permits through more than one cycle a day, though the actual cycle times were not checked;
- the *FCM's* arrangements with its customers, governed by its customer agreements and by rules not read here, while Rules 814 and 820 bind clearing members;
- *residual default risk*: prices move between cycles, performance bond may fall short, and losses then pass along the waterfall of Rule 802 to CME's contribution, the other members' prefunded contributions and, within the caps, assessments on them.

Daily settlement bounds the exposure that can build between cycles; it does not eliminate credit risk continuously. Harris notes that futures are popular with hedgers and speculators because positions can be taken without posting large margins [alpha-harris, chap. "Index and Portfolio Markets", sec. 23.5, p. 492].

Practice. The legacy statement that the broker or FCM "manages client margin calls, liquidation rights, and cash balances" describes the customer side, which was not checked; the clearing-house rules above are the member side. *A textbook account, generic rather than CME's rules (slice 31).* Hull illustrates the customer side with a broker's margin account [s7-hulloptions, chap. 2, "Futures Markets and Central Counterparties", sec. 2.4, pp. 51–54]: an initial margin is deposited, the account is settled to the day's gain or loss each day (daily settlement or marking to market), and the daily flow between longs and shorts is variation margin. Below a maintenance margin, "somewhat lower than the initial margin" and usually about 75 percent of it, the trader receives a margin call to restore the initial level or is closed out; excess over the initial margin can be withdrawn; most brokers pay interest on balances, and Treasury bills (at about 90 percent of value) or shares (about 50 percent) can usually meet initial but not later calls. Clearing members, by contrast, face a maintenance margin equal to the initial margin, margin computed on members' net positions, possible intraday calls, and a guaranty fund; margin is usually set so that the clearing house is "about 99% certain" it covers losses on a member's default (pp. 53–54). Hull's gold example is checked: two contracts of 100 ounces at $1,750 lose $200 \times 9 = \$1{,}800$ on a $9 fall, and the day-7 balance of $7,980 prompts a call of $4,020 back to $12,000 (p. 51; Table 2.1, p. 52; p. 53). These levels, ratios and haircuts are Hull's illustrations, not rules of any clearing house; minimum margins are set by each clearing house and brokers may demand more (p. 53).

Pitfalls. The legacy mechanism concluded that credit risk "is therefore controlled continuously rather than only at final settlement". It is controlled at each settlement cycle, at least daily, with performance bond for the moves in between and a mutualised default waterfall for what remains; "continuously" overstates it. The legacy taxonomy's items were checked as follows:

- "fungible by expiry and specification": contract terms were not read;
- "margin and default management built into the market": Rules 802, 814 and 820;
- "leverage": Harris p. 492;
- "hedging link to cash baskets, ETFs, options and swaps": only partly, since Rule 538 allows cash and OTC-derivative related positions; ETFs were not checked.

See also. CME Globex trading conventions (daily settlement prices); futures block trades and EFRPs (acceptance after first settlement variation); arbitrage risks (financing of losing legs).

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/futures-execution/`.

Index. futures margin; performance bond; initial margin; variation margin; settlement variation; settlement cycle; clearing house; clearing member; FCM; guaranty fund; default waterfall; mutualisation; credit risk; leverage

Status. Verification: partly checked 2026-10-15 (against CME Rulebook chap. 8, Rules 802.A, 802.B, 814 and 820, as retrieved on 15 October 2026, 802 read in full in the slice-19 follow-up, and `alpha-harris`, 2002 Oxford printing, p. 492; in slice 31, `s7-hulloptions`, 11th Global edition, chap. 2, pp. 51–54, for the generic account). Stability: draft. Currency: CME rules as of 15 October 2026; Hull's generic account as of 2021.

Evidence 9.42 Price discovery between index futures and their cash and ETF markets: the E-mini's leadership in 2000, and what explains leadership

Statement. Hasbrouck studies price discovery across U.S. equity index markets at up to one-second resolution, using vector error-correction models and information shares, for 1 March to 31 May 2000. He finds:

- for the S&P 500 and Nasdaq-100, whose index markets comprised an ETF, a floor-traded futures contract and an electronic E-mini, most price discovery occurred in the E-mini, whose prices generally led the regular futures and the ETFs; the ETFs' contributions were statistically significant but small;
- for the S&P 400 MidCap, with only an ETF and regular futures, both contributed substantially and the ETF appeared to dominate;
- the S&P 500 sector ETFs contributed little to price discovery in the index.

[`ex-hasbrouck-2003`, NYU working paper FIN-00-046, draft of 16 November 2000, abstract and pp. 4–5, 13 and 34]. The earlier literature he reviews found that the dominant influence ran from futures to cash, with weaker but measurable effects in the reverse direction (p. 5).

Conditions. The evidence is for three U.S. indexes in three months of 2000, when the regular contracts traded on the CME floor during regular hours and on Globex otherwise, and the E-minis on Globex (p. 6). It is from a preliminary working-paper draft. The catalogue describes the published 2003 version's S&P 400 result as shared between the regular futures and the ETF; the published version was not read, so the difference is recorded, not resolved. The theoretical treatment of futures–cash price discovery the pathway assigns, Garbade and Silber (1983, `w5-garbade-silber`), was not available and is unread.

Practice. Harris's reason for index-market leadership is that index traders price only index risk, while traders of the components must price every risk of their securities, so index markets discover the price of index risk faster (portfolio trading, p. 489).

Pitfalls. The legacy mechanism said futures lead "when they are cheaper to trade, more leveraged, or open during information arrival", and cash markets lead "when constituent-level information dominates". Harris's argument supports only the general direction for index risk. The specific conditions are plausible hypotheses not tested in the passages read, and the working paper does not attribute the E-mini's leadership to cost or leverage. Leadership in information shares is a statistical measure of where innovations first appear, not a causal claim about who is informed.

See also. portfolio trading, on why index markets discover the price of index risk first, owned by the section overview `execution-trading-costs`; ETF creation and redemption; arbitrage (the basis and delivery arbitrage).

Further reading. The legacy block it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/futures-execution/`. Garbade and Silber (1983) for the theory.

Index. price discovery; lead–lag; index futures; E-mini; ETF; information share; vector error correction model; cointegration; basis; futures leads cash; S&P 500; Nasdaq-100; S&P 400

Status. Verification: partly checked 2026-10-15 (against `ex-hasbrouck-2003` in its NYU working-paper draft of 16 November 2000, abstract and pp. 4–5, 13 and 34, text layer, not the Journal of Finance version; Garbade and Silber unread). Stability: draft. Currency: evidence for March–May 2000.

9.14 Fixed-income and OTC trading

Definition 9.43 Fixed-income market structure: segments, trading protocols, venues and dealer risk-bearing, kept apart (U.S., as surveyed in 2019)

Statement. Except for a few segments, fixed-income instruments trade in dealer-oriented over-the-counter (OTC) search markets that differ significantly from equity markets [wide-bsv-fi-survey, sec. I, p. 1]. Three things that the word "market" runs together are kept apart here.

- *Segments.* U.S. Treasury securities are "in many ways the simplest" fixed-income instruments: low default risk, high volume, standardised contracts and generally liquid markets (p. 1). Corporate bond trading is fragmented because each issue of an issuer is a distinct contract, not fungible with the others; individual issues did not trade on 52 percent of days in Edwards, Harris and Piwowar's sample, and in 2016 9.3 percent of TRACE bonds never traded and 1.5 percent traded every day (pp. 5 and 14). Structured products split between to-be-announced (TBA) agency MBS, forward contracts on any deliverable pool, the most actively traded structured securities, and specified pools and other structured bonds, where most MBS and ABS did not trade at all in one 21-month sample (pp. 5 and 13–14). Municipal bond trading is also highly fragmented, with over 1.5 million distinct bonds (p. 16).

- *Trading protocols*, the rules by which a client obtains a price. Dealers broadcast indicative bids and offers ("runs"); clients negotiate by phone or chat, or send bid- and offer-wanted lists; in an electronic request for quote (RFQ), as Hendershott and Madhavan describe it on MarketAxess, the client chooses how many dealers to query for a given size and a deadline, sees all quotes at the deadline and may trade with the best (pp. 2–3 and 7–8). Fixed-income quotes are typically indicative, not firm (p. 3).

- *Venues*, the systems on which trading happens: interdealer brokers, about 70 percent of U.S. Treasury interdealer trading being on electronic limit-order-book platforms where principal trading firms also take part; RFQ platforms for customers; and newer all-to-all venues on which institutions trade with each other (pp. 8 and 11).

- *Dealer risk-bearing.* Outside Treasuries and TBA MBS, trading is dominated by customer–dealer OTC trades, traditionally on a principal basis, the dealer absorbing the order into inventory. Dealers have reduced capital commitment in corporate bonds, relying more on prearranged trades in which the dealer has already located the other side, and on shorter intermediation chains; nonbank dealers' share of customer–dealer corporate bond volume rose from 2.4 percent before the 2007–2009 crisis to 12.5 percent in 2014–2016 (p. 4).

Conditions. The survey describes U.S. markets with data to about 2017–2018; its electronic-share figures come from industry reports: an industry report published in July 2016 estimated that 19 percent of U.S. investment-grade corporate bond trading was electronically facilitated, mostly by RFQ; the observation period behind that estimate is not given in the passage (p. 2). Hendershott and Madhavan find that periodic one-sided electronic auctions are "a viable and important source of liquidity even in inactively traded instruments", a compromise between bilateral search and continuous double auctions [wide-hendershott-madhavan, abstract]; in the survey's account RFQ use is greatest for recently issued, investment-grade, large-issue bonds (p. 3). O'Hara and Zhou, combining MarketAxess and TRACE data, find electronic RFQ trading "fairly small and segmented" but with wide-ranging effects on transaction costs and execution quality in electronic and voice trading and the interdealer market [wide-ohara-zhou-electronic, abstract]. Both are abstract-level checks.

Practice. Interpretation: the legacy "dealer balance sheet as venue" is a metaphor. A balance sheet is not a venue; the dealer's willingness to hold inventory is a source of liquidity, reached through a protocol (voice, RFQ) on a venue or bilaterally. The survey links the reduction in dealers' capital commitment to post-crisis capital requirements and the Volcker Rule, and notes that counting only bank-affiliated dealers' capital may overstate the effect on markets because nonbank dealers, electronic intermediation and customers' liquidity provision through prearranged trades respond (pp. 4 and 32–33). The generic inventory mechanics are in dealer inventory risk and dealer quote adjustment, from equity and floor examples.

Pitfalls. The legacy taxonomy's "government bonds: most standardised and electronically tradable" holds for U.S. Treasuries in the survey, whose electronic share is interdealer; customer-to-dealer Treasury trading relies on electronic communication for a large share but was not measured as electronic execution (p. 3). "Credit derivatives: standard indices trade more centrally than bespoke risk" and "swaps ... often remain dealer-mediated because contracts differ" were not checked: the survey mentions only that most standardised derivatives, including interest rate swaps, must now be centrally cleared, and a study linking post-trade transparency to lower CDS trading costs (pp. 30 and 32). The legacy principle's "RFQ dominance" is not what the survey's figure shows for U.S. investment-grade corporate bonds, of which 19 percent was electronically facilitated, mostly by RFQ, in the estimate of a July 2016 industry report (p. 2); RFQ here is one protocol among voice negotiation, runs and lists. Its claim that products trade by RFQ "because" they are less standardised and less active is not

adopted as a finding: the survey treats why fixed-income trading remains dealer-oriented and OTC as an open question (pp. 37–38). One corporate-bond sample does not describe every fixed-income market.

See also. bond transparency rules; bond transparency and trading costs; dealer inventory risk; ETF creation and redemption (corporate bond ETFs and dealer inventories); portfolio trading.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/otc-fixed-income-trading/`.

Index. fixed-income market structure; over-the-counter market; search market; Treasury market; corporate bond market; municipal bonds; structured products; TBA MBS; specified pools; dealer runs; indicative quotes; request for quote; RFQ; bid wanted in competition; interdealer broker; all-to-all trading; dealer capital commitment; prearranged trades; riskless principal; nonbank dealers; Volcker Rule; electronic bond trading

Status. Verification: partly checked 2026-10-15 (against `wide-bsv-fi-survey`, published JFQA version, pp. 1–5, 7–8, 11, 13–16, 29–33 and 37–39, text layer; for `wide-hendershott-madhavan` and for `wide-ohara-zhou-electronic`, the abstracts only, from the publishers' records via OpenAlex and RePEc, with the full texts unread). Stability: draft. Currency: U.S. market structure as surveyed to about 2018.

Rule 9.44 Bond transparency rules: U.S. TRACE reporting and dissemination, and EU post-trade deferrals (as of October 2026)

Statement. Transparency rules separate *reporting* to a regulator or its facility from *public dissemination*, and *post-trade* publication of completed trades from *pre-trade* publication of quotes or orders.

- *U.S. reporting.* Under FINRA Rule 6730 as displayed on 15 October 2026, each member that is a party to a transaction in a TRACE-eligible security must report it "as soon as practicable, but no later than within 15 minutes of the Time of Execution", with stated exceptions, some of which allow 60 minutes [`finra-rule-6730`, Rule 6730(a), text retrieved 2026-10-15].

- *U.S. dissemination.* Under Rule 6750, FINRA disseminates information on all transactions in TRACE-eligible securities, Rule 144A transactions included, immediately upon receipt, except that certain collateralised mortgage obligation trades are disseminated in aggregate weekly and monthly; individual trades in on-the-run nominal coupon Treasuries are disseminated at the end of the day; and other Treasuries, foreign sovereign debt, CMBS, CDOs, some CMO trades and certain affiliate, list-offering and takedown transactions are not disseminated, FINRA being allowed to publish aggregated statistics [`finra-rule-6750`, Rule 6750(a)–(d) and .01, text retrieved 2026-10-15].

- *EU post-trade deferrals.* ESMA's manual of 12 May 2026 gives worked examples of the bond regime that applies from 2 March 2026 under amended MiFIR Article 11 and RTS 2 Article 8a. For a trade in a group-1 (most liquid sovereign) bond on a trading venue, a medium-size trade in category 1 is published within 15 minutes, and a large one in category 3 has its price published within one trading day and its volume within one week; competent authorities may add supplementary deferrals for sovereign bonds, omitting volume for up to six months beyond the initial deferral, or publishing weekly aggregates [`esma-transparency-manual-2026`, secs. 4.2.8 and 4.2.8.2, pp. 262–272].

Conditions. Dates and status:

- *History, United States.* TRACE reporting of corporate bonds began in July 2002 within 75 minutes, with immediate public dissemination only for about 500 large investment-grade and 50 high-yield bonds; dissemination was widened in 2003–2005, and the reporting lag cut to 45 (October 2003), 30 (October 2004) and 15 minutes (July 2005), with all registered corporate bonds disseminated by January 2006. ABS and MBS reporting began in May 2011 without dissemination, which was phased in for TBAs (2012), specified pools (2013), Rule 144A bonds (2014), ABS (2015) and CMOs (2017). Treasury reporting began in July 2017, with no public dissemination at the time [`wide-bsv-fi-survey`, Table 2, p. 9]. Disseminated sizes were capped, for example "5MM+" for investment-grade and "1MM+" for high-yield corporate trades (p. 10, footnote 10); the current caps were not checked.

- *Changes and proposals, United States.* FINRA's filing SR-FINRA-2024-004 proposed reducing the 15-minute reporting limit to one minute, with exceptions for firms with de minimis activity and for manual trades; SR-FINRA-2025-008, by its title, maintains "the currently effective 15-minute outer limit" for the securities covered by the earlier filing, and the rulebook lists both as effective 8 June 2026 [`finra-sr-2024-004`, filing pages, retrieved 2026-10-15]. The approval orders were not read. SR-FINRA-2026-009, approved by the SEC on 24 July 2026, expands the indicator that suppresses from dissemination qualifying same-day, same-price, same-security transactions between a member and its *non-member* affiliate to transactions between *member* affiliates, renaming it the Affiliate–Principal Transaction indicator; FINRA said it would set an implementation date at least 12 months after approval and allow voluntary earlier use [`sec-34-105991`, pp. 47909–47911]. The rulebook lists it as effective 2 August 2027. Until then the presently effective exclusion is the non-member-affiliate one; the displayed text of Rule 6750(d)(1),

which already reads "Affiliate—principal transaction indicator", is the future wording. The proposal of the SEC's Fixed Income Market Structure Advisory Committee to delay dissemination of very large trades by 48 hours was a proposal, not a rule (survey, p. 29).

- *European Union.* Fixed-income securities traded without pre- or post-trade transparency in Europe until MiFID II applied in January 2018, which required reporting of bond transactions to a national regulator and real-time publication for liquid, non-block trades, and pre-trade publication of bids and offers for small trades in a limited set of about 200 sovereign bonds (survey, p. 30; historical). On 10 October 2025 ESMA stated that the Commission's amending RTS 1 and RTS 2 would apply most revised bond transparency requirements from 2 March 2026 following publication in the Official Journal, which had not yet happened, and invited market participants to anticipate them [esma-mifir-review-statement-2025, p. 4]. The amended MiFIR text (Regulation (EU) 2024/791) and the current bond pre-trade regime were not read: EUR-Lex returned an empty page to the request.

- *United Kingdom.* The U.K. regime is separate from the EU's and was not checked.

Practice. Interpretation: a TRACE report is a regulatory obligation of the dealer; what the public sees is set by Rule 6750, so the reported and the disseminated sets of U.S. trades differ, most visibly for Treasuries. In the EU the deferral category, not only the instrument, decides when price and volume become public. Trade-size caps and deferrals let dealers "hedge or sell a portion of [their] position before disclosure", the reason the survey gives for U.S. regulators' initial delays (p. 29).

Pitfalls. The legacy rule said TRACE "reports U.S. corporate bond trades after execution". TRACE is FINRA's reporting facility for all TRACE-eligible securities, of which corporate bonds are one class; reports go to FINRA, and public dissemination is a separate step with exceptions. "Improving post-trade transparency" is an evidence claim (bond transparency and trading costs). "Similar transparency pushes in Europe aim to make fixed-income liquidity easier to measure without forcing all products into a CLOB" was not checked against any EU statement of aims. The 2002–2017 history is from a survey, not from the rule filings themselves.

See also. fixed-income market structure; bond transparency and trading costs; best execution.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/otc-fixed-income-trading/`.

Index. TRACE; FINRA Rule 6730; FINRA Rule 6750; trade reporting; public dissemination; reporting timeframe; dissemination cap; Treasury TRACE reporting; post-trade transparency; pre-trade transparency; MiFID II; MiFIR; RTS 2; deferred publication; supplementary deferral; sovereign bonds; ESMA

Status. Verification: partly checked 2026-10-15 (FINRA Rules 6730 and 6750 as displayed on finra.org; the FINRA filing pages for SR-FINRA-2024-004 and SR-FINRA-2025-008, titles and summaries only; the SEC approval order for SR-FINRA-2026-009, rechecked in a follow-up the same day; the ESMA public statement of 10 October 2025, p. 4; the ESMA manual of 12 May 2026, pp. 24–29 and 258–272; `wide-bsv-fi-survey`, pp. 8–10 and 29–30). Stability: draft. Currency: U.S. rules as displayed on 15 October 2026; EU bond deferrals as described for application from 2 March 2026; EU pre-trade and U.K. rules unchecked.

Evidence 9.45 Bond transparency and trading costs: U.S. post-trade transparency evidence, the open question of large trades, and what data allow clients to measure

Statement. Edwards, Harris and Piwowar estimate average transaction costs by trade size for every U.S. corporate bond that traded more than nine times between January 2003 and January 2005, from a complete record of OTC secondary trades. Costs fall significantly with trade size; highly rated, recently issued and near-maturity bonds cost less; costs are lower for bonds with transparent trade prices and drop when TRACE starts to disseminate their prices; and the results "suggest that public traders benefit significantly from price transparency" [wide-edwards-harris-piwowar, abstract]. In the survey's account, their estimates fall from 75 basis points for $5,000 trades to under 10 basis points for $1 million or more, the reverse of the equity pattern [wide-bsv-fi-survey, p. 5]. Three studies with different designs, a stratified random sample of BBB bonds phased in in April 2003 (Goldstein, Hotchkiss and Sirri), the disseminated-against-undisseminated comparison of Edwards, Harris and Piwowar, and a difference-in-differences on insurance-company trades (Bessembinder, Maxwell and Venkataraman), all conclude that public reporting through TRACE is associated with significant reductions in customer execution costs; the last also suggests a liquidity externality to bonds not yet reported. Similar associations are reported for CDS, TBA MBS and Rule 144A bonds (survey, p. 30).

Conditions. These are associations in U.S. corporate bond, CDS, TBA and Rule 144A samples from the phase-in periods, as summarised by a survey; only Edwards, Harris and Piwowar's abstract was read among the primary studies. The survey states the limits: much of the evidence concerns execution costs of completed trades, which may reflect order splitting and do not capture the cost and difficulty of executing large orders; the optimal degree of post-trade transparency "has not been established"; dealers, and in some cases large buy-side investors, are often critical of greater transparency; and the regime optimal for one fixed-income market may not be optimal for another (pp. 38–39). Back, Liu and Teguia's model, as the survey reports it, has trade

disclosure conveying the dealer's information and reducing its rents when it disposes of inventory (p. 29), a model prediction. Pre-trade transparency in bonds is a matter of open questions and proposals in the survey, such as the public display of customer interest advocated by Harris (2015), not of evidence (p. 37).

Practice. What data let clients measure, as of the survey: TRACE and MSRB transaction data; an enhanced TRACE file for researchers with masked dealer identities and actual sizes; aggregators that combine trades, dealer runs and platform quotes for buy-side institutions (pp. 7–8 and 10). Fixed-income markets have no analogue of SEC Rules 605 and 606 on execution quality and order routing, and without consolidated quotations many participants cannot see the best available prices or identify trade-throughs; FINRA and the MSRB adopted best-execution rules (pp. 31–32). Since 14 May 2018 FINRA Rule 2232(c) requires a confirmation to show the mark-up or mark-down when a member trades as principal in a corporate or agency debt security with a non-institutional customer and makes offsetting trades of at least that size on the same trading day, looking through non-arm's-length affiliate trades; (d) exempts functionally separate trading desks and same-day fixed-price-offering sales [`finra-rule-2232`, paras. (c)–(f), text retrieved 2026-10-15]; [`finra-rn-17-08`, effective date and MSRB paragraph]. FINRA's notice reports a parallel MSRB rule for municipal securities, with the same effective date; MSRB's text was not read. Measure costs by trade size, rating and segment, and state which trades are visible: completed trades only, capped sizes, deferred or undisseminated classes.

Pitfalls. The legacy caveat said publishing pre-trade interest in illiquid bonds "can reveal inventory needs and increase dealer risk", so policy "must balance" monitoring against exposing liquidity. The sources read give this as dealers' argument and an open question, and the survey's reason for U.S. reporting delays; no source read measures the effect of pre-trade transparency on bond liquidity, and the claim is not adopted as a finding. The legacy note's "electronification changes execution quality and governance of trading desks" is not established causally: O'Hara and Zhou report effects of electronic RFQ trading in their abstract, and nothing read concerns desk governance. Transparency is not shown here to help or to harm liquidity in general.

See also. bond transparency rules; fixed-income market structure; spread measures; best execution; transaction cost analysis.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/otc-fixed-income-trading/`.

Index. bond transaction costs; TRACE dissemination; post-trade transparency; corporate bond trading costs by trade size; liquidity externality; large-trade dissemination; pre-trade transparency in bonds; enhanced TRACE; markup disclosure; bond best execution; consolidated quotations; trade-through

Status. Verification: partly checked 2026-10-15 (`wide-bsv-fi-survey`, published JFQA version, pp. 5, 7–10, 29–32 and 37–39, text layer; for `wide-edwards-harris-piwowar` and `wide-ohara-zhou-electronic`, the abstracts only, from the publishers' records via OpenAlex and RePEc, with the full texts unread; FINRA Rule 2232 and Regulatory Notice 17-08, in a follow-up the same day). Stability: draft. Currency: U.S. evidence from 2002–2017 samples; Rule 2232 as displayed on 15 October 2026; other regulatory facts as of the survey (2019).

Chapter 10

Commodity futures: storage, curves, hedging and options

Pathways drawn on: `commodity-hedging` (two draft entries from the migration: futures hedges, basis and basis risk, and the minimum-variance hedge ratio, from Hull's 11th Global edition; none of the pathway's readings, from Working and Ederington to the evidence on why firms hedge, has been read). The entries came from a legacy file of Hull notes mapped to the rates overview and to this pathway.

10.1 Hedging with futures

Definition 10.1 Futures hedges and basis: short and long hedges, the basis, basis risk, cross hedging, contract choice and stack and roll

Statement. Hull treats *hedge-and-forget* strategies: a futures position is taken at the start and closed at the end, with no adjustment in between [s7-hulloptions, chap. 3, "Hedging Strategies Using Futures", p. 70]. A *short hedge*, a short futures position, suits a hedger who owns or will own an asset and expects to sell it; a *long hedge* suits one who will buy an asset and wants to lock in a price now (sec. 3.1, pp. 71–72). The *basis* is "spot price of asset to be hedged − futures price of contract used"; an increase is a strengthening, a decrease a weakening (sec. 3.3, p. 76). With spot S_i, futures F_i and basis $b_i = S_i - F_i$ at the start t_1 and end t_2, the effective price obtained or paid with a hedge is

$$S_2 + F_1 - F_2 = F_1 + b_2$$

for both short and long hedges (p. 77); F_1 is known at t_1, and the uncertainty in b_2 is *basis risk*. *Cross hedging* uses a futures contract on a different asset (p. 77; sec. 3.4, p. 79); with S_2^* the price at t_2 of the asset underlying the futures, Hull writes the hedged price $S_2 + F_1 - F_2$ as

$$F_1 + (S_2^* - F_2) + (S_2 - S_2^*),$$

whose two last terms are "the two components of the basis": $S_2^* - F_2$ is the basis that would exist if the hedged asset were the futures' underlying (the futures-underlying basis), and $S_2 - S_2^*$ is "the basis arising from the difference between the two assets" (p. 77). Both expressions assume a hedge ratio of 1.0, one futures unit per unit of exposure, which Hull calls natural when the two assets are the same and "not always optimal" in a cross hedge (p. 79). Interpretation, derived here: with a hedge ratio h, the size of the futures position per unit of exposure taken in the direction that offsets it (short for a seller, long for a buyer), the effective price received or paid per unit is

$$S_2 + h(F_1 - F_2),$$

which reduces to $F_1 + b_2$ at $h = 1$; $h = 0$ is no hedge, and a negative h is a futures position in the same direction as the exposure (long futures against a long exposure). Whether that adds or removes risk depends on how the two prices co-move, not on the sign of h: when $\mathrm{Cov}(\Delta S, \Delta F) < 0$ the minimum-variance ratio of the next entry is negative and reduces variance. Prices are per unit of each asset, and h is in futures units per unit of exposure, so it carries any conversion between the two assets' units (none is needed in Hull's jet-fuel and heating-oil example, both in gallons).

Conditions. Hedges are imperfect because the hedged asset may differ from the futures' asset, the date of purchase or sale may be uncertain, and the futures may have to be closed before its delivery month (pp. 75–76). Basis changes can help or hurt: an unexpected strengthening improves a short hedger's position and worsens a long hedger's (p. 77). Hull defines the basis as spot minus futures but notes the reverse definition is sometimes

used, especially for financial futures (p. 76, n. 2); the sign convention must be stated. As a rule of thumb, Hull chooses the delivery month closest to, but after, the hedge's end, because basis risk grows with the gap and prices can be erratic in the delivery month; the rule assumes enough liquidity (p. 78). When the hedge outlasts all liquid contracts, *stack and roll* closes one contract and opens a later one, repeatedly; Metallgesellschaft's rolled hedges in the early 1990s produced cash outflows before the offsetting gains on its supply contracts, a liquidity problem (sec. 3.6, pp. 89–90).

Practice. Checked with Hull's examples: an oil producer shorting August futures at \$49 against a spot of \$50 realises about \$49 a barrel whether spot ends at \$45 (a \$4 futures gain) or \$55 (a \$6 loss) (p. 71); $S_1, F_1, S_2, F_2 = 2.50, 2.20, 2.00, 1.90$ give $b_1 = 0.30$, $b_2 = 0.10$ and an effective price of 2.30 (p. 77); the yen hedge gives $1.0800 + (-0.0050) = 1.0750$ cents and \$537,500 for 50 million yen (Example 3.1, pp. 78–79); the oil purchase gives $48.00 + 0.90 = 48.90$ and \$978,000 (Example 3.2, p. 79); the rolled oil hedge earns $0.80 + 0.50 + 0.40 = 1.70$ a barrel against a \$3 fall in spot (p. 90).

Pitfalls. The legacy table added a long-hedge column beside Hull's short-hedge oil example, with a loss of 4 and a gain of 6; Hull's oil example is a short hedge only, and his long-hedge example is copper (pp. 71–72). The mirror-image column is arithmetically consistent (a buyer paying the August spot and losing or gaining on a long hedge also ends near \$49) but is not Hull's. The legacy's three causes of imperfection are Hull's (pp. 75–76). Whether hedging adds value to a firm, and time-varying hedge ratios, are outside these passages (backlog).

See also. minimum-variance hedge ratio; derivative contracts (convergence); cost-of-carry forward price (why futures and spot differ before delivery); arbitrage (Harris's basis).

Further reading. The pathway's readings on why firms hedge and on dynamic hedge ratios are named in its backlog. The legacy file is kept verbatim in `notes/artefacts/volatility-options/`.

Index. hedge-and-forget; short hedge; long hedge; basis; strengthening of the basis; weakening of the basis; basis risk; effective price; cross hedging; choice of delivery month; stack and roll; rolling hedges; Metallgesellschaft

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 3, pp. 70–79 and 89–90, printed page = PDF page −1; five examples recomputed; the basis components and hedge-ratio wording rechecked 2026-10-16, pp. 77–79; the general-h expression derived here; the sign convention for negative h restated 2026-10-16). Stability: draft. Currency: concepts timeless; contract examples as of 2021.

Model 10.2 Minimum-variance hedge ratio: derivation, hedge effectiveness, number of contracts, daily settlement and tailing, and beta hedges with index futures

Statement. The *hedge ratio* is the size of the futures position over the size of the exposure [s7-hulloptions, chap. 3, "Hedging Strategies Using Futures", sec. 3.4, p. 79]. Ignoring daily settlement, with ΔS and ΔF the changes in spot and futures prices over the life of the hedge and $\Delta S = a + b\,\Delta F + \epsilon$ approximately linear, a hedger long the asset and short h futures per unit holds $\Delta S - h\Delta F = a + (b-h)\Delta F + \epsilon$, whose variance is minimised at $h = b$ (p. 80):

$$h^* = \rho \frac{\sigma_S}{\sigma_F}, \qquad N^* = \frac{h^* Q_A}{Q_F},$$

with σ_S, σ_F the standard deviations of $\Delta S, \Delta F$, ρ their correlation, Q_A the units hedged and Q_F the units per contract (eqs. (3.1)–(3.2), pp. 80–81). Hedge effectiveness, the share of variance removed, is the regression R^2, equal to ρ^2 (p. 81).

Notation. $\Delta S, \Delta F$ price changes per unit, in the same price units; h dimensionless; N^* contracts. Interpretation, derived here: for a hedged position $V(h) = \Delta S - h\Delta F$, $\operatorname{Var} V(h) = \sigma_S^2 - 2h\rho\sigma_S\sigma_F + h^2\sigma_F^2$, a convex quadratic minimised where $d \operatorname{Var} V(h)/dh = 0$, at $h^* = \rho\sigma_S/\sigma_F = \operatorname{Cov}(\Delta S, \Delta F)/\operatorname{Var}\Delta F$, the OLS slope; the minimum is $\sigma_S^2(1-\rho^2)$. A short futures position is $h > 0$ against a long exposure; a hedger short the asset buys h^*. The sign of h^* is the sign of ρ: with $\operatorname{Cov}(\Delta S, \Delta F) < 0$, $h^* < 0$ is a long futures position against a long exposure, and it still reduces the variance to $\sigma_S^2(1-\rho^2)$. Checked: with $\sigma_S = \sigma_F = 1$ and $\rho = -0.6$, $h^* = -0.6$ leaves a variance of 0.64 against 1 unhedged, while $h = +0.6$ raises it to 2.08 (`negative-hedge-ratio-check.py` in `notes/artefacts/volatility-options/`, analytic and simulated).

Conditions. The estimate assumes the future resembles the past; ρ, σ_S, σ_F are usually estimated from equal non-overlapping intervals, ideally as long as the hedge (p. 81). The $R^2 = \rho^2$ result is in-sample: it describes the estimation period and assumes the linear relation and its parameters hold over the hedge. *Daily settlement.* Futures make a series of one-day hedges; with $\hat\sigma_S, \hat\sigma_F, \hat\rho$ for percentage one-day changes, the hedge from a regression of percentage changes is $\hat h = \hat\rho\hat\sigma_S/\hat\sigma_F$ and $N^* = \hat h V_A/V_F$, with $V_A = SQ_A$ and $V_F = FQ_F$; equivalently the one-day price-change hedge is $\hat\rho\hat\sigma_S S/(\hat\sigma_F F)$ (eq. (3.3), p. 83). Price-change and return regressions therefore differ by the ratio S/F and must not be mixed. *Tailing*: allowing for interest on daily settlement, Hull divides N^* by 1.05 for a hedge with one year left at 5 percent (p. 83); interpretation: the

general factor is the growth of \$1 over the remaining life at the relevant rate, which equals $1 + 0.01r$ only for one year at simple rate r percent.

Practice. *Checked*, recomputing Hull's Example 3.3 from his Table 3.2 data: $\sigma_F = 0.0313$, $\sigma_S = 0.0263$, $\rho = 0.928$, $h^* = 0.778$ and $N^* = 0.778 \times 2{,}000{,}000/42{,}000 = 37.0$, so 37 contracts (p. 82). The same h^* results from minimising the sample variance of $\Delta S - h\Delta F$ numerically and equals the OLS slope; the variance falls to 13.8 percent of unhedged, so 86.2 percent is removed, equal to ρ^2; a ratio of 1 leaves 20.9 percent (Figure 10.1). Hull's one-day example gives $0.8 \times 2{,}200{,}000/54{,}600 = 32.2$ contracts (p. 83). *Index futures*: to hedge a portfolio of value V_A and beta β, short $N^* = \beta V_A/V_F$ contracts, assuming the futures matures close to the hedge; to move beta from β to β^*, short $(\beta - \beta^*)V_A/V_F$ if $\beta > \beta^*$, otherwise go long $(\beta^* - \beta)V_A/V_F$ (eq. (3.5), pp. 86 and 88). Hull's example: $1.5 \times 5{,}050{,}000/252{,}500 = 30$ contracts, gaining $30 \times (1{,}010 - 902) \times 250 = \$810{,}000$ if the index falls to 900; the hedged position grows at about the risk-free rate (Table 3.4, pp. 86–87).

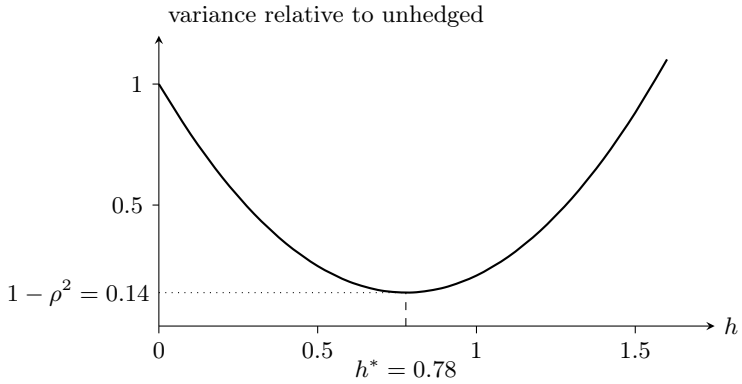


Figure 10.1: Variance of $\Delta S - h\Delta F$ relative to the unhedged variance for Hull's jet-fuel and heating-oil data (Table 3.2, p. 82), computed for this entry; the minimum is at h^*, as in Hull's schematic Figure 3.3 (p. 81).

Pitfalls. The legacy tailing rule "$N^*/(1 + 0.01r)$" holds only for one year remaining, as in Hull's example. The legacy "standard deviation is minimised by setting $h = b$" needs ϵ uncorrelated with ΔF, which OLS guarantees in sample. Hull's index-hedge table assumes a predictable dividend yield, a constant risk-free rate and perfect correlation between portfolio and index returns, so "in practice ... the hedge works rather less well" (p. 87, n. 5); CAPM supplies an expected return there, not a guarantee. Hull's reasons for index hedging (confidence in stock selection, short-term protection without selling) are motivations, not evidence that such hedges add value (pp. 87–88).

See also. futures hedges and basis; factor risk model (beta as a factor exposure); duration and convexity.

Further reading. Time-varying and GARCH hedge ratios, and the evidence on hedging's value, are the pathway's readings (backlog). The legacy file and its original figure are kept verbatim in `notes/artefacts/volatility-options/`.

Index. hedge ratio; minimum-variance hedge ratio; optimal hedge ratio; hedge effectiveness; R-squared; correlation; regression slope; number of contracts; daily settlement; percentage changes; tailing the hedge; stock index futures; beta; changing beta; cross hedge

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `s7-hulloptions`, 11th Global edition, chap. 3, pp. 79–90; Table 3.2 recomputed, the variance minimised numerically and the curve plotted; the derivation is the entry's own; a negative-covariance case computed 2026-10-16). Stability: draft. Currency: timeless; data as of Hull's example.

Chapter 11

Crypto settlement, venues and derivatives

Pathways drawn on: `crypto-perpetuals` (draft entries from the migration: crypto venue and instrument structure, perpetual funding with one venue's rules, and the layers of crypto trading risk, from Makarov and Schoar's 2018 discussion paper, Ackerer, Hugonnier and Jermann's and He, Manela, Ross and von Wachter's preprints, and Deribit's rulebook and funding specification as retrieved on 15 October 2026; none of the pathway's readings yet studied). The entries came from a legacy execution file whose map named `crypto-mm-microstructure`; its blocks concern cross-venue structure, funding and margin rather than quoting one order book, so they were routed here.

11.1 Crypto venues, perpetuals and trading risk

Definition 11.1 Crypto venue and instrument structure: fragmented executable prices, reference prices and best execution, and spot, dated futures, perpetuals (linear, inverse, quanto), options and pools

Statement. Crypto assets trade on many independently owned venues. Makarov and Schoar, using tick data from 34 exchanges in 19 countries, find that individual exchanges mostly run centralised order books like equity markets, but that, unlike regulated equity markets, the crypto market "lacks any provisions to ensure that investors receive the best price", contrasting the U.S. NBBO requirement [`s6-makarov`, discussion-paper version of December 2018, pp. 1–2 and footnote 1]. Kept apart here:

- *executable prices*: the bids and offers on each venue, in its own quote currency, custody and fee terms;
- *reference prices*: indices a venue or benchmark administrator computes from several venues, against which derivatives are marked or settled (Deribit marks positions with a Mark Price and computes funding against a Deribit Index, perpetual funding);
- *fair-value models*: a trader's own estimate of value, which need not equal either;
- *legal best-execution obligations*: duties under a jurisdiction's rules (best execution), which Makarov and Schoar found absent across crypto venues as a market-wide price-protection rule (p. 1).

Instruments. *Spot* trades exchange one asset for another; on Deribit spot trades are brokerage services settled and delivered immediately [`gapclose-deribit-rules`, Rulebook v2.0, Rules 1.2(a) and 11.3(a)]. *Dated futures* expire, and convergence to spot at expiry limits the basis [`roadmap-perpetual-pricing`, arXiv v2, p. 2]. *Perpetual futures* have no expiry and are anchored by periodic funding payments (perpetual funding). Ackerer, Hugonnier and Jermann distinguish three contract types (pp. 2–3):

- *linear*: contract size in units of the base asset (1 BTC), margined and settled in the quote currency (USD, or in practice a stablecoin such as USDT, p. 7);
- *inverse*: the base asset itself (BTC) is the margin and settlement asset, and the contract size is in the quote currency (for example 10,000 USD);
- *quanto*: a third currency (for example ETH) is used for margining and settlement.

Options: Deribit lists European-style cash-settled options and linear options; its European options are priced in BTC or ETH and cannot be exercised before expiry (Rulebook, Appendix A, 2(d)). *On-chain pools* (automated market makers) were not checked in this slice.

Conditions. The fragmentation evidence is historical. Over December 2017 to February 2018 the daily average price ratio between the United States and Korea exceeded 15 percent and reached 40 percent on several

days; Japan–U.S. differences were about 10 percent and U.S.–Europe about 3 percent, while differences within a country typically did not exceed one percent on average (pp. 1–2). The authors estimate at least $2 billion of potential arbitrage profit over that period from executed trades (p. 2, footnote 2). Spreads co-moved across countries and opened during rapid appreciation; between cryptocurrencies on the same exchanges spreads were an order of magnitude smaller, and pairwise capital-control tightness was positively related to the correlation of spreads, which the authors read as evidence that capital controls contribute (pp. 2–3). These are 2017–2018 findings; current spreads were not measured.

Practice. Interpretation: a crypto price is a price on a venue, in a quote currency, with that venue's custody, collateral and fees. Quote-currency differences (USD, USDT, USDC, KRW) and settlement currency (linear against inverse) change the units of P&L and margin. Cross-references: fragmentation without a consolidated tape in spot FX is in spot FX market structure, routing across venues in smart order routing, limits to arbitrage in arbitrage risks.

Pitfalls. The legacy principle said "there is no universal consolidated tape, so fair value and best execution are venue-dependent". The sources read support the absence of a market-wide best-price rule in 2018 (Makarov and Schoar) but say nothing on consolidated tapes; "fair value is venue-dependent" conflates venue prices with fair value. Its taxonomy's "perpetual futures: leveraged swaps kept near spot through funding" overstates the anchoring (perpetual funding); "options: volatility exposure concentrated in fewer venues and strikes" and "on-chain pools: automated market makers with inventory curves and gas costs" were not checked and are stubbed for `crypto-options-numeraire` and `amm-lending`. Deribit's terms are one venue's.

See also. perpetual funding; crypto trading risk layers; best execution; spot FX market structure.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/crypto-mm-microstructure/`.

Index. crypto market fragmentation; cross-exchange arbitrage; Kimchi premium; capital controls; reference price; index price; mark price; best execution in crypto; spot crypto; dated crypto futures; perpetual futures; linear perpetual; inverse perpetual; quanto perpetual; crypto options; automated market maker; quote currency; settlement currency

Status. Verification: partly checked 2026-10-15 (`s6-makarov`, LSE FMG Discussion Paper 782 of December 2018, abstract and pp. 1–3, text layer; `roadmap-perpetual-pricing`, arXiv v2 of 3 September 2024, pp. 2–3 and 7; `gapclose-deribit-rules`, Rulebook version 2.0 applicable 13 August 2026, retrieved 15 October 2026, Rules 1.2 and 11.3 and Appendix A). Stability: draft. Currency: fragmentation evidence 2017–2018; Deribit terms as of October 2026.

Mechanism 11.2 Perpetual funding: premium and interest terms, sign convention, accrual and settlement, damper and caps (Deribit as one venue), and why convergence is not guaranteed

Statement. A perpetual future has no expiry, so it cannot rely on convergence at maturity; instead, longs and shorts exchange periodic *funding* payments. Ackerer, Hugonnier and Jermann describe the payment as generally the sum of a *premium* part $\kappa_t(f_t - x_t)$, paid by longs when the perpetual price f_t is above spot x_t, and an *interest* part $\iota_t x_t$ set by the exchange to reflect the interest-rate differential between the two currencies; for a linear contract the long's cash flow over one period is

$$(f_{t+1} - f_t) - \kappa_t(f_t - x_t) - \iota_t x_t$$

in the quote currency [`roadmap-perpetual-pricing`, arXiv v2, sec. 3, pp. 7–8, eqs. (4)–(6)]. Funding is commonly computed every 8 hours (p. 2). In their frictionless no-arbitrage model the exchange can choose the interest factor so that the perpetual price equals spot for any sufficiently large premium rate (p. 4; Corollary 1, p. 12). That is a model result. He, Manela, Ross and von Wachter state that perpetuals "are not guaranteed to converge to the spot price", derive bounds with trading costs, and find empirical deviations larger than in traditional currency markets, co-moving across currencies and diminishing over time [`roadmap-perpetual-fundamentals`, arXiv v7 of 17 September 2026, abstract].

Conditions. *One venue's rule: Deribit, Funding Specifications page edited 27 July 2026, retrieved 15 October 2026* [`gapclose-deribit-rules`, Funding Specifications]:

- *sign*: when the funding rate is positive, longs pay shorts; when negative, shorts pay longs; Deribit charges no fee on funding, so payments net to zero between holders;
- *rate*: premium rate P = (Mark Price − Deribit Index)/Deribit Index; a damper sets funding to zero for $|P| \le 0.025\%$ and otherwise moves P towards zero by 0.025%; a cap then bounds the rate, ±0.5% for BTC, ±1.0% for ETH and ±5.0% for the USDC- and USDT-margined perpetuals:

$$\text{rate} = \min\big(c,\ \max\big(-c,\ \max(0.025\%, P) + \min(-0.025\%, P)\big)\big);$$

- *units and cadence*: the rate is expressed per 8 hours but accrues continuously; the payment is rate × position size × (period / 8 hours), in the settlement currency of the contract: BTC in Deribit's BTC

example, where a USD 100,000 position is 1 BTC, an inverse contract in Ackerer, Hugonnier and Jermann's sense (Interpretation), and USDC for a USDC-margined linear one;

- *settlement*: accrued funding shows at once in account equity (realised session P&L) and can be used for trading, but moves to the cash balance only at the daily settlement at 08:00 UTC, after which it can be withdrawn (also Rulebook v2.0, Rule 11.4).

Practice. Worked example (Deribit's, recomputed by the script `funding_checks.py` beside the artefact): BTC perpetual mark USD 100,075 against index USD 100,000 gives $P = 0.075\%$ and a rate of 0.05% per 8 hours; a long of USD 100,000 (1 BTC) pays $0.05\% \times 1 \text{ BTC} \times 1/480 = 0.000001041667$ BTC per minute and 0.0005 BTC over 8 hours. A USDC-margined SOL perpetual at mark 100.10 against index 100 gives 0.075%, so a USDC 100,000 long pays 0.15625 USDC per minute and 75 USDC over 8 hours. The script also checks the printed max/min formula against the prose rule on a grid of premia and the other examples (damper, caps, offsetting minutes). Interpretation: funding is a carry cost or income of holding the perpetual against spot, in the contract's settlement currency; whether it signals crowding was not checked.

Pitfalls. The legacy definition said funding is "the periodic payment that links perpetual futures to spot", with longs paying when it is positive, and that "it is both a carry term and a crowding signal". The sign convention matches Deribit's; "links ... to spot" holds as a design intent and a model result, not a guarantee; "crowding signal" is not supported by the sources read and is stubbed (`ev:crypto-funding-crowding`). Venue rules differ: other venues' cadence, interest term, clamps and index construction were not checked, and Deribit's own page contains slips in its ETH examples (an index "at USD 10,000" and "a 1 BTC long position" in the ETH column of Example 3; 7.5% inside the ETH cap formula of Example 5), whose stated results are nevertheless correct.

See also. crypto venue and instrument structure; crypto trading risk layers; futures margin and settlement; lending fees and rebates (financing legs).

Further reading. The legacy block it replaces, verbatim, is in `derivations.tex` in the folder `notes/artefacts/crypto-mm-microstructure/`.

Index. perpetual futures; funding rate; funding payment; premium rate; interest rate differential; anchoring; mark price; Deribit index; funding damper; funding cap; 8-hour funding rate; continuous funding accrual; daily settlement; linear perpetual funding; inverse perpetual funding; perpetual basis

Status. Verification: partly checked 2026-10-15 (`roadmap-perpetual-pricing`, arXiv v2 of 3 September 2024, pp. 2, 4, 7–8 and 12, text layer; `roadmap-perpetual-fundamentals`, arXiv v7 of 17 September 2026, abstract; `gapclose-deribit-rules`, Funding Specifications edited 27 July 2026 and Rulebook v2.0 Rule 11.4, retrieved 15 October 2026 through the help-centre API; formula and examples recomputed by script). Stability: draft. Currency: Deribit terms as of October 2026; model results timeless.

Definition 11.3 Crypto trading risk layers: liquidation, insurance fund, socialised loss and auto-deleveraging at one venue, kept apart from custody, stablecoin and network-settlement risk

Statement. The losses a crypto trader can bear beyond market risk come in layers that are governed by different documents and fail in different ways:

- *liquidation*: forced closing of positions when margin falls short. Under Deribit's rulebook, if a member fails to meet maintenance margin, Deribit may close positions, cancel orders, delta-hedge with futures and perpetuals, apply auto-deleveraging where applicable, or close the whole account, together the Liquidation Process, and may charge a liquidation fee [`gapclose-deribit-rules`, Rulebook v2.0, Rules 10.1 and 10.5];

- *insurance fund*: if an account's margin balance in a currency is negative after liquidation, Deribit transfers assets from its insurance fund, funded mainly by liquidation fees and excess collateral from liquidations, to bring it to zero; if the fund in that currency is depleted, Deribit may, in its absolute discretion, convert another currency of the fund (Rules 10.6–10.7; definition of Insurance Fund);

- *socialised loss*: Deribit may, in its absolute discretion, trigger the Socialised Loss Mechanism if the insurance fund is depleted in all currencies, or if it is depleted in the bankrupt account's currency and Deribit decides not to convert other currencies (Rule 10.8), so the mechanism can be triggered while other currencies remain in the fund, and need not be triggered when the fund is exhausted. Once it is, Deribit may deduct from all members with positive profit and loss in the same day's session, in any currency, their pro-rata share of the negative balance, halt trading, cancel orders and close positions at the mark price; members agree to this and may dispute only a manifest error in their share (Rules 10.9–10.11);

- *auto-deleveraging*: a step in the liquidation waterfall in which, once liquidity-support providers' capacity is used, eligible positions of a distressed account are transferred to a deleveraged counterparty at a preferential price and closed against that member's existing positions (definition of Auto-Deleveraging; the full waterfall is in the Exchange Membership Terms, clause 12, not read);

- *custody*: Deribit holds members' assets in its own wallets or through a third-party custodian of its choice, or acts as limited agent over a member-chosen custodian for margining, liquidation and settlement (Rule 2, guidance);

- *stablecoin, network and bridge risk*: redemption and depeg of stablecoin collateral, chain congestion, reorganisations and transfer delays. Not checked in this slice (stub).

Conditions. All rules above are Deribit FZE's, a virtual-asset service provider regulated by Dubai's VARA, under Rulebook version 2.0 applicable 13 August 2026 (Rule 1.1); other venues' waterfalls were not checked and may differ. Venue insolvency is a separate legal process from a venue's socialised-loss mechanism; neither insolvency outcomes nor customer-asset treatment in insolvency were checked.

Practice. Interpretation: keep apart a trader's own liquidation (its positions closed), a loss to the venue's backstop (insurance fund), a loss charged to other members (socialised loss), a position closed against a counterparty's will (auto-deleveraging) and a loss of assets held at the venue (custody, insolvency). Cross-venue strategies add transfer and settlement paths between venues, during which capital cannot respond. Margin and default waterfalls in a regulated futures clearing house are in futures margin and settlement.

Pitfalls. The legacy taxonomy listed venue risk ("outage, withdrawal halt, auto-deleveraging, or insolvency"), liquidation risk, stablecoin risk, bridge and settlement risk, and fragmentation ("basis and price gaps across venues when capital cannot move quickly"). Auto-deleveraging and liquidation are checked only for Deribit; outages and withdrawal halts were not checked; fragmentation is a market-structure fact (crypto venue and instrument structure), linked by Makarov and Schoar to capital controls rather than to slow crypto transfers. Stablecoin and bridge risks are stubbed for their own pathways.

See also. perpetual funding; crypto venue and instrument structure; futures margin and settlement.

Further reading. The legacy blocks it replaces, verbatim, are in `derivations.tex` in the folder `notes/artefacts/crypto-mm-microstructure/`.

Index. crypto liquidation; maintenance margin; liquidation fee; insurance fund; bankrupt account; socialised loss; auto-deleveraging; ADL; liquidation waterfall; crypto custody; third-party custodian; stablecoin depeg; bridge risk; network settlement; venue insolvency; Deribit; VARA

Status. Verification: partly checked 2026-10-15 (`gapclose-deribit-rules`, Rulebook version 2.0 applicable 13 August 2026, page updated 15 September 2026, retrieved 15 October 2026 through the help-centre API: Rules 1.1, 2 (guidance on custody), 10.1–10.11 and the definitions of Auto-Deleveraging and Insurance Fund; the Exchange Membership Terms not read; Rules 10.6–10.11 rechecked against the same version on 15 October 2026, correcting the socialised-loss trigger and its discretion). Stability: draft. Currency: Deribit as of October 2026.

Notes awaiting migration

Chapter 12

Multi-asset and systematic macro

Global macro is the strategy of identifying economic regime shifts early and expressing them through the most liquid instruments. The primary objective is surviving and profiting from crises, rather than maximising returns in stable periods [s7-gliner].

12.1 Regime and World State

12.1.1 Survey of Global Macro Landscape

Definition 12.1.1. *Global Macro*
Global macro is a top-down investment strategy that trades economic regimes, not securities.
The core idea is simple but powerful: macro forces dominate asset prices when they shift, and those forces operate across countries, asset classes, and policy domains. Global macro traders

i. Form views on growth, inflation, monetary policy, and geopolitics
ii. Express those views via equities, rates, FX, and commodities
iii. Are unconstrained by geography, asset class, or direction (long/short)

Global macro is often considered the most flexible and opportunistic hedge fund strategy, due to the scope of traded products and the number of markets it covers. Its aim is to preserve capital, using stringent risk management to limit drawdowns.

Remark 12.1.2. *Existence of Global Macro as a Strategy*
Macro exists because economic regimes are unstable, but markets price them slowly and asymmetrically.
Three structural reasons:

i. Policy is discrete, markets are continuous: Central banks and governments move in jumps (rate decisions, QE, capital controls). Markets must reprice entire distributions after the fact.
ii. Capital is sticky: Large allocators (pensions, insurers, sovereign funds) cannot reposition quickly. Macro traders exploit this inertia.
iii. Cross-asset linkages are under-arbitraged: Many investors specialise in a single asset class. Macro traders operate between silos.

Hence macro is a strategy of structural inefficiency, not informational advantage.

Remark 12.1.3. *Types of Global Macro Strategies*
The four broad approaches are:

i. Discretionary Macro
Relies on trader experience, intelligence, and knowledge to take subjective bets on global markets. Requires serious organization and data processing skills. Uses top-down analysis of risks and opportunities across industries, sectors, countries, and the macroeconomic situation.
Discretionary traders can execute *directional trades* (betting on asset direction) or *relative value trades* (pairing assets to capture value differentials). Strong in crises and regime breaks where human judgment outperforms models.
ii. Systematic Macro
Model-driven, employing quantitative factors to produce trading positions that remove human emotion. Prides itself on stringent process, strong back-tests, and ability to operate solely on quantitative analysis.

Over long periods, systematic funds produce more consistent returns than discretionary strategies; however, in periods of high volatility, they tend to underperform (e.g., 2008).
The ability to trade multiple liquid asset classes allows these funds to scale to very large AUM.

iii. High-Frequency Macro
Uses sophisticated technology to trade very short-term (millisecond to hours) dislocations around macro events. Processing speed is paramount. Limited scalability compared to discretionary and systematic.

iv. CTAs (Commodity Trading Advisors)
Primarily futures-based, momentum-driven trend-following strategies. Position sizing methodology originated with the Turtle Traders. Perform well over longer periods but subject to large drawdowns.

It is wise to allocate to both discretionary and systematic macro in a balanced manner. Discretionary is negatively correlated during stress, while systematic provides predictable allocation with back-tested drawdowns.

Remark 12.1.4. *Trade Types in Macro*
Discretionary macro traders express views through two primary trade structures:

i. Directional Trades: Betting on an asset moving in a particular direction. Example: going long copper on a bullish commodities view.

ii. Relative Value Trades: Pairing or grouping assets to capture the relative value differential between them. Example: during European crisis, short Italian 5-year BTPs vs. long German Bobls, betting Italy will see yields rise relative to Germany.

iii. Thematic Trades: Longer-term positions based on structural macro views.
Examples: "fiat money debasement", hence long gold; "Japan debt unsustainability", hence short JGBs.
Thematic trades should be sized smaller than short-term trades because they can take years to play out and may incur negative carry.

Remark 12.1.5. *Return Profile Across Cycles*
Global macro's defining feature is return asymmetry across market regimes. Empirical characteristics:

i. Lower average volatility than equities

ii. Comparable or higher long-run returns

iii. Low or negative correlation to equities during crises

iv. Higher Sharpe ratio than most hedge fund strategies

Macro tends to underperform in bull markets (when staying long works), but outperform during dislocations, recessions, and policy shifts. This asymmetry exists because macro traders can go short, can hold cash, can trade volatility, rates, and FX directly (flexibility that equity-focused strategies lack).

Remark 12.1.6. *Behaviour in Economic Crisis*
In stress regimes:

i. Equity managers are trapped long (mandates prevent shorting or holding cash)

ii. Credit liquidity vanishes (bids disappear, spreads gap wider)

iii. Correlations converge toward 1 (diversification fails)

Macro funds, by contrast, can:

i. Reduce gross exposure rapidly

ii. Short risk assets directly

iii. Express views via rates, FX, or volatility instead of equities

iv. Profit from the dislocation rather than merely survive it

Remark 12.1.7. *Global Macro Correlation Properties*
Global macro's real value is conditional correlation:

i. In normal times: low correlation to equities

ii. In crises: negative correlation to equities

This makes macro an effective portfolio diversifier and tail-risk hedge. Combined with high liquidity (trading the most liquid markets globally), macro is the most popular hedge fund allocation by pension funds.

Remark 12.1.8. *Liquidity Advantage*
One of most liquid hedge fund strategies because it trades the most liquid underlying markets:

i. G10 FX and major EM currencies
ii. Government bond futures (Treasuries, Bunds, JGBs, Gilts)
iii. Equity index futures (S&P 500, Euro Stoxx, Nikkei, etc.)
iv. Major commodity futures (crude oil, gold, copper, etc.)

This liquidity allows macro funds to:

i. Scale to very large AUM without market impact
ii. Enter and exit positions quickly during stress
iii. Offer better redemption terms than illiquid strategies

12.1.2 Role of Central Banks in Global Macro

Remark 12.1.9. *Monetary Policy Regimes*
Central banks define the macro regime more than any other actor. Markets do not trade rates, they trade policy intent under constraints. Regimes fall into following types:

i. Conventional Policy Regime: Policy rate is the main tool. Yield curve responds normally. Transmission is through rates to credit, to growth, then to inflation.
ii. Unconventional Policy Regime: Policy rate loses effectiveness (zero lower bound). Balance sheet becomes the primary tool. Transmission shifts to asset prices, expectations, and FX.
iii. Crisis/Emergency Regime: Stability is more important than inflation targeting. Rules are suspended. Liquidity provision dominates all other objectives.

Remark 12.1.10. *Objectives of Central Banks*
Central banks typically have mandates that include:

i. Price Stability: Most central banks target inflation around 2%
ii. Full Employment: Explicit in Fed's dual mandate; implicit elsewhere
iii. Financial System Stability: Lender of last resort function; systemic risk monitoring
iv. Currency Stability: Explicit for some (SNB, PBOC); implicit for others

Under stress, these objectives may conflict. Central banks must choose which objective to sacrifice. Understanding these trade-offs is critical for macro trading.

Remark 12.1.11. *Conventional Central Bank Tools*

i. Policy Rate: Set by changing overnight rate at which banks lend and borrow to meet reserve requirements. The primary tool in normal times.
ii. Reserve Ratio: Raising reserve ratio decreases money multiplier and contracts money supply.
iii. Open Market Operations (OMO): Buying and selling government bonds in local currency. Repos are short-term agreements where the central bank agrees to repurchase securities within a specified time.
iv. Currency Intervention:
    1. Sterilised intervention: Trading own currency via FX reserves without changing monetary base
    2. Unsterilised intervention: Allows for changes in monetary base

Remark 12.1.12. *Taylor's Rule*
Maps macroeconomic deviations to policy stance:

$$i_t = r^* + \pi_t + \alpha(\pi_t - \pi^*) + \beta(y_t - y_t^*)$$

where r^* is the neutral real rate (unobservable, typically estimated 2–3%), π_t is current inflation, π^* is the inflation target, $(y_t - y_t^*)$ is the output gap, and α, β are response coefficients (typically 0.5).
The rule states that the nominal policy rate must rise more than one-to-one with inflation (Taylor Principle); otherwise real rates fall, inflation becomes self-reinforcing, and expectations de-anchor.
Central banks treat Taylor's rule as a benchmark, communication tool, and political shield.

Remark 12.1.13. *The Impossible Trinity*
A country cannot simultaneously maintain:

i. Fixed exchange rate

ii. Free capital flows

iii. Independent monetary policy

This constraint explains currency crises (ERM 1992, Asian crisis 1997), forced rate hikes or cuts, FX pegs breaking, and why EM central banks behave "irrationally" under stress. Countries must choose two of three.

Remark 12.1.14. *Central Bank Balance Sheet*

i. Monetary Base: Sum of currency in circulation and reserves held at the central bank. Can only be controlled via open market operations.

ii. Money Supply: Changes in money supply are leading predictors of GDP growth and inflation. The Quantity Theory of Money:
$$M \times V = P \times Y$$
where M is money supply, V is velocity of money (turnover rate), P is price level, Y is real output.

Post-2008, central bank balance sheets expanded dramatically. The Fed releases weekly H.4.1 report showing sources and uses of reserves.

Remark 12.1.15. *Types of Money Supply*

i. $M0$: Physical currency in circulation (notes and coins)

ii. $M1$: $M0$ + demand deposits, checking accounts, traveler's checks

iii. $M2$: $M1$ + savings deposits, money market accounts, small time deposits, retail money market funds

iv. $M3$: $M2$ + large time deposits, institutional money market funds, repos, eurodollars

Central banks primarily control $M0$; broader measures depend on banking system credit creation.

Remark 12.1.16. *Zero Lower Bound and the Liquidity Trap*
When central bank rate is at or near zero:

i. Rate cuts lose marginal effectiveness

ii. Expectations dominate outcomes

iii. Forward guidance becomes critical

iv. Fiscal policy starts to matter more

Yield curves flatten naturally; carry trades distort; volatility is suppressed artificially.
Some central banks have gone to negative rates, testing the effective lower bound.

Remark 12.1.17. *Quantitative Easing (QE)*
Central banks purchase financial assets (government bonds, MBS, corporate bonds), creating electronic reserves to expand the balance sheet. QE works through multiple channels:

i. Portfolio rebalancing (investors pushed into riskier assets)

ii. Risk premia compression (term premium, credit spreads)

iii. Signalling effects (commitment to low rates)

iv. Exchange rate channel (currency weakening)

Bank lending does not mechanically follow reserve creation. Transmission depends on credit demand and bank capital. Effectiveness diminishes with successive rounds.
Fed research shows QE1 lowered 10-year yields by 100bp; QE2 by only 14bp.

Remark 12.1.18. *Hawks and Doves*
Central bankers have individual biases that shape their policy views:

i. Hawks: Wary of inflation, favour tighter policy, defend currency value

ii. Doves: Favour growth and employment, tolerate higher inflation, support looser policy

When a known hawk makes dovish statements (or vice versa), it signals a significant shift and moves markets more than statements consistent with known bias.
Tracking individual central banker positions is essential for anticipating policy shifts.

Remark 12.1.19. *Central Bank Communication*
Central bank communication is a policy tool in itself:

i. Statements: Released after each meeting; market moves on hawkish/dovish tone changes
ii. Minutes: Released with lag (2 to 3 weeks); reveal vote splits and internal debates
iii. Press Conferences: Chair/President Q&A; tone and body language matter
iv. Projections: Dot plots (Fed), staff forecasts; reveal expected rate path
v. Speeches: Individual members signalling; watch for coordinated messaging

Example: BOE June 2012 minutes revealed 5–4 vote against QE (vs 8–1 in May), correctly predicting QE expansion in July.

Remark 12.1.20. *Major Central Banks*

i. Federal Reserve (Fed): Dual mandate (price stability, full employment). FOMC meets 8 times/year with 12 voting members (7 governors, NY Fed, 4 rotating).
Most important central bank globally due to USD reserve status.
ii. European Central Bank (ECB): Price stability mandate (HICP $< 2\%$). Governing Council of 25 members. Key rates: Main Refinancing Operations (MRO), deposit facility rate. Tools include LTRO, TLTRO, OMT, APP.
iii. Bank of Japan (BOJ): Targets 2% inflation (chronically undershoots). Policy Board of 9 members. Pioneer of QE, yield curve control (YCC), negative rates.
iv. Bank of England (BOE): 2% inflation target. MPC of 9 members with individual accountability. First major central bank to publish inflation forecasts.
v. Swiss National Bank (SNB): Inflation target $< 2\%$. Notable for 2011–2015 EUR/CHF floor at 1.20 and subsequent abandonment. Massive FX reserve accumulation.
vi. People's Bank of China (PBOC): Multiple objectives including growth, stability, FX management. Manages CNY within trading band. Uses reserve requirements actively.

Remark 12.1.21. *ECB-Specific Tools*

i. LTRO/TLTRO: Long-Term Refinancing Operations provide multi-year funding to banks against collateral. 3-year LTROs in 2011 to 2012 injected over EUR1 trillion.
ii. OMT (Outright Monetary Transactions): Announced September 2012. Unlimited bond purchases for countries in EFSF/ESM programs. Never actually used but stabilised spreads.
iii. APP (Asset Purchase Programme): QE-equivalent. Purchases of government bonds, covered bonds, corporate bonds, ABS.
iv. TARGET2: Payment system showing intra-Eurozone imbalances. German TARGET2 claims rose to EUR700B+ during crisis, reflecting capital flight from periphery.

Remark 12.1.22. *Fed Crisis-Era Programmes*
During and after 2008, the Fed deployed unprecedented tools:

i. Central Bank Swap Lines: USD liquidity to foreign central banks to prevent LIBOR spikes
ii. TAF (Term Auction Facility): 28–84 day unsecured lending without discount window stigma
iii. TSLF (Term Securities Lending Facility): Treasury lending against lower-quality collateral
iv. CPFF (Commercial Paper Funding Facility): Commercial paper purchases to prevent market freeze
v. TALF (Term ABS Loan Facility): Lending against consumer ABS to support credit markets
vi. Agency MBS Purchases: \$1.25 trillion in QE1 alone

Understanding these tools helps anticipate future crisis responses.

12.1.3 Economic Data Releases and Demographics

Remark 12.1.23. *Data Releases and Market Impact*
Data releases are central bank inputs, not just statistics. Markets price expectations before the release; the move comes from the surprise, not the level. Key principles:

i. Surprise Direction: Above/below consensus drives immediate price action. Magnitude of surprise matters more than absolute level.

ii. Revision Risk: Prior releases are frequently revised. Initial release moves markets; revisions move markets again if material.

iii. Hierarchy: Not all data is equal. NFP and CPI dominate in the US; PMIs dominate globally due to frequency and forward-looking nature.

iv. Regime Dependence: What data matters depends on the regime. In low-inflation regimes, growth dominates. In high-inflation regimes, inflation dominates everything.

Back-testing asset responses to past data releases (queries) can inform trade sizing and direction. However, false positives are common. If wrong, cut quickly.

Remark 12.1.24. *Growth Indicators*
Growth data sets the directional bias for risk assets, but markets trade changes, not levels.

i. GDP (Headline, QoQ, YoY): Measures total production in current or constant prices (nominal, real GDP). Released quarterly with preliminary, revised, and final prints. Backward-looking but sets narrative.

ii. Purchasing Managers' Indices (PMIs): Survey business managers on employment, new orders, inventories, production, supplier deliveries, prices, backlogs, imports, and exports on scale of better/same/worse vs prior month. Above 50 = expansion; below 50 = contraction. Most important leading indicator.

iii. Industrial Production: Measures growth in manufacturing, utilities, and mining. More cyclical than services. Released monthly. Leading indicator for GDP forecasting.

iv. Trade Volumes, Baltic Dry Index (BDI): Shipping costs for dry bulk cargo. Leading indicator on global trade activity since it cannot be speculated (no futures market).

PMIs and surveys move markets more than GDP because they are higher frequency and forward-looking.
Growth surprises affect equities (earnings expectations), commodities (demand), FX (via growth differentials).

Remark 12.1.25. *Inflation*
Inflation data is primarily a policy variable. It constrains or enables central bank action.

i. Consumer Price Index (CPI)/Core CPI: Prices paid by consumers for basket of goods and services. Core excludes food and energy (volatile). Central banks target core.

ii. Producer Price Index (PPI): Average change in selling prices received by domestic producers. Leads CPI by 1–3 months as input costs pass through.

iii. PCE (Personal Consumption Expenditures): Fed's preferred inflation measure. Differs from CPI in weights and methodology.

iv. Inflation Expectations: Market-based (breakevens from TIPS) or survey-based (Michigan, NY Fed). Central banks watch expectations closely. De-anchoring is a regime change.

Inflation surprises directly affect real yields, curve shape, FX via real rate differentials.
Inflation matters most when central banks are near constraints (ZLB, credibility questioned).

Remark 12.1.26. *Employment Data*
Labor data bridges growth, inflation, and central bank reaction function.

i. Non-Farm Payrolls (NFP): Number of workers in US excluding farms, private households, nonprofits, self-employed, and military. Released first Friday of each month. Among highest-volatility releases for rates, FX, equities.

ii. Unemployment Rate: Unemployed as percentage of labor force. Headline number; can be distorted by participation changes.

iii. Participation Rate: Labor force as percentage of civilian population. Structural decline (aging) vs cyclical decline (discouraged workers) matters.

iv. Wage Growth (Average Hourly Earnings): More important than job count for inflation outlook.

v. JOLTS (Job Openings and Labor Turnover Survey): Shows job openings, hires, quits. Quits rate indicates labor market tightness and worker confidence.

Wage growth drives services inflation, which is stickier than goods inflation.
Fed's dual mandate makes employment data directly policy-relevant.

Remark 12.1.27. *External Accounts*
External balances define structural currency pressure and financing needs.

i. Current Account: Net exports + net foreign income + transfer payments. Deficit countries need capital inflows or currency depreciation; surplus accumulate reserves and have structurally stronger currencies.

ii. Capital Account/Financial Account: Portfolio flows, FDI, banking flows. Can finance current account deficits but creates vulnerabilities (sudden stop risk).

iii. Trade Balance: Exports minus imports. Incremental changes in exports or imports can have big impact on trade balance as percentage of GDP.

iv. Terms of Trade (ToT): Export prices divided by import prices. Improves when export prices rise relative to imports. Currency depreciation typically worsens ToT.

v. FX Reserves: Central bank holdings of foreign currency and gold. Countries with large reserves can intervene; watching reserve changes indicates intervention.

Persistent current-account deficits require higher yields to attract capital or eventual currency adjustment.

Remark 12.1.28. *Government/Fiscal Indicators*
Fiscal data determines long-term sustainability, not short-term price action.

i. Budget Balance: Revenues minus expenditures. Deficit = borrowing required. Deficit/GDP ratio matters more than absolute level.

ii. Debt-to-GDP: Stock measure of accumulated borrowing. Matters more for EM and Euro periphery than for reserve currency issuers.

iii. Primary Balance: Budget balance excluding interest payments. Shows underlying fiscal stance; positive primary balance means debt/GDP can stabilise if growth exceeds interest rate.

iv. Debt Sustainability: Interest payments as % of revenue. Rising share means deteriorating sustainability.

Fiscal deterioration matters when markets doubt financing ability or central bank independence.
Sovereign risk appears first in bond spreads, CDS, FX (especially EM).

Remark 12.1.29. *Consumption Indicators*
Consumption drives developed-market growth (70%+ of US GDP).

i. Retail Sales: Consumer demand for finished goods. Durable vs non-durable breakdown matters.

ii. Consumer Confidence: Survey of expected financial situation. Conference Board and Michigan surveys in US. Leading indicator, confidence leads spending, but noisy.

iii. Personal Income and Spending: Income growth enables spending growth. Savings rate = income minus spending.

iv. Savings Rate: Higher savings = lower current consumption but more sustainable. US savings rate has declined structurally; Asian savings rates remain high.

Consumption data is most relevant for equity indices, cyclical sectors, domestic currencies.
US consumer accounts for $\sim 15\%$ of global GDP directly through consumption.

Remark 12.1.30. *Housing Indicators*
Housing is cyclically important and credit-sensitive.

i. Building Permits: Lead housing starts since permits are required before construction. Most forward-looking housing indicator.

ii. Housing Starts: Counted when construction begins. Gauge of housing demand and construction activity.

iii. Existing/New Home Sales: Volume of transactions. Prices vs volumes can diverge.

iv. Home Price Indices: Case-Shiller, FHFA. Lag actual market conditions.

v. Household Formations: Structural driver of housing demand. Post-crisis decline reduced demand.

Housing matters for rate sensitivity, consumer wealth effects, and financial stability.

Remark 12.1.31. *Business and Industry Indicators*
These indicators show where growth is coming from and where it's heading.

i. Manufacturing vs Services PMI: Manufacturing is more cyclical and globally linked. Services dominate GDP but move slower.

ii. New Orders: Most forward-looking PMI subcomponent. Rising new orders = future production.

iii. Inventories: High inventories relative to sales = future production cuts. Inventory cycles matter for commodities and industrial equities.

iv. Capacity Utilisation: Actual output relative to potential. Over 100% = overheating, inflation risk. Well under 100% = slack, deflation risk.

v. Durable Goods Orders: Capital expenditure intentions. Core capital goods orders (ex-aircraft, ex-defence) is key business investment indicator.

ISM New Orders minus Inventories spread is a reliable leading indicator of manufacturing cycles.

Remark 12.1.32. *Demographics*
Slowest-moving macro variable, but the hardest constraint. Demographics cannot be QE'd.

i. Working-Age Population (15–64): Drives potential growth. Many developed countries have peaked or will peak soon.

ii. Prime Saving Age (35–69): Peak saving years. Countries past this peak see declining savings and current accounts.

iii. Dependency Ratio: Non-working population (young + old) divided by working population. Higher ratio = more fiscal pressure, lower savings.

iv. Median Age: Summary statistic for demographic structure. Europe/Japan: 40s; US: 38; EM: 20s–30s.

v. Retirement Age vs Life Expectancy: Gap determines pension system sustainability. Life expectancy rising; retirement ages barely moving.

Demographics affect long-term growth, savings rates, current accounts, equilibrium real rates.

Remark 12.1.33. *Demographic Implications by Country*
Peak working-age populations determine structural trajectories:

i. Already peaked: Germany (2006), Japan (2016), Netherlands (1989), UK (2009), US (2007)

ii. Peaking soon: China (2032), Korea (2023), Russia (2025), Spain (2022)

iii. Long runway: India (2050), Indonesia (2050), Nigeria (2050), Philippines (2050)

Countries past peak will see declining current accounts, rising dependency ratios, pressure on fiscal positions. Jean-Claude Trichet: "Current account balance is an important summary indicator that signals losses of competitiveness and emerging imbalances."

Remark 12.1.34. *Building Queries for Data Events*
Queries back-test asset behaviour around past occurrences of specific events:

i. Identify the event (e.g., BOE QE announcement, Fed rate hike, NFP beat)

ii. Find historical dates with similar conditions

iii. Calculate asset returns +1 day, +2 days, +1 week, +1 month after event

iv. Compute hit rates (percentage of times asset moved in expected direction)

v. Assess risk/reward: average return vs standard deviation

Queries are powerful for short-term event trading and understanding portfolio exposure to upcoming releases. Caveat: False positives are common. If position moves against you immediately after event, cut quickly.

Remark 12.1.35. *Key Data Release Calendar (US)*
Monthly cadence of major releases:

i. First Friday: Employment Report (NFP, unemployment rate, wages)

ii. Mid-month: CPI, Retail Sales, Industrial Production

iii. Third week: Housing Starts, Building Permits

iv. Late month: GDP (quarterly), PCE, Durable Goods

v. First business day: ISM Manufacturing PMI

vi. Third business day: ISM Services PMI

FOMC meetings (8 per year) and ECB meetings (6 per year) often dominate all other releases.
Economic calendars (Bloomberg ECO, Reuters, Trading Economics) are essential tools.

12.2 Transmission and Cross-Asset Mapping

This section services as a primer for four basic asset classes and their usage in global macro trading.

12.2.1 Building Blocks of Global Macro

Remark 12.2.1. *Macro Strategy Product Groups*
The four core product groups in global macro are:

i. Fixed Income (FI) (Policy and Growth Sensor): encodes growth expectations, inflation expectations, central bank credibility. Yield curves summarise the entire macro narrative; steepening means growth/reflation; flattening or inversion means slowdown/recession. Fixed income tells what the market thinks will happen before it happens.

ii. Foreign Exchange (FX) (Adjustment Mechanism): FX reflects relative, not absolute, macro conditions. Sensitive to rate differentials, capital flows, balance of payments, risk-on/risk-off regimes. FX absorbs and redistributes macro shocks across countries.

iii. Equities (Growth Expression): equities are long growth, long liquidity, short uncertainty. Equity indices aggregate earnings expectations, discount rates, risk appetite. Volatility (VIX) is a second-order macro signal, not just an equity signal. Equities express the confidence (or fear) embedded in the macro outlook.

iv. Commodities (Inflation and Real-Economy Link): driven by physical supply and demand, inventories, geopolitics. Highly sensitive to growth cycles, inflation shocks, currency moves (especially USD). Commodities anchor macro views in real economic constraints.

These are the most liquid markets globally, which allows macro strategies to scale to large AUM while maintaining execution quality.

Remark 12.2.2. *Relationship Between Assets*
These relationships are regime-dependent, not permanent laws:

i. Growth Shocks: rates rise, equities rally, cyclicals outperform, commodities rise. FX favours growth-linked currencies (AUD, CAD, EM FX).

ii. Recession Shocks: rates fall, curves flatten/invert, equities sell off, USD strengthens (flight to safety), commodities weaken (except gold).

iii. Inflation Shocks: nominal rates up, real rates determine equity impact. Commodities outperform. FX punishes low-credibility central banks.

iv. Liquidity Shocks: USD strengthens sharply, credit spreads widen, correlations spike toward 1, volatility explodes. Cross-currency basis blows out.

A single macro thesis should have expressions across multiple asset classes, each with different risk profiles and convexity characteristics.

Remark 12.2.3. *Asset Correlation Dynamics*
Asset correlations change under stress. Diversification fails when needed most, where assets that appear uncorrelated in calm periods may collapse together in crises.
Macro traders profit when they anticipate correlation breakdowns and position for nonlinear regime shifts. This is why macro portfolios use convex instruments (options), size conservatively, and focus on liquidity.
Traders must know what assets they are trading in aggregate. If entire portfolio has assets with high correlation to one another, they can be stopped out of all positions more easily if the market turns against them.

Remark 12.2.4. *Position Sizing Framework*
Position sizing is arguably more important than knowing what to buy or sell. The first rule of investing is to preserve capital and avoid loss.
Position sizing aims to adjust each position for volatility. This normalises all positions such that losses are more predictable and have the same probability of occurring:

i. Calculate N = 20-day exponential moving average of True Range

ii. True Range = max(High − Low, High − Previous Close, Previous Close − Low)

iii. Use $2N$ for stop-loss calculation (two standard deviations) to avoid being stopped out by normal volatility

iv. With $2N$, there is approximately 2% chance of being stopped out on any given day (vs. 16% with $1N$)

This methodology, developed by the Turtle Traders, allows traders to take positions in a broad range of markets while monitoring risk consistently.

Remark 12.2.5. *Risk/Reward Discipline*
Given human error and transaction costs, traders can only hope to be right about 50% of the time. Having good risk/reward on each trade will make or break long-term success.
As a baseline, always strive for a 1:3 risk-to-reward ratio or higher:

i. For every trade risking 25 basis points, expect to make at least 75 basis points
ii. Set targets before the trade is executed to avoid biases of irrational human decision making
iii. Log reasons for each trade in a trade journal
iv. If original thesis no longer holds true, take trade off, but do not create excuses for taking trades off early

Remark 12.2.6. *Gap Risk*
Gap risk is one of the biggest risks when sizing positions. Gap risk occurs when prices change without any trading in between (e.g., overnight, over weekends). This risk is especially acute in:

i. EM currencies and assets (less liquid, more event-driven)
ii. Positions held over weekends or holidays
iii. Periods of elevated geopolitical risk
iv. Earnings/data release windows

Remark 12.2.7. *Risk Utilisation*
Risk-taking is nonlinear. Even with 50% conviction, should not run 50% of risk utilisation.

i. Base case: set a maximum VaR limit (e.g., 5% daily VaR at 95% confidence)
ii. If up 5% on the year, allow VaR limit to increase to 6%
iii. If down 2.5% on the year, decrease VaR limit to 4%
iv. At the start of each new year, reset to 0% return regardless of prior year performance

The intention is to create steady returns while preserving capital and avoiding blow-up risk.

Remark 12.2.8. *Stress Testing*
Stress testing aims to go beyond VaR to find potential worst-case scenarios:

i. Run historical back-test on P&L for previous 500 trading days ($\approx$ 2 years)
ii. Sort from best to worst days; examine the average of the 5 worst days (worst 1%)
iii. In periods of low volatility, also test against extreme historical events: October 1987 crash, 1998 LTCM/Russian default, 2008 financial crisis, 2011 Japan earthquake, 2015 CNY devaluation, 2020 COVID crash

Knowing maximum loss outside of VaR helps traders understand exposure to tail events.

Remark 12.2.9. *Performance Metrics*
Key metrics for evaluating macro trading performance:

i. Sharpe Ratio: $\frac{E[R]-R_f}{\sigma}$, risk-adjusted return; higher is better. Assumes normal distribution.
ii. Sortino Ratio: Sharpe with only downside volatility, differentiating between good and bad volatility.
iii. Maximum Drawdown: peak-to-trough decline; critical for understanding worst-case capital loss.
iv. Value at Risk (VaR): 95% or 99% daily VaR (expected loss exceeded on 1 in 20 or 100 days).

When evaluating strategies, test Sharpe ratios rather than raw returns. Break down P&L by asset class, duration, and conviction level to identify strengths and weaknesses.

Remark 12.2.10. *Macro View Thesis*
A macro view expressed in only one asset class is incomplete. A single macro thesis should have multiple expressions, each with different risk profiles:

i. Growth view: rates (duration), equities (beta), FX (pro-cyclical currencies), commodities (cyclicals)
ii. Inflation view: commodities (energy, agriculture), breakevens, FX (commodity currencies), TIPS
iii. Crisis/Deflation view: bonds (duration), USD/JPY/CHF (havens), volatility (VIX, swaptions), gold
iv. Policy divergence view: FX (rate differentials), yield curve spreads, relative equity indices

Having multiple expressions allows for confirmation across asset classes and provides natural hedging if one leg underperforms.

12.2.2 Foreign Exchange in Global Macro

FX plays three special roles simultaneously:

i. Macro shock absorber

ii. Macro transmission channel

iii. Macro expression instrument

Remark 12.2.11. *FX Market Structure*
The FX market is the world's largest and most liquid market, and operates 24 hours per day, 5.5 days per week across global financial centers (London, New York, Tokyo, Singapore, Hong Kong).
Key participants include: central banks (intervention, reserve management), commercial banks (market-making, proprietary trading), corporations (hedging trade flows), asset managers (currency overlay, alpha generation), hedge funds (macro speculation, carry trades), and retail traders.
The market is predominantly OTC, with spot, forwards, and swaps as the main instruments. FX futures trade on exchanges (CME) but represent a small fraction of total volume.

Remark 12.2.12. *Role of the US Dollar*
Global macro is de facto USD-centric. USD is the global reserve currency, most commodities are USD-priced, global funding markets clear in USD, and cross-border leverage is USD-denominated.
The USD is involved in approximately 85% of all FX transactions globally. Many "local" macro trades are actually USD liquidity trades in disguise.
Stress often appears first in US Dollar Index (DXY), cross-currency basis, EM FX before local assets.
The DXY Index measures USD against a basket of six major currencies: EUR (57.6%), JPY (13.6%), GBP (11.9%), CAD (9.1%), SEK (4.2%), CHF (3.6%). The DXY is not trade-weighted and over-represents EUR.

Remark 12.2.13. *Special Drawing Rights (SDRs)*
SDRs are international reserve assets created by the IMF to supplement member countries' official reserves. The SDR basket is reviewed every five years. SDRs serve as a unit of account for IMF transactions and provide insight into official reserve diversification trends.

Remark 12.2.14. *Currency Regimes*
Currency regimes matter more than models. FX does not behave uniformly. Regime dominates signal.

i. Free-floating: price discovery happens in FX

ii. Managed/pegged: pressure builds elsewhere such as in rates, reserves, capital controls

iii. Hard pegs/Currency boards: FX looks stable until it breaks violently

FX valuation only works within regime constraints. Ignore the regime and "cheap" can stay cheap forever.

Remark 12.2.15. *Central Bank Intervention*
Central banks intervene in FX markets for multiple objectives:

i. Defend a peg or managed float band

ii. Smooth excessive volatility

iii. Counter speculative attacks

iv. Accumulate or deploy reserves

Intervention effectiveness depends on: reserve adequacy, policy credibility, coordination with monetary policy, and market liquidity conditions. Intervention against fundamental pressures typically fails. Watch for reserve depletion as a leading indicator of forced adjustment.

Remark 12.2.16. *Valuation Techniques*

i. Market Implied and Risk-Appetite Signals

   1. Equity Index Price Performance: strong local equities (vs peers) attract inflows and support currency; sharp equity underperformance foreshadows FX weakness via risk-off deleveraging and foreign selling.
   2. Credit/Sovereign Risk: widening sovereign spreads/CDS signals default/policy credibility/funding stress, weakening FX or raising devaluation risk premia. Also a regime detector for EM FX.
   3. Sentiment/Positioning: call-put skew on three-month risk reversals. Acts as a multiplier. Can create overshoots/undershoots around fair value in crowded carry or crisis episodes.

ii. Trade and Competitiveness Fundamentals

1. Trade Balance/External Account: structurally strong trade position often supports FX; persistent deficits can make the currency reliant on financing.
2. Trade-Weighted Index (TWI): reflects currency's effective competitiveness against trading partners; the currency basket that matters for the real economy.
3. Economic Activity: strong activity can support FX via higher expected returns and inflows, but it can also worsen the trade balance.
4. Export Partner Growth: mapping who buys exports and whether those buyers are expanding/contracting. Export partner growth leads exporter's FX through expected trade receipts.

iii. Macro Sustainability Indicators

1. Debt-to-GDP: high public debt with FX liabilities or weak domestic savings can increase risk premia and currency vulnerability, as it tightens the policy constraint set (inflation vs austerity vs default).
2. Current Account Balance: persistent deficits require capital inflows; sudden stops trigger sharp FX adjustment.
3. FX Reserves: adequacy (months of import cover, short-term debt coverage) determines policy space.

iv. Long-Term Valuation Anchors

1. Purchasing Power Parity (PPP): the idea that exchange rates should adjust so that identical goods cost the same in different countries. In practice, PPP convergence can take years or decades.
2. GDP Per Capita: structural development/productivity proxy. Richer, higher productivity economies tend to sustain stronger real exchange rates over time (Balassa-Samuelson effect).

v. Carry Indicators

1. OIS Differential: compares respective policy rates of two countries. Carry is the interest rate differential between two currencies.
2. LIBOR/SOFR Differential: reflects unsecured interbank rates. Can be decomposed into risk-free rate + credit spread component. Using relative spreads gives a sense of direction; interest rate differential drives demand for a currency.
3. Risk-Adjusted Carry: adjusts carry for volatility. Outperforms standard carry trades and is useful in cross-currency analysis:

$$\text{Carry-to-Risk Ratio} = \frac{\text{3M Carry (Rate Differential)}}{\text{30D Realized Volatility}}$$

Within high-yielding currency pairs, one would ideally want the highest carry-to-risk ratio. Volatility and carry components are always in flux, so adjustments should be made frequently.

Remark 12.2.17. *Carry Trade Mechanics*
Carry strategies are typically executed by borrowing in lower-yielding currencies and buying higher-yielding currencies. The carry trade is a crowded trade; in large risk-off markets, the strategy suffers significant losses as positions unwind simultaneously.
Carry is the return earned on holding a currency, assuming the exchange rate is held constant. In practice, exchange rate moves often dominate carry returns. Historically, carry strategies exhibit positive returns on average but with significant negative skewness (small gains, occasional large losses).

Remark 12.2.18. *Carry Trade Risk Factors*
Key risks to carry trades include:

i. Crowding Risk: carry trades are popular, leading to violent unwinds during risk-off episodes.
ii. Gap Risk: EM currencies can gap significantly on weekends or during crises, bypassing stop-losses.
iii. Regime Shifts: central bank policy changes can rapidly alter rate differentials.
iv. Correlation Breakdown: in stress, all carry currencies tend to sell off together against funding currencies.
v. Liquidity Risk: EM FX can become illiquid precisely when exits are needed most.

Finding the carry-to-risk ratio is useful, as is comparing all currency pairs on a relative basis. Measuring the standard error of the sample and finding the most attractive pairs helps in mean-reverting carry strategies.

12.2.3 Equities in Global Macro

In macro, a trader should understand equity in terms of market behaviour in the country being traded, the sectors that make up those indices, and correlation risk.

Remark 12.2.19. *Top-Down Macro Approach to Equities*
Many macro traders take a top-down approach to trading equities:

i. Start with a global view
ii. Narrow to countries expected to outperform
iii. Select the best sectors within those countries
iv. Optionally, select best companies within sectors

If the investor has a knack for knowing where the next opportunity lies, the most reward typically comes from concentrated exposure rather than diversified index holdings.

Remark 12.2.20. *Major Equity Indices by Region*

i. United States: S&P 500, Nasdaq, DJIA, Russell 2000, Wilshire 5000
ii. Europe: Euro Stoxx 50, DAX, CAC 40, FTSE 100, FTSE MIB, IBEX 35, AEX
iii. Asia Ex-Japan: Shanghai Comp, Hang Seng, KOSPI, ASX 200, NZX 50
iv. Japan: Nikkei, Topix, JASDAQ
v. Emerging Markets: Bovespa, NIFTY, Micex/RTS, Mexico IPC, TOP 40/JALSH, BIST 30/100 etc.

Indices can be traded directionally or as relative value trades (long one index, short another). Different indices have different sector exposures: Bovespa is more commodity-sensitive than NIFTY; S&P 500 has more financial exposure than Nasdaq (technology-weighted).

Remark 12.2.21. *Index Construction Methods*
Index construction methodology affects interpretation:

i. Price-Weighted: considers only stock price, ignores market cap. A single large stock move can dominate.
ii. Market-Cap Weighted: weighted by market capitalisation. Large-cap stocks dominate performance.
iii. Equal-Weighted: each constituent has equal weight regardless of size.

Understanding weighting structure is essential for accurate interpretation of index moves.

Remark 12.2.22. *Sector Rotation Across Economic Cycles*
Different sectors outperform in different economic regimes:

i. Recession/Contraction: Consumer Staples, Utilities, Healthcare (defensive sectors)
ii. Recovery: Financials, Consumer Discretionary, Real Estate
iii. Expansion: Industrials, Materials, Technology
iv. Slowing Growth/Late Cycle: Energy, Materials (inflation hedges)

To best position an equity portfolio, it is critical to know whether the economy is in contraction, recovery, expansion, or slowing growth.

Remark 12.2.23. *Equity Derivatives Overview*
Key equity derivatives for macro trading:

i. ETFs: fast, cheap exposure to indices, sectors, countries, commodities, FX, FI. Trade like stocks.
ii. ADRs (American Depositary Receipts): foreign stocks listed on U.S. exchanges in USD. Subject to FX risk since underlying shares are held locally.
iii. Index Futures: liquid, leveraged exposure to equity indices.
iv. Options: provide asymmetric payoffs for hedging and directional views.

Remark 12.2.24. *The Volatility Index (VIX)*
The VIX measures implied volatility of S&P 500 options one month out. Key properties:

i. Negatively correlated with equity returns (volatility rises when equities fall)
ii. During large selloffs, VIX can move from mid-teens to 30+ in days

iii. VIX options provide asymmetric hedges for equity portfolios

iv. VIX futures typically trade in contango; during sharp selloffs, shift to backwardation

Volatility is measured relative to $\sqrt{252} \approx 16$. A VIX of 16 implies approximately 1% expected daily move; VIX of 24 implies 1.5% daily move ($24/16 = 1.5\%$).
V2X (Euro Stoxx), VXN (Nasdaq), VXD (DJIA), RVX (Russell 2000), GVZ (Gold), OVX (Crude Oil).

Remark 12.2.25. *Variance Swaps*
Variance swaps are OTC products providing pure variance exposure. Advantages over options:

i. Pure volatility exposure without delta hedging

ii. Priced with realised volatility (usually lower than implied)

iii. No interest rate or dividend risk

Payoff structure is convex and nonlinear. Long variance benefits disproportionately from large moves in either direction. Quoted by strike (reference realised volatility), vega notional, variance units, maturity.
Variance units $= \frac{\text{Vega Notional}}{2\times\text{Strike}}$, Payoff $= (\sigma^2_{\text{realised}} - K^2) \times$ Variance Units

Remark 12.2.26. *Dividend Swaps*
An OTC or exchange-traded product that allows one to take a view on dividends of an index to be higher, or lower, than a fixed amount.

Remark 12.2.27. *Equity Valuation Techniques for Macro*

i. Price-to-Book (P/B): P/B $> 2.5\times$ generally overbought; P/B $< 1.5\times$ oversold. Useful for detecting bubbles and value opportunities.

ii. Dividend Yield: dividend/price. Holder gets paid while waiting (long carry). Useful for relative value across markets.

iii. Price-to-Earnings (P/E): price per \$1 of earnings. Best for comparable analysis across countries, sectors, stocks. Forward P/E uses analyst EPS estimates.

iv. Free Cash Flow Yield: FCF/Price. Cash flow from operations minus capital expenditures. Graham and Dodd value investors use this as top screen.

v. Market Cap to GDP: useful for locating potential bubbles. Above 100% warrants caution; rapid rises to 150%+ indicate bubble conditions (Japan 1989).

Remark 12.2.28. *Leading Indicators for Equities*
Useful macro indicators that lead equity performance:

i. PMI (Purchasing Managers' Index): leading indicator scored 0-100. Sharp slope changes and deviations from 50 provide strong signals. ISM below 50 typically indicates U.S. recession.

ii. Baltic Dry Index: shipping prices for raw material dry bulk. Supply is fixed (ships take years to build), demand is inelastic. Leading indicator for raw material demand.

iii. CDX High Yield Index: CDS spread on high-yield corporates. Highly correlated with VIX; inversely related to equity prices. Rising spreads signal risk-off.

iv. Consumer Confidence: in the U.S., consumption accounts for $> 70\%$ of GDP. Higher confidence leads to higher spending and GDP growth.

v. Commodity Prices (YoY): rising commodity prices are inflationary, typically met by central bank tightening, which has bearish equity implications.

Remark 12.2.29. *AUD Volatility as Risk Barometer*
The Australian dollar is a main risk-on currency due to Australia's commodity exports and reliance on Asian economies. AUD moves in tandem with equity prices.
Three-month AUD/USD implied volatility is highly correlated with VIX and inversely correlated with S&P 500. In times of stress, the rate of change in AUD volatility shifts suddenly and aggressively.
Using VIX and AUD volatility combined on an absolute basis smooths out error as a more useful measure.

12.2.4 Fixed Income in Global Macro

Fixed income is the lifeblood of the global financial system. Bank credit and debt capital markets fund governments, corporations, and consumers. Understanding fixed income is essential for macro trading because yields encode growth expectations, inflation expectations, and central bank credibility.

Remark 12.2.30. *Fixed Income Universe*
The fixed income market can be decomposed into:

i. Money Markets: Short-term instruments (maturity < 1 year). Includes T-bills, commercial paper, repos, Fed Funds, SOFR.
ii. Government Bonds: Sovereign debt across the curve (2Y, 5Y, 10Y, 30Y). Risk-free rate benchmark.
iii. Inflation-Linked Bonds: TIPS in US, Linkers in UK, OATi in France. Index to CPI.
iv. Corporate Bonds: Investment grade (IG) and high yield (HY). Credit spread over risk-free rate.
v. Municipal Bonds: State and local government debt, often tax-exempt.
vi. Mortgage-Backed Securities (MBS): Agency and non-agency. Prepayment risk.
vii. Asset-Backed Securities (ABS): Auto loans, credit cards, student loans.

Fixed income is the largest of the four product groups by notional outstanding.

Remark 12.2.31. *Money Markets and Funding*
Money markets allow short-term capital flow between lenders and borrowers:

i. Treasury Bills (T-Bills): Short-term US government obligations (4-week, 13-week, 26-week, 52-week). Risk-free benchmark.
ii. Commercial Paper: Unsecured short-term corporate debt, typically 90–270 days. Lower rates than bank borrowing.
iii. Repurchase Agreements (Repos): Sale of securities with agreement to repurchase. Critical for short-term funding and leverage.
iv. Certificates of Deposit (CDs): Bank time deposits, 30 days to 5 years.
v. Banker's Acceptances: Promissory notes issued by firms to banks, typically for trade finance.

Money market stress (rising repo rates, LIBOR-OIS widening) often precedes broader financial stress.

Remark 12.2.32. *Key Reference Rates*
Reference rates anchor pricing across the fixed income universe:

i. Fed Funds Rate: Unsecured overnight lending rate between depository institutions.
ii. SOFR (Secured Overnight Financing Rate): Overnight Treasury repo rate as USD benchmark.
iii. LIBOR (Legacy): London Interbank Offered Rate. Historically the most important rate in global finance. Phased out post-2021 due to manipulation scandals.
iv. EONIA/EURSTR: Euro overnight rates. EURSTR replaced EONIA in 2022.
v. SONIA: Sterling overnight rate for GBP.

The transition from LIBOR to risk-free rates (SOFR, SONIA, EURSTR) is one of the largest market structure changes in decades.

Remark 12.2.33. *Overnight Index Swaps (OIS)*
OIS is a fixed-for-floating interest rate swap indexed against an overnight rate (Fed Funds effective in US, EONIA/EURSTR in Europe):

i. Excellent tool to hedge or speculate on central bank action
ii. Typically maturity less than two years
iii. Customisable dates for precise central bank meeting targeting
iv. Less volatile than LIBOR, especially in periods of stress

Example: If BOE rate is 0.50% and December MPC OIS trades at 0.625%, the market prices 50% chance of a 25bp hike.

Remark 12.2.34. *LIBOR-OIS Spread*
The LIBOR-OIS spread is a key gauge of funding stress:

i. LIBOR is unsecured interbank lending; OIS is effectively risk-free overnight rate

ii. Spread widens during credit stress as banks demand higher compensation for counterparty risk

iii. 2008 financial crisis: LIBOR-OIS spiked as institutions stopped trusting each other

iv. European crisis: Spread widened as European banks faced USD funding stress

A widening spread does not automatically indicate crisis. Serves as useful barometer for funding market pressure.

Remark 12.2.35. *Eurodollar Futures*
Eurodollar deposits are USD time deposits at banks outside Fed jurisdiction. Eurodollar futures are the most heavily traded futures contracts in the world:

i. Contract on 3-month LIBOR (transitioning to SOFR futures)

ii. Contract size: $\$1,000,000$ notional

iii. Price: $100 -$ 3-Month LIBOR Yield

iv. Tick value: 1 basis point = \$25

v. Months: March (H), June (M), September (U), December (Z)

vi. Colour codes by year: White (Y1), Red (Y2), Green (Y3), Blue (Y4), Gold (Y5)

Eurodollar futures used to hedge LIBOR exposure, manage weighted average maturity of short-term liabilities.

Remark 12.2.36. *Interest Rate Swaps*
Interest rate swaps allow exchanging one set of interest payments for another:

i. Fixed-for-Floating: Most common. Exchange fixed rate for floating rate (SOFR, EURSTR).

ii. Receiver: Receives fixed, pays floating. Profits if floating rates fall.

iii. Payer: Pays fixed, receives floating. Profits if floating rates rise.

iv. Swap curve driven by future rate expectations; increasingly exchange-traded post-Dodd-Frank.

Interest rate swaps are the largest derivatives market by notional. Used for hedging, speculation, and transforming asset/liability profiles.

Remark 12.2.37. *Forward Rate Agreements (FRAs)*
FRAs are OTC contracts to exchange a reference rate (SOFR, formerly LIBOR) for a fixed rate:

i. More customisable than Eurodollar futures (pick specific dates)

ii. Notation: "3×6" means 3-month forward, 3-month rate

iii. Fixing date, settle date, and maturity date are key terms

iv. Used for hedging or speculating on short-term rate movements

FRAs allow precise targeting of rate exposure around central bank meetings or data releases.

Remark 12.2.38. *US Treasury Futures*
Treasury futures are standardised contracts for trading various maturities:

i. 2-Year (TU): Contract size $\$200,000$, tick \$15.625

ii. 5-Year (FV): Contract size $\$100,000$, tick \$7.8125

iii. 10-Year (TY): Contract size $\$100,000$, tick \$15.625

iv. 30-Year (US): Contract size $\$100,000$, tick \$31.25

v. Ultra-Long (WN): 25+ years, contract size $\$100,000$, tick \$31.25

Treasury futures are physically settled (cheapest-to-deliver mechanics). Most traders roll positions before delivery. Futures curve typically in backwardation due to term premium.

Remark 12.2.39. *German Bund Futures*
German government bonds (Bunds) are the European benchmark:

i. Schatz (DU): 1.75–2.25 year maturity

ii. Bobl (OE): 4.5–5.5 year maturity

iii. Bund (RX): 8.5–10.5 year maturity

iv. Buxl (UB): 24–35 year maturity

Bunds trade as "risk-free" for the Eurozone. Spreads to Bunds (e.g., BTP-Bund spread for Italy) are key measures of peripheral sovereign risk.

Remark 12.2.40. *Yield Curve Trades*
Curve trades express views on the shape of the yield curve (done DV01 neutral):

i. Steepener: Long short-end, short long-end. Profits from curve steepening.
ii. Flattener: Short short-end, long long-end. Profits from curve flattening.
iii. Bull Flattener: Rates fall, long end falls more than short end.
iv. Bear Flattener: Rates rise, short end rises more than long end (often Fed hiking).
v. Bull Steepener: Rates fall, short end falls more than long end (often Fed cutting).
vi. Bear Steepener: Rates rise, long end rises more than short end (inflation fears).

The 2s10s spread (10Y yield minus 2Y yield) is a classic recession indicator. Inversion has preceded every US recession since 1970.

Remark 12.2.41. *Carry and Rolldown*
Fixed income positions have carry and rolldown components:

i. Carry: Coupon income − financing cost. + for steepeners, − for flatteners in normal yield curve.
ii. Rolldown: Return from bond "rolling down" a positively sloped yield curve as it approaches maturity.
iii. In DV01-neutral trades, carry differential between legs must be accounted for.
iv. Rolldown is deterministic if curve shape unchanged; carry is known at trade inception.

Total return = Price change + Carry + Rolldown. Carry and rolldown dominate in low-volatility environments.

Remark 12.2.42. *Treasury Inflation-Protected Securities (TIPS)*
TIPS are US government bonds indexed to inflation (CPI-U NSA):

i. Principal adjusts with CPI; coupon is fixed rate applied to adjusted principal
ii. 5Y, 10Y, and 30Y maturities issued quarterly
iii. Floor at par: At maturity, receive greater of par value or inflation-adjusted principal
iv. 3-month lag between CPI and index calculation

TIPS allow investors to take views on real rates or inflation. Short-end TIPS trade like food and energy.

Remark 12.2.43. *Breakeven Inflation*
Breakeven inflation rate is the implied inflation rate from TIPS:

$$\text{Breakevens} = \text{Nominal Treasury Yield} - \text{TIPS Real Yield}$$

For example, if 10Y Treasury yields 4.0% and 10Y TIPS yields 1.5%, 10Y breakeven is 2.5%.

i. Breakevens reflect market-implied inflation expectations
ii. Can be traded directly via breakeven swaps
iii. Rising breakevens = market expects higher inflation
iv. Fed monitors breakevens for inflation expectations anchoring

Negative real rates (TIPS yield < 0) indicate investors accepting guaranteed loss of purchasing power for safety.

Remark 12.2.44. *Sovereign Credit Risk*
Sovereign debt is issued by national governments. Key concepts:

i. Reference Entity: Legal name for CDS purposes (e.g., Hellenic Republic for Greece)
ii. Curve Inversion: When short-term yields exceed long-term yields, market prices default risk
iii. Credit Events: Failure to pay, restructuring, moratorium
iv. Recovery Rate: Typically assumed 40% for sovereigns; historical average 55% with high variance

Peripheral European spreads (Italy, Spain, Portugal, Greece) to German Bunds are key risk indicators.

Remark 12.2.45. *Credit Default Swaps (CDS)*
CDS provides insurance-like protection against credit events:

i. Protection Buyer: Pays spread (premium), receives par if credit event occurs

ii. Protection Seller: Receives spread, pays par minus recovery if credit event occurs

iii. Premium quoted as annual spread, paid quarterly (March, June, September, December)

iv. Can buy protection without owning underlying bond ("naked CDS")

CDS provides asymmetric payoff: pay small premium for potentially large payout. Attractive risk/reward for tail hedging. Counterparty risk is critical.

Remark 12.2.46. *CDS Credit Events*
Credit events trigger CDS settlement:

i. Failure to Pay: Sovereign/corporate misses interest or principal payment

ii. Restructuring: Postponement, reduction, or deferral of obligations

iii. Bankruptcy: Corporate only (sovereigns cannot file bankruptcy)

For sovereigns, restructuring is not automatic trigger. Both buyer and seller have right (not obligation) to trigger CDS. Greece 2012 required ISDA determination committee ruling.

Remark 12.2.47. *Historical Sovereign Defaults*
Recent sovereign defaults and restructurings:

i. Russia 1998: Devaluation and default on local currency obligations, $73B

ii. Argentina 2001: Largest sovereign default at time, $82B, 70% haircut

iii. Uruguay 2003: Contagion from Argentina, 5-year maturity extension

iv. Greece 2012: PSI restructuring, largest sovereign default in history, $200B+

v. Argentina 2020: Ninth default, $65B restructured

Recovery rates vary widely (20%–80%). "Orderly" restructurings tend to have higher recovery.

Remark 12.2.48. *Fixed Income ETFs*
Fixed income ETFs provide liquid, diversified exposure:

i. Treasury: SHY (1–3Y), IEF (7–10Y), TLT (20+Y), TBT (2x inverse 20+Y)

ii. TIPS: TIP (broad TIPS)

iii. Investment Grade: LQD (corporate IG), AGG (aggregate bond)

iv. High Yield: HYG, JNK (high yield corporate)

v. Emerging Markets: EMB, PCY (EM sovereign debt)

vi. Municipals: MUB (muni bonds)

vii. MBS: MBB (mortgage-backed)

ETFs provide quick and cheap exposure to markets otherwise requiring large minimums or dealer relationships.

Remark 12.2.49. *Duration and DV01*
Duration measures sensitivity to interest rate changes:

i. Modified Duration: Approximate percentage price change for 1% yield change

ii. DV01 (Dollar Value of 01): Dollar change for 1bp yield change $= \frac{\text{Duration} \times \text{Price}}{10000}$

iii. Longer maturity = higher duration = more rate sensitivity

iv. Curve trades are structured DV01-neutral to isolate curve shape from level

A 10Y bond with duration 8 loses approximately 8% if yields rise 100bp.

Remark 12.2.50. *Convexity*
Convexity measures how duration changes as yields change:

i. Positive convexity: Bond gains more when yields fall than it loses when yields rise (most bonds)

ii. Negative convexity: MBS, callable bonds. Gains are capped due to prepayment/call risk

iii. Convexity matters more for large yield moves

iv. Long convexity = long volatility; benefits from rate moves in either direction

Convexity adjustment:

$$\Delta P \approx -\text{Duration} \times \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

Remark 12.2.51. *Curve Inversion as Recession Indicator*
Yield curve inversion (short rates exceeding long rates) has predicted every US recession since 1970:

i. 2s10s inversion: 10Y yield below 2Y yield
ii. 3m10s inversion: 10Y yield below 3-month T-bill
iii. Typical lead time: 6–18 months before recession
iv. Mechanism: Markets expect Fed to cut rates in response to coming weakness

Curve inversion is necessary but not sufficient for recession. Duration and depth of inversion also matter.

Remark 12.2.52. *Fixed Income in Different Macro Regimes*

i. Growth Shock (positive): Rates rise, curve steepens, credit spreads tighten
ii. Recession: Rates fall sharply, curve bull steepens then flattens, credit spreads widen
iii. Inflation Shock: Nominal rates rise, real rates may fall initially, curve bear steepens, TIPS outperform nominals
iv. Deflation/Disinflation: Nominal rates fall, breakevens collapse, duration rallies
v. Financial Crisis: Flight to quality (Treasuries rally), credit spreads explode, funding markets seize

Fixed income tells what the market thinks will happen before it happens. The curve summarises the entire macro narrative.

12.2.5 Commodities in Global Macro

Commodities are driven by physical supply and demand, inventories, and geopolitics. They are highly sensitive to growth cycles, inflation shocks, and currency moves (especially USD). Commodities anchor macro views in real economic constraints.

Remark 12.2.53. *Contango and Backwardation*
The shape of the futures curve reveals market structure:

i. Contango: Near-term futures trade cheaper than further-dated contracts. Exists due to storage costs, financing costs, and opportunity cost of capital. Normal state for storable commodities like gold.
ii. Backwardation: Near-term futures trade at premium to further-dated contracts. Occurs during supply shortages or immediate demand spikes.
iii. Agricultural seasonality: Curves can exhibit both. Contango after harvest (abundant supply, storage costs), backwardation before harvest (depleted stocks, tight supply).

Contango creates negative roll yield; backwardation creates positive roll yield.

Remark 12.2.54. *Commodity Indices*
Major commodity indices for benchmarking and trading:

i. CRB (Commodity Research Bureau) Index: 22 commodities, broad representation of prices.
ii. S&P GSCI: Goldman Sachs Commodity Index. Production-weighted, heavily energy-exposed ($> 50\%$).
iii. Bloomberg Commodity Index (BCOM): Diversified, liquidity-weighted.
iv. ETFs: DBC (Deutsche Bank commodities, $> 50\%$ energy), DBA (agriculture only, includes corn, soybeans, sugar, cattle, coffee, cocoa).

Caution for ETFs: contango and roll costs can cause significant performance drag versus spot prices.

Remark 12.2.55. *Commodity Risk and Return Profile*
Commodities have distinct risk characteristics:

i. Higher volatility than FX, equities, and fixed income
ii. Historically attractive Sharpe ratio as a relative comparison
iii. Prior to 2008, exhibited negative correlation with equities (diversification benefit)
iv. Post-2008, correlations with risk assets increased significantly

v. Supply shocks can cause extreme price moves (geopolitics, weather, production disruptions)

Commodities are particularly useful as inflation hedges and for expressing views on global growth.

Remark 12.2.56. *Crude Oil Fundamentals*
Oil is the most important energy commodity, accounting for 33%of global energy demand:

i. Quality measures: API gravity (higher = lighter) and sulphur content (lower = sweeter)
ii. Light sweet crude: Most desirable (WTI, Brent). Heavy sour crude least desirable (Venezuela, Mexico)
iii. Products: Gasoline (cars), kerosene (diesel, jet fuel), heating oil, LPGs
iv. Global consumption approximately 90 million barrels per day

Remark 12.2.57. *Major Oil Benchmarks*
Key oil benchmarks and their characteristics:

i. WTI (West Texas Intermediate): Light sweet crude. Trades on NYMEX (CL). Delivery at Cushing, Oklahoma. Contract: 1,000 barrels. ETF: USO.
ii. Brent Crude: North Sea light sweet crude. Trades on ICE (CO). Prices more than half of world's internationally traded crude. Delivery at Sullom Voe, Scotland.
iii. WTI-Brent Spread: Historically WTI traded at premium (higher grade), but infrastructure constraints at Cushing and Middle East supply sensitivity pushed Brent to premium.

Remark 12.2.58. *OPEC and Oil Supply*
OPEC controls significant portion of global oil supply:

i. 12 member countries controlling $> 35\%$ of global production, 73% of reserves
ii. Saudi Arabia has largest spare capacity. "The Call on OPEC" typically means Saudi increases production
iii. Key NOCs: Saudi Aramco (largest company by value, 1/6 of global reserves), NIOC (Iran), PDVSA (Venezuela), INOC (Iraq), KOC (Kuwait)
iv. OPEC quota system attempts to coordinate production to stabilise prices

Political risk in OPEC members is critical for oil price forecasting.

Remark 12.2.59. *Oil Chokepoints and Infrastructure*
Critical infrastructure for global oil flows:

i. Strait of Hormuz: Most important chokepoint. 17 million bbl/d (20% of global traded oil). Between Iran and Oman. Geopolitical flashpoint.
ii. Strait of Malacca: 14 million bbl/d. Connects Indian and Pacific Oceans. Critical for Asian supply.
iii. Mandab Strait/Suez Canal: 3 million and 2 million bbl/d respectively. Piracy risk.
iv. Pipelines: Druzhba (Russia to Europe, 2,300 miles, 1.4 million bbl/d). Keystone (Canada to US).

Supply disruption at chokepoints causes immediate price spikes, especially in Brent.

Remark 12.2.60. *Crack Spread*
The crack spread measures refining margins:

i. Spread between crude oil input and refined product output (gasoline, heating oil)
ii. $3:2:1$ Crack Spread: 3 barrels crude vs. 2 barrels gasoline + 1 barrel heating oil (most common)
iii. High crack spread: refiners profitable, incentive to increase production
iv. Low/negative crack spread: margins squeezed, refiners slow production
v. Factors: Cushing supply, driving demand, refinery shutdowns, hurricanes

Conversion: gasoline/heating oil in cents/gallon, crude in dollars/barrel. 1 barrel = 42 gallons.

Remark 12.2.61. *Natural Gas*
Natural gas is the cleanest fossil fuel (30–50% less CO2 than gasoline/coal):

i. US and Russia account for 38% of global production
ii. Shale revolution: Horizontal drilling and hydraulic fracturing transformed US from importer to exporter
iii. Flaring: Burning unwanted natural gas, common in US due to regulatory requirements

iv. LNG: Liquefied at $-260°$F for transport, 600x less volume than gas state

v. Uses: Power generation, heating, fertiliser (ammonia), transportation

Natural gas futures (NG) trade on NYMEX. Contract: 10,000 MMBtu, delivery at Henry Hub, Louisiana. Quoted in \$/MMBtu. ETF: UNG (subject to contango drag).

Remark 12.2.62. *Gold*
Gold is both precious metal and monetary asset. Currency symbol: XAU.

i. Jewelry: 40%+ of demand. Investment: second largest demand driver

ii. Central banks hold 31,920 tonnes. US largest holder (8,134 tonnes, 73%of reserves)

iii. China overtook South Africa as largest producer

iv. India largest consumer (50% used for weddings)

v. Real rates: Gold inversely correlated with real interest rates. Negative real rates support gold.

Trading: GLD ETF (1/10 oz per share), gold futures (GC) on COMEX (100 troy oz contract).

Remark 12.2.63. *Silver*
Silver has both monetary and industrial properties. Currency symbol: XAG.

i. Highest electrical conductivity of all metals

ii. Industrial applications: 44% of demand (batteries, electronics, solar panels)

iii. Jewelry/silverware: 22% of demand

iv. Mexico and China account for 47% of global production

v. More volatile than gold; often trades as leveraged gold play

Trading: SLV ETF (1 oz per share), silver futures (SI) on COMEX (5,000 troy oz). "Triple 9s" (99.9% purity) is deliverable grade.

Remark 12.2.64. *Platinum*
Platinum is extremely rare with concentrated supply:

i. South Africa holds 80% of global reserves

ii. Autocatalyst (catalytic converters): 38% of demand

iii. Jewelry: 31% of demand

iv. Almost twice as heavy as gold (density 21 vs. 19)

v. Play on emerging market growth and automotive demand

Platinum futures (PL) trade on COMEX (50 troy oz). Platinum/gold ratio is a key relative value indicator.

Remark 12.2.65. *Copper*
Copper is considered a leading economic indicator:

i. Chile dominates production (27%); China dominates consumption (43%), stockpiles are closely watched

ii. Uses: Building construction (33%), infrastructure (15%), equipment manufacturing (52%)

iii. Major input for urbanising economies, plays into EM growth story

iv. Recycled copper ("secondary copper") represents significant market share

Trading: LME (LMCADS03, \$/ton, 25 tons/contract) or NYMEX (HG, cents/lb, 25,000 lbs/contract).

Remark 12.2.66. *Aluminium*
Aluminium is the most abundant metal in Earth's crust:

i. China accounts for 40%+ of both production and consumption

ii. Lightweight, durable, does not corrode

iii. Uses: Construction, transportation (less energy needed to move lighter vehicles), packaging

iv. One of few commodities where China is self-sufficient

v. Energy-intensive to produce (aluminium smelters locate near cheap power)

Trading: LME (LMAHDS03, \$/ton, 25 tons/contract).

Remark 12.2.67. *Agricultural Commodities Overview*
Key characteristics of agricultural trading:

i. Supply determined by weather, planting decisions, yields
ii. Demand relatively inelastic (food consumption stable)
iii. US is largest player in most agricultural products
iv. Best agriculture traders employ full-time meteorologists
v. Crop calendars define sensitive periods for each commodity
vi. Planting, flowering/silking, and harvest are critical stages

Remark 12.2.68. *Corn*
Corn is the largest cereal crop globally:

i. US accounts for 35%+ of production, 30%+ of consumption, largest exporter
ii. Western Corn Belt (Iowa, Illinois, Nebraska) produces 47% of US corn
iii. Uses: Livestock feed, ethanol (27% of US demand, growing), food
iv. Ethanol mandate: US Energy Policy Act 2005 requires ethanol blending, creating structural demand
v. China demand growth: feedstock demand from 0% (2007) to 5%+ of imports

Trading: CBOT (C), cents/bushel, 5,000 bushels/contract. 1 bushel = 56 lbs.
Delivery months: H, K, N, U, Z. Critical period: June–August (silking).

Remark 12.2.69. *Wheat*
Wheat has been cultivated for $10,000+$ years:

i. EU, China, India are largest producers; US is fourth largest but largest exporter
ii. Classifications: season (winter/spring), gluten content (hard/soft), colour (red/white/amber)
iii. Uses: Bread, flour, noodles, beer, feedstock
iv. Close substitute for corn. Prices move together
v. Russia export bans have caused supply shocks

Trading: CBOT (W), cents/bushel, $5,000$ bushels/contract. 1 bushel = 60 lbs.
Critical period: heading stage.

Remark 12.2.70. *Soybeans*
Soybeans are a critical oilseed with high protein content:

i. US largest producer (31%), Brazil second largest but largest exporter
ii. China largest consumer (crush) and importer (63% of global imports)
iii. Products: Soybean meal, vegetable oil, soy milk, tofu, edamame
iv. 90%+ of soybeans are genetically engineered (lower pesticide costs)
v. Brazil/Argentina weather critical, neighbouring countries, correlated weather risk

Trading: CBOT (S), cents/bushel, 5,000 bushels/contract. 1 bushel = 60 lbs. Most vulnerable during flowering.
Delivery months: F, H, K, N, Q, U, X.

Remark 12.2.71. *Cotton*
Cotton is the most important textile fiber:

i. China, India, US are largest producers (65% of production)
ii. China largest producer and consumer (textile industry)
iii. US third largest producer but largest exporter (30% of exports)
iv. Classifications: character (strength), grade (colour/purity), staple (fibre length)
v. More drought-resistant than other crops

Trading: ICE (CT), cents/lb, $50,000$ lbs/contract (≈ 104 bales). 1 bale = 480 lbs.
Delivery months: H, K, N, V, Z.

Remark 12.2.72. *Coffee*
Coffee has distinct supply characteristics:

i. Two types: Arabica (60% of production, higher grade, higher altitude) and Robusta (stronger taste, more caffeine, lower altitude)

ii. Brazil largest producer/exporter (37%). Vietnam second (mostly robusta)

iii. EU and US are largest importers (46% and 23% respectively)

iv. Biennial cycle: Coffee trees alternate high and low yield years

v. Tree life cycle approximately 20 years

Trading: ICE (KC), cents/lb of arabica, $37,500$ lbs/contract. Industry measures in bags (60 kg = 132 lbs). Critical period: blooming (water is critical).

Remark 12.2.73. *Ending Stocks and Supply/Demand Balance*
Ending stocks (inventory) are the most important driver of commodity prices:

i. Ending Stocks = Beginning Stocks + Production − Consumption

ii. Low stocks relative to consumption (stocks-to-use ratio) = price support

iii. High stocks = price pressure

iv. USDA WASDE reports provide monthly supply/demand estimates

v. Small changes in production or consumption can cause large price moves when stocks are tight

Remark 12.2.74. *Weather Risk in Agriculture*
Weather is the primary supply shock driver in agriculture:

i. Drought during critical growth stages (flowering, silking) is most damaging

ii. Livestock cascade: high grain prices → expensive feed → early livestock slaughter → short-term meat supply up, future supply down

iii. El Niño/La Niña cycles affect global weather patterns

iv. Monitor: NOAA drought monitors, crop condition reports, weather forecasts for key growing regions

Remark 12.2.75. *Commodities in Different Macro Regimes*

i. Growth Shock: Commodity demand rises, prices increase, especially energy and industrial metals

ii. Recession: Demand destruction, prices fall (except gold as safe haven)

iii. Inflation Shock: Commodities outperform as inflation hedge; energy and agriculture lead

iv. USD Weakness: Commodity prices rise (inverse correlation with dollar)

v. Geopolitical Crisis: Energy spikes on supply fears; gold rises on safe-haven demand

Commodities are the inflation and real-economy link in the macro framework.

12.3 Practitioner Frameworks

This section distills operational frameworks from leading global macro practitioners, drawn primarily from Steven Drobny's interviews [s7-drobny]. Focus is on actionable principles rather than historical narratives.

12.3.1 Portfolio Construction and Risk Frameworks

Remark 12.3.1. *Systematic Risk Premia Allocation (Leitner Framework)*

i. Equal 20% risk allocation across: Equities (value screens), Fixed Income (5x leveraged short-duration), FX (carry + vol premia), Commodities (roll yield), Real Estate

ii. Additional 10% for 1–2 high-conviction opportunities annually

iii. Anchor to data first, then narratives; actively seek disconfirming evidence

iv. Not being invested = foregoing risk premia; short positions pay away premia

Remark 12.3.2. *Convexity and Optionality (Siva-Jothy/Anonymous Currency Manager)*

i. Maintain long gamma; never short gamma in macro portfolios

ii. 50–60% options (3–6 months); 75% of price moves occur 10–15% of time around data releases

iii. Systematically long options to capture regime shifts; leverage is function of correct macro analysis

iv. Never forced seller; liquidity-only mandate (G10 only)

v. Timing: buy options when consensus says inevitable and market at extreme

Remark 12.3.3. *Trade Construction Discipline (Drobny/Wadhwani)*

i. Views without actionable trades are useless; compare fundamentals to market perception

ii. Look for positioning extremes (2–5 year timeframes); CA deficit currency rising = longs building

iii. Minimum risk/reward $4:1$; after two stop-outs on same thesis, move on

iv. 3–4 major opportunities annually; overtrading as damaging as running losers

v. Positioning and sentiment dominate fundamentals short-term

vi. Stop-loss most critical; exit immediately when thesis invalidated

Remark 12.3.4. *Fixed Income Specific (Porter)*

i. Time horizon arbitrage: institutional short-termism creates medium-term opportunity

ii. Focus on one-year, one-year forwards (most volatile, greatest overshooting)

iii. Interest rate cycles 16–24 months; maximise income per VaR via front-end

iv. Curve steepener $\approx$ 90% bullish bonds; flattener $\approx$ bearish

12.3.2 Asset-Class Specific Frameworks

Remark 12.3.5. *Commodity Framework (Anderson)*

i. Pure microeconomics: supply/demand, cost curves, competitive positioning

ii. Fundamentals over price: ignore day-to-day moves, focus on economics

iii. Mean reversion: margins revert to mean, not prices

iv. Contrarian timing: front-page hysteria = reversal indicator (sell euphoria, buy panic)

v. Micro-economic inevitability: stress-test thesis against adverse macro scenarios. Long = assume low growth; short = assume high growth

vi. On-the-ground research: observe anecdotal evidence, see fundamental change before markets price it

vii. Only invest low-cost producers; high-cost assets can't be bought cheap enough

viii. Reject VaR for concentrated portfolios (assumes diversification); use volatility-adjusted sizing

Remark 12.3.6. *Equity Macro Framework (Bessent)*

i. Express macro views through equities: 50% of stock move = market, 30% = sector, 20% = stock-specific

ii. Bottom-up research for macro confirmation: micro insights reveal macro picture

iii. Entry points critical: good entry means rarely stopped out

iv. Concentrated portfolio: 8–14 large positions, not equally weighted

v. Paradigm shifts = major profits: identify when game changes, not how to play game better

Remark 12.3.7. *FX Framework (Anonymous Currency Manager)*

i. FX is tail of credit curve: Fed funds → short rates → govts → credit → equities → FX

ii. Currencies reflect relative sentiment/credit spreads between economies

iii. Three currency blocs: USD, EUR, Asia

iv. Directional concentration: 3–4 macro calls maximum (bonds, equities, USD vs blocs)

v. Five-month time horizon for thesis validation

vi. Basic law: bullish = long or flat, bearish = short or flat. Never relative value

Remark 12.3.8. *Emerging Markets Crisis Framework (Dimitrijević)*

i. Country implosion acts like company collapse: all instruments correlate

ii. Strict limits: 10% global macro fund, 15% EM fund per country

iii. Best EM trades after panics/disasters

iv. Clean sheet valuation: ignore yesterday's price. Ask: "would we invest at current price with clean sheet?"

v. Stop-loss mechanics: work in continuous markets, dangerous in distressed/value plays

vi. Internal stops only (counterparties invariably hit external stops)

vii. One-month cool-off after stop-out

viii. Hedge mismatch: short rich asset class as hedge to long attractive asset class within same country

12.3.3 Execution and Flow Analysis

Remark 12.3.9. *Flow and Cross-Market Linkages (Harris)*

i. Flow matrix: mental model tracking global money flows seeking highest risk-adjusted return

ii. Money craves stability; fear greater than greed drives capital flight

iii. Cross-market linkages: geopolitical events ripple through asset classes (e.g., Brazilian real devaluation → sell soybeans as farmers hedge USD revenues)

iv. Slower Fool Theory: exit before discontinuous leap into illiquidity

v. Risk management: 80% trades wrong, 20% profitable must exceed losses

vi. Small probes (reconnaissance) test thesis before scaling

vii. Never pray trades back to breakeven; clear books, start fresh

viii. Best at 3–4 concentrated positions; avoid overtrading

Remark 12.3.10. *Contrarian Indicators*

i. Concept exhaustion: magazine covers = fully discounted

ii. Gestalt shifts: volatility spikes when protection buying signals crowded positioning

iii. "Frozen ice theory": strong trades at edges, crowded in middle on proxies

iv. Central banks often counter-indicators in market timing

Chapter 13

Market microstructure and market making

Notes awaiting migration. Entries already migrated from this chapter are in Chapter 8; the record of each block's destination is in `notes/migration.md`.

Market microstructure studies how trading rules, intermediaries, and order flow transform latent trading demands into transactions, prices, and allocations. The central departure from frictionless asset pricing is that the trading mechanism is not neutral: priority rules, transparency, inventory, information, and immediacy affect the prices at which trades occur.

This chapter treats a market as an engineered trading system rather than a single supply-demand crossing. The practical questions are: who wants immediacy, who absorbs the other side, what information is revealed by the order, how the venue ranks competing messages, and how much of the final execution cost comes from spread, market impact, delay, adverse selection, and operational frictions.

The source base is front-loaded here to avoid repeated citations throughout the section. Demsetz provides the canonical transaction-cost view of why immediacy has a price [demsetz-1968]. O'Hara and Madhavan frame the theoretical links among information, strategic trading, dealer behavior, and price discovery [catalogue-0496]; [madhavan-2000]. Harris supplies the practitioner taxonomy of traders, orders, markets, and trading costs [alpha-harris]. Hasbrouck anchors the empirical treatment of trades, quotes, order flow, and high-frequency data [s6-hasbrouck]. Lehalle and Laruelle connect these ideas to fragmented electronic markets, smart routing, latency, and modern execution practice [alpha-laruelle].

The section therefore moves from market organization to order mechanics, then to information and spreads, empirical order-flow analysis, and market making; algorithmic execution and asset-class-specific microstructure, which followed in a separate legacy chapter, have since been migrated to Chapter 9 and other chapters. The common thread is implementation: every model is useful only insofar as it explains how real orders become real fills under a particular market design.

13.1 Foundations of Trading and Market Organization

13.1.1 Roles of Financial Markets

Principle 13.1.1. *Core Market Functions*
A financial market is useful when it performs four linked functions:

i. *liquidity provision*: converts desired portfolio changes into executable trades;

ii. *price discovery*: aggregates dispersed beliefs and information into observable prices;

iii. *risk transfer*: reallocates exposures to agents more willing or able to bear them;

iv. *capital allocation*: directs savings toward issuers and projects with attractive risk-adjusted uses of capital.

Microstructure studies the frictions inside these functions rather than assuming they occur automatically.

Principle 13.1.2. *Immediacy Has a Price*
Asynchronous order arrival makes immediacy costly. The bid-ask spread compensates traders who absorb inventory, funding, processing, and adverse-selection risks.

Proposition 13.1.3. *Zero-Sum Trading Accounting*
For a trade at P with common terminal value V, buyer and seller gains sum to zero: $(V - P) + (P - V) = 0$. Frictions make trading negative-sum unless it creates information, liquidity, risk-transfer, or allocation value.

Proof. Buyer gain plus seller opportunity gain is zero before costs; costs subtract. □

Taxonomy 13.1.4. *Trading Benefit Segmentation*

i. *private benefits*: accrue to traders through exchange, hedging, funding, immediacy, or speculation;

ii. *public benefits*: accrue broadly when liquid markets and informative prices improve allocation decisions.

Definition 13.1.5. *Liquidity Provision*
Liquidity is the ability to trade size quickly, cheaply, and with limited price impact.
Supplied by limit orders, dealers, market makers, internalisers, and crossing systems, and consumed by marketable orders and urgent traders.

Mechanism 13.1.6. *Price Discovery*
Trades, quotes, cancellations, and depth transmit information about value, urgency, and risk-bearing capacity. Price discovery is strong when prices update quickly without being dominated by transient imbalances.

Mechanism 13.1.7. *Risk Transfer*
Markets separate cash-flow ownership from risk-bearing. Hedgers shed unwanted exposure; speculators, dealers, and market makers accept it for compensation.

Principle 13.1.8. *Capital Allocation*
Primary markets move capital from savers to issuers. Liquid and informative secondary markets provide exit prices, lower financing costs, and discipline poor uses of capital.

Mechanism 13.1.9. *Informative Prices and Allocation*
Primary prices screen new projects; secondary prices help investors judge managers, compensate performance, and reallocate control. Noisy prices weaken all three channels.

Principle 13.1.10. *Mechanism Relevance*
In Walrasian abstraction, the mechanism disappears after supply and demand clear. In real markets, sequencing, asymmetric information, costly immediacy, and priority rules make the mechanism part of price formation.

Caveat 13.1.11. *Microstructure Trade-Off*
Market design rarely improves every function at once. Transparency, spread size, venue opacity, and priority rules trade off liquidity provision, execution cost, information leakage, and public price discovery.

13.1.2 Market Participants and Trading Motives

Model 13.1.12. *Trading Game View*
A trading mechanism is a game in which participants submit orders under venue rules. The players include customers, brokers, dealers, market makers, and traders acting for their own account. Some demand liquidity, some supply liquidity, some carry information, and some process orders.
Intuition: a trader's best order depends on what others may know, how fast they can react, and how the venue ranks competing messages.

Notation 13.1.13. *Long and Short Positions*
A trader with a *long position* owns the asset and benefits when its price rises. A trader with a *short position* has sold an asset not owned, or has equivalent negative exposure, and benefits when its price falls. Closing a short position is called *covering*.

Taxonomy 13.1.14. *Motive Segmentation*

i. *utilitarian*: solve an outside problem;

ii. *profit-motivated*: expect trading profits above costs;

iii. *futile*: expect profit without enough information, skill, discipline, or agency control.

Motive is inferred from context, horizon, and exposure, not order side alone.

Caveat 13.1.15. *Multiple Trader Identities*
Traders often combine motives. A pension fund may invest, hedge FX, and time factors; an HFT may make markets in one book and arbitrage another.

Principle 13.1.16. *Patience Defines the Liquidity Side*
Impatient traders demand liquidity by initiating trades. Patient traders supply liquidity by waiting, posting, quoting, or making themselves discoverable.
The same institution can demand liquidity for one order and supply it for another.

Taxonomy 13.1.17. *Buy-Side Motives*
Buy-side traders use markets to solve problems outside the market itself:

i. investors move wealth across time;

ii. borrowers move wealth from the future to the present;

iii. hedgers reduce business or portfolio risk;

iv. asset exchangers acquire assets they value more than what they tender;

v. gamblers trade for entertainment or payoff convexity rather than investment efficiency.

These traders buy liquidity services rather than sell them.

Principle 13.1.18. *Instrument Choice Follows the Problem*
Investors and borrowers care about after-cost return and funding access. Asset exchangers need the right asset. Hedgers need correlation, maturity, and liquidity. Gamblers care about payoff shape.

Taxonomy 13.1.19. *Hedge-Vehicle Fit*

i. *forwards*: customise asset, date, size, and delivery, but create search, credit, and exit frictions;

ii. *futures*: standardise terms and clearing, improving liquidity while introducing basis and delivery mismatch;

iii. *options*: cap downside for a premium, preserving convex upside or nonlinear exposure;

iv. *swaps*: exchange cash-flow risks over time, especially interest-rate, currency, credit, or inflation exposures.

The best hedge removes the intended risk without adding larger liquidity, basis, or counterparty risks.

Taxonomy 13.1.20. *Information-Motivated Traders*
Informed traders trade when prices lag information. Value traders use fundamentals; news traders use recent information; arbitrageurs use relative mispricing.
Their orders improve prices only as the market absorbs the information.

Taxonomy 13.1.21. *Informed Trader Liquidity Role Segmentation*

i. *value traders*: supply liquidity when uninformed demand pushes price away from fundamental value;

ii. *news traders*: demand liquidity when new information has not yet been incorporated into price;

iii. *arbitrageurs*: supply or demand liquidity across related instruments to close relative mispricing.

The same trader can switch roles. The useful distinction is whether the order corrects stale prices, absorbs non-value flow, or links related prices.

Mechanism 13.1.22. *Arbitrageur Economic Function*
Arbitrageurs trade relative value. They connect buyers and sellers across instruments, venues, maturities, currencies, or risk forms. Their activity enforces the law of one price only after costs, funding, hedge error, and execution risk are small enough for the trade to be worth doing.

Taxonomy 13.1.23. *Liquidity Supplier Segmentation*

i. *market makers*: supply immediacy for small to moderate size at narrow spreads, but lose to informed flow and adverse inventory moves;

ii. *block dealers*: supply depth to known clients by warehousing risk, but lose when client information or unwind cost is misread;

iii. *value traders*: supply depth and resiliency when price deviates from fundamental value, but lose to better-informed value traders or financing pressure;

iv. *precommitted traders*: post aggressive limit orders to lower the cost of trades they already want to do, but risk non-execution or adverse selection;

v. *arbitrageurs*: move liquidity across related markets, but lose to faster arbitrageurs, legging risk, funding shocks, and stale hedges.

Treating all passive flow as "market making" hides the source of depth.

Taxonomy 13.1.24. *Profit-Motivated Trader Segmentation*

i. *informed speculators*: profit when prices move toward information they already understand;

ii. *order anticipators*: profit by predicting trades other participants must make;

iii. *bluffers and manipulators*: profit only if others misread false signals;

iv. *dealers*: sell immediacy and inventory capacity rather than directional views.

Informed speculators aid price discovery; dealers supply liquidity. Middle categories mainly redistribute surplus.

Principle 13.1.25. *Speculation Requires Comparative Advantage*
Profitable speculation requires more than being intelligent or informed in isolation. The trader must have an advantage over the marginal counterparty after spread, fees, financing, market impact, and operational error.

Taxonomy 13.1.26. *Trading Edge Segmentation*

i. *information*: faster, cleaner, or more exclusive interpretation of value-relevant facts;

ii. *model*: better mapping from data to price, hedge, or risk;

iii. *execution*: lower leakage, better routing, queue skill, or lower impact;

iv. *risk bearing*: ability to hold inventory or convergence risk when others cannot;

v. *discipline*: sizing, stopping, and avoiding trades outside the edge.

An edge that cannot survive trading costs is not a trading edge.

Caveat 13.1.27. *Skill and Luck Look Similar Ex Post*
Large realised profits may come from skill, favourable exposure, or simple luck.
A useful skill claim must explain why the decision was right before the outcome was known, why the edge should persist, and why it is not already competed away.

Proposition 13.1.28. *Past Profit Alone Does Not Identify Skill*
Let future excess profit be $\Pi_{t+1} = S_{t+1} + L_{t+1}$, where S is persistent skill and L is zero-mean luck.
If S_{t+1} is independent of past realised profits conditional on current information, then past profit has no incremental predictive value for Π_{t+1}.

Proof. Conditional on current information $\mathcal{F}_t$, $\mathbb{E}[\Pi_{t+1} \mid \mathcal{F}_t, \Pi_t] = \mathbb{E}[S_{t+1} \mid \mathcal{F}_t] + \mathbb{E}[L_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\Pi_{t+1} \mid \mathcal{F}_t]$. □

Corollary 13.1.29. *Profit-Motivated Flow Needs Non-Profit Motives*
Speculators and dealers cannot all profit by trading only with each other. Their gains require traders who pay for immediacy, risk transfer, information, or entertainment.

Proof. Let G_i be trader i's gross trading gain in a closed set of profit-motivated traders, and let $C_i \geq 0$ be trading costs. By zero-sum accounting, $\sum_i G_i = 0$, so aggregate net profit is $\sum_i (G_i - C_i) = -\sum_i C_i \leq 0$.
Thus the group cannot earn positive aggregate net trading profit from itself; positive profits require counterparties with offsetting trading losses or non-trading service value. □

Corollary 13.1.30. *Turnover Drag for Active Management*
Let R_M be the gross market return, a_i active manager i's gross active excess return, $c_i \geq 0$ the manager's trading and fee drag, and w_i value weights. If active managers aggregate to the market portfolio, $\sum_i w_i a_i = 0$, so aggregate net active excess return is $\sum_i w_i(a_i - c_i) = -\sum_i w_i c_i \leq 0$.
A single active manager beats a passive fund with drag c_P only if $a_i > c_i - c_P$.

Proof. Active deviations clear against each other in aggregate, giving $\sum_i w_i a_i = 0$. Subtracting costs gives the aggregate inequality. Relative to passive exposure, manager i must cover the incremental drag $c_i - c_P$. □

Definition 13.1.31. *Noise and Liquidity Traders*
Noise traders trade for non-value reasons: rebalancing, cash needs, index changes, behavioural demand, or errors. They mask informed flow and give liquidity suppliers less toxic orders.

Taxonomy 13.1.32. *Futile Trader Segmentation*

i. *pseudo-informed*: trade stale or already-public information as if it were private;

ii. *inefficient*: have a plausible strategy but execute, size, or control risk poorly;

iii. *victimised*: rely on advisers, brokers, or agents whose incentives are misaligned;

iv. *rogue*: hide losses or risks because internal controls fail.

Futile traders matter because their losses can fund informed traders and dealers, while their distress can still damage clients, firms, or market confidence.

Mechanism 13.1.33. *Dealers and Market Makers*
Dealers trade for their own account. Market makers quote prices or otherwise stand ready to trade, often holding inventory only until an offsetting order arrives. Their compensation comes from spreads, rebates, information, or client relationships; their risks are inventory, adverse selection, funding, and operational failure. They are paid for immediacy: the client trades now, while the market maker manages the uncertainty of unwinding later.

Taxonomy 13.1.34. *Sell-Side Intermediaries*
Brokers arrange trades for clients without intending to own the position. Dealers trade as principal. Broker-dealers combine both functions, sometimes arranging agency trades and sometimes committing their own capital. The sell side exists because buy-side traders pay for search, immediacy, execution expertise, clearing access, and balance-sheet intermediation.

Taxonomy 13.1.35. *Broker Service Segmentation*

i. *search*: find counterparties, dealers, venues, or natural block liquidity;
ii. *representation*: submit, monitor, and negotiate client orders;
iii. *access*: connect clients to exchanges, dealers, ECNs, and clearing members;
iv. *post-trade*: manage confirmation, clearing, settlement, custody, margin, and records;
v. *information control*: expose enough order interest to trade without revealing the full parent order.

Taxonomy 13.1.36. *Block-Trading Role Segmentation*

i. *block initiator*: trader with a parent order too large for normal displayed liquidity;
ii. *block liquidity supplier*: trader, dealer, or institution willing to fill the block;
iii. *block dealer*: principal that facilitates the block and then unwinds inventory;
iv. *block broker*: agent that assembles counterparties without intending to own the risk;
v. *upstairs market*: relationship and telephone market used to find size without public exposure.

Block trading is mostly a search and information-control problem, not a different asset.

Principle 13.1.37. *Block Dealers vs Block Brokers*
Dealers are useful when immediacy matters and balance sheet can absorb the position.
Brokers are useful when the block is too large or information-sensitive for one principal, so assembling natural counterparties is cheaper than warehousing risk.
Broker-dealers can be credible when they are willing to participate in the trade they recommend.

Taxonomy 13.1.38. *Block Trader Skill Segmentation*

i. *latent demand discovery*: know who might trade if contacted;
ii. *motive audit*: infer whether the initiator is uninformed, informed, honest, or strategic;
iii. *exposure control*: reveal size only to counterparties who will not trade ahead;
iv. *transaction-cost analysis*: price facilitation, unwind risk, and likely market impact;
v. *reputation management*: make future counterparties willing to participate.

Caveat 13.1.39. *Broker Agency Conflict*
Brokers are agents, but commissions, soft dollars, payments for order flow, inventory, and affiliated dealers can pull routing decisions away from pure client execution quality.

Practitioner Note 13.1.40. *Institutional Investors and HFTs*
Institutional investors often trade large parent orders that must be hidden, sliced, or negotiated to reduce market impact. High-frequency traders use speed, automation, and short holding periods; they may act as market makers, arbitrageurs, statistical traders, or order anticipators. Their role depends on strategy.

Taxonomy 13.1.41. *HFT Identification Markers*

i. automated order generation, routing, execution, and cancellation;
ii. low-latency infrastructure, co-location, and direct market-data feeds;
iii. short holding periods and low overnight inventory;
iv. high message counts relative to executed trades;
v. market making, cross-venue arbitrage, or very short-horizon statistical trading.

These markers describe technology and horizon, not whether the strategy is beneficial.

Mechanism 13.1.42. *HFT as Fragmentation Link*
Fast market makers connect fragmented venues by quoting related prices, arbitraging stale quotes, and hedging inventory across books. Their compensation is the spread, rebates, and short-lived information advantages; their cost is technology, inventory risk, and adverse selection. They act like moving bridges between books: useful when they keep prices aligned, fragile when those bridges are cancelled under stress.

Caveat 13.1.43. *Tighter Spread, Different Cost*
An HFT can improve the displayed spread by stepping inside an older quote, but investor execution cost may not fall by the same amount. Smaller trades, more prints, and higher routing complexity can shift fixed and reporting costs toward slower investors.

Caveat 13.1.44. *Investor Size Asymmetry*
The uncertainty created by meeting a fast counterparty is not borne equally. A small investor sees high variance in a small number of trades; a large institution diversifies this uncertainty across many child orders but still pays aggregate spread, impact, and signalling costs.

Caveat 13.1.45. *Blurring of Roles*
Fragmentation blurs the distinction between liquidity takers and providers. The same agent may consume liquidity in one venue, supply it in another, and intermediate price formation through routing speed.

13.1.3 Trading Venues and Market Architecture

Principle 13.1.46. *Venue as Rule System*
A market is a rule system that governs access, order submission, visibility, execution, clearing, and settlement.

Taxonomy 13.1.47. *Market-Structure Segmentation*

i. *session*: when orders may enter and trades may occur;
ii. *execution*: who arranges the match and which rule prices the trade;
iii. *information*: who observes orders, quotes, trades, and when;
iv. *access*: how traders send orders, cancels, and reports;
v. *price grid*: the tick sizes and admissible price points.

Market structure matters because it defines both tradability and information access.

Taxonomy 13.1.48. *Access Form Segmentation*

i. *physically convened*: traders meet on a floor or pit;
ii. *distributed*: traders negotiate by phone, message, or bilateral networks;
iii. *screen-based*: orders, quotes, and trades are submitted through electronic interfaces.

Taxonomy 13.1.49. *Market-Centre Segmentation*

i. *market centre*: any venue, dealer, broker system, or crossing mechanism that can arrange trades;
ii. *consolidated market*: traders for same instrument meet on one dominant centre or tightly linked book;
iii. *fragmented market*: the same instrument trades across several centres with shared information or routing;
iv. *segmented market*: traders select centres because order size, information, patience, access, or credit differs.

Fragmentation is not automatically bad; it is costly when the links among centres fail.

Definition 13.1.50. *Exchanges*
Exchanges centralise order interaction under membership, listing, trading, reporting, and clearing rules. They standardise access and make execution more reliable.

Mechanism 13.1.51. *Exchanges as Search Engines*
An exchange lowers bilateral search cost by collecting trading interest, ranking it under common rules, and publishing enough information for traders to find the best available counterparty.
Its economic service is not only matching; it is making liquidity searchable.

Definition 13.1.52. *OTC Markets*
OTC markets use bilateral or dealer-mediated negotiation rather than a central book. They fit bonds, swaps, FX, bespoke derivatives, and heterogeneous contracts.

Taxonomy 13.1.53. *Electronic Communication Networks (ECNs) and Alternative Trading Systems (ATSs)*
ECNs organise electronic matching outside traditional exchange structures. ATSs include dark pools, crossing systems, and specialised venues competing on anonymity, fees, block handling, or lower information leakage.

Taxonomy 13.1.54. *Dark Pools, Internalizers, and Crossing Networks*
Dark pools hide pre-trade interest; internalizers match against broker-dealer inventory or internal flow; crossing networks match natural buyers and sellers, often at lit-market reference prices. These venues reduce signalling but can weaken public price discovery.

Taxonomy 13.1.55. *Order-Flow Arrangement Segmentation*

i. *internalisation*: broker-dealer fills client orders as principal or against affiliated liquidity;
ii. *preferencing*: broker routes orders to a selected dealer or venue by arrangement;
iii. *payment for order flow*: liquidity supplier pays broker for routed orders;
iv. *internal crossing*: broker matches client orders against each other away from public books.

These arrangements lower search and commission costs when flow is uninformed, but fragment order interaction and create best-execution conflicts.

Taxonomy 13.1.56. *Brokered Transaction Segmentation*

i. *order flow*: broker receives client orders and matches, routes, or exposes them;
ii. *block*: broker searches quietly for the other side of a large trade;
iii. *offering*: broker distributes new or seasoned securities to investors;
iv. *corporate-control*: broker advises or introduces parties in merger and acquisition trades.

Taxonomy 13.1.57. *Request for Quotation (RFQ) vs Central Limit Order Book (CLOB)*
RFQ markets request quotes from dealers, often bilaterally. A CLOB exposes standing interest multilaterally; RFQ protects relationships and discretion, while CLOBs improve comparability and public interaction.

Taxonomy 13.1.58. *Designated Market Maker Segmentation*

i. *single specialist*: one member carries quoting, auction, and order-book duties for an instrument;
ii. *multiple DMM*: several assigned dealers share liquidity obligations and compete for flow;
iii. *third-market dealer*: quotes away from the listing exchange, usually with weaker formal duties;
iv. *pure CLOB*: no assigned dealer; public orders and voluntary market makers supply liquidity.

Design choice is whether mandatory liquidity support is worth granting privileged access or order-flow rights.

Taxonomy 13.1.59. *Market Information Systems*

i. *collection*: capture orders, quotes, trades, and participant reports;
ii. *distribution*: publish or sell feeds, tapes, and reference data;
iii. *routing*: transmit orders, cancels, acknowledgements, and fills;
iv. *presentation*: display books, quotes, trades, indications, and alerts;
v. *storage*: preserve audit trails and historical market data.

Mechanism 13.1.60. *Trading Systems and Technology*
Communications technology lowers the cost of knowing where liquidity is; computing lowers the cost of acting on it. Modern trading systems therefore compete not only on matching rules, but on market-data quality, order-routing speed, conditional order handling, and auditability.

Mechanism 13.1.61. *Automated Reporting Advantage*
When orders and trades are born digital, quotes, executions, order status, and timestamps can be reported faster and with fewer transcription errors. Floor systems need manual entry or support systems to reach comparable audit and market-data quality.

Caveat 13.1.62. *Stale Quotes and Vanishing Liquidity*
RFQ quotes can become stale before acceptance; displayed CLOB liquidity can vanish before an order arrives. In fragmented electronic markets, observed liquidity is an estimate, not a guarantee.

Process 13.1.63. *Post-Trade Infrastructure*
Trading does not end at execution:

i. *clearing* matches buyer and seller records and validates trade terms;

ii. *settlement* exchanges cash and securities;

iii. *custody* holds assets on behalf of clients;

iv. *depositories* immobilise or dematerialise securities so ownership can transfer by book entry.

Clearinghouses may become buyer to every seller and seller to every buyer, reducing counterparty risk through margin and default management.

Proposition 13.1.64. *Brokered Credit Compression*
With n traders, direct bilateral trading can require $n(n-1)/2$ credit relationships.
If clients trade through B clearing brokers, the relationship count falls to $n + B(B-1)/2$ when each client relies on one broker and brokers check each other.

Proof. The direct market must check every unordered trader pair. In the brokered market, each client checks one broker, and the remaining checks are among broker pairs. □

Practitioner Note 13.1.65. *Architecture Choice*
Venue choice depends on order size, urgency, information sensitivity, instrument standardisation, and counterparty risk. Small urgent orders favour visible and immediate venues; large or information-sensitive orders often favour dark, negotiated, or sliced execution.

Mechanism 13.1.66. *Liquidity Bridges*
Fragmented venues require routing, arbitrage, and market-making links between books. These links reduce price dispersion, but their compensation is paid through spreads, fees, latency races, or information leakage.

Practitioner Note 13.1.67. *Straight-Through Processing*
Straight-through processing sends trade details from execution to clearing, settlement, custody, and reporting with minimal manual intervention.

13.1.4 Auction and Dealer Market Structures

Taxonomy 13.1.68. *Execution-System Segmentation*

i. *quote-driven*: dealers supply liquidity through tradable quotes and balance sheets;

ii. *order-driven*: rules match public orders under priority and pricing rules;

iii. *brokered*: brokers search for counterparties and reveal interest selectively;

iv. *hybrid*: venue rules combine books, dealers, auctions, and negotiation.

Taxonomy 13.1.69. *Trading Session Segmentation*

i. *continuous*: trades can occur whenever compatible orders arrive;

ii. *call*: orders batch until a scheduled or triggered uncrossing;

iii. *rotation*: instruments are called sequentially;

iv. *extended*: trading occurs outside the regular session, usually with thinner liquidity.

Mechanism 13.1.70. *Open-Outcry Oral Auctions*
Open-outcry markets arrange trades through public bids, offers, and acceptances on a floor or pit. The economic idea is still useful electronically: trading interest should be exposed enough for competing traders to react, while the rule book decides whose interest has precedence.

Rule 13.1.71. *Public Exposure Rule*
An auction is open when bids, offers, and acceptances are announced to the crowd before a trade is final. Public exposure supports fairness and price competition, but it also reveals trading interest.

Taxonomy 13.1.72. *Floor vs Automated Auction Segmentation*

i. *operational fairness*: automated rules are easier to apply uniformly and audit;

ii. *fair access*: floors privilege physical presence; screens distribute access through networks;

iii. *capacity*: oral auctions are bounded by human attention; matching engines scale with system capacity;

iv. *speed*: electronic systems record and match standard orders faster;

v. *information exchange*: floor brokers can exchange soft information useful for blocks;

vi. *cost*: floor systems carry labour and real-estate costs; automated systems carry fixed technology and reliability costs.

Neither form dominates every use case; the valuable question is which trading problem the structure solves.

Proposition 13.1.73. *Fast-Market Capacity Condition*
If N traders submit messages at average rate λ and the auction mechanism processes messages at rate μ, orderly real-time handling requires $N\lambda < \mu$.

Proof. The aggregate arrival rate is $N\lambda$. When arrivals exceed processing capacity, the backlog grows on average, so quotes, order reports, and time precedence become unreliable. □

Taxonomy 13.1.74. *Order-Driven Rule Stack*

i. *precedence rules*: rank orders when more than one trader wants the same trade;
ii. *matching rules*: decide which buy and sell orders can trade;
iii. *pricing rules*: determine the transaction price once a match is feasible;
iv. *access rules*: decide who may submit, observe, cancel, or route orders.

Mechanism 13.1.75. *Continuous Auctions*
Orders execute whenever they arrive and match resting interest. Continuous trading provides immediacy but exposes prices to asynchronous order-flow imbalances.

Mechanism 13.1.76. *Call Auctions*
Orders accumulate, then execute at a single clearing price. Opening and closing calls concentrate liquidity when uncertainty and imbalance are high.

Mechanism 13.1.77. *Fixing Auctions*
Fixing auctions are call auctions used to create reference prices.
They are valuable when investors need one public price for index valuation, NAV computation, derivative settlement, opening risk transfer, or end-of-day inventory adjustment.

Principle 13.1.78. *Reference Price Monopoly*
Even when continuous trading fragments, primary exchanges often retain the auction flow that defines official reference prices. This makes auction volume structurally different from continuous-session volume and gives the primary market a separate source of liquidity concentration.

Taxonomy 13.1.79. *Auction Session Segmentation*

i. *opening call*: rebuilds the book after overnight information and sets the first tradable price;
ii. *intraday call*: concentrates liquidity around scheduled events or low-liquidity periods;
iii. *closing call*: sets the end-of-day reference price and absorbs benchmark-driven flow;
iv. *volatility call*: pauses continuous trading when price moves or imbalances breach venue rules.

Mechanism 13.1.80. *Dealer Markets*
Prices form through dealer quotes rather than a single public book. Dealers absorb timing gaps, manage inventory, and revise quotes as information and inventory change.

Mechanism 13.1.81. *Brokered Markets*
Brokers create liquidity by search rather than inventory commitment. This fits blocks, unique assets, and illiquid instruments where public standing orders would reveal too much or attract too little interest.

Caveat 13.1.82. *Floor Information Exchange*
Floor and brokered systems can reveal who might trade, whether more size remains, and whether the contra side appears informed. This helps blocks and relationship trading, but weakens equal access and creates audit, exposure, and allocation risk.

Taxonomy 13.1.83. *Hybrid Markets*
Hybrid markets combine order books, dealers, auctions, hidden orders, and off-book negotiation. They mix public order interaction with intermediary support.

Taxonomy 13.1.84. *Specialist Role Segmentation*

i. *dealer*: trades as principal when public liquidity is absent or imbalanced;
ii. *broker*: represents routed orders and searches for contra-side interest;

iii. *auctioneer*: conducts openings, rotations, and imbalance-resolution calls;

iv. *exchange official*: helps maintain orderly markets and disseminated quotes.

The same agent may protect public orders, trade against them, and observe order flow first.

Rule 13.1.85. *Priority and Access Rules*
Priority, tick size, order type, display, halt, and access rules determine how trades are resolved. The same latent supply and demand can produce different prices, quantities, and costs under different rules.

Case Study 13.1.86. *Gold Fixing as Sequential Auction*
The London gold fixing resembles a call process: participants submit net interest, revise after tentative prices, and stop when net demand and supply balance. It produces a clearing price, not continuous immediacy.

Practitioner Note 13.1.87. *Why Large Orders Like Calls*
Calls reduce the need to reveal urgency sequentially. A large order can interact with accumulated opposite interest at one price, often with less signalling than chasing liquidity through the continuous book.

Caveat 13.1.88. *Auction Transparency Trade-Off*
Pre-auction imbalance messages help traders converge to a clearing price, but they can also invite probing, cancellation, and last-second repricing. More transparency improves price discovery only when participants cannot cheaply game the signal.

13.1.5 Market Quality and Efficiency

Definition 13.1.89. *Market Quality*
Market quality measures how well trading interest becomes low-cost execution and informative prices. Include liquidity, tightness, depth, resiliency, transparency, volatility control, and informational efficiency.

Definition 13.1.90. *Liquidity*
Liquidity is the ability to trade quickly, in size, at low cost, and with limited price impact. Spreads, depth, volume, turnover, fill rates, and price impact each capture only one face of it.

Mechanism 13.1.91. *Liquidity as Bilateral Search*
Liquidity is the object of a bilateral search: buyers search for sellers, sellers search for buyers, and both sides decide how much interest to reveal. Markets, brokers, dealers, and routers are useful when they reduce this search cost without leaking too much information.

Proposition 13.1.92. *Search Stopping Rule*
Let B be the expected improvement from one more search attempt, c the direct cost of searching, and L the expected loss from the current best match disappearing while the trader searches. Continue searching iff $B > c+L$.

Proof. The incremental expected gain from another search is the improvement B less the search cost c and lost-option cost L. Searching is optimal exactly when this net gain is positive. □

Taxonomy 13.1.93. *Liquidity Search Strategy Segmentation*

i. *active search*: cross the spread, call dealers, expose interest, or route aggressively;

ii. *passive search*: post limits, show indications, wait for contra-side interest, or reveal only to brokers;

iii. *selective search*: ask trusted counterparties first, then broaden exposure if cost remains acceptable.

Active search buys speed with leakage. Passive search buys discretion with non-execution risk.

Mechanism 13.1.94. *Public Benefits of Liquid Markets*
Liquid markets lower the cost of exchange, hedging, risk sharing, and informed trading. They can benefit even non-traders through cheaper goods, lower capital costs, and better prices.

Taxonomy 13.1.95. *Four Liquidity Variables*

i. *traded volume*: realised liquidity, where traders actually agreed to trade;

ii. *quoted quantity*: offered liquidity at the best limits or across the visible book;

iii. *bid-ask spread*: price of immediacy for small trades;

iv. *volatility*: uncertainty faced by liquidity providers and liquidity takers.

Past volume is not enough: a market can trade actively and still have poor offered liquidity for the next order.

Definition 13.1.96. *Liquidity Risk*
Liquidity risk is uncertainty about the cost of turning a desired position into an executed trade. It includes spread risk, depth risk, market-impact risk, timing risk, and the risk that displayed liquidity disappears.

Taxonomy 13.1.97. *Tightness, Depth, and Resiliency*
Tightness is cost of an immediate small round trip, proxied by bid-ask spread. Depth is the quantity available at or near current prices. Resiliency is the speed with which liquidity returns after a shock or large trade.

Definition 13.1.98. *Price Continuity*
Price continuity is the absence of unnecessary jumps and reversals for ordinary order flow.
It is useful when jumps reflect missing liquidity rather than new information; forcing continuity against informed flow only transfers losses to liquidity suppliers.

Taxonomy 13.1.99. *Search-Output Liquidity Dimensions*

i. *immediacy*: time required to arrange the trade at acceptable terms;
ii. *width*: cost of trading a given size, including spread, fees, and price concession;
iii. *depth*: size available at a given cost or price concession;
iv. *resiliency*: speed with which prices and depth recover after order-flow shocks.

For small trades, width is often the spread. For large trades, depth and information leakage dominate.

Proposition 13.1.100. *Depth-Width Duality*
For a fixed horizon, let $C(q)$ be the nondecreasing all-in cost of trading size q, and let $D(c) = \max\{q : C(q) \leq c\}$ be executable depth at cost budget c. Then $q \leq D(c)$ iff $C(q) \leq c$.

Proof. If $C(q) \leq c$, then q is feasible under the cost budget, so $q \leq D(c)$.
If $q \leq D(c)$, monotonicity gives $C(q) \leq C(D(c)) \leq c$. □

Taxonomy 13.1.101. *Displayed, Undisclosed, and Latent Liquidity*

i. *displayed*: firm quotes and exposed limit orders that can be taken immediately;
ii. *reserve*: hidden size behind a displayed slice or venue-level hidden interest;
iii. *indicated*: IOIs, RFQs, dealer axes, or broker-distributed interest that must be asked for;
iv. *latent*: buy-side, value-trader, or dealer interest that appears only after contact, price movement, or relationship search.

Displayed liquidity is easiest to measure; latent liquidity is often decisive for blocks.

Definition 13.1.102. *Book Size*
Book size is the quantity available at the best bid and ask, or within a specified number of price levels.
For small-tick instruments, several limits must often be summed; for large-tick instruments, the first limits may capture most executable size.

Caveat 13.1.103. *Book Size Is a Lower Bound*
Displayed book size understates tradable quantity when hidden, reserve, midpoint, or fast-replenishing liquidity exists. It also overstates certainty when posted quantities are short-lived.

Mechanism 13.1.104. *Market Impact and Decay*
Market impact is the price movement caused by trading pressure. Temporary impact decays when other traders disagree with the move; permanent impact remains when the order flow reveals information.

Principle 13.1.105. *Resiliency as Synchronisation*
Liquidity is resilient when opposing trading interests arrive before a price move becomes self-reinforcing. Poor synchronisation turns temporary imbalances into persistent price pressure.

Mechanism 13.1.106. *Value-Trader Resiliency*
Value traders make markets more resilient when they buy after uninformed selling pushes price below value, or sell after uninformed buying pushes price above value. Their balance sheet is a delayed liquidity buffer: useful only when valuation confidence, funding, and risk limits survive the shock.

Taxonomy 13.1.107. *Liquidity State Drivers*

i. *trader identity*: uninformed flow is easier to absorb than informed flow;
ii. *instrument attention*: widely followed instruments attract more latent counterparties;

iii. *valuation uncertainty*: unclear fundamentals make suppliers demand larger concessions;

iv. *timing*: closed markets, holidays, and event windows raise search cost;

v. *same-side imbalance*: one-sided displayed interest makes the other side expensive;

vi. *counterparty quality*: reputable traders find liquidity because others expect fairer interaction.

These drivers explain why a market can look liquid in normal prints but become expensive for a specific order.

Principle 13.1.108. *Informational Efficiency*
Informational efficiency means prices incorporate available information quickly and with limited noise. Visibility, update speed, and uninformed flow determine how much information enters prices.

Mechanism 13.1.109. *Bid-Ask Bounce*
Transaction prices alternate between bid and ask even when the efficient price is stable.
A simple bounce statistic compares alternations with continuations: if N^a is the number of price alternations and N^c the number of same-direction continuations, then $\eta = \frac{N^c}{2N^a}$.
Lower η means alternations dominate continuations, so more observed price movement is microstructure noise rather than information.

Caveat 13.1.110. *Transparency Paradox*
More shared information can improve price formation, but each trader prefers to reveal as little as possible before execution. Market design balances public price discovery against private information leakage.

Taxonomy 13.1.111. *Transparency Segmentation*

i. *pre-trade*: orders, quotes, and depth visible before execution;

ii. *post-trade*: prices, sizes, venues, and trade reports after execution;

iii. *top-of-book*: best bid and ask only;

iv. *depth-of-book*: multiple price levels or order-level detail;

v. *timeliness*: real-time, delayed, batched, or end-of-day publication.

Caveat 13.1.112. *Latency and Book State*
When order books update faster than an observer can receive data, decide, and route an order, the displayed book is stale. Available liquidity becomes probabilistic rather than deterministic.

Definition 13.1.113. *Nyquist-Shannon Latency Bound*
For an order-book update frequency f, the latency bound is $\tau^\star = \frac{1}{2f}$. If round-trip observation and routing time exceeds this scale, deterministic use of the displayed book is no longer reliable.

Caveat 13.1.114. *Quality Trade-Offs*
Market quality criteria conflict. Displayed depth, tick size, dark trading, and transparency shift costs across liquidity provision, queue priority, information leakage, and price discovery.

Empirical Finding 13.1.115. *Intraday Liquidity Seasonality*
Volume, volatility, spreads, and book size are time-of-day conditional. Opens tend to have high volume, high volatility, and wide spreads; the middle of the day is closer to steady state; the close attracts benchmark flow, higher volume, and often larger best-limit size.

Principle 13.1.116. *Spread and Depth Substitute Imperfectly*
When the spread cannot narrow further because the tick binds, liquidity providers compete by increasing size at the best bid and ask. A larger book can then play a role similar to a smaller spread, but only for orders small enough to fit inside displayed depth.

Mechanism 13.1.117. *Spread-Volatility Link*
Market makers ask for wider spreads when volatility rises because adverse selection and inventory risk increase. The relation is cleaner after controlling for tick size, trading frequency, and volatility per trade.

Definition 13.1.118. *Flow Reactivity*
Flow reactivity measures how strongly trade counts rise with high-frequency volatility. A resilient venue keeps market makers active when uncertainty rises; a fragile venue loses displayed liquidity when it is most valuable.

Empirical Finding 13.1.119. *Volume Is Not Directly Comparable*
Reported volume depends on architecture. A direct match may count once, while dealer intermediation may count multiple legs, so cross-venue comparisons need the counting convention.

Caveat 13.1.120. *Visible Activity vs Economic Importance*
Visible markets are not always the largest exposures. Stocks attract attention, while real estate, bonds, currencies, swaps, and futures may dominate capital value, risk transfer, or notional activity.

13.1.6 Regulatory and Structural Evolution

Principle 13.1.121. *Regulation as Market Design*
Regulation is market design by law. Good rules support communication, performance, investor protection, competition, and access; bad rules protect incumbents or raise trading costs.

Taxonomy 13.1.122. *Market-Design Question Segmentation*

i. *consolidation*: central book versus fragmented venues;
ii. *intermediation*: order-driven, quote-driven, brokered, or hybrid;
iii. *access*: internalisation, preferencing, linkages, and who may trade;
iv. *transparency*: order-book visibility, trade reporting, and market-data ownership;
v. *controls*: tick sizes, price limits, halts, and listing permissions.

Process 13.1.123. *Welfare Evaluation*

i. identify feasible market structures;
ii. state the objective function and trade-offs;
iii. estimate effects on liquidity, information, volatility, and distribution;
iv. choose the rule with the best expected social welfare.

Taxonomy 13.1.124. *Policy Objective Segmentation*

i. *utilitarian traders*: protect the users whose needs make markets exist;
ii. *public benefits*: promote informative prices and low-cost liquidity;
iii. *profit-motivated traders*: support them only when they supply liquidity or information;
iv. *predatory activity*: discourage manipulation, bluffing, and front-running.

Rule 13.1.125. *Insider-Trading Restrictions*
Rules restrict trading or tipping on material nonpublic information obtained through duty, confidence, or misappropriation. Insiders may still trade under disclosure, prearranged plans, trading windows, and approvals.

Taxonomy 13.1.126. *Insider-Trading Control Segmentation*

i. *disclosure controls*: broad public release and selective-disclosure limits;
ii. *ownership reports*: public filings of insider holdings and changes in ownership;
iii. *prearranged plans*: good-faith trading plans adopted before the trader has material nonpublic information;
iv. *blackout windows*: firm-level restrictions around earnings, deals, and other sensitive events;
v. *short-swing rules*: disgorge short-horizon gains by statutory insiders;
vi. *surveillance and audit trails*: link abnormal trading to information access and communications.

The controls separate legitimate insider liquidity needs from trades that monetise privileged information.

Process 13.1.127. *Insider-Trading Enforcement Logic*

i. detect abnormal price, volume, option activity, or order flow before a material event;
ii. identify the public event and the set of people who knew the information beforehand;
iii. match suspicious traders to access, relationships, communications, and beneficial owners;
iv. test alternative explanations such as public analysis, hedging, or unrelated liquidity demand;
v. pursue disgorgement, penalties, injunctions, or criminal referral when evidence supports intent.

Surveillance starts from market data, but proof usually requires the information path.

Taxonomy 13.1.128. *Insider-Trading Debate Segmentation*

i. *fairness and confidence*: restrictions can make outsiders more willing to supply capital;
ii. *liquidity*: restrictions remove a class of toxic informed flow and can narrow spreads;
iii. *corporate control*: restrictions reduce incentives for managers to hide information or front-run their firms;
iv. *price efficiency*: permissive trading can move prices toward value faster;
v. *enforcement cost*: detecting remote tippees and slow trading is expensive and imperfect;

vi. *incentives*: allowing some insider gains can resemble compensation for entrepreneurial information.

The welfare question is not whether insider trading is informative, but whether the information benefit exceeds liquidity, control, confidence, and enforcement costs.

Proposition 13.1.129. *Insider-Trading Policy Trade-Off*
Let L be the liquidity and confidence gain from restricting insider trading, C the corporate-control gain, E the enforcement cost, and D the cost of delayed price discovery.
A restriction improves market welfare when $L + C > E + D$.

Proof. The policy's net welfare change is the benefit from lower adverse selection and better control, $L + C$, less enforcement and information-delay costs, $E + D$.
The change is positive exactly when $L + C > E + D$. □

Taxonomy 13.1.130. *Exposure-Protection Rule Segmentation*

i. *time precedence*: protects displayed orders from later traders at the same price;
ii. *minimum price increment*: makes stepping ahead costly enough to preserve queue value;
iii. *undisplayed orders*: let traders make firm commitments without full pre-trade exposure;
iv. *audit trail*: deters broker front-running, dual-capacity abuse, and allocation manipulation;
v. *delayed block reporting*: can protect large trades while they are being completed.

These reduce exposure costs, but each also trades off public transparency, price discovery, or queue competition.

Rule 13.1.131. *Decimalization*
Decimalization reduced U.S. equity tick sizes from fractions to cents. Smaller ticks narrowed quoted spreads but reduced displayed depth and queue stability.

Rule 13.1.132. *Reg NMS*
Reg NMS links U.S. equity venues through protected quotes and routing obligations. It protects displayed top-of-book prices, while depth, fees, latency, and hidden liquidity still differ across venues.

Rule 13.1.133. *MiFID and MiFID II*
MiFID broke exchange monopolies by allowing trading outside historical exchanges; MiFID II tightened best-execution, transparency, and reporting duties. The combined effect was more competition, more venue types, and greater reliance on routing, reporting, and post-trade data quality.

Principle 13.1.134. *Best Execution After Fragmentation*
Best execution is not only the best displayed price. In fragmented markets it also depends on fees, likelihood of execution, latency, hidden liquidity, venue access, order size, and the investor's horizon.

Mechanism 13.1.135. *Fragmentation*
Fragmentation means the same instrument trades across multiple venues. It can lower fees and specialise execution, but splits liquidity and complicates best execution.

Proposition 13.1.136. *Order-Flow Externality*
Let venue i give trader utility $U_i = s_i - c_i + \ell(F_i)$, where s_i is service value, c_i is access and trading cost, F_i is order flow, and ℓ is increasing. A technically better entrant with $s_e - c_e > s_m - c_m$ will still fail to attract a trader from incumbent m when $\ell(F_m) - \ell(F_e) > (s_e - c_e) - (s_m - c_m)$.

Proof. The trader chooses the entrant only if $U_e > U_m$.
Rearranging $s_e - c_e + \ell(F_e) > s_m - c_m + \ell(F_m)$ gives the stated threshold. □

Taxonomy 13.1.137. *Levels of Fragmentation*

i. *operators*: new market operators compete with incumbents, then often merge as fixed costs bite;
ii. *venues*: one operator may run lit books, dark books, periodic auctions, and internalisation channels;
iii. *order books*: one venue may maintain several books with different access, transparency, or priority rules;
iv. *orders*: large parent orders are split through time and across compatible venues;
v. *technology*: protocols, market data feeds, and native gateways become part of venue competition.

Mechanism 13.1.138. *Structural Evolution of Fragmentation*
Post-MiFID fragmentation first reduced the monopoly share of historical exchanges, then consolidated around large pan-European operators. Complexity did not disappear: liquidity was segmented into lit books, dark pools, broker crossing systems, systematic internalisers, and specialised order books.

Taxonomy 13.1.139. *Post-MiFID Phases*

i. *entry*: new MTFs competed for order flow, often with bank sponsorship or imported technology;

ii. *convergence*: successful venues moved toward pan-European offers and common functionality;

iii. *dark segmentation*: broker crossing networks, dark MTFs grew around first-hand liquidity, lower signalling;

iv. *consolidation*: fixed technology, data, and connectivity costs pushed operators toward mergers.

Caveat 13.1.140. *Early MiFID Failures*
Turquoise and NASDAQ OMX Europe show that technology and sponsorship were not sufficient. Turquoise gained relevance only after its liquidity agreement ended and it became part of the LSE group; NASDAQ OMX's European MTF and router never reached durable scale.

Caveat 13.1.141. *Liquidity Segmentation*
Venue competition separates natural investor liquidity from opportunistic market-making liquidity. This can improve matching for specific order types, but the trader now pays for aggregation through routing logic, data, connectivity, and information leakage.

Taxonomy 13.1.142. *Market-Segmentation Motives*

i. *size*: small orders want broad exposure; blocks want controlled exposure;

ii. *information*: informed flow wants anonymity; uninformed flow wants identification and price improvement;

iii. *patience*: impatient traders buy immediacy; patient traders prefer orders with time precedence;

iv. *access*: members, floor traders, prime-brokered clients, and retail accounts do not see the same facilities;

v. *credit and trust*: bilateral markets exclude weak counterparties or make them trade through guarantors.

These differences explain why several market structures can survive for the same instrument.

Taxonomy 13.1.143. *Technology Fragmentation*

i. *application protocols*: venue APIs and market-data formats such as ITCH, XDP, EMDI, or RWF;

ii. *network protocols*: multicast UDP, TCP, replay channels, and venue-specific recovery logic;

iii. *compression protocols*: FAST, ZLIB, or uncompressed feeds.

Even when trading rules converge, access remains fragmented through protocols, feeds, and operational detail.

Rule 13.1.144. *Transparency and Dark Trading*
Pre-trade transparency reveals resting interest; post-trade transparency reveals completed transactions. Dark trading reduces information leakage, so regulators often pair it with caps, waivers, or stronger reporting.

Mechanism 13.1.145. *Broker Crossing Networks*
Broker crossing networks internalise client flow before exposing it to lit markets.
They can protect large investors from signalling, but they also raise conflict questions when broker prop flow, market-making flow, and client flow meet in the same pool.

Rule 13.1.146. *MiFID II Dark Caps*
MiFID II constrained dark trading through double volume caps: a venue-level cap and an aggregate market-level cap. The intent is to preserve price formation on lit venues while still allowing limited dark execution.

Taxonomy 13.1.147. *MiFID Venue Statuses*
European transactions may occur on regulated markets, multilateral trading facilities, systematic internalisers, or OTC channels. Regulated markets and MTFs use multilateral rule books; systematic internalisers trade bilaterally against client flow; OTC trading keeps negotiation bilateral but still faces reporting obligations.

Rule 13.1.148. *Tick Size Reforms*
Tick size rules balance price competition against queue discipline. Large ticks widen spreads but support displayed queues; small ticks narrow spreads but encourage queue jumping, cancellation, smaller displayed size.

Caveat 13.1.149. *Tick Size as Market Share Tool*
In fragmented markets, venues can use smaller ticks to undercut incumbent quotes and attract flow. This may improve displayed prices in the short run, but can also start a race to economically meaningless price increments.

Rule 13.1.150. *Sub-Penny and Tick-Size Wars*
Sub-penny rules restrict displayed price improvement, but off-exchange venues can still internalise flow at tiny increments. In Europe, early post-MiFID venues used different tick tables to compete until harmonisation reduced tick-size competition.

Taxonomy 13.1.151. *Who Gains From Tick Changes*

i. *liquidity takers*: may gain from narrower spreads on small trades;

ii. *liquidity providers*: lose queue rents when stepping ahead becomes cheap;

iii. *large investors*: may face less displayed depth and higher search or slicing costs;

iv. *venues*: can gain share by offering finer price grids or specialised order types.

Principle 13.1.152. *Tick Policy Design*
Tick tables should vary with price and liquidity, stay stable enough for adaptation, and be revised using spreads, depth, order-to-trade ratios, and standard-size execution costs.

Caveat 13.1.153. *Cross-Market Precedence Failure*
Time precedence protects a displayed order only inside the venue that enforces it.
In fragmented markets, quote matchers can bypass an older queue by trading elsewhere unless tick, routing, and trade-through rules restore economic precedence across venues.

Rule 13.1.154. *Circuit Breakers and Structural Controls*
Halts, limit moves, short-sale constraints, margin rules, and position limits do not remove volatility. They change when and how traders express demand, supply, and leverage.

Taxonomy 13.1.155. *Circuit Breaker Scope*

i. *local*: halt one instrument or one venue;

ii. *related-instrument*: halt derivatives, ETFs, or baskets tied to the halted instrument;

iii. *market-wide*: halt a broad market after index-level stress;

iv. *dynamic bands*: reject or pause trades outside a price corridor;

v. *static thresholds*: trigger from a fixed reference price or previous close.

Taxonomy 13.1.156. *Extreme-Volatility Control Segmentation*

i. *trading halts*: stop trading until information, order imbalance, or system stress is resolved;

ii. *price limits*: permit trading only inside a daily or intraday price band;

iii. *transaction taxes*: raise the cost of trading, especially high-turnover strategies;

iv. *margin and position limits*: cap leverage and forced-loss exposure;

v. *collars*: restrict specific order types or program trades under defined market moves.

The same rule can damp transitory volatility while delaying fundamental repricing.

Principle 13.1.157. *Circuit Breaker Design*
Circuit breakers should stop disorderly liquidity demand, not ordinary price discovery. Useful triggers combine price movement with liquidity, volatility, correlation, or related-instrument stress.

Proposition 13.1.158. *Halt Value Depends on Volatility Source*
A halt improves price quality only if the expected reduction in transitory noise and operational stress exceeds the cost of delayed fundamental price discovery and pent-up order flow.

Proof. Let benefit B be the avoided transitory trading error and system overload. Let cost C be stale prices, delayed hedging, and accumulated imbalance.
The halt improves expected price quality exactly when $B > C$. □

Caveat 13.1.159. *Gravitational Effect*
If traders expect a halt or limit to prevent later execution, they may submit orders earlier. The rule can therefore pull liquidity demand forward and increase volatility near the trigger.

Caveat 13.1.160. *Propagation Problem*
Halting one instrument can push pressure into substitutes. If ETFs, futures, options, or highly correlated names remain open, the halt may move price discovery elsewhere instead of calming it.

Mechanism 13.1.161. *Price Limits and Margin Waterfall*
Price limits can slow default losses by forcing losing traders to post margin over several sessions instead of one jump. They also delay liquidation, so unresolved losses and pent-up orders may reappear when trading resumes.

Taxonomy 13.1.162. *Trading-Restriction Incidence*

i. *informed traders*: restrictions slow incorporation of value-relevant information;

ii. *dealers*: higher costs reduce supplied immediacy and widen spreads;

iii. *order anticipators*: restrictions may reduce front-running and forced-flow extraction;

iv. *bluffers*: higher manipulation cost can reduce transitory volatility;

v. *utilitarian traders*: hedgers, investors, and borrowers may lose useful risk-transfer access;

vi. *gamblers*: discouraging stale-information trading can reduce noise.

Good policy discriminates by harmful behaviour, not by trading frequency alone.

Rule 13.1.163. *Post-Flash-Crash Controls*
After the 2010 Flash Crash, U.S. market structure moved toward single-stock pauses, limit-up/limit-down bands, consolidated audit trail reporting, stub-quote restrictions, and tighter controls on naked market access.

Caveat 13.1.164. *Stub Quotes*
Stub quotes satisfy a formal two-sided quoting duty at prices far from the market. They are not real liquidity; executions against them reveal missing protection in the trading mechanism.

Principle 13.1.165. *Consolidated Tape as Resilience*
Helps fragmented markets retain a shared post-trade reference. Without it, stress or outages can leave participants unsure whether price discovery is happening in one market or several disconnected ones.

Taxonomy 13.1.166. *Regulatory Institutions*
Market oversight is shared across:

i. governmental regulators, such as securities and futures commissions;

ii. self-regulatory organisations, such as exchanges, clearinghouses, and trader associations;

iii. private standard setters, such as accounting, performance-reporting, and professional bodies;

iv. international organisations, such as IOSCO and the World Federation of Exchanges.

This division creates expertise, jurisdictional overlap, and regulatory competition.

Taxonomy 13.1.167. *Regulatory Political-Economy Risks*

i. *regulatory risk*: officials adopt visible rules after crises because inaction is blamed more than costly action;

ii. *regulatory capture*: rules favour incumbents over diffuse market users;

iii. *measurement risk*: policy is adopted before transitory and fundamental volatility can be cleanly separated.

Post-crash rules should therefore be judged by mechanism, not only by public reassurance.

Caveat 13.1.168. *Unintended Consequences*
Rules change incentives. Taxes may move liquidity abroad, bans may move risk into substitutes, protected quotes may increase routing complexity, and exchange rules may protect incumbents.

Taxonomy 13.1.169. *Regulatory-Service Segmentation*

i. *private services*: connectivity, routing, order management, clearing access, account tools paid by users;

ii. *public services*: surveillance, audit trails, price continuity, disclosure, and manipulation controls.

Competition supplies private services readily.
Public services usually need exchange rules, regulation, or industry coordination.

Mechanism 13.1.170. *Competition to Merger*
Opening trading to competition first produces innovative venues and lower explicit costs. Once practices converge, scale becomes decisive: mergers reduce fixed technology costs, expand connectivity, and preserve distinct order books under larger operating groups.

Practitioner Note 13.1.171. *Electronification Beyond Equities*
Electronification does not imply one architecture. Equities move to lit books and dark pools; fixed income and derivatives retain RFQ, dealer networks, and central clearing because contracts and counterparty risks differ.

13.2 Information, Price Formation, and Spreads

13.2.1 Price Discovery

Mechanism 13.2.1. *Fragmentation and Price Formation*
Fragmentation can improve prices when venues remain synchronised by routing, arbitrage, and consolidated information. It can damage price formation when venues lose a common reference, when primary quotes become stale, or when market participants pause routing during stress.

Mechanism 13.2.2. *Fast Liquidity and Price Discovery*
Fast market makers help price discovery when they update quotes and arbitrage stale venues. They hurt it when speed investment mainly earns queue priority or fee rebates without adding durable liquidity. Fee schedules therefore affect the price formation process, not only venue revenue.

Caveat 13.2.3. *Outage Price Discovery*
If alternative venues keep trading while primary data or primary matching is unavailable, prices may temporarily split into parallel formation processes rather than one consolidated market price.

13.2.2 Volatility and Liquidity Interaction

Mechanism 13.2.4. *Liquidity Spiral*
A large order can consume depth, widen spreads, and raise volatility. Higher volatility makes liquidity providers reduce size or cancel quotes, forcing the next order to trade through less liquidity and amplifying the move. The state variable is confidence in replenishment: once traders doubt the book will refill, everyone demands more price concession at the same time.

Case Study 13.2.5. *Flash Crash Mechanism*
The 2010 Flash Crash illustrates a liquidity spiral: a large automated futures sell program met thin liquidity, high-frequency traders recycled inventory quickly, aggressive flow migrated into cash equities, spreads widened, and some routers or venues paused.
The trigger was an order; the crash mechanism was fragile liquidity under linked-market stress.

Model 13.2.6. *Depth-Driven Flash Crash Toy Model*
On one side of the book, let F_t be the market-order arrival rate, $\bar{F}$ the normal refill rate, $r_t = F_t - \bar{F}$ the overflow, and $I_t = 1/D_t$ inverse depth. A compact depth-impact dynamic is $\frac{dI_t}{I_t} = \sigma_I dW_t + r_t dt$.
Here D_t is market depth, I_t is impact, σ_I is liquidity-shock volatility, and W_t captures random liquidity shocks. Positive overflow r_t makes inverse depth compound upward, so each additional trade has larger price impact than the last.

Mechanism 13.2.7. *Hot-Potato Amplification*
If a fraction ρ of traders follows volume mechanically, initial trading rate r_0 can amplify roughly as $r_t \simeq \frac{1-\rho^t}{1-\rho} r_0$. When ρ is high, overflow consumes depth faster than passive liquidity can refill it. The loop is mechanical: trading creates volume, volume targets force more trading, and depth absorbs the loop until it fails.

Taxonomy 13.2.8. *Flash-Crash Control Segmentation*

i. *rate controls*: cap participation rates or stop volume-following feedback;
ii. *depth controls*: require or incentivise fast refill near the touch;
iii. *price controls*: pause trading after large price moves;
iv. *cross-market controls*: coordinate futures, ETFs, cash equities, and options;
v. *data controls*: detect overflow, quote withdrawal, and broken references in real time.

Caveat 13.2.9. *Systemic Risk Link*
Fragmentation is not systemic risk by itself. It becomes dangerous when market participants react to stress in the same direction, when related venues desynchronise, or when liquidity withdrawal in one instrument propagates to hedges, ETFs, futures, and correlated stocks.

Taxonomy 13.2.10. *Shock Propagation Channels*

i. *cross-asset hedging*: futures, ETFs, options, and cash baskets transmit inventory pressure;
ii. *routing feedback*: routers chase fading liquidity or pause when fills become unreliable;
iii. *quote withdrawal*: market makers widen or cancel when volatility exceeds their risk budget;
iv. *reference-price dependency*: pegged and benchmark orders inherit stale or broken references;

v. *correlated constraints*: margin, risk limits, and kill switches cause simultaneous de-risking.

Empirical Finding 13.2.11. *Trade Count as HFT Proxy*
Rising trades per day can proxy for higher high-frequency participation when trade sizes shrink and message intensity rises. It is not sufficient alone: the same count can reflect genuine liquidity, excessive slicing, or stress-induced churn.

13.3 Order Flow and Empirical Microstructure

13.3.1 Feature Engineering for Microstructure Alpha

Taxonomy 13.3.1. *Relative-Value Feature Segmentation*

i. *basis*: raw spread, carry-adjusted spread, z-score, percentile, and distance to arbitrage bounds;
ii. *hedge quality*: rolling beta, cointegration residual, factor residual, and hedge-ratio instability;
iii. *convergence clock*: time to delivery, expiry, fixing, rebalance, event date, or funding reset;
iv. *carry*: funding rate, borrow fee, dividend, storage, roll yield, and collateral return;
v. *execution synchrony*: leg liquidity, queue priority, fill correlation, and latency between venues;
vi. *stress capacity*: margin usage, drawdown to liquidation, short recall risk, and borrow availability.

These features separate cheapness from tradability.

Model 13.3.2. *Factor-Residual Arbitrage View*
For instruments i and j, write $r_i = \beta_i^\top f + \epsilon_i$ and $r_j = \beta_j^\top f + \epsilon_j$. A relative-value signal should explain whether the spread move came from common factors f, hedge-ratio error $\beta_i - \beta_j$, or residual shocks $\epsilon_i - \epsilon_j$.

Principle 13.3.3. *Alpha Needs Comparative Advantage*
A feature is useful only if it captures information or execution capacity that competitors do not fully exploit. A high in-sample score without an economic reason is usually a selection result, not an edge.

Taxonomy 13.3.4. *Microstructure Alpha Evaluation Segmentation*

i. *forecast*: relation to future midprice, spread, fill, or volatility;
ii. *tradability*: spread, queue loss, borrow, fees, and venue access;
iii. *capacity*: decay of performance with size and participation;
iv. *stability*: sensitivity to regime, tick size, latency, and participant mix;
v. *crowding*: whether many agents can observe and monetize the same state.

The feature is only the first filter; the executable strategy is the object being evaluated.

Principle 13.3.5. *Arbitrage Labels Need Path Information*
A spread that later converges is not automatically a good label.
The trade may still be infeasible if the interim drawdown breaches margin, borrow, inventory, or risk limits.
Label both terminal convergence and maximum adverse excursion.

Taxonomy 13.3.6. *Arbitrage Event Label Segmentation*

i. *entry*: spread crosses estimated arbitrage bound after costs;
ii. *confirmation*: flow suggests uninformed pressure or slow common-factor adjustment;
iii. *failure*: specific news, borrow loss, deal break, or hedge-ratio regime change;
iv. *exit*: convergence, stop-out, forced liquidation, expiry, or capital reallocation.

13.3.2 Microstructure Data Engineering

Taxonomy 13.3.7. *Market-Data Depth Segmentation*

i. *L1*: best bid, best ask, sizes at the touch, and last trade;
ii. *L2*: visible depth by price level, usually aggregated by side and venue;
iii. *L3*: order-level messages, allowing queue reconstruction and cancellation analysis;
iv. *trades*: prints with price, size, time, venue, and often trade condition codes.

The empirical question should choose the shallowest feed that identifies the event cleanly.

Taxonomy 13.3.8. *Trading-Message Segmentation*

i. *order entry*: new, cancel, replace, reject, and execution-report messages;
ii. *market data*: public trades, quotes, status, and depth updates;
iii. *drop copy*: private execution and routing records for audit and reconciliation;

iv. *clearing feed*: allocations, positions, margin, fees, and settlement status.

FIX standardises much of order-entry layer. Venues still differ in native gateways, tags, throttles, recovery rules.

Caveat 13.3.9. *Ticker Is Not an Instrument Key*
Ticker symbols are display mnemonics. Robust microstructure data keys need venue, currency, instrument type, expiry, strike, option right, corporate-action history, and identifier validity dates.

Definition 13.3.10. *Microstructure Proportion*
A microstructure proportion is the share of events satisfying a binary condition: trade at the bid, execution on a venue, hidden-liquidity fill, HFT-tagged message, or cancellation before queue execution.

Model 13.3.11. *Bernoulli Proportion Estimator*
For independent indicators $X_1, \ldots, X_n \in \{0, 1\}$ with success probability p,
$\hat{p} = \frac{1}{n} \sum_{\ell=1}^{n} X_\ell$, $\mathbb{E}[\hat{p}] = p$, $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$. The usual large-sample interval is $\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.
Here X_ℓ is the indicator for event ℓ, n is the number of observations, p is the true success probability, $\hat{p}$ is the sample proportion, α is the error probability, and $z_{1-\alpha/2}$ is the corresponding standard-normal quantile.
For a 95% interval, $z_{1-\alpha/2} \approx 1.96$.

Caveat 13.3.12. *Normal Approximation Boundary*
The normal interval is fragile when the sample is small or the estimated proportion is close to zero or one. A practical check is $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$; otherwise use an exact or adjusted binomial interval.

Taxonomy 13.3.13. *Sampling Unit Segmentation*

i. *trade count*: estimates the probability that the next print has a property;
ii. *traded value*: estimates where economic turnover is concentrated;
iii. *message count*: estimates quoting, cancellation, or latency behaviour;
iv. *time bucket*: estimates realised activity inside a comparable clock segment;
v. *security bucket*: estimates whether the result is driven by large caps, small caps, sectors, or venues.

Changing the sampling unit changes the estimand, even when the same symbol $\hat{p}$ is used.

Model 13.3.14. *Simple Linear Regression Baseline*
For observations (x_i, y_i), a simple linear regression writes $y_i = a_0 + a_1 x_i + \epsilon_i$, $1 \le i \le n$.
Here y_i is the dependent variable, x_i is the explanatory variable, a_0 is the intercept, a_1 is the slope, ϵ_i is the residual, and n is the sample size.
Ordinary least squares estimates $\hat{a}_1 = \left(\sum_i (x_i - \bar{x})(y_i - \bar{y})\right) / \left(\sum_i (x_i - \bar{x})^2\right)$, $\hat{a}_0 = \bar{y} - \hat{a}_1 \bar{x}$. Here $\bar{x}$ and $\bar{y}$ are sample means, and $\hat{a}_0, \hat{a}_1$ are the estimated intercept and slope.

Definition 13.3.15. *Regression Fit Quality*
For fitted values $\hat{y}_i$ and residuals $\hat{\epsilon}_i = y_i - \hat{y}_i$, $R^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2}{\sum_i (y_i - \bar{y})^2}$.
Here $\hat{y}_i$ is the fitted value, $\hat{\epsilon}_i$ is the residual, and R^2 is explained variability relative to total variability. It measures fit, not causality or trading usefulness.

Model 13.3.16. *Mean Excess-Return Test*
For excess observations $x_1, \ldots, x_n$, sample mean $\bar{x}$, and sample standard deviation s_x, the statistic is $t = \frac{\bar{x}}{s_x / \sqrt{n}}$.
It asks whether realised average excess performance is large relative to the noise in the same sample. In trading data, the hard part is defining x_i so it is comparable, independent enough, and not selected after the fact.

Proposition 13.3.17. *Power Is a Signal-to-Noise Problem*
For independent observations with true mean μ and standard deviation σ, the expected t-statistic is approximately $\mu\sqrt{n}/\sigma$. Detecting a small edge therefore requires n of order $(\sigma/\mu)^2$.

Proof. The standard error of the sample mean is $\sigma/\sqrt{n}$.
Dividing the true mean by this standard error gives the signal-to-noise ratio $\mu\sqrt{n}/\sigma$. □

Caveat 13.3.18. *More Ticks Are Not More History*
Sampling more frequently can estimate intraday mechanics more accurately, but does not create new independent market regimes. Serial correlation, order splitting, common shocks, and stale prices reduce effective sample size.

Caveat 13.3.19. *Regression Is an Empirical Approximation*
Linearity is often a local summary for spread, volatility, turnover, and impact relations. Check residuals, non-stationarity, heteroskedasticity, and whether a log or event-time transformation gives a more stable relation.

Taxonomy 13.3.20. *Performance-Inference Bias Segmentation*

i. *survivorship*: failed strategies, funds, venues, or symbols disappear from the dataset;
ii. *selection*: the sample is chosen because it already looked good;
iii. *multiple testing*: the best result among many trials is treated as a single unbiased test;
iv. *look-ahead*: timestamps, corporate actions, or benchmark constituents use future information;
v. *fillability*: simulated fills assume liquidity that was not actually reachable.

Most backtest errors are sampling errors before they are modelling errors.

Caveat 13.3.21. *Regression to the Mean*
Selecting the best recent performer raises the starting estimate. If the original outperformance mixed skill and luck, subsequent performance usually falls toward the population mean even when the strategy has not changed.

Caveat 13.3.22. *Distributional Shape Matters*
Fat tails, asymmetric payoff profiles, return smoothing, rare devaluations, and fraud-like reporting can make normal-based tests overconfident. Inspect drawdowns, skew, jump exposure, stale marks, and the economic source of returns before trusting a high t-statistic.

Caveat 13.3.23. *Averaging Effect*
An average intraday profile is not a typical realised path. Averaging volatility, volume, or spread across days can hide event days, close auctions, and isolated liquidity shocks.

Model 13.3.24. *Average of Regressions vs Regression of Averages*
For an index $I_t = \sum_{k=1}^{N} w_k S_t^k$, spreads aggregate linearly as $\psi^I = \sum_{k=1}^{N} w_k \psi^k$, but volatility generally does not satisfy $\sigma^I = \sum_{k=1}^{N} w_k \sigma^k$. Here w_k is the weight of stock k, S_t^k its price, ψ^k its spread, and σ^k its volatility. Regression coefficients estimated on index quantities therefore need not equal weighted averages of single-stock regression coefficients.

13.4 Market Making and Liquidity Provision

13.4.1 Economics of Market Making

Definition 13.4.1. *Dealer*
A dealer trades as principal: buying from clients that want to sell and selling to clients that want to buy. The service sold is immediacy; the compensation is spread, relationship value, or order-flow value.

Taxonomy 13.4.2. *Dealer Problem Segmentation*

i. *quote*: set bid, ask, and sizes;
ii. *order flow*: attract clients without buying toxic flow;
iii. *inventory*: keep positions near target levels;
iv. *information*: infer value from trades, quotes, news, and client behaviour.

Taxonomy 13.4.3. *Designated Market Maker Obligation Segmentation*

i. *affirmative*: quote meaningful two-sided markets and trade when public liquidity is absent;
ii. *negative*: yield to public orders and avoid trading ahead of customer liquidity;
iii. *auction*: organise openings or calls so orders receive orderly representation;
iv. *capital*: maintain enough balance sheet to supply liquidity in stressed conditions.

Obligations buy price continuity; privileges pay for the obligation.

Taxonomy 13.4.4. *Specialist Privilege Segmentation*

i. *order-flow information*: see routed orders, floor interest, or book imbalance earlier;
ii. *quote control*: set the disseminated quote within exposure rules;
iii. *participation option*: step in when public orders are thin or unattractive;
iv. *brokerage access*: earn fees or relationship value from represented flow.

Each privilege must be evaluated against the liquidity supplied when no one else wants to trade.

Taxonomy 13.4.5. *Dealer Quote Segmentation*

i. *two-sided*: quote both bid and ask; *one-sided*: quote only the desired side;
ii. *firm*: executable up to a stated size; *soft*: indication that may be revised;
iii. *displayed*: visible to the market; *request-only*: revealed after an RFQ or negotiation;
iv. *size*: maximum quantity available at the quoted price.

Taxonomy 13.4.6. *Dealer Spread vs Outside Spread*

i. *valuation target*: dealers quote around current market value; value traders trade on fundamental value;
ii. *holding period*: dealers try to turn inventory quickly; value traders wait for convergence;
iii. *risk source*: dealers bear inventory, client-selection risk.
Value traders bear valuation, financing, winner's-curse risk;
iv. *scale*: dealers can earn small spreads repeatedly; value traders need fewer, larger opportunities;
v. *profit capture*: dealers may earn a full round-trip spread.
Value trader often earns only the move from entry back toward value.

This is why dealer spreads is tight even when fundamental investors require a wide outside spread before trading.

Taxonomy 13.4.7. *Order-Flow Acquisition Segmentation*

i. *price*: quote aggressively enough to win trades;
ii. *size*: show capacity for larger trades;
iii. *service*: provide research, execution support, credit, and settlement reliability;
iv. *relationship*: cultivate clients and brokers;
v. *inducement*: pay or reciprocate for routed order flow.

Caveat 13.4.8. *Preferencing and Payment for Order Flow*
Preferencing can lower search costs when dealers compete for relationships, but it can also route orders toward economic arrangements rather than best all-in execution.

Practitioner Note 13.4.9. *Trading with Dealers*
Experienced clients often ask for a two-sided quote before revealing direction. A wide quote is information: the dealer may suspect informed flow, inventory pressure, or poor unwind conditions.

13.4.2 Optimal Quoting and Inventory Control

Model 13.4.10. *Specialist Participation Option*
Let V_{step} be the expected value of stepping ahead or trading with the book, $C_{\text{obligation}}$ the expected loss from required liquidity provision, and K the compliance and capital cost. A specialist assignment is privately attractive when $V_{\text{step}} > C_{\text{obligation}} + K$. Regulators should ask the public version: whether the price-continuity benefit exceeds the rents transferred through the privilege.

Caveat 13.4.11. *Tick Size and Participation*
Small ticks reduce the cost of stepping ahead of public orders, making specialist or DMM participation easier and public queue value weaker. Larger ticks protect queue rents but can widen quoted spreads.

13.4.3 Latency and Queue Competition

Principle 13.4.12. *Latency as Queue Option*
Speed buys queue priority. It is most valuable when queues are thin, stale, or information-sensitive.

Model 13.4.13. *HFMM and Investor Race Model*
Split participants into high-frequency market makers and longer-horizon investors.
A market maker earns spread surplus $s\Delta$ net of maker fee c_M; an investor earns value surplus Γ net of spread and taker fee c_T: $\pi_M = s\Delta - c_M$, $\pi_T = \Gamma - s\Delta - c_T$, $\bar{c} = c_M + c_T$.
Here s is spread in ticks, Δ is tick size, Γ is investor gross value gain, c_M, c_T are maker/taker fees, π_M, π_T are net per-trade gains, and $\bar{c}$ is venue revenue per trade. The model asks who should be subsidised to invest in speed, not whether speed is socially free.

Mechanism 13.4.14. *Race Intensity*
Let μ_m be market maker m's arrival intensity and τ_n investor n's arrival intensity, with averages $\bar{\mu}$ and $\bar{\tau}$.
If technology costs are quadratic, $\Pi_M(m) = \frac{\mu_m}{\bar{\mu}}\pi_M\mathcal{R} - \frac{1}{2}\beta\mu_m^2$, and $\Pi_T(n) = \frac{\tau_n}{\bar{\tau}}\pi_T\mathcal{R} - \frac{1}{2}\gamma\tau_n^2$.
Here $\mathcal{R}$ is trade rate, Π_M, Π_T are total expected gains, and β, γ are speed-cost coefficients. The ratios $\mu_m/\bar{\mu}$ and $\tau_n/\bar{\tau}$ act like race shares; lower cost raises optimal speed because winning one more race becomes cheaper.

Taxonomy 13.4.15. *Latency-Cost Segmentation*

i. *market-maker cost*: co-location, feed handling, quoting engines, and hedge automation;
ii. *investor cost*: broker connectivity, routing latency, algo responsiveness, and monitoring;
iii. *venue fee split*: maker fee or rebate versus taker fee;
iv. *surplus split*: spread capture for makers versus value capture for investors;
v. *competition level*: number of fast makers relative to investor flow.

Definition 13.4.16. *Efficient Maker Fee*
In the stylised fee-choice model, the efficient maker fee satisfies $c_M^\star = s\Delta - \frac{\Gamma - \bar{c}}{1+\left(\frac{M}{N}\frac{\gamma}{\beta}\right)^{1/3}}$, $\bar{c} = c_T^\star + c_M^\star$.
Here M, N are the numbers of market makers and investors.
Negative $c_M^\star$ is a maker rebate; negative $c_T^\star$ is a taker rebate.

Caveat 13.4.17. *Fee Rebates Are Comparative*
Maker rebates fit scarce or costly market-making; taker rebates fit crowded makers and costly investor access.

Practitioner Note 13.4.18. *Queue Competition Diagnostic*
Do not classify a fee schedule as good or bad in isolation. Segment by spread size, maker count, investor urgency, queue stability, latency cost, and whether rebates improve displayed liquidity or merely pay for fleeting quotes.

13.4.4 Cross-Asset and Cross-Venue Market Making

Principle 13.4.19. *Arbitrageurs as Cross-Sectional Dealers*
Dealers use inventory to connect buyers and sellers across time. Arbitrageurs use hedge portfolios to connect buyers and sellers across instruments, venues, locations, maturities, and risk forms. Both sell immediacy, but arbitrageurs often sell a transformed exposure rather than the same instrument.

Taxonomy 13.4.20. *Cross-Asset Arbitrage Segmentation*

i. *cash-futures*: buy cash and sell futures, or reverse, after carry and delivery costs;
ii. *ETF-basket*: create or redeem ETF shares against the underlying basket;
iii. *ADR-local*: trade depositary receipts against local shares after FX, fees, and settlement frictions;

iv. *FX triangular*: compare direct exchange rates with synthetic cross rates;

v. *option conversion*: manufacture stock, cash, or volatility exposure from options and underlying hedges;

vi. *merger spread*: trade target and acquirer legs against deal terms and completion probability.

Each case answers: what synthetic instrument does hedge portfolio manufacture, and what costs bound basis?

Model 13.4.21. *Triangular FX No-Arbitrage*
For currencies A, B, C, no triangular arbitrage requires $S_{A/C} = S_{A/B}S_{B/C}$ after bid-ask spreads, funding, and settlement costs. A tradable deviation must exceed the cost of executing all three legs before quotes move.

Mechanism 13.4.22. *Derivative Conversion*
Derivatives let arbitrageurs repackage risk. A market maker who buys an option and delta-hedges with the underlying is converting option demand into stock trading, funding, and volatility exposure. The hedge is dynamic because delta, gamma, vega, and funding costs change with market state.

Caveat 13.4.23. *Legging and Synchronisation Risk*
Multi-leg arbitrage is most fragile while the hedge is incomplete. Market orders reduce legging risk but pay spread and impact; limit orders reduce cost but may leave one leg filled and the other missing. Execution logic therefore matters as much as fair-value logic.

Process 13.4.24. *Balanced Arbitrage Execution*

i. estimate fair basis, costs, and hedge ratios;

ii. choose leg order from liquidity, volatility, and unwind optionality;

iii. submit synchronised orders or hedge the most liquid leg last;

iv. monitor partial fills, basis movement, and residual exposure;

v. cancel, complete, or rebalance before residual risk dominates the spread.

Caveat 13.4.25. *Arbitrage Capital Constraint*
The widest spreads often appear when balance sheet is scarce.
If arbitrageurs cannot finance losses, meet margin, borrow securities, or carry inventory, forced liquidation can widen the basis precisely when the forward-looking trade is most attractive.

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